



UNDERSTANDING **MODERN** WARFARE

DAVID JORDAN, JAMES D. KIRAS,
DAVID J. LONSDALE, IAN SPELLER,
CHRISTOPHER TUCK and C. DALE WALTON

SECOND EDITION

UNDERSTANDING MODERN WARFARE

Understanding Modern Warfare has established itself as the leading introduction to the issues, ideas, concepts and context necessary to understand the theory and conduct of warfare in the twentieth and twenty-first centuries. It is an invaluable text for military professionals and students of military history. Key features include: incisive coverage of the debates surrounding contemporary and future warfare; accessible, yet sophisticated, discussion across the land, sea and air environments; and coverage of contemporary topics such as drones, cyber warfare and hybrid warfare. The book makes extensive use of text boxes to explain key concepts and to reference extended examples, annotated guides to further reading and key questions to promote the reader's further thinking. This second edition has been fully revised and updated to take into account new debates and recent events in Syria, Iraq and Ukraine, and also restructured to further improve its usefulness as a teaching tool.

David Jordan is a Senior Lecturer in the Defence Studies Department, King's College London, based at the Joint Services Command and Staff College, Shrivenham. His previous publications include *Battle of the Bulge* (2003) and *The Fall of Hitler's Reich: Germany's Defeat in Europe, 1943–45* (2004).

James D. Kiras is an Associate Professor of Strategic Studies at the School of Advanced Air and Space Studies, Air University, Maxwell Air Force Base, Alabama. He is also an Associate Fellow of the Joint Special Operations University, Tampa, Florida. His first book was titled *Special Operations and Strategy: From World War II to the War on Terrorism* (2006).

David J. Lonsdale is Director of the Centre for Security Studies, University of Hull. His publications include *The Nature of War in the Information Age: Clausewitzian Future* (2003), *Alexander the Great: Lessons in Strategy* (2007) and *Understanding Contemporary Strategy* (2012).

Ian Speller is Senior Lecturer in Military History in the Department of History and is the Director of the Centre for Military History and Strategic Studies at Maynooth University, National University of Ireland. He also lectures at the Irish Military College and has lectured on maritime strategy and naval warfare at the UK Defence Academy and at NATO headquarters in London and Naples. He is the author of *Understanding Naval Warfare* (2014) and *The Role of Amphibious Warfare in British Defence Policy 1945–56* (2001), and is co-author/editor of *Small Navies: Strategy and Policy for Small Navies in War and Peace* (2014), *Amphibious Warfare: Strategy and Tactics from Gallipoli to Iraq* (2014) and *The Royal Navy and Maritime Power in the Twentieth Century* (2005).

Christopher Tuck is Senior Lecturer in the Department of Defence Studies, King's College London, based at the Joint Services Command and Staff College, Shrivenham. He is the author of *Understanding Land Warfare* (2014), and *Confrontation, Strategy, and War Termination* (2013), co-author of *Amphibious Warfare: Strategy and Tactics from Gallipoli to Iraq* (2014) and co-editor of *British Propaganda and Wars of Empire: Influencing Friend and Foe* (2014).

C. Dale Walton is an Associate Professor of International Relations at Lindenwood University in St Charles, Missouri. He has published three books to date, including *Grand Strategy and the Presidency: Foreign Policy, War and the American Role in the World* (2012), *Geopolitics and the Great Powers in the Twenty-first Century* (2007) and *The Myth of Inevitable U.S. Defeat in Vietnam* (2002).



UNDERSTANDING MODERN WARFARE

SECOND EDITION

DAVID JORDAN, JAMES D. KIRAS
DAVID J. LONSDALE, IAN SPELLER
CHRISTOPHER TUCK AND
C. DALE WALTON



CAMBRIDGE
UNIVERSITY PRESS

CAMBRIDGE
UNIVERSITY PRESS

University Printing House, Cambridge CB2 8BS, United Kingdom

Cambridge University Press is part of the University of Cambridge.

It furthers the University's mission by disseminating knowledge in the pursuit of education, learning and research at the highest international levels of excellence.

www.cambridge.org

Information on this title: www.cambridge.org/9781107134195

© David Jordan, James D. Kiras, David J. Lonsdale, Ian Speller, Christopher Tuck, C. Dale Walton 2016

This publication is in copyright. Subject to statutory exception and to the provisions of relevant collective licensing agreements, no reproduction of any part may take place without the written permission of Cambridge University Press.

First published 2008

Second edition 2016

Printed in the United Kingdom by TJ International Ltd. Padstow Cornwall

A catalogue record for this publication is available from the British Library

ISBN 978-1-107-13419-5 Hardback

ISBN 978-1-107-59275-9 Paperback

Additional resources for this publication at www.cambridge.org/jordan2

Cambridge University Press has no responsibility for the persistence or accuracy of URLs for external or third-party internet websites referred to in this publication, and does not guarantee that any content on such websites is, or will remain, accurate or appropriate.



CONTENTS

List of illustrations	page ix
List of boxes and tables	xii
Preface to the second edition	xv
<i>Ian Speller</i>	

Introduction to the second edition	1
<i>Ian Speller</i>	

Part I Strategy

David J. Lonsdale

1 The study and theory of strategy	21
Introduction	21
The study of strategy	22
Strategic theory	27
Conclusion	35
2 Strategy defined	39
Introduction	39
Strategy defined	40
The levels of strategy	42
Why is strategy difficult?	45
Conclusion	56
3 The practice of strategy	61
Introduction	61
The use of force	62
Strategy in the contemporary environment	66
Conclusion	75

Part II Land warfare

Christopher Tuck

4 Concepts of land warfare	83
Introduction	84
The land environment	85
The attributes of land forces	86
Key developments in land warfare	88
The principles of land warfare	92
Conclusion	97

5	Modern land warfare	101
	Introduction	101
	The First World War and the emergence of modern tactics	102
	The Second World War and after: the operational art	111
	Conclusion	123
6	Future land warfare	128
	Introduction	128
	The Revolution in Military Affairs	129
	New wars	132
	Hybrid warfare	134
	Conclusion	151

Part III Naval warfare

Ian Speller

7	Concepts of naval warfare	159
	The maritime operating environment	160
	Attributes of naval forces	162
	Maritime strategy	164
	Alternative approaches	173
	Conclusion	176
8	The evolution of naval warfare	180
	From fighting sail to steam, steel and shellfire	181
	Into a new century	185
	The test of war, 1914–18	186
	From arms control to World War, 1918–45	189
	Navies in the nuclear age	194
	Conclusion	199
9	Naval warfare in the twenty-first century	202
	Naval warfare: changing roles?	203
	Modern navies and post-modern navies	207
	Naval warfare: changing forms?	209
	Plus ça change plus c'est la même chose	212
	Contemporary naval policy	214
	Conclusion	218

Part IV Air and space warfare

David Jordan

10	Concepts and characteristics of air and space warfare	227
	Introduction	227
	The air environment	230
	Concepts	237

Information dominance	243
Joint operations	243
Command and control	244
11 The evolution of air and space power	250
The evolution of control of the air	251
Contemporary aspects of control of the air	254
Strategic air power	255
Theory, practice and controversy	257
Coercive air power	259
A loss of joint effort?	264
Joint air power in the nuclear age	266
Space power	267
The development of space power	268
Conclusions	269
12 Air and space power in the contemporary era: 1990–2030	273
An air power revolution?	275
Theory into practice	275
Air power after 9/11	282
The expansion of space power	286
Air and space power – ongoing challenges and concluding thoughts	291
Conclusions	292

Part V Irregular warfare

James D. Kiras

13 Key concepts and terms of irregular warfare	301
Introduction	301
Definitions	305
14 The historical practice of irregular warfare	319
Understanding modern irregular warfare through the lens of the past	319
15 Current irregular warfare	344
Insurgent strategic after Mao	346
Developments in counterinsurgency	347
Why is modern irregular warfare so difficult?	357
SOF and the golden age of counterterrorism?	365
Conclusion	368

Part VI Weapons of mass destruction

C. Dale Walton

16 Weapons of mass destruction: radiological, biological and chemical weapons	379
Introduction: defining weapons of mass destruction (WMDs)	379

	Radiological, biological and chemical weapons: a brief primer	382
	Why states acquire and retain chemical and biological weapons today	399
	Conclusion: CBWs in the present and future	401
17	Weapons of mass destruction: nuclear weapons	405
	Introduction: the 'nuclear club'	405
	Nuclear weapons	406
	The nightmare of nuclear terrorism	412
	Nuclear deterrence	413
	The First and Second Nuclear Ages: past and future possibilities for conflict	417
	Conclusion: nuclear weapons in the present and future	428
	Part VII Conclusion	
	Christopher Tuck	
18	The future of warfare	437
	The future of war	438
	Warfare and technology	442
	The operational level of war and operational art	447
	Joint warfare	449
	What is victory?	452
	Conclusion	456
	Glossary	461
	Index	472



ILLUSTRATIONS

(i)	US Admiral Jonathan Greenert addresses an international audience at the Malaysian Armed Forces Staff College in February 2014. (Courtesy of US Navy; photo by Chief Mass Communication Specialist Peter D. Lawlor)	page 5
(ii)	A Lockheed Martin F-35A Lightning II. (Courtesy of US Air Force; photo by Alex R. Lloyd)	7
(iii)	A Hospital Corpsman assigned to a Female Engagement Team patrols with the US Marine Corps in Helmand province (Afghanistan) in 2010. (Courtesy of US Marine Corps; photo by Corporal Lindsay L. Sayres)	9
1.1	Hiroshima – the first use of nuclear weapons. (Courtesy of US Air Force)	24
2.1	President Barack Obama and Vice President Joe Biden meet with members of the National Security Council, 2014. (White House photo by Pete Souza; courtesy of US Department of Defense)	41
3.1	An MQ-9 Reaper unmanned aerial attack vehicle descends into an airfield in Afghanistan. (Courtesy of US Air Force; photo by Staff Sergeant Brian Fergusson)	71
4.1	US Marines refuel an M1A1 Abrams tank. (Courtesy of US Marine Corps; photo by Staff Sergeant Brian A. Lautenslager)	91
5.1	US soldiers fire their 120mm mortars, Afghanistan, 2014. (Courtesy of US Army; photo by Captain John Landry)	103
5.2	US Army Private, First Class, Robert Parker (2nd from left) provides fire support for his squad members, Iraq, 2010. (Courtesy of US Department of Defense; photo by Petty Officer 2nd Class Ted Green)	110
6.1	The Stryker armoured vehicle on exercise. (US Army photo by Gertrud Zach/Released 3 January 2012, Picture 120103-A-HE359-023)	139
6.2	British troops land in Normandy, 6 June 1944. (Courtesy of the UK Ministry of Defence under Open Government Licence)	144
6.3	Norwegian tanks participate in a NATO exercise, 2014. (Courtesy of US Air Force; photo by Technical Sergeant Burt Traynor)	150
7.1	Combined Task Force 521 conducts convoy escort operations with a large natural gas tanker during an international mine countermeasures exercise in 2013. (Courtesy of US Navy; photo by Mass Communication Specialist 2nd Class Bryan Blair)	162
7.2	The Republic of Korea diesel-electric submarine <i>Chang Bogo</i> (SSK 61). (Courtesy of US Navy; photo by Photographer's Mate 1st Class David A. Levy)	171

7.3	American troops disembarking from a ship onto small boats near Cavite, Philippines in 1898 or 1899. (Courtesy of Naval History and Heritage Command, US Navy)	173
8.1	Officers from the USS <i>Monitor</i> relax on deck after the engagement with the CSS <i>Virginia</i> at the battle of Hampton Roads, 1862. (Courtesy of Naval History and Heritage Command, US Navy)	183
8.2	Ships of the US Third Fleet and the British Pacific Fleet off the coast of Japan in August 1945. (Naval Historical and Heritage Command, 80-G-339360)	193
8.3	The aircraft carrier USS <i>George H. W. Bush</i> and the amphibious assault ship USS <i>Makin Island</i> , 2014. (Courtesy of US Navy; photo by Lt Juan D. Guerra)	197
9.1	The Singapore navy frigate RSS <i>Formidable</i> and the Indian Navy frigate INS <i>Brahmaputra</i> conducting exercises in the Bay of Bengal in 2007. (Courtesy of US Navy)	208
9.2	The US Navy's experimental unmanned X-47B lands aboard the aircraft carrier USS <i>Theodore Roosevelt</i> in August 2014. (Courtesy of US Navy)	211
9.3	The guided-missile destroyer USS <i>Arleigh Burke</i> launches Tomahawk cruise missiles to conduct strikes against Islamic State targets in September 2014. (Courtesy of US Navy; photo by Mass Communication Specialist 2nd class Carlos M. Vazquez II)	217
10.1	F-22 Raptor. (Courtesy of US Air Force)	238
10.2	A US Air Force B-1B Bomber. (Courtesy of US Air Force)	240
11.1	B-29 Superfortress. (Courtesy of US Air Force)	253
11.2	F-105D Thunderchief. (Courtesy of US Air Force)	260
11.3	A Delta II Rocket launch. (Courtesy US Air Force/US Space Command; photo by Joe Davila)	270
12.1	An X-47 unmanned combat air vehicle and F/A-18F Super Hornet. (Courtesy of US Navy; photo by Mass Communications Specialist Alex Millar)	274
12.2	Tornado GR4 strike aircraft of 12 Squadron. (David Jordan)	277
12.3	F-15 Eagle launching an ASM-135 anti-satellite missile (ASAT). (Courtesy of US Air Force)	290
13.1	Philippine Scout Ranger Carmen Binasing Ambrocia, Jr, a member of the Philippine Citizen Armed Forces Geographical Unit (CAFGU) at Harvest City, Basilan. (Courtesy of the US Department of Defense; VIRIN: 020324-N-MJ265-001)	303
14.1	Marines fight rebellious Boxers outside the Peking Legation. (Courtesy of the National Archives, Washington, DC; NWDNS-127-N-515039)	327
14.2	Insurgent dead near Santa Ana (Courtesy of the National Archives, Washington, DC; NWDNS-111-AG-3(22))	331

-
- | | | |
|------|--|-----|
| 15.1 | A French Foreign Legionnaire patrols along the dry rib of a rice paddy in Indochina. (Courtesy of the National Archives, Washington, DC; NWDNS-306-PS-54(10014)) | 349 |
| 15.2 | Native Moros, Taluk Samgay, Zamboanga Province, Mindanao. Governor Captain Finley in the centre. (Courtesy of the National Archives, Washington, DC; NWDNS-111-RB-1633) | 352 |
| 15.3 | ISAF Afghanistan 'spaghetti chart' attempting to capture the complexity of the operating environment. (Courtesy of the US Department of Defense) | 355 |
| 15.4 | An Iraqi waits while interpreters and a US Army soldier process him into a laptop in Multaka, Iraq, 13 Nov. 2007. (Courtesy of US Army; photo by Specialist Laura M. Buchta) | 356 |
| 15.5 | Four Vietnamese and three Americans were killed, and dozens of Vietnamese buildings were heavily damaged during a Viet Cong bomb attack against a multi-storey US officers billet in Saigon, 04/01/1966. (Courtesy of the National Archives, Washington, DC; NWDNS-306-MVP-5(3)) | 361 |
| 15.6 | Navy Special Warfare Combatants move a watercraft using an Army MH-47G Chinook helicopter during training in Moses Lake, Washington, 21 May 2014. (Courtesy of US Army; photo by Sergeant Christopher Prows) | 366 |
| 16.1 | US Navy sailors during an exercise in which yellow smoke is used to simulate exposure to chemical, biological or radiological weapons. (Courtesy of US Navy; photo by Photographer's Mate 3rd Class Todd Frantom) | 381 |
| 16.2 | Italian firefighters participating in a WMDs exercise. (Courtesy of US Navy; photo by Journalist 3rd Class Stephen P. Weaver) | 388 |
| 16.3 | The guided-missile destroyer USS <i>Boxer</i> performs a test of the ship's countermeasure water wash-down system. (Courtesy of US Navy; photo by Photographer's Mate Airman Rudy Polach) | 398 |
| 17.1 | The YB-52, a prototype version of the B-52 bomber, lands on a dry lake at Edwards Air Base, 1953. (Courtesy of US Air Force) | 410 |
| 17.2 | A Standard Missile Three (SM-3) is launched from the US Navy Aegis cruiser USS <i>Lake Erie</i> . (Courtesy of US Navy) | 411 |



BOXES AND TABLES

Boxes

(i) Defining war	<i>page 3</i>
(ii) Interstate wars, intra-state wars and non-state wars	8
(iii) The comprehensive/integrated approach	10
(iv) The principles of war	12
(v) The levels of war	14
1.1 Strategic theory	35
2.1 Strategy defined	40
2.2 The levels of strategy	42
2.3 Why is strategy difficult?	45
2.4 Gray's dimensions of strategy	47
3.1 The many uses of force	66
3.2 Just war	74
4.1 The US principles of war	93
5.1 The reality of blitzkrieg	116
5.2 The operational level of war	122
5.3 Manoeuvre warfare	123
6.1 The army as a system	131
6.2 Globalisation	132
6.3 Fourth-generation warfare	133
6.4 Hybrid warfare	135
6.5 Future missions of the US military	140
7.1 Attributes/characteristics of maritime power and naval forces	163
7.2 Alfred Thayer Mahan, 1840–1914	165
7.3 Sir Julian Corbett, 1854–1922	166
7.4 Wolfgang Wegener, 1875–1956	168
8.1 'Crossing the T'	184
8.2 The battle of Jutland, 31 May 1916	187
8.3 Convoys	188
8.4 The battle of the Coral Sea and battle of Midway, 1942	192
8.5 The 1980s 'Tanker War'	195
9.1 Sea basing	204
9.2 Contemporary piracy	207
9.3 Anti-access/area-denial (A2/AD)	210
10.1 Air power defined	229
10.2 Air power and security	241
10.3 Thoughts on air power and joint operations	244

11.1	John Boyd and John A. Warden III	262
12.1	A precision revolution?	278
12.2	Stealth aircraft	279
13.1	Definitions of irregular warfare	309
14.1	What is doctrine?	322
15.1	SOF: force of choice	367
16.1	The Ebola virus and bioterrorism	390
16.2	Cheating on arms control: the Soviet Union and the Biological Weapons Convention (BWC)	392
16.3	'Super-BWs'?	393
17.1	Strategic culture and nuclear weapons	407
17.2	The first-strike quandary and the Cuban missile crisis	415
17.3	The 'Libya quandary'	424
17.4	Different leaders, different goals	427
18.1	The future operating environment	440
18.2	The sources of future peace	441
18.3	The key elements of globally integrated operations	451
18.4	Stabilisation defined	454
18.5	Strategic success	455

Tables

(i)	'The Spectrum of Conflict and Operational Themes'	10
5.1	1914 infantry company deployed in 'column of platoons'	105
5.2	Late First World War German defence in depth	107
5.3	A modern infantry squad	112
5.4	Deployment of a modern infantry squad	113
5.5	Deep operations	115
6.1	The RMA system-of-systems argument	130
6.2	Full spectrum operations	138
7.1	Sea control and sea denial	171
9.1	The span of naval tasks	205
11.1	Boyd's OODA loop	262
11.2	Warden's 'five ring' model	263
13.1	Types of insurgencies	307
13.2	Attributes of types of irregular warfare	310



PREFACE TO THE SECOND EDITION

This is a book about warfare. It focuses on the conduct of war in the twentieth and twenty-first centuries. It is designed to be read by anyone with an interest in the subject and the reader requires no specialist prior knowledge. Indeed, the first edition of this book had its origins in an unsuccessful search by the authors to find a source that could provide an authoritative yet accessible introduction to the theory and practice of modern warfare. There are numerous good books devoted to an examination of aspects of modern warfare and even more that address its history over the centuries. Despite this we found none that offered the reader, in a single volume of manageable proportions, a thorough grounding in the critical issues, ideas, concepts and vocabulary necessary to develop and articulate an understanding of the conduct of war in its various forms and in its different operating environments. *Understanding Modern Warfare* was intended to fill this gap and was written accordingly.

The first book was well received but, inevitably, some aspects of the analysis have now been overtaken by events and a work that was up to date in 2008 now requires some revision. Thus, in this second edition we have updated the original text in order to accommodate developments in military thought and practice and to reflect the march of events. As has always been the case in the study of warfare, some things have changed a little, some have changed a lot and others have remained broadly the same and this is reflected within the text. In addition, and based on feedback on the first edition, we have restructured the book in order to make it more accessible. Whereas previously we had six large chapters we now have six Parts, which correspond to the previous chapters, and each Part now has two or three shorter chapters devoted to specific aspects of these subjects. A seventh and final Part constitutes the book's Conclusion. All this should make the book easier to use while also providing a useful vehicle for the overall analysis. If this encourages readers to engage with the subject then it will have served its purpose. That this remains important is sadly all too evident given the continued propensity for peoples to engage in, or prepare for, warfare of one form or another.

The contributors to this book have all benefited from experience teaching students at civilian universities and military academies. We owe these students, uniformed or otherwise, a debt of gratitude for the stimulus and challenge that they have provided over the years. Exposure to their different ideas and innumerable questions has forced each of us individually to explain and defend our views and opinions many times over. This book is therefore partly a result of such debates and, hopefully, will provoke and inform many more in the future. An additional debt is owed to our colleagues in the various departments and

institutions in which we work. They are too numerous to mention individually but, without their assistance and, sometimes, their forbearance, works such as this one would not be possible. Particular mention must be made of Dr David Murphy and Dr Deborah Sanders for their comments and assistance. Michael Watson and the editorial staff of Cambridge University Press provided the perfect balance between enthusiasm and encouragement on one hand and firm guidance on the other. We would also like to thank the anonymous reviewers of the manuscript who provided a number of keen insights that improved the quality of the work. Finally, it should be noted that the analysis, opinions and conclusions expressed in this book are those of the authors and do not necessarily represent the views of the US Department of Defense, the United States Air Force, the UK Ministry of Defence, the UK Defence Academy or any other organisation. Any mistakes are, of course, our own.



Introduction to the second edition

Ian Speller

Contents

What is war?

What is modern warfare?

How to use this book

This book focuses on modern warfare. It examines the conduct of war in its different environments and forms and provides an introduction to the issues, ideas, concepts, context and vocabulary necessary to develop an understanding of the subject. It is not a history book, although relevant historical examples are used throughout to illustrate the analysis. Rather, the book is designed to equip the reader with a sophisticated introduction to the concepts, issues and debates that will help them to understand current concerns and future possibilities and also to unpick past campaigns.

This subject is an important one. War and warfare have had a pervasive and often a pernicious influence on human affairs throughout history. Optimistic claims that we are evolving towards a less violent international system do not appear to be entirely borne out by recent events. Despite this, the conduct of war as an academic field of enquiry is not a subject that everyone is comfortable with. It requires one to study a phenomenon that many disapprove of and to think about things that some prefer to ignore. That such study is often encouraged or supported by armed forces eager to derive ‘lessons’ intended to improve future performance has done little to endear it to liberal academics. Insofar as they study war at all most Western universities prefer to focus on ‘war and society’, examining the impact that war has had on wider society rather than focusing explicitly on warfare. The result has been a demilitarisation of the topic within much of academia, what Michael Howard called a ‘flight to the suburbs’.¹ This is not a suburban book. It self-consciously focuses on the central activity of armed forces and on the urban centre of the subject, on warfare. It does so in recognition that this does not address the totality of war, which is about more than just warfare, but is based on the notion that one cannot understand modern war unless one also understands modern warfare.

The book is also based on the judgement that such understanding is important. To paraphrase Sun Tzu, the conduct of war is of such importance, quite literally the province of life and death, it is vital that it be studied carefully. It

should never be forgotten that wars always result in death, destruction, waste and human suffering, all too frequently on a truly staggering scale. Unfortunately, ignoring the phenomenon is unlikely to make it go away and to do so fosters ignorance of something that has had, and continues to have, a major impact on human affairs. Whether one wishes to avoid warfare, to mitigate its impact or to prepare to conduct it more efficiently (and these are not mutually exclusive positions) it is important that it be studied. Indeed, one might suggest that in a democracy in the twenty-first century it is particularly important that as wide a range of people as possible should understand the nature of modern warfare in order that they are equipped to make intelligent judgements about the way in which their own governments seek to employ military force. The requirement for military personnel to understand warfare should be too obvious to require further elaboration given the historical correlation between ignorance and military incompetence.

What is war?

According to the *Oxford English Dictionary*, war is a 'hostile contention by means of armed forces, carried on between nations, states, or rulers, or between parties in the same nation or state'.² More succinctly, Hedley Bull, the renowned scholar of international relations, described war as 'organised violence carried on by political units against each other'.³ Other definitions distinguish between armed conflict and war. These tend to focus on the scale and degree of violence employed, suggesting that 'wars' are characterised by large numbers of combatants, heavy casualties and/or high-intensity fighting. The Correlates of War project, for example, defines war as involving sustained combat by organised armed forces resulting in a minimum of 1,000 battle-related deaths between combatants in any given year. The Uppsala Conflict Database Project, on the other hand, focuses on 'armed conflict' defined as a 'contested incompatibility' involving at least twenty-five battle-related deaths within a year.⁴ The collection and use of quantifiable data clearly poses some methodological challenges and the shortcomings of either definition are readily apparent. Is it really the case that a terrorist campaign that kills twenty-five people within twelve months is indicative of an armed conflict but one that kills twenty-four is not? Does a conflict really cease to be a war simply because the casualty figures slip below 1,000 in any given year? Such definitions tell us little about the nature or the character of war; they are best recognised for what they are, simply a means of gathering and handling data.

Other approaches focus on legal issues, identifying 'war' as a state of law that regulates armed conflict between groups, usually states. These reflect the conventional understanding of war, as articulated in international law and treaties, as organised rule-bounded violence conducted between the uniformed representatives of states. They tend to exclude the activities of sub-state groups whose use of violence is not given the credibility or legitimacy associated with the

Box (i) Defining war

Carl von Clausewitz (1832): 'War is not merely an act of policy but a true political instrument, a continuation of political intercourse, carried on with other means.'¹

L. F. L. Oppenheim (1906): 'War is a contention between two or more states through their armed forces, for the purpose of overpowering each other and imposing such conditions of peace as the victor pleases.'²

Mao Zedong (1938): 'Politics is war without bloodshed while war is politics with bloodshed.'³

Quincy Wright (1942): '[War is] a legal condition which equally permits two or more hostile groups to carry on a conflict by armed force.'⁴

Hedley Bull (1977): '[War is] organised violence carried on by political units against each other'.⁵

Colin Gray (2005): 'War is organised violence threatened or waged for political purposes.'⁶

US Joint Doctrine (2013): 'War is socially sanctioned violence to achieve a political purpose.'⁷

¹ Carl von Clausewitz, *On War*, ed. and trans. Michael Howard and Peter Paret (Princeton University Press, 1976 [1832]), p. 99.

² Quoted in Lawrence Freedman, 'Defining War', in Julian Lindley-French and Yves Boyer (eds.), *The Oxford Handbook of War* (Oxford University Press, 2012), p. 19.

³ *Selected Military Writings of Mao Tse-Tung* (Beijing: Foreign Languages Press, 1967), pp. 226–7.

⁴ Quincy Wright, *A Study of War*, vol. II (University of Chicago Press, 1942), p. 685.

⁵ Hedley Bull, *The Anarchical Society* (Columbia University Press, 1977), p. 178.

⁶ Colin S. Gray, *Another Bloody Century: Future War* (London: Weidenfeld & Nicolson, 2005), p. 30.

⁷ Joint Publication 1, *Doctrine for the Armed Forces of the United States* (2013), pp. 1–2.

term. Given that, thus far into the twenty-first century, most armed conflict has occurred either between sub-state groups or between such groups and conventional militaries, such definitions, while logically coherent, may be too restrictive. For the purposes of this book Bull's more expansive approach seems more satisfactory (see [Box \(i\)](#)).

Contemporary Western approaches to war are profoundly influenced by the work of Carl von Clausewitz whose posthumous *magnum opus*, *On War*, published in 1832, stated that war is 'an act of force to compel our enemy to do our will . . . a clash of major interests, which is resolved by bloodshed'. Famously, Clausewitz emphasised the political nature of war, stressing that 'war is not merely an act of policy but a true political instrument, a continuation of political intercourse, carried on with other means', a statement that is as widely known as it is frequently misquoted. Far from advocating war as an alternative to politics he intended to show that war was driven by politics and could only be understood in that context. The means of waging war can never be considered in isolation from the aim that they are designed to support. As Clausewitz put it, '[w]ar may have its own grammar, but not its own logic. The logic is determined by the political aim.'⁵ Logic and grammar may collide, often with unfortunate consequences.

Not everyone views war in this way, a point that was emphasised by Anatol Rapoport in his introduction to an abridged edition of Clausewitz's work published by Penguin. Rapoport identified that not all people or groups viewed war as political or approached it in the rational and instrumental manner described in *On War*.⁶ More recently, authors such as Mary Kaldor have argued that Clausewitz's approach is less useful within the context of so-called 'new wars', where there may be no formal government, no clear political objectives and where many of the combatants may fight because they enjoy it or because it gives them access to the things that they value; they may have no interest in military victory in the traditional sense.⁷ On the other hand, one might argue that even self-serving warlords have aims, even if these are associated with the continuation of the current conflict. For them violence is indeed a tool of policy, even if that policy bears little relation to those traditionally adopted by states. It may well be the case that, as Kaldor and others have argued, some of those engaged in war are motivated by cultural imperatives or even perhaps by a non-rational urge to fight, but this does not mean that the conflict in which they participate is not directed towards some end. These ends may involve the achievement of wealth, security or prestige; they may be profoundly influenced by culture, identity, religion and/or ideology; and may be poorly defined or even barely understood, but they do exist. It seems fair to suggest that war remains political, in the sense that it is pursued in support of some or other policy, even if that policy involves the pursuit of goals that appear non-political in the narrowest sense of that word.

As will already be apparent, the study of war clearly requires an investigation that is not limited to the battlefield. Even works, such as this one, which focus explicitly on the conduct of military operations, must recognise that these occur within a broader context in which other factors intrude. War is a duel and this imposes a certain discipline (the enemy gets a say in how things are done); sub-optimal approaches chosen for cultural reasons usually incur some cost. However, the purposes for which wars are fought and the manner in which they are conducted necessarily reflects the character of those who wage them. For this reason the character of each war is different and to understand this character one must engage with a number of disciplines. A sophisticated understanding of war requires one to engage with (at least) political, social, cultural, economic and technological matters and this is reflected in the curricula of many military colleges (see [Figure \(i\)](#)). Any attempt to examine and explain war in its broadest sense is, therefore, a huge undertaking. As Azar Gat explained, '[w]ith war being connected to everything else and everything else being connected to war, explaining and tracing its development in relation to human development in general almost amounts to a theory and a history of everything'.⁸ This book does not attempt to provide this. It is informed by an understanding of this complexity, and of the interdisciplinary nature of war studies, but the specific focus of this book is on the conduct of warfare.



Figure (i) US Admiral Jonathan Greenert addresses an international audience at the Malaysian Armed Forces Staff College in February 2014. Colleges such as this reflect an understanding that armed forces must invest in the education of their personnel if they are to make appropriate decisions in an increasingly complex environment.

What is modern warfare?

US joint doctrine defines warfare as ‘the mechanism, method, or modality of armed conflict against an enemy. It is “the how” of waging war’. More succinctly, British doctrine defines it as ‘the conduct of war.’⁹ The study of warfare, therefore, is a subset within the study of war. Warfare revolves around the use, and the threat to use, violence. Its study implies a particular focus on combat, the basic hard currency of war and the *raison d’être* for armed forces. This does not mean that political, social, cultural, economic and technological factors are not relevant; nothing could be further from the truth. Such factors set the context within which warfare is conducted and all play a part in determining how different societies or organisations approach the conduct of war. For this reason it is important to remember that the character of war and of warfare continues to change, as does the character of those societies that wage war and, by extension, there will be more than one characteristic form of war (and of warfare) in existence at any given time.

This rather begs the question as to what we actually mean by ‘modern’ warfare. Michael Sheehan has defined it as ‘the forms of warfare shaped by and

reflecting the modern era of human history.’¹⁰ Sadly this shifts but does not solve the definitional problem. What is the ‘modern era of human history’? Returning to the dictionary suggests that ‘modern’ means ‘of or relating to the present and recent times’ or ‘being in existence at this time’.¹¹ This may not help us much as there is little consensus on either point. Different authors have interpreted ‘present and recent times’ in different ways. The starting point of Theodore Ropp’s 1959 examination of *War in the Modern World*, for example, was 1415.¹² Some authors begin with the age of Marlborough or Frederick the Great in the eighteenth century, the French Revolution of 1789 or the defeat of Napoleon in 1815. Others focus on the period from the outbreak of the American Civil War in 1861, from the start of the twentieth century, the end of the First World War and so on.

There is no more agreement on what ‘being in existence at this time’ might mean. Some might suggest that this implies a focus on the latest technology, most recent doctrine and evolving ‘best practice’ amongst the most advanced militaries (see [Figure \(ii\)](#)). Others might argue that some of the most effective practitioners of modern warfare have been sub-state groups able to exploit low-technology weapons and asymmetric tactics in support of their aims. Still more might claim that the idea of modern warfare is now rather passé and that we have entered a new era of ‘post-modern warfare’, ‘fourth-generation warfare’, ‘hybrid warfare’, of ‘war amongst the people’ or some other apparently new form of war. For example, it has become fashionable in recent years to conclude that future conflict will be predominantly asymmetric, that it will occur between irregular combatants or between these and regular armed forces and that interstate conventional warfare is a thing of the past. Such analysis may describe a particular moment in time when intra-state conflict and asymmetric warfare was the dominant concern. It is dangerous to assume that this will last forever.

The use of and the threat to use armed force is a recurrent feature of contemporary international relations. The Heidelberg Institute for International Conflict Research identified that there were twenty-one wars and twenty-five limited wars in 2014 in addition to 177 violent crises. The majority of these were intra-state conflicts, with just one limited war and eleven violent crises identified as being between states. None of the latter escalated into war but a number did involve sporadic clashes and an apparent readiness to employ lethal force.¹³ While interstate wars may have declined recently in terms of size, intensity and import, this may not represent as radical a change as is sometimes claimed (see [Box \(ii\)](#)). Further, it would seem premature to assume that there will never again be cause for one armed force to face another in a conventional conflict. As US hegemony is replaced by multi-polarity, great powers may once again find cause to fight each other. New technology or new techniques may give lesser powers the ability to defend themselves without having to resort, in the first instance, to asymmetric tactics that see their institutions toppled and their cities occupied. It may be the case that the deterrent effect of nuclear weapons may wane with



Figure (ii) As an advanced multi-role fifth-generation fighter the Lockheed Martin F-35A Lightning II represents the cutting edge of modern military technology. Weapons such as these reflect a Western emphasis on high-technology systems. Others engaged in 'modern warfare' may exploit much cheaper low-tech solutions out of choice or necessity.

further proliferation or with the development of more reliable ways of shooting down incoming missiles. All of these issues (and more) are discussed in this book. The only thing that can be said with certainty is that the future is uncertain.

It is important also to recognise the impact created by the potential to prevail in conventional combat. Military force does not have to be used to be useful. When a Vietnamese coastguard vessel gives way to its Chinese counterpart (without firing a shot) in a dispute in the South China Seas it may do so because it recognises that to open fire is to invite a conflict that it can only lose. The superior Chinese navy may not be actively engaged but is clearly relevant to the outcome. Similarly, Russian tanks may not have fought battles against their Ukrainian counterparts in spring 2014, but their perceived ability to do so helped to set the context within which Ukraine was inhibited from dealing with separatists in Crimea, leading to the secession of that region. Conventional military force was far from the only factor at play, but it is spurious to suggest

Box (ii) Interstate wars, intra-state wars and non-state wars

Across the centuries intra-state wars (i.e. wars within states) have occurred more frequently than interstate wars (those conducted between different states). The Correlates of War project identified that there were a total of ninety-five interstate wars between 1816 and 2007, of which thirty-eight occurred after the Second World War. Of the latter, nine came in the period after the end of the Cold War (i.e. since 1990). Over the same period (1816–2007) there were over 330 intra-state wars in addition to 62 non-state wars (where neither side was a state) and over 160 extra-state wars (where a state fought a non-state entity beyond its own borders). There were also many minor armed conflicts that did not meet the project's threshold for consideration. As a corrective to those who believe that interstate conflict now occurs much less frequently than before, it may be worth noting that in the first twenty-five years of the study (1816–41) there were only two interstate wars, twenty-four non-state wars and over thirty intra-state ones. There were more interstate wars in the last twenty-five years of the study than in the first twenty-five. On the other hand it would be fair to say that recent interstate conflicts have lacked the scale and the impact of many of those before 1945. There has also been a reduction in the number of battle-related casualties (as wars have tended to be shorter and less intense). This has not necessarily led to a reduction in overall civilian casualties, and these almost always surpass those of combatants. For example, there were around 145,000 battle-related deaths during the war in the Democratic Republic of Congo from 1998 to 2001 but the war overall may have resulted in as many as 2.5 million deaths.¹

¹ Correlates of War Project (www.correlatesofwar.org/). Bethany Lacina and Nils Peter Gleditsch, 'Monitoring Trends in Global Combat: A New Dataset of Battle Deaths', *European Journal of Population*, 21 (2005), 145–66.

that it was irrelevant. The 2008 Russo-Georgian War, in which Russian joint forces engaged and defeated their Georgian counterparts (who were attempting to suppress pro-Russian separatists in South Ossetia and Abkhazia), provides an illustration of the costs of thinking otherwise. Similarly, the United States and its allies may have spent a decade fighting insurgencies in Afghanistan and Iraq, typical of 'war amongst the people', but one should not forget why their enemies adopted asymmetric tactics; they had no other choice. Had the Taliban or Saddam Hussein been able to prevail in conventional warfare the United States and its allies would not have faced insurgencies in Iraq and Afghanistan; they could not have been there in the first place. Conventional superiority (across all domains) provided the context within which the enemies of the USA were forced to try something else.

In light of events in recent conflicts (notably the war in Afghanistan and the 2006 Israel–Hezbollah conflict) it is often now argued that modern warfare is increasingly 'hybrid' in nature, with adversaries employing a variety of techniques and procedures, both conventional and unconventional, within the same battlespace and tailored to exploit enemy weaknesses (see [Figure \(iii\)](#)). The



Figure (iii) A Hospital Corpsman assigned to a Female Engagement Team patrols with the US Marine Corps in Helmand province (Afghanistan) in 2010. Experience in Afghanistan demonstrated the difficulty of operating in a complex environment against an enemy adept at exploiting the challenges of ‘war amongst the people’ and where civil and political factors had a major impact on military possibilities.

concept of ‘hybrid warfare’ has gained significant traction and is often repeated within academic, professional and official literature. One might reasonably ask whether there has ever been a war that was not hybrid to some degree but, banal though it may be, the concept does at least focus attention on the complexity of current military operations and may help to foster an understanding that, in order to achieve a satisfactory outcome in these, one requires a comprehensive/integrated approach that brings together both military and civilian agencies (see [Box \(iii\)](#)). It also supports the idea that, rather than concentrating on only one apparently dominant form of warfare, armed forces may be required to operate across the spectrum of warfare (see [Table \(i\)](#)).

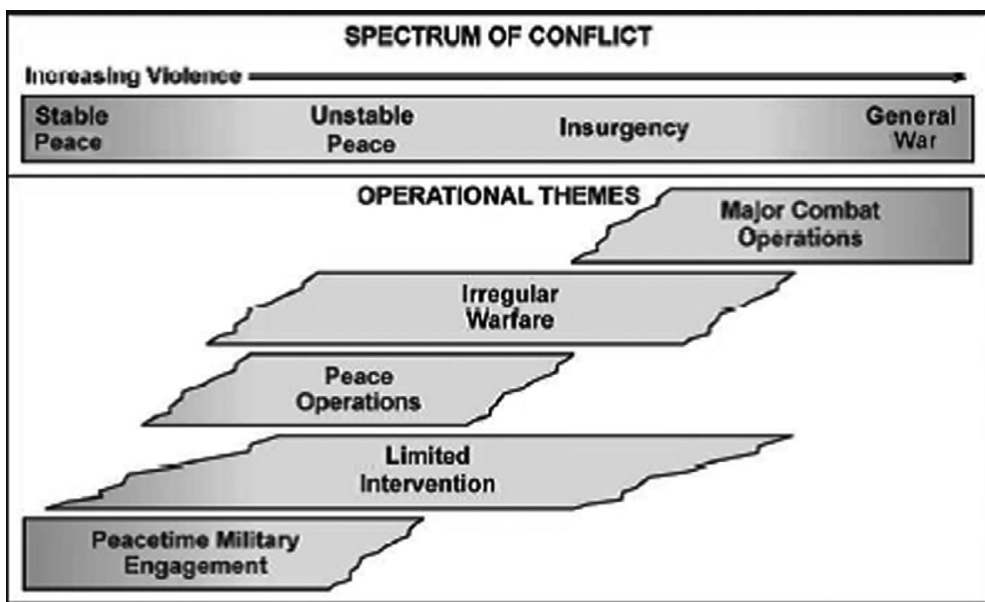
These issues are addressed throughout this book. The focus is on the kind of warfare that is ‘in existence at this time’ and our aim is to examine activities across the spectrum of warfare involving both conventional and unconventional forces. In order to examine these in a satisfactory fashion this requires also an examination of military concepts, organisations and activities in ‘recent times’. We have tried to avoid being overly prescriptive about what this might actually mean. The degree to which one needs to look back in order to understand

Box (iii) The comprehensive/integrated approach

In the 2000s, and in light of experience in Afghanistan, the Balkans (in the 1990s) and also in Iraq, NATO developed the idea that success in crisis-management operations depended on the adoption of a comprehensive military, political and civilian approach. Military means, while essential, could not themselves provide solutions to such crises. Recently the UK armed forces have talked of the need for an 'integrated approach' within the context of crisis management and stabilisation, emphasising the need for many different agencies to work together, with the military just one element within a complex whole. While it may always have been the case that military, political and civilian agencies have had to co-operate to achieve useful results in such operations (e.g. see French operations in North Africa in the nineteenth century), the explicit articulation of ideas such as these makes formal and obvious that which was previously often only dimly understood.¹

¹ For example, see the web page of the UK Ministry of Defence 'Stabilisation Unit' (www.stabilisationunit.gov.uk/component/docman/cat_view/163-thematic/190-comprehensive-integrated-approach.html?Itemid=230).

Table (i) *'The Spectrum of Conflict and Operational Themes', Dept. of the [US] Army. FM 3-0, Operations (2008)*



the present varies depending on the subject. To understand current strategy, for example, one must deal with concepts and theories that were first articulated centuries ago. On the other hand, an understanding of air power theory implies a particular focus on the period since the Wright brothers' pioneering flight in 1903 (although, even here, apparently new theory is often grounded in concepts that are centuries old). We focus primarily on warfare in the twentieth and twenty-first centuries. This allows us to chart the evolution of thought and practice over an interesting, relevant and dynamic period. Where necessary it also draws on issues, events and concepts that pre-date this timescale. Sadly there is not the space in this one volume to provide a detailed examination of strategic history across the centuries and we have not tried to do so. The analysis is informed by the authors' historical understanding and examples are used throughout in order to illustrate and explain the conceptual analysis. The list of recommended reading at the end of this chapter provides guidance on a number of sources that do focus explicitly on relevant history and familiarity with the concepts discussed in this book will help the reader to gain the best from such works.

How to use this book

The book is not intended to provide a history of modern warfare. Many such books exist already and there is not the space in this slim volume to provide a satisfactory historical narrative. Rather, the book examines the critical issues, ideas and concepts that one must engage with in order to develop and articulate an understanding of the conduct of war in its various forms and in its different operating environments. While it does not attempt to provide a history of modern warfare the book does seek to introduce the reader to both the *theory* and the *practice* of modern warfare. It is important to engage with both as one cannot properly be understood without reference to the other. The practice of warfare – what armed forces do and the way that they do it – is influenced by theoretical concepts and constructs or, where these are not expressed formally, by preconceived ideas that themselves represent an informal (sometimes an unconscious) equivalent to theory. Today such concepts are often expressed officially through military doctrine, adopting and adapting theoretical constructs in a manner designed to foster understanding amongst practitioners. Thus, one of the aims of this book is to examine and explain the concepts and ideas that underpin modern doctrine and that inform current practice.

Academic commentators, professional think-tanks and military-doctrine writers have generated a plethora of templates, ideas and rules on the methods through which warfare can be conducted successfully. The manner in which these are often used is reflected in the way in which many militaries have articulated formal 'principles of war' to guide the effective application of military power (see [Box \(iv\)](#)). As this book makes clear, however, the conduct of war is an art and not a science. It cannot be made subject to immutable rules. Knowledge of 'principles' is less important than an understanding of the challenges

Box (iv) Principles of war: an illustration

Soviet Union	United States	United Kingdom
Offensive Action	Objective	Selection and maintenance of the aim
Manoeuvre and initiative	Offence	Concentration of force
Concentration of force	Mass	Economy of effort
Economy of force	Economy of force	Maintenance of morale
Surprise and deception	Manoeuvre	Offensive action
Momentum	Unity of command	Flexibility
Annihilation	Security	Co-operation
Reserves	Surprise	Security
Co-operation	Simplicity	Surprise
	Restraint	Sustainability
	Perseverance	
	Legitimacy	

faced when trying to apply them in any given context. A century ago Julian Corbett emphasised this point, writing that ‘nothing is so dangerous in the study of war as to permit maxims to become the substitute for judgement’.¹⁴ Knowledge of principles and concepts is not a substitute for judgement or understanding, merely a means of fertilising both. We hope, therefore, that in addition to informing the reader this book will act as a stimulus for thought and reflection on the subject.

For ease of use the book has been broken down into six main parts that examine strategy ([Part I](#)), land warfare ([Part II](#)), naval warfare ([Part III](#)), air warfare ([Part IV](#)), irregular warfare ([Part V](#)) and weapons of mass destruction ([Part VI](#)). Each of these parts consists of individual chapters that each addresses a different aspect of the topic. The six parts have been written in such a way as to enable them to be read individually; they need not necessarily be read sequentially. [Part I](#), relating to strategy, does provide a useful foundation that is relevant to all other areas and thus it makes sense for the reader to first acquaint themselves with the concepts and issues discussed there, particularly if they do not already have a sound grounding in strategic thought. The first chapter in each of the six parts will introduce core concepts and these are then discussed and developed in the following chapters. The final chapter in this book ([chapter 18](#)) asks the question ‘what will the future look like?’ and addresses key issues that highlight the connection, and also the tension, that exists between theory and practice. All of the chapters make full use of subheadings in order to ‘sign-post’ important topics for the reader and all include a guide to further reading.

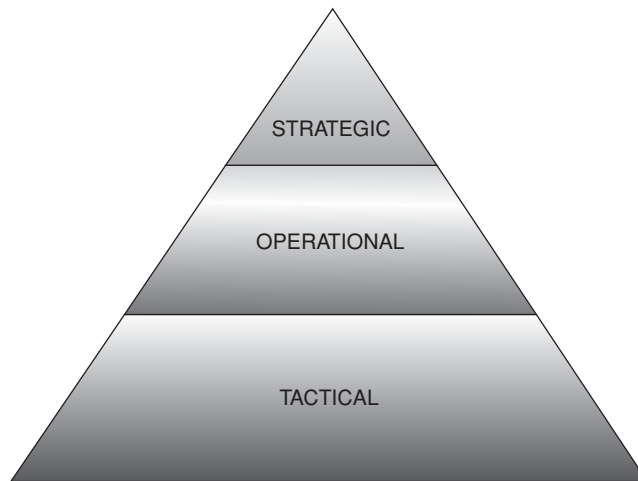
It should be remembered that the breakdown discussed above is rather artificial. This is particularly true of the division of chapters into separate environments (land, sea and air). In truth these are all interconnected and

operations in one usually depend on supporting activities in the others. This has been reflected in the particular emphasis placed by many armed forces on proficiency in joint operations, meaning those that involve co-operation between the different services. Such operations are notoriously difficult, requiring as they do competence across environmental (and thus institutional) boundaries. Nevertheless, it is now widely accepted that to prevail in warfare one must integrate activities across all relevant war fighting domains. The US military currently list five such domains: land, sea, air, space and cyber. The first three of these are discussed in Parts II to IV of this book, space is examined within [Part IV](#) under the general examination of air power and the importance of cyberspace is discussed throughout the book.

The importance of joint operations represents a common theme across the various chapters in this book. Another key theme is that modern warfare is more of an evolutionary phenomenon than it is revolutionary. Many of the important features of warfare in the twenty-first century have their origins in the gradual development of concepts and structures from earlier periods. A third theme is the importance of thinking about warfare in terms of different, but inter-related, levels of war (see [Box \(v\)](#)). These provide a framework within which one can categorise and rationalise military activity and are usually reflected in formal military command structures. One should note that the boundaries between the levels are often blurred and that each influences the other in both an upward and a downward direction. Thus, decisions at the strategic level will influence activity at the operational and tactical level, while actions at the tactical level can have impact on the levels above (indeed, there is not much point if they don't). Major General Charles Krulak's discussion of the 'strategic corporal' provided a good illustration of how, in complex environments, decision-making can devolve lower and lower down the command chain, and also how decisions made at the tactical level (i.e. by corporals) can have a strategic impact.¹⁵

While [chapters 1 to 3](#) demonstrate that strategy is always of importance, the subsequent chapters illustrate that different environments and forms of warfare have seen continuity and change that manifest themselves in different ways at different levels of war. For example as the chapters on naval warfare explain, changes at the tactical level of war do not necessarily invalidate established principles at the operational and strategic levels. Another recurrent theme is that it is evidently rarely easy to distinguish between continuity and change. Consequently, militaries have often found it difficult to predict accurately the future of warfare.

This last theme, in particular, is an important one. Adapting successfully to the future depends in part upon having a grasp of the past and the present. The future is not fixed, nor is it easily predicted. As the eighteenth-century philosopher David Hume noted 'nothing that we imagine is absolutely impossible'. While *knowledge* of the past and present of warfare may provide part of the foundation for coping with the future, more important is developing an *understanding*. We hope that this book is a useful contribution to that process.

Box (v) The levels of war (according to US joint doctrine)

Tactical: the level of war at which battles and engagements are planned and executed to achieve military objectives.

Operational: the level of war at which campaigns and major operations are planned, conducted and sustained to achieve major objectives within theatres of other operational areas.

Strategic: the level of war at which a nation, often as a member of a group of nations, determines national or multinational strategic security objectives and guidance, then develops and uses national resources to achieve those objectives.¹

¹ Joint Doctrine Publication 1-02, *Department of Defense: Dictionary of Military and Associated Terms* (August 2012).

FURTHER READING

Angstrom, Jan and J. J. Widen, *Contemporary Military Theory: The Dynamics of War* (New York, NY: Routledge, 2015). This book offers an introduction to classic and modern military theory, focusing specifically on 'theories' rather than on the context or practical application of such theory.

Black, Jeremy, *Rethinking Military History* (London: Routledge, 2004). A bold critique of much contemporary military history that includes a survey of key themes within the subject from 1500 to the present.

Black, Jeremy, *War: A Short History* (New York, NY: Continuum, 2009). Black provides a brief examination of the history of war since ancient times, offering valuable insight on a range of issues that remain relevant today. In particular he challenges the Euro-centric bias evident in some other works.

- Gat, Azar, *War in Human Civilization* (Oxford University Press, 2006). In this thought-provoking book Gat combines many disciplines in an attempt to examine why people engage in war.
- Gray, Colin S., *Another Bloody Century: Future Warfare* (London: Weidenfeld & Nicolson, 2005). A challenging look at war and at warfare from one of the world's leading experts on the subject.
- Lindley French, Julian and Yves Boyer (eds.), *The Oxford Handbook of War* (Oxford University Press, 2012). This edited volume includes a series of short articles written by academics and practitioners that address a broad range of topics relating to the causes, conduct and consequences of war.
- Millett, Alan, Peter Maslowski and William B. Feis, *For the Common Defense: A Military History of the United States from 1607 to 2012* (New York, NY: The Free Press, 2012). An excellent introduction to the military history of the United States.
- Parker, Geoffrey (ed.), *The Cambridge History of Warfare* (Cambridge University Press, 2005). A collection of essays that examines the history of warfare since ancient times, with a particular emphasis on an apparent 'Western way of war'.
- Townshend, Charles, *The Oxford History of Modern War*, 2nd edn (Oxford University Press, 2005). Written by some of the leading experts in the field, this book examines various aspects of war and warfare from the seventeenth to the twenty-first century.

ONLINE RESOURCES

- There are numerous think-tanks and pressure groups that study the causes, conduct and consequences of war. The following all contain useful and interesting material.
- Correlates of War project, www.correlatesofwar.org/. Initiated by the University of Michigan, this is an academic project that seeks to collect and disseminate reliable statistics relating to international relations, and primarily to war.
- The RAND organisation, www.rand.org/. RAND is a major US think-tank that undertakes research in a number of areas, including in international relations, diplomacy and security. The website includes links to numerous blogs, reports and articles.
- The Stockholm International Peace Research Institute, www.sipri.org/. As the title suggests, this institute studies issues relating to war and peace and publishes useful annual reports relating to conflict, the arms trade etc.
- The Uppsala Conflict Data Program, www.pcr.uu.se/research/ucdp/. Based at a Swedish university this programme seeks to provide data on active armed conflicts. The website includes numerous datasets and links to other publications.

NOTES

1. Michael Howard, 'Military History and the History of War', in Williamson Murray and Richard Hart Sinnreich (eds.), *The Past as Prologue: The Importance of History to the Military Profession* (Cambridge University Press, 2006), p. 17.
2. <http://dictionary.oed.com>.
3. Hedley Bull, *The Anarchical Society* (Columbia University Press, 1977), p. 178.
4. See Correlates of War Project (www.correlatesofwar.org/) and Uppsala Conflict Database Program (www.pcr.uu.se/research/ucdp/definitions/).
5. See von Clausewitz, *On War*.

6. Anatol Rapoport, 'Introduction', in Carl von Clausewitz, *On War* (London: Penguin, 1968). Unfortunately the Penguin edition of *On War* is poorly translated and badly edited; it is best avoided.
7. See Mary Kaldor, *New & Old Wars: Organized Violence in a Global Era*, 2nd edn (Stanford University Press, 2007).
8. Azar Gat, *War in Human Civilisation* (Oxford University Press, 2006), p. ix.
9. Joint Publication 1, Doctrine for the Armed Forces of the United States (2013), pp. 1–4. Joint Doctrine Publication 0–01, *British Defence Doctrine*, 4th edn (2011), p. 2–2.
10. Michael Sheehan, 'The Evolution of Modern Warfare', in John Baylis, James Wirtz and Colin Gray (eds.), *Strategy in the Contemporary World*, 4th edn (Oxford University Press, 2013), pp. 39–59.
11. <http://dictionary.oed.com>.
12. Theodore Ropp, *War in the Modern World* (Duke University Press, 1959).
13. Heidelberg institute for International Conflict Research, *Conflict Barometer 2014*. No. 23. (2015) (www.hiik.de/de/konfliktbarometer/pdf/ConflictBarometer_2014.pdf).
14. Sir Julian Corbett, *Some Principles of Maritime Strategy* (Annapolis, MD: Naval Institute Press, 1988 [1911]), p. 10.
15. Charles Krulak, 'The Strategic Corporal: Leadership in the Three Block War', *Marine Corps Gazette*, January 1999 (www.au.af.mil/au/awc/awcgate/usmc/strategic_corporal.htm).





Part I

Strategy

David J. Lonsdale



1

The study and theory of strategy

Contents

Introduction

The study of strategy

Strategic theory

Clausewitz

Jomini

Sun Tzu

Modern theory

Conclusion

KEY THEMES

- Strategy is neither war nor politics, and thus requires its own intellectual tradition.
- Strategy requires serious study removed from moral, political, operational and intellectual fashions.
- Strategic theory can help the practitioner cope with complexity, but offers no guarantees.

Introduction

War is practised in many different environments and contexts. While this has always been the case, the modern period has witnessed a growing complexity to warfare. In the twentieth century the air, space and cyberspace environments took their place as theatres of war alongside the traditional environments of land and sea. With each new environment of warfare come distinct features, challenges and opportunities for the commander to deal with. In addition to these new arenas, the invention of nuclear weapons presented a severe challenge for those involved in using military force in the service of policy objectives. However, the twentieth century was not just about the 'new'. Older forms of warfare also underwent substantial developments. Irregular warfare witnessed a degree of theoretical and practical maturation through the works and careers of men such as Mao and Robert Thompson. At the same time, regular forms of warfare have had to adapt to new technology and an increasingly joint approach.

Finally, political and social developments have added to the challenges faced by commanders. In an age of ubiquitous social media, legal and moral restrictions on the use of force are more readily applied and upheld.

This book will explore the significance of the above changes to the character of war. Yet, despite the seeming novelty of warfare in the modern period, the very essence of war has remained the same. Across time and place, although the character of war has altered, its nature has remained constant. At the heart of that nature is politics. War is a violent political activity. Regardless of the specific context, each war has a guiding policy rationale. Ensuring that military means serve policy ends effectively is the realm of strategy. Each particular form of warfare, whether acting in isolation or operating in concert with others, must be guided by strategy: the process that ensures war has dynamic purpose.

In our quest to understand strategy, this opening chapter begins by identifying the need for an academic approach to the subject. Having explored the requirement for, and development of, the discipline of Strategic Studies, the work focuses upon the role of theory in strategy. To that end, the chapter provides a summary of the two great works of universal theory, Carl von Clausewitz's *On War* and Sun Tzu's *The Art of War*. Mention will also be made of Baron Antoine Henri de Jomini's *The Art of War*, and the most important works of modern theory.

The study of strategy

In the last fifteen years the study of strategy has enjoyed something of a renaissance. The spread of conflict in the post-9/11 world has fired a growing interest in the subject. This is reflected in the university sector in the proliferation of related degree programmes and the growth in student numbers to fill them. However, it has not always been thus. Since its inception during the Cold War, Strategic Studies has enjoyed mixed fortunes. During periods of perceived heightened threat (e.g. from terrorism or nuclear weapons) interest in strategy has flourished. However, changes in the international security environment, evident failures in strategy or moral repugnance towards war can undermine the perceived validity, desire and/or requirement for Strategic Studies.

It has also been difficult to maintain a unified, coherent approach to the study of strategy. Like many other academic disciplines, especially those tied so closely to the policy world, Strategic Studies suffers from fads and fashions. The subject tends to respond quite vigorously to the changing security environment. While this is understandable, and to some degree desirable – Strategic Studies should help inform the policy process – the tendency to 'trend' invariably dissects the discipline and leaves certain areas neglected. Thus, the 1990s were dominated by peacekeeping and revolutions in military affairs; whereas study of the post-9/11 world is primarily concerned with irregular warfare. Throughout this entire post-Cold War era nuclear strategy has been seriously neglected. This is troublesome, since nuclear weapons have not been dis-invented.

To clarify, in order to make sense of the complex strategic environment we need a vibrant, unified discipline of Strategic Studies. Richard Betts is convincing when he states that university ‘faculties should decide what to cover on the basis of long-term evidence of what mattered in world politics rather than recent events, intellectual fads, or moral hopes.’¹ The study of strategy is essential to give military activity meaning beyond the battlespace.² Without an understanding of strategy we cannot construct meaningful theories of victory. In this sense, victory cannot be defined merely in military terms. Rather, a strategic theory of victory is concerned with the achievement of policy objectives. Strategic Studies enables us to conceptually make this jump from the battlespace, and appreciate the true value of military actions. Thus, in the first instance this chapter seeks to outline the need for the continued development of Strategic Studies as an academic discipline, acting in the service of those who practise strategy in the real world.

The need for Strategic Studies is perhaps at its most intense in response to the complexity of strategy. This complexity, and the challenges it produces, is amply demonstrated in the ongoing conflicts in Afghanistan, Iraq and Syria. Indeed, we can conclude that the complexity of strategy is such that achieving a satisfactory end state at reasonable cost, and within a reasonable time-frame, is often more elusive than ill-informed comment would suggest. As a result, we must have realistic expectations of what is possible from both the theorist and the practitioner. As David Jablonsky notes, ‘a true scientific product is not possible from the study of strategy’.³ Strategy does not tolerate formulas for success; rather it calls for an approach more suited to art than science. Every strategic context is unique, and therefore requires its own unique mixture and application of strategic assets. In order to appreciate fully the challenges faced by those practising strategy, and to provide a common set of definitions and concepts, the first three chapters of this book seek to dissect the art of strategy. In undertaking this task, the work is by default asking the question: ‘why is strategy so difficult?’, and seeking means to alleviate the challenges faced.

In many respects, the development of modern Strategic Studies can be attributed to one man and one invention. The man is Bernard Brodie; the invention is the nuclear weapon (see [Figure 1.1](#)). The unique characteristics of nuclear weapons, and the absence of a history of use – beyond the attacks on Hiroshima and Nagasaki at the end of the Second World War – opened the door for civilian strategists to lead debates about strategy during the Cold War. The rise of the civilian strategic analyst is illustrated by Alain Enthoven, who made the following comment during a discussion on strategic plans: ‘General, I have fought just as many nuclear wars as you have.’⁴ Brodie’s contribution was to call for a more systematic, academic approach to the subject. Ironically, Brodie’s call somewhat backfired. In outlining the need for a more scientific approach to strategy, he inadvertently instigated an approach (systems analysis) that attempted to reduce the complexity of strategy to quantifiable mathematical phenomena.

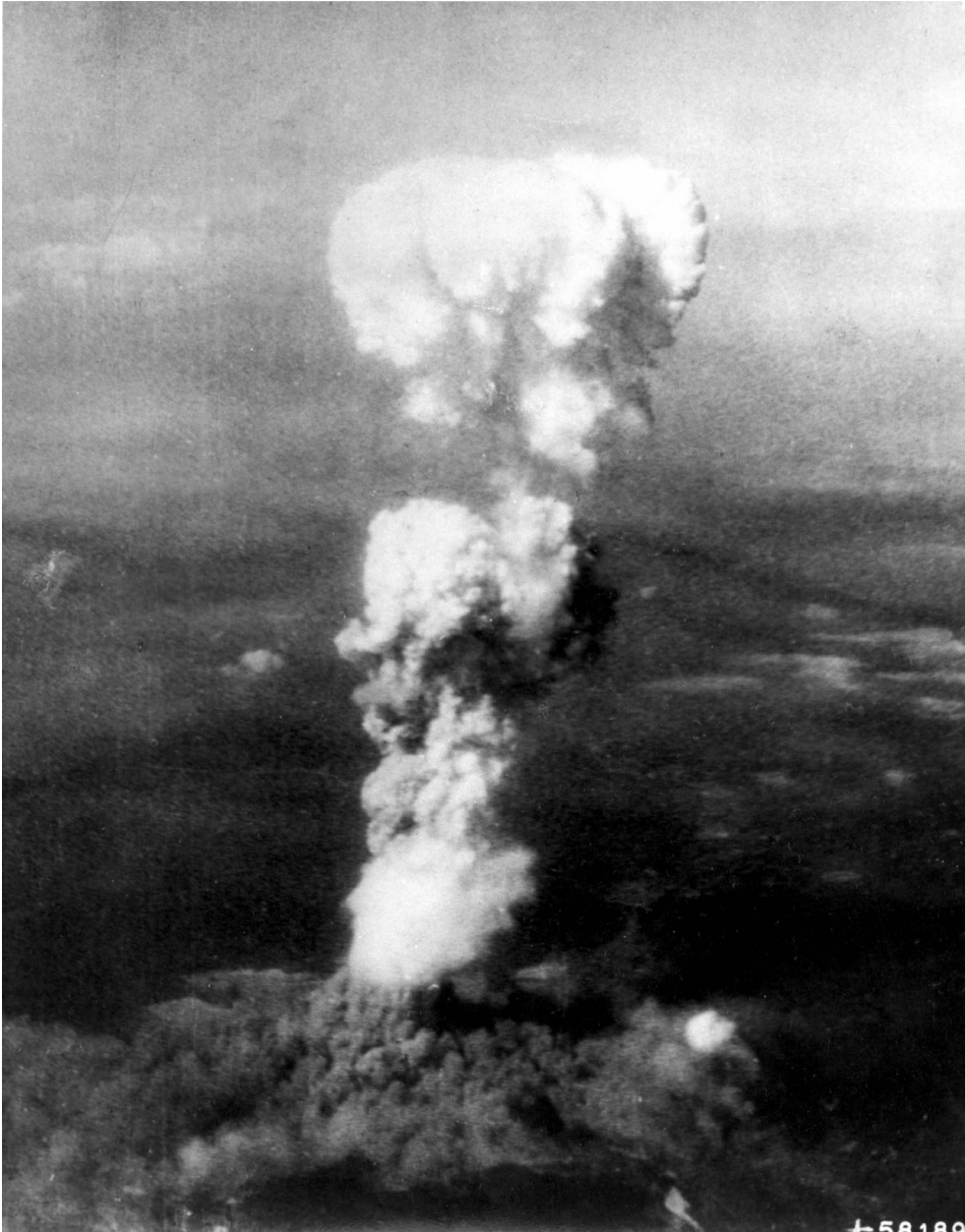


Figure 1.1 The invention of nuclear weapons, used first at Hiroshima (depicted here) and Nagasaki in August 1945, played a key part in the development of the academic discipline of strategic studies.

In response, Brodie called for the reintegration of politics and history into the subject.

Since strategy is a process that sits uncomfortably between two worlds occupied by politics and the military, it has historically suffered from the absence of an intellectual tradition of its own. Political theory has a long tradition stretching back to ancient Greece. Whereas, although strategic theory has been blessed with outstanding works such as Clausewitz's *On War* and Sun Tzu's *The Art of War*, the development of theory relating to military matters has tended to be limited to a focus on tactical and operational issues. Indeed, Brodie bemoans the lack of any real successors to the two great theorists. Since 1949 a number of commendable theorists have emerged – Brodie amongst them – yet his comments regarding the anti-intellectual bias of the military still ring true: 'Soldiers usually are close students of tactics, but only rarely are they students of strategy and practically never of war!'⁵ This seemingly odd statement can be explained by the following comment by Gray, citing historian Peter Browning: 'War is a relationship between belligerents; it is the whole context for warfare. Warfare is defined as "the act of making war".'⁶ Understanding these two related, but distinct, concepts requires very different approaches. Worryingly, Brodie's lament that military staff colleges focus primarily, and intensively, on training, at the expense of true analytical reflection, could easily be an observation of the current situation. In modern military colleges strategic theory often receives significantly less attention than subjects such as defence management and procurement. The result of such an approach is that the military tend towards an over-reliance on simplistic principles concerned with tactical and operational issues, such as the Principles of War. It also misunderstands the role of theory. As Gray notes, 'strategic theory is about education, not training or doctrine.'⁷ As Brodie again notes, within the cultural confines of the military, there simply is neither the inclination nor time to engage in detailed analysis of strategic issues.⁸ One might say that because there is no inclination, sufficient time is not created.

This should not necessarily be taken as an outright condemnation of the military profession's attitude to study. Nor should it be regarded as a call for an abstract, purely theoretical pedagogic approach. In many respects, it is understandable that a profession so rightly focused on practical considerations and results should neglect deep analytical ruminations. Also, since strategy is so bound up with intangible forces, such as morale and will, judgement, rather than endless analysis, is often the key to success. However, these two considerations, while somewhat understandable, do not excuse the neglect of academic analysis. Clausewitz, both a soldier and a scholar, was aware of the value of strategic theory: 'Theory exists so that one need not start afresh each time sorting out the material and ploughing through it, but will find it ready to hand and in good order. It is meant to educate the mind of the future commander, or, more accurately, to guide him in his self-education, not accompany him to the battlefield.'⁹ In this sense, theory helps us to understand what matters in strategy, and gives us insight into how the many dimensions of strategy interact.

The relationship between theory and practical strategy is a complex one. An overly enthusiastic fixation on the practical can lead to analytical stagnation, and an obsession with today's problem. A theory constructed and developed beyond immediate policy concerns may therefore have more lasting value, and may be able to illustrate a range of tomorrow's policy issues. However, at the same time, theory devoid of policy relevance will not further the goal of better strategy, which must be an objective for Strategic Studies; bad strategy kills. It is therefore understandable that, as Raymond Aron observes, strategic theory often reflects the strategic issues of the day. Again, Betts summarises the relationship well: 'Neither theory nor policy can be optimised apart from each other. Central theoretical insights often flow from grappling with concrete questions rather than *a priori* constructs.'¹⁰ Or, as Gray puts it: 'Most practitioners need to be educated to recognise the relevance of theory to their search for workable solutions to today's problems, whereas theorists must never forget that their labours ultimately only have meaning and value for the world of strategic behaviour.'¹¹ Ultimately, the strategist has to act, usually, if not always, without a complete understanding of the situation. At such times judgement and moral courage come to the fore. Nevertheless, the further development of strategic theory can only help us to improve our understanding of the complex relationships amongst the many dimensions in strategy.

In the absence of an intellectual strategic tradition within the military, Strategic Studies has had to develop its own identity within the civilian sector, within universities, think-tanks and research institutes. Initially centred in the latter, which resulted in a focus on current policy demands, Lawrence Freedman notes that the subject failed to establish a scholarly framework. Again, akin to its relationship with the military, academic analysis finds little empathy in the world of policy-makers, who want solutions, not further elucidation of the problems and complexities facing them. As Freedman notes, 'Policy-makers became impatient with those qualities that academics believe to be the most valuable: long-term thinking, stretching the bounds of the possible, and taking complexity as a challenge rather than an excuse for not going into too much detail.'¹² Thus, alongside the need for a more academic approach in staff colleges, the maintenance of Strategic Studies in universities is equally critical for the development of a genuine analytical framework and tradition. Within such an environment the strategic analyst is afforded the freedom and time to develop the interdisciplinary, analytical approach required to enhance understanding of strategy.

Again, the above comments are not designed to denigrate the more practical aspects of military education and training, such as doctrine. However, while accepting the need for doctrine, all those concerned with the development of strategy should also embrace a theoretical approach in the Clausewitzian tradition, rather than a slightly more sophisticated version of Jomini's work (based as it is in principles and operational-level issues). Too often, the significance of Clausewitz's work is acknowledged, but only through the citation of an overused maxim. Clausewitz's real value exists in the intellectual development one

experiences when dealing with the whole complexity of his theory. Despite the many instances when this chapter will extol the virtues of Clausewitz's work, it should not be taken as an advertising campaign for *On War*. We must heed Brodie's advice, and go beyond Clausewitz. Strategic Studies, and thereby our understanding of strategy, cannot develop if all we do is reiterate Clausewitz's ideas. Clausewitz is a great starting point, but we must develop theory further. This can only be achieved by theorising in an intellectually permissive environment.

In light of the above discussion, a fair question to pose is: do we need civilian strategists? Could not the military and policy-makers construct an effective understanding of strategy between themselves? Richard Betts thinks not: 'If civilian strategists are not to decide along with the professional military, either ignorant civilians will do it, disjoining political and military logic, or the military will do it alone.'¹³

Since strategy is the meeting point of two different worlds, its complexity cannot be underestimated. Thus, we must resist the temptation to side-line Strategic Studies as an academic discipline, favouring instead an overly practical military or political take on the subject. The real challenge for Strategic Studies is how to remain useful in the service of policy in a timely fashion, without forgoing the detailed analysis necessary to understand the complexities of the subject.

Strategic theory

Efforts to understand the complexities of strategy have been aided by a number of key theorists. Although each particular form of warfare has its own dominant works of theory, there are universal theorists who grapple with the nature of war and the broader aspects of strategy. For the practitioner these works help to identify and illuminate the various elements of strategy. In addition, they provide some advice, if limited, on how to cope with the challenges of strategy. However, the more prescriptive a work, the less universal it tends to be. This is particularly the case with Jomini's work, which suffers in comparison with Clausewitz as a result.

Although these universal works provide enlightenment on strategy, the limits of theory must be kept in mind. As a general proposition, Wylie's assertion that no general theory can guarantee success should be treated as the first and most important thought on this issue.¹⁴ In this vein, Clausewitz recognised that theory could only ever be second best to what military genius does: 'What genius does is the best rule, and theory can do no better than show how and why this should be the case.'¹⁵ Yet as we have already seen, he does value theory as a means to ensure practitioners avoid reinventing the wheel.¹⁶ Indeed, theory has often performed a very practical function within strategy. One of the most striking examples of this is the relationship between the theory and practice of strategic bombing. The influence of theory was particularly evident during

the interwar years, when the central theoretical tenets of both precision and area/morale bombing were formulated in the works of Douhet, Trenchard and Mitchell. Williamson Murray notes how pervasive the influence of theory can be, reaching into areas such as doctrine and force composition: 'The theories of Douhet and other early airpower advocates . . . have exercised a great influence on the development of air forces since that time.'¹⁷ In a similar manner, the theorists of nuclear strategy during the Cold War were said to have 'wielded enormous influence, not only over the way an entire generation's thoughts about military issues were shaped but also over the formulation of defence policy in the nuclear-weapon states'.¹⁸

In order to be useful to the practitioner theory must coincide with reality. To this end, Murray and Grimsley's declaration that 'strategy is the art of the possible' is an important warning to those who would construct naively optimistic theories.¹⁹ Again Clausewitz is persuasive when he warns: '[Theory's] purpose is to demonstrate what war is in practice, not what its ideal nature ought to be.' In a similar vein, Brodie argues that: 'Above all, strategic theory is a theory for action.' Echoing Brodie's wise counsel, Daniel Moran stipulates that 'The goal of theory in any field is to improve our understanding of reality, and our ability to act effectively.'²⁰ Finally, Clausewitz summarises both the value and limits of theory: '[Theory] is meant to educate the mind of the future commander, or, more accurately, to guide him in his self-education, not to accompany him to the battlefield.'²¹ Later on in the work he elaborates on these thoughts:

Theory cannot equip the mind with formulas for solving problems, nor can it mark the narrow path on which the sole solution is supposed to lie by planting a hedge of principles on either side. But it can give the mind insight into the great mass of phenomena and of their relationships, then leave it free to rise into the higher realms of action.²²

Clausewitz

Of the universal theorists, the first amongst equals is the aforementioned Carl von Clausewitz. Having served in the Prussian Army against Napoleon's forces, Clausewitz's major work, *On War*, was posthumously published in 1832. *On War* is arguably the most influential work of strategic theory to date. Brodie is unequivocal about the value of Clausewitz's work, which he describes as 'not simply the greatest but the only truly great book on war'. Similarly, Gray, one of the most respected of modern strategic thinkers, describes Clausewitz as his constant companion.²³ The practitioners of strategy are no less impressed by the Prussian's credentials. The USMC's doctrinal publication *Warfighting* stipulates that Clausewitz's *On War* is 'the definitive treatment of the nature and theory of war'.²⁴

Despite his prominence in modern strategic affairs, Clausewitz's work is often misunderstood and criticised. Much of this is due to the unfinished nature of

the work, and the fact that Clausewitz chose a philosophical framework that, although popular in his day, is now rarely utilised. *On War* was published as a single volume after Clausewitz's wife gathered together elements of the unfinished manuscript. Indeed, in a book of 123 chapters, Clausewitz only considered the first to be fully complete. In addition, Clausewitz's use of the dialectical approach has led to some confusion. Rather than merely present one perspective, this philosophical method arrives at a synthesis after discussing the thesis and antithesis. Specifically, Clausewitz first presents a vision of war that is absolute, escalatory and unrestricted. He spends much of the rest of the book reining in this idea with the various means by which war is limited. So, for example, absolute war can never be achieved because it is limited by policy concerns and the play of friction, to name just two restraining effects. On this basis Clausewitz distinguishes between absolute war and real war. Unfortunately, some commentators have failed to grasp his approach, and have thus criticised Clausewitz for advocating unrestricted forms of war in the early sections of the book.

Clausewitz has come under fire from some of the most prominent writers in modern Strategic Studies. Martin van Creveld, perhaps in an attempt to bolster his view that war is becoming increasingly irregular in character, regards Clausewitz's work as being stuck in a state-centric view of the world, and therefore narrow in its utility. The military historian John Keegan likewise regards Clausewitz's view of war as being too narrow. Keegan argues that rather than being merely an instrument of policy, war in fact is a cultural phenomenon.²⁵ These narrow interpretations of *On War* do a disservice to Clausewitz, whose work is far more universal. Clausewitz's focus on policy as the guiding force in war is anything but restrictive. It certainly is not limited to state-based war fought for purely rational and secular motives. The following statement from Clausewitz illustrates this point perfectly: 'Policy, of course, is nothing in itself; it is simply the trustee for all these interests against other states . . . we can only treat policy as representative of all interests of the community.'²⁶ The emphasis here is on the broad interpretation of the concept of policy, and the use of the term community. Thus, policy may refer to any objective for which war is waged. This can include religious issues, territorial disputes, resources or indeed as an expression of culture.

Clausewitz's vision of war is fairly complex and dense. Although it is not possible to fully express his ideas in just one chapter of a book, the main concepts can be summarised. However, it is worth noting that there is simply no substitute for spending time analysing *On War*. Amongst strategic thinkers Clausewitz comes closest to defining the nature of war. For Clausewitz war is a complex phenomenon, composed of rational and non-rational forces. This is expressed most clearly in the 'Trinity', which describes how the nature of war is composed of three interactive forces: 'primordial violence, hatred, and enmity', 'the play of chance and probability' and 'its element of subordination, as an instrument of policy, which makes it subject to reason.'²⁷ The true nature of any particular war

is to be found in the dynamic interplay of these three forces. Much emphasis is given to war's rational element as a tool of policy. However, this subjection to reason is tempered by non-rational forces that make war uncertain, chaotic and non-linear. For Clausewitz war is infused by chance, emotions and significant levels of friction that prevent war from reaching the higher levels of operational efficiency.

Although Clausewitz spends much of his time in *On War* discussing the various complexities of strategy, he does offer a number of means through which to deal with the associated problems. In a work that places so much emphasis on the human element, it is not surprising that the primary factor that can mitigate the complexities of strategy is to be found in man. Indeed, Clausewitz's theory of war cannot be fully understood without reference to the role of the commander or, to use the correct terminology, the 'military genius'. Winston Churchill, a man who had to wrestle with strategy at the highest levels, is of the same mind: '[war's] highest solution must be evolved from the eye and brain and soul of a single man . . . Nothing but genius, the demon in man, can answer the riddles of war.'²⁸

Although Clausewitz fought against Napoleon, it is clear that his notion of the 'military genius' was largely based upon the Corsican. Although the term genius tends to suggest that Clausewitz was referring to a few gifted individuals, we can utilise his concept to discuss the attributes of good command more generally. Thus, good command requires certain cognitive skills, certain moral qualities, and an understanding of the human condition. More specifically, Clausewitz identifies a number of characteristics which a commander should possess. These include physical and moral courage, incisiveness, presence of mind, strength of will and character and an ambitious nature. Particular prominence is given to a general's intuitive ability, his *coup d'oeil*, and a determined character to pursue his decisions through to conclusion. Clausewitz was also conscious of the significance of leadership within the chaotic and stressful environment of war. Finally, and rather obviously, a Clausewitzian general must understand the process of strategy by which war is used for political effect.²⁹

Important though the commander is, command as a whole cannot be reduced simply to the attributes of the individual. As Gray postulates, an astute commander may be unavailable when the next war occurs. Thus, a process of command must be designed to compensate for the potential lack of genius: 'War cabinets, general staffs, and chiefs of staff committees were invented to function as a surrogate for individual strategic genius.'³⁰ There are a number of examples of command systems that are able to produce consistently good results. One prominent example, often cited, is the Prussian General Staff. T. N. Dupuy goes as far as to suggest that the success of the Prussian/German General Staff can be attributed to the fact that 'it enabled men who individually lacked the qualities of a genius to perform institutionally in a manner that would provide results ordinarily achievable only by genius'.³¹ Thus, we can conclude that the complexity of strategy can be somewhat unravelled by a judicious combination

of the commander's qualities, the command structure and an appropriate command ethos.

In many respects, the main utility of Clausewitz's work for the practitioner may simply be the fact that he identifies the complexities of strategy. To be able to deal with a problem, one must first acknowledge and understand it. However, *On War* goes further. Aside from an important, if not always practical, focus on the military genius, Clausewitz provided a few conceptual aides. Most noteworthy in this respect is 'centre of gravity'. This concept dominates much of modern military doctrine. Centres of gravity are defined by Clausewitz as: 'particular factors can often be decisive . . . One must keep the dominant characteristics of both belligerents in mind. Out of these characteristics a certain center of gravity develops, the hub of all power and movement, on which everything depends. That is the point against which all our energies should be directed.'³²

Wylie incorporates centre of gravity into his own theory regarding the search for strategic success: 'The basic problem facing the strategist throughout all these situations, through any war, any time, any place, is this: Where shall be the center of gravity of the war? . . . Control of the strategic weights or centers of gravity in any war, large or small, limited or unlimited, is a basic advantage that should be sought by any strategist. It is the fundamental key to the conduct of warfare.'

Wylie's above statement identifies that there may be more than one centre of gravity for each belligerent, and therefore a number of routes to victory. Indeed, it is possible to identify centres of gravity at all of the levels of strategy. An airfield may act as the operational-level centre of gravity. At the same time, an individual leader or the public will may be the centre of gravity at the grand strategic level. Although Clausewitz himself recognises that there may be various centres of gravity, he does conclude that it is preferable to 'trace them back to a single one'. Clausewitz's candidates for the centre of gravity are: the army, capital city, public will, alliance cohesion or personalities of the leadership. Nonetheless, reflecting his emphasis on battle, Clausewitz concludes that 'the defeat and destruction of [the enemy's] fighting force remains the best way to begin, and in every case will be a very significant feature of the campaign.'³³

In this respect, Clausewitz's work is noteworthy for the emphasis it places on battle. While this may seem rather obvious in a work about war, it is a refreshing counterpoint to the modern tendency to undervalue engagement with the enemy. Clausewitz was certainly aware of the limits of force, but he also stressed the dominance of the destructive principle and the value of defeating the enemy's forces. The latter is obviously not an end in itself but, as noted earlier, is usually an important step on the road to achieving policy objectives. Battlefield victory over the enemy destroys his ability to resist, thereby opening the door to control. Whether that control is enforceable may depend on effective utilisation of the non-violent tools of grand strategy. Nonetheless, success in battle will at least present a more permissive security environment.

Clausewitz's emphasis on battle is wise counsel in an age when we may be guilty of overthinking strategy in our search for overly complex and subtle

answers to strategic dilemmas. Indeed, simplicity, one of the modern principles of war, is often overlooked; yet Clausewitz regarded it as an essential ingredient of his theory of war: 'It seems to us that this is proof enough of the superiority of the simple and direct over the complex . . . rather than try to outbid the enemy with complicated schemes, one should, on the contrary, try to outdo him in simplicity.'³⁴ And for those who regard a direct, battle-orientated approach as intellectually immature, Clausewitz comments: 'The maximum use of force is in no way incompatible with the simultaneous use of the intellect.'³⁵

Regardless of what constitutes the centre(s) of gravity in any particular war, the concept does act as a very useful analytical and planning tool. Nonetheless, as with all theory, it does not offer a guaranteed route to victory. Therefore, in order to fully appreciate the relative value of centres of gravity, we must ask a number of questions in each circumstance. Does the enemy actually have a centre of gravity? Can the enemy's centre of gravity be identified? The intelligence and analytical challenge posed by this last question should not be underestimated. For example, a debate still rages over what constituted the centre of gravity within the German war economy during the Second World War. This is not surprising when one considers the size of the task involved in trying to undertake an accurate macro-level analysis of the German economy at that time. Robert A. Pape acknowledges that the required information was simply not available.³⁶ Even with the benefits of historical research and hindsight, historians still disagree over which component of the German economy, at which period in the war, represented its key vulnerability. Understanding the functioning of the German economy at the time must have been even more of a challenge. These difficulties are compounded by the fact that often the intelligence acquired is based on a peacetime analysis of the enemy, rather than when they have mobilised their economy for war.

Even if a centre of gravity has been correctly identified, is it always possible to put it under sufficient pressure? As Beaufre argues: 'the enemy's vulnerable points must be set against our own capabilities'. For example, if the Wehrmacht represented one of Nazi Germany's centres of gravity, it was beyond the reach of certain countries at certain times. Until the D-Day landings the Wehrmacht could not be sufficiently engaged by British and American military power. Furthermore, in 1939, although German forces were in contact with the Poles, the latter lacked the required capabilities to put the former under significant pressure. With such a mismatch in capabilities and opportunities it matters little that you have identified the enemy's key attribute; defeat will still occur. Finally, the Soviet Union, although able to eventually defeat the Wehrmacht, could only do so at great cost. Most countries would not have been able to sustain such losses and deprivations.

There is much more that could be written about *On War*. For now, we must satisfy ourselves with this brief discussion of Clausewitz's significance to Strategic Studies. Nonetheless, Clausewitzian concepts will appear constantly throughout this book. Most obviously, the following chapter will discuss in detail the important concept of 'friction.'

Jomini

Before moving on to the work of Sun Tzu, the other great universal theorist, it is important to mention the work of Baron Antoine Henri de Jomini. Despite being a contemporary of Clausewitz, who also participated in the Napoleonic Wars, Jomini offers a distinctly different vision of war to the Prussian. In contrast to Clausewitz's emphasis on military genius overcoming the complexities of a chaotic world, Jomini provides the reader with a number of principles, which if followed should lead to victory. Indeed, rather than describing war as a non-linear activity in which intangible forces dominate, Jomini's approach to war is rather geometric. In particular, Jomini stresses the advantages of interior lines of communication. Jomini's experience with the general staff may have been instrumental in shaping his planning-based approach to the subject of war. This may help to explain his greater emphasis on concepts such as lines of operation.

It is interesting to note that two strategic thinkers from the same period, and sharing some of the same experiences, can produce such radically different theories. In broad philosophical terms both Jomini and Clausewitz display a tendency for the Enlightenment's propensity towards rational analysis. Jomini in particular apes Newton in his quest to discover the fundamental principles underpinning the activity of war. However, whereas Jomini is criticised for failing to escape the rationalism of the eighteenth century, Clausewitz managed to create a synthesis of the enlightenment's rationality and the non-rational approach of German Romanticism with its greater emphasis on the psychological, emotional, metaphysical and intuitive. Despite his apparent failings, Jomini proved particularly popular during the nineteenth century. However, over time the limitations of the prescriptive nature of Jomini's work have become evident.

Sun Tzu

The second great universal theorist pre-dates the Napoleonic period by over 2,000 years. The Chinese strategist Sun Tzu offers yet another different vision of war. Sun Tzu's work is influenced by the Taoist tradition with its emphasis on non-material force-multipliers. Thomas Cleary explains that Taoism can be understood as 'the ancient tradition of knowledge'.³⁷ In this sense, we may conclude that for Sun Tzu war in its ideal form becomes an intellectual and metaphysical exercise rather than a physical one. For Sun Tzu war is a necessary evil that represents a departure from cosmic harmony.

In contrast to Clausewitz's emphasis on violent clashes of arms, Sun Tzu seeks attainment of policy objectives, preferably, without the application of violence. For Sun Tzu 'to win one hundred victories in one hundred battles is not the acme of skill. To subdue the enemy without fighting is the acme of skill'.³⁸ In order to achieve this, or indeed to gain victory at minimal cost should fighting occur, Sun Tzu puts great emphasis on the role of information and knowledge. Indeed, much of Sun Tzu's work is built around the need to acquire and use

intelligence about factors such as the enemy, environment and oneself. With intelligence at the heart of his theory, Sun Tzu understandably highlights the significance of deception. His statement, 'All warfare is based upon deception', is revealing, since in order to deceive the enemy effectively, one must exert control over information.

Whereas Clausewitz regards a more direct, battle-orientated approach as the most profitable, Sun Tzu is famed for the indirect approach. Rather than highlighting the gains to be made from destroying the enemy, Sun Tzu extols the virtues of absorbing the enemy's resources into one's own: 'Your aim must be to take All-under-Heaven intact. Thus your troops are not worn out and your gains will be complete.'³⁹ From the above it is understandable why the ancient Chinese strategist has become popular in the modern period. In many respects his approach is ideally suited to a Western, information-age approach to strategy that seeks minimal costs, and champions the manoeuvrist or effects-based approach to warfare.

Whereas Clausewitz's work may be criticised for being too dense and sometimes confusing, Sun Tzu suffers from the opposite problem. Although there is little to argue with his ideas concerning the value of acquiring information, overall his work gives the impression that war is a controllable and predictable activity. Lawrence Freedman goes as far as to suggest, 'Sun Tzu believed that perfect knowledge could be obtained.'⁴⁰ In the chapter 'Energy', Sun Tzu provides a picture of combat that appears chaotic, but in fact 'there is no disorder'.⁴¹ Organisation and good communications are the means by which order is produced from seeming disorder. Rather than Clausewitz's gritty 'real war', Sun Tzu provides something of an ideal vision of how strategy can be conducted. Taking Sun Tzu too seriously could leave the practitioner naively optimistic of what is possible in strategy.

While Sun Tzu and Clausewitz offer very different approaches to strategy, their ideas can be effectively combined. This has been most obviously and successfully done in Maoist strategic thinking. Mao's three-stage theory of revolutionary war utilises the indirect approach in the early stages; switches to Clausewitzian battle in the final stage; and throughout is infused by a clear adherence to the idea that war is very much a political instrument. Indeed, Mao was extremely conscious of the fact that his forces were political actors as well as being military units.⁴²

Modern theory

Taken together, the works of Clausewitz and Sun Tzu provide the practitioner with a solid theoretical basis for his future undertakings (see [Box 1.1](#)). However, Gray is right in his assertion that we need more than just the classics. At the level of general theory, three works suggest themselves. These are Luttwak's *Strategy: The Logic of War and Peace*, primarily for its identification of the paradoxical logic of strategy, although its discussion on the harmony of the levels of strategy is also a worthwhile and beneficial read; Wylie's *Military*

Box 1.1 Strategic theory**Clausewitz**

- War is a continuation of political intercourse.
- The nature of war is composed of rational and non-rational forces.
- Wars may differ in their character, but are all things of the same nature.
- Friction, which distinguishes war on paper from war in reality, impedes efficacy. War is violent, uncertain and competitive.
- There are many potential centres of gravity, but defeating the enemy's forces is usually the best place to start.

Sun Tzu

- The acquisition of knowledge is the key to success in strategy.
- Deception has a central role in war.
- More is gained through victory without battle/violence.
- Victory may be achieved through either a direct or an indirect approach.

Strategy: A General Theory of Power Control, which is distinguished for its universally useful concept of control, and the related ideas concerning 'the man on the scene with a gun'; and Gray's *Modern Strategy*, which is chosen for its identification of the many dimensions of strategy, its broad scope covering the whole spectrum of strategic matters and its concept that strategy represents a unified, practical undertaking. The core theory presented in *Modern Strategy* has been developed considerably in Gray's recent *Strategy* trilogy: *The Strategy Bridge*, *Perspectives on Strategy* and *Strategy and Defence Planning*.

Conclusion

As a consequence of the fact that strategy is neither politics nor war – rather it is the process that links one to the other – it is not endowed with a strong intellectual tradition of its own. That being the case, the discipline of Strategic Studies, based primarily in civilian institutions, plays an essential role in keeping the theoretical flame alive. Strategic theory, however, is not a self-justifying activity. In the strategic realm theory plays an important role in educating the practitioner. To reiterate Brodie's important observation, 'strategic theory is theory for action'.

Although steadily growing, universal strategic theory is still thin on the ground. Thankfully, what exists is of high quality. Alongside important modern works (including Gray, Wylie and Luttwak), the theories of Clausewitz and Sun Tzu stand out. Together, these two works provide invaluable insights into the nature of war and the challenges of strategy. For those responsible for the practice of strategy, the great works of theory act as important tools in the difficult task of understanding this most complex and important of subjects. To paraphrase Sun Tzu, strategy is of such import that it should be thoroughly studied.

RESEARCH QUESTIONS

1. Why is the study of strategy important?
2. In what ways can theory aid the practitioner of strategy?
3. Does Clausewitz deserve his pre-eminence in modern strategy?
4. Is Sun Tzu's *The Art of War* especially well suited to the modern age?
5. How well does modern strategic theory build upon the classics?

FURTHER READING

- Betts, Richard K., 'Should Strategic Studies Survive?', *World Politics*, 50 (October 1997). Provides a compelling argument on why we need to keep the Strategic Studies flame alive.
- Brodie, Bernard, 'Strategy as a Science', *World Politics*, 1/4 (July 1949), 467–88. A vastly important article that outlines the need for an intellectual development of Strategic Studies.
- Clausewitz, Carl von, *On War*, trans. Michael Howard and Peter Paret (London: David Campbell Publishers Ltd, 1993). Quite simply the best work of strategic theory ever written. Clausewitz gets closer than anyone to defining the true nature of war.
- Freedman, Lawrence, *Strategy: A History* (Oxford University Press, 2015). A masterly and exhaustive exploration of the theory and practice of strategy.
- Gray, Colin S., *Modern Strategy* (Oxford University Press, 1999). Probably the greatest modern work of strategic theory. Gray's analysis is comprehensive and thought-provoking.
- Handel, Michael I., *Masters of War: Classical Strategic Thought*, 2nd rev. edn (London: Frank Cass, 1996). A great introduction to the classic works of strategic theory.
- Jomini, Baron Antoine Henri de, *The Art of War* (London: Greenhill Books, 1996). A flawed classic; Jomini is still worth reading to see how a 'principles' approach to strategy can underestimate the challenge of complexity.
- Luttwak, Edward N., *Strategy: The Logic of War and Peace* (Cambridge: The Belknap Press of Harvard University Press, 1987). A modern classic which introduces the paradoxical logic and discusses harmonisation of the levels.
- Mao Tse-Tung, *Selected Military Writings of Mao Tse-Tung* (Beijing: Foreign Languages Press, 1963). Mao is arguably one of the most important theorists and practitioners of irregular warfare. He also manages to combine the theories of Clausewitz and Sun Tzu.
- Schelling, Thomas C., *Arms and Influence* (New Haven: Yale University Press, 1966). An outstanding analysis of how military force can be used for strategic effect.
- Sun Tzu, *The Art of War*, trans. Samuel B. Griffith (London: Oxford University Press, 1971). A rich work of strategic theory, which considers military strategy in its grand strategic context.
- Watts, Barry D., *Clausewitzian Friction and Future War*, McNair Paper 52 (Washington, DC: Institute for National Strategic Studies, National Defense University, October 1996). The best analysis of Clausewitzian friction to date.

Wylie, J. C., *Military Strategy: A General Theory of Power Control* (Annapolis: Naval Institute Press, 1967). Often overlooked, Wylie's little book provides an excellent discussion of strategic theory and introduces the central concept of 'control'.

ONLINE RESOURCES

www.clausewitz.com: a site full of interesting papers on this most important of strategic thinkers.

www.strategicstudiesinstitute.army.mil/: the website of the Strategic Studies Institute, which contains lots of in-depth analysis of important strategic issues.

www.strategicstudiesinstitute.army.mil/pubs/display.cfm?pubID=947: an important article by Colin S. Gray on strategic education.

www.strategicstudiesinstitute.army.mil/pubs/display.cfm?pubID=641: *Strategic Theory for the 21st Century*, Harry Yarger.

www.rand.org/topics/military-strategy.html: offers a wide selection of papers on strategy.

www.infinityjournal.com/: an online journal devoted to strategy.

www.au.af.mil/au/awc/awcgate/awc-thry.htm#gen: a vast array of papers on strategy from the United States Air War College

NOTES

1. Richard K. Betts, 'Should Strategic Studies Survive?', *World Politics*, 50 (October 1997), 41.
2. The term 'battlespace' refers to the environment in which warfare occurs. The term is used in place of the more traditional 'battlefield' to indicate the increasingly complex, and joint, operational environment. Hence, battlespace includes the traditional environments of land, sea and air. It also includes space, cyberspace, and the electromagnetic spectrum more generally.
3. David Jablonsky, 'Why Is Strategy Difficult?', in Joseph R. Cerami and James F. Holcomb, Jr (eds.), *US Army War College Guide to Strategy* (February 2001) (<http://permanent.access.gpo.gov/lps11754/00354.pdf>).
4. Quoted in Fred Kaplan, *The Wizards of Armageddon* (New York: Simon and Schuster, 1983), p. 254.
5. Bernard Brodie, *War and Politics* (London: Cassell, 1973), p. 11.
6. Colin S. Gray, 'From Principles of Warfare to Principles of War: A Clausewitzian Solution', in Colin S. Gray, *Strategy and History: Essays on Theory and Practice* (London: Routledge, 2006), p. 82.
7. Colin S. Gray, 'Introduction', in Colin S. Gray, *Strategy and History: Essays on Theory and Practice* (London: Routledge, 2006), p. 2.
8. Bernard Brodie, 'Strategy as a Science', *World Politics*, 1(4) (July 1949), 486–7.
9. Carl von Clausewitz, *On War*, trans. Michael Howard and Peter Paret (London: David Campbell Publishers Ltd, 1993), p. 141.
10. Betts, 'Should Strategic Studies Survive?', p. 37.
11. Gray, 'Introduction', p. 5.

12. Lawrence Freedman, 'The Future of Strategic Studies', in John Baylis *et al.* (eds.), *Strategy in the Contemporary World*, 2nd edn (Oxford University Press, 2007), p. 360.
13. Betts, 'Should Strategic Studies Survive?', 3.
14. J. C. Wylie, *Military Strategy: A General Theory of Power Control* (Annapolis: Naval Institute Press, 1967).
15. Clausewitz, *On War*, p. 136.
16. *Ibid.*, p. 141.
17. Williamson Murray, *The Luftwaffe 1933–45: Strategy for Defeat* (Washington, DC: Brassey's, 1996), p. xxiv.
18. John Baylis and John Garnett, 'Introduction', in John Baylis and John Garnett (eds.), *Makers of Nuclear Strategy* (New York: St Martin's Press, 1991), pp. 1–2.
19. Williamson Murray and Mark Grimsley, 'Introduction: On Strategy', in Williamson Murray, MacGregor Knox and Alvin Bernstein (eds.), *The Making of Strategy: Rulers, States, and War* (Cambridge University Press, 1994), p. 22.
20. Daniel Moran, 'Strategic Theory and the History of War', in John Baylis *et al.* (eds.), *Strategy in the Contemporary World: An Introduction to Strategic Studies* (Oxford University Press, 2002), p. 17.
21. Clausewitz, *On War*, p. 141.
22. *Ibid.*, p. 578.
23. Colin S. Gray, *Modern Strategy* (Oxford University Press, 1999).
24. H. T. Hayden (ed.), *Warfighting: Manoeuvre Warfare in the US Marine Corps* (London: Greenhill, 1995), p. 43.
25. John Keegan, *A History of Warfare* (London: Pimlico, 2004).
26. Clausewitz, *On War*, p. 606.
27. *Ibid.*, p. 89.
28. Winston S. Churchill, quoted in M. Carver, 'Montgomery', in John Keegan (ed.), *Churchill's Generals* (London: Weidenfeld & Nicolson, 1991), p. 148.
29. Clausewitz, *On War*, pp. 100–12.
30. Gray, *Modern Strategy*, pp. 53, 108.
31. T. N. Dupuy, *A Genius for War: The German Army and General Staff, 1807–1945* (London: MacDonald and Jane's, 1977), p. 307.
32. Clausewitz, *On War*, pp. 595–6.
33. *Ibid.*, p. 596.
34. *Ibid.*, p. 272.
35. *Ibid.*, p. 75.
36. Robert A. Pape, *Bombing to Win: Air Power and Coercion in War* (Ithaca, NY: Cornell University Press, 1996), p. 275.
37. Thomas Cleary, 'Translator's Introduction', in Sun Tzu, *The Art of War*, trans. Thomas Cleary (Boston: Shambhala, 1988), p. 2.
38. Sun Tzu, *The Art of War*, trans. Thomas Cleary (Boston: Shambhala, 1988), pp. 77–9.
39. *Ibid.*
40. Lawrence Freedman, *The Revolution in Strategic Affairs*, Adelphi Paper 318 (Oxford University Press, 1998), p. 60.
41. Sun Tzu, *The Art of War*, pp. 92–3.
42. Mao Tse-Tung, *Selected Military Writings of Mao Tse-Tung* (Peking: Foreign Language Press, 1966).



2

Strategy defined

Contents

Introduction

Strategy defined

The levels of strategy

Why is strategy difficult?

 Multidimensional nature

 Disharmony amongst the levels

 Paradoxical logic

 Nature of war

 Friction

 Human complications

 Polymorphous character of war

Conclusion

KEY THEMES

- Strategy is a process that connects military power with policy effect.
- Strategy has various levels that must be in harmony.
- Strategy is a complex, challenging activity.

Introduction

Now that we have identified the need to develop an analytical approach to the subject, we must begin our exploration with some key definitions of strategy and its various levels. Having illuminated the essence of strategy, this chapter will analyse the various factors that make it difficult. In particular, the chapter will explore strategy's multidimensional nature; disharmony amongst the levels; the paradoxical logic; nature of war; friction; human participation; and war's polymorphous character. It is hoped that by the end of this chapter the reader will have a better understanding of the challenges involved in strategy, and how these can be dealt with so that the use of military force can best serve policy objectives.

Box 2.1 Strategy

The process that converts military power into policy effect

Strategy defined

Extant strategic literature contains various definitions of strategy. For Clausewitz, it can be understood as ‘the use of engagements for the object of the war.’¹ Similarly, Gray defines strategy as ‘the use that is made of force and the threat of force for the ends of policy’;² whereas Andre Beaufre highlights the dynamic interaction between belligerents: ‘the art of the dialectic of two opposing wills using force to resolve their dispute.’³ This work defines strategy as *the process that converts military power into policy effect* (see [Box 2.1](#)). Importantly, this definition identifies the key relationship in strategy: that between policy and the military instrument. Furthermore, it acknowledges strategy as process. The latter is recognition of that fact that strategy is not tangible, but is dynamic and does have manifest effect.

As noted, the core relationship within strategy is that between military force and the policy objective. Gray describes this relationship as a bridge that links the military and political worlds.⁴ Alternatively, Eliot Cohen described this relationship as an ‘unequal dialogue’ (see [Figure 2.1](#)).⁵ In this sense, we can regard strategy as a process by which military force creates political effect. Although the supremacy of policy is established and well understood, and thereby military force must serve policy, it is not simply a case of the political leadership demanding what it requires from the military instrument. Indeed, Clausewitz notes that although policy must remain the supreme consideration in the conduct of war, ‘[t]hat, however, does not imply that the political aim is a tyrant. It must adapt itself to its chosen means.’⁶ Thus, the military and political leaderships must engage in discussions concerning what policy requires and, just as importantly, what the military instrument can deliver. Hence, the significance of dialogue becomes clear in the quest for positive strategic effect. However, the dialogue lacks equity because the military instrument must ultimately serve policy goals.

A constructive dialogue may avert an obvious danger in such a relationship; namely, that policy-makers may not understand the instrument at their disposal. As Gray notes, this danger can manifest itself in two ways: ‘There is always a danger either that out of ignorance or incompetence policy will ask more of its military instrument than that instrument should be expected to deliver, or that policy will ask less of its “sword” than could, and perhaps should, be secured.’⁷ To mitigate the potential for such failures, some modicum of education in strategic studies for policy-makers would be welcome, but still would not guarantee success. Policy-makers may conduct poor strategy, even in the face



Figure 2.1 President Barack Obama and Vice President Joe Biden meet with members of the National Security Council, including Defense Secretary Chuck Hagel, centre right, and Army Gen. Martin E. Dempsey, chairman of the Joint Chiefs of Staff, in the Situation Room of the White House, 10 September 2014. Strategy serves political goals and acts as a bridge between the military and political world.

of good strategic theory, for a host of different reasons. For example, domestic political considerations may limit the choices available to policy-makers. Note, for example, the House and Senate votes against President Bush's decision to increase troop numbers in Iraq in early 2007. Aside from bolstering enemy morale and will, such a move creates an atmosphere that is not conducive to freedom of strategic choice. There is clearly an important but difficult balance to be made. Strategic freedom must be protected, but not at the expense of core democratic principles.

This suggests that the relationship between policy and the military is seemingly at its most complex in a modern democratic state. In such an environment the political and military leaderships often have little experience or understanding of their opposite number. In his excellent study of Field Marshal Sir Douglas Haig, John Terraine notes that within Haig's generation there was a substantial amount of mistrust and misunderstanding between the military and civil establishments. In addition, Terraine correctly concludes that Haig's 'absolute lack of interest in politics . . . cannot be considered as other than a serious defect'.⁸

Box 2.2 The levels of strategy

- Policy: the objective(s) sought/the guiding rationale for war.
- Grand strategy: the co-ordination of the state's resources towards the attainment of the policy objective(s).
- Military strategy: the use of military force to achieve the policy objective.
- Operational: the conceptual and/or geographical linking of tactical actions to military strategy.
- Tactical: actions in the battlespace in the face of the enemy.

The problems that emanate from the political–military relationship are not restricted to modern democratic states, however. Although on the surface this relationship seems much simpler for the likes Alexander the Great or Napoleon, problems still existed. In these instances the unification of the roles of both the political and military leaderships creates its own problems. Without an external dialogue to assess the strategic rationale of decisions, there may be a corresponding absence of any effective check on the strategic efficacy of certain actions. In the case of Alexander this problem intensified over time as he became more confident, paranoid, and increasingly sure of his divine destiny. By the time he had reached India his campaigns had lost a degree of strategic focus and subtlety of method. The result of this deterioration in his strategic performance was a long, bloody campaign that eventually provoked a mutiny in the army. Thus, the complexities of the military–policy relationship seem to be less a problem of command organisation (although this can exacerbate the difficulties), as both modern democracies and imperial dictators alike struggle to bring together these two worlds effectively. The problem is more conceptual; it is about means and ends, which at times can be very different in their nature and may even sometimes appear contradictory. Nonetheless, as difficult as this relationship often is, it is the beating heart of war and the very driving force of strategy.

The levels of strategy

Successful strategic performance requires far more than merely overcoming the challenges involved in the process of matching means to ends. It also requires competent performance across all of the levels of strategy. It will be instructive at this point to define the levels of strategy and comment on how they interact (see [Box 2.2](#)). The highest level in the taxonomy of strategy is *policy*. Policy is simply the overall objective that is sought. The range of policy objectives is essentially infinite, as Clausewitz so ably describes: ‘Policy, of course, is nothing in itself; it is simply the trustee for all these interests against other states . . . we can only treat policy as representative of all interests of the community.’⁹

Policy objectives should influence actions within the other levels of strategy; indeed, more specifically, they should strongly influence the methods used in the campaigns. However, this is easier said than done. Clausewitz's comment that the policy objective should guide the military instrument often creates problems when the political and military objectives are incompatible. Consider, for example, counterinsurgency campaigns, especially those with a strong nation-building element. How does one use a blunt military instrument to achieve subtle policy objectives? As is often the case, Clausewitz provides us with the answer. He notes that when the policy and military aims are not the same, a military aim should be chosen that leads to the fulfilment of the policy objective indirectly.¹⁰ For example, you should endeavour to give yourself a strong military position from which to negotiate a peace. Similarly, a strong security environment facilitates development projects in support of nation-building activities.

We should remember that politics, and therefore war, is about power. Perhaps at times we are guilty of over-thinking Clausewitz, and have tried to establish too close a relationship between policy and war. War cannot always itself directly lead to a policy objective, but it can alter the power relationships, through which policy can be pursued against the enemy. A more nuanced reading of Clausewitz would note that, while strategy is about the use of military force to achieve policy objectives, this relationship is not always straightforward or direct. In the first instance strategy is about the acquisition and use of power. Once gained, an imbalance in power can be used to achieve policy objectives. Thus the attainment of power represents the hidden step in Clausewitz's concept of strategy. Although Clausewitz does not explicitly outline this step in terms of power acquisition, it is evident in his work when he notes that 'war is thus an act of force to compel our enemy to do our will'.¹¹ This statement represents a definition of power that is remarkably similar to that offered by Raymond Aron, the theorist of international relations: 'the capacity of a political unit to impose its will upon other units'.¹²

With sufficient power, one can engage in the many subtle activities that achievement of the policy objective calls for: nation-building, civic works, reconstruction, providing security, to name just four. If, however, the enemy still retains sufficient power himself, he too can influence events, for example by coercing the population. In this sense, he may be able to challenge one's influence and authority. The relationship between military force and power is central to Admiral J. C. Wylie's often-overlooked work *Military Strategy: A General Theory of Power Control*. Wylie's basic assumption is that 'the aim of war is some measure of control over the enemy'.¹³ This is achieved by controlling the pattern of the war, which in turn is achieved by manipulating the enemy's centres of gravity. The means by which this occurs is the physical presence of armed men: 'The ultimate determinant in war is the man on the scene with the gun. This man is the final power in war. He is control.' Control is not an absolute condition. Rather, Wylie notes that it can be 'direct, indirect,

subtle, passive, partial or complete.’ Interestingly for the purposes of understanding the role of military force in the acquisition of power, when describing the more indirect examples of control, Wylie is happy to use the term ‘influence’ instead. Thus, when thinking about strategy, we should remember that although the relationship between war and policy objectives is not always direct or obvious, the former relates to the latter through the medium of power.

Once a policy objective has been decided upon, a *grand strategy* must be devised through which to pursue it. Grand strategy encompasses all the instruments at the state’s disposal: diplomatic, intelligence, military and economic. Basil H. Liddell Hart succinctly outlines the function of grand strategy: ‘to coordinate and direct all the resources of a nation, or band of nations, towards the attainment of the political object of the war – the goal defined by fundamental policy.’¹⁴ One of the main challenges in grand strategy is choosing the right instrument, or the right balance of instruments, to fulfil the particular policy objectives sought. This decision should be arrived at after taking into account a number of factors, including the policy itself, available resources, nature of potential enemies, strategic culture and the geopolitical environment, to name just five.

The aspect of grand strategy of primary interest to us here is military strategy. Once it has been established how military force will be used to serve the ends of policy, the strategy must be put into practice. This is achieved through actions at the *tactical* and *operational* levels. The former is concerned with actions in the battlespace in the face of the enemy. Therefore, for example, tactics includes how forces are deployed, how they engage the enemy and how various units interact with one another. In essence, tactics is about the details of combat. Each contact with the enemy represents a tactical event that occurs in a distinct time and place. At first, one could be forgiven for thinking that success at the military strategic level stems from a victorious outcome in every tactical event. However, successful military strategy requires a more developed understanding of the significance of tactical events. Thus, it is possible that tactical ‘failure’ can make an effective contribution to the attainment of a military strategic end state. A striking example of the latter is the Tet offensive launched by the Viet Cong in 1968. After initial, but short-lived, success, Tet was characterised by a series of tactical failures. However, the fact that the Viet Cong was able to launch such a range of tactical actions across South Vietnam seriously undermined morale within the United States. In this respect, it is the interactions amongst the different tactical actions (in the case of Tet, their cumulative effect on public and political perception in the United States) and their relationship with the military strategic level that matter most. It is at this point that the *operational* level performs its role. The operational level links tactical engagements with the overall military strategy. Edward Luttwak aptly describes it thus: ‘this operational level governs the consequences of what is done and not done tactically.’¹⁵ Put simply, in and of themselves, tactical actions may have little significance. Victory or defeat

Box 2.3 Why is strategy difficult?

- The relationship between war and policy is difficult and awkward.
- Strategy is multidimensional (including such dimensions as society, culture, economics and logistics, strategic theory and doctrine, command and time) and success requires a level of competence to be achieved in them all.
- Success requires harmony of actions amongst the levels.
- Strategy has a paradoxical logic. The enemy will attempt to offset one's advantages. What worked today may not work tomorrow.
- The nature of war is constant and must be respected, but puts pressure on the commander and his forces, especially in relation to modern norms and social values.
- Friction prevents war from achieving its maximum operational efficiency.
- War is a human activity imbued with non-rational, emotional forces.
- War has a polymorphous character. It can take many forms, each of which requires a subtly different approach. Often, a particular war will exhibit a mixture of different forms operating simultaneously.

at this lowest level is insignificant without reference to its military strategic implications.

The operational level has both conceptual and material elements. Conceptually, the operational level provides a framework within which tactical actions can be linked together so that they serve military strategy. So, for example, a series of tactical aerial bombing missions needs to be considered as a whole so that they produce a decisive coercive effect on the enemy. Materially, we can think in terms of a geographic area of operations, within which the operational level commander moves his forces from objective to objective. From a geographical standpoint there are a number of factors that the operational commander must consider in his quest for success. These include logistics and lines of communication, movements of the enemy, and decisive points in the theatre of operations such as cities and key terrain features. The above brief, linear description of the levels of strategy, valid though it is, does not reflect the complexity and non-linear nature of strategy. This chapter will now analyse those features of strategy that make it so complex and such a difficult activity to get right.

Why is strategy difficult?

The complexity of strategy is the product of a number of different factors, which taken together produce substantial challenges for the practitioner. The aforementioned awkward relationship between policy and military force is but one such factor. There are many other considerations that must be dealt with by the practitioner in his quest for strategic success (see [Box 2.3](#)).

Multidimensional nature

The notion of ‘dimensions’ helps us to understand the many different factors that impinge upon, or play a part in strategy. Depending upon which theorist you choose to read, the number of dimensions in strategy varies. Clausewitz identified five dimensions of strategy: moral, physical, mathematical, geographical and statistical. In the modern era Michael Howard, in his 1979 article, ‘The Forgotten Dimensions of Strategy’, identified four dimensions. The four dimensions in Howard’s taxonomy are technological, operational, social and logistical. The relative prominence of each dimension is dependent upon circumstance.¹⁶ More recently, in *Modern Strategy*, Gray produced a considerably more developed theory of the dimensions, and by doing so has dissected the art of strategy considerably further than either Clausewitz or Howard. In total, Gray identifies seventeen dimensions, which he organises into three categories: ‘People and Politics’, ‘Preparation for War’ and ‘War Proper’. Within these categories the dimensions include society, culture, economics and logistics, strategic theory and doctrine, command, and time, to name just six. Is seventeen the final and correct number of dimensions? As Gray himself notes, ‘the precise number [of dimensions] does not matter so long as everything of importance is properly corralled.’¹⁷

For our purposes, the significance of identifying the dimensions of strategy, and Gray’s analysis in particular, is that it highlights the substantial range of factors that must be considered and dealt with (see [Box 2.4](#)). The strategic waters are muddled further by the fact that complex interactions occur amongst the dimensions. And, as Gray notes, despite the range of dimensions and their complex interactions, success requires the practitioner to achieve a reasonable degree of competence in each of them. This challenge is further intensified by the fact that an enemy may have the advantage in one very significant dimension. For example, the English Channel gives Britain a substantial advantage in the geographical dimension over Continental foes. These significant dimensions, depending on the context, do not necessarily provide an automatic advantage, although they may provoke a degree of natural friction for the enemy. Rather, it is down to the practitioners themselves to exploit them to their advantage. Thus, the English Channel did not prove a substantial obstacle for the Roman or Norman invading forces. However, once the inhabitants of the British Isles had developed competent naval forces, the significance of the geographical dimension was substantially magnified.

Disharmony amongst the levels

The strategic practitioner has to achieve a degree of competence in each of the levels of strategy, as described above. However, he must also harmonise his actions amongst these levels. Actions at the tactical level must be in accord with those at the higher levels of strategy. Although this seems to be fairly

Box 2.4 Gray's dimensions of strategy

- People
- Society
- Culture
- Politics
- Ethics
- Economics and logistics
- Organisation
- Military administration
- Information and intelligence
- Strategic theory and doctrine
- Technology
- Military operations
- Command
- Geography
- Friction
- Adversary
- Time

straightforward advice, the realisation of it is far from easy. Clausewitz's following general point about strategy is especially relevant when considering harmony amongst the levels: 'Everything in strategy is very simple, but that does not mean that everything is very easy.'¹⁸ Indeed, Edward Luttwak notes that the normal state of affairs is measured disharmony. In his work *Strategy: The Logic of War and Peace*, Luttwak paints a disturbing picture for the strategist; one in which the challenges of achieving harmony are substantial. To illustrate, Luttwak cites the Japanese attack on Pearl Harbor. Although at the tactical and operational levels the attack was relatively successful, from the grand strategic perspective this success proved problematic. The scale of the attack compelled the United States to pursue a policy of unconditional surrender. From this position it can be argued that Japan was always destined to lose the resulting war. Thus, harmonisation of the levels of strategy is not necessarily concerned with achieving success within all of the said levels. Rather, it is about ensuring that actions within each level support, rather than undermine, efforts in the other levels of strategy. More limited, less aggressive Japanese action at the tactical and operational levels, although almost certainly aggravating the United States, may have provoked a more limited response, which may not have ended in total defeat for the empire.

Harmony also demands that both policy and grand strategy create a conducive and permissive environment for military strategy to play its part. The dangers inherent in getting this wrong are evident in the case of Germany in

the Second World War. Germany's overly ambitious policy objectives and flawed grand strategy ensured that tactical and operational prowess could not be fully exploited. Germany simply acquired too many powerful enemies, who, as a result of the nature of the Third Reich and its objectives, were beyond negotiation. Much like in the case of Japan, the enemies of Germany were set on a course of unconditional surrender. Despite substantial battlefield success for Germany, their enemies were constantly able, and willing, to put fresh forces in to the field.

To illustrate further we can also look to the Second Punic War, fought between Carthage and the Roman Republic. The Carthaginian commander, Hannibal, was astute at the tactical and operational levels. Thus, he was able to inflict a series of stunning battlefield defeats on Rome during his invasion of Italy. The most notable of these were the battles of Cannae and Lake Trasimene. At the former Hannibal destroyed two Roman armies, killing 60,000 men in a single day. However, he failed to translate these successes into a favourable policy outcome. In a curious decision Hannibal chose not to attack Rome itself. Thus, the Republic survived and was able to continually exploit its vast manpower resources. Taking into account the nature of his enemy, Hannibal's outstanding successes on the battlefield were out of kilter with the situation at the strategic and grand strategic levels. The effects of this disharmony were magnified by Rome's strategy, masterminded by Fabius Maximus, of avoiding battle under anything but the most favourable circumstances. This Fabian strategy gave Rome the time it required to mobilise its resources. Thus, we can appreciate Luttwak's notion that the challenge of harmonising the levels of strategy is further complicated by the actions of the enemy, which ensures that a complex series of interactions is taking place within and amongst the levels of strategy.

Paradoxical logic

Beyond the notion of harmony, Luttwak identified another central feature of strategy, which he terms the 'paradoxical logic'. This pervasive feature operates in many different ways, and can perhaps best be explained through a number of examples, which illustrate that the expected does not always occur, strength leads to weakness, and advantage does not always flow from a linear progression of events. Much of this is the result of interaction with a thinking enemy. For example, the paradoxical logic is at play when a manoeuvre that takes the more difficult, less logical route provides success. This may be the result of the enemy being caught by surprise. Alternatively, the paradoxical logic suggests that a successful military operation may begin to produce diminishing returns over time. This may occur as advancing forces move further from their sources of supply, or the enemy may simply adapt and subsequently counter a successful course of action. Such a situation is evident in the above example of Rome's adoption of the Fabian strategy, or more recently in Iraq once Saddam's regular forces had been defeated.

The war in Iraq offers a prime example of the paradoxical logic in action. As in the 1991 Gulf War, US-led forces easily overcame their Iraqi counterparts. However, in 2003 the defeat of the Iraqi regular forces merely represented the first stage in an ongoing struggle. Those wishing to challenge the presence of US-led forces sought to offset the conventional superiority of their enemies, and opted for irregular warfare, blending together elements of insurgency and terrorism. In this sense, the paradoxical logic began to operate at different levels for the US-led coalition. At the tactical and operational levels roadside bombs could inflict casualties on US forces and their allies, while not allowing their conventional superiority to be brought to bear. In addition, the insurgents could avoid contact with the coalition forces entirely by targeting civilians with car and truck bombs. Such tactics, and the associated casualties and destruction, created pressure at the higher levels of strategy within the coalition. Understanding that they could not militarily defeat the coalition, the insurgents attempted to break the will of political communities in the coalition, thus forcing a withdrawal.

Although it is certainly possible to overlay the significance of the paradoxical logic, interaction with the enemy is worthy of continued attention. This point is well made by Luttwak: 'Although each separate element in the conduct of warfare can be very simple . . . the totality of these simple things can become enormously difficult when there is a live enemy opposite, who reacts to undo everything being attempted, with his own moves and his own strength.'¹⁹ The same point is more starkly made by General George Pickett who, when asked why the Confederates lost at Gettysburg, replied: 'I think the Union Army had something to do with it.'²⁰

Nature of war

Those seeking to achieve policy objectives through the use of military force also have to contend with the nature of war. The nature of war is different from its character. The character of war is a constantly changing phenomenon. It changes depending upon a number of factors, including the geography of the battlespace, the belligerents and their level of technological development, to name just three. In contrast, the nature of war is unchanging across time and place. Like strategy, the nature of war is complex. It is composed of elements that are always present, but fluctuate in their relationships with each other. There are various conceptual means by which to describe the nature of war. As discussed in [chapter 1](#), Clausewitz's 'fascinating trinity' is one such conceptual device that encompasses rational and non-rational forces (policy, emotion, chance). The relative strength or influence of each of these forces varies with context. Thus, the nature of each particular war is unique, and can be found floating somewhere between these forces. From this position, Christopher Bassford concludes that the key to understanding the nature of war is appreciating the dynamics that occur within the trinity, and between the trinities of the various belligerents in a conflict.²¹ For Alan Beyerchen non-linearity is central to

the nature of war. This non-linearity emanates from the many complex interactions that occur and the moral forces that infuse war. Within such a non-linear world small fluctuations can have significant consequences.²²

In addition to non-linearity and the interplay of rational and non-rational forces, the nature of war includes other, ever present, aspects. Of particular significance are 'uncertainty' and 'violence'. In addition, war is of course characterised by human participation, and the many complexities this produces. These additional elements in the nature of war are reflected in another Clausewitzian concept, 'the climate of war', which the Prussian theorist describes as being composed of danger, exertion, uncertainty and chance.²³

Uncertainty in war comes from a number of sources. For Clausewitz, who declared that everything in war is uncertain, unreliable and/or incomplete information produces a metaphorical fog of war over the battlespace. In addition, uncertainty emanates from interactions between the opposing actors in war. These interactions produce non-linear results that are difficult to predict and challenging in terms of understanding their implications. In addition, human participation infuses war with powerful moral forces of an unquantifiable nature. Uncertainty may be further enhanced by the geography of the battlespace, especially in dense environments such as forests and urban areas. Often, such environments are used by irregular forces, who themselves add to the uncertainty since they rarely assume regular formations or wear uniforms. Finally, as information has to be handled and interpreted by human users, uncertainty may be the product of human error or bias.

Violence, or what Clausewitz describes as the 'dominance of the destructive principle', is inherent to the nature of war: 'War is a clash between major interests, which is resolved by bloodshed – that is the only way in which it differs from other conflicts.'²⁴ Although the levels of violence vary in war, it is always present or waiting in the wings. As Clausewitz notes: 'it is inherent in the very concept of war that everything that occurs *must originally derive from combat*'²⁵ (emphasis in the original). Although there are those who promote less violent forms of war, its presence and influence can never be eradicated. This is primarily because war has a dialectic nature. As Clausewitz recognised, the competitive aspect of war creates an escalatory dynamic as each belligerent attempts to seek an advantage. Thus, even if one belligerent in a conflict extols a less bloody method, the enemy may introduce higher levels of violence to gain the upper hand. The intensification of the destructive principle may seek to debilitate the enemy's capabilities, and/or break his will through the application of violence.

Strategy itself may require that enemy forces be physically destroyed, rather than just coerced or out-manoeuvred. In order to pursue the policy of unconditional surrender against Nazi Germany, as well as to leave post-war Germany militarily impotent, the Combined Chiefs of Staff issued the following directive to the commander Eisenhower: 'You will enter the continent of Europe and, in conjunction with the other Allied Nations, undertake operations aimed at the

heart of Germany and the destruction of her armed forces.' Eisenhower goes on to note, 'This purpose of destroying enemy forces was always our guiding principle.'²⁶

Violence complicates strategy because it provokes strong moral forces and has an attritional effect on the forces involved. Objectives must still be achieved despite the debilitating effect violence has on the commander, his forces and possibly also on the will of the domestic population and political community. Thus, those involved in strategy must hope, and indeed endeavour, to maintain sufficient will and capabilities to pursue objectives in the face of destruction and human suffering.

The significance of this violent, uncertain and competitive nature must not be underestimated. Those who do submit to this error run the risk of being at a disadvantage. As Clausewitz warns us: 'kind-hearted people might of course think there was some ingenious way to disarm or defeat an enemy without too much bloodshed'; he continues: 'If one side uses force without compunction, undeterred by the bloodshed it involves, while the other side refrains, the first will gain the upper hand.' In a similar vein Clausewitz warned: 'The fact that slaughter is a horrifying spectacle must make us take war more seriously, but not provide an excuse for gradually blunting our swords in the name of humanity. Sooner or later someone will come along with a sharp sword and hack off our arms.'²⁷

While this advice cannot be ignored without serious consequence, in the same instance war cannot be waged without reference to the social norms of the day. In a modern setting certain actions in war are simply not possible from an ethical standpoint. As noted by Gray, '[war] should be waged militarily in such a manner as not to sabotage its political goals.'²⁸ A particularly brutal form of war may cause such a backlash that the policy objectives could be lost amongst the condemnation. As with so much in strategy there is simply no straightforward solution to this dilemma. Instead, the strategist must rely upon his judgement to create the required balance. Again, Clausewitz is instructive: 'We can thus only say that the aims a belligerent adopts, and the resources he employs, must be governed by the particular characteristics of his own position; but they will also conform to the spirit of the age and to its general character. Finally, they must always be governed by the general conclusions to be drawn from the nature of war itself.'²⁹

Friction

The conduct of strategy is further complicated by the play of chance and the existence of friction. Again, this is an essential ingredient of Clausewitz's theory of war: 'No other human activity is so continuously or universally bound up with chance.' Friction can be understood as 'Countless minor incidents – the kind you can never really foresee – [that] combine to lower the general level

of performance, so that one always falls short of the intended goal.’ For Clausewitz friction is an essential ingredient in the reality of war: ‘Friction is the only concept that more or less corresponds to the factors that distinguish real war from war on paper.’³⁰ The various factors that make up friction, and thereby complicate the practice of strategy, have found theoretical form in Clausewitz’s ‘unified concept of general friction,’ as identified by Barry D. Watts. In his excellent study on future war Watts identifies the following taxonomy for the unified concept of general friction: danger; physical exertion; uncertainties and imperfections in information; friction in the narrow sense of resistance within one’s own forces; chance events; physical and political limits on the use of military force; unpredictability stemming from interaction with the enemy; and disconnects between ends and means in war.³¹

To complicate matters further, friction can occur as a result of, and be exacerbated by, the paradoxical logic. Paradoxical courses of action, for example to surprise an enemy, may forgo the normal, complete and logical procedures. As Luttwak states, these actions ‘will further increase friction and therefore the risk of organisational failure.’³² Naturally, certain elements of friction are also enhanced by the effects of interaction with the enemy. Losses of men and equipment in battle deplete the capabilities of a force, which in turn can hinder future operations. In addition, the enemy may actively seek to complicate one’s actions. For example, the pace of an advance may be severely curtailed by the enemy blowing a bridge.

Despite the potency of friction, and its potential to wreck military operations, its effects can be mitigated. Having identified this debilitating force in war, Clausewitz discusses how to deal with it. In particular, he suggests that human characteristics such as determination can go some way towards overcoming friction. Gray develops this argument and suggests a number of practical steps can be taken to help limit the influence of friction on performance. These include good and ample equipment, high morale, rigorous training, imaginative planning, historical education, combat experience and sensitivity to potential problems.³³ However, despite the ability to reduce the influence of friction, there is still much wisdom in the following advice from the USMC’s doctrine manual *Warfighting*: ‘the greater requirement is to fight effectively within the medium of friction.’³⁴

Human complications

At its heart war is a human activity: ‘in war and strategy people matter most.’³⁵ Thus, as Clausewitz reminds us war is infused with human emotions and concerns: ‘The art of war deals with living and moral forces.’³⁶ If we return to Clausewitz’s climate of war we see that much of it is concerned with the human dimension. Uncertainty is compounded by human perceptions, and friction partly occurs as a result of human actions. The final two elements of the climate of war, danger and exertion, relate directly to human participation. Within the

stress and exertion of war, the human actor, whether a general or a foot soldier, may be negatively affected in relation to their ability to think and act effectively. This complicates the task of the strategist by restricting his options and freedom of action. In addition, as Gray notes, 'war is usually so traumatic an experience that generally it follows a course and has consequences quite unintended by the belligerents. To go to war is to roll the iron dice.'³⁷ Thus, we can regard the effects of danger and physical effort as contributory factors to friction in war. The reality of human limitations, in both a physical and psychological sense, restricts the potential efficacy of armed forces.

Polymorphous character of war

War does not take a single regular form, for which the strategist can plan with certainty. Indeed, the reverse is actually true. In this sense, war can be said to have a polymorphous character. Historically, war has always taken a number of different forms. However, from an operational perspective the modern era has undoubtedly produced a more complex environment. Technological developments in the twentieth century facilitated the exploitation of the air, space and cyberspace environments. In addition, the century also witnessed further developments in the theory and practice of insurgency and terrorism. Finally, perhaps the greatest challenge for strategy has been the arrival of nuclear weapons. The extreme destructive nature of these weapons makes them difficult tools of policy. Each of these various areas of strategy will be discussed in detail in the following chapters. For now we must recognise that many of them overlap, creating new opportunities and challenges in the different environments. For example, maritime strategy has had to incorporate both air power and nuclear weapons. Thus, commanders in each geographical environment, as well as those running an entire campaign, have to deal with a more complex battlespace.

The polymorphous character of war is further complicated by the pace of technological developments. For some, the 1991 Gulf War heralded a so-called Revolution in Military Affairs (RMA), or in current parlance a military transformation. When an RMA occurs the advantage is said to go to those who fully exploit and operationalise the latest technological developments. Since the RMA debate remains of such import, within both academic and military circles, it is worth exploring the validity of this concept. In addition, by doing so we can assess the relative value of technology in strategy and gain insight into the relationship between the permanent and malleable aspects of war.

The development of blitzkrieg during the twentieth century offers a prime example of a candidate RMA, and thus represents an important test case for the RMA hypothesis. In the early twentieth century a range of new technologies appeared that could be utilised in the art of warfare. The internal combustion engine made possible the development of the tank and related logistics and

infantry vehicles. Taken together, these vehicles enabled more rapid manoeuvres in the theatre of operations. The command and control of these manoeuvres was facilitated by the advent of wireless radio. All told, this produced forces that could move rapidly in a co-ordinated manner and thereby achieve a higher tempo of operations.

However, merely having advanced technological capability is not enough. An adequate doctrine is required to fully exploit the potential of new technologies. The significance of this latter point is evidenced by the fact that the various European powers utilised these new forces differently in the early stages of the Second World War. The French, with their defensive and conservative outlook, dispersed their armoured forces throughout their existing infantry formations, essentially treating them as mobile artillery. This was hardly a revolutionary step forward. In contrast, Nazi Germany developed the operational concept of blitzkrieg. Within this new doctrine, concentrated armoured forces acted as the spearhead of rapid offensive formations. The immediate origins of blitzkrieg can be found towards the end of the First World War, when both sides created self-contained formations that could conduct deep and rapid advances along a narrow axis of attack. These concepts were brought to maturity by men such as Heinz Guderian.

Although tanks are powerful and rapid-moving instruments, when operating on their own they are vulnerable to infantry anti-tank weapons. To guard against this the Germans developed the Panzer division. This new formation contained tanks, infantry, artillery and support troops. This range of capabilities gave the Panzer division a degree of self-sustainability, and ensured that the different elements could offer each other mutual support and protection.

As evidenced in their invasions of Poland (1939), France (1940) and the Soviet Union (1941), blitzkrieg initially proved remarkably successful for Germany. However, as the war wore on the Germans began to reap diminishing returns from blitzkrieg. This can be attributed to a number of factors. The weather and geographic depth of the Soviet Union neutralised many of the advantages conferred by blitzkrieg. In addition, due to the substantial resources of their enemies, Germany was unable to translate tactical and operational successes into a war-winning outcome. Finally, Germany's enemies adapted to blitzkrieg, either imitating it themselves, or discovering ways to offset its advantages. Thus, the case of blitzkrieg demonstrates that the advantages conferred by operational innovation, great though they may be, can be offset. The complexities faced by the strategist do not often permit straightforward, operational answers to complex strategic questions.

As is evident from the RMA debate, it is all too easy to overestimate the significance of technological developments and their associated operational implications. The noted military historian J. F. C. Fuller is guilty of such a mistake: 'Tools or weapons, if only the right ones can be discovered, form ninety-nine per cent of victory.'³⁸ A military transformation methodological approach is also prone to see efficacious novelty where none exists, whereas a closer look

at history reveals that many seemingly new operational developments, such as joint operations, manoeuvre warfare or effects-based operations, have historical precedent. There is little, if anything, genuinely new in strategy. For example, Alexander the Great's military method is very reminiscent of the characteristics of blitzkrieg; albeit at a tactical rather than at the operational level. Christopher Tuck's chapters on land warfare describe how the key to German operational performance in the Second World War was not new technology per se, but the integration of units into an effective combined arms force. The resulting forces were thus able to manoeuvre more rapidly, deliver shock, penetrate the enemy's lines on a narrow front and finally exploit the gap to strike deep into the rear of the enemy's forces. We see a similar approach taken by Alexander in his three great battles against the Persian main army. Indeed, Alexander was an extremely astute exponent of combined arms warfare. His rapid tactical manoeuvres were led by a spearhead of heavy cavalry, which punched through the enemy's lines along a narrow axis. This lead unit, the Companion Cavalry, was supported by an evolved combined-arms formation, which included various types of infantry and troops with ranged weapons. Such an approach was devastating against the more limited tactical formations of Greek hoplite warfare. It also proved to be decisive against the more evolved forces of the Persian Empire. Like Alexander, the Persians put into the field a mixed army comprised of cavalry, infantry and missile troops. However, akin to the Germans in the Second World War, much of Alexander's advantage came from superior doctrine and training.

Thus, when reading the following chapters, which will discuss the different forms of warfare and how they have evolved, it is important to remember that the fundamentals of strategy never change, regardless of time or place.

The remainder of this book will discuss the range and complexity of forms war can take. In this sense, the character of any particular war can include weapons of mass destruction (WMD), regular conflict between uniformed and similarly organised foes and/or various forms of irregular warfare including terrorism and insurgency. The challenge posed by such a variety of forms concerns both the preparation and conduct of war. Each particular form of warfare requires a somewhat distinct set of capabilities, which in turn may require varied types of forces, equipment, doctrine and/or training. Charles Callwell, the Victorian theorist of irregular warfare, notes how 'The art of war, as generally understood [for regular forms of warfare] must be modified to suit the circumstances of each particular case. The conduct of small wars is in fact in certain respects an art in itself.'³⁹ Almost one hundred years after Callwell, Gray cautions against complacency in the face of the polymorphous character of war: 'Small wars can detract from the readiness of regular forces to take the field against other regular forces.'⁴⁰ For the strategist the challenge posed by this range of war forms is that he must endeavour to achieve a degree of competence across the spectrum of warfare. In this sense, one may require competence in a particular form of war, without being too specialised so that one cannot easily adapt to another form.

The task of the strategist is further complicated by the fact that particular wars often display a complex mix of different forms. Thus, the following counsel from Clausewitz is valid, but not always easy to achieve: 'The first, the supreme, the most far-reaching act of judgement that the statesman and commander have to make is to establish by that test the kind of war on which they are embarking.'⁴¹ Wars such as the US war in Vietnam and the war of American Independence contain elements common to both regular and irregular warfare. However, despite the fact that each particular form of warfare presents its own unique challenges, they are all still forms of war. In which case, they take place within the same complex nature discussed above. In particular, they are still violent, competitive conflicts acting in the service of policy. Thus, the battle of the wills is often paramount, and a theory of victory must exist. It is all too easy to give too much significance to the challenges of a particular form of warfare. This is nowhere more evident than in counterinsurgency (COIN).

Modern COIN practice, at least partially based upon seminal works such as Thompson's *Defeating Communist Insurgency*, tends to be regarded less as a form of war, and more as a security challenge, with popularity and legitimacy being the key to achieving the desirable end state. While there is certainly value in the hearts-and-minds aspects of COIN doctrine, we must never lose sight of the fact that COIN is still a form of war. In fact, an essential ingredient of COIN is inflicting serious military setback on the insurgents. This not only restricts their ability to undermine security in the contested territory, it also promotes a sense of authority for the local government and their allies. On this point Callwell is clear: 'Prestige is everything in such warfare.' Similarly, Julian Paget, in the classic work *Counter-insurgency Campaigning*, concludes that in order to win the support of the local population 'the Government must demonstrate its determination and its ability to defeat the insurgents.'⁴² Finally, Steven Metz notes: 'it is less an assessment of a preferred future that drives insurgents or insurgent supporters than an assessment of who will prevail – the insurgents or the regime.'⁴³

The successful British COIN campaign in Malaya provides a telling example. The campaign took twelve years, and included some fairly draconian methods. The latter included curfews, the mass forced relocation of 400,000 people, detention without trial and execution for possession of firearms or explosives. In addition, the insurgents were subjected to a protracted war of attrition, during which their numbers were slowly eroded, at the cost of 500 British deaths. In summary, rather than being a competition in popularity, COIN is rather a competition in authority. Once again, the contest over power is paramount.

Conclusion

Strategy, *the process that converts military power into policy effect*, is a multilevel complex activity. This complexity emanates from its multidimensional nature;

disharmony amongst the levels; the nature of war; the polymorphous character of war; the involvement of humans; the existence of an intelligent enemy; the play of friction; and the fact that it is the meeting point between the two different worlds of politics and the military. In the face of myriad difficulties, the strategist must strive to control the process of strategy. As indicated in the [previous chapter](#), in this daunting task he is aided by strategic theory, which orientates his mind towards strategic understanding. With such understanding in place, the practitioner can begin to use force in various ways to pursue the desired policy objectives. The proceeding chapter will explore these various uses of force conceptually, while also focusing on the modern strategic landscape.

RESEARCH QUESTIONS

1. How does strategy function as a process?
2. Is success at the strategic level more important than success at the tactical and operational levels?
3. Why is strategy so complex?
4. Can friction be overcome?
5. How do you maximise the chances for success in strategy?

FURTHER READING

- Aron, Raymond, *Peace and War: A Theory of International Relations* (trans. Richard Howard and Annette Baker Fox) (New York: Anchor Press/Doubleday, 1973). One of the best discussions of war and power in international politics.
- Baylis, John *et al.* (eds.), *Strategy in the Contemporary World: An Introduction to Strategic Studies* (Oxford University Press, 2002). An excellent introduction to the theory and practice of strategy in the contemporary world.
- Beaufre, Andre, *An Introduction to Strategy: With Particular Reference to the Problems of Defence, Politics, Economics, and Diplomacy in the Nuclear Age* (London: Faber and Faber, 1965). A classic work of strategic theory from the Cold War, which contains many insights into the complexities of the subject.
- Callwell, Charles E., *Small Wars: A Tactical Textbook for Imperial Soldiers* (London: Greenhill Books, 1990). An insightful analysis of British imperial strategy with many pertinent observations for the contemporary environment.
- Cohen, Eliot A., *Supreme Command: Soldiers, Statesmen, and Leadership in Wartime* (New York: The Free Press, 2002). An interesting study of the relationship between policy and the military instrument. The work benefits from some good historical case studies.
- Kane, Thomas M. and David J. Lonsdale, *Understanding Contemporary Strategy* (London: Routledge, 2012). A comprehensive introduction to strategy in all of its forms.
- Liddell Hart, Basil H., *Strategy: The Indirect Approach* (London: Faber and Faber, 1967). An interesting updating of Sun Tzu's main ideas.

Lonsdale, David J., *Alexander the Great: Lessons in Strategy* (London: Routledge, 2007).

Alexander the Great is still the best example of how to overcome the complexities of strategy and achieve outstanding success.

Summers, Jr, Harry G., *On Strategy: A Critical Analysis of the Vietnam War* (Novato: Presidio, 1982). An insightful and sobering analysis of what happens when strategy is not given its due respect.

ONLINE RESOURCES

Jablonsky, David, 'Why Is Strategy Difficult?', in Joseph R. Cerami and James F. Holcomb, Jr (eds.), *US Army War College Guide to Strategy* (February 2001), <http://permanent.access.gpo.gov/lps11754/00354.pdf>. This work enables the reader to begin to understand the complexities of strategy.

www.au.af.mil/au/awc/awcgate/awc-thry.htm#gen: a vast array of papers on strategy from the United States Air War College.

www.clausewitz.com: a site full of interesting papers on this most important of strategic thinkers.

www.infinityjournal.com/: an online journal devoted to strategy.

www.rand.org/topics/military-strategy.html: offers a wide selection of papers on strategy.

www.strategicstudiesinstitute.army.mil/: the website of the Strategic Studies Institute, which contains lots of in-depth analysis of important strategic issues.

www.strategicstudiesinstitute.army.mil/pubs/display.cfm?pubID=947: an important article by Colin S. Gray on strategic education.

www.strategicstudiesinstitute.army.mil/pubs/display.cfm?pubID=641: *Strategic Theory for the 21st Century*, Harry Yarger.

NOTES

1. Carl von Clausewitz, *On War*, trans. Michael Howard and Peter Paret (London: David Campbell Publishers Ltd, 1993), p. 177.
2. Gray, Colin S., *Modern Strategy* (Oxford University Press, 1999), p. 17.
3. Andre Beaufre, *An Introduction to Strategy: With Particular Reference to the Problems of Defence, Politics, Economics, and Diplomacy in the Nuclear Age* (London: Faber and Faber, 1965), p. 22.
4. Colin S. Gray, *War, Peace, and Victory: Strategy and Statecraft for the Next Century* (New York: Simon and Schuster, 1990).
5. Eliot A. Cohen, *Supreme Command: Soldiers, Statesmen, and Leadership in Wartime* (New York: The Free Press, 2002), p. 208.
6. Clausewitz, *On War*, p. 87.
7. Colin S. Gray, 'New Directions for Strategic Studies? How Can Theory Help Practice?', in Colin S. Gray, *Strategy and History: Essays on Theory and Practice* (London: Routledge, 2006), p. 43.
8. John Terraine, *Douglas Haig: The Educated Soldier* (London: Cassell & Co., 2000), pp. 31, 53.

9. Clausewitz, *On War*, p. 606.
10. *Ibid.*, p. 81.
11. *Ibid.*, p. 75.
12. Raymond Aron, *Peace and War: A Theory of International Relations*, trans. Richard Howard and Annette Baker Fox (New York: Anchor Press/Doubleday, 1973), p. 44.
13. J. C. Wylie, *Military Strategy: A General Theory of Power Control* (Annapolis: Naval Institute Press, 1967), p. 66.
14. Basil H. Liddell Hart, *Strategy: The Indirect Approach* (London: Faber and Faber, 1967), pp. 335–6.
15. Edward N. Luttwak, *Strategy: The Logic of War and Peace* (Cambridge: The Belknap Press of Harvard University Press, 1987), p. 69.
16. Michael Howard, 'The Forgotten Dimensions of Strategy', *Foreign Affairs*, 57 (1979), 976–86.
17. Gray, *Modern Strategy*, p. 24.
18. Clausewitz, *On War*, p. 178.
19. Luttwak, *Strategy: The Logic of War and Peace*, p. 7.
20. Quoted in R. L. DiNardo and Daniel J. Hughes, 'Some Cautionary Thoughts on Information Warfare', *Airpower Journal*, 9(4) (1995), 76.
21. Christopher Bassford, 'Primacy of Policy versus the Trinity', paper presented at the 'Clausewitz in the 21st Century' Conference, Oxford Leverhulme Programme on the Changing Character of War (Oxford, 21–23 March 2005).
22. Alan D. Beyerchen, 'Clausewitz and the Nonlinear Nature of Warfare', paper presented at the 'Clausewitz in the 21st Century' Conference, Oxford Leverhulme Programme on the Changing Character of War (Oxford, 21–23 March 2005).
23. Clausewitz, *On War*, p. 104.
24. *Ibid.*, p. 149.
25. *Ibid.*, p. 108.
26. Dwight D. Eisenhower, *Crusade in Europe* (London: William Heinemann Limited, 1948), p. 247.
27. Clausewitz, *On War*, pp. 83–4, 718.
28. Gray, 'From Principles of Warfare', p. 86.
29. Clausewitz, *On War*, p. 718.
30. *Ibid.*, pp. 119–21.
31. Barry D. Watts, 'Clausewitzian Friction and Future War', McNair Paper 52, Washington, DC: Institute for National Strategic Studies, National Defense University, October 1996, p. 32.
32. Luttwak, *Strategy: The Logic of War and Peace*, p. 14.
33. Gray, *War, Peace, and Victory*, pp. 107–8.
34. H. T. Hayden (ed.), *Warfighting: Manoeuvre Warfare in the U.S. Marine Corps* (London: Greenhill, 1995), p. 38.
35. Gray, *Modern Strategy*, p. 97.
36. Clausewitz, *On War*, p. 86.
37. Gray, 'From Principles of Warfare', p. 87.
38. J. F. C. Fuller, *Armament and History: A Study of the Influence of Armament on History from the Dawn of Classical Warfare to the Second World War* (London: Eyre and Spottiswoode, 1946), p. v.

- 39. Charles. E. Callwell, *Small Wars: A Tactical Textbook for Imperial Soldiers* (London: Greenhill Books, 1990), p. 23.
- 40. Gray, *Modern Strategy*, p. 279.
- 41. Clausewitz, *On War*, p. 88.
- 42. Julian Paget, *Counter-insurgency Campaigning* (London: Faber and Faber, 1967), p. 176.
- 43. S. Metz and R. Millen, 'Insurgency and Counterinsurgency in the 21st Century: Reconceptualising the Threat and Response', Strategic Studies Institute, US Army War College, PA, November 2004 (www.au.af.mil/au/awc/awcgate/ssi/insurgency21c.pdf), p. 5.



3

The practice of strategy

Contents

The use of force

- Defence
- Deterrence
- Compellence
- Posturing
- Offence
- Miscellaneous

Strategy in the contemporary environment

- A complex reality
- Cyberwar
- Targeted killing
- Strategic ethics

Conclusion

KEY THEMES

- Military force is a flexible instrument of policy.
- The contemporary strategic environment is complex and requires astute strategy.
- Certain trends, such as cyberwar and military ethics, interact in complex and often unpredictable ways.

Introduction

Strategy may be complex and difficult. And yet, perhaps with the aid of theory, experience and ideally military genius, the strategist can, indeed must, utilise the military instrument in the service of policy. This can be done in a number of ways. Highlighting the fact that military power is a far more flexible instrument than many assume, this chapter explores the various uses of force in the modern world. These will be divided into six categories: defence, deterrence, compellence, posturing, offence and miscellaneous. It should of course be noted that the actual use of force will often simultaneously cover a number of these

categories. It is important to remember that the categorisation of anything rarely reflects the complexities of reality in an absolute sense.

Often, discussion is limited to the first four categories. The exclusion of the fifth and sixth categories may reflect a philosophical and/or ethical shift away from regarding military force as a useful, flexible or legitimate instrument of policy. However, the conflict-ridden international security environment suggests that this is clearly a naive philosophical approach. Take, for example, the recent history of Afghanistan and Iraq. Regardless of the subsequent challenges encountered by the counterinsurgency campaigns in these two wars, it is undeniable that the offensive use of military power removed the Taliban and Ba'athist regimes.

Having discussed the use of force in general conceptual terms, the chapter will conclude by analysing the current and future state of strategy. In particular, the work will explore the interplay amongst various developments in the strategic landscape: military transformation, irregular warfare, nuclear strategy, cyberwar, targeted killing and military ethics.

The use of force

Defence

Generally speaking, defending the state/community is the primary function of military force. The defensive use of force has two functions: to repel an attack and to limit the damage caused should an attack occur. What actually constitutes an act of self-defence has become something of a debate since the terrorist attacks of 9/11, and the Bush administration's response to it. Traditionally, it has been generally assumed that self-defence can only occur in the face of an armed attack, or if such an attack is imminent. However, the Bush administration's National Security Strategy declared, in response to more insidious threats, that both pre-emptive attacks and preventative war must be considered as forms of the defensive use of force. Of course, the problem with this position is that an overlap begins to develop between the defensive and offensive use of force. While this may be of import for international lawyers and academics, if a preventative war achieves its motivating policy objectives, perhaps it matters little how it is categorised. Facts on the ground are more compelling than theoretical or legal arguments. This argument, however, only holds if legitimacy is not a necessary concern. If observations of legitimacy matter, then how an act of force is perceived will have strategic bearing.

Deterrence

The defensive use of force is a physical act. In contrast deterrence, compellence and indeed posturing are dependent upon the psychological effects of military power. Deterrence 'has the negative object of persuading an adversary not to

take an action he might otherwise have done'.¹ This can be achieved through the threat of punishment and/or denial. Punishment seeks to deter an actor by threatening to respond in such a way as to make the costs of the enemy's actions outweigh the potential benefits. Deterrence by denial seeks to persuade an actor that his goals cannot be achieved.

Deterrence will be discussed in detail in C. Dale Walton's chapter on nuclear strategy. In the context of this chapter it is worth noting that deterrence as strategy has its own challenges. Increasingly, the reliability of deterrence is being questioned. By default, the efficacy of deterrence is difficult to prove, in that one cannot prove a negative. To have a chance of being successful a strategy of deterrence must be credible. To achieve credibility deterrence must fulfil three criteria: capability, commitment and communication. To deter, an actor must possess the capability and commitment to fulfil the threatened punishment or denial. Just as importantly, these must be communicated to the enemy, otherwise the basis for the deterrent effect remains hidden. Deterrence can apply to all forms of military power. Nonetheless, it is most closely associated with nuclear strategy. Depending upon the context, when nuclear weapons are involved the commitment criterion is problematic. For example, during the Cold War, when mutually assured destruction (MAD) had become a reality, the United States had to persuade the Soviet Union that it was willing to commit suicide in order to defend Western Europe from Soviet attack.

In the complex world of strategy, fulfilling the three above criteria is not sufficient to guarantee success. As Lawrence Freedman notes, 'deterrence works best when the targets are able to act rationally, and when the deterrer and the deterred are working within a sufficiently shared normative framework'.² Cultural differences regarding rational choices may cause a mismatch in the deterrence relationship. It is also plausible that some actors are beyond deterrence. This may be the case for some within radical terror groups. However, even radical terrorist organisations may be deterred from attacking certain targets if they are well protected enough. After all, failure to destroy the target can have negative strategic consequences for a terrorist campaign. To further illustrate, Gray notes that the recruitment of terrorist foot soldiers becomes more difficult if the cause appears to be losing momentum.³ In both cases deterrence by denial seems to have a role to play in deterring certain terrorist organisations, who initially seem beyond deterrence.

In the final analysis, despite the challenges of deterrence, it has always been, and will remain, an important instrument of strategy. Problems arise when too much is expected of it. Like all forms of strategy, deterrence has limitations. Thus, deterrence should be considered as merely one possible instrument of grand strategy: 'deterrence does not offer a self-contained strategic relationship but is part of a wider set of relationships'.⁴ In the final analysis, it must be recognised that the success or failure of deterrence will be decided by the actor to be deterred.

Compellence

Along the same lines as deterrence, compellence aims 'to be able either to stop an adversary from doing something that he has already undertaken or to get him to do something that he has not yet undertaken'.⁵ Like deterrence, compellence relies upon the psychological effects of force. In this sense, the situation is not completely under one's control: 'The adversary must still have the capacity for organised violence but choose not to exercise it.'⁶ Thomas Schelling, one of the most influential writers on the subject describes this difference: 'There is a difference between taking what you want and making someone give it to you.'⁷ Although compellence can be achieved without the application of violence, it is the threat of violence that still compels. Indeed, Schelling argues that the power to hurt is actually most effective when held in reserve. In this sense, it is the threat of more pain to come that compels. Compellence is particularly suited to pursuing limited objectives, or when the attacker wishes to engage in some form of post-conflict positive relationship with an enemy. Compellence may also prove useful for those with limited resources or without the will to sustain a long and costly attritional struggle. To this end, a carefully targeted application of force against valued assets of the enemy may bring success at reasonable cost.

Of course, the entire notion of compellence rests on one's ability to understand what the enemy values, and the amount of pain that will compel him to alter his behaviour. Sun Tzu's advice to know your enemy is especially useful in such circumstances. However, estimating the enemy's pain threshold is notoriously difficult. For example, during the 1999 Kosovo conflict elements within the US leadership predicted that a mere three days of bombing would bring Serbia back to the negotiating table. In the end, the campaign lasted seventy-eight days. And, although the coercive air campaign did have an effect on Serbia's decision-making process, other factors were also significant. Particularly worthy of mention are the withdrawal of Russian support from its traditional Serbian ally, and increasing discussion and preparations for a NATO ground offensive.

Like deterrence, compellence clearly has its limitations. However, it is a vital form of strategy. In fact, few wars end with the absolute overthrow of the enemy. Thus, most conflicts cease when one side has been coerced. In this sense, the compellent use of force is central to strategic effect. However, curiously the ability to inflict pain for strategic effect is often overlooked. Schelling bemoans this lack of recognition: 'It is extraordinary how many treaties on war and strategy have declined to recognise that the power to hurt has been, throughout history, a fundamental character of military force and fundamental to the diplomacy based on it.' This problem, which Schelling noted in the 1960s, has only intensified in more recent times. It seems plausible that 'the power to hurt' has been increasingly undermined by social, ethical and legal concerns. Inflicting pain to achieve policy objectives does not sit well with prevailing liberal values.

Nonetheless, unpalatable though it might be, much like the nature of war, the power to hurt remains an important ingredient in strategy.

Posturing

Often indirectly supportive of deterrence and compellence is 'swaggering' or 'posturing'. This use, or more specifically this 'display' of military capabilities, is designed to enhance the reputation of an actor, without necessarily being aimed at any other actor in particular. In this sense it has a general rather than a specific objective in mind. Posturing can be seen in such actions as military parades, or in the purchase of advanced capabilities. It can also be supported by 'defence diplomacy', whereby a naval port visit not only greases the wheels of diplomacy, but can also be used to impress upon the hosting nation the power of the visiting force.

Offence

As noted, often missing from discussion on the use of force is the offensive use of military power. Again, as in the case of the power to hurt, Schelling recognises the many functions that an offensive use of force can perform: 'penetrate and occupy, seize, exterminate, disarm and disable, confine, deny access'. Specifically, in an offensive manner military force can seize or destroy resources, force political change (which may include occupation) and degrade or eradicate an enemy. To this end, military force can be the instrument by which 'control' is exercised in a much more direct and physical sense than would be the case with deterrence or compellence. However, it must be remembered that the different uses of offensive force have varying strategic effect, dependent upon context. To take one example that reveals the benefits and limitations of differing offensive uses of force, US drone strikes in recent years have seriously degraded the capability of Islamist insurgency groups, but they have not contributed significantly to the achievement of lasting solutions. The latter are equally, if not more dependent upon security provided by ground forces, in addition to the workings of broader grand strategic instruments.

Miscellaneous

In addition to the above, military force has many other, less headline-grabbing, uses (see [Box 3.1](#)). Falling within the miscellaneous category is a vast range of activities, including, but not restricted to, policing, humanitarian aid, disaster relief, ceremonial, counter-smuggling operations and garrison duties. Some of these activities are often corralled under the heading of Military Assistance to Civil Authorities (MACA). The latter can include providing a stand-by fire-fighting force in the event of industrial action, or helping local civil authorities and water companies during floods. A larger MACA commitment can be

Box 3.1 The many uses of force

- Defence: to repel an attack or limit the damage should one occur.
- Deterrence: to dissuade an adversary from taking certain actions through the threat of punishment or denial.
- Compellence: to persuade an adversary to act in a certain way through the threat of, or actual, infliction of pain.
- Posturing: to enhance the strategic reputation of an actor through a display of military power.
- Offence: many uses, including occupation, seizure, extermination, disarmament and confinement.
- Miscellaneous: includes policing, humanitarian aid, disaster relief, ceremonial, counter-smuggling operations and garrison duties.

witnessed in the use of the British Army in support of police operations in Northern Ireland. As the above discussion indicates, military force is far from being a blunt and limited instrument. Rather, it represents a very flexible tool of policy, albeit one that is underpinned by the ability to deliver violence to exert influence and power. This of course is not the case in relation to some MACA activities, which instead are based upon the military's range of capabilities and skills.

Strategy in the contemporary environment

A complex reality

As the Cold War came to an end, a certain optimism arose in the defence communities of the West. The Revolution in Military Affairs/military transformation appeared to promise a leap forward in the efficiency of warfare. With increasing levels of information available to commanders, the fulfilment of concepts such as Dominant Battlespace Knowledge and Total Situational Awareness seemed possible. Linked to precision-guided munitions, assured kills became a distinct possibility. It was thought that under such conditions a networked RMA force would be dominant, to the point at which the acquisition of information itself could decide the outcome of a conflict. The possible futures that emanate from such visions promised to revolutionise the very act of war. Attritional forms of warfare would become anachronistic, to be replaced by effects-based operations. The cohesion of the enemy could theoretically be broken by a few well-placed attacks, severely reducing casualties and levels of destruction. In its most extreme form, the future promised 'humane warfare', perhaps without any significant levels of violence to speak of.

Much of the optimism upon which the RMA hypothesis was founded emanated from the experience of the 1991 Gulf War. In response to the Iraqi

invasion of Kuwait, an American-led coalition defeated the world's fourth-largest army at the cost of less than 500 coalition fatalities. Further validation appeared to come with the Kosovo conflict in 1999. Despite some significant problems of target identification and destruction, NATO forces did finally coerce Serbia to halt its campaign of ethnic cleansing and withdraw its forces from the province. Most remarkably, this was achieved without any NATO combat fatalities. The promise of low-cost, tightly regulated, humane warfare seemed to be within reach. However, the theorists and commentators of the RMA had largely restricted their discussions to the tactical and operational levels. They committed the error of regarding victory purely in military terms. A theory of victory must be constructed with reference to the policy objective. Although military success often acts as the basis for policy attainment, in the complex world of strategy it is rarely sufficient on its own.

The RMA debate also suffered from the fact that it tended to focus on regular warfare. 9/11 was a stark reminder of the polymorphous character of war. The al-Qaeda attacks on New York and Washington brought back into sharp focus a form of war that had received little attention amongst the euphoria of the RMA debate. This is not to suggest that there had not been some warnings of the growing threat from irregular forms of war. A section of the Strategic Studies community had been debating for some time the rise of asymmetric forms of war, and in particular the potential for super-terrorist acts. Indeed, a growing body of literature had warned that the very success of the RMA would force enemies of the West to opt for asymmetrical strategies. Stephen Metz defines asymmetric warfare as 'acting, organizing and thinking differently from opponents to maximize relative strengths, exploit opponents' weaknesses or gain greater freedom of action. It can be political-strategic, military-strategic, operational or a combination, and entail different methods, technologies, values, organizations or time perspectives.'⁸

Somewhat understandably, in the wake of 9/11 Western defence communities adopted a vision of the future which was dominated by the war on terror. However, they did not completely abandon military transformation. Indeed, they tried to adapt the concept to the new threat, with disastrous results in Iraq. Under the guidance of Donald Rumsfeld as secretary of defence, the invasion and occupation of Iraq was undertaken with a relatively small force, on the premise that military transformation allowed you to do more with less. However, occupation and counterinsurgency have traditionally been manpower-intensive activities, relying less on high-tech weaponry and more on 'the man on the scene with a gun'. Just as limited-war theory was found wanting in Vietnam, so too military transformation looked inadequate in the face of the polymorphous, complex character of war as revealed in Iraq.

Just as worrying as the attempt to shoehorn military transformation into areas where it does not belong, is the debate surrounding the future of large-scale war. The contemporary security environment has fostered concepts such as fourth-generation warfare and hybrid warfare, along with the dubious claim

that large-scale war has become a thing of the past. Understandably, defence communities tend to focus on the immediate problem to be faced, sometimes at the expense of preparation for an alternative, complex future. This can have serious debilitating effects on future performance. For example, at the close of the First World War the British Army was unprepared to return to its duties of imperial policing. The obvious, though challenging, answer to this dilemma is not to assume that the future will be dominated by one particular form of war. From a practical perspective this means that despite the temptation to focus all of one's attentions on the immediate strategic challenge, an eye must be kept on the future polymorphous character of war. Alternative capabilities must be kept active.

It is prudent to work on the basis that the future, akin to the past, will be characterised by a mixture of styles of warfare. It is only in our minds that one period of history is dominated by any one form of war. Unfortunately, such an approach can become manifest in the world of war preparation. For example, prior to Vietnam the United States Air Force (USAF) had a doctrinal focus on the delivery of nuclear payloads in preparation for a war with the Soviet Union. Clearly, this left the service initially unprepared for its role in Vietnam.

Our picture of the future is further complicated by the proliferation of WMD. Although discussion of non-proliferation has retained a certain degree of prominence, nuclear strategy has received little attention since the end of the Cold War. Yet, it is essential to keep the strategic flame alive on this most challenging of aspects of the art. University undergraduate and postgraduate programmes are currently dominated by students pursuing studies on terrorism and insurgency. It is rare to find one interested in nuclear strategy. This negligence of the subject is also evident in the policy world. The Obama administration, along with recent British governments, has neglected nuclear strategy to some degree. Although both the United States and Britain have retained a nuclear capability, they have committed themselves to the goal of disarmament. Increasingly, policy discourse on the subject is light on discussion of nuclear strategy, yet heavy on further disarmament. One is left with the distinct impression that the United States and Britain are reluctant nuclear powers. This raises questions about the commitment of the West's primary nuclear powers to nuclear use. As noted, this is one way in which the credibility of deterrence can fail.

In contrast to prevailing policy attitudes, we need an energised debate on nuclear strategy, one that constructs a meaningful place for these instruments in the modern strategic environment. Undoubtedly, nuclear weapons represent a significant challenge for the strategic thinker and practitioner. However, as was noted earlier, their appearance was responsible for the great classic works of theory in the Cold War. Modern Strategic Studies, in both its military and civilian spheres, must devote more attention to nuclear strategy. If it does not, we will have to rely upon an ad hoc approach when the real challenges appear. To reference Herman Kahn, it is time to rethink the unthinkable.

Cyberwar

In contrast to the relative dearth of thinking on nuclear strategy, 'cyberwar' continues to attract significant amounts of attention. The spectre of the great powers being crippled by an unseen virtual enemy continues to grip security discourse: 'Individuals can now thwart authority and conduct asymmetrical attacks that can paralyze an entire infrastructure and stall communications, and the weakest systems can now threaten the security of the greatest of nations.'⁹ These concerns have been echoed in the highest policy circles. President Obama reflects the consensus view when he declared: 'It's now clear this cyber threat is one of the most serious economic and national security challenges we face as a nation.'¹⁰

Although there is much hyperbole concerning cyberwar, there is considerable empirical evidence to support growing concern over the cyber threat. From the 1994 Rome Labs incident to the discovery in 2014 of the Careto operation, vulnerability to cyber-attack remains extant.¹¹ From a general security perspective, the statistics concerning cyber-attacks are compelling, and make sober reading for information-age societies. It is now well understood that socio-economic wellbeing is dependent upon a functioning critical information infrastructure. Most critical services are reliant upon computer networks and E-commerce now totals over \$10 trillion.¹² It is thus of great concern that the software security firm McAfee identified 20 million new pieces of malware in 2010.¹³ By 2013 McAfee were discovering one new malware every second.¹⁴ Just as worrying, US military networks are probed and scanned approximately 250,000 times every hour.¹⁵ In terms of the damage that can be inflicted, the Love Bug virus presents a pertinent example. Developed by a lone hacker in the Philippines, Love Bug caused an estimated \$15 billion worth of damage.¹⁶

In the face of this threat governments around the world are investing considerable amounts of energy and resources. In the fiscal year 2010, major executive branch agencies in the United States spent \$12 billion on cyber-security.¹⁷ That same year, US Cyber Command became operational. Across the Atlantic, in its 2010 National Security Strategy, the British government classified hostile cyber-attacks as one of four tier-one security threats.¹⁸ In response, the government has enacted a National Cyber Security Programme, backed by £650 million over four years.¹⁹ In March 2014, in support of its National Cyber Security Strategy, the United Kingdom launched CERT-UK, the UK's first Computer Emergency Response Team.²⁰

In terms of national security, with an eye to the tactical level of analysis, cyber threats certainly are worthy of attention. Critical infrastructures are undoubtedly vulnerable to cyber-attack. However, once the analysis is elevated to the strategic level (acting in the service of policy) cyberwar looks far less impressive. Considered as an instrument of strategy, there are eight prominent factors that retard cyberwar's efficacy: reductionism, geography, intelligence, the enemy, the polymorphous character of war, ethics, lack of commitment and friction.

Akin to the strategic bombing literature, cyberwar discourse overly simplifies the complex, multidimensional, polymorphous activity of strategy. It assumes too readily that strategic success can be reduced to tactical performance in one field of activity. Furthermore, said tactical success is often taken for granted. In doing so, cyberwar enthusiasts underestimate the difficulties of conducting any form of military operation. They often ignore the play of friction, the actions of the enemy, the influence of ethical and political concerns and the intelligence challenge of conducting major cyber-attacks. Cyberwar analysis has a tendency also to ignore the dominant role played by physical geography in human affairs. This is not to ignore the significance of the virtual environment. Rather, it is merely recognition of the fact that the drama of human politics is primarily played out on solid terrain. Finally, even when tactically potent, cyberwar lacks the statement of political commitment associated with physical troop deployment. In the final analysis, we are left to conclude that cyberwar deserves its place alongside the established tools of policy, but in most cases it is likely to act as the supporting rather than the supported arm. In terms of independent actions, cyber activities are best suited to small raids (Stuxnet) and espionage.

Targeted killing

Alongside cyberwar, the contemporary security environment remains awash with irregular warfare. At the time of writing, the conflicts in Afghanistan, Iraq and Syria continue to rumble on. Indeed, the latter two have become linked and have evolved into a more developed form of conflict between ISIS and its enemies. The growth of ISIS is indicative of how irregular forms of conflict often develop into more substantial security threats. With echoes of Mao's stages in revolutionary war, albeit somewhat more complex, the threat from militant Islamists in the region has developed from sporadic uprising, through insurgency, to a more regular territorially based threat.

Interestingly, having fought Islamists on the ground in protracted counter-insurgency campaigns since 2001, Western powers have turned increasingly to air power as their lead instrument. This is not a wholly new approach. In the latter stages of Vietnam the United States provided air support for Army of the Republic of Vietnam (ARVN) ground forces. However, by this stage of the war the conflict had become far more regular in character, with the North Vietnamese Army (NVA) providing the majority of communist ground forces. The contemporary variant of using kinetic air power in irregular warfare covers a range of air power missions: close air support, interdiction and targeted killing. Close air support and interdiction are well-established uses of air power. Their use in irregular warfare strongly mirrors that seen in regular warfare. Specifically, these missions seek to degrade the mobility and capability of enemy forces, while acting as a force multiplier to friendly ground forces – the latter often being lightly armed infantry units. Targeted killing represents a rather more



Figure 3.1 An MQ-9 Reaper unmanned aerial attack vehicle descends into an airfield in Afghanistan. The Reaper is able to carry both precision-guided bombs and missiles and represents the kind of technology often associated with theories of an RMA.

innovative use of force in irregular warfare, and is therefore worthy of special attention.

Targeted killing, ‘premeditated acts of lethal force employed by states in times of peace or during armed conflict to eliminate specific individuals outside their custody’, has grown in prominence in recent years, owing largely to the activities of Israel and the United States.²¹ The former used this form of operation extensively as part of its campaign against Hamas and other groups during the Second Intifada (2002–6), although there were precedents going back to the 1970s.²² Similarly, the United States has increasingly turned to targeted drone strikes in its war against al-Qaeda and its affiliates (see [Figure 3.1](#)). A study by the New America Foundation revealed that between 2009 and January 2013 the United States conducted 291 drone strikes, killing between 1,299 and 2,264 militants. As the Israeli example reveals, targeted killing can come in many forms: airstrikes, snipers, even tank fire. However, it is the aerial variety that has garnered most attention in recent years. Due to the global nature of America’s war on terror, and the geographic location of many of the terrorist targets, the United States relies heavily upon airstrikes for targeted killing.²³

Targeted killing has proven to be controversial, with many commentators and political actors questioning its legality. To some, since targeted killing primarily occurs in the legal grey area of irregular warfare, it is no more than extrajudicial killing by the state. In response, those conducting such operations have argued that they are legitimate, representing pre-emptive or preventative acts of self-defence. Indeed, officially the United States conducts targeted killing under the 2001 Authorization for the Use of Military Force, which authorises military action against those responsible for the 9/11 attacks. Despite claiming a legal basis for its actions, the Obama administration has sought to shore up the

legitimacy of these operations by issuing detailed guidelines and greater oversight of targeted killing.²⁴

Targeted killing can be used for three main strategic purposes: destruction of a target en route to an attack; the killing of key individuals to degrade enemy capability (e.g. taking out key terrorist bomb makers); and killing enemy leadership targets as an act of coercion or to undermine the political stability of the target organisation. From an air power perspective, the third category appears to sit comfortably within Warden's five-ring model of modern air power.²⁵ Avi Kober also claims, somewhat convincingly, that targeted killing fits well with the West's proclivity for post-heroic forms of warfare. Although there are distinct advantages (primarily intelligence) to be gained from apprehending a terrorist, there are substantial risks when such operations are conducted in hostile territory. Thus, targeted killing, especially from the air, represents a relatively low-risk method of taking the fight to the terrorist enemy.²⁶

As this first section of the book is concerned with strategy, the obvious question to be asked is: are targeted killings an effective method in the pursuit of policy? A full answer to this question is dependent upon each specific context – namely, the precise policy objectives sought. Nonetheless, in general terms it is reasonable to assert that counterterrorism campaigns seek to reduce terrorist violence while not undermining attempts at long-term political solutions to a conflict. In the following analysis targeted killing will be assessed on this basis.

There is some evidence to suggest that targeted killing may lead to an increase in terrorist violence as target organisations seek reprisal for the killing of key members of their group.²⁷ That being said, over time protracted targeted killing campaigns do appear to degrade the military capability of terrorist groups. As Daniel Byman argues persuasively, 'Contrary to popular myth, the number of skilled terrorists is quite limited. Bomb makers, terrorism trainers, forgers, recruiters, and terrorist leaders are scarce; they need many months, if not years, to gain enough expertise to be effective. When these individuals are arrested or killed, their organizations are disrupted.'²⁸ Targeted killings also complicate terrorist operations and their command and control. Under the threat of an effective targeted killing campaign, key terrorists have to restrict their movements and communications. All of this helps to degrade terrorist capabilities and may subsequently lead to a decrease in terrorist violence. This certainly seems to have been the case during the Second Intifada. Although it is difficult to disassociate the effects of targeted killing from other security measures, such as the building of the controversial security fence, targeted killing appears to have played its part in helping to reduce the scale of Palestinian violence against Israel.²⁹

A reduction in terrorist violence certainly is important, and as noted may be a policy objective. Nonetheless, it is equally important to track the impact of military action on broader political considerations and objectives. Do targeted killings contribute positively to the achievement of longer-term policy objectives? Alternatively, are they likely to complicate or undermine the

political process, and thereby represent poor strategy? Targeted killings have certainly attracted much criticism, have led to escalations in violence in the near-term, and have complicated relations amongst states. Some of the criticisms are moral in character, while others relate to issues of sovereignty. In relation to the latter, US drone strikes have complicated relations with countries in which the strikes occur, such as Pakistan. Since Pakistan is an important ally in the war on terror, in this respect targeted killing may have negative strategic effect.

Despite negative reactions, it seems that targeted killing can produce positive strategic outcomes. During the Second Intifada, Israel's protracted and intense targeted-killing campaign against the Hamas, Fatah and Islamic Jihad leaderships appears to have successfully coerced Israel's enemies to suspend hostilities.³⁰ The campaign also satisfied Israeli-population demands for action against Hamas, and rebuilt faith in the government's ability to provide security. Additionally, although the danger of collateral damage is ever present, precision strikes with limited means offer a potentially lower-risk instrument.³¹ In the final analysis, as with any method of pursuing strategy, the efficacy of targeted killing is dependent upon context and normally represents just one part of a larger campaign. However, if the situation is right, targeted killing offers the counterterrorist or counterinsurgent a reasonably low-risk, low-casualty, precise method to stop terrorist attacks, degrade enemy capability and coerce terrorist leadership. Nonetheless, a successful targeted-killing campaign relies heavily upon an effective intelligence capability and responsive command and control process.

Strategic ethics

As illustrated by the discussions surrounding targeted killing, moral concerns play an increasingly prominent role in strategy. Modern military ethics, built upon the foundation of the Just War tradition and corralled under the War Convention, present the strategist with both challenges and opportunities. This final section of the chapter outlines the challenge posed by the War Convention, before discussing how the strategist can respond to ensure that military ethics become more of a help than a hindrance to strategic best practice.

Before we go any further in our discussion, it is important to note that moral concerns are an integral part of strategy, not some foreign interloper sent to impose an alien concept on the rational pursuit of strategy. Politics has a moral dimension (in a general sense, with differences evident over time and place). Thus, it follows that strategy likewise has a moral component. This is the case even if the strategist is engaged in a struggle with the resultant ethical demands. The point to be made is that strategy cannot operate in a moral and ethical vacuum.³² This is not to say that there exists a universally acknowledged and constantly applied set of ethics. At first glance, this acceptance of a moral dimension appears to sit uncomfortably with some traditional realist interpretations

Box 3.2 Just War

Jus ad bellum

- Just cause
- Right intention
- Last resort
- Legitimate authority
- Proportionality
- Reasonable prospect of success

Jus in bello

- Proportionality
- Discrimination

of the subject: 'states in anarchy cannot afford to be moral'.³³ Indeed, the master strategic theorist Clausewitz warns against the inclusion of moral concerns in the practice of strategy: 'If one side uses force without compunction, undeterred by the bloodshed it involves, while the other refrains, the first will gain the upper hand.'³⁴ However, although fairly blunt in their tone, these statements should not be seen as the abject dismissal of the existence of ethics in strategy. Rather, they represent a normative approach, one which will be modified in this chapter.

Strategy is practised within a moral framework best described as the 'War Convention'. This is described by Michael Walzer as 'the set of articulated norms, customs, professional codes, legal precepts, religious and philosophical principles, and reciprocal arrangements that shape our judgements of military conduct'.³⁵ Since this is a rather broad set of criteria, which interact with strategy in a complex and challenging manner, military ethics is often delineated within the Just War tradition. The latter is a philosophical device, albeit one with an eye on the practical, to bring together moral concerns with the necessity to wage war. In essence, Just War seeks to codify the conditions under which political violence is necessary and legitimate, and strives to limit the suffering and damage caused when war occurs.

The Just War tradition, which can be traced back to ancient Greece, Rome and early Christian theologians, is broken down into two components: *jus ad bellum* (justification for going to war) and *jus in bello* (the justice of how war is waged) (see Box 3.2). The former is subdivided into six categories, all of which must be met for the recourse to war to be regarded as just: just cause, right intention (a better peace), last resort, legitimate authority, proportionality and reasonable prospect of success (to prevent unnecessary suffering). *Jus in bello* is divided into discrimination (between combatants and non-combatants) and proportionality. It will be immediately apparent that many of these categories are open to subjective interpretation. Even discrimination, which on

the surface appears straightforward, is not always so. For example, how does one distinguish between combatants and non-combatants in complex irregular environments? Do civilians working in armament factories forgo their non-combatant immunity? These, and other issues, suggest that military ethics is prone to judgement, differences of opinion and misunderstandings. Nonetheless, the Just War tradition does provide a conceptual framework within which to understand the interplay between ethics and the demands of strategy.

Although Just War provides important insight and guidance into the tense relationship between military necessity and ethical obligations, for the strategist it is a largely prohibitive approach. There is a sense that ethical considerations 'limit' strategic options. This perspective is growing as ethical and legal demands play an increasingly prominent role in strategic affairs. However, a more positive approach to military ethics is possible: 'strategic ethics'. This new approach, which remembers that war is politics (with its attendant ethical dimension), incorporates ethics as a core component of strategy. This broader conception is not interested in 'military necessity', but rather concerns itself with 'strategic necessity'. The latter puts policy considerations first (as should all strategic activity), and in doing so ensures that ethical demands are considered fully and within proper context. With the emphasis on strategy (using force to achieve policy objectives), moral and political legitimacy have a heightened profile. At the same time, a focus on achieving victory (obtaining the policy objective) ensures that moral demands do not overwhelm the process of strategy. It is hoped that strategic necessity will provide a more balanced and rational approach to the subject of military ethics.

Conclusion

As the following chapters in this book illustrate, war is a varied activity characterised by constant evolution. In order to be successful, the practitioner must achieve tactical and operational proficiency in each of the environments in which war occurs. This represents a constant challenge because the technological and doctrinal landscape is ever-changing. However, proficiency at the lower levels is simply not enough. Every use of the military instrument must be guided by strategy: the first-order consideration. Strategy, that process by which military force serves policy objectives, must be the dominant thought in the mind of the commander. What an admiral does with his fleet, or what an insurgent does with his AK 47, must be guided by the policy objective.

Even with a good education in strategic theory as his aid, the strategist must ultimately rely upon his own judgement to ensure that his use of force works towards the desired policy objectives. In this sense, the strategist must translate theory into practice in a vast range of different, indeed unique, contexts. To this end, he can use force in a variety of ways and for a variety of purposes. Sometimes military force may be used in a direct manner in a defensive or offensive way. In addition, military force is often used in a more limited and

indirect fashion, whereby the effects are felt more at the psychological level in the realms of deterrence, compellence and posturing. Finally, the flexibility of military force endows it with a host of other miscellaneous uses, from disaster relief to counter-narcotic operations.

With the above fully acknowledged, the modern practitioner of strategy is better equipped to deal with the complex interplay of extant strategic challenges. Military transformation, cyberwar and targeted killing provide seemingly less violent, more precise forms of military power. This may be especially important in the modern security environment, in which moral concerns play an increasingly prominent role. At the same time, the threat of regular war, WMD proliferation and certain forms of irregular warfare imply that extreme levels of violence are still a contemporary possibility. That being the case, the strategist is best served by adopting an approach grounded in 'strategic necessity', in which policy attainment remains paramount, while still taking full account of the ethical context. Regardless of how it is used, it must always be remembered that military force is ultimately premised on the ability to deliver violence. Military culture and our social attitudes regarding force must be faithful to that notion.

RESEARCH QUESTIONS

1. In what ways can military force serve policy in the contemporary environment?
2. Is the offensive use of force legitimate?
3. Is cyberwar changing the very nature of war?
4. Should strategists account for moral concerns in the practice of strategy?
5. Is targeted killing an especially ethical use of force?

FURTHER READING

- Arquilla, John and David Ronfeldt (eds.), *In Athena's Camp: Preparing for Conflict in the Information Age* (Santa Monica, CA: RAND, 1996).
- Art, Robert J., 'The Four Functions of Force', in Robert J. Art and Robert Jervis (eds.), *International Politics: Enduring Concepts and Contemporary Issues* (New York: Longman, 2003).
- Bellamy, Alex, *Just Wars: From Cicero to Iraq* (Cambridge: Polity Press, 2006).
- Betz, D. J. and T. Stevens, *Cyberspace and the State: Toward a Strategy for Cyber-power* (New York: Routledge, 2011).
- Freedman, Lawrence, *Deterrence* (Cambridge: Polity, 2004).
- Gray, Colin S., *Another Bloody Century* (London: Weidenfeld and Nicolson, 2005).
- Lonsdale, David J., *The Nature of War in the Information Age: Clausewitzian Future* (London: Frank Cass, 2004).
- Schelling, Thomas C., *Arms and Influence* (New Haven: Yale University Press, 1966).
- Walzer, Michael, *Just and Unjust Wars: A Moral Argument with Historical Illustrations* (Harmondsworth: Penguin Books, 1980).

Whetham, David (ed.), *Ethics, Law and Military Operations* (Basingstoke: Palgrave MacMillan, 2011).

ONLINE RESOURCES

- Arquilla, John, 'Cyberwar is Already upon Us: But Can it Be Controlled', *Foreign Policy*, March/April 2012 (www.foreignpolicy.com/articles/2012/02/27/cyberwar_is_already_upon_us).
- Byman, Daniel, 'Do Targeted Killings Work?' (www.foreignaffairs.com/articles/61513/daniel-byman/do-targeted-killings-work).
- Dipert, Randall R., 'The Ethics of Cyberwarfare', *Journal of Military Ethics*, 9(4) (2010) (www3.nd.edu/~cpence/ewt/Dipert2010.pdf).
- Dworking, Anthony, *Drones and Targeted Killing: Defining a European Position* (www.ecfr.eu/page/-/ECFR84_DRONES_BRIEF.pdf).
- Gray, Colin S., *Hard Power and Soft Power: The Utility of Military Force as an Instrument of Policy in the 21st Century* (www.strategicstudiesinstitute.army.mil/pdffiles/PUB1059.pdf).
- Gray, Colin S., *Maintaining Effective Deterrence* (www.strategicstudiesinstitute.army.mil/pdffiles/PUB211.pdf).
- International Society for Military Ethics (<http://isme.tamu.edu/>) (Great links to past conferences and papers).
- Libicki, Martin, *Cyberwar and Cyberdeterrence* (Santa Monica, CA: RAND, 2009) (www.rand.org/pubs/monographs/MG877.html).
- Lord, Kristin M. and Travis Sharp (eds.), *America's Cyber Future: Security and Prosperity in the Information Age*, vols. I–II (Washington, DC: Center for a New American Security, 2011) (www.cnas.org/publications/reports/america-s-cyber-future-security-and-prosperity-in-the-information-age#.VFIEPPmsXWg).
- Masters, Johnathan, 'Targeted Killings' (www.cfr.org/counterterrorism/targeted-killings/p9627).

NOTES

1. Colin S. Gray, *Maintaining Effective Deterrence* (Carlisle: Strategic Studies Institute, 2003), p. 25.
2. Lawrence Freedman, *Deterrence* (Cambridge: Polity Press, 2004), p. 5.
3. Gray, *Maintaining Effective Deterrence*.
4. Freedman, *Deterrence*, p. 4.
5. Robert J. Art, 'The Four Functions of Force', in Robert J. Art and Robert Jervis (eds.), *International Politics: Enduring Concepts and Contemporary Issues* (New York: Longman, 2003), p. 155.
6. Daniel Byman and Matthew Waxman, *The Dynamics of Coercion: American Foreign Policy and the Limits of Military Might* (Cambridge University Press, 2002).
7. Thomas C. Schelling, *Arms and Influence* (New Haven: Yale University Press, 1966), p. 2.
8. Dr Stephen Metz and Dr Douglas Johnson II, in *Asymmetry and U.S. Military Strategy: Definition, Background, and Strategic Concepts* (Carlisle, PA: US Army Strategic Studies Institute, January 2001), p. 8.

9. Jody R. Westby, 'Introduction', in Hamadoun I. Touré and the Permanent Monitoring Panel on Information Security (eds.), *The Quest for Cyber Peace* (Geneva: International Telecommunication Union & World Federation of Scientists, 2011), p. 1.
10. President Barack Obama, 'Remarks by the President on Securing Our Nation's Cyber Infrastructure', The White House, 29 May 2009 (www.whitehouse.gov/the-press-office/remarks-president-securing-our-nations-cyber-infrastructure).
11. 'Kaspersky Lab Uncovers "The Mask": One of the Most Advanced Global Cyber-espionage Operations to Date Due to the Complexity of the Toolset Used by the Attackers' (www.kaspersky.com/about/news/virus/2014/Kaspersky-Lab-Uncovers-The-Mask-One-of-the-Most-Advanced-Global-Cyber-espionage-Operations-to-Date-Due-to-the-Complexity-of-the-Toolset-Used-by-the-Attackers).
12. P. W. Singer and Allan Friedman, *Cybersecurity and Cyberwar: What Everyone Needs to Know* (Oxford University Press, 2014), p. 15.
13. Kristin M. Lord and Travis Sharp, 'Introduction', in Kristin M. Lord and Travis Sharp (eds.), *America's Cyber Future: Security and Prosperity in the Information Age* (Washington, DC: Center for a New American Security, 2011), vol. I, p. 22.
14. Singer and Friedman, *Cybersecurity and Cyberwar*, p. 60.
15. Lord and Sharp, 'Introduction', p. 13.
16. Joseph S. Nye, Jr, 'Power and National Security in Cyberspace', in Kristin M. Lord and Travis Sharp (eds.), *America's Cyber Future: Security and Prosperity in the Information Age* (Washington, DC: Center for a New American Security, 2011), vol. II, p. 13.
17. Lord and Sharp, 'Introduction', p. 34.
18. *A Strong Britain in an Age of Uncertainty: The National Security Strategy* (London: HMSO, 2010).
19. *Securing Britain in an Age of Uncertainty: The Strategic Defence and Security Review* (London: HMSO, 2010).
20. Chris Gibson, *The Launch of CERT-UK*, Cabinet Office, 31 March 2014 (www.gov.uk/government/speeches/chris-gibson-speech-at-launch-of-cert-uk).
21. United Nations General Assembly Human Rights Council, *Report of the Special Rapporteur on extrajudicial, summary or arbitrary executions* (www2.ohchr.org/english/bodies/hrcouncil/docs/14session/A.HRC.14.24.Add6.pdf).
22. Daniel Byman, 'Do Targeted Killings Work?', *Foreign Affairs* (March/April 2006).
23. This is not to say that it relies entirely on drone strikes. The United States also conducts hundreds of kill/capture raids each year using special operations forces. Johnathan Masters, 'Targeted Killings', *Council on Foreign Relations* (www.cfr.org/counterterrorism/targeted-killings/p9627).
24. *Ibid.*
25. Avi Kober, 'Targeted Killing during the Second Intifada: The Quest for Effectiveness', *The Journal of Conflict Studies* (<http://journals.hil.unb.ca/index.php/JCS/article/view/8292/9353>).
26. *Ibid.*
27. Kaplan, quoted *ibid.*
28. Byman, 'Do Targeted Killings Work?'
29. See Byman, 'Do Targeted Killings Work?' and Kober, 'Targeted Killing during the Second Intifada'.
30. Kober, 'Targeted Killing during the Second Intifada'.

-
31. Byman, 'Do Targetted Killings Work?'
 32. Colin S. Gray, *Perspectives on Strategy* (Oxford University Press, 2013).
 33. Art and Waltz, quoted in Jack Donnelly, 'Realism', in Scott Burchill *et al.* (eds.), *Theories of International Relations* (Basingstoke: Palgrave Macmillan, 2005), p. 31.
 34. Carl von Clausewitz, *On War*, ed. and trans. Michael Howard and Peter Paret (Princeton University Press, 1976 [1832]), pp. 83–4.
 35. Michael Walzer, *Just and Unjust Wars: A Moral Argument with Historical Illustrations* (Harmondsworth: Penguin Books, 1980), p. 44.





Part II

Land warfare

Christopher Tuck



4

Concepts of land warfare

Contents

Introduction

The land environment

Political importance

Variety

Friction

Opacity

The attributes of land forces

Complexity

Versatility

Persistence

Decisiveness

Key developments in land warfare

Scale

Firepower

Command and control

Logistics

Joint operations

The principles of land warfare

Attack and defence

Manoeuvre and attrition

Consolidation and exploitation

Concentration and dispersal

Centralisation and decentralisation

Conclusion

KEY THEMES

- Land, in the form of the ground that warfare is fought on, gives land warfare certain unique characteristics.
- These characteristics in turn shape the nature of the forces that fight upon land.
- Land warfare is complex: its prosecution requires navigating a wide array of competing trade-offs.

Introduction

Warfare on land has been pivotal to military outcomes throughout history. This is because human beings live on land; therefore, the capacity to seize and control territory often carries with it decisive political consequences. As the strategist Colin Gray has noted, 'the inherent strength of land warfare is that it carries the promise of achieving decision'.¹

The next three chapters explore the key ideas, concepts, principles and debates associated with conventional, high-intensity land warfare in the twentieth and twenty-first centuries. At the heart of these chapters lies the idea of the so-called 'modern system' of land warfare. During the twentieth century, armies faced a range of problems resulting from the interaction between various forces for change, including the effects of increasing firepower and the problem of moving, feeding and supplying larger armies. Incrementally, armies found potential solutions for these problems by manipulating some of the core areas of continuity in land warfare, not least the nature of the land environment itself and the basic characteristics of armies. These solutions created a dominant set of themes in the conduct of land warfare: dispersal, combined arms, fire-and-manoeuvre, depth and close co-operation with air and maritime forces (joint operations). Collectively, these themes constitute modern-system land warfare.

As with the later parts of this book on maritime and air warfare, our discussion of land warfare begins with an exploration of some of the key concepts that lie at the heart of the subject. This provides an essential background to the development of the modern system of land warfare both in terms of explaining the problems facing armies at the beginning of the twentieth century but also how important continuities, such as the effect of terrain and the flexibility of armies, have shaped potential solutions. Building on the concepts explored in this chapter, [chapter 5](#) then explores how and why the modern system evolved during the twentieth century. Whereas the modern system is essentially an evolutionary and adaptive development, more recent debates have often focused on the potential for revolutionary changes in the conduct of land warfare. [Chapter 6](#) develops this idea and examines the debates on the future of land warfare.

Like all warfare, land warfare is shaped by the nature of the environment in which it is fought. Although this may seem like a statement of the obvious, it is important to understand that one of the key differences between warfare on land and the other environments is land itself. The modern system has been shaped decisively by the way that land (in terms of terrain) can be used to mitigate the effects of firepower. The air and sea environments are fundamentally different in this respect. This means that there is no equivalent of a modern system in air and maritime warfare. It also means that, whereas the key developments in land warfare during the period have been (to use the 'levels of strategy' outlined in [chapter 2](#), p. 42) at the tactical and operational level, in the air and maritime environments the focus is much more on the operational and strategic levels.

The land environment

The land environment exhibits a number of attributes, including: its political importance; variety; friction; and opacity.

Political importance

Human beings live upon land, which gives land a profound tangible and intangible significance. Control of territory is central to the physical capacity and viability of states, and to the credibility and legitimacy of governments.² Thus, a capacity to take and hold territory is often a crucial index of success or failure in major combat operations: Adolf Hitler was not defeated until Berlin fell; Saddam Hussein was not defeated until Baghdad was taken; France surrendered in 1940 because it no longer had the capacity to prevent the occupation of its territory.

Variety

Different combinations of elevation, global position, climate and population density can create great varieties in operating environment even in relatively small areas. Land is therefore a complex environment in which to operate.³ In jungles, for example, moving and sustaining heavy forces is difficult; problems with visibility and navigation make co-ordination a challenge. In the desert, excessive heat and sand pose problems for the reliability of equipment and the stamina of personnel. With few obvious landmarks, navigation is problematic if access to global positioning systems is unavailable. Snow and sub-zero temperatures can likewise inhibit the capacity of military organisations to function effectively.

Friction

As has been noted in [chapter 2](#) (pp. 51–2) general friction is inherent in war. The land environment generates additional specific friction in such areas as movement and physical exertion because land is difficult to move over. The simple act of moving land forces can create all kinds of difficulties, even where there is no actual fighting. It is, moreover, relatively difficult to move over land the sort of bulk goods that are often essential for the logistics of an army, such as ammunition and fuel. Indeed, in land warfare, the problems of actual combat may be dwarfed by the challenges of moving and sustaining troops in the field. Humans have long made attempts to reduce this friction by using rivers and creating such man-made communications routes as roads, canals and railways. Although these can reduce the friction involved in movement, they can also channel movement, reducing flexibility and creating such vulnerabilities as the possible interdiction of transport nodes.

Opacity

Land is rarely truly flat; as Stephen Biddle notes, even on the 'flat' North German Plain, typically 65 per cent of the terrain within 1,000 yards is invisible to an average weapons position.⁴ The various obstructions provided by variations in elevation, vegetation and urbanisation can block or inhibit the capacity of firepower and sensors to function. This opacity creates opportunities for cover and concealment. Exploiting terrain gives opportunities to mitigate the effects of sensors and firepower through such techniques as using cover, camouflage and dispersal to reduce the ability of the enemy either to find targets or to apply combat power against them. Guerrilla warfare, for example, is to an extent an exercise in exploiting the cover and concealment provided by land. While the opacity of land can be an important enabler, it also creates challenges. Opacity can make communication and co-operation between friendly forces more difficult, as well as promoting in combat personnel such negative psychological effects as feelings of isolation.

Land is thus a politically vital medium: taking and holding it (or threatening credibly to do so) will often be an end in war as well as a means. The challenges of operating on land are shaped by the variety of different land environments and also by the fact that, as an opaque medium, land can be used as a mitigating factor in war. Land shapes, channels, facilitates and mitigates elements in the conduct of warfare. The interaction between the four factors outlined above means that some land matters more than another; varieties in land can create important avenues of advance, fields of fire, points of observation or such obstacles as steep slopes, rivers or towns. Understanding and manipulating 'ground' (the lie of the land) is often crucial to success in land warfare because terrain can provide not only an important obstacle to operations but also a means of reducing the effect of enemy firepower.

The attributes of land forces

Land forces have a number of fundamental attributes shaped by the demands of land as a medium. These attributes include: complexity, versatility, persistence and decisiveness.

Complexity

Land forces are complex; not necessarily in terms of technology (which may vary depending upon the army in question), but in terms of numbers of 'moving parts'. Whereas, put crudely, air forces and navies 'man the equipment', armies 'equip the man'.⁵ While fighting 'platforms' (such as tanks) have become more important since the First World War, armies still place a premium on having a larger number of smaller combat elements.⁶ This is necessary as a means to control ground and also to manipulate the advantages of terrain through dispersal and camouflage. One key expression of this is the continued

importance of infantry. This complexity is multiplied by the growing role of specialists in land warfare. Land forces encompass not just combat elements, such as tanks and infantry, but also combat support, combat service support and command support elements. Combat support provides such indirect support as fire support, air defence, reconnaissance and combat engineering, and includes artillery, engineers and air defence units. Combat service support encompasses logistics, supply, administration and medical services. Staff officers, and intelligence and communications personnel provide the command support. While complexity is a functional requirement for armies, it multiplies friction in warfare, particularly in relation to command and control.

Versatility

Because land forces are complex and personnel-intensive, they are adaptable; they are relatively less dependent on technology to define capabilities than air forces or navies. By refocusing such non-technological variables as training and ethos, armies can solve many military problems through new techniques and approaches. One reflection of this is the general utility of land forces across the whole spectrum of war, from peace support operations through to major combat operations. Armies, however, are not infinitely versatile. Land forces are likely to be optimised for particular kinds of warfare and particular physical environments; success or failure in major combat operations is likely to relate in part to the relative capacity of belligerents to adapt to the reality of local conditions.

Persistence

Land forces can remain in place for long periods of time. This presence allows land forces to apply continuous fire and surveillance. Land forces can hold ground, whereas air and naval forces cannot. General Norman Schwarzkopf has noted that 'There is not a military commander in the entire world who would claim he had taken an objective by flying over it.'⁷ Physical presence is often a requirement for political control and for the attainment of such broader objectives as providing security for civilians. Without ground forces, military effects can be ephemeral. As one Iraqi pilot noted in 2014 in relation to his attacks against Islamic State forces: 'When we go to bomb a place the ground troops don't accompany us. We bomb a place and kill a few, then Isis disperses but they regroup later.'⁸ There is often no substitute for the presence of land forces. For example, although precision-guided munitions (PGMs) are effective in delivering firepower, self-evidently they cannot patrol the streets of a town. However, while the presence of land forces can be militarily essential and send a powerful political signal, it may also create vulnerabilities. Persistent presence in a locale can make land forces more exposed to attack, and the larger political footprint of armies may cause longer-term political problems; for example, it may create friction with the local population.

Decisiveness

Land power is 'the strongest among equals. It alone can achieve the greatest strategic and political effect.'⁹ Air and naval attacks cannot, by themselves, defeat land forces. Land forces may be made irrelevant; for example, in the latter stages of the Pacific campaign in the Second World War sea control allowed the United States to bypass some Japanese island garrisons, simply allowing them to 'wither on the vine'. Firepower delivered from the air or from maritime assets may also inflict considerable damage on and disruption to land forces and their operations. Ultimately, however, only land forces can physically defeat other land forces. Numerous ways exist in which land forces can mitigate the effects of remotely delivered firepower, from greater dispersal and deception through to more effective anti-platform weapons, such as surface-to-air missiles (SAMs).

In combination, the characteristics of the land environment and the attributes of land forces have been fundamental to the development of modern land warfare. The political significance of land means that performance in land warfare matters. Yet there are basic difficulties in the effective conduct of land operations. Armies are difficult to move. They are difficult to sustain. They are problematic both to command and to control. On the other hand, these attributes and characteristics also provide features that provide great utility. Armies can occupy and control; they are flexible and adaptable; terrain can be exploited to great advantage.

Key developments in land warfare

If the characteristics and attributes outlined above provide a baseline for understanding some of the enduring issues in the conduct of land warfare, the reasons for the evolution of the modern system of land warfare lie in the interaction between those features and a range of key developments that became of decisive importance in the twentieth century.

Scale

The first development was the growing scale of war. The period from the mid-nineteenth century through to 1914 saw an enormous growth in the size of armies. Reflecting broader political, social and economic changes, the emergence of 'industrialised people's war' led to a massive expansion in the size of armies.¹⁰ By 1914, European armed forces collectively stood at around 20 million men. One consequence was the prospect of war without flanks: war in which the armies disposed by belligerents were so large that the formation of a continuous front across the whole theatre of battle was possible. Unless a victory could be won quickly, before the enemy's forces had been fully deployed, land warfare was likely to become an exercise in frontal assaults. Once this had become the case by 1915, the onus on the military was to develop ways in which

to facilitate frontal breakthroughs of the enemy. Another consequence of the increasing scale of war was the increasing irrelevance of what has been termed the 'Napoleonic paradigm of war': the view that war could be won quickly in a single decisive battlefield clash between two main armies. With mass armies, the effect of tactical actions was unlikely to be decisive in isolation, because belligerents possessed more than one army. Thus, the means to win wars would focus more and more on the co-ordination and sequencing of multiple tactical actions in order to produce a broader effect. This would shape the emergence of operational art.

Firepower

Another key influence on land warfare since 1900 was the growth in firepower. Indeed, this is a point of continuing relevance. Between 1900 and 1990, average artillery ranges increased by a factor of more than twenty; small-arms rates of fire increased three or four times; and the weapons payload and un-refuelled range of ground-attack aircraft increased by more than six times.¹¹ The machine guns, quick-firing artillery and magazine-loaded rifles of the early twentieth century have been augmented by tanks, aircraft and, latterly, PGMs such as laser-guided bombs. The growing lethality of firepower challenged the ability of armies to operate in close formations and in the open. In response, armies moved to consider ways in which to mitigate the effects of enemy firepower. Classically, these included recourse to cover and concealment, and to suppressive fire. The extended reach of firepower 'deepened' the battlefield, allowing distinctions to be made between 'deep' and 'close' operations. Depth opened up opportunities to use firepower at extended ranges to attack targets behind the enemy front line – to attack reserves, for example, and such critical infrastructure as transport nodes, command-and-control facilities and rear artillery positions. Because of the greater range and accuracy of firepower, military forces were capable of providing support over greater distances if properly co-ordinated. This permitted additional, if operationally challenging, opportunities for combinations of close and deep attack and also provided a permissive element in allowing greater depth in defence. War is of course adversarial, so these opportunities were also, paradoxically, vulnerabilities, since these developments were open to the enemy as well. For example, during the Cold War the doctrines of NATO and the Warsaw Pact both focused on the use of new technology to add greater range and firepower in the attack, while also attempting to incorporate such measures as greater dispersal to reduce their vulnerability to one another.

Command and control

Developments have also taken place in the command and control of armies. Command and control is, and always has been, fundamental to the effective

performance of armies. Command and control has been complicated by a number of developments. One has been the growing complexity of modern armies in terms of organisation, specialisation, size, mobility and dispersal. These developments have made land forces increasingly hard to control and to co-ordinate. Another development has been the multiplication in the technological aspects of command and control, such as communications and data acquisition and processing, through such technology as television, computers, mobile phones, data links and remote sensors. Taken together, these two issues have created the problem of how these systems should be organised and managed effectively, given the increasing volumes of information available.¹² In general, the systems for the command and control of armies have grown in size and complexity over time. 'Personalised' approaches to command and control, which have traditionally placed an emphasis on the activities and decisions of a small number of very senior commanders, have been replaced with permanent staff systems with formalised structures and procedures. In addition to being more costly, developments in precision firepower have also made command-and-control systems more vulnerable to enemy attack, not least because headquarters have become larger and command-and-control systems more integrated.

Logistics

Logistics have a crucial impact on the conduct of military operations. Effective logistics are vital to sustaining the tempo, momentum, duration and intensity of land warfare. The British General Sir Archibald Wavell noted that 'It takes little skill or imagination to see *where* you would like your army to be and *when*; it takes much more knowledge and hard work to know where you can place your forces and whether you can maintain them there.'¹³ The logistics of land warfare became increasingly complex in the twentieth century. In part, this was because of the growing scale of war. But it was also a reflection of the growing complexity of military organisations and their increasing consumption of the materiel of war, especially ammunition. By 1918, for example, the German Army was expending 300 million rounds of ammunition a month. Railways provided an imperfect solution, because although they could supply static forces on railheads, they could not support mobile operations. Once forces left their railheads, they continued to rely on horse-drawn supply. Rail supply was thus inflexible. It was the internal combustion engine that promised to bridge the gap between a mobile army and its railhead. Yet, as experience during and after the Second World War illustrated, motor transport also creates its own logistic problems. Generating, sustaining, maintaining and operating the number of motor transport vehicles required for a modern army is difficult and comes with associated costs for the flexibility and mobility of an army as a whole (see [Figure 4.1](#)).

Logistic constraints continue to impinge on the capacity of modern military systems to realise their theoretical potential in terms of such things as



Figure 4.1 Modern mechanised warfare has created new opportunities but also additional challenges, not least in the sphere of logistics. Here US Marines refuel an M1A1 Abrams tank in Helmand province, Afghanistan, 2011. Depending upon conditions, the Abrams can consume 300 gallons of fuel every eight hours.

mobility. Paradoxically, modern logistics, which have been a crucial enabler, are also an important source of friction in land warfare. During the 2003 Iraq War, for example, the single longest delay in the coalition advance towards Baghdad was imposed by logistic problems rather than the actions of Iraqi conventional forces.

Joint operations

Joint, that is multi-service, operations are not new in land warfare; land operations have often had important riverine or amphibious elements. However, air power has had an increasingly pervasive influence on land warfare. Air power can provide an important means to mass firepower quickly at particular points and at ranges beyond that of land systems. Combinations of air and land forces can produce important synergies because the means used by enemy forces to escape the effects of air power, such as dispersion, are often those things that make them more vulnerable to land attack and vice versa. Joint operations between land and air forces can be controlled according to a spectrum running from simple deconfliction at one end through to full integration. Historically, however, joint warfare (or ‘jointery’ as it is sometimes termed) has often proved

to be problematic. This often stems from a difference in perspective: soldiers have a tendency to see air power as a shock instrument for immediate and visible close air support; airmen take a theatre-wide view – many crucial airpower roles are not immediately visible to land forces.

These key developments have raised recurrent problems for the conduct of land warfare. How does one attack successfully in the face of increasing fire-power? How does one manoeuvre swiftly, given the complexity of logistics? How does one maintain control over large, complex army organisations under the stress of combat? Fundamentally, how can one win?

The principles of land warfare

The question of how armies should deal with the consequences of the developments outlined above has been a recurrent challenge in the twentieth and twenty-first centuries. Based on an assessment of past experience, armies have developed sets of principles designed to express concepts fundamental to the successful application of military art on land. An indicative selection is provided by the United States' 'principles of war' (actually better termed 'principles of warfare') (see [Box 4.1](#)). Whatever their theoretical merit, however, applying these principles in the real world is often very difficult. The value and application of these principles is a matter of context. For example, surprise can be a key force enabler and has been used routinely throughout history as a way of multiplying the effectiveness of military operations. However, failed attempts to achieve surprise may leave friendly forces far more vulnerable than they would otherwise have been. In the 1944 Battle of the Bulge, for example, the choice of the difficult terrain of the Ardennes as the avenue of attack contributed to achieving surprise but at a cost of making the subsequent exploitation much more difficult because the lines of advance were narrow and separated, and because the terrain created defensive choke-points.

Indeed, a central problem facing armies in responding to the key agents of change in the modern period has been that land warfare embodies competing demands. Prosecuting land warfare effectively involves important trade-offs that comprise competing perspectives on how to relate time, space, fire, movement and protection.¹⁴ The value of these may be entirely context-specific: techniques that work in one context may invite disaster in another. Some of the key trade-offs during the period in question include those between: attack and defence; manoeuvre and attrition; consolidation and exploitation; concentration and dispersal; and centralisation and decentralisation.

Attack and defence

Victory in land warfare requires offensive operations. Even where an army conducts prolonged defensive operations in order to wear down the opposition,

Box 4.1 The US principles of war

Objectives: military operations should be directed towards a defined, decisive and attainable objective that contributes to the broader political and military objectives.

Offensive: military operations should be proactive rather than reactive. Success in land warfare is more likely if a focus is placed on aggressive operations designed to seize and hold the initiative.

Mass: military power should be concentrated at the decisive time and place. Success is more likely if a preponderance of force can be gathered for the most significant battles and not dispersed at less important points.

Economy of force: forces allocated to secondary objectives or positions should be minimised in order to facilitate the concentration of forces at the decisive point.

Manoeuvre: friendly forces should use movement to place the enemy at a disadvantage. This might involve trying to move onto the enemy's flanks as a way of avoiding a frontal assault, or using combinations of fire and manoeuvre to suppress the enemy and advance on a position.

Unity of command: the forces allocated to a specific operation or purpose should be allocated a single overall commander so that the activities of different elements are properly co-ordinated.

Security: friendly forces should be protected from enemy action that might disrupt or harm them. This principle encompasses a wide range of activities, from providing land forces with adequate anti-aircraft protection through to information-security techniques, such as establishing secure communications and aggressive patrolling to inhibit enemy reconnaissance efforts.

Surprise: military forces should be used in a manner, time or place that is unexpected by the enemy.

Simplicity: unnecessary complexity should be avoided in preparing, planning or conducting operations. Those factors that create general friction in warfare, such as human involvement, fear, chance and uncertainty, mean that the more complex a military operation is, the more likely it is that something will go wrong.

a return to offensive operations is likely to be required in order to exploit this attrition. At the strategic level, the growing scale of war has generated economic and political costs that often place a premium on ending wars as quickly as possible. At the tactical and operational levels, a force that does not attack is likely to cede the initiative to the enemy. In the air and maritime environments, attack is often the stronger form because the physical capacity of air and maritime forces to absorb the effects of modern firepower is limited. This means that victory will often go to the side that finds and attacks the enemy first.

However, traditionally in land warfare it is often more difficult to attack than it is to defend. The frictions associated with such issues as supply, command and

control, co-ordination, integration and the application of firepower tend to be magnified in attack because attacking generally requires that forces manoeuvre. Manoeuvre is challenging because armies are complex and because land is difficult to move over and inhibits communication. Defending forces can also use the terrain to multiply the effectiveness of their forces. Defending troops can be concealed or placed in prepared positions as ways of reducing the effect of the attack. The problems associated with offensive operations have been worsened by developments in firepower that have increased the weight and distance of the defensive firepower to which attackers are likely to be exposed. Thus, while armies cannot win without attacking, they are also often most vulnerable when engaging in offensive operations.

Manoeuvre and attrition

Manoeuvre constitutes the movement of forces relative to the enemy to achieve an advantageous position. It is sometimes viewed as the opposite of attrition, the process of undermining the enemy by killing personnel and destroying materiel, because of the association between attrition and static positional warfare, such as that which characterised the First World War. In reality, attrition and manoeuvre are related concepts. High rates of movement do not necessarily translate into effective manoeuvre. Effective manoeuvre involves placing one's forces in positions of advantage relative to the enemy. This relational element has a spatial and a temporal dimension: it is about being in the required place at the appropriate time. Attrition may enable manoeuvre by opening gaps that friendly forces can move through; manoeuvre may enable favourable attrition by placing forces in positions to fire upon the enemy more effectively or as a means of gaining surprise.

The dynamic qualities inherent in manoeuvre give it the potential for decisive results. Classic forms of manoeuvre include those that focus on placing forces, or threatening to place forces, on an enemy's flanks or rear – namely, turning movements (passing around a flank), envelopments (passing around the flank and threatening to move to the rear) and encirclements (trapping forces by blocking movement to the flanks or rear), creating what the Germans termed a *Kesselschlacht* or 'cauldron battle'.

Some of the developments since 1900 have strengthened the capacity for manoeuvre: the internal-combustion engine; air mobility, which has created the capacity for 'vertical envelopment'; and the shift from such fixed communications as field telephones to wireless radios and digital communications equipment. Against these developments must be set such others as the growing scale of war and, in particular, increases in firepower that have made manoeuvring forces potentially more vulnerable. While methods exist to reduce the effect of firepower on manoeuvring forces, such as dispersal and the use of cover, these

countermeasures pose two challenges: first, they reduce the speed of manoeuvre, which may undermine its effectiveness; secondly, they are demanding techniques in terms of training, motivation and command and control. Paradoxically, then, while manoeuvre may be the means to avoid unfavourable attrition, it may also, if poorly executed, have entirely the opposite effect by exposing moving forces to the full effect of enemy firepower.

Consolidation and exploitation

A successful operation leads to a choice between exploitation, pushing on to maximise the initial victory, and consolidation, pausing to reconstitute the attacking forces before moving on at a later point in time. Exploitation is a key means of translating local and temporary successes into broader and more significant gains. The initial victory may leave the enemy disorientated and unbalanced. However, the effects of demoralisation, disorder and shock, while powerful, are often temporary. Exploitation is a means to sustain and magnify these effects. It is one enabler for a high tempo of operations. Tempo describes the rate of activity relative to the enemy. This relational component is significant. Sustaining a high tempo of operations is an important way of keeping the initiative, maintaining surprise and invalidating much of the enemy's decision-making. Indeed, Richard Simpkin defines initiative as 'constantly creating new situations to be exploited'.¹⁵ Thus, exploitation can be a key way of encouraging the broader systemic collapse of a portion of the enemy forces. While consolidation is a passive activity, exploitation is active.

Despite its apparent effectiveness as a tool, exploitation can be a double-edged sword. First, successful exploitation is challenging and requires, amongst other things, an ability to identify opportunities and avenues for exploitation. Exploitation requires timely activity. While planned exploitation is important, opportunistic exploitation can be one of the most effective means of exploitation because of its flexibility and speed. Exploitation will often require fresh reserves or echeloned forces. Exploitation can be facilitated by simultaneity: seeking to overload the enemy's decision-making so that the enemy cannot establish effective priorities. This can be achieved through multiple penetrations, or penetrations whose objectives are unclear and that might threaten several targets. However, multiple penetrations will create significant challenges for the command, control and co-ordination of attacking forces.

Secondly, exploitation can bring risks. One purpose of rapid exploitation is to pre-empt battle and avoid counter-attacks. However, in land warfare the relationship between attack and defence is a fluid one. Generally, one of the purposes of defensive operations will be to create the right conditions for the offensive. Defensive operations can permit economy of force in one area to enable the creation of mass in another; they can allow a pause for the reorganisation or reconstitution of reserves; they can also inflict attrition on the

attacking forces and create the conditions for counter-attack. Because exploiting forces will often be advancing rapidly, on a narrow front and without the organised logistic and combat support available for pre-planned assaults, they are vulnerable to aggressive, or 'active', defences, particularly those based upon depth (area defence) or mobile counter-attack (mobile defence). The successful German offensive at Kharkhov in 1943 was a strike at Russian forces over-extended by exploitation. If consolidation is passive, it is a much easier and less risky task to perform than exploitation.

Concentration and dispersal

Concentrating one's forces has long been recognised as a desirable element in war. Clausewitz notes, for example, that one key constituent of success in war is 'always *to be very strong*; first in general, and then at the decisive point'.¹⁶ Dispersal can create difficulties in command and control and can make forces vulnerable to being swept away by larger enemy forces. Indeed, differential concentration, accepting weakness in some areas to achieve the necessary mass at the decisive point, has been used routinely as a way of trying to achieve breakthroughs. However, the logic of the dispersed battlefield makes concentration problematic. Progressively greater dispersal has occurred precisely because mass has become increasingly vulnerable to firepower. Moreover, mass cannot substitute reliably for deficiencies in skill. In 1973, the Syrian attack on the Golan achieved a 6:1 superiority over the Israelis by massing on a narrow front three echelons (or waves) totalling five divisions and four brigades. However, the difference in tactical skill between the two sides stalled the Syrian attack, making the Syrian concentration an effective exercise in Israeli economy of effort.¹⁷ In addition dispersal, in the form of multiple echelons, or lines of forces, deployed successively can bring many advantages. In the attack, for example, the rear echelons, or 'follow-on forces', provide the means to exploit quickly any initial breakthroughs. They allow success to be reinforced. In defence, multiple echelons give great 'elasticity'. The attackers are worn down as they encounter each defensive echelon, and uncommitted echelons provide the means for the defender to counter-attack and regain lost ground.

Centralisation and decentralisation

Land warfare involves many trade-offs; many of the crucial ones involve time. Hasty attacks, for example, trade mass for time; prepared assaults do the reverse. Rapidity can be a telling force multiplier, magnifying the effects of manoeuvre, surprise and firepower. However, the growing complexity of land warfare, in terms of the nature of armies (size, specialisation, potential mobility) and the volume of information available, creates important trade-offs between time and certainty. For example, co-ordination is crucial in land warfare. Most land

warfare involves elements of firepower, movement and decision-making, and intelligence-gathering, processing and dissemination by a multitude of different combat, command and support elements. One way of responding to the uncertainties of complexity is to centralise command and control, concentrating certainty at the top. This 'directed command' approach allows commanders to impose more unity of effort and co-ordination on armies. However, centralisation often trades certainty and order for time. In particular, centralisation may slow decision-making processes down, as decisions need to be referred upwards, and may reduce the initiative displayed by local commanders. This tension can manifest itself in many ways. The complexity of modern logistics, with the associated premium that is often placed on detailed planning, is also at odds with more modern concepts of opportunistic exploitation and mobility. The ability to exploit local successes with reserves is likely to be undermined by centralised decision-making.

Decentralised decision-making is an obvious alternative. Termed by the Germans *Auftragstaktik* (mission command), this mode of command and control places an emphasis on instilling in subordinates a clear understanding of what needs to be achieved but giving them latitude as to how it is to be done. By devolving decision-making downwards, the tempo of operations is increased and the best use can be made of opportunities as they arrive – opportunities for exploitation, for example. Decentralisation accepts less certainty than centralisation but attempts to deal with this by increasing the ability of an army to act faster and thus shape events. Despite its theoretical advantages, however, decentralised command and control carries potential penalties. These include a loss of focus in operations, problems in combined arms and jointery, and over-extension in exploitation operations. Effective decentralisation is a fundamentally demanding mode of command and control that requires excellent training, effective communications, a supporting philosophy of command and a significant degree of junior initiative. Whatever its theoretical advantages, therefore, decentralised command and control may create risks and simply be inappropriate for many armies.

Conclusion

Modern land warfare has been shaped by the existence of a range of inter-related challenges. Some of these derive from the land environment itself, such as the impact of terrain on operations. Some derive from the characteristics of armies, such as their complexity. Others have emerged because of such key developments in technology, politics and economics, as the growing scale of war and the increasing lethality of firepower. Responding to these challenges has been complicated by the wide range of trade-offs that exist in land warfare which allow for the creation of many different and often conflicting responses. In the [next chapter](#), we examine the consequences of these factors and the

development of the modern system in land warfare as the dominant solution to the challenges of land warfare that have been outlined.

RESEARCH QUESTIONS

1. In what ways does the land environment differ from the air and the maritime?
2. What, in relation to the air and maritime arms, are the strengths and weaknesses of land forces?
3. How useful are principles of war?
4. Why is joint warfare often so problematic to execute successfully?
5. Looking at the trade-offs outlined above, which approaches seem best to characterise contemporary Western approaches to land warfare, and why?

FURTHER READING

- Angstrom, Jan and J. J. Widen, *Contemporary Military Theory: The Dynamics of War* (London: Routledge, 2015), chs. 5, 6 and 7 provide an illuminating discussion on the principles of war, joint operations and the interplay between manoeuvre and attrition.
- Biddle, Stephen, *Military Power: Explaining Victory and Defeat in Modern Battle* (Princeton University Press, 2004). Biddle explains the development of the modern system of land warfare and its significance as an antidote to such developments as increases in weapon firepower.
- Griffin, Stuart, *Joint Operations: A Short History* (London: Training Specialist Services HQ, 2005). Through the use of a range of historical case studies, Griffin provides an excellent grounding in the challenges associated with joint warfare.
- Holmes, Richard, *Acts of War: The Behaviour of Men in Battle* (London: Cassell, 2004). An essential introduction to the experience of land warfare through the eyes of the soldier.
- Johnson, Rob, Michael Whitby and John France, *How to Win on the Battlefield: The 25 Key Tactics of all Time* (London: Thames and Hudson, 2013). An easily digestible introduction to such relevant concepts as shock action, attack, envelopment and concentration.
- Shamir, Eitan, *Transforming Command: The Pursuit of Mission Command in the U.S., British and Israeli Armies* (Stanford University Press, 2011). Investigates the relationship between military change and approaches to command and control.
- Strachan, Hew, *European Armies and the Conduct of War* (London: Allen & Unwin, 1983). An excellent overview of European warfare from the eighteenth century through to 1945.
- Tuck, Christopher, *Understanding Land Warfare* (London: Routledge, 2014). Chapter 1 includes an extended discussion of the issues covered here, and identifies, in particular, the weaknesses inherent in land power.
- Van Creveld, Martin, *Supplying War: Logistics from Wallenstein to Patton* (Cambridge University Press, 1977). An engaging and incisive study of the development of military logistics.

ONLINE RESOURCES

For a balanced analysis of the strengths and weaknesses of land power, see Milevski, Lukas, 'Fortissimus Inter Pares: The Utility of Landpower in Grand Strategy', *Parameters* (Summer 2012). *Parameters*, the journal of the US Army War College, also has many other relevant articles, <http://strategicstudiesinstitute.army.mil/pubs/parameters/articles/2012summer/milevski.pdf>.

The US Army War College has produced *Landpower: A Selected Bibliography* which contains a large selection of doctrinal and other publications, many of which are available online: www.carlisle.army.mil/library/bibs/landpower12.pdf.

The US National Defense University publishes *Joint Force Quarterly*, <http://ndupress.ndu.edu/JFQ.aspx>, which contains many relevant articles. See, for example, Kevin D. Stringer and Katie M. Sizemore, *The Future of U.S. Landpower*.

Military doctrine can provide important insights into land warfare from the practitioner's perspective. Useful Australian and British analyses can be found here: *Land Warfare Doctrine 1: The Fundamentals of Land Warfare* (Australian Army, 2014), ch. 1, www.army.gov.au/Our-future/Publications/Key-Publications/Land-Warfare-Docctrine-1.

Army Doctrine Publication, *Operations* (2010), www.gov.uk/government/uploads/system/uploads/attachment_data/file/33695/ADPOperationsDec10.pdf.

NOTES

1. 'Land Warfare Doctrine 1: The Fundamentals of Land Warfare' (Australian Army, 2002), ch. 1, p. 1.
2. Colin S. Gray, *Modern Strategy* (Oxford University Press, 1999), p. 213.
3. See Patrick O'Sullivan, *Terrain and Tactics* (London: Greenwood, 1991).
4. Stephen Biddle, *Military Power: Explaining Victory and Defeat in Modern Battle* (Princeton University Press, 2004), p. 36.
5. Gray, *Modern Strategy*, p. 213.
6. Lukas Milevski, 'Fortissimus Inter Pares: The Utility of Landpower in Grand Strategy', *Parameters* (Summer 2012), 11.
7. Quoted in David Lonsdale, *The Nature of War in the Information Age: Clausewitzian Future* (London: Frank Cass, 2004), p. 50.
8. 'Iraq Voices Anger as US Air Force Defends Irbil – But Not Baghdad', *The Observer*, 17 August 2014.
9. Milevski, 'Fortissimus Inter Pares', p. 15.
10. See Stig Forster and Jorg Nagler (eds.), *On the Road to Total War: The American Civil War and the German Wars of Unification, 1861–1871* (Cambridge University Press, 2002).
11. Biddle, *Military Power*, p. 30.
12. Martin van Creveld, *Command in War* (London: Harvard University Press, 1985), pp. 1–10.
13. *Ibid.*, pp. 231–2.
14. Jan Angstrom and J. J. Widen, *Contemporary Military Theory: The Dynamics of War* (London: Routledge, 2015), p. 111.

15. Quoted in J. P. Kiszely, 'The Contribution of Originality to Military Success', in Brian Holden Reid (ed.), *The Science of War: Back to First Principles* (London: Routledge, 1993), p. 26.
16. Carl von Clausewitz, *On War*, ed. and trans. Michael Howard and Peter Paret (Princeton University Press, 1976), p. 61.
17. See Michael J. Eisenstadt and Kenneth M. Pollack, 'Armies of Snow and Armies of Sand: The Impact of Soviet Military Doctrine on Arab Militaries', in Emily O. Goldman and Leslie C. Eliason (eds.), *The Diffusion of Military Technology and Ideas* (Stanford University Press, 2003).



5

Modern land warfare

Contents

Introduction

The First World War and the emergence of modern tactics

Defence in depth

Infiltration techniques

The Second World War and after: the operational art

Operational-level mobile warfare

Operational defence in depth

The operational level of war

Conclusion

KEY THEMES

- Despite debates on the revolutionary impact of such concepts as blitzkrieg, modern land warfare actually has had a strong evolutionary dynamic.
- Modern tactics emerged during the First World War as a result of the need to cope with dramatic increases in firepower.
- Building on theories developed during the interwar period, the concept of operational art emerged in practice during the Second World War as a means of connecting tactics to strategy.

Introduction

The parameters of how we commonly think about modern land warfare were established by developments in the First and Second World Wars. These developments cover the shift from a mode of warfare characterised in the early part of the twentieth century by linear battle with a focus on tactics to a kind of warfare characterised towards the end of the twentieth century by a focus on mobile, combined-arms warfare at the operational level. These developments have been identified by Stephen Biddle as constituting the emergence of a definable 'modern system of war'. This modern system comprises 'a tightly interrelated complex of cover, concealment, dispersion, suppression, small-unit

independent maneuver, and combined arms at the tactical level, and depth, reserves, and differential concentration at the operational level of war'.¹

By and large, the basis of the modern tactical system of warfare was established by the end of the First World War, as the belligerents struggled to develop the means to cope with the inadequacy of existing tactical concepts in the face of the effects of modern firepower. In the wake of the First World War, this system was refined by belligerents, but the same basic precepts informed tactics and operations in the Second World War and after. Operational art emerged as a response to the demonstrated challenges of converting the tactical-level successes attainable using modern tactics into broader successes at the higher levels of war. The conscious and unconscious focus on operational-level solutions informed Soviet theories on operational art and German blitzkrieg, and formed the basis of operational techniques developed during the Cold War.

The First World War and the emergence of modern tactics

The events of 1914 cruelly exposed the limitations of pre-war concepts on the conduct of land warfare. The severity of the mismatch between the theory and the reality of land warfare in 1914 might encourage the belief that it was the result of incompetence and wilful ignorance with regard to the effects that change, particularly in the scale of war and the increases in battlefield firepower, would have on the character of land warfare. This is not true. Pre-war armies were alive to the new, emerging trends in war; debates on the implications of change were widespread, erudite and rational. It was long recognised, for example, that increases in firepower posed a challenge (see [Figure 5.1](#)). Increases in firepower created an enormous problem in crossing the open ground between two forces, what the French Colonel (later Marshal) Ferdinand Foch (1851–1929), a key pre-war theorist, referred to as the 'zone of death'. The invention of machine guns, quick-firing artillery, magazine-loaded rifles and smokeless propellants in the late nineteenth century had extended and intensified the 'zone of death' between armies. The solutions discussed were rational and, indeed, form a key part of modern tactics. One was dispersal: allowing troops to spread themselves out and to make use of local cover, so making them more difficult targets. Another was a focus on suppressive fire as a way of beating down the defence and preventing them from bringing their firepower to bear. However, both carried their own challenges. Lacking the technology for effective tactical wireless radio communications, dispersed troops were much more difficult to command, control and motivate, threatening a loss of momentum in the attack; and suppressive fire was difficult to co-ordinate between units, particularly between advancing infantry and their supporting artillery. These problems were multiplied by the general focus placed at the time upon offensive action. Since industrialised mass warfare was believed to be unsustainable in the long term, success would have to be achieved quickly, and this could only be done through offensive operations.²



Figure 5.1 Modern land warfare has been conditioned by increases in battlefield firepower. While machine guns have had a pervasive impact, mortars, especially, have also proved to be a major source of infantry casualties in war. Here, US soldiers fire their 120mm mortars during a live-fire exercise on Forward Operating Base Lightning in Afghanistan's Paktia province, 17 January 2014.

The solution was a focus on the moral component of war as a means of compensating for technology. Victory was to be attained by dominating the battlefield through a focus on the human will: seizing the initiative and exploiting it through close offensive action to break the enemy's spirit. Defensive operations were viewed as a temporary mode in preparation for the attack, or the means of holding in some areas of the battlefield to concentrate the resources for the offensive in the decisive sector. Under such theorists as Foch and Colonel Louis Loyzeau de Grandmaison (1861–1915), offensive action became a cardinal principle in land warfare. Grandmaison, for example, argued that 'the French army, returning to its traditions, no longer knows any other law than the offensive . . . All attacks are to be pushed to the extreme . . . to charge the enemy with the bayonet to destroy him . . . Any other conception ought to be rejected as contrary to the very nature of war.'³

However, the tactical expression of this approach was more sophisticated than might be supposed: an emphasis was placed on both firepower and shock action. Firepower was essential in order to suppress and weaken the defence; this would prepare the way for the decisive act – closing with the enemy. Infantry would thus infiltrate forwards, using cover and concealment; they

would then use small-arms in combination with support from quick-firing artillery and machine guns to weaken the defence before closing with cold steel. For example, French infantry regulations for 1895 prescribed that the

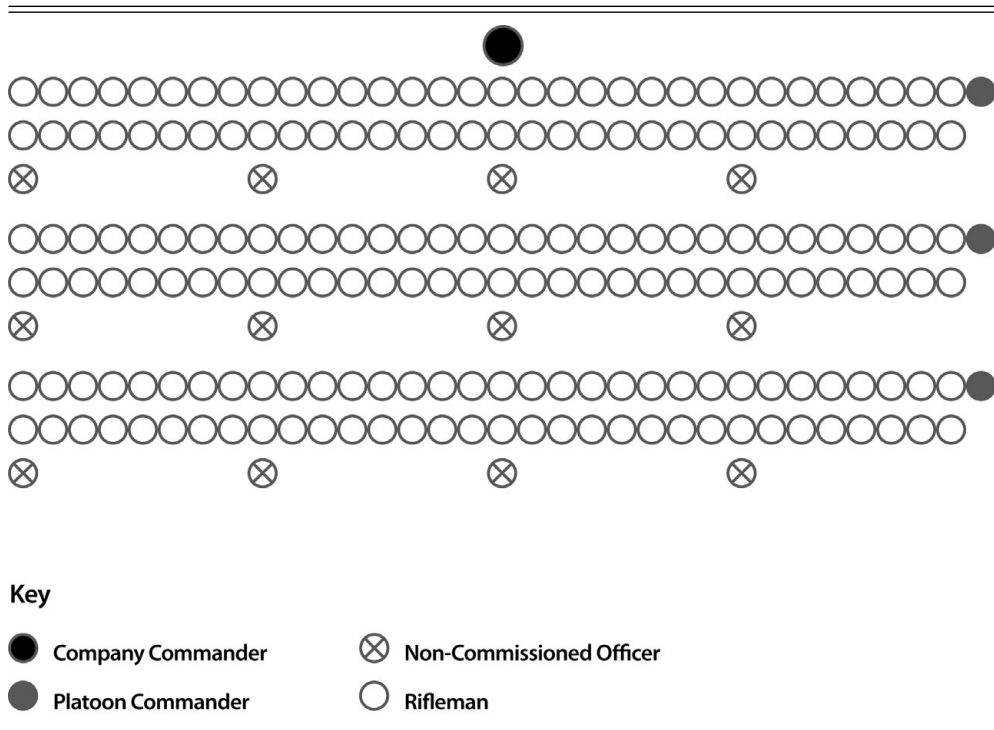
advance is made by successive rushes followed by a quick fire of short duration. The fighting line reinforced by the reserves . . . gradually reaches to within 150 or 200 metres of the enemy. At this distance magazine fire is commenced, and all available reserves . . . close up for the assault. At a signal from the Colonel the drums beat, the bugles sound the advance and the entire line charges forward with cries of '*en avant, à la baionette*'.⁴

The tactical dilemmas facing the armies of the period were evident in their responses to the Boer War (1899–1902). The heavy casualties inflicted on the more closely ordered British troops by their Boer adversaries encouraged experimentation by European armies with looser 'Boer tactics'. But these tactics proved impractical for mass warfare. The German Army found, for example, that approaches based on dispersing troops into skirmish order would spread an infantry battalion over a frontage of 3,000 metres, making it impossible to control effectively.⁵ By 1903, the German Army had returned to methods in which the basic tactical unit was an infantry company of similarly armed riflemen, deployed in a fashion that would allow its officers to maintain close control over the troops (see [Table 5.1](#)).

It was recognised that an offensive approach to land warfare was likely to be costly. This bloody sacrifice was regarded as being the necessary price to pay for achieving rapid and decisive success. Thus, the lessons learned from the conflicts up to 1914, which were contradictory, tended to be used to reinforce existing orthodoxy. A British observer of the Russo-Japanese War (1904–5), Sir Ian Hamilton (later the British land commander at Gallipoli), noted in 1910 that 'War is essentially the triumph, not of the Chassepot over a needlegun, not of a line of men entrenched behind wire entanglements and fireswept zones over men exposing themselves in the open, but of one will over another weaker will.'⁶ By 1914, then, what had emerged was, in a sense, 'anti-modern' warfare, a misinterpretation of some of the implications of social and technological change and a greater focus on enduring human elements of warfare, such as morale, discipline and offensive spirit.

The first year of the war demonstrated the inapplicability of these approaches. Modern firepower made frontal assaults by infantry brutally ineffective. In August 1914, German forces encountered the British Expeditionary Force at Mons. A British soldier noted that the Germans 'advanced in companies of quite 150 men in files five deep . . . The first company were simply blasted away to Heaven by a volley at 700 yards and, in their insane formation, every bullet was bound to find two billets.'⁷ The official German account noted that 'Well-entrenched and completely hidden, the enemy opened a murderous fire . . . casualties increased, the rushes became shorter and finally the whole

Table 5.1 1914 infantry company deployed in 'column of platoons'*



* Bruce I. Gudmundsson, *Stormtroop Tactics: Innovation in the German Army, 1914–1918* (Westport, CT: Praeger, 1995), p. 9.

advance stopped. With bloody losses the attack gradually came to an end.⁸ Yet the increasing scale of war resulted in the creation of a continuous front that made the obvious alternative, flanking movements, impossible.

Some important adjustments were made relatively quickly. By March 1915, artillery, not infantry, had become the 'Queen of Battle'; the tactical mantra became 'the artillery conquers, the infantry occupies'. By the Somme offensive, which opened on 1 July 1916, artillery was employed in staggering quantities: 1,437 artillery pieces fired 1.5 million shells over the first seven-day period.⁹ However, the utility of artillery as an arm of decision was fundamentally undermined by developments in defensive tactical methods – the adoption of the concept of defence in depth – which mitigated the effects of this offensive firepower.

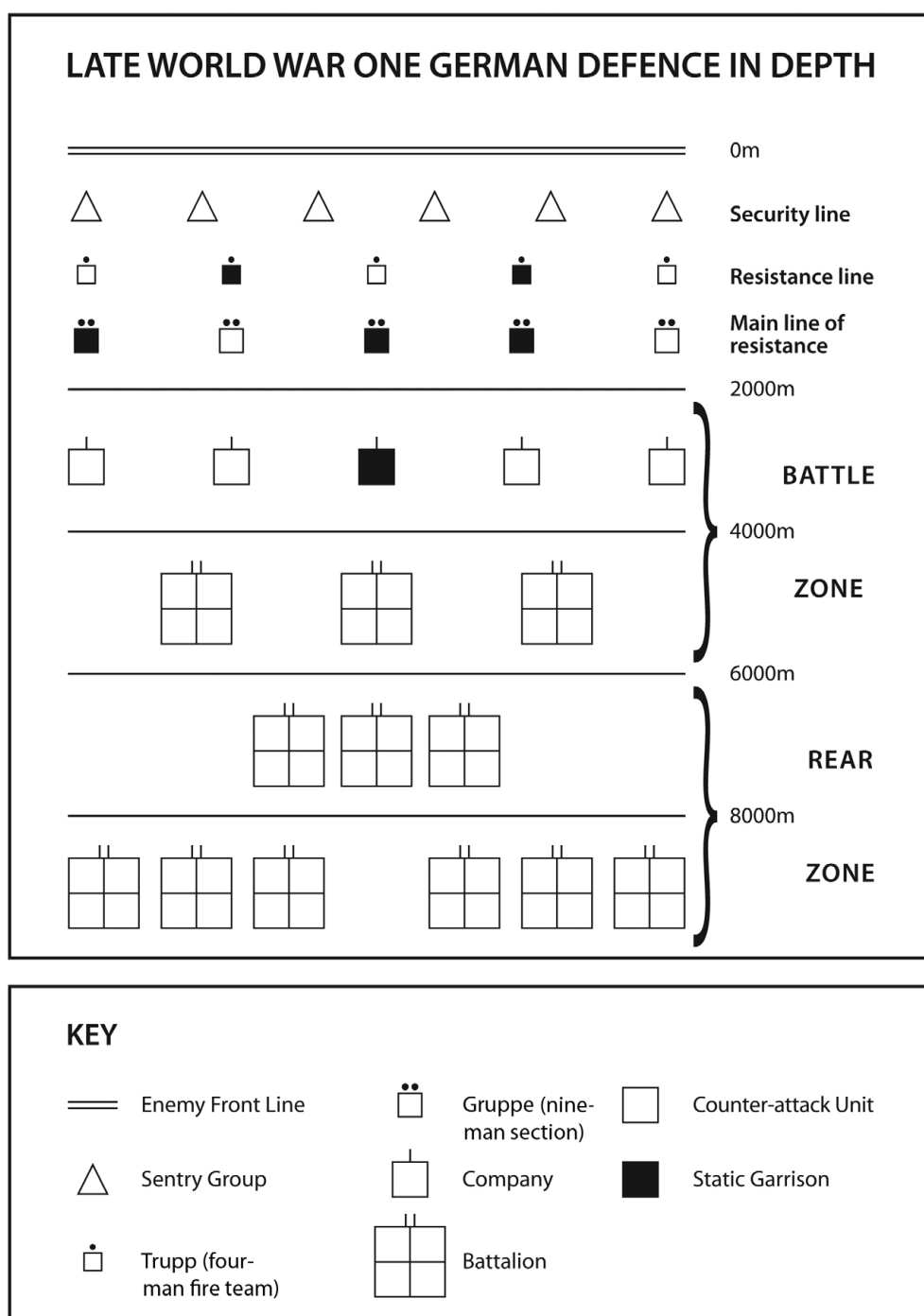
Defence in depth

Attached to a principle of holding ground through forward defence, the German High Command was appalled at the attrition inflicted on their troops by the battles on the Somme, which seemed to presage the emergence of *die Materialschlacht*, 'the battle of materiel': warfare decided purely by which side

could endure attrition the longest. The German Army set about developing the tactical means to counter the firepower of attacking artillery, and the conclusions were reflected in the doctrinal publication *The Principles of Command in the Defensive Battle in Position Warfare*, published 1 December 1916. Four pillars underpinned the new doctrine: depth, increasing infantry firepower, elasticity and initiative:

- Depth was achieved by having multiple defensive zones, creating a 'web defence'. By 1917, this might consist of up to five successive defensive lines, mixing multiple trench lines and strong points sited for cover and all-round defence. These would include an 'outpost zone' of 500–1,000 metres in depth, which acted as 'tripwire'; a 'battle zone' of 2 kilometres or more; and an artillery protection zone (see [Table 5.2](#)). Ground was used to mitigate enemy firepower; for example, where possible, defensive lines were placed on the reverse slope of hills to make enemy observation and direct fire more difficult. Forward strong points were often just linked networks of shell holes, making them more difficult to spot. Depth performed a number of functions: it allowed the trading of space for time; it imposed disorder and attrition on the advancing enemy; and it allowed the forward zone, which was the zone that would be saturated by enemy artillery, to be lightly held.
- Infantry firepower was enhanced by deploying new technology, including light machine guns; by utilising smaller, more flexible sub-units using fire-and-manoeuve to utilise the advantages of cover; and by improved integration of the artillery into the defence plan. This increased firepower allowed fewer forces to be deployed in the forward areas and yet still allowed them to impose meaningful costs on the advancing enemy.
- Elasticity expressed the focus in the new doctrine on rapid counter-attacks. Attackers were always likely to achieve some initial successes, especially against the weaker outpost zone. On the principle that attacking forces were likely to be most vulnerable soon after their attack had broken into the outer defences, defensive units were echeloned for immediate counter-attacks. Through the effective use of terrain, enhanced tactical firepower and local strong points designed to provide economy of force, a larger portion of the defensive manpower could be placed in reserve and allocated to counter-attacking roles.
- In order to work effectively, all of these ideas required a greater exercise of tactical initiative by defending forces. Isolated elements in the outpost zones needed to have the motivation to keep on fighting even when cut off; counter-attacks needed to be launched quickly to take full advantage of the disorganisation inflicted on the enemy through the depth and firepower of the defence. This would be achieved by devolving command responsibility and decision-making downwards. Local commanders were given command over all of the forces in their area, allowing them to initiate counter-attacks without waiting for orders from above.¹⁰

Table 5.2 *Late First World War German defence in depth*



The development of defence in depth worsened already existing challenges for the attacker and created for offensive operations a fundamental difficulty: exploitation. The central problem for much of the First World War was not that tactical success was impossible. Far from it: in general, with effective artillery support infantry could succeed in taking the first line of the enemy's defences. However, the reliance in the attack on extended artillery barrages could deliver only limited gains. Heavy bombardments destroyed the ground over which the attacking infantry would have to advance and also gave away the point of attack, allowing the enemy to mass reserves in preparation for counter-attacks. Moreover, even where troops achieved initial successes, other problems would then manifest themselves. Communications relied on field telephones connected by cables. These were static. It therefore proved difficult to co-ordinate with their reserves the activities of assaulting troops once they had left their trenches (usually disorganised and enervated by the attack), with other friendly forces on their flanks and with artillery support. Difficulties in communication also meant that higher headquarters often had little idea of how an attack was developing – even of as basic an issue as where the front line of the advance might be. Once defence-in-depth techniques became disseminated amongst the other belligerents, however imperfectly, the problem of translating a local break-in into a more general breakthrough and then breakout became even more challenging.

Infiltration techniques

The answer to the predominance of the tactical defence did not lie in new technology. For example, while both aircraft and tanks assumed greater significance during the war, the technology of both was still too immature to exert a decisive influence on the conduct of operations; light payloads and command-and-control difficulties mitigated the effect of the former; unreliability and vulnerability the latter. Instead, solutions were found in the use of new tactical techniques. Again, it was the German Army that innovated more comprehensively. In part, German adaptation reflected, amongst other things, the need in 1917 to find the tactical means to exploit the transient strategic advantage provided by the surrender of Russia and the consequent transfer of German troops to the west. But the German Army was also simply a more effective learning institution than its adversaries, facilitated, amongst other things, by its lack of a strong tradition of centralised doctrine, an effective system for learning lessons, a robust learning culture and devolved training.¹¹

Reflected in such documents as *The Training Manual for Foot Troops* of December 1916 and January 1918 (for battalions and below), and *The Attack in Position Warfare* of January 1918 (for regiments and above), the new German offensive techniques placed a priority on two things. First, the objective of offensive operations was to be the disruption of enemy forces rather than their destruction. By focusing on the disruption of the enemy military system,

the offensive would paralyse the enemy's ability to respond appropriately. Secondly, the revised German methods were based upon flexible infiltration. Referred to as *Flachen und Lücken*, or 'surfaces and gaps', attacking forces would avoid enemy strength (surfaces) and instead seek to flow through gaps in the defence.

Achieving these two effects depended upon realising a number of principles:

- One was neutralisation of the enemy. For example, rather than extended, destructive artillery barrages the focus was on heavy but sudden and unexpected attacks facilitated by more scientific gunnery techniques, such as pre-registering fire by map. The technique of 'map shooting' had been developed on the Eastern Front by Colonel Georg Bruchmüller. Targeting focused less on enemy entrenchments than on the broader infrastructure required for the enemy to function and respond, including headquarters, command posts, transport nodes and enemy artillery positions. As Bruchmüller noted, 'We desired only to break the morale of the enemy, pin him to his position.'¹²
- A second requirement was depth in the attack. The assault required multiple echelons. The first echelon would bypass centres of resistance, leaving them to be mopped up by follow-on forces. The first echelon would ignore flanks and push on, following routes of tactical success. They would not halt to be relieved. This facilitated continuous offensive action. The speed of offensive action and the depth in the assault maximised the shock against the enemy, provided protection to the vulnerable flanks of the attack and helped to ensure that the defenders did not regain the initiative. Jointery was also a feature, with aircraft used to help interdict the enemy rear areas, adding yet more depth to the attack. Another important principle was effective combined arms. Although, in general, the control of heavy artillery was centralised, pyrotechnics were used to control a creeping artillery barrage, the *Feuerwalze* or 'fire wall', which would facilitate more flexible artillery support for the infantry.
- A further means was a focus on independently manoeuvring small-unit infantry teams. The new doctrine eschewed a rigid and inflexible linear advance in favour of greater flexibility and a focus on initiative at lower levels (see Figure 5.2). Specialist 'storm battalions' were organised around assault teams with light machine guns, mortars, flamethrowers and light artillery. These units were used as the cutting-edge of the attack. The basic infantry sub-unit became much smaller: the *Gruppe*, a section with riflemen and a light machine gun. The section manoeuvred around the light machine gun, which acted as a firebase. To enable these changes required improved training and small-unit leadership. In January 1917, for example, the Germans established an infantry division whose sole purpose was to experiment with, and to train in, the new tactics.¹³

The first major use of these tactics on the Western Front occurred in March 1918 as part of the German spring offensive. They achieved impressive tactical



Figure 5.2 A US soldier provides fire support for his squad members during a live-fire exercise, Iraq, 2010. Squad-level fire and manoeuvre lies at the heart of modern infantry tactics. Fire suppresses the enemy and enables manoeuvre; manoeuvre allows fire to be brought to bear on the enemy more effectively.

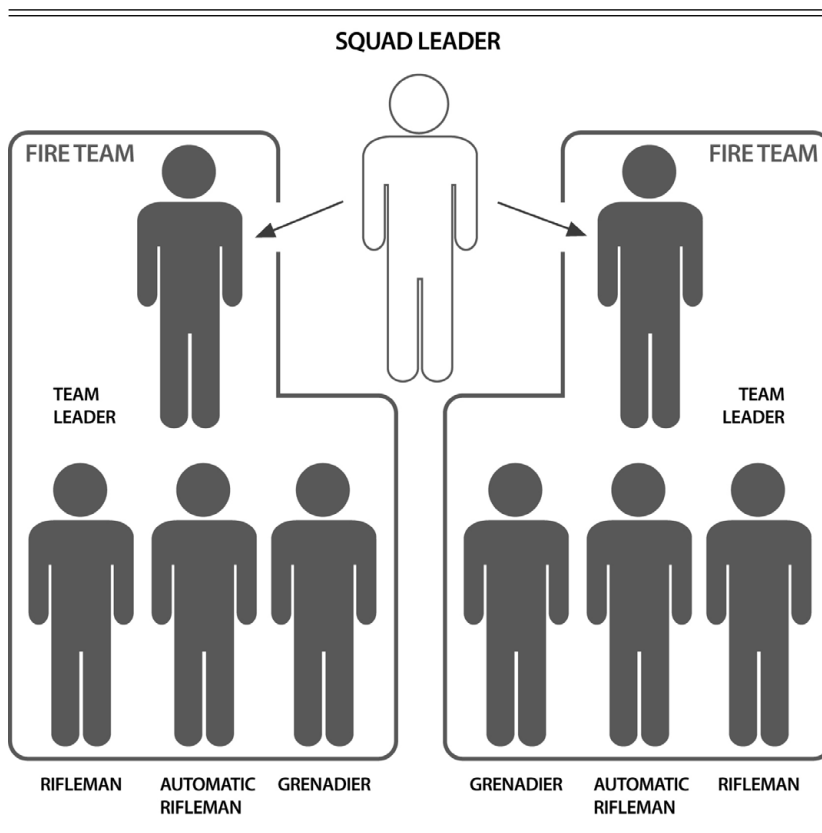
successes, and the techniques were adapted and used by the Allies. In the last year of the war, these techniques appeared to have restored the superiority of offensive over defensive operations: the Germans overran Allied defences three times – at the second battle of the Somme, the battle of Lys River and at the Chemin des Dames. The Allies used similar techniques in their later offensives that ended the war, including the second battle of the Marne in July, Amiens in August and the 100 Days’ offensive of September to November 1918. That mobility did not depend upon tanks was illustrated by the great success of General Allenby in Palestine: at Megiddo in 1918, Allenby achieved both breakthrough and exploitation with cavalry in concert with air power. It is important to note, however, that these offensives still involved attrition. The 1918 offensives cost the Germans a million men. British losses from August to November 1918 were greater than in the Passchendaele offensives of 1917. Still, by the end of the war, these new ‘modern’ tactical techniques of land warfare had restored an element of manoeuvre to the battlefield and had established the basis of techniques that would later be used in the Second World War.

Thus, the period from 1900 to 1918 saw a shift from what can crudely be termed 'anti-modern' warfare to a form of land warfare that was recognisably modern. In 1900, the response to the material conditions of land warfare had been to place undue emphasis on the moral component of war, expressed through a concentration on linear, close shock action. By 1918, the response was to place more emphasis on the conceptual dimension of warfare, finding new ways to use existing resources, not least combined arms, flexible sub-unit manoeuvre, infiltration tactics and tactical depth. The German Army pioneered these techniques, but they spread rapidly, if with varying degrees of success, to the other combatants through a 'bloody process of mutual education'. The war had produced, as Stephen Biddle notes, a 'stable and essentially transnational body of ideas on the methods needed to operate effectively in the face of radically lethal modern weaponry'.¹⁴ New technology, such as mines and anti-tank guns, would complicate, but not change, the basic tactical principles developed in 1917 and 1918. These principles, including decentralisation, combined arms, fire and movement, dispersal, cover and concealment and small-unit manoeuvre, continue to be the foundations of tactics today. For example, contemporary infantry tactics are founded on the activities of the infantry squad or section, which has its own automatic weapons and manoeuvres independently, in a dispersed fashion, combining the activities of its sub-elements, the fire teams (see [Tables 5.3 and 5.4](#)).

The Second World War and after: the operational art

The period after the First World War is commonly perceived as one dominated by the development and execution of blitzkrieg, a form of warfare that for many writers constitutes a definable military revolution. Taking this view, we could argue that, while states other than Germany were mired after 1918 in First World War thinking, German post-First World War techniques represented a 'paradigm shift' that provides the bridge between the static attrition of the Great War and modern mobile land warfare. But the reality is much more complex. As the preceding discussion has demonstrated, warfare had already become more mobile by the end of the First World War, and infiltration tactics provided the basis of tactics during the Second World War and after. German successes of 1939–41 did not derive from a revolutionary tactical superiority but from other advantages: but what advantages? The problem experienced in 1918 was how to translate the success of tactical actions into something broader and more decisive. The German spring offensive of that year, for example, had mauled two British armies, but this tactical success had not resulted in broader operational and strategic success: the offensive could not be sustained logistically, and the German High Command lost control over the offensive after five or six days.¹⁵ How could this gap between tactical success and strategic effect be bridged?

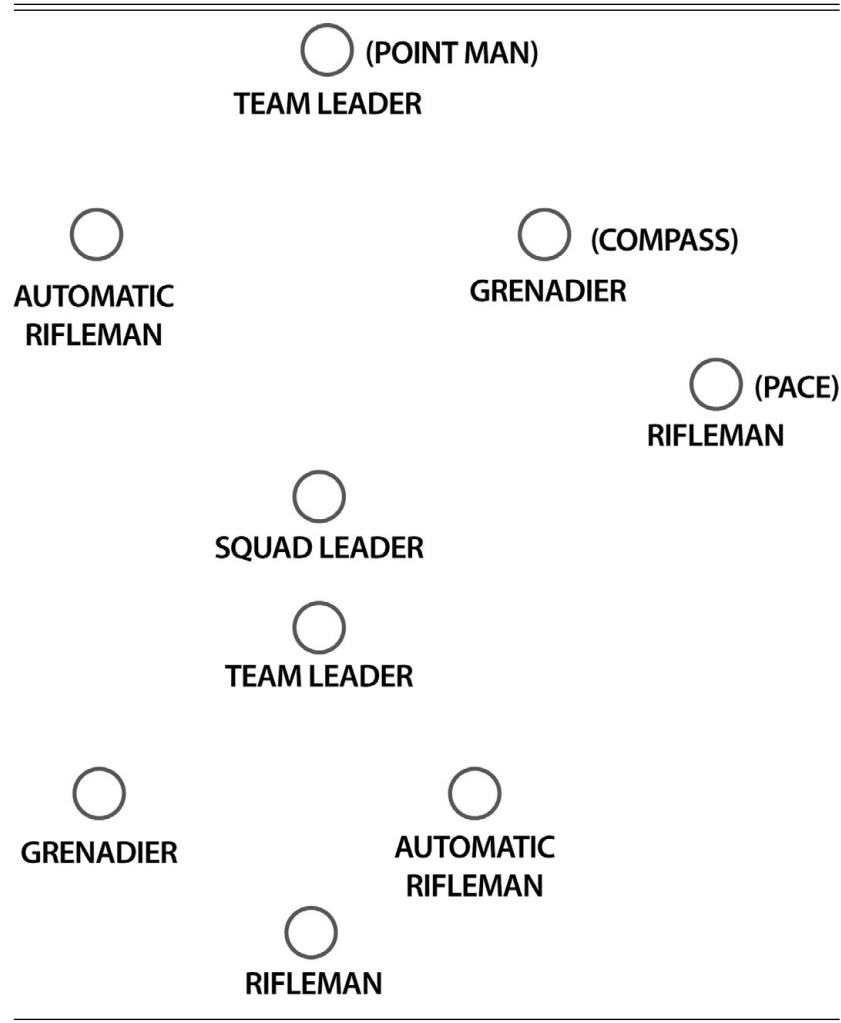
Table 5.3 *A modern infantry squad. The squad is broken down into two fire teams, allowing it to advance by bounds of alternate fire and movement**



* Adapted from Field Manual 7–8, *The Infantry Platoon and Squad* (Headquarters of the Army, 22 April 1992).

Traditional military thought had identified two levels of war: strategy and tactics. Strategy encompassed all those activities tasked with bringing forces to the battlefield with maximum advantage; tactics covered the conduct of forces on the battlefield itself.¹⁶ What was most significant about the period after 1918 was not blitzkrieg itself, but the fact that it expressed the conscious and unconscious reflection of the importance of an intermediate level of war, between strategy and tactics, of which the ideas and practices that informed what later became known as blitzkrieg were a manifestation. The emergence of this level, an operational level that would require the practice of operational art, resulted from material and cognitive developments. In material terms, the growing scale of modern war meant that strategic success could not be delivered through a single decisive battle, but only through the co-ordination in time and space of multiple tactical actions. As noted in [chapter 1](#), at a cognitive level some recognition of this had already begun to occur: Jomini, for example, used the term ‘grand

Table 5.4 *Deployment of a modern infantry squad**



* Compare this with Table 5.1. Adapted from Field Manual 7–8, *The Infantry Platoon and Squad* (Headquarters of the Army, 22 April 1992).

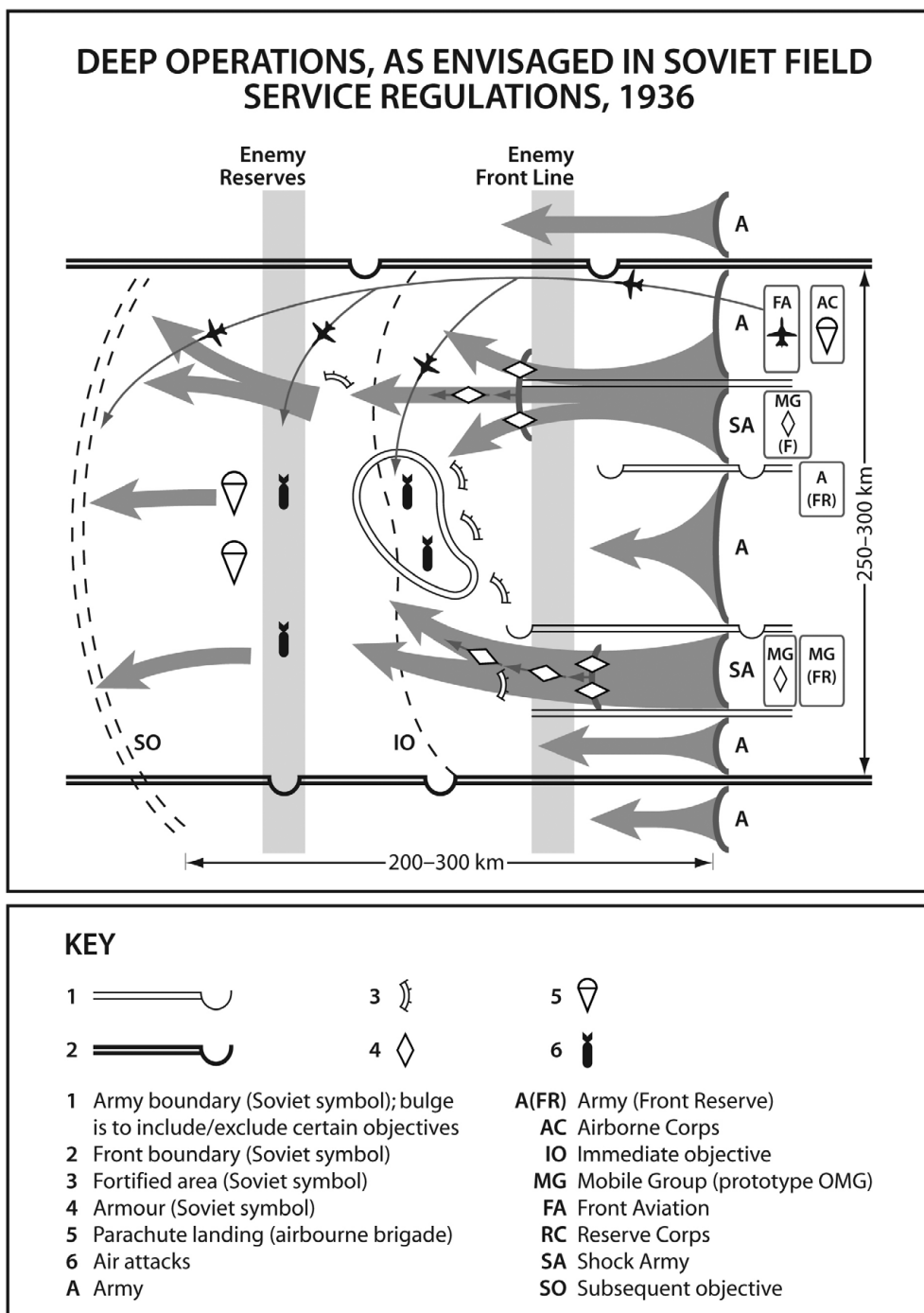
tactics’ to refer to this intermediate level, and the Prussian/German Army had begun to refer to a lower level of strategy using such terms as *Kriegsoperationen* (‘war operations’) and *operativ* (loosely, ‘operational’). The pre-First World War military mindset, with its focus on decisive battle, was not conducive to more systematic thinking about the subject. Von Schlieffen, for example, envisaged his plan for war in 1914 as a giant Cannae, and his attempt to treat the whole massive operation of 1.5 million men as a single choreographed tactical action exceeded the command and logistic realities of the time.¹⁷

Operational-level mobile warfare

The most systematic thinking in the interwar period regarding this intermediate level of war emerged not from the Germans but from the Red Army. Concluding that, in modern war, one could not defeat the enemy with one tactical blow, Soviet thinkers began to develop the conceptual framework for a mode of war that would allow the delivery of co-ordinated, successive attacks against the enemy that would enable local tactical successes to be exploited to produce a larger effect. This focused on operational art (*Operativnoye Iskusstvo*): the theory and practice of preparing for and conducting combined and independent operations by large units (armies and groups of armies). As A. A. Svechin (1878–1938), one of the key Soviet theorists, noted, ‘tactics make the steps from which operational leaps are assembled; strategy points out the path.’¹⁸ For Svechin, ‘Operational art governs tactical creativity and links together tactical actions into a campaign to achieve the strategic goal.’¹⁹

The Soviet perspective emerged from their recognition of the growing importance of depth in the practice of warfare. Experiences in the Russian Civil War (1918–24) had demonstrated how cavalry corps and cavalry armies could be used to engage in deep penetration well beyond the enemy front line. After the First World War, the maturing technologies of mechanisation and wireless radio seemed to provide new means with which to realise this concept. Mechanised mobile forces would allow ‘deep battle’ and ‘deep operations’. ‘Deep battle’ comprised the ambition tactically to penetrate the enemy to their rear artillery line. This would be followed by ‘deep operations’: exploitation beyond an individual battlefield into the depth of the theatre of operations. Enshrined in the Field Service Regulations of 1936, this doctrine envisaged infantry, cavalry and mechanised forces operating with air support in a two-part battle (see [Table 5.5](#)). First, using artillery fire to suppress the enemy, an echeloned, combined-arms attack would be launched on a narrow front to penetrate the enemy line. Secondly, ‘mobile groups’ consisting of cavalry and mechanised forces would exploit the initial breakthrough, striking deep and outflanking the enemy defences. These mobile groups would encircle the enemy to a depth of 200 or 300 kilometres. Also enshrined in this doctrine was the importance of simultaneity: influencing at the same time the entire depth of the enemy positions by using smoke, deception and air attack. The aim, according to the regulations, was the ‘violent development of tactical success into operational success with the aim of the complete encirclement and destruction of the enemy’.

The possibilities promised by new technology were also recognised by other armies. In May 1918, J. F. C. Fuller, a British staff officer, conceived ‘Plan 1919’, a massive armoured offensive that he hoped would end the war. Plan 1919 combined tanks and aircraft in a massive joint effort on a narrow front that would rupture the enemy line. Using the mobility of armour and the reach of aircraft, this force would then engage in what was in effect a ‘deep battle’, exploiting quickly, overrunning enemy command and control nodes and completely paralysing the German capacity to respond. Another British military

Table 5.5 *Deep operations, as envisaged in Soviet Field Service Regulations, 1936*

Box 5.1 The reality of blitzkrieg

The term 'blitzkrieg' provides a convenient short-hand for operational-level mobile, combined-arms warfare. It is important to note, however, that blitzkrieg never existed as a formal part of German doctrine. The term blitzkrieg was rarely used in the German Army; indeed, in November 1941, when Hitler first heard it, he referred to it as 'a very stupid word'. Moreover, the German principles for mobile combined-arms warfare were not regarded by the German Army as revolutionary. The scale of the success experienced in France in 1940 was thus wholly unexpected, Hitler referring to the result as 'a miracle, an absolute miracle'.

The ideas that we commonly associate with blitzkrieg continued to be resisted by significant elements in the German Army and Luftwaffe. Even in the Polish campaign in 1939, tensions existed between such radicals as Heinz Guderian (author of *Achtung – Panzer!*) and conservatives over the proper use of armour and mobile forces. Many in the German Army continued to view armour as primarily an infantry-support weapon. Thus, as late as 1937 and 1938 scarce tanks were allocated to form two infantry-support tank brigades. The Luftwaffe continued to prioritise tasks other than close air support. Not until 1940 did the German Army's regulations focus on the operational-level use of armour. Even afterwards it remained far from being a template that was applied consistently throughout the army.

theorist, Basil H. Liddell Hart, argued for a future similar to that of Fuller's; however, whereas Fuller focused exclusively on tanks, Liddell Hart saw the need for a force of accompanying mobile infantry – in effect, mechanised infantry. In 1927, Britain created the Experimental Mechanised Force (EMF), consisting of a mixed force of tanks, motorised infantry and artillery with air support. By 1939, the United Kingdom had the only fully motorised army in the world. The United States also recognised the possibilities. In 1928–9, the United States created the Experimental Armoured Force, which included small tanks and lorried infantry.²⁰

However, only Germany was as comprehensive in its focus on combined-arms exploitation as the Soviets. Moreover, since the Red Army was thrown into turmoil by political purges in 1937–8, the German Army was left as the pre-eminent exponent of a form of operational art, even if this was not formally codified in the way that it was in the Soviet Union. Under the direction from 1919 to 1926 of the Commander of the Army, Hans von Seeckt, the German Army sought to compensate for the restrictions placed upon it by the Versailles Treaty by creating a military machine with superior mobility and flexibility. Despite the popular tendency to focus on the revolutionary character of blitzkrieg (see [Box 5.1](#)), the principles from which it was derived were evolutionary and stemmed from the tactical principles extant at the end of the First World War. German approaches applied technologies for mobile warfare through a conscious adaptation of the tactical concepts developed in the latter stages of that war. Thus, the Reichswehr's first field manual, the 1921 'Leadership and Combat of Combined Arms' (*Fuehrung und Gefecht der*

verbundenen Waffen) and the later *Die Truppenführung*, or 'troop leadership' of 1933, married the infantry assault unit tactics and effective artillery doctrines of the First World War with close air support and the use of tanks.²¹ The new doctrine continued to stress flexibility, initiative and exploitation, with a focus on either narrow breakthroughs or encirclement. The essence of the German techniques used in the early part of the war was a surprise attack on a narrow front designed to ensure a penetration of the enemy front line. Swift exploitation would then become the priority, with a focus on the demoralisation of the enemy as much as their physical destruction: multiple columns would exploit the initial penetration by moving to the rear of the enemy, bypassing concentrations and acting as pincers to encircle the enemy. Aircraft would act as flying artillery and, in concert with rapidly moving panzer divisions, would deepen the battle, helping to create operational-scale systemic shock.²²

Despite the apparent importance of panzer divisions to the German techniques, blitzkrieg was not primarily a tank phenomenon. Indeed, by 1941 the German Army was reducing the amount of armour in panzer divisions and trying to increase the allocation of motorised and mechanised infantry. Combined arms, rather than tanks, was one of the keys to German success. For example, anti-tank guns would be used to protect the vulnerable flanks of the breakthrough. Engineers would be on hand to increase mobility. Combined arms was facilitated by flexible training and command-and-control structures that allowed the use of all-arms battlegroups to suit the situation. Blitzkrieg was thus the result of training and doctrine in combined arms, tied to a focus on penetration and manoeuvre.

A key enabler was the recognition of the importance of radio communications in land warfare. Widespread use of wireless radios enabled the effective command and control of more complex, mobile and dispersed forces. From the mid 1920s, for example, the German Army fitted new vehicles with radio mounts, and efforts were made to develop a full range of radios including short-range sets for vehicles and high-power/low-frequency sets for headquarters. There was also recognition of the importance of joint operations. Even before the re-creation of the Luftwaffe in 1933, the German Army maintained a doctrine on co-operation with the air. This co-operation was reflected also in the development of specific joint assets, especially the JU-87 Stuka dive bomber. The practical reflection of these ideas was the creation of panzer divisions – mobile all-arms formations, focused around the tank but including infantry, artillery and other supporting arms. By 1939, two-thirds of the German tank force was concentrated in six panzer divisions.

The German Army achieved startling successes in France and Poland and in the first stages of the attack on Russia. In the battles against France, at least, the operational-level application of concepts of deep battle seemed vindicated.²³ Five days into the attack, for example, German forces were already conducting deep penetration and exploitation, wreaking havoc on Allied command and control and morale. Deep operations, realised through such principles as combined arms, mobility, rapidity, fire-and-manoevre, massing at decisive points,

decentralised command and control and rapid exploitation, seemed to have restored the capacity to deliver decisive outcomes in land warfare, even if these principles were still being applied in a rather ad hoc fashion.

Operational defence in depth

Yet, despite these extraordinary successes, the Second World War came to exhibit a progressively more positional and attritional character. Why did blitzkrieg cease to deliver remarkable victories? One reason for this was that the early opposition facing the German Army provided to an extent a 'best-case' test for operational-level mobile warfare, because armies such as those of the Poles and French were badly deployed and had inadequate reserves. Operation Barbarossa was launched against a Russian opponent suffering from the mutually reinforcing problems of political purges, reorganisation, expansion and inexperienced commanders. Moreover, the Soviet leadership forced ill-chosen offensive and defensive operations on Russian commanders. If Barbarossa had been launched four years earlier, or one year later, it might well have been much less successful.

Logistics imposed other emerging limitations on operational-level manoeuvre. Mobile armoured warfare may have helped solve some tactical problems, but it created new logistical ones. Operational-level mobile warfare required intensive, complex, mobile logistic support if it were not quickly to lose momentum. Whereas a 1914 infantry division required 100 tons of supplies a day, a panzer division required 300 tons and a US division by 1944 required 600 tons.²⁴ In the Polish campaign of 1939, no German unit advanced more than 150 miles into Poland, yet it still proved difficult to keep front-line units in supply. In Russia, with supply lines extending up to 900 miles, the problems of supplying the needs of high-intensity mobile warfare often proved insurmountable. Even operating the rail lines was problematic, given freezing weather conditions and incompatible rail gauges; linking railheads to mobile forces was even more difficult, with the German Army's dearth of motor transport and the severe operating conditions. For example, by November 1941 some of the lead German infantry divisions were operating 90 miles from their railheads, a distance twice that from which they could be supplied effectively.

Exploitation posed a related challenge. In 1940, only one-tenth of the German Army was made up of panzer or mechanised divisions. The rest was reliant on horses for its transport; indeed, the German Army used nearly twice as many horses in the Second World War (some 2.7 million) as it did in the First.²⁵ Panzer exploitation could often only take place once infantry and logistics had caught up. It is notable that the average daily rate of advance for panzer divisions in Poland was only 11 miles. Quick 'leaps' forward were accompanied by long pauses while the means to consolidate gains caught up. In Russia, this 'mobility gap' imposed delays that reduced the tempo of advance and provided opportunities for Russian counter-attacks. Operational-level mobile warfare also

therefore required broad-based mechanisation if effective mobile combined arms was to be achieved. In that sense, the German Army of the Second World War was only 'semi-modern'.

Another difficulty was the vulnerability of attacking forces. Operational-level mobile warfare relied on causing the systemic collapse of the enemy in order to prevent the enemy from launching counter-attacks against vulnerable exploitation forces. If the enemy did not collapse quickly, then the costs of a risk-based exploitation approach were likely to mount. In the Polish campaign, German tank losses amounted to 10 per cent of the initial force. Many units also lost up to 50 per cent of their motor vehicles. The cutting edge of blitzkrieg, the panzer divisions, suffered heavy attrition even in the early stages of the campaign in Russia: from June to November 1941 the German Army lost two-thirds of its motor vehicles and 65 per cent of its tanks. The attrition in motor vehicles compounded the larger logistic problems. From November 1941 to March 1942, the German Army lost another 75,000 vehicles but could only find 7,500 replacements.²⁶

However, the central difficulty experienced by Germany was the relational nature of warfare: in essence, its enemies adapted. The essence of blitzkrieg comprised the pushing of a mechanised combined-arms force through the enemy front line so that it could penetrate the enemy far into the depth of their defences, causing their systemic collapse. The answer to this lay in the rational development of concepts developed in the First World War, in this case the principle of 'defence in depth', applied at an operational and not just a tactical level. This was not a new idea. The Russian deployment prior to Barbarossa was in essence a defence in depth, but it was poorly executed. By the middle of the Second World War, as combatants became more experienced, defence in depth had reasserted the strength of the defence. New technology, such as anti-tank guns, complicated combined arms but it also made defence in depth potentially more effective by improving the effectiveness of non-tank forces in fighting armour. At Kursk, for example, the Soviet defences were in places 110 miles deep. Defences this deep were designed not just to inflict losses on the attacker but to disorganise them and to break down the capacity for combined arms by separating armour from supporting infantry. At points of expected German attack, the defences might have 1,500 to 2,000 mines and twenty-five to thirty anti-tank guns per kilometre of frontage. The defence-in-depth concept also harnessed armoured and mechanised forces. The Soviets held the Steppe Front of five armies in reserve for counter-attack. Depth helped to mitigate surprise and gave opportunities for counter-attack.²⁷ The Germans themselves elevated the mobile counter-attack to an art form through the use of 'manoeuvre on interior lines', using mobile forces based on a central position to counter-attack enemy encirclement attempts, defeating them sequentially in detail.

This did not mean that the tactical and operational offensive concepts developed since 1917 were irrelevant; far from it. Offensive successes were still possible. For example, in Operation Bagration, which began in June 1944, the

Russians smashed Germany's Army Group Centre, destroying seventeen divisions and reducing another fifty to below half strength. The offensive pushed the Germans back 350 miles. Soviet concepts developed from 1943 were illustrative of the techniques elaborated in an attempt to overcome defence in depth. Reconnaissance elements were strengthened in order to promote the chances of a breakthrough, transforming time-consuming reconnaissance into the first echelon of the attack designed to seize outposts and 'recce' by fire. Attack groups were made more flexible by dividing them into reconnaissance, blocking, fire and attack subgroups to promote greater flexibility in combined arms. Use was made of tailored assault groups and specialist formations, such as combat engineers. Great emphasis was put on deception to promote surprise and to facilitate massing at critical points. In order to increase the pace of exploitation, the Soviets promoted decentralised command and control: relatively junior officers were encouraged to use initiative and seize targets of opportunity. The size of forward elements and their distance from the main body was increased: by 1944, forward elements for a tank corps might operate some 50 miles in front of parent corps, and might include a tank brigade and field and anti-aircraft artillery, assault guns and engineers. For important points of exploitation, forward air controllers would accompany lead elements, and dedicated air assets would be allocated for close air support to speed the advance.

Nevertheless, neither Bagration nor other examples of rapid advances, such as General Patton's advance in the Normandy campaign of 400 miles in twenty-six days, induced the general systemic collapse of the enemy forces as blitzkrieg had in 1939 and 1940. These were not the victories envisaged by Fuller and Liddell Hart. Operations from late 1942 onwards were, in some respects, reminiscent of the First World War, in their focus, for example, on set-piece attacks, particularly those conducted by Britain and the Soviets. Eventual Allied success depended upon broad, protracted attrition at the strategic level, in which elements of more mobile warfare were interspersed with long periods of positional fighting. In a general sense, this could still be decisive: in two and half years of war, the Soviets drove the Germans from the Volga River to the Elbe. However, Allied breakthroughs depended on attrition first rather than lightning breakthroughs, and this attrition might be as bad as or worse than that of the First World War. The combined casualties at Verdun in that war were 750,000 suffered over a period of ten months; in Normandy, 637,000 casualties were suffered in eighty days. The breakout from Normandy was enabled by two months of heavy, static fighting that had inflicted significant losses on the Germans. For example, most German infantry units suffered more than 100 per cent casualties during the campaign.²⁸

The operational level of war

German and Soviet thinking had focused on the importance of operational art. The operational level of war did not exist for them as a formal concept, and

instead lay at whatever level operational art was practised. Perhaps the most significant post-war developments were (a) the spread of concepts of operational art to English-speaking armies, and (b) the formal doctrinal recognition by the United States in the 1980s of a distinct operational level of war.

The essentially evolutionary nature of the development of modern land warfare was sustained during the Cold War despite the great uncertainty caused by the impact of such new technology as nuclear weapons, helicopters and precision-guided munitions (PGMs) and by such new strategic ideas as deterrence and limited wars. Nuclear weapons posed a fundamental challenge to existing concepts of land warfare. As the eminent civilian strategist Bernard Brodie noted, 'Thus far the chief purposes of our [the United States'] military establishment has been to win wars. From now on its chief purpose must be to avert them.'²⁹ In the Soviet Union, less emphasis was placed on operational art; in the United States, the solution was to focus on nuclear armed units, emphasising firepower over manoeuvre. By the 1960s, however, it was clear that nuclear war was not inevitable, and that conventional land warfare might still have a crucial role to play in future conflict.

In the period from the 1960s through to the 1980s, Soviet military doctrine returned to an emphasis on familiar themes of operational art, albeit modified to take into account developments in technology and US doctrine. Soviet doctrine noted that 'the possibilities of defeating the enemy in the entire depth of his operation combat formations have increased'.³⁰ New technology, for example, provided the means to supplement ground operations in 'land-air' battle with a 'vertical echelon' delivered by air-mobile forces. Nevertheless, Soviet doctrine continued to stress the need for surprise, attacking the enemy throughout its depth, superiority in numbers and firepower at decisive points, mobility, continuous operations and the importance of maintaining large reserves.

In the United States, however, defeat in Vietnam prompted a reassessment of existing doctrine.³¹ One conclusion reached was that the defeat there was partly attributable to an over-reliance on firepower and a managerial approach to war that focused on systems and technology. Other factors also influenced this reassessment. For example, the Arab-Israeli Yom Kippur/Ramadan War of 1973 demonstrated the impact of PGMs: in three weeks of fighting, the losses of armour and artillery exceeded in numbers the entire US inventory at the time. Modern war appeared to be fought at a high tempo in a fast-moving, complex, all-arms environment which required excellent command and control. The result of this process of reflection was the emergence of a new doctrinal concept, 'AirLand battle', introduced formally in a new edition of Field Manual 100-5, *Operations* in 1982. AirLand battle tried to weave into a whole, close operational interaction between air-ground forces, combined arms, fire support, electronic warfare, deception, intelligence and manoeuvre. AirLand battle tried to marry firepower with manoeuvre by attaching importance to four concepts: initiative, depth, agility and synchronisation. The essence of AirLand battle was 'deep battle' and developing the capacity to slow Soviet second-echelon

Box 5.2 The operational level of war

'The operational level links the tactical employment of forces to national and military strategic objectives. The focus at this level is on the design, planning, and execution of operations.'

Joint Publication 3-0, Joint Operations, 11 August 2011, I-13

'Operational art is the use of creative thinking by commanders and staffs to design strategies, campaigns, and major operations and to organize and employ military forces.'

Joint Publication 3-0, Joint Operations, 11 August 2011, II-13

The focus of the operational *level* is the campaign, a series of tactical actions that should be linked by time, geography and a common purpose, so that they serve broader military and grand strategic objectives. Operational *art* is the employment of military forces at this level.

forces while pinning down and destroying the first-echelon forces. Manoeuvre and firepower were treated as inseparable elements.³² As part of this new approach, the new doctrine identified for the first time a formal operational *level* of war (indeed, it did not mention operational art at all). This operational level of war was defined as:

planning and conducting campaigns. Campaigns are sustained operations designed to defeat an enemy force in a specific place and time with simultaneous and sequential battles. The disposition of forces, selection of objectives, and actions to weaken or outmaneuver the enemy all set the terms for the next battle and exploit tactical gains. They are all part of the operational level of war.³³

Subsequent doctrinal developments since then have formally codified the concept of operational art in the doctrine of the United States, and that of their allies, as well as retaining the notion of a formal operational level of war (see [Box 5.2](#)).

The formal integration into Western military doctrines from the 1980s onwards of the operational art was accompanied, and indeed supported, by the parallel emergence of the concept of manoeuvre warfare. Responding to criticisms that the US army was too attritional in its approaches, advocates of manoeuvre warfare argued for an emphasis on attacking the conceptual and moral components of an enemy's fighting power rather than simply the physical (see [Box 5.3](#)). This required a focus on undermining the enemy's will, cohesion and decision-making capacity, rather than destruction in a crude attritional struggle. Manoeuvre warfare has been characterised as the capacity to cycle through OODA loops (Observe, Orientate, Decide, Act) faster than the opponent, thus invalidating their decision-making, robbing them of the initiative and promoting systemic collapse (the OODA loop concept is discussed in more depth in [chapter 11](#), p. 262).³⁴ As exponents of manoeuvre warfare

Box 5.3 Manoeuvre warfare

Manoeuvre warfare is a war-fighting philosophy that seeks to shatter the enemy's cohesion through a variety of rapid, focused and unexpected actions which create a turbulent and rapidly deteriorating situation with which the enemy cannot cope (US Marine Corps publication *MCDP1 Warfighting*, p. 73).

Manoeuvre warfare places an emphasis on speed, focus, surprise, disruption and exploitation. It is a demanding philosophy that requires of armies initiative, flexibility, risk-taking, lateral thinking and an ability to cope well with uncertainty. Manoeuvre warfare was reconceptualised by the British in the mid 1990s into the 'manoeuvrist approach', a philosophy of unorthodoxy and adaptability that was applicable to all forms of warfare.

acknowledged, there was nothing strictly new in this approach; instead, it reflected a systematisation of historical 'best practice'. This evolutionary approach was reflected in a resurgence in Western armies of the study of the Soviet methods of 1943–5 and also of German operational art. Some of the key tools associated with manoeuvre warfare drew on German approaches adopted and utilised before and during the Second World War, including *Auftragstaktik* ('mission command'), *Schwerpunkt* ('focus of effort') and *Flachen und Lücken* ('surfaces and gaps').

Conclusion

The preceding discussion demonstrates that there is a strong evolutionary dynamic to the way in which land warfare has developed over time. The infiltration tactics of 1917–18, blitzkrieg and manoeuvre warfare form much more of a continuum of development than they do radical discontinuities. Modern tactics evolved on the battlefields of the First World War; they embodied the principles of decentralisation, combined arms, fire-and-movement, dispersal, cover and concealment and small-unit manoeuvre. Modern tactics enabled offensives to succeed even under the conditions of the radical increases in the theoretical lethality of battlefield firepower. Yet, given the growing scale of warfare, there were limits to the extent to which tactics could solve the problem of translating local successes into larger-scale breakthroughs. Modern operational art emerged as a solution. Building on the capabilities provided by mechanisation and radio communications, new concepts emerged to guide the sequencing of tactical actions in time and space in order to meet strategic objectives.

Successful mastery of the modern system of land warfare has enabled armies to mitigate some of the consequences of industrialised people's war, such as firepower, and to harness others, such as new weapons platforms and means of mobility. The use of this system explains why casualty rates have not expanded in relation to theoretical lethality or the speed of modern weapons. Despite

increases in firepower technology, in the period between 1900 and 1990 average casualty rates decreased by more than 60 per cent. It also explains why, despite the power of the defence demonstrated in the initial part of the First World War, modern armies have been able to conduct successful offensive operations. But will the modern system of warfare continue to apply in the future? Moreover, if the modern system of warfare is so effective, can we assume that it is the only way of fighting land warfare? To answer these questions we turn now to the final chapter on land warfare.

RESEARCH QUESTIONS

1. In the period 1870–1913, what key factors shaped the nature of the debates on the impact of technology on land warfare?
2. Why did the German army adapt more quickly to battlefield conditions than its adversaries during the First World War?
3. What were the strengths and weaknesses of Soviet operational art during the Second World War?
4. Why did it take so long for operational art to be codified in US military doctrine?
5. How useful is the concept of a distinct operational level of warfare?

FURTHER READING

- Corum, James S., *The Roots of Blitzkrieg: Hans von Seeckt and German Military Reform* (University Press of Kansas, 1992). This analyses the sources and nature of the German interwar process of reflection and adaptation.
- Freiser, Karl-Heinz, *The Blitzkrieg Legend: The 1940 Campaign in the West* (Annapolis, MD: Naval Institute Press, 2005). This book discusses the essentially evolutionary origins of German doctrines for operational-level mobile warfare.
- Glantz, David, *Soviet Military Operational Art: In Pursuit of Deep Battle* (London: Frank Cass, 1991). Glantz charts the development of Soviet thinking on operational art from its origins in the 1920s to the doctrines of the Cold War.
- House, Jonathan, *Combined Arms Warfare in the Twentieth Century* (Lawrence, KS: Kansas University Press, 2001). A detailed survey of developments in land warfare from the late nineteenth century to the 1990s.
- Leonhard, Robert, *The Art of Maneuver: Maneuver-Warfare Theory and AirLand Battle* (New York: Ballantine, 1991). An introduction to US operational art and manoeuvre warfare and its application to the doctrine of AirLand battle.
- Lupfer, Timothy, *The Dynamics of Doctrine: The Changes in German Tactical Doctrine during the First World War* (Kansas, KS: Combat Studies Institute, July 1981). Lupfer explains the development of German infiltration tactics.
- Olsen, John Andreas and Martin Van Creveld (eds.), *The Evolution of Operational Art: From Napoleon to the Present* (Oxford University Press, 2011). A comprehensive assessment of the theory and development of operational art across different nations.

Sheffield, Gary, *Forgotten Victory: The First World War – Myths and Realities* (London: Headline, 2001). Highlights convincingly how far land warfare had developed by 1918.

ONLINE RESOURCES

The *Journal of Military Operations*, www.tjomo.com/issues/, is a free to register online journal that carries relevant pieces. See, for example, Long, Gerry, 'Storming Back to the Future: Why We Wrestle with the Basics', 2(3) (Summer 2014) and Kiszely, John, 'Where to for "the Operational?" An Answer', 1(4) (Spring 2013).

A large collection of US Army Field manuals can be found at www.globalsecurity.org/military/library/policy/army/fm/. See, in particular, FM 3.21-8, *The Infantry Rifle Platoon and Squad* and FM 3.21-10, *The Infantry Company* for an examination of the principles of modern tactics.

The US Combat Studies Institute has a wide range of relevant publications, available at <http://usacac.army.mil/cac2/CSI/CSIPubs.asp>.

The US Joint Electronic Library has a huge collection of contemporary doctrinal material, including access to the journal *Joint Force Quarterly*, www.dtic.mil/doctrine/.

For a look at manoeuvre warfare at the tactical level, see www.fs.fed.us/fire/doctrine/genesis_and_evolution/source_materials/MCDP-1-3_tactics.pdf.

For some rather more visual material on modern tactics, see www.youtube.com/watch?v=9rLW9f6l4Tg. Focusing on the battle of Amiens in 1918, this opening episode of the BBC's *Twentieth Century Battlefields* looks at military developments, especially in relation to combined arms, and demonstrates the continuity between the First World War and today.

www.youtube.com/watch?v=Kz0a_QGifPM. This is an Allied film from the Second World War designed to educate soldiers in German tactics.

NOTES

1. Stephen Biddle, *Military Power: Explaining Victory and Defeat in Modern Battle* (Princeton University Press, 2004), p. 3.
2. Colin McInnes, *Men, Machines and the Emergence of Modern Warfare, 1914–1945* (Camberly: Strategic and Combat Studies Institute, 1992), p. 4.
3. Quoted in Jay Luvaas, *The Military Legacy of the Civil War: The European Inheritance* (University Press of Kansas, 1988), p. 167.
4. *Ibid.*, p. 167.
5. Bruce I. Gudmundsson, *Stormtroop Tactics: Innovation in the German Army, 1914–1918* (Westport, CT: Praeger, 1995), p. 1.
6. McInnes, *Men, Machines and the Emergence of Modern Warfare*, p. 8.
7. Michael Glover, *Warfare from Waterloo to Mons* (London: Book Club Associates, 1980), p. 248.
8. *Ibid.*
9. Williamson A. Murray, 'The West at War', in Geoffrey Parker (ed.), *The Cambridge History of Warfare* (Cambridge University Press, 2005), p. 294.
10. Robert T. Foley, 'A Case Study in Horizontal Military Innovation: The German Army, 1916–1918', *Journal of Strategic Studies*, 35(6) (December 2012), 808–12.

11. *Ibid.*, 812–20.
12. Timothy T. Lupfer, *The Dynamics of Doctrine: The Changes in German Tactical Doctrine during the First World War* (Lawrence, KS: Combat Studies Institute, July 1981), p. 45.
13. *Ibid.*, p. 41.
14. Biddle, *Military Power*, p. 28.
15. See David T. Zabecki, *The German 1918 Offensives: A Case Study in the Operational Level of War* (London: Routledge, 2009).
16. On the relationship between strategy and tactics see Beatrice Heuser, *The Evolution of Strategy: Thinking War from Antiquity to the Present* (Cambridge University Press, 2010), pp. 3–35.
17. It was at Cannae that Hannibal achieved a successful, and since widely studied, envelopment of a Roman army in 216 BC. See also ch. 1, p. 48.
18. Quoted in John Kiszley, ‘Thinking about the Operational level’, *RUSI Journal* (December 2005), 38.
19. Jacob W. Kipp, ‘The Tsarist and Soviet Operational Art: 1853–1991’, in John Andreas Olsen and Martin Van Creveld (eds.), *The Evolution of Operational Art: From Napoleon to the Present* (Oxford University Press, 2011), p. 66.
20. For a discussion on interwar theorists, see Allan R. Millett and Williamson Murray (eds.), *Military Effectiveness, II: The Interwar Period* (Cambridge University Press, 2010).
21. Milan Vego, *Joint Operational Warfare: Theory and Practice* (Newport, RI: US Naval War College, 2009), p. I-22.
22. Dennis Showalter, ‘Prussian-German Operational Art, 1740–1943’, in John Andreas Olsen and Martin Van Creveld (eds.), *The Evolution of Operational Art: From Napoleon to the Present* (Oxford University Press, 2011), p. 51.
23. There is still debate as to whether the Polish campaign was really blitzkrieg at all, given that the use of mobile armoured forces was still overwhelmingly tactical in conception.
24. Martin Van Creveld, *Technology and War: From 2000 to the Present* (London: Brassey’s, 1991), p. 181.
25. Karl-Heinz Freiser, *The Blitzkrieg Legend: The 1940 Campaign in the West* (Annapolis, MD: Naval Institute Press, 2005), p. 27.
26. For details of the huge logistic problems see David Glantz, *Barbarossa Derailed: The Battle for Smolensk, 10 July–10 September 1941*, vol. I (Solihull: Helion and Company, 2010).
27. see Lloyd Clarke, *Kursk: The Greatest Battle* (London: Headline, 2012), pp. 207–12.
28. Total casualties (including replacements) as compared with initial strength.
29. Quoted in Michael Evans, *The Primacy of Doctrine: The United States Army and Military Innovation and Reforms, 1945–1995*, Army Occasional Paper No. 1 (Canberra: Directorate of Army Research and Analysis, Department of Defence, August 1996), p. 3.
30. David Glantz, *Soviet Military Operational Art: In Pursuit of Deep Battle* (London: Frank Cass, 1991), p. 38.
31. Richard Lock-Pullan, ‘“An Inward Looking Time”: The United States Army, 1973–76’, *The Journal of Military History*, 69(2)(April 2003), 498–9.

-
32. Evans, *The Primacy of Doctrine*, p. 20.
 33. Justin Kelly and Mike Brennan, *Alien: How Operational Art Devoured Strategy* (Carlisle, PA: Strategic Studies Institute, 1992), p. 61.
 34. William S. Lind, *The Maneuver Warfare Handbook* (Boulder, CO: Westview, 1985), p. 6.



6

Future land warfare

Contents

Introduction

The Revolution in Military Affairs

New wars

Hybrid warfare

 The future land warfare debate

 The paradigm army

 Context and the future of land warfare

 Western culture and the future of land warfare

Conclusion

KEY THEMES

- The future form of land warfare is far from certain.
- For some, the future is a technologically focused Revolution in Military Affairs; for others it is a future of new wars, brutal, local and low-tech; still others see a future marked by hybrid warfare threats that mix conventional and unconventional techniques.
- History suggests that, in the future, many different forms of land warfare are likely to co-exist because land warfare is shaped by different political, economic, social and cultural contexts.

Introduction

In this final chapter on land warfare, we turn our attention to the future. The two preceding chapters have highlighted the evolutionary underpinnings of the 'modern system' approach to land warfare. Can we assume similarly that future land warfare will look much like that of the past? This is clearly a crucial question for land forces around the world. Governments do not have the luxury of perpetual analysis: given the time taken to generate effective military power, choices have to be made now about the sorts of land forces that will be required for the future – their size, structure, equipment, training and doctrine. But these choices carry risks: while we need to know that we are preparing for the right

future, history suggests, as the [previous chapter](#) has illustrated, that we often do not get the sorts of wars that we expect.

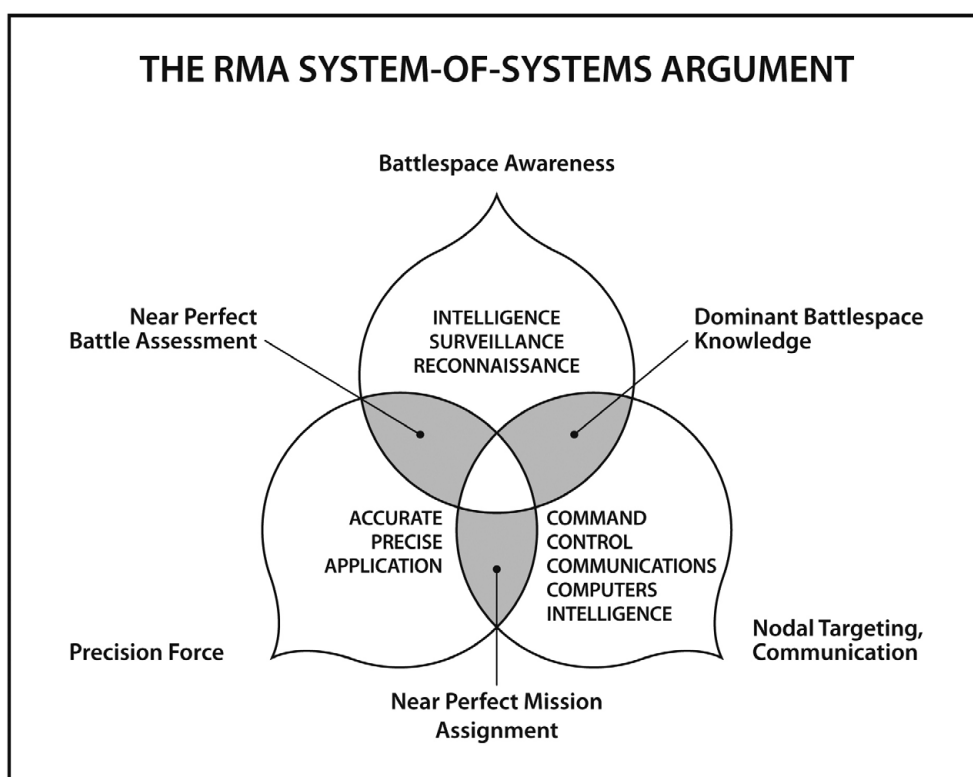
Sadly, for those tasked with generating future land power there is no consensus on what the future of land warfare holds. This chapter looks at the issue of future land warfare through the lens of three of the key contending schools of thought on the issue: the Revolution in Military Affairs (RMA); new wars; and hybrid warfare. Each takes the view that the character of warfare is changing; each has implications for the kind of land forces required to fight it; but each is also the subject of important critiques.

The Revolution in Military Affairs

The Gulf War of 1991 cast a long shadow over subsequent debates on the character and future of land warfare. The then US secretary of defense, Richard ('Dick') Cheney, argued that the Gulf War 'demonstrated dramatically the new possibilities of what has been called the "military-technical revolution in warfare".¹ The rapid, decisive coalition victory achieved in 1991 at minimal cost strengthened the hand of those who argued that the developments of the 1970s and 1980s could be taken further: that in the post-Cold War period, the United States had a unique opportunity to leverage the possibilities inherent in emerging technologies to produce a dramatic leap forward in military effectiveness. US thinking in the early and mid 1990s was led by the Pentagon's Office of Net Assessment and by such individuals within the military as Admiral William Owens, who became the vice chairman of the Joint Chiefs of Staff. The fruits of this new thinking were evident in a number of key Pentagon and Congressional documents, including *Joint Vision 2010*, produced in 1997, *Joint Vision 2020*, produced in 2000, and the Quadrennial Defense Reviews of 1997 and 2001. The basis of this thinking was founded on the perception that technology was providing new means for the conduct of war:

- new sensor technology, such as aerial and space-based systems (including Global Positioning System, or GPS), and optical, thermal and laser systems which would dramatically improve the ability to find enemy targets.
- improved means for delivering firepower, through such developments as stealth technology, cruise missiles and Joint Direct Attack Munitions (JDAMS, essentially GPS kits that could be attached to conventional bombs) that would allow those targets to be destroyed more quickly and with more precision.
- unprecedented networking, enabled by new digital technologies for collecting and processing information, that would connect the sensors to the 'shooter' systems.

Through greater integration and the creation of a 'system of systems', the combination of precision attack, sensors and networking promised to create what were

Table 6.1 *The RMA system-of-systems argument*

termed variously 'guided-munitions networks' or 'reconnaissance-strike complexes' (see Table 6.1). Cumulatively, therefore, new technology would succeed in creating a paradigm shift in warfare, a new military revolution: an entirely new way of fighting that would invalidate many of the verities of the past. Leveraging this RMA would allow the United States to attain 'full-spectrum dominance', a decisive superiority in military effectiveness at every level of war. However, there were two rather different conclusions on what this might mean for land forces: one that focused on the relative importance of land warfare, and the other focusing on the form that land warfare might take.

For some, this new RMA was essentially an air power revolution. This perspective took the view that the Gulf War, reinforced by subsequent conflicts in Kosovo in 1999, in Afghanistan in 2001 and Libya in 2011, illustrated the subsidiary nature of land warfare as against the devastating new capabilities demonstrated by air power. To paraphrase an old adage, this argument took the line that in the future 'air power conquers, land forces occupy'. Proponents of this view asserted that 'Simply (if boldly) stated, air power won the Gulf War' and that 'air power was the decisive factor'.² Victory in the Gulf War, Kosovo, Afghanistan and Libya was conditioned by air attacks that inflicted attrition; undermined enemy morale; disrupted enemy command, control and

Box 6.1 The army as a system

Generically, a system is a set of interrelated elements that collectively form a whole. Armies are made up of many different systems: myriad units, organisations, command arrangements, multiple communications nets, logistic structures and so on. Co-operation between these elements is required for effective command and control, movement, fighting and supply. Contemporary thinking about armies as systems has led to the emergence of two concepts: the 'system of systems' and 'systemic shock'.

A 'system of systems' is created through intense networking between systems, utilising new technology (especially digital communications) and associated procedures to create a more unified whole out of the distinct elements. This allows information to flow much more quickly and efficiently, which, in theory, dramatically increases the speed at which decisions can be made and the tempo at which operations can be conducted.

Conversely, it is possible to render enemy forces ineffective by destroying their capacity to function as a system, even if the enemy combat elements are still intact. This can be done by inducing 'systemic shock' in the enemy: paralysing the ability of the individual elements in an army to function together. This can be achieved through a variety of related means, including disrupting enemy communications, attacking command-and-control infrastructure, undermining enemy decision-making through high-tempo operations, denying the enemy information and undermining their morale.

communications; interdicted enemy logistics; and constrained the enemy's ability to operate in the open. Expressed in various forms through such concepts as 'nodal targeting', 'shock and awe' and 'effects-based operations' (EBO), the air-orientated RMA perspective argued that air power could defeat an enemy by inducing their systemic collapse, achieved through attacks against critical political, economic and military targets (see [Box 6.1](#)).

However, a second perspective saw the RMA in a different way. Here the argument was that the successes since 1991 were not those of air power, but of information. Theorists argued that the US military had triumphed in 1991 because it had achieved 'dominant battlespace awareness': a decisive information advantage that had allowed the United States to manoeuvre and to apply massive force, rapidly and with great precision, while preventing the enemy from doing so. Similarly, Afghanistan in 2001 was seen as a demonstration of the potential power of 'netwar'. The future, therefore, was network-centric warfare (NCW) – success in war would be founded increasingly on information dominance: 'One's ability to collect, process, and disseminate an uninterrupted flow of information while exploiting or denying an enemy's ability to do the same.'³ By developing new ways of accumulating information, and by developing highly integrated networks to allow this information to be exchanged across the whole force, militaries of the future would give themselves a decisive advantage over non-RMA adversaries in relation to speed of decision-making, tempo of operations, decentralised co-ordination and the dispersed massing of firepower.

Box 6.2 Globalisation

'Globalization refers to a process – still ongoing – through which the world has in many respects been becoming a single place.'¹

There is often a tendency to see globalisation as an economic phenomenon reflected, for example, in the growing connectedness of international financial markets. But globalisation also has powerful political, societal and cultural dimensions that can have contradictory effects: it can facilitate integration and fragmentation, globalisation and localisation, stability and instability.

¹ John Baylis and Steve Smith (eds.), *The Globalization of World Politics: An Introduction to International Relations* (Oxford University Press, 1999), 19.

In either case, an RMA future is one that would require radically different land forces. If advocates of an air-centric RMA are right, then the future would be one in which land forces would play an auxiliary role. In the 'Afghan model' of warfare, small, elite ground forces would be required to help direct air attacks. The remaining land contribution could be composed of constabulary forces designed to hold ground. A net-centric future also has important implications for land warfare. Organisationally, land forces would need to comprise of even smaller combined-arms manoeuvre teams, organised in a modular pattern, with even greater joint co-operation. In terms of concepts, Network-Centric Warfare would replace a traditional focus on such principles as concentration, mass and sequential operations with dispersal, agility and simultaneous activity. It would certainly be a future that was not conducive to traditional tank and mechanised warfare.

New wars

However, a second view on future warfare, the 'new wars' perspective, reached very different conclusions. For the 'new wars' school, the future was not the Gulf War, Kosovo, Afghanistan or Iraq in 2003; it was, instead, the conflicts in the former Yugoslavia in the first part of the 1990s. Reflected in the writings of such authors as Mary Kaldor, Herfried Munkler and General Sir Rupert Smith,⁴ the 'new wars' school saw a future that is in many respects the polar opposite of the RMA vision: instead of shiny, technocratic, high-intensity conventional war, these writers saw a future that was almost neo-medieval in character: future warfare will be brutal, local and decidedly low-tech. Contrasting 'New' and 'Old' wars, the 'new wars' school saw future warfare as having a number of distinguishing characteristics.

For the 'new wars' school, the driving force in shaping future warfare is not military technology but the processes of globalisation (see [Box 6.2](#)). This globalised context has democratised power, so it is argued, by empowering non-state actors, allowing them access to sources of financial, technological and

Box 6.3 Fourth-generation warfare (4GW)

In his book *The Sling and the Stone*, Colonel Thomas X. Hammes argues that the contemporary conflicts in Iraq and Afghanistan are symptomatic of a new 'fourth generation' in warfare. First-generation warfare was horse-and-musket warfare, exemplified by the Napoleonic Wars. Second-generation warfare was the rifle-and-railway warfare that evolved from the American Civil War to the First World War. Third-generation warfare encompassed blitzkrieg and manoeuvre warfare.

Fourth-generation warfare (4GW), however, represents an evolution in insurgency. Fourth-generation warfare is marked by a focus on the higher-level political decision-making of the enemy. Using political, economic and social networks, as well as military action, 4GW seeks to undermine the enemy's political will to fight.

It is worth noting, however, that this 'generational' approach to the development of warfare is a contested one, with some writers arguing that it imposes on the phenomenon of military change a structure that is misleading.¹

¹ Antulio J. Echevarria II, 'Beyond Generations: Breaking the Cycle', in Karl Erik Haug and Ole Jorgen Maaø (eds.), *Conceptualising Modern War* (London: Hurst and Company, 2011), pp. 49–64.

military backing, and enhancing their means to mobilise political support and influence international opinion through such media as the internet. Whereas old wars were fought between states, new wars increasingly will involve non-state actors: insurgents, terrorists, militias, warlords, clans or private security contractors (PSCs). In terms of objectives whereas 'old wars' were fought for reasons of state, or over differing ideologies, or for resources and territory, new wars will be 'identity wars' fought because of real or imagined national, ethnic or communal differences. Future conflicts will be waged through strategies that were once unwelcome by-products of war – terror, for example, and the deliberate targeting of civilians.

Moreover, new wars increasingly will blur the lines between war and criminality, with conflicts waged on the foundations of, and to defend or obtain, illicit resources. Thinking back to Clausewitz's trinity, discussed in chapter 1, new war, fundamentally, will be 'post-Clausewitzian', in the sense that the state will be less and less significant as an actor. New wars, or 'war amongst the people'⁵ as it has also been termed, are already here, so it is argued: in the continuing violence in Libya, post-2011, with the struggle between various tribal groupings; in Syria, from 2011, in the struggle between a government supported by Alawites and Christians against a predominantly Sunni opposition; and in Iraq, from 2006, in the escalating and shifting conflict between Shia and Sunni communities. In a 'new wars' future, armies orientated towards high-technology conventional warfare would find themselves increasingly irrelevant: instead, where military capabilities were required, they would be more akin to those required for counterinsurgency and peace support operations (see Box 6.3).

Hybrid warfare

But there is yet another key school of thought on the future. If the RMA and 'new wars' perspectives appear to be at either ends of a spectrum, then there exists a third important school of thought that might sit somewhere in between: the hybrid warfare perspective. Instead of a focus on the Gulf War or the conflicts in Bosnia, the hybrid warfare perspective looks to the 2006 war in Lebanon between Israel and Hezbollah as the marker for the future. In 2006, Israel launched a thirty-four-day military campaign against Hezbollah, a campaign which did not go according to plan. In its fight against the Israeli Defence Forces (IDF), Hezbollah seemed to utilise a new approach to warfare. Though they were an irregular non-state actor, Hezbollah seemed to mix familiar aspects of unconventional warfare, such as a decentralised, cellular structure, and tactics that utilised mobility, evasion and ambush, with elements of regular warfare. The latter included a focus on training and discipline; an effective system of command and control; an ability when required to contest ground, even to counter-attack; and the use of advanced technology, including night vision equipment and anti-tank guided weapons (ATGW).

Looking at the Lebanon War, and the difficulties experienced by the IDF in adapting to the threat, such authors as Frank Hoffman argued that Hezbollah represented the next development in warfare: a future that would be marked by a fusion of conventional and unconventional warfare. Whereas Hoffman noted that guerrilla and conventional warfare had often historically been used in parallel, he argued that they were used in parallel by separate forces: in the Vietnam War, for example, North Vietnam had Viet Cong irregulars to fight the unconventional war in the south, and regular forces that launched conventional assaults from the north in 1972 and 1975. In contrast to these old 'compound wars' Hoffman argued that the hybrid warfare would feature the use of regular and irregular warfare by the same military organisation, noting that 'Instead of separate challengers with fundamentally different approaches (conventional, irregular, or terrorist), we can expect to face competitors who employ all forms of war and tactics, perhaps simultaneously.'⁶

Box 6.4 contains some other definitions of hybrid warfare. Proponents of the hybrid warfare view could also point to the techniques used by Chechen forces in Russia, and by pro-Russian Ukrainian separatists in 2014.⁷

The future land warfare debate

The difficulty facing those tasked with generating future land power is that there is no consensus on which of these three perspectives, if any, is accurate.

Proponents of an RMA future face a number of criticisms. In relation to the air power-centric RMA, for example, the empirical evidence suggests that air power is a crucial component in modern warfare, but not that it is

Box 6.4 Hybrid warfare

[T]he diverse and dynamic combination of regular forces, irregular forces, terrorist forces, criminal elements, or a combination of these forces and elements all unified to achieve mutually benefitting effects.

US Army Doctrine Publication 3-0, *Unified Land Operations* (Headquarters, Department of the Army, October 2011), p. 4.

[C]onflict involving a combination of conventional military forces and irregulars (guerrillas, insurgents, and terrorist), which could include both state and non-state actors, aimed at achieving a common political purpose.

Peter Mansoor, 'Introduction: Hybrid Warfare in History' in Williamson Murray and Peter Mansoor (eds.) *Hybrid Warfare: Fighting Complex Opponents from the Ancient World to the Present* (Cambridge University Press, 2012), p. 2.

independently decisive. Campaigns from the Gulf War onwards suggest that long-standing methods through which the effects of air power can be mitigated are still effective. In Afghanistan, early successes against inexperienced Taliban fighters progressively gave way to tougher fights against more experienced al-Qaeda personnel. Realising their vulnerability to US air power, al-Qaeda fighters began successfully to emphasise techniques designed to counter the effects of air power, including the use of cover, dispersal, camouflage, deception and 'hugging' coalition forces. By mitigating the effects of air power, al-Qaeda forced the coalition to re-emphasise the use of larger quantities of better-quality ground forces with a stress on such traditional skills as fire-and-manoeuve. Whether Kosovo Liberation Army forces in 1999, Northern Alliance troops in Afghanistan in 2001 or Kurdish Peshmerga and Iraqi troops in 2014 in the struggle against the Islamic State insurgents, ground forces continue to play a central role in warfare. Moreover, such air-centric concepts as effects-based operations have become increasingly discredited amidst arguments that they are impossibly complex to implement in the real world. In 2008, for example, the head of the US Marine Corps, General James Mattis, announced that EBO would no longer be used as a planning tool, 'the reaction,' notes one author, 'of a military practitioner . . . confronted with a high-flying, half-baked, quasi-intellectual war-fighting concept'.⁸

But network-centric visions of the RMA have equally been pilloried. Critics argue that this is too technophile a vision of the future – that it fails, for example, to take into account human limitations on our ability to evaluate information; and it doesn't consider the likely impact of enemy countermeasures.⁹ Nor do important aspects of Network-Centric Warfare appear to work in practice. In operations in Iraq in 2003, for example, technical difficulties caused routine crashes of the network systems; commanders at the tactical level often lacked

key information on the position of enemy forces; and the 'just in time' logistics system 'functioned barely above subsistence level' leaving front-line troops short of food, fuel and medical supplies.¹⁰ Indeed, Network-Centric Warfare may risk becoming self-defeating: the more integrated a system becomes the more likely it is that shocks to one part will be transferred to other parts. Since, as chapter 1 has noted, war is the realm of chance and uncertainty, critics argue that a reliance on intense networking risks self-induced systemic shock when the inevitable frictions of war take hold.¹¹ More generally, do such wars as those against Iraq in 1991 and 2003 constitute a genuine military test of RMA-type systems? For some, the Iraqis were 'the perfect enemy' – poorly led and poorly organised.¹² Were the victories won against them the result of the leveraging of an emerging military revolution, or were they simply the result of the superiority of coalition forces in strategy and the basics of modern system warfare?

As an alternative, the hybrid warfare concept has gained considerable traction. The latest iteration of US Army doctrine identifies hybrid enemies as a potent future threat. But the hybrid warfare concept is littered with difficulties. There is, for example, very little definitional rigour to the concept. Looking at [Box 6.4](#), the first definition of hybrid warfare is that of the US Army. But by the US Army's definition many existing land forces, including the US Army itself, would be an exponent of hybrid warfare, combining as it does a capacity for conventional and unconventional activities. The second definition in [Box 6.4](#) encompasses what Hoffman would term compound wars: taking that perspective, hybrid warfare is a point of continuity in warfare, not a paradigm shift. Most previous wars would be hybrid by this definition, a point that some theorists on hybrid warfare accept explicitly, noting that 'much of the present only echoes the past'.¹³ Indeed, for some critics of the hybrid warfare concept, the importance attached to it is as much the result of politics and military culture as it is of any rigorous theorising. The hybrid warfare concept taps into a post-Afghanistan military *zeitgeist*, in which Western armies fear the degree to which a focus on irregular operations and nation-building have undermined their ability to perform core conventional war competencies. Hybrid warfare, so it is argued, justifies a renewed focus on these conventional and orthodox competencies within a conceptual framework that still appears to embrace the notion of change.

Many criticisms have also been levelled at the 'new wars' perspective. Some argue, for example, that the 'new wars' approach is based on faulty analysis of the historical evidence: in essence, much that is supposedly new about new wars is not. The deliberate targeting of civilians and the application of terror are, sadly, recurrent features throughout history. The Thirty Years War, for example, killed between 25 and 40 per cent of the population of the local German states; China's Taiping rebellion of 1850–64 killed 30 million people. Others note that, as evidenced by the Gulf War of 1991, the Afghanistan War of 2001 and the Iraq War of 2003, interstate wars are still a feature of the international system. For such states as India, China, Pakistan and Russia (and, one can argue given its pivot

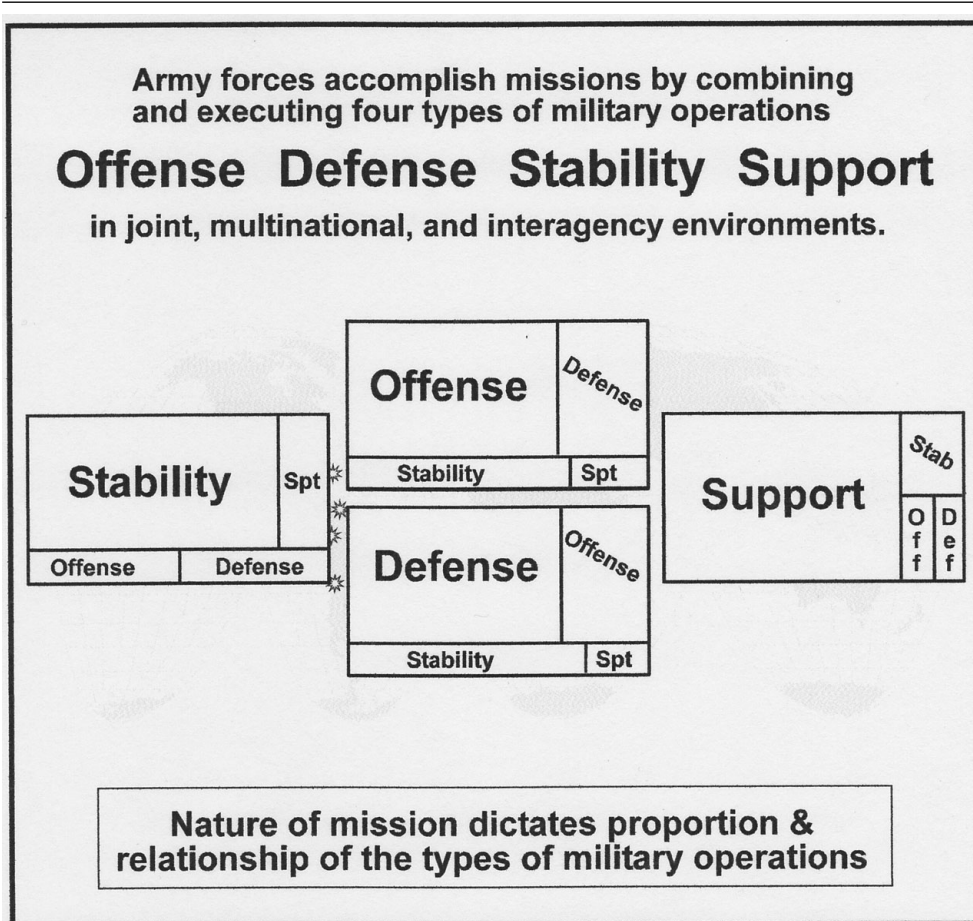
towards the Asia-Pacific, the United States), interstate war remains an important planning focus. Moreover critics also argue that the whole notion of 'Post-Clausewitzian' war is a mangling of the Prussian theorist's ideas. Clausewitz's trinity, as noted in chapter 1, is not founded upon the people, the army and the state but on human passion, chance and probability and war's subordination to policy. Whether used by a state or a non-state actor, war is still purposeful; it is designed to achieve a set of goals; it is still Clausewitzian in the sense that it can be understood through the lens of Clausewitz's theories on war.¹⁴

The paradigm army

So which of these visions have land forces found most persuasive? It is useful to touch briefly on the experiences of the US Army because, for many, the US Army is a 'paradigm army': a model for emulation, thanks to its operational experience and the depth and breadth of its analysis on the needs of future war.

For much of the period since the end of the Cold War, US Army development took place under the banner of the term 'transformation', a term first used during President Bill Clinton's tenure in office (1992–2001). It was a programme pursued with enthusiasm under President George W. Bush, especially by his secretary of state for defense, Donald Rumsfeld, who created in 2002 the Office for Force Transformation. In practice, transformation was in many respects a network-centric RMA view on the demands of future conflict. Organisationally, the focus lay in the modularisation of the army. Standardised brigades (Brigade Combat Teams, or BCTs) replaced divisions as the key formation. Doctrinally, 2001 saw the publication of Field Manual 3-0, *Operations*. Building on earlier doctrinal concepts, FM 3-0 introduced in particular the concept of 'full spectrum operations' (see Table 6.2), a concept that focused on the importance not just of offensive and defensive tasks but also of stability and support activities so as to enable the army to succeed at every level of conflict. FM 3-0 also embraced effects-based operations as a means to shape the most appropriate balance between violent ('kinetic') and non-violent ('non-kinetic') activities. All of this was underpinned by investment in new technology, especially in information systems and in procuring a Future Combat System (FCS): a line of manned and unmanned networked vehicles, including the Stryker (see Figure 6.1), that would provide an alternative to heavy armoured forces. Finally, transformation also focused on improving even further joint co-operation through such measures as improving joint training and investing in joint equipment. Overall, transformation aimed to produce lighter, more mobile, networked forces that would be able to leverage the agility, rapidity and flexibility of network-centric combat power.

This US vision of future land warfare had a much wider impact. The global influence of 'transformation' was illustrated by a US Government Accountability Office report of 1997. This indicated that over one hundred nations, ranging from Poland to Malaysia, were then planning military modernisation.

Table 6.2 *Full spectrum operations**

Full spectrum operations embodied the idea that the US Army could be effective across the whole spectrum of war, from high-intensity conventional conflict to support for the civil authority, by focusing on developing its expertise in four areas and then employing these, in different combinations, as dictated by local conditions.

Offence: destroy or defeat an opponent to impose US will and achieve decisive victory

Defence: defeat, delay, economise in forces. Creating conditions for later success

Stability: developmental, co-operative and coercive activities in response to a crisis

Support: assist foreign or domestic civil authorities to mitigate crises and relieve suffering

* FM 3-0, *Operations* (Headquarters Department of the Army, June 2001), pp. 1-14 to 1-16.



Figure 6.1 The Stryker armoured vehicle on exercise. Designed to improve the deployability of US ground forces, the Stryker provides infantry with protected mobility as part of a medium-weight, networked force.

Significantly, despite the diversity of these states, their military modernisation proceeded according to principles that accorded with the United States' transformation agenda, with a focus on improved networking, strategic mobility, tailoring and modularity of forces; joint and international connectivity; and the versatility to operate over the whole spectrum of warfare.¹⁵ For example, the United Kingdom, a key US ally, pursued a modernisation agenda that echoed transformation. This agenda included a more modest version of Network-Centric Warfare, entitled Network Enabled Capability; a less prescriptive version of Effects-Based Operations, entitled the Effects-Based Approach to Operations (EBAO); and a programme of organisational adaptation and equipment procurement in order to shape land forces that were more joint, expeditionary and modular.¹⁶ But even China, in many respects one of the United States' chief strategic competitors, has embraced a transformation-style programme of modernisation. Embracing enthusiastically the idea of an ongoing *xin junshi geming* ('new military revolution'), the Chinese People's Liberation Army (PLA) has been developing armed forces that will be able to prosecute *xinxi hua*, or 'informationalised warfare'. This agenda features a range of very familiar themes: the importance of joint and integrated operations; the requirement to invest in advanced technology; the need to improve networking

Box 6.5 Future missions of the US military

Future missions of the US Military

- Counterterrorism and irregular warfare
- Deter and defeat aggression
- Counter weapons of mass destruction
- Defend the homeland and provide support to civil authorities
- Project power despite anti-access/area denial challenges
- Operate effectively in cyberspace
- Operate effectively in space
- Maintain a safe, secure and effective nuclear deterrent
- Provide a stabilising presence
- Conduct stability and counterinsurgency operations
- Conduct humanitarian assistance, disaster relief and other operations

between platforms and units; the need to develop capabilities to fight information warfare; and the requirement for better trained and more educated personnel.¹⁷

Russia and India have also espoused themes consonant with those of the United States. Russia began a 'New Look' defence reform process in 2008. The emphasis has been on reducing the size of the armed forces and improving their quality, reorganising them into heavy, medium and light brigade-sized formations. There has been a move to reduce the reliance on conscripts and to improve service conditions for professional soldiers. The future vision is of a more agile, mobile army able to operate in a network-centric environment. In April 2004 the Indian Army introduced a new doctrine known colloquially as 'Cold Start'. This doctrine also contained many familiar elements: a desire to reshape the structures of the Indian Army from clumsy army corps to smaller 'integrated battlegroups'; to create forces capable of a faster and less predictable manoeuvre warfare-style approach to operations; and a desire to see much better joint integration.

However, by the period of the Barack Obama administration, the term 'transformation' had largely fallen out of favour. Indeed, the Office of Force Transformation had already been shut in 2006. In recent years, the US Army has laid out a less ambitious vision of future development. The *Army Strategic Planning Guidance* of 2013 maintained a focus on such themes as equipment modernisation, improving cyber and space capabilities, improving capabilities for joint warfare and improving its abilities for operations across a wide range of missions (see Box 6.5).¹⁸ The reasons for this return us to debates covered in the first half of this chapter. While such grand ideas, or 'meta-narratives' as the RMA, transformation, or new or hybrid wars are interesting and important concepts (not least as a focus for intellectual debate) they have proved difficult to apply to the

complex realities that have faced armies over the past decades. Indeed, the US Army's experiences have raised a range of compelling questions relevant to the general debate on the future of land warfare.

First, can theoretical future warfare concepts actually be realised in practice? One of the many difficulties with transformation was that the vision seemed to run ahead of the ability to create the doctrines and equipment necessary to deliver it. Effects-Based Operations was one example of this, but another was the Future Combat Systems programme, a programme beset by technical difficulties and cost inflation that was eventually labelled 'a very expensive failure' and which was cancelled in 2009.¹⁹ Likewise the British Effects-Based Approach to Operations concept was dropped because it was too amorphous for officers to understand; and much of the British parallel to the Future Combat System (the Future Rapid Effects System, or FRES) was cancelled in 2008.

Secondly, do we unduly privilege change while neglecting the role of continuity? One obvious point is that many of the precepts of transformed land warfare are hardly new: jointery, combined arms, manoeuvre, dispersal, surprise, flexibility, disruption, simultaneity and tempo, for example, are core themes in the evolution of modern warfare. The 1991 Gulf War and the 2003 Iraq War were both very conventional victories in this sense. In the end, even the latter war was a twentieth-century fight, albeit with improved jointery, tempo, precision firepower and intelligence. In 2003, only a single division, the 4th Infantry Division, was fully digitised, and even that was not involved in the initial attack. The first Stryker brigade was not available for operations until December. Moreover, as has already been noted, many of the key 'transformative' systems relating to command, communications and logistics did not work effectively when subjected to the friction of real combat. Is the story of US post-Cold War successes in conventional warfare really a tale of the victory of transformed US forces over untransformed adversaries, or is it simply, to parallel the German experiences of 1939 and 1940, the proper application by the United States of evolutionary modern system warfare techniques against poor-quality opposition?

Thirdly, can a single vision for the future prepare an army effectively for the full spectrum of warfare? The belief that a transformed military might be able to achieve rapid, decisive victory was built on the assumption that war-winning and peace-winning were the same things. This was evident in the debates prior to Operation Iraqi Freedom over the size of forces necessary for the operation. While Secretary of Defense Rumsfeld argued that the operation might be concluded successfully with as few as 60,000 troops, General Eric Shinseki, then Chief of Staff of the Army, argued, using the example of Bosnia, that it might require nearly half a million troops. In practice, the campaigns in Iraq and Afghanistan since 2001 have demonstrated the very real tensions between technologically focused paradigms for delivering rapid, decisive conventional success, and the complex, long-term, manpower-intensive, and highly political demands of counterinsurgency and nation-building scenarios. In fact, this is hardly a new lesson: in April 1941, Germany defeated Yugoslavia in a week,

losing only 151 men. Three years later, 350,000 German troops and their allies were still there trying to pacify the country.

Context and the future of land warfare

There remains a final, powerful criticism of future war ‘meta-narratives’. Here we return to the argument made in chapter 2 concerning war’s multifarious contexts. The [previous chapter](#) outlined the evolutionary development of the modern system of land warfare. The implication that might be taken from this is that the modern system of warfare is monolithic: that the modern system reflects the way in which all land warfare is fought. Indeed logically there should be an important imitative dynamic in warfare. War is a serious business. Failure can often be catastrophic for a state. There should be every incentive, therefore, for armies to imitate the structures, values and practices of other armies that have a demonstrated history of effectiveness. Even where states are not actually at war, examining the experiences of other states that are at war may provide important lessons. Over time, particular armies, through noted success in war, may acquire the status of ‘paradigm armies’ – armies that appear to embody so many of the standards of military effectiveness that they are widely copied. ‘Modern war’ is therefore ‘Western war’ because in the twentieth and twenty-first centuries the Western way of warfare, the key tactical and operational features of which have been outlined in the preceding discussion, appears to be uniquely effective, at least in terms of the prosecution of major conventional high-intensity land combat.

However, the processes that have shaped the way states develop and use military power on land (and in other environments, for that matter) are complex. The range of potential responses to military change is broad:

- innovation, or the introduction of new techniques or ideas
- resuscitation, or the repair of existing institutions that have fallen into decay
- adaptation, or the contextualisation of imported values/ideas
- imitation, or the importation and recreation of values and ideas
- stasis, or no change at all.

The last of these may be a perfectly rational option, given that the reasons for defeat are often not self-evident and, as chapter 2 has illustrated, may lie outside issues relating to the effectiveness of an army itself and instead with such factors as poor political choices or mistaken grand strategy. Crucially, direct imitation is rarer than might be imagined, even amongst European armies. Generally, context is crucial in determining whether, when and how states respond to the perceived lessons of the battlefield. Context is also crucial in determining fundamental benchmarks in land warfare, such as a state’s judgement regarding what ‘effectiveness’ means. This context may include a state’s factor endowments (such as resources, demographics or geography), its political culture, civil–military relations, the organisational culture of its armed forces, the desired

goals of the government, the outlook of its people and so on. This context, rather than strict military necessity, may play a decisive role in shaping how states fight land warfare. For example, the issue of whether or not a state has a professional or a conscript army may have an important bearing on how it conducts land warfare; yet the choice between the two may have as much to do with broader political and ideational objectives, such as nation-building, as with military-centred definitions of efficiency.

Same paradigm, different responses

For example, despite the development and diffusion during the First World War of many common military practices, the interwar period was marked by significant differences in approaches to land warfare. There were many generic challenges to the 'modernisation' of armies during the period leading up to the Second World War. There was a strong anti-war bias in the politics of many countries during the period, which tended to marginalise the military; economic conditions were often unfavourable to heavy military investment and additional calls on those resources that were available were made by air forces. There was also the effect of peace; notwithstanding the Spanish Civil War (1936–9), new concepts and new technology could not be extensively tested and validated. Institutional factors also shaped views on modern warfare; for example, the 'cavalry outlook' that had a strong representation in some armies tended to shape the view on the potential uses of mobile forces. Political ideologies, from the revanchism of Nazism to the modernist outlook of Marxism, were also relevant. The point is that these generic political, economic, historical and bureaucratic conditions influenced each country differently. The evidence demonstrates that even where countries have close military interaction, there may still be significant differences in the way in which they organise and practise land warfare. Even during wartime, these differences may persist. During the Second World War, for example, Britain adopted consistently a more centralised and methodical approach to operations than Germany, with a great emphasis on artillery support. This approach was conditioned by such specific factors as Britain's progressive shortages in manpower.²⁰

Moreover, even where armies might plan to fight in a modern-system fashion, political necessity may force armies to fight in conditions that are unexpected and/or not conducive to the application of assumed norms of land warfare (see [Figure 6.2](#)). As Lord Kitchener commented, 'We make war as we must, not as we would like.' There is often a tendency for armies to focus on the needs of symmetrical, high-intensity war, because these wars pose the greatest risk to the state and because this task may reinforce accepted norms of military professionalism. As the costs of such wars have grown (and bearing in mind the potential, in some cases, for nuclear escalation), the probability of their occurrence has diminished. After 1945, the number of interstate conflicts declined and the number of intrastate conflicts increased. The need to adapt techniques to meet what, from a Western military point of view, might be regarded as 'peripheral'



Figure 6.2 Adaptation in war: British troops are seen here landing on the beaches of Normandy, France on the 6 June 1944. Joint land–sea training had languished during the interwar period. Circumstances dictated that these skills and capabilities would have to be regenerated if the Allies hoped to open up a second front against Germany.

tasks is complicated by the inherently asymmetric nature of warfare: opponents will tend to adapt their techniques to mitigate their own weaknesses and exploit their strengths. Martin van Creveld notes, for example, that an inverse relationship often seems to exist between the modernity of an army (defined in terms of technology and organisation) and its effectiveness at unconventional operations.²¹

In the Indo-China War (1946–54), Viet Minh strengths in irregular jungle warfare led the French to adopt elements of static, positional attrition warfare. Used most famously at Dien Bien Phu in 1954, the concept of the *base aéro-terrestre*, establishing air-supplied fortified camps on Viet Minh supply routes, was adopted because it was believed, erroneously as it transpired, that French firepower would ensure that the attrition would be inflicted disproportionately on their opponent. During the Korean War of 1950–3, troop shortages, difficult terrain and enemy tactics forced the UN to drop defence in depth in favour of a thin line of discontinuous strong points on high ground, supported by air power and often huge quantities of artillery fire.

Similarly, the doctrinal focus of the Russian Army in the 1990s was conventional warfare, rooted in the Soviet thinking of the Cold War. Even in 2000, the focus of doctrine was still on high-technology war against external threats. Yet the major combat experience of the Russian Army consisted of urban operations in Grozny in 1994–5 and 1999–2000 as part of the war in Chechnya. Even if we were to assume, then, that many armies make similar general assumptions about the way to conduct land warfare, we can conclude that there are still likely to be important differences in how operations are actually conducted, not least because political necessity and adaptive enemies may force armies to fight according to unfamiliar rules.

Multiple paradigms

Variety in forms of land warfare is also likely because land warfare, like all warfare, cannot be divorced from its wider context. The modern system is difficult to implement because it carries with it a variety of political, social and economic trade-offs. Effective combined-arms and joint capabilities are expensive and may be beyond the abilities of states to procure. Devolved command and control may be at odds with more authoritarian political systems. This means that states may have modern equipment but not conduct ‘modern’ warfare, at least by the definition of contemporary Western militaries. This contingent effect of varying context has been extended by some writers into the realms of culture. For example, Victor Davis Hanson has argued that the ‘Western way of war’ has demonstrable superiority and that, at its heart, this superiority stems from Western culture. Features of this culture, such as individualism, consensual government and civic militarism, provide the foundation for specific military advantages like organisation, discipline, morale, initiative and flexibility. Williamson Murray has also identified a ‘Western way of war’ built on ‘finance, technology, eclecticism, and discipline.’²² However, cultural explanations for military behaviour remain controversial, not least because of the difficulty of defining culture and identifying exactly how it shapes actions.

It is not difficult to find evidence of the way that local context has shaped the ability of states to fight land warfare. One example would be the Iran–Iraq War of 1980–8, in which the character of the fighting was reminiscent of the middle element of the First World War, with a focus on artillery and infantry assaults in the context of static, attritional warfare. Both sides laboured under great difficulties. Command, control and intelligence were a recurring challenge. In part, this stemmed from difficulties at the grand strategic level that cascaded downwards. In Iran, military operations were subject to interference from the mullahs, and serious divisions existed between the army proper and the *Pasdaran*, or Revolutionary Guards. In Iraq, a climate of fear and a focus by Saddam Hussein on loyalty rather than competence stifled enterprise and initiative. There were also basic technical, doctrinal and organisational problems: major failings in intelligence-gathering, a lack of effective and secure communications and rigid and slow command chains. There was also a lack of training, leadership

and organisation for combined-arms warfare. Reliance on conscripts and high attrition of combat-effective elements reduced the capacity of both sides to formulate, disseminate and implement demanding alternatives. Neither side had the logistic capacity to sustain large-scale mobile warfare; even small increases in logistic lines or relatively minor movements of forces often resulted in disorganisation and disruption.

However, we should resist the tendency to see local context and efficiency as automatic trade-offs. There is a political context. The purposes of armies vary. Some are not designed to fight major combat operations. For example, although Latin American armies have acquired much of the panoply of war for modern major combat operations, in the period we are examining they have been deployed predominantly on internal operations. In other states, considerations of military effectiveness may be traded off deliberately against domestic political priorities; in Saudi Arabia, for example, parallel military structures, consisting of an army and a national guard, have been created to help forestall coups. Jeremy Black argues that Western attempts to develop taxonomies of military change fail to understand the role played by social and political circumstances in shaping the development of armed forces. Many of the things that we regard as essential for military efficiency, such as a focus on professional officers, discipline and training, are culturally conditioned. Thus, Black comments that 'Modernity, defined in a Western fashion and as a Westernising project, emerged in large part through military forces operating under Western systems of control and discipline.' As he goes on to argue, even where organisation and weapons might converge with Western practices, this does not imply 'convergence in terms of the cultural suppositions affecting war, especially understandings of victory, suffering and loss.'²³ Cultural perspectives argue that the legitimacy of a particular paradigm of land warfare cannot be defined solely in terms of military efficiency or organisational interests.

For example, the Yom Kippur/Ramadan War of 1973 might be construed as the victory of a Western-style army, the Israeli Defence Force (IDF), featuring a military system based on integration, trust and professionalism, over fundamentally less competent, Soviet-orientated Arab opposition. In reality, the IDF had developed a style of land warfare that was somewhat at odds with Western norms of war, not least because the IDF placed an emphasis on offensive air and armour operations at the expense of true combined arms because this had worked in previous Arab-Israeli wars. In fact, in the early part of the 1973 war, the IDF forces were often poorly handled, an Egyptian general noting that 'their sole tactic remains the cavalry charge'. The Syrian and Egyptian Armies certainly suffered from challenges relating to the poor performance of junior officers, problems with combined arms and a lack of flexibility. Many of these issues derived from a deeply embedded culture of conformity, deference and avoidance of shame that often resulted in passivity on the part of officers and a lack of initiative. Recognising these challenges, Egypt deliberately set out to compensate, its president, Anwar Sadat, noting, 'We will simply have to use our

talents and our planning to compensate.’ The improvement in the effectiveness of the Egyptian Army in 1973 came from methods that might be regarded as ‘anti-modern’ in terms of the conduct of contemporary land warfare: rigid planning; the exhaustive practising of a very limited number of missions; a focus on consolidation, not exploitation; and an emphasis on defence, not attack. From an Egyptian perspective, the 1973 war was far from being a defeat, and in the sense that Egypt achieved its political objectives it was indeed a successful use of military power.

Another example of the potent impact of differing contexts was the methods used by the Chinese People’s Liberation Army (PLA) forces in Korea. The PLA response to encountering the conventional forces of the United Nations encompassed elements of adaptation and compensation. Adaptation stemmed from a basic lesson: large-scale modern land warfare could not be fought against UN forces without adequate logistics and firepower. The Korean War saw the evolution of the PLA from a light, guerrilla war-orientated army to a mirror in some ways of the US Army, with an emphasis on firepower, especially artillery, and on air power, professionalism and logistics. Where they could not adapt, the Chinese compensated tactically. The Chinese often advanced at night, using low ground to bypass strong points and infiltrate. This allowed the Chinese to attack from unexpected directions and to threaten the artillery positions to the rear that were so vital for the support of company positions. If assaults had to be made on positions, the Chinese rarely used crude human-wave tactics. Instead, they would use machine guns and mortars to suppress and pin down the defenders before assault groups crept forward searching for weak points. In order to counter US air and artillery support, Chinese forces would try and ‘hug’ US positions.

The history of land warfare in the twentieth century was filled with examples that combine elements of continuity and important elements of variation. The war between Somalia and Ethiopia from 1977 to 1978 over the Ogaden involved tanks, armoured personnel carriers and heavy artillery, and featured the use of tank spearheads, combined arms and air-mobile operations. Yet it was also conditioned by the local context – such as the interaction between guerrilla and conventional warfare, between foreign advisors and indigenous forces and between regular forces and militias – into something very different from the Western European experience of war. Similarly, the 1987 war between Chad and Libya featured Chadian ‘Toyota War’, using light vehicles to inflict heavy losses on the heavier and less mobile Libyans. The Indo-Pakistan War of 1971 has been characterised by Robert M. Citino as ‘high-tempo manoeuvre warfare’ but under South Asian conditions.²⁴ Victory was achieved very quickly by an Indian Army lacking a significant degree of mechanisation; and rapidity was achieved by capitalising on local advantages, such as support from the population in East Pakistan (who provided intelligence and logistic support), by the use of infiltration tactics and by exploiting the passivity of the Pakistan defences. In the Sino-Vietnam War of 1979, the Vietnamese engaged

the Chinese with local and regional forces, manipulating the difficulties in terrain and the Chinese reliance on roads to launch infantry-based irregular attacks while main-force regular units were generally held in reserve.

The complex and variable character of modern land warfare is evident when we look at armies today. Even those forces that advocate transformation-style themes display marked differences in how those ideas are realised in practice. Thus, how armies might look on paper might say little about how they are actually used or perform. In India, military reform has been held back by many different factors: inter-service rivalries, sclerotic equipment procurement procedures, tensions in civil–military relations, budgetary problems and shortages in suitably qualified officers as a result of social changes that have seen the prestige fall of a career in the military. Despite continuing tensions with Pakistan, and the increase since 2006 of tensions with China, the focus of India's army has also been absorbed by the significant demands of such internal problems as the Maoist Naxalite insurgency.²⁵ The Indian Army is thus much more orthodox in its approach to land warfare than its doctrine might suggest. In Russia, the focus on containing NATO has been accompanied increasingly with preparations by the army for the possibility of intervention operations in areas of the former Soviet Union. The problems associated with Russian military capabilities were illustrated by the 2008 war against Georgia, where continuing deficiencies were shown in speed of mobilisation, command and control and jointery. Political imperatives have meant that the most significant intervention since then, the Russian intervention in Ukraine, has been conducted in a very unconventional fashion, with Russian ground forces deployed piecemeal unofficially since June 2014 in support of pro-Russian separatist forces. In addition to continuing problems linked to corruption, the armed forces have suffered from Russia's severe demographic shrinkage leading to continuing difficulties in obtaining the number and quality of personnel. Even China's PLA continues to exhibit a pronounced gap between the rhetoric of military modernisation and its reality. China has recognised that the PLA's goal of 'building informationalised armed forces and being capable of winning informationalised wars' is an unlikely goal until the middle of this century.²⁶ Faced with land forces that are likely to continue to feature a great deal of older technology, Chinese thinking has focused on the ways in which creative adaption and compensatory methods can be used to defeat high-technology enemies.²⁷ These asymmetric measures include targeting an enemy's resolve through cyber-attack, for example, or by deliberately embracing attrition.

Similarly, irregular adversaries can display complex adaptive behaviour depending upon local conditions. The 30,000 or so Islamic State (IS) militants in Syria and Iraq in 2014 were able to obtain equipment taken from retreating Iraqi forces, including Abrams tanks, armoured fighting vehicles and artillery. In its operations, IS demonstrated the complex fluidity of modern land warfare. In mid September 2014, for example, with air strikes being launched against IS forces in Iraq but not Syria, IS was fighting two kinds of war: in

Syria, free from the threat of air attack, the assault on Kobani in Kurdish Syria was launched using tanks, rockets and artillery. Simultaneously, in Iraq, IS adopted more traditional irregular methods to lessen the impact of enemy air power.

Western culture and the future of land warfare

The idea, then, that modern land warfare has existed in a single template is false, as is the idea that only one approach can be effective. There is no reason to presume that this will change in the future. Why, then, is there so often a tendency for Western analyses of future warfare to be guilty of ‘presentism’:²⁸ to assume that the immediate Western experience in Iraq, Afghanistan or Syria reflects the whole of the debate on contemporary and future land warfare?

One reason is the tendency to ignore the ‘contingent character of warfare’ outlined in chapter 2. The challenges of Western land warfare, expressed in such terms as the political problems associated with foreign intervention, an aversion to casualties (on either side) and a desire for ever greater precision are related in part to Western policies, particularly the use of land forces in ‘wars of choice’. These policies may change. In contrast to Afghanistan in 2001 and Iraq in 2003, the civil war in Syria from 2011 and the crisis in Iraq from 2014 have both demonstrated an unwillingness on the part of Western states to put ‘boots on the ground’. Moreover, for most states in the world, Iraq and Syria are not even the present, let alone the future, of land warfare. Reflecting the spirit of the debates in the previous chapters of this book, Western experiences and outlooks may be irrelevant to the armies of other states around the world because they may be faced with very different contexts (see Figure 6.3).

For some at least, the real difficulty with contemporary Western thinking on future land warfare is that, because the future is unknowable, inevitably we fill the vacuum with ideas based on our respective beliefs, values and culture. For this reason, current debates on the future of land warfare have also been criticised as essentially ethnocentric: they fail to recognise the extent to which Western views on future land warfare may be an expression of unspoken assumptions or desires on the part of Western states regarding how they would *like* war to be fought. One consequence is that our assumptions regarding future land warfare may be an unconscious expression of how we see ourselves. As US General John R. Galvin has commented, ‘We arrange in our minds a war we can comprehend on our own terms, usually with an enemy that looks like us and acts like us.’²⁹ The RMA-type military has become the ‘paradigm army’ of the twenty-first century: a model of a ‘modern’ army that serves to legitimise militaries in their own eyes and in the eyes of others. The fact that so many diverse states have accepted this model does not automatically mean that it is related to objective conditions (or indeed that it is militarily effective); rather, it may say more about how military professionals see themselves and how they want others to see them.



Figure 6.3 Norwegian tanks participate in a NATO exercise in 2014. While some states have focused their efforts on meeting unconventional threats, rising tensions with Russia have for many of its near-neighbours reinforced the continued utility of maintaining capabilities for high-intensity conventional warfare.

For this reason, contemporary debates on land warfare, with their focus on the rapid application of overwhelming military power, on manoeuvre rather than attrition and on disruption rather than destruction are expressions of the sorts of war the West is more comfortable grappling with. Western debates on the future of warfare often stress the role of technological solutions to political and social problems; this ‘technism’ or ‘cyborgism’ has led some to argue that technocratic, information-based, ‘post-modern war’ is as much a cultural phenomenon or an ideology as it is an objective reflection of reality. US strategic culture, for example, has according to some a tendency to emphasise technology, decisiveness and efficiency.³⁰ Thus, future war has been couched as ‘virtual war’ or ‘post-heroic war’ in which warfare can be conducted at a distance without much of its traditional savagery. Warfare, as Western societies would like it to be, becomes a form of ‘de-bellicised’, ‘victimless war’ with an emphasis on disruption rather than destruction. As part of this, Jeremy Black detects a RAM, a Revolution in Attitudes to the Military, that includes not just Western societies’ unwillingness to accept military casualties but an unease at causing enemy casualties (reflected in the hand-wringing over Kosovo in 1999) and even unease at causing enemy military casualties (evidenced by the disquiet caused by images of the ‘Road of Death’ during the 1991 Gulf War).³¹ Indeed, Robert R. Tomes ponders on the existence in Western strategic culture of a ‘rapid dominance *zeitgeist*’: an affinity for ‘temporal escalation’ and decisiveness that leads

Western militaries to focus on sudden brilliance as the acme of military skill rather than such concepts as protraction or attrition.³² There may be a real danger that armies that focus on smaller, more mobile, information-based militaries with manoeuvre warfare solutions fail to understand a lesson from the Second World War: unless the disparity between the quality of the protagonists is very marked (and the war is therefore won very quickly), modern warfare has a tendency to return to attrition.

Conclusion

In the end, then, the future of conventional land warfare is far from certain. The future may be marked by the dominance of an information-focused Revolution in Military Affairs; or it may be characterised by brutal and low-technology new wars fought between communities; or it may be a blend simultaneously of hybrid conventional and unconventional techniques. Whatever the future of land warfare, it is unlikely to be a homogenous one; the future, like the past, is almost certain to be marked by many different forms of warfare. Processes of innovation, adaptation and compensation shaped by differing political, economic, social and cultural contexts are likely to lead different political actors to prosecute land warfare in different ways. If there are recurrent themes in the conduct of land warfare expressed in relation to the tactical and operational-level manifestations of the modern system, they will not always be realised in the same way.

The political significance of land as an environment outlined in [chapter 4](#) is likely to continue to persist, and so land warfare is highly likely to remain important in the future. But while the land environment will almost certainly remain crucial in wars of the future, it may be that predicting the future is less important than building into land forces the capacity for flexibility: the capacity to adapt to the actual conditions of land warfare faster than an adversary. Amongst other things, this requires creating within armies a culture of flexibility. As the historian Sir Michael Howard has noted, 'No matter how clearly one thinks, it is impossible to anticipate precisely the character of future conflict. The key is to be not so far off the mark that it becomes impossible to adjust once that character is revealed.'³³

RESEARCH QUESTIONS

1. Which of the three perspectives on future land warfare, if any, seems to you to be the most compelling?
2. To what extent can valid lessons for future land warfare be drawn from the most recent wars in Iraq and Afghanistan?
3. What are the challenges associated with military imitation?

4. What factors have shaped China's approaches to military modernisation in the last fifteen years?
5. What are the characteristics of military flexibility? What are the issues associated with realising these characteristics?

FURTHER READING

- Adams, Thomas K., *The Army after Next: The First Post-industrial Army* (London: Praeger, 2006). A critical analysis of the application of the concepts of military transformation and the RMA.
- Black, Jeremy, *War since 1945* (London: Reaktion, 2004). An antidote to Eurocentric, Cold War-focused analyses of warfare.
- Farrell, Theo and Terry Terriff (eds.), *The Sources of Military Change: Culture, Politics, Technology* (Boulder, CO: Lynne Reiner, 2002). Using historical case studies, this book examines the manifold influences that shape how military organisations change.
- Finkel, Meir, *On Flexibility: Recovery from Technological and Doctrinal Surprise on the Battlefield* (Stanford University Press, 2011). Finkel provides a detailed analysis of the nature and importance of military flexibility in coping with unexpected threats.
- Hoffman, Frank, *Conflict in the 21st Century: The Rise of Hybrid Wars* (Arlington, VA: Potomac Institute for Policy Studies, 2007). Hoffman makes a compelling case for the future importance of hybrid warfare.
- Kaldor, Mary, *New and Old Wars: Organized Violence in a Global Era*, 3rd edn (Cambridge: Polity Press, 2012). This updates Kaldor's original 'new wars' thesis to take into account the wars in Iraq and Afghanistan.
- Osinga, Frans and Theo Farrell (eds.), *A Transformation Gap: American Innovations and European Military Change* (Stanford University Press, 2010). This book examines the challenges experienced by European states in absorbing US concepts on future war.
- Pollack, Kenneth M., *Arabs at War: Military Effectiveness, 1948–1991* (Lincoln, NE: University of Nebraska Press, 2002). A fascinating case study of the challenges of absorbing Western norms in land warfare.
- Scales, Robert H., Jr, *Yellow Smoke: The Future of Land Warfare for America's Military* (Lanham, MD: Rowman and Littlefield, 2006). A Western-centric, RMA-inspired view on the future character of land warfare.
- Shimko, Keith L., *The Iraq Wars and America's Military Revolution* (Cambridge University Press, 2010). An excellent assessment of the relevance of the RMA in the context of the wars in Iraq and Afghanistan.
- Sondhaus, Lawrence, *Strategic Culture and Ways of War* (London: Routledge, 2006). Explores cultural contexts and their impact on militaries.

ONLINE RESOURCES

For the US Army's views on its role in future warfare, see *The U.S. Army Operating Concept: Win in a Complex World*, TRADOC Pamphlet 525–3–1 (7 October 2014), www.tradoc.army.mil/tpubs/pams/TP525-3-1.pdf.

As a British counterpoint, see *Transforming the British Army – Force 2020* (July 2013), www.army.mod.uk/documents/general/Army2020_Report.pdf.

For a discussion of the origins, meaning and development of the RMA and transformation, see Michael G. Vickers and Robert C. Martinage, *The Revolution in War* (Center for Strategic and Budgetary Assessments, 2004), www.csbaonline.org/publications/2004/12/the-revolution-in-war, and Barry D. Watts, *The Maturing Revolution in Military Affairs* (Center for Strategic and Budgetary Assessments, 2011), www.csbaonline.org/publications/2011/06/the-maturing-revolution-in-military-affairs/.

For an assessment of transformation, its achievements and problems see Paul K. Davis, 'Military Transformation: Which Transformation and What Lies Ahead,' www.rand.org/content/dam/rand/pubs/reprints/2010/RAND_RP1413.pdf.

For discussion of hybrid warfare and new wars, see Frank G. Hoffman, 'Hybrid Warfare and Challenges,' *Joint Force Quarterly*, 52 (1st Quarter 2009), www.kcl.ac.uk/sspp/departments/warstudies/people/pubs/HybridWarfareChallengesHoffman.pdf.

Infinity Journal, www.infinityjournal.com/, is free but requires readers to register. It has many future warfare related articles. See, for example, Dan G. Cox, Thomas Brusino and Alex Ryan, 'Why Hybrid Warfare Is Tactics Not Strategy: A Rejoinder to "Future Threats and Strategic Thinking"', Article 5, vol. 2, issue 2.

The Strategic Studies Institute of the US Army War College has an online presence with a wide range of relevant articles, <http://strategicstudiesinstitute.army.mil/>. See, for example, Stephen Biddle and Jeffrey A. Friedman, *The 2006 Lebanon Campaign and the Future of Warfare: Implications for Army and Defense Policy* (Carlisle, PA: Strategic Studies Institute), www.strategicstudiesinstitute.army.mil/pdffiles/pub882.pdf.

For a brief overview of Mary Kaldor's arguments see <http://understandingempire.wordpress.com/2013/05/31/new-and-old-wars-organized-violence-in-a-global-era/>.

Mary Kaldor provides a defence of her 'new wars' argument in *Stability: International Journal of Security and Development*, 2(1), Article 4, pp. 1–16, <http://dx.doi.org/10.5334/sta.at>.

Mary Kaldor outlines new wars on Youtube, www.youtube.com/watch?v=eGhI58kjmB0.

On the debates surrounding future developments, see 'How Will the US Field a Futuristic Army?', www.usnews.com/opinion/blogs/world-report/2013/05/03/two-conflicting-visions-for-the-future-of-the-us-army.

Victor Davis Hanson and John Arquilla debate the merits of 'Netwar', www.youtube.com/watch?v=N0D8tAAc3ow.

NOTES

1. Quoted in Eliot A. Cohen, 'The Mystique of U.S. Airpower,' *Foreign Affairs*, 73(1) (January/February, 1994), 110.
2. Richard Hallion and the Gulf War Air Power Survey respectively, quoted in Daryl G. Press, 'The Myth of Air Power in the Persian Gulf War and the Future of Warfare,' *International Security*, 26(2) (Fall 1991), 10.
3. Milan Vego, *Joint Operational Warfare: Theory and Practice* (Newport, RI: US Naval War College, 2009), XIII-4.
4. Herfried Munkler, *The New Wars* (Cambridge: Polity, 2005); Rupert Smith, *The Utility of Force: the Art of War* (London: Allen Lane, 2005).

5. Smith, *The Utility of Force*, pp. 267–9.
6. Frank G. Hoffman, 'Hybrid Warfare and Challenges', *Joint Force Quarterly*, 52 (1st Quarter 2009), 35.
7. See, for example, 'Russia's Hybrid War in Ukraine', Institute for War and Peace Reporting, 11 June 2014 (<http://iwpr.net/report-news/russias-hybrid-war-ukraine>).
8. Torgeir E. Saeveraas, 'Effects-based Operations: Origins, Implementation in US Military Doctrine and Practical Usage', in Karl Erik Haug and Ole Jorgen Maaø (eds.), *Conceptualising Modern War* (London: Hurst and Company, 2011), p. 191.
9. Vego, *Joint Operational Warfare*, XIII-4.
10. Gregory Fontenot, E. J. Degen and Tommy Franks, *On Point: The United States Army in Operation Iraqi Freedom* (Annapolis, MD: Naval Institute Press, 2005), p. 408.
11. See Chris C. Demchak, 'Creating the Enemy: Global Diffusion of the Information-Technology Based Military Model', in Emily O. Goldman and Leslie Eliason (eds.), *The Diffusion of Military Technology and Ideas* (Stanford University Press, 2003), pp. 307–47.
12. John Mueller, 'The Perfect Enemy: Assessing the Gulf War', *Security Studies*, 5(1) (Autumn 1995), 79, 111.
13. Williamson Murray, 'Conclusions', in Williamson Murray and Peter Mansoor (eds.), *Hybrid Warfare: Fighting Complex Opponents from the Ancient World to the Present* (Cambridge University Press, 2012), p. 291.
14. Colin M. Fleming, 'New or Old Wars? Debating a Clausewitzian Future', *The Journal of Strategic Studies*, 32(2) (April 2009), 233.
15. See, for example, Deborah Sanders, 'Ukraine's Military Reform: Building a Paradigm Army', *The Journal of Slavic Military Studies*, 21(4) (2008), 599–614.
16. Theo Farrell, 'The Dynamics of British Military Transformation', *International Affairs*, 84(4) (2008), 784–5.
17. Jacqueline Newmyer, 'The Revolution in Military Affairs with Chinese Characteristics', *The Journal of Strategic Studies*, 33(4) (August 2010), 491–4.
18. *Sustaining U.S. Global Leadership: Priorities for 21st Century Defense* (Department of Defense, January 2012), pp. 4–6.
19. Andrew Krepinevich, quoted in BreakingDefense.com, 'Total Cost to Close Out Cancelled Army FCS Could Top \$1 billion', 19 June 2012 (<http://breakingdefense.com/2012/06/19/total-cost-to-close-out-cancelled-army-fcs-could-top-1-billion/>).
20. See Stephen Ashley Hart, *Colossal Cracks: Montgomery's 21st Army Group in North-west Europe, 1944–45* (Mechanicsville, PA: Stackpole, 2007).
21. Martin van Creveld, 'Technology and War II: Postmodern War?', in Charles Townshend (ed.), *The Oxford Illustrated History of Modern War* (Oxford University Press, 1997), pp. 311–12.
22. Victor Davis Hanson, *Why the West Has Won* (New York: Faber and Faber, 2002); Williamson Murray, 'The Future of Western Warfare', in Geoffrey Parker (ed.), *The Cambridge History of Warfare* (Cambridge University Press, 2005), p. 417.
23. Jeremy Black, 'Military Organisations and Military Change in Historical Perspective', *The Journal of Military History*, 62(4) (October 1998), 886.
24. Robert M. Citino, *Blitzkrieg to Desert Storm: The Evolution of Operational Warfare* (Lawrence, KS: University of Kansas, 2004), pp. 209–11.

-
25. Shashank Joshi, 'India's Military Instrument: A Doctrine Still Born', *Journal of Strategic Studies*, 36(4) (2013), 512–40, 512.
 26. See Dennis J. Blasko, "'Technology Determines Tactics': The Relationship between Technology and Doctrine in Chinese Military Thinking", *Journal of Strategic Studies*, 34(3) (June 2011), 355–81.
 27. *Ibid.*, p. 363.
 28. Colin S. Gray, 'The Twenty-first Century Security Environment and the Future of War', *Parameters*, 38(4) (Winter 2008–9), 23.
 29. Quoted in Brian McAllister Linn, *The Echo of Battle: The Army's Way of War* (Harvard University Press, 2009), p. 4.
 30. See Russell F. Weigley, *The American Way of War: A History of United States Strategy and Policy* (Hoboken, NJ: John Wiley, 1977).
 31. Jeremy Black, *War: Past, Present, and Future* (New York: St Martin's Press, 2000), pp. 245–6.
 32. Robert R. Tomes, 'Schlock and Blah: Counter-Insurgency Realities in a Rapid Dominance Era', *Small Wars and Insurgencies*, 16(1) (March 2005), 41.
 33. Quoted in *Strategic Trends Programme: Future Character of Conflict* (Shrivenham Development, Concepts and Doctrine Centre, February 2010), p. 1.





Part III

Naval warfare

Ian Speller



7

Concepts of naval warfare

Contents

The maritime operating environment

Attributes of naval forces

Maritime strategy

Command of the sea

Decisive battle

Offensive action and the concentration of force

Fleet-in-being

Blockade

Sea control and sea denial

Cover

Exploiting sea control

Alternative approaches

Raoul Castex, 1878–1968

Conclusion

KEY THEMES

- Navies tend to have distinctive attributes that result from the unique nature of the maritime operating environment.
- The nature of this environment means that many of the concepts and theories pertaining to military operations on land or in the air do not apply at sea. To understand naval warfare, therefore, one must engage with a distinct body of thought known as maritime strategy.
- Many of the key ideas about maritime strategy today have their roots in ideas first articulated in their current form over a century ago.
- Despite the many changes that have occurred over the years, these ideas continue to inform thinking about maritime strategy and thus, by extension, about naval warfare today.

This chapter examines concepts and theories associated with naval warfare. It begins with a discussion of the nature of the maritime operating environment, and the notion that navies have particular attributes that they derive from this.

It then introduces traditional theories of maritime strategy, focusing first on the dominant 'Anglo-American' tradition before examining some alternative approaches. The discussion here sets the context for the examination of the conduct of naval warfare over the past century, in [chapter 8](#), and the analysis of current practice and future possibilities in [chapter 9](#).

The particular characteristics of the sea and of naval forces, discussed below, have led to the development of a series of concepts and principles peculiar to naval warfare. In order to understand naval warfare and, in particular, to understand the thought processes that lie behind much modern naval activity, it is necessary to be conversant with distinctly naval concepts and principles. This, in turn, requires familiarity with the work of the traditional maritime strategists who first articulated and popularised such concepts. The impact of the most notable of these, Alfred Thayer Mahan, was reflected in the 1940s by the former US secretary of war, Henry Stimson, when he complained that the Navy Department frequently 'seemed to retire from the realm of logic into a dim religious world in which Neptune was God, Mahan his prophet and the United States Navy the only true Church'.¹ While this may not now be a fair reflection of Mahan's impact on the US Navy it is true to say that the concepts that he and others advanced have continued to influence naval thought and practice through to the present day.

It could be suggested that a conceptual approach derived primarily from an analysis of naval warfare during the age of sail, as was that of Mahan and many of his contemporaries, might find itself increasingly challenged given the dramatic changes in ship design and armament and in the tactical conduct of naval operations that have occurred since the nineteenth century. The contention of these chapters is that, while change is a recurring feature at the tactical level of naval warfare, at the strategic and operational levels the established concepts retain much utility. A key reason for this is that the concepts of maritime strategy and power are derived from the nature of the maritime environment and the possibilities associated with a capacity to exploit this environment. Technology may change, and this has an important impact on tactics, but the nature of the environment itself is enduring

The maritime operating environment

The maritime environment or domain (the term employed in US doctrine) includes the oceans, seas, bays, estuaries and coastal areas in which navies operate. Today it must also include the airspace above them. It is important to remember that the maritime domain is not limited to the sea but also includes that portion of the land that can be used to have a direct bearing on activity at sea and that may be subject directly to influence from the sea. This is commonly known as the 'littoral region'. As humans live on land and not generally at sea it is inevitable that much naval activity will focus on this region, where land and sea meet. Notwithstanding this important point, it would be fair to say

that it is the nature of the maritime domain beyond the low water mark (i.e. out at sea) that gives navies their particular characteristics and that distinguishes naval warfare from its counterpart on land.

The most obvious feature of the sea is its size, covering around 70 per cent of the planet. The size of the oceans, and the ability of ships to traverse them, mean that battles and campaigns can occur across hundreds and even thousands of miles. Distance works differently at sea than on land. For example, the battle of Trafalgar in 1805 was preceded by a British pursuit of the French that took Nelson from the Mediterranean, across the Atlantic to the Caribbean and back again before he finally met and defeated Admiral Villeneuve off Cadiz. The main American base at Pearl Harbor in Hawaii was around 4,000 miles from Japan but this distance offered no protection against a surprise attack launched by the Imperial Japanese Navy in December 1941. In the 1982 Falklands/Malvinas campaign the British Royal Navy successfully conducted an unexpected campaign over 8,000 miles from home and 3,500 miles from the nearest friendly airfield. The requirement for admirals to 'think big', to consider manoeuvre across hundreds or thousands of miles, has led to the suggestion that they tend to be at home at the strategic level.²

Equally important is that fact that most of this area is connected. Furthermore, and in stark contrast to the land and the airspace above land (which are criss-crossed by innumerable physical and political barriers) the sea provides a medium that is freely available to all in peacetime and one that is difficult to deny to a superior navy in war. Distant lands that are bounded by the sea are thus connected by what Alfred Mahan famously called a 'great highway . . . a wide common over which men may pass in all directions.'³ This has important implications for military operations. It offers navies the capacity for manoeuvre on a global scale while simultaneously acting as a barrier to those unable to use the sea. It is also critical for economic activity as the connectivity of the sea, allied to the unrivalled bulk transport capacity of ships, ensures that around 96 per cent of global trade (measured by weight) travels by sea (see [Figure 7.1](#)).⁴ The need to protect maritime trade adds an important economic dimension to naval activity.

In addition to being vast, the seas are also inhospitable. One cannot live on them or travel across them without recourse to special platforms designed for this purpose. Naval warfare, therefore, tends to be about the use and the denial of the use of such platforms. These platforms tend to be expensive, meaning that there are relatively few of them, and they are usually hard to replace without delay. Sadly, they can sometimes be destroyed rather quickly and not always by enemy action. Even today it is true that the sailor must first fight and survive the environment before they can engage any human opponent. The inhospitable nature of the seas also means that there is no indigenous population. There is no one to remark on the passing of warships or to object to their presence. Equally, there is no resident population to defend, and very little fixed agriculture or industry. The consequences of declining battle at sea are thus much less serious



Figure 7.1 Combined Task Force 521 conducts convoy escort operations with a large natural gas tanker during an international mine countermeasures exercise in 2013. The globalised world economy depends on the free flow of maritime trade and the need to protect this flow is reflected in naval theory and practice.

than on land, making it easier for an inferior maritime force to avoid a superior opponent. Usually the enemy cannot occupy the deserted battlefield in any meaningful way. This means that naval warfare focuses on the use and the denial of use of the sea rather than the physical control of a medium that can be neither occupied nor fortified.

Geography and topography can be important in naval warfare. The presence or otherwise of shore bases can enable or constrain operations. Narrow straits may create ‘choke-points’ that can inhibit operations and ships and submarines cannot travel across land. The particular features of a coastline may offer advantage to one side or another, and the presence of offshore islands, shoals and reefs can have a notable impact on operations. Nevertheless, in comparison to the land, the sea is essentially featureless. There are fewer natural barriers to observation and there is usually no terrain that can be exploited to the advantage of the defence. A weaker navy has less ability to mitigate qualitative or quantitative inferiority than does its equivalent on land.

Attributes of naval forces

It is often claimed that naval forces have distinctive characteristics or attributes that are derived from the forces themselves and from the medium in which

Box 7.1 Attributes/characteristics of maritime power/naval forces

Indian Navy, 'characteristics of naval forces':

- Mobility, Versatility, Sustained Reach, Resilience, Lift Capacity, Poise, Leverage, Cost Effectiveness, Maintenance & Refit.

NATO 'characteristics of maritime power':

- Poise, Readiness, Flexibility, Self-Sustainment, Mobility.

Royal Australian Navy, 'characteristics and attributes of sea power':

- Mobility in Mass, Readiness, Access, Flexibility, Adaptability, Sustained Reach, Poise & Persistence, Resilience, Transience, Indirectness, Speed.

UK Royal Navy, 'attributes of maritime power':

- Access, Mobility, Lift, Sustained Reach, Versatility, Poise, Resilience, Leverage.

they operate. Different navies have expressed this in slightly different ways (see [Box 7.1](#)).⁵

What all of these approaches have in common is the notion that the nature of the maritime environment gives navies an opportunity for sustained manoeuvre that they can exploit because of the flexibility and versatility possessed by most warships. This can provide useful options in wartime where the sea can become key manoeuvre space, a medium for global connectivity or an impenetrable barrier. It also has important consequences in situations short of all-out war. It means that ships can be deployed overseas without having to cross any territorial boundaries. They can be poised in international waters without infringing any state's sovereignty and without the need to negotiate basing rights. While at sea they are far less visible than their land-based equivalents, both in a literal and a figurative sense. This means that the deployment of naval forces can be less provocative than that of air forces or armies. The routine deployment of individual ships, task groups or even entire fleets into particular regions is generally less likely to excite adverse comment than the deployment of a squadron of land-based aircraft or of ground forces, both of which require a footprint ashore that can become the focus for political dissent. Unsurprisingly, there is a considerable body of literature devoted to the analysis of naval diplomacy, examining the way in which navies can exploit these attributes in situations of armed peace or limited war. This was addressed in a number of classic works written by the British diplomat James Cable, has continued to be examined by a number of twenty-first-century commentators and is reflected in the published doctrine of numerous contemporary navies.⁶

The US Navy does not discuss 'attributes' as such but rather it identifies 'core capabilities' that represent enablers for the navy's strategic missions. The four established capabilities of Forward Presence, Deterrence, Sea Control

and Power Projection have, in recent years, been joined by two new capabilities, Humanitarian Assistance and Disaster Relief; and Maritime Security.⁷ All of these capabilities are derived from the particular nature of the maritime domain. Forward Presence, for example, is enabled by the connectivity provided by the sea and by the ability to operate in an environment that is not subject to territorial control or political jurisdiction; Sea Control is profoundly different from control on land because the nature of the environment is so dissimilar.

Maritime strategy

The ability to take advantage of these attributes or capabilities does not come automatically with proximity to the sea. It depends upon a capacity to exploit the maritime environment. Naturally, much of the literature relating to naval warfare is focused on the ways and means of doing this. There is a rich history of such literature dating back to classical times and from many different countries. However, it would be fair to say that the foundations of modern maritime strategy were laid in the last decades of the nineteenth century and the first of the twentieth, with pioneering work undertaken by a number of commentators, the most significant and influential of whom were an American, Alfred Thayer Mahan, and a Briton, Sir Julian Corbett (see [Boxes 7.2](#) and [7.3](#)).

Mahan and Corbett were the most prominent writers on maritime strategy in their day and they remain the most influential but they were not the only ones. Many others contributed to what was a vibrant field of study, including Lieutenant-Captain Berezin (Russia), John and Philip Colomb (Britain), Stephen B. Luce (USA), Jurien de la Graviere (France), Stepan Makarov (Russia) and Curt von Maltzhan (Germany).⁸ In Japan Akiyama Saneyuki and Sató Tetsutaró laid the foundations for an approach that was deeply influenced by Mahan.⁹ Thus, there were a considerable number of individuals who wrote on maritime strategy and history and who may, despite their various differences, be said to have formed something of a 'blue water' school of thought regarding the use of major navies. Focused on the advantages to be gained by securing, maintaining and exploiting command of the sea, and usually also emphasising concentration of the battle fleet in pursuit of a decisive engagement, they helped to establish the key concepts around which much subsequent debate has focused. Many of these concepts remain relevant today and are referenced in current doctrine and debate;¹⁰ the key points are introduced below.

Command of the sea

The sea itself cannot be commanded in the same way that the land can be. It is not an environment open to physical control. Command of the sea has therefore tended to be defined as the freedom to use the seas and to deny that use to an enemy. This is frequently portrayed as the most important task for navies in war. Philip Colomb, for example, argued that 'The primary aim of naval war is the

Box 7.2 Alfred Thayer Mahan, 1840–1914

Born in 1840 at West Point, the son of a professor of civil and military engineering at the US Military Academy, Mahan was to become one of the most prolific and influential writers on maritime strategy. In 1856 he joined the US Naval Academy at Annapolis and remained a naval officer until 1896 when he retired with the rank of captain. Unusually for his time and profession he was not keen on service at sea and he is remembered not for his exploits as a commander but rather for a written output that amounted to twenty books and 137 articles before he succumbed to heart failure in December 1914.

Mahan sought to demonstrate the influence of 'sea power' upon history and to affirm that there were enduring principles of maritime strategy that remained valid despite changing technology. Such principles could be discovered through an examination of the past, which revealed them in 'successes and failures, the same from age to age'. Unfortunately his interpretation of the past often lacked the accuracy or detachment of an impartial historian. He wrote to promote the education and instruction of fellow naval officers and his focus on principles must be seen within this context.

The key point of Mahan's work was that maritime power had played a decisive role in the course of history and that strength at sea was key to a state's material prosperity and success in war. He believed that the past showed that maritime preponderance was the key to prosperity and success as a great power, focusing in particular on the experience of the British from the seventeenth century. To Mahan national prosperity, and through it the means to wage war, were dependent on sea-borne trade. Such trade required the protection of a navy. There was therefore a close and beneficial relationship between trade and naval strength. Maritime trade encouraged the development of a navy to protect it and also provided the raw materials (trained manpower, ship-building and repair facilities etc.) to support such a navy. A strong navy could lead to dominance at sea, at the expense of rival countries, which in turn encouraged more trade.¹

¹ See Jon Sumida, *Inventing Grand Strategy and Teaching Command: The Classic Works of Alfred Thayer Mahan Reconsidered* (Baltimore, MD: Johns Hopkins University Press, 1997).

command of the sea.'¹¹ Both Mahan and Corbett recognised the importance of this concept, although Mahan tended not to employ the phrase. Some commentators have portrayed command in absolute terms but most (including Mahan, Colomb and Corbett) recognised that the extent to which one could control activity at sea varied according to circumstance and that total command was unusual. Colomb, for example, wrote of levels that ranged from 'indifference', to 'disputed command' and then 'assured command'.¹² Corbett argued that disputed command was a normal state of affairs, at least in the early stages of any conflict, and that general command of the sea was not a necessary precursor to all operations. Indeed, he suggested that there might well be circumstances when general command of the sea was not the first and overriding priority and

Box 7.3 Sir Julian Corbett, 1854–1922

Born in 1854 and trained as lawyer, Corbett taught at the Royal Naval War College at Greenwich from 1900. Like Mahan, Corbett used an examination of the past to identify the important role that maritime power had played in British military and economic success. While recognising the strengths of maritime power, Corbett placed a greater emphasis than Mahan on its limitations. In contrast to Mahan, whose focus tended to be on activity at sea, Corbett emphasised the importance of joint operations involving the co-ordination of land and sea forces. His work also placed less emphasis on the need to seek battle, something that his critics described as 'sea heresies'. The British Admiralty explicitly disassociated itself from such ideas in their official history of the First World War at sea, inserting a disclaimer to the effect that they did not agree with the author (Corbett). Corbett remained unrepentant asking, in a post-war public lecture, 'what material advantage did Trafalgar give that Jutland did not give?'¹ It is a question that students of strategy today would do well to ponder. In retrospect it would be fair to say that he got more right than did his critics. His historical method was certainly more satisfactory than Mahan's and his conclusions are more nuanced. Nevertheless, he did get some things wrong; most obviously in standing against the introduction of convoys for trade defence during the First World War.

¹ N. A. M. Rodger, 'Naval Strategy in the Eighteenth and Nineteenth Centuries', in Geoffrey Till (ed.), *The Development of British Naval Thinking: Essays in Memory of Bryan Ranft* (London: Routledge, 2006), p. 31. For a recent analysis of Corbett's work see J. J. Widen, *Theorist of Maritime Strategy: Sir Julian Corbett and his Contribution to Military and Naval Thought* (Farnham: Ashgate, 2012).

that a navy might first seek to exploit a local command to secure some military or political objective.

It is important to stress that, as both Mahan and Corbett identified, command of the sea means command of communications at sea. It does not imply physical control of the environment, as might be the case on land, but rather control of enemy activity. Unlike road and rail communications on land, lines at sea exist only in a notional sense. They can be neither attacked nor defended. To secure sea lines of communication, therefore, one does not protect an actual line (as none exists). One protects or attacks those things that attempt to use the lines of communication; usually those things are ships.

Decisive battle

For Mahan the favoured means of achieving command of the sea was through a decisive battle in which the main enemy fleet was defeated or, better still, destroyed. Examples of such battles might be Trafalgar (1805), Navarino (1827), Tsushima (1905) or Leyte Gulf (1944) all of which, to a greater or lesser degree, removed the main threat posed by the enemy fleet. In truth, however, it is relatively rare for a single battle to be decisive. Often identified as the shining

example of a decisive battle, Trafalgar was actually just one of a series of British victories over their French (in this case Franco-Spanish) opponents. Similarly, at Leyte Gulf the US Navy virtually extinguished the threat posed by the Japanese surface fleet, but this was only the last in a series of impressive American victories. Most frequently decision is a result of a series of encounters.

It is important also to question the extent to which even a dramatic victory is decisive. A battle may have a significant impact on the war at sea without necessarily having an immediately decisive effect on events ashore. British victory at Trafalgar, for example, could not stop Napoleon from defeating their Austrian and Russian allies at Austerlitz six weeks later. That battle had more immediate significance for the balance of power in Europe than did Nelson's victory, even if British command of the sea (enabled by victories such as Trafalgar) did set the conditions that allowed for eventual French defeat. Of course, had the result of Trafalgar been reversed and the British defeated comprehensively then the impact may have been more decisive as it could have enabled a French invasion of Britain (although, in truth, the obstacles to this would still have remained significant). The point here is that a victory is decisive because of the things that it enables. A narrow success that can be exploited may be of more value than a spectacular rout that cannot. Equally, the potential fruits of victory must be weighed against the costs of any defeat.

Offensive action and the concentration of force

Mahan's belief in the central importance of gaining command of the sea led him to stress the need for offensive action and the value of concentrating force in pursuit of the defeat of the enemy. In his examination of the various wars between Britain and France he praised the British for recognising this and criticised the French for their failure to consistently focus on the enemy battle fleet; it was a point that many others were keen to emphasise. The German Curt von Moltzhan, for example, argued that battle was the keystone to the whole system of naval warfare and stressed the importance of the offensive, describing a reluctance to take the offensive and to risk the loss of ships in battle as being a 'cankerworm of decay'.¹³ According to this approach battle was at the heart of naval warfare; it proved a seductive concept for navies.

In practice an imperative towards offensive action might not actually be helpful. The Imperial German Navy, in particular, has often been criticised for its stubborn belief in the centrality of battle and for failing to realise that their main rival, the British, did not need to win a battle in order to prevail. By closing off the exits from the North Sea and the English Channel Britain could dominate the key maritime communications without seeking battle and without needing to institute a close blockade. German failure to recognise this, or to devise a suitable response once the policy became apparent during the First World War, was criticised (during the war and after) by Wolfgang Wegener (see [Box 7.4](#)).

Box 7.4 Wolfgang Wegener, 1875–1956

Wolfgang Wegener served in the German Navy before, during and after the First World War, retiring in 1926 as a vice admiral. He criticised German policy during that war, arguing that deference to the German Army's ingrained concepts of land warfare had led the navy to prepare for battle without thinking enough about strategy and, in particular, about the importance of geography and maritime communications. To Wegener the exploitation of maritime power required a combination of a suitable *fleet* and *position*. Bottled up in the Helgoland Bight the Imperial German Navy had had the former but not the latter and it was therefore unable to engage the Royal Navy on anything like favourable terms. He favoured the seizure of Denmark (he later added Norway) as one means of improving the navy's strategic position. These ideas were criticised by another German, Herbert Rosinski, who argued that Wegener paid insufficient attention to the issue of material superiority without which victory was unlikely no matter where it occurred. It is ironic that in the Second World War, when Germany secured all the territory that Wegener wanted, and more, it had the position but lacked a fleet strong enough to truly exploit it.¹

¹ Vice Admiral Wolfgang Wegener, *The Naval Strategy of the World War*, ed. and trans. Holger Werwig (Annapolis, MD: Naval institute Press, 1989 [1929]). Herbert Rosinski, 'German Theories of Sea Warfare', in B. Mitchell Simpson, III (ed.), *The Development of Naval Thought: Essays by Herbert Rosinski* (Newport, RI: Naval War College Press, 1977).

Corbett offered a more balanced view on the role of battle and the utility of concentration. He understood the value of both but recognised that sometimes it might not be possible to force battle on an opponent and it also might not be necessary. In some situations it was possible to dominate key maritime communications without necessarily having to fight a battle, as was the case for Britain in 1914. In such circumstances it made no sense to fight on anything other than favourable terms. In any case, a weaker fleet was unlikely to accept battle against the concentrated might of its enemy. They would have to be forced to fight in order to protect something that mattered to them. This might imply some level of diversion to tempt them out or the conduct of raids or expeditions that, by threatening a vital interest, would require a response. In addition to this, assets would have to be diverted away from the battle fleet in order to undertake the myriad duties that the navy would have to attend to while it waited for the opportunity to deal with the main enemy force. Concentration could be more difficult in practice than it was in theory.

Fleet-in-being

One of the main problems with a policy that revolved around seeking out and destroying the enemy in main battle is the fact that weaker opponents are usually loath to let this happen. It is easier for a navy to decline battle than it is for

an army as at sea there are generally no fixed points to be protected, no population or territory to defend. A weaker navy might simply avoid battle or remain within a protected anchorage, maintaining their 'fleet-in-being' on the assumption that its mere existence would constrain enemy choices and thus serve a useful strategic purpose even without achieving command of the sea.

Colomb illustrated this concept with reference to the actions of the English Admiral Torrington who, in 1690, faced a superior French fleet in the Channel. Torrington declined to engage it in battle, reasoning that the mere presence of his undefeated force would inhibit the French from conducting an invasion of England as they could not do so without running an intolerable risk. Unfortunately his strategy was not understood ashore and his hand was forced by his superiors who demanded an attack. Torrington had no choice but to comply and was defeated at the battle of Beachy Head. French indecision and poor weather were all that remained to save England from invasion.

Colomb, who approved of Torrington's initial caution, tended to emphasise the impact that even a markedly inferior fleet-in-being could effect and, by extension, to stress the value in defeating such a force whenever the opportunity presented itself. Mahan disagreed, arguing that a bold French commander would not have been deterred by the mere existence of Torrington's fleet; his preferred solution remained the defeat of the enemy in battle and he stressed that war could not be fought without taking risks. Corbett's conclusion was that a fleet-in-being could dispute command of the sea by adopting a defensive attitude, and avoiding a decisive battle, if it actively sought every opportunity for a counter-strike.¹⁴

Blockade

Blockade was a common response to an adversary that maintained a fleet-in-being, with an enemy force confined to harbour or unable to sortie without the risk of encountering superior force. Close blockade involves the stationing of forces close inshore, able to intercept the movement of any ships in or out of harbour. While this approach has clear advantages it can be very difficult to sustain and, given the confluence of threats that exist within littoral waters, may be particularly difficult to employ today. Distant blockade, sometimes described as observation blockade or choke-point control, does not attempt to stop the enemy from leaving harbour; rather it aims to deny them access to particular areas. As such, they do not require the main force to remain close inshore, although some capacity to observe enemy movements is obviously helpful. Distant blockades have the additional advantage that, in allowing an enemy to sortie, they may give the blockader the opportunity to intercept and to defeat their foe. Today, with the development of remote surveillance and stand-off weaponry the distinction between the two types could be less obvious. It may be possible for a superior navy to create the effect of a close blockade while keeping most assets at some distance.

Sea control and sea denial

Command of the sea is often portrayed in rather absolute terms. In their published maritime doctrine the Royal Australian Navy, for example, suggests that 'command of the sea implies that dominance has been achieved to such a degree that the risk to one's own forces from enemy action is negligible'.¹⁵ Dominance such as this is not the norm historically, at least not in a conflict against a peer rival. There may be some circumstances when positive use of the sea can be achieved without meeting such exacting requirements. Command is usually limited in time, place, extent and consequence. Equally to the point, one cannot actually 'command' the sea in the same way that one can control land; it is not open to territorial conquest. As has been discussed, gaining command of the sea simply means gaining control of sea communications. Defeating an enemy fleet is not an end in itself but rather it is a means of dominating the use of those communications. This is reflected in the modern concept of sea control, which can be defined as the 'condition that exists when one has freedom of action to use an area of sea for one's own purposes for a period of time and, if necessary, deny its use to an opponent'.¹⁶ This concept, which owes much to Corbett's analysis, reflects a realistic understanding of the degree of control that can be achieved and of the role of sea control as an enabler and not an end in itself.

It may often be the case that a navy will not seek control in a particular area, but instead will focus on the more limited task of denying such control to an adversary. Sea denial tends to require a lower level of capability than does control and was described by the US Admiral Stansfield Turner as being akin to 'guerrilla warfare at sea'. Navies will usually find themselves balancing the need to exert control or to achieve denial, with the emphasis on one or the other differing in time and place. Turner expressed this in the form of a diagram (see [Table 7.1](#)), with navies sometimes emphasising activity to the right of the figure, and sometimes to the left. It is worth noting that the assertion of sea control will also deny control to an enemy (control cannot be shared), but denial of control does not imply that control passes to friendly forces (control can remain in dispute) (see [Figure 7.2](#)).¹⁷

Cover

An important function of sea control in one region might be to provide cover to weaker forces or detached units to enable them to conduct their operations unhindered by the main force of the enemy. In this sense the main function of a powerful naval force in one region might be to enable friendly activity hundreds or even thousands of miles away by providing the appropriate cover. In this way the British Grand Fleet at Scapa Flow, by neutralising its German counterpart, provided vital cover for all other Allied activities at sea during the First World War.

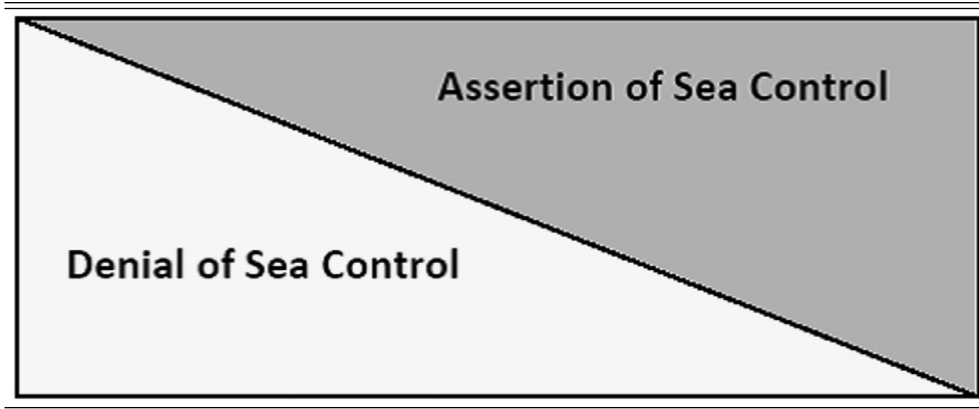
Table 7.1 *Sea control and sea denial*

Figure 7.2 The Republic of Korea diesel-electric submarine, *Chang Bogo* (SSK 61). Vessels such as these are far more capable than the submarines used in the two World Wars but they share many basic characteristics. They share an ability to attack shipping and to deny the use of the sea without necessarily having to achieve positive control themselves.

Exploiting sea control

As has been discussed, sea control is relevant only insofar as it enables other things. There are numerous ways in which it can be exploited to good effect. Both Mahan and Corbett laid great emphasis on the value of an economic blockade, founded on control of the sea. This would enable the superior navy to drive enemy trade from the seas, undermining their industry, impoverishing their finances and perhaps also leading to shortages of food and raw materials. At the same time control would provide a necessary foundation on which trade defence could be built. It supplies the necessary cover to allow light forces to protect friendly trade and to hunt down enemy raiders who, operating without control, might sometimes be a nuisance but, it was argued, would never be able to achieve the slow strangulation that was associated with a successful blockade.

Another way in which sea control could be exploited was through the direct projection of power ashore, either through shore bombardment or by landing troops. Corbett paid particular attention to such operations, regarding joint army–navy co-operation as being central to British success in a series of wars. In 1911 in his most famous work, *Some Principles of Maritime Strategy*, he articulated this in the following way:

Since men live upon the land and not upon the sea, great issues between nations at war have always been decided – except in the rarest cases – either by what your army can do against your enemy’s territory and national life, or else by fear of what the fleet makes it possible for your army to do.¹⁸

That this now seems to be a statement of the obvious is a reflection both of the perspicacity of Corbett’s work but also of the degree to which his ideas continue to influence modern naval thinking. Views such as this were discomfiting for a naval audience in the 1900s whose preferred focus was on battle at sea.

A number of Corbett’s contemporaries examined the conduct of amphibious operations in some detail. Most notable amongst these was Sir Charles Callwell, an Anglo-Irish officer in the British Army who wrote two impressive studies on the importance of military operations to activity at sea and also on the value of maritime command to land campaigns. Unfortunately, these do not appear to have had much impact in their day. Too often navies succumbed to the tendency to concentrate primarily on operations at sea, and on fighting other navies, and paid scant attention to the challenges presented by the type of joint operations that Corbett and Callwell favoured (see [Figure 7.3](#)).¹⁹

The concepts outlined above were derived largely from an examination that focused mainly on the challenges and opportunities facing major ‘sea powers’. They have been described as representing an Anglo-American tradition in maritime strategy, although many that contributed to it were neither British nor American.²⁰ For many commentators this tradition was validated by experience in two World Wars. Thus, for example, after 1918 Herbert Richmond

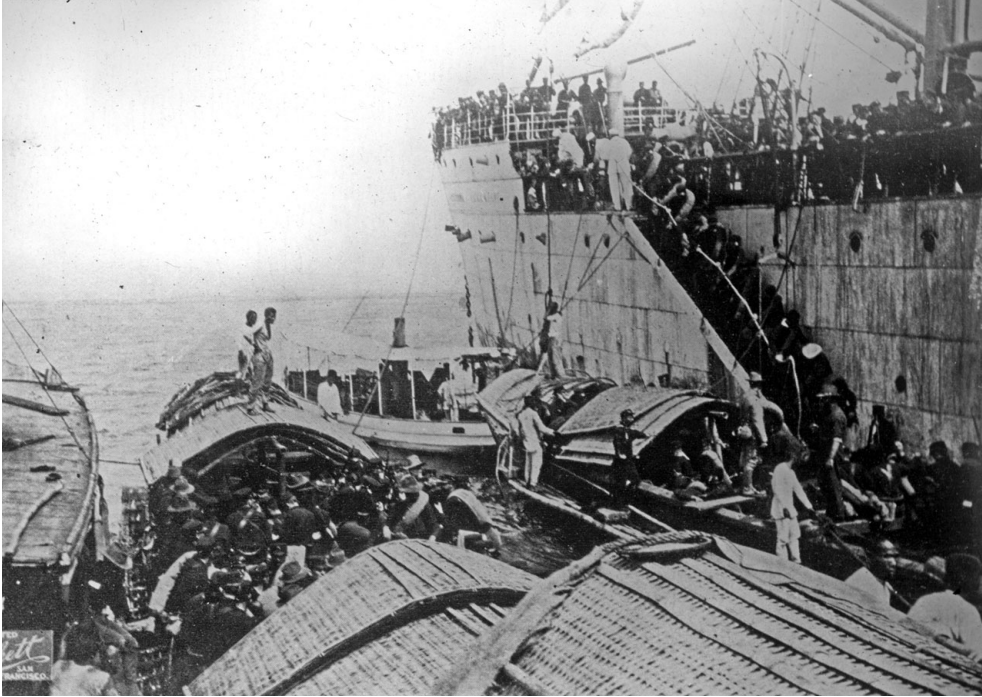


Figure 7.3 An undated photo showing American troops disembarking from a ship onto small boats near Cavite, Philippines in 1898 or 1899. The capacity to exploit control of the sea, including by landing troops ashore, has always been important even if it was sometimes rather neglected by those writers whose focus tended to be on activities *at* sea rather than those *from* the sea.

offered analyses that reflected Corbett while Mahan continued to influence many navies, including those of Japan and the USA. Post-1945 Bernard Brodie, Stephen Roskill, Stansfield Turner, Edward Wegener and James Wylie were amongst those who offered perspectives that broadly reflected traditional assumptions about maritime strategy and sea control, updated also to include an understanding of the impact of air power on the established concepts.²¹ From the other side of the iron curtain the commander-in-chief of the Soviet Navy, Admiral Sergey Gorshkov, supported the dramatic expansion of his navy from the 1960s with an analysis that (once one strips out the communist rhetoric) was grounded in the same logic as that of Corbett.²²

Alternative approaches

It could be argued that the 'Anglo-American' tradition discussed above was not particularly well suited to countries that had an inferior navy or that were ill-placed to exploit maritime power in the traditional way. In such situations it

might make more sense to devote limited resources to denying the enemy the unfettered use of the sea rather than seeking to gain positive use for oneself. This is particularly true as sea denial can often be achieved with small and relatively unsophisticated forces that today might typically include mines, submarines, anti-ship missiles and land-based aircraft.

Even as Mahan was formulating his ideas an alternative school of thought in France was articulating the utility of an asymmetric strategy of sea denial in any future war with Britain. Known as the *Jeune École* (Young School), the influence of this group spread beyond France and it had an impact upon all of the major European navies in the 1880s and 1890s, raising important questions about the utility of traditional strategies based upon securing command of the sea with large, expensive battleships. The School advocated adopting an asymmetric response to British sea control. Abandoning any attempt to match the British battleship for battleship, they argued instead for relying on coastal defence and small, fast and inexpensive torpedo launches to deny the British the ability to institute a close blockade, releasing fast steam-driven cruisers into the Atlantic to prey on British trade in a ruthless '*guerre de course*' designed to force Britain to sue for peace. The *Jeune École* had fallen from favour in France by the end of the century, not least because the strategy that they advocated was unlikely to offer any useful military options in a war against their most likely future opponent, Germany. In any case, the technological developments that made big ships vulnerable to small craft were soon matched by countermeasures that somewhat evened the balance, including the provision of more small craft to defend the battleships.²³

The *Jeune École* may have been eclipsed but this did not necessarily invalidate their central argument. Within a few years a new form of technology, that of the submarine, offered a different means of achieving the same result. The unrestricted U-boat campaigns undertaken by Germany in both World Wars were grounded in the same logic as that expressed by the *Jeune École*. Rather than attempting to gain positive use of the sea, the U-boat campaigns sought to exploit a particular technology in order to deny its use to Britain and its allies. Thus, a number of interwar German commentators, including Hugo von Waldeyer Hartz and Kurt Assmann, emphasised economic warfare over the battle fleet, based on the logic that Britain was dependent on maritime communications and could be forced onto the defensive, perhaps even defeated, by a weaker navy not so encumbered.²⁴ That the German submarine campaigns both failed does not necessarily mean that the idea was not well grounded. For a country able to devote only limited resources to maritime power, and facing an opponent reliant on using the sea, a strategy that focuses on sea denial rather than sea control may indeed make more sense.

An emphasis on local sea denial was also apparent in the coastal-defence theories developed by the New School that emerged in the Soviet Union in the late 1920s and early 1930s. Reacting to the paucity of traditional maritime

assets, and in light of the experience of maritime power projection being applied against the nascent Soviet state in 1919, this school of thought advocated an integrated system of minefields, coastal artillery, submarines and motor torpedo boats instead of a more traditional, balanced fleet. This approach was, however, abandoned by the Soviet Union in the 1930s when greater resources and increased ambition led the navy to emphasise a more traditional approach to maritime power that could extend beyond coastal waters.²⁵ Communist China went through a similar process emphasising the value of ‘mosquito fleets’ of small defensive coastal craft in the 1950s before moving towards a more traditional interpretation of the role of the fleet once greater resources became available.²⁶ Some Western navies, including those of Denmark, Norway and Sweden, emphasised coastal defence during the Cold War. The removal of the threat from the Soviet Union has seen these navies changing their policies and exploring ways of making more positive use of the sea. This does not mean that writings on coastal defence by, for example, Jacob Borresen, do not still have utility for other small navies who may face a threat from the sea.²⁷

Raoul Castex, 1878–1968

While achieving, maintaining and exploiting sea control might represent the main aim for dominant maritime powers, and coastal defence and commerce raiding might be the favoured option of their weaker opponents, it is not immediately clear whether either approach is suitable for a major navy that is still inferior to its main rival. To address this problem it is helpful to refer to the work of a French naval officer, Raoul Castex, a prolific writer, publishing eighteen major works and over fifty journal articles. His major contribution was the enormous five-volume *Théories Stratégiques* (Strategic Theories) published between 1929 and 1935.²⁸ He recognised the value of gaining what he called ‘mastery of the sea,’ defined as control of sea communications. He also recognised the centrality of battle to its achievement. He was, however, very aware of the problem of achieving this with inferior forces. For those not yet in a favourable position of strength there were a range of things that could be done to improve the situation. At the heart of his theory was the concept of ‘strategic manoeuvre.’ Such manoeuvre implied exploiting the possibilities of activity such as attacks on commerce, naval raids, amphibious operations or blockades to force an opponent to divert resources. This also implied a diversion of friendly resources, something that Mahan warned against. However, Castex suggested that it was possible to use space and distance to concentrate more rapidly than an opponent, thereby using strategic manoeuvre as a vehicle for creating local superiority of numbers and thus the conditions under which battle might be successfully entertained. To Castex the *guerre de course*, including submarine operations, could be a useful way of diverting enemy forces but

it could not be decisive on its own. Rather, it had to be integrated into a system of operations whose eventual aim was to contest mastery of the sea. The main target remained the enemy fleet, and the chance of defeating a fleet with inferior numbers was ‘almost nil’ but one need not passively accept an inferior position.

Conclusion

This chapter has argued that navies tend to have distinctive attributes that result from the unique nature of the maritime operating environment. For this reason the theory and practice of naval warfare is different from its equivalents on land and in the air. The key concepts of maritime strategy, discussed above, were established over a hundred years ago and were often derived from an examination of the practice of naval warfare over the previous centuries. They are far from new but they continue to have an impact on thought and practice today. This is particularly true of Corbett, whose emphasis on joint operations provides a close fit with the priorities of many contemporary navies. On the other hand one tends not to find overt positive references to the *Jeune École* or German U-boat theorists in public policy statements today; few organisations are willing to admit to be preparing to launch ruthless attacks on civilian ships. Of course, it would be foolish to assume that there could never again be a sustained *guerre de course*; a wise student of strategy soon learns to be wary of the word ‘never’. It is certainly the case that many navies today are focused on sea denial at a local, a regional and (perhaps for some) a global level. Given this, and the constant references within naval doctrine and both professional and academic commentary, it seems reasonable to suppose that the traditional concepts retain some currency. The following chapters will test this assumption with an examination of the evolution of naval warfare over the last century ([chapter 8](#)) and of the nature of naval warfare in the twenty-first century ([chapter 9](#)).

RESEARCH QUESTIONS

1. Are the traditional concepts of maritime strategy still relevant in the twenty-first century?
2. How does the nature of the maritime environment influence the characteristics and capabilities of naval forces?
3. Have political, military, economic and/or technological developments eroded the importance of the sea as a ‘great highway’ over the course of the last century?
4. Do the ideas of the *Jeune École* offer a useful guide to action for inferior navies today?
5. Is ‘command of the sea’ the primary aim of naval warfare?

FURTHER READING

- Hattendorf, J. N. (ed.), *Mahan on Naval Strategy: A Selection of Essays by Alfred Thayer Mahan* (Annapolis, MD: Naval Institute Press, 1991). This book offers excerpts from Mahan's major works, providing a valuable summary of his ideas.
- Heuser, Beatrice, *The Evolution of Strategy: Thinking War from Antiquity to the Present* (Cambridge University Press, 2010). Heuser's study of strategic thought includes five chapters that provide a useful insight into maritime strategy.
- Roksund, Erne, *The Jeune École* (Leiden: Brill, 2007). Roksund provides a detailed and authoritative examination of the ideas developed by the *Jeune École*.
- Speller, Ian, *Understanding Naval Warfare* (London: Routledge, 2014). This book addresses traditional concepts of naval warfare and also analyses contemporary practice. It examines the issues raised in this chapter in greater depth and detail.
- Sumida, Jon, *Inventing Grand Strategy and Teaching Command: The Classic Works of Alfred Thayer Mahan Reconsidered* (Baltimore, MD: Johns Hopkins University Press, 1997). Sumida provides a clear, accessible and authoritative introduction to Mahan's ideas.
- Till, Geoffrey (ed.), *The Development of British Naval Thinking: Essays in Memory of Brian Ranft* (London: Routledge, 2006). This collection of essays provides an examination of the evolution of British naval thinking from the eighteenth century until the end of the twentieth.
- Till, Geoffrey, *Seapower: A Guide for the Twenty-first Century*, 3rd edn (London: Routledge, 2013). Written by one of the most respected commentators on contemporary maritime strategy, this book provides a detailed and wide-ranging examination of the subject.

It is, of course, always useful to refer to the original texts. Of these the following are of particular interest.

- Callwell, Charles E., *Military Operations and Maritime Preponderance: Their Relations and Interdependence* (Annapolis, MD: Naval Institute Press, 1996) with an introduction and notes by Colin S. Gray.
- Castex, Raoul, *Strategic Theories*, ed. and trans. Eugenia C. Kiesling (Annapolis, MD: Naval Institute Press, 1994).
- Colomb, Philip, *Naval Warfare: Its Ruling Principles and Practice Historically Treated* (Annapolis, MD: Naval Institute Press, 1990) with an introduction and notes by Barry M. Gough
- Corbett, Julian S., *Some Principles of Maritime Strategy* (Annapolis, MD: Naval Institute Press, 1988) with an introduction and notes by Eric J. Grove.
- Wegener, Vice Admiral Wolfgang, *The Naval Strategy of the World War*, ed. and trans. Holger Herwig (Annapolis, MD: Naval Institute Press, 1989).

ONLINE RESOURCES

Original versions of Alfred Mahan's major works, including *The Influence of Seapower upon History, 1660–1783* (1890) and *The Influence of Seapower upon the French Revolution, 1793–1812* (1892), and many works of classic naval history and maritime

strategy written by other authors (including Corbett and Colomb) are available free online at <https://archive.org/>.

NOTES

1. Philip Crowl, 'Alfred Thayer Mahan: The Naval Historian', in Peter Paret (ed.), *Makers of Modern Strategy from Machiavelli to the Nuclear Age* (Princeton University Press, 1986), p. 444.
2. Admiral Lord West of Spithead, 'Maritime Warfare and the Importance of Sea Control', in Julian Lindley-French and Yves Boyer (eds.), *The Oxford Handbook of War* (Oxford University Press, 2014), p. 437.
3. Alfred Thayer Mahan, *The Influence of Sea Power Upon History 1660–1783* (Boston, MA: Little Brown, 1890), p. 25.
4. US Navy, *Naval Doctrine Publication 1, Naval Warfare* (March 2010), p. 14.
5. NATO AJP 3.1, *Allied Joint Maritime Operations* (2004), pp. 1-3, 1-4. JDP 0-01, *British Maritime Doctrine* (2011), pp. 2-1-2-6. *Australian Maritime Doctrine: RAN Doctrine 1* (2010), pp. 86–94. Ministry of Defence (Navy), *INBR8 Indian Maritime Doctrine* (2004), pp. 97–9.
6. In particular see James Cable, *Gunboat Diplomacy 1919–1979: Political Applications of Limited Naval Force*, 3rd edn (Basingstoke: Macmillan, 1999). Also see Peter Hayden, 'Naval Diplomacy: Is it Relevant in the 21st Century?', in Andrew W. Tan (ed.), *The Politics of Maritime Power: A Survey* (Abingdon: Routledge, 2007); Christian Miere, *Maritime Diplomacy in the Twenty-first Century: Drivers and Challenges* (Abingdon: Routledge, 2014).
7. US Navy, *Naval Doctrine Publication 1. Naval Warfare* (2010).
8. Geoffrey Till, *Seapower: A Guide for the Twenty-first Century*, 3rd edn (London: Routledge), ch. 3.
9. Sadao Asado, *From Mahan to Pearl Harbor: The Imperial Japanese Navy and the United States* (Annapolis, MD: US Naval Institute Press, 2012), ch. 2.
10. For example, see Robert C. Rubel, 'Command of the Sea: An Old Concept Resurfaces in a New Form', *US Naval War College Review*, 65(4) (Autumn 2012), 21–33.
11. Philip Colomb, *Naval Warfare: Its Ruling Principles and Practice Historically Treated* (Annapolis, MD: Naval Institute Press, 1990 [1891]), p. 1.
12. *Ibid.*, 204–12.
13. Curt von Maltzhan, *Naval Warfare*, trans. John Combe Miller (London: Longmans, 1908), p. 46.
14. John B. Hattendorf, 'The Idea of a "Fleet-in-Being" in Historical Perspective', *Naval War College Review*, 67(1) (Winter 2014), 43–60.
15. *Australian Maritime Doctrine* (2010), p. 71.
16. *British Maritime Doctrine*, p. 289.
17. Stansfield Turner, 'The Naval Balance: Not Just a Numbers Game', *Foreign Affairs*, 55(2) (January 1977) and 'Missions of the US Navy', *Naval War College Review* (March–April, 1974). Also see Christofer Waldenstrom, 'Sea Control through the Eyes of the Person Who Does It: A Theoretical Analysis', *US Naval War College Review*, 66(1) (Winter 2013), 77–100.
18. Julian S. Corbett, *Some Principles of Maritime Strategy* (Annapolis, MD: Naval Institute Press, 1990 [1911]), p. 16.

19. Sir Charles Callwell, *The Effect of Maritime Command on Land Campaigns since Waterloo* (London: Blackwood, 1897) and *Military Operations and Maritime Preponderance: Their Relations and Interdependence* (London: Blackwood, 1905). Also see Sir George Aston, *Letters on Amphibious Wars* (London: John Murray, 1911) and George Furse, *Military Expeditions Beyond the Seas*, 2 vols. (London: William Clowes, 1897).
20. See Speller, *Understanding Naval Warfare*, ch. 1.
21. Bernard Brodie, *A Guide to Naval Strategy* (New York, NY: Praeger, 1965); Sir Herbert Richmond, *Seapower in the Modern World* (Oxford: Bell, 1934); Stephen Roskill, *Strategy of Seapower* (London: Collins, 1962); Edward Wegener, 'Theory of Naval Strategy in the Nuclear Age', *US Naval Institute Proceedings* (May 1972); James Wylie, *Military Strategy: A General Theory of Power Control* (Annapolis, MD: Naval Institute Press, 1989 [1967]).
22. Sergei Gorshkov, *Navies in War and Peace* (Annapolis, MD: Naval Institute Press, 1974) and *The Seapower of the State* (London: Pergamon, 1979).
23. See Eric Roksund, *The Jeune École* (Leiden: Brill, 2007).
24. B. Mitchell Simpson, III (ed.), *The Development of Naval Thought: Essays by Herbert Roskinski* (Newport, RI: Naval War College Press, 1977), pp. 15–17. Herger H. Holwig, 'The Failure of German Sea Power, 1914–1945: Mahan, Tirpitz, and Raeder Reconsidered', *The International History Review*, 10(1) (February 1988), p. 95.
25. Bryan Ranft and Geoffrey Till, *The Sea in Soviet Strategy* (London: Macmillan, 1983), pp. 85–8.
26. Bernard D. Cole, 'The History of the Twenty-first Century Chinese Navy', *Naval War College Review*, 67(3) (Summer 2014).
27. See Jacob Borresen, 'Coastal Power: The Sea Power of the Coastal State and the Management of Maritime Resources', in Rolf Hobson and Tom Kristiansen (eds.), *Navies in Northern Waters, 1721–2000* (London: Frank Cass, 2004), pp. 249–75.
28. Raoul Castex, *Strategic Theories*, ed. and trans. Eugenia Kiesling (Annapolis, MD: Naval Institute Press, 1994).



8

The evolution of naval warfare

Contents

From fighting sail to steam, steel and shellfire

Into a new century

The test of war, 1914–18

From arms control to World War, 1919–45

Navies in the nuclear age

Conclusion

KEY THEMES

- The techniques of naval warfare have been subject to many changes over the past two centuries.
- Technology has also changed and this has had an impact on all forms of naval activity. However, to understand technology one must place it within an appropriate context. It is not an independent variable.
- Despite such changes there have also been some notable continuities and these suggest that the concepts articulated in the [previous chapter](#) retain some relevance.

The [previous chapter](#) suggested that there has been a considerable degree of continuity in thinking about naval warfare; concepts and principles articulated in the nineteenth century continue to be employed in the twenty-first. Over the same period there have been many obvious changes in the conduct of naval warfare, particularly in the tactics adopted and in the technology used. No navy today maintains a fleet of wooden sailing ships similar to those employed by Admiral Nelson in 1805 nor do any possess pre-dreadnought battleships akin to those used to good effect by Admiral Togo a century later. No modern commander would deploy their fleet in battle in the same way as did either Nelson or Togo and to do so would be to invite disaster. It is legitimate therefore to question whether principles articulated in Togo's era, which were often derived from an examination of Nelson's, are still useful today when platforms, weapons and sensors have changed so much and where the general strategic context has been transformed. This chapter will address that question.

Unfortunately there is not the space here to provide a detailed history of the evolution of naval warfare over the past century, still less can we examine the very rich history before this period. The reader is fortunate that there are numerous good books that already do precisely this and some of these are introduced in the 'Further reading' guide at the end of this chapter. The aim here is not to present a comprehensive history but rather to illustrate some of the ways in which naval warfare has changed over time, and also some of the continuities. There will be a particular focus on the past 150 years, roughly the time at which steam, steel and shellfire replaced the 'wooden ships and iron men' of Nelson's era. This should offer some context within which the concepts and principles articulated in the [previous chapter](#) can be assessed and will also provide a useful foundation for the [next chapter](#) which addresses naval warfare in the current century.

From fighting sail to steam, steel and shellfire

In the century between the battles of Trafalgar (1805) and Tsushima (1905) there were dramatic changes in naval technology and also in the conduct of naval warfare at the tactical level. The most obvious technological changes revolved around the transformation of warships from wooden sailing vessels armed with short-range cannon firing solid shot to armoured steel vessels powered by steam engines and equipped with much longer-ranged heavy weapons firing armour-piercing shells. The adoption of steam propulsion was particularly significant and brought costs and benefits. It provided for greater speed and offered ships the ability to manoeuvre without relying on the vagaries of the wind. This was particularly useful when operating on rivers or inshore areas, places where sailing ships had been at a disadvantage. Steam power also increased the potential to provide reliable logistic support to overseas expeditions. On the other hand, whereas sailing ships had enjoyed almost unlimited endurance, steam power relied on coal (later oil) and the requirement to restock with this fuel limited the endurance of warships, making possession of overseas coaling stations of particular importance. This factor favoured the British who possessed a string of bases at key points throughout the world. They also gained strategic advantage from their dominance of the network of under-sea telegraph cables which, in the second half of the century, began to make possible strategic direction of forces in different parts of the world.

The wooden three-decked ship of the line was eventually replaced by steam-powered ironclads and then by modern-style battleships that were steam powered, built of iron (later steel), armoured, armed with heavy guns mounted in revolving turrets on the centre-line of the ship and with numerous medium- and small-calibre guns added as a secondary armament. HMS *Victory* and HIJMS *Mikasa*, Nelson's and Togo's flagships at Trafalgar and Tsushima, are preserved at Portsmouth and Yokosuka (respectively) and reflect the technological advances of a century. However, the obvious differences between these vessels

do not disguise an important element of continuity. They were both designed to do broadly the same thing, namely to fight and defeat other warships in battle. They were not the only warships of their time but they were understood to be the most powerful vessels, the key measure of naval strength and the ships that would determine victory in battle. To use a twentieth-century term, they were the 'capital ships' of their eras and the major navies (and some minor ones) were willing to expend vast sums on their construction and upkeep.

Even as the battleship emerged as the new capital ship a number of developments promised to threaten their ascendancy. These included the explosive sea mine, first employed during the Crimean War (1854–5) and used to good effect by the Confederacy in the US Civil War (1861–5). Such weapons posed a serious threat to even the most powerful vessels and the Union Ironclad USS *Cairo* was the first warship to be sunk by a mine, in 1862. A generation later, mines accounted for the loss of three battleships and a number of cruisers in the Russo-Japanese War (1904–5). They were used on a large scale by all sides in the First World War, and to notably good effect by Ottoman Turkey during their defence of the Dardanelles in 1915.

The self-propelled torpedo, first developed in the 1870s by Robert Whitehead, was another relatively cheap weapon that could sink the mightiest warship if it could be delivered accurately. The threat posed to big ships by these weapons and by the small fast torpedo boats that carried them led to the development of a new kind of ship, the torpedo-boat destroyer (later known simply as destroyers) to protect the bigger ships and, if possible, to launch their own torpedo strikes. Pioneering work with submarines resulted, by the turn of the century, in vessels that had considerable military potential. Powered on the surface by diesel engines and by electric motors when submerged and armed with torpedoes (and usually also a light gun on deck), submarines added yet another dimension to naval warfare. Such weapons, in addition to long-range shore-based artillery, promised to make enemy coastal waters a dangerous place to be.¹

Inevitably in a period of technological change, the second half of the nineteenth century saw a considerable degree of experimentation in ship design and also in tactics. In the age of sail the effective range of the cannon mandated that battle would usually be decided at distances of 200 metres or less, with the British in particular favouring encounters at point-blank range where their superior gunnery would usually prevail. If ships could be laid alongside each other boarding operations were possible, providing an opportunity to capture vessels that were notoriously difficult to sink using solid shot.² Explosive shells, deployed at sea from the 1830s, made wooden ships easier to sink but armour redressed the balance, to the point where the USS *Monitor* and CSS *Virginia* were unable to sink each other in the first ironclad-vs-ironclad engagement at Hampton Roads (1862) (see [Figure 8.1](#)). Experience during the US Civil War, and Austrian success in ramming and thus sinking the Italian flagship at the battle of Lissa (1866), prompted an interest in a rather ancient weapon, the

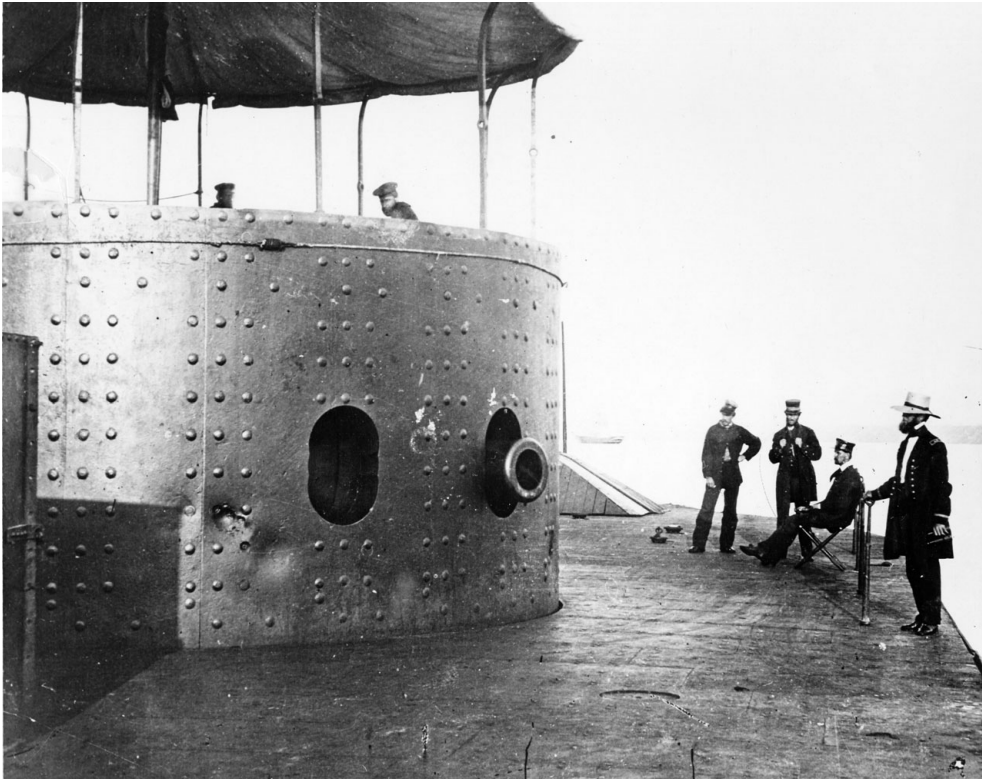
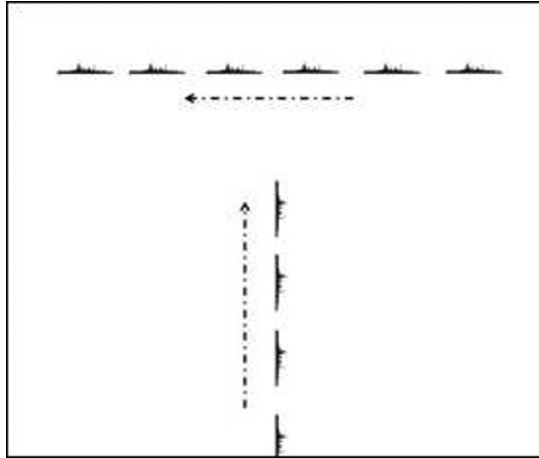


Figure 8.1 Officers from the USS *Monitor* relax on deck after the engagement with the CSS *Virginia* at the battle of Hampton Roads, 1862. Visible are one of the ship's two 11-inch Dahlgren guns and dents in the ship's armoured turret caused by enemy fire.

naval ram, until improvements in the range, accuracy and lethality of naval guns ensured that combat would no longer take place at such close range. At the battle of the Yalu, fought between the Japanese and Chinese in 1894, most fire occurred at a range of around 2,000 metres. Ten years later, at the battle of the Yellow Sea, the Japanese opened fire at 11,000 metres and both they and their Russian opponents were fully engaged at 9,000 metres.³ In such circumstances the long-range heavy guns of the battleship were the dominant weapon and tactics involved manoeuvring the heavy ships into a superior firing position and, where possible, launching torpedo attacks with cruisers and flotilla craft (see [Box 8.1](#)).

While the tactics and technology of naval warfare certainly changed over the course of the century, there were some notable continuities at the operational and strategic level. There had been numerous major naval battles in the decades leading up to Trafalgar; there were few in the century that followed. Those battles that did occur, including for example at Navarino (1827), Sinope (1854), Lissa (1866), the Yalu River (1894) and Santiago de Cuba (1898), tended

Box 8.1 'Crossing the T'



Crossing the T was a tactic in naval warfare, where one fleet manoeuvred to cross in front of the line of enemy vessels, allowing its ships to concentrate fire from guns at the front and the rear of each vessel while only receiving fire from the forward guns of the leading enemy ships. In essence this placed the friendly fleet in the position of the losers at Trafalgar, with the enemy bearing down on them in line ahead although the increased range, accuracy and lethality of naval gunfire meant that the result would likely be rather different. The Japanese successfully 'crossed the T' at Tsushima (1905) and the British did so twice at Jutland (1916). In the last-ever battleship-vs-battleship engagement, the battle of the Surigao Strait (1944), US battleships crossed the Japanese T and devastated their opponents with accurate radar-guided fire at long range.

to be on a smaller scale and did not feature clashes between the leading navy (the British) and its main rivals. Some, such as the battles of Hampton Roads (1862) or Iquique (1879), revolved around duels between individual ships. This did not diminish the potential significance of victory or defeat. The battles for sea control in the War of the Pacific (1879–83), for example, were on a much smaller scale than the Anglo-French wars of the previous century but the consequences remained familiar. Thus, the Peruvian victory at Iquique lifted the Chilean blockade of that town but subsequent defeats at the second battle of Iquique and the battle of Angamos (also in 1879) gave Chile control of the sea and opened up Peru and Bolivia to invasion and eventual defeat. The battles may have been small, the consequences were not.⁴

Steam propulsion enhanced a navy's ability to project power ashore, through expeditionary operations or through riverine operations such as those undertaken during the First Anglo-Burmese War (1824–6) and the First Opium War (1838–42). Protected by armour, powered by steam and equipped with heavy

guns, rockets and/or mortars, navies could also conduct effective strike operations against coastal fortifications. This was illustrated in a variety of actions, and was forcefully demonstrated by successful maritime power projection during the Crimean War (1854–5) and the US Civil War (1861–5). The land could fight back, using mines and coastal fortifications to protect vulnerable targets and exploiting the potential for rail links (and much later also motor transport) to rapidly concentrate forces to contain any landings. Nevertheless, evidence in these two wars suggested that the advantage was, for the time being at least, tipped in favour of the maritime force.⁵

In the early years of the new century the battle of Tsushima (1905) provided a classic example of an engagement in which the main force of one side (Russia) was decisively defeated by another (Japan). For some this reaffirmed the centrality of battle in naval warfare. However, the dramatic nature and scale of that victory should not hide the reality that the war revolved around more than one battle and that, while battle at sea was important, this was only one element in eventual Japanese success. Japanese torpedo boats and mines, for example, sank or damaged a number of Russian ships. Critically, of course, the war was joint in nature and the Imperial Japanese Navy was required to land and support the army ashore and also received support from that army (which destroyed Russian ships hiding in besieged Port Arthur). Thus, while Tsushima was taken by many as an affirmation of the importance of battle, others, such as Corbett, reflected on the importance of joint co-operation for eventual success.⁶

Into a new century

The launch by the British in 1906 of HMS *Dreadnought* revolutionised battleship design. Heavily armed and armoured, the 21,850-ton *Dreadnought* was notable for new steam turbine engines that provided an impressive top speed of 21 knots and for an all-big-gun armament of ten 12-inch guns. Making all existing battleships obsolete, *Dreadnought* lent its name to a new generation of capital ships designed along similar lines, albeit in progressively greater size and strength. The British also pioneered a new type of vessel, the battlecruiser. These ships carried dreadnought-type guns but sacrificed heavy armour for improved speed. The number and quality of dreadnought battleships was the measure by which navies judged themselves prior to 1914 and rival dreadnought construction set the terms for the pre-war Anglo-German naval race. The prestige associated with possession of such ships, and their potential to dominate any battle at sea, led to the construction or purchase of dreadnoughts by all the major navies and by many navies of second or third rank, including those of Argentina, Brazil and Chile.

The all-big-gun armament adopted by dreadnought battleships improved the accuracy of long-range gun fire. Fire control was much easier if all the guns were the same calibre, firing in salvos that could be corrected until the enemy ship was straddled and then hit by shell fire. By 1910 continuous aim fire and

director control had replaced local gun-laying, further improving long-range accuracy (and making effective fire-control equipment and procedures as important as superior guns). Battle could now occur at very long range. At Jutland in 1916 the engagement began when the battlecruisers of either side opened fire at around 16,000 yards (14,600 metres). In the last-ever battleship-vs-battleship engagement, at the Surigao Strait in October 1944, US gunfire was aided by radar-guided fire control and the battle line opened fire at 26,000 yards (23,700 metres), over twice the range at Tsushima forty years earlier.⁷

Battleships may have been the ‘capital ships’ of the age but they were not the only vessels that navies required. By the end of the nineteenth century the major fleets were made up of a mix of vessels that included battleships designed to secure control of the sea by engaging and defeating other warships, cruisers able to exercise that control by protecting or attacking trade and projecting power ashore through gunfire or naval landing parties, and flotilla craft required to conduct inshore work, carry despatches, land troops and, increasingly, to protect larger vessels from attack by other flotilla craft. Successful fleet tactics involved co-ordinating the activity of multiple ships of different types and could be considered to be analogous to the combined-arms tactics of land warfare.

It was very difficult to control a large fleet containing many ships with different roles and capabilities. Communications remained dependent on flag signalling until the 1900s when wireless telegraphy began to be deployed at sea. The latter did not solve the problem, and in some senses it served to compound it, tempting some commanders to centralise too much control in their own hands and stifling initiative amongst their subordinates. Use of radio did provide a means for dispersed ships to communicate, greatly assisting the process of scouting and providing shore-based admiralties a means to support (or interfere with) commanders at sea. Unfortunately, however, radio transmissions could be detected, betraying the presence of enemy vessels and, if your codes were broken, much useful information might be gathered by the enemy. For this reason ships at sea often maintained radio silence, continuing to communicate by flag signals and searchlight flashes.⁸

The test of war, 1914–18

The dangers facing battleships in enemy coastal waters led the Royal Navy to adopt a distant rather than a close blockade of Germany once war broke out in 1914. This robbed the Germans of the chance that they had been counting on, to whittle down British battleship strength to the point at which the smaller German fleet might engage them with some prospect of success. As a result, the German High Seas Fleet spent the war imprisoned in the Baltic and North Seas. On occasion it was able to bombard towns on the east coast of England and to threaten the Scandinavian convoys and, once, at the battle of Jutland in May 1916, to assault its jailor (see [Box 8.2](#)). The realisation of how close the High Seas Fleet had come to annihilation in the latter case led its leadership to be more

Box 8.2 The battle of Jutland, 31 May 1916

There were a number of naval battles during the war, notably at Helgoland Bight (28 August 1914), Coronel (1 November 1914), the Falkland Islands (8 December 1914) and Dogger Bank (24 January 1915). However the caution exercised by both sides in the use of their big ships and their reluctance to fight when outnumbered meant that the main fleets clashed only once, at the battle of Jutland, on 31 May 1916. Typically, the battle occurred as a result of a German attempt to tempt out and then ambush a detachment of the British fleet. Forewarned by signals intelligence the British, on the other hand, sought to exploit a rare opportunity to engage the main enemy force. With 250 vessels employed and no significant contribution from either submarines or aircraft, the resulting engagement stands as the largest surface-only engagement of modern times and the only major battle of the dreadnought era. The much-anticipated clash proved to be an unsatisfactory experience for the British, who suffered a greater material loss than the Germans. The latter managed to avoid annihilation through a mixture of skill, luck and the caution of their enemy and arrived home badly battered but largely intact. Despite this, the battle merely confirmed the defeat already being inflicted on a navy that could find no way to use its surface fleet to achieve meaningful strategic effect. That the German high command recognised this was reflected in the shift of priority away from the big ships and towards their submarines.¹

¹ Robert K. Massie, *Castles of Steel: Britain, Germany and the Winning of the Great War at Sea* (London: Jonathan Cape, 2004), esp. pp. 658–84.

circumspect in future. In the meantime the British blockade cut off German trade, isolated its colonies and slowly strangled its people. By neutralising the High Seas Fleet the dreadnoughts of the British Grand Fleet set the conditions for Allied sea control beyond the North Sea. The cautious way in which these vessels were employed should not be allowed to hide the importance of this role. The control provided by the Grand Fleet allowed other vessels to engage in trade protection, amphibious operations and a variety of other tasks unmolested.

Reliance on the sea represented a vulnerability for Britain, a country whose industry was dependent on the import of raw materials and whose people would starve without imported foodstuffs. The Germans, like the French before them, recognised this vulnerability. Pre-war plans had envisaged the use of surface ships in a traditional *guerre de course*. In the event a small number of German light cruisers and converted merchant ships attacked British trade before being sunk or interned in neutral ports. Despite causing considerable alarm and disruption their effect on the British war effort was limited. The submarine campaign against shipping heading towards British ports had a far greater impact. Once German U-boats (submarines) were released by their government from the requirement to operate within the bounds of international law, and were allowed to sink ships without warning and with no consideration for the safety

Box 8.3 Convoys

The use of convoy, meaning that ships travel to their destination in groups under naval guidance or protection, was a traditional way of protecting trade. Unfortunately convoys are a very inefficient way of moving ships. They take time to assemble, sail at the pace of the slowest vessel and may swamp the reception facilities of a port once they arrive. They can also be difficult to maintain, with some merchant ships struggling to keep station. They have the additional problem of placing rather a lot of 'eggs in one basket', raising fears that a raider might overpower the escort and then devour all in the convoy. They do, however, have numerous advantages. Convoys may be protected by escorting warships; with appropriate intelligence they may be re-routed to avoid known danger areas; and by gathering ships together into one group they are much harder to find in the vastness of the oceans. If they are found and attacked then the escort may drive off the attacker or, if it is unable to do so, may at least buy time for the merchant ships to flee. Convoy was often employed during the age of sail and was to prove equally effective in both World Wars despite the ill-founded misgivings expressed by Corbett. Indeed, not only did it serve a vital defensive function in protecting ships from attack, it also served an offensive function; the best place to find and then attack a raider was usually in the vicinity of its prey.

of civilian passengers and crew, British losses mounted alarmingly. Given the primitive state of submarine detection equipment and anti-submarine weapons it was fortunate that the first such exercise of unrestricted U-boat warfare, in 1915–16, was curtailed due to the international outrage that it caused. The second attempt, begun in February 1917, came close to success before the British adopted suitable countermeasures, primarily through the introduction of convoys (see [Box 8.3](#)). Unfortunately for Germany the campaign contributed to the US decision to declare war in 1917, adding the American fleet to the Allied inventory. The U-boat campaign demonstrated the vulnerability of a state dependent on seaborne communications to a sea-denial strategy but (as Mahan would have expected) the impact that it had on British life and economy failed to match that of the Allied blockade on Germany.⁹

The absence of any dramatic victory at sea during the First World War and the magnitude of the horror of the land campaigns can mislead some into neglecting the importance of naval activity in that war. Allied sea control provided a necessary precondition for eventual victory over the Central Powers. Sea control kept Britain safe from invasion. It enabled Britain to deploy its armies to France and to sustain them there through four harrowing years of war. It allowed Britain and France to mobilise the resources of their global empires, to conduct operations and campaigns overseas and to seize isolated German colonies. It set the context within which Britain could meet the U-boat challenge and eventually enabled the United States to deploy its armed forces across the Atlantic. It also allowed the Allies to trade with the rest of the world, giving them access to

resources that were denied to the Germans. With sea control Allied victory was possible; without it defeat would have been inevitable.¹⁰

One area where the Allies were not able to exploit sea power as effectively as they would have wished was that of power projection. Indeed, their failure to exploit fully the potential for maritime power projection denied the Allies the full fruits of their naval superiority. They did conduct numerous shore bombardments and the British even launched small-scale carrier airstrikes against targets ashore. A small number of amphibious operations were tried, including a failed expedition to Tanga in German East Africa (1914) and the raids on Zeebrugge and Ostend (1918). The largest and most famous operation, the assault on Gallipoli in 1915, succeeded only in securing a tiny lodgement that was later abandoned in the face of determined opposition and severe logistical problems. The advantage, it seemed, had shifted in favour of the defender. Ironically, while the initial assault in April 1915 and a subsequent landing in August were characterised by muddle and lost opportunities, the evacuation, staged between December 1915 and January 1916 and conducted under the noses of the Ottoman Army, was a masterpiece of careful planning and effective joint co-operation. The Germans had rather more success in landings conducted to seize islands in the Gulf of Riga in 1917, exploiting local sea control to good effect although the circumstances there were more benign than they had been at Gallipoli.¹¹

From arms control to World War, 1918–45

The end of the war in 1918, and the scuttling of the captive High Seas Fleet at Scapa Flow the following year did not bring an end to naval rivalry. In the years that followed there were, however, attempts to manage this through arms-control agreements that sought to limit the size and number of major warships. The Washington Treaty of 1922 was successful in halting, or at least delaying, a potential American, British and Japanese naval arms race but this, and the later London Treaties of 1930 and 1936 could not long remove the root causes of naval rivalry. By the limits that they set, the treaties influenced the design and construction of warships, and led to the creation of a new type of vessel, the heavy cruiser of up to 10,000 tons. The 'building holiday' that was imposed on capital ship construction hit the British hardest as the Royal Navy was generally equipped with older vessels than its rivals. Limits on battleship size and armament also worked against the British who, unlike their future adversaries, abided by these limitations. The 1930 London Treaty established limits on the numbers of cruisers that were determined by the relative strength of the major navies and not by their likely tasks, again disadvantaging the British who were unique in having a global empire to protect. Typically, restrictions on the use of submarines against merchant ships, contained in the London Treaties and also in the 1936 Anglo-German Naval Agreement, proved to be worthless in wartime.¹²

Despite the attempts at arms control, navies continued to try to adapt techniques and technologies for future conflict. Notable technological advances included the development of ASDIC (now called sonar), a device that used sound waves to detect submerged submarines and also of radar that, when deployed at sea, could be used to detect enemy vessels/aircraft and assist in fire control. Improved sensors, signals intelligence and the ability to conduct reconnaissance by air reduced the opacity of the sea somewhat, leading to the evolution of what one commentator has described as 'picture centric warfare'. Having a superior picture might mean more than having superior guns and weapons were worthless unless information could be gathered and then processed in a timely and efficient manner. The need to link sensor to shooter, to manage multiple friendly forces and to counter numerous enemies simultaneously saw the development of manual and then automated means of gathering and managing information, evolving in the Second World War into an early form of the current Combat Information Centre/Operations Room, the nerve centre of a modern warship.¹³

Despite the relative lack of success of amphibious operations during the First World War the Americans, British and Japanese all studied ways of undertaking such operations in the future. A lack of priority retarded British progress, although much useful theoretical work was undertaken and prototype landing craft developed. With more obvious use for amphibious forces in a conflict between themselves, the Japanese and Americans made better progress. Japanese capabilities were in advance of all others, prompted by the use of amphibious forces in their war with China.¹⁴ However, it was to be American and British techniques and capabilities that would, during the Second World War, solve the problems of conducting landings on an enemy-held shore and by 1943–4 offer the Allies the ability to land hundreds of thousands of troops, plus all of their supporting equipment and logistics, over open beaches in the face of enemy opposition. Amphibious forces had, finally, come of age.

One area where technologies developed significantly was in the use of aircraft. The potential for aircraft to have an impact on war at sea was evident even prior to the First World War. During that conflict both fixed-wing heavier-than-air aircraft and lighter-than-air rigid and non-rigid airships played a role in reconnaissance, anti-submarine operations and bombing. After years of experimentation the Royal Navy commissioned HMS *Argus* in September 1918, the first true aircraft carrier (with a full flight deck from bow to stern and a hangar below deck for the carriage and maintenance of aircraft). By this time the British were planning to launch a strike by carrier-based aircraft against the German High Seas Fleet in harbour although the end of the war brought the cancellation of such plans. Twenty-two years later they would demonstrate the ability of carrier-based aircraft to strike at ships in harbour when just twenty-one Swordfish biplanes flying from HMS *Illustrious* sank one Italian battleship and damaged two others at Taranto on the night of 11–12 November 1940. A year later, on 7 December 1941, the Imperial Japanese Navy improved on this,

sinking four battleships and damaging many other warships in a pre-emptive strike on the US Pacific Fleet at Pearl Harbor.

Although expected initially to play a role subordinate to the guns of the battle fleet, in the Second World War carrier-based aircraft were to prove themselves to be the decisive arm in naval warfare. Carrier-based aircraft and, when in range, land-based aircraft could fulfil a variety of roles at sea, including reconnaissance, spotting for gunfire, bombing and torpedo attack, air defence and anti-submarine operations. They could act in support of traditional surface-orientated action, as at the battle of Cape Matapan in the Mediterranean in March 1941. They could also threaten naval vessels independently. Allied experience off the coast of Norway and at Dunkirk in 1940 and in the Mediterranean from 1941 demonstrated the vulnerability of warships to attack by land-based aircraft and showed the need for appropriate anti-aircraft weapons, tactics and, above all else, fighter cover. Too often land-based air forces proved unable to provide the latter, leaving their naval colleagues dangerously exposed. The most radical developments occurred in the Pacific War, where the true potential of carrier-based aircraft was shown in the attack on Pearl Harbor and in subsequent operations. At the battles of the Coral Sea in May 1942 and again, on a greater scale, at the battle of Midway in June that year the US Navy and the Imperial Japanese Navy fought major battles in which their surface vessels were never within sight of each other or within range of their big guns. The outcome was decided in both cases by the action of carrier-based aircraft resulting, at Midway, in a major defeat for the Japanese (see [Box 8.4](#)).

In contrast to the First World War, where the vital contribution of maritime power to Allied success was overshadowed by the dramatic events on land, the struggle at sea in the Second World War was obviously central to the eventual outcome. Within Europe, Allied naval power set the limits to Axis expansion, sustained Britain during the dark days of 1940–2 and eventually provided the means for a counter-offensive across the Mediterranean, into Italy and France and then the Third Reich itself. Once again German surface raiders and U-boats attempted to cut Allied seaborne communications, this time aided by long-range aircraft and German possession of Norway and French Atlantic ports. Once again the Allies were able to overcome the threat, with a combination of convoys, escorts and, of increasing importance, air support. Aircraft, either shore- or sea-based, proved to be one of the most effective means of defeating submarines. The anti-submarine campaign in the Atlantic also illustrated the importance of intelligence and of good operational analysis. The ability of US industry to replace lost shipping was also critical. On the other side of the world, from 1942 the United States initiated a submarine campaign against Japanese shipping that was as ruthless as its German counterpart but that, given woeful Japanese countermeasures, had, by 1945, destroyed Japan's merchant fleet.¹⁵

It should be noted, of course, that the war in Europe was not won at sea but, as Corbett would have been quick to comment, by armies on land. In

Box 8.4 The battles of the Coral Sea (4–8 May 1942) and Midway (4–7 June 1942)

The battle of the Coral Sea is significant as it was the first engagement between fleets at sea decided entirely by air power. Indeed, the surface ships of both sides were never even close to being within visual or weapons range of each other. The battle resulted in one US aircraft carrier sunk and another damaged in exchange for one small Japanese carrier sunk and two others damaged. In all cases the damage was inflicted by enemy carrier aviation. Although tactically indecisive the battle resulted in the Japanese abandoning an attempted invasion of Port Moresby in New Guinea. The battle of Midway resulted from a Japanese attempt to draw out and destroy the remaining US carriers by attacking the US base at Midway atoll. Forewarned by successful intelligence, and assisted by a degree of luck, the US fleet was able to surprise the larger Japanese force, sinking all four Japanese carriers present for the loss of only one of their own. Once again neither fleet was ever within range of the other's guns. Despite their losses the Japanese still possessed a large superiority of gun-armed surface forces but, in the face of US control of the skies, were compelled to withdraw. The battles illustrated the growing importance of aircraft carriers in naval warfare.¹

¹ Samuel Eliot Morison, *History of US Naval Operations in World War II: Coral Sea, Midway and Submarine Actions, May 1942–August 1942* (Boston, MA: Little Brown & Co., 1949); Craig L. Symonds, *The Battle of Midway* (Oxford University Press, 2011).

the case of the Western allies, these armies were landed from the sea and, given their continued reliance on seaborne supplies, may justly be regarded as an expression of maritime power. In the east the Soviet Union absorbed, contained and then destroyed the main German armies. They were aided in this process by the diversionary effect of Allied maritime power, ensuring that the Germans could never concentrate their entire strength against them, and also by the provision of vital supplies by convoys fighting their way through to Murmansk.

The Pacific War, fought over a vast expanse of ocean was, of course, primarily maritime in nature notwithstanding land campaigns in China and Burma that depleted Japanese strength. After an initial surge of success that was halted at Midway in June 1942 the progressive destruction of the Japanese fleet, and in particular its aircraft carriers and naval aviators, set the conditions for Allied victory. In the process the US Navy grew into a force of unprecedented size and capability, dwarfing all of its rivals and relegating the British, for the first time in centuries, to a navy of the second rank. In a series of engagements fought across the Central Pacific between 1942 and 1945 the US Navy shattered the Imperial Japanese Navy opening up Japanese-held islands to assault by ground forces (see [Figure 8.2](#)). To the south, naval forces supported operations by US and Australian troops that liberated New Guinea and the Philippines. By 1944 the Japanese position had become so desperate that they resorted to using kamikaze



Figure 8.2 Ships of the US Third Fleet and the British Pacific Fleet off the coast of Japan in August 1945. In the foreground is the battleship USS *Missouri*, in company with the battleships HMS *Duke of York*, HMS *King George V* and USS *Colorado* and a number of cruisers and destroyers.

attacks, attempting to fly explosive-laden aircraft directly into ships. This seriously complicated the problems of air defence, as it was now no longer enough to distract or deter a pilot; the aircraft had to be destroyed. In this sense the kamikazes presaged the problems that would face navies during the missile era. In any case, US maritime power had, by summer 1945, brought Japan to its knees. The atomic attacks on Hiroshima and Nagasaki, with bombs transported by sea and dropped by aircraft operating from islands seized from the sea, forced

the Japanese leadership to accept this fact. They also threatened to pose a fresh set of problems for navies.¹⁶

Navies in the nuclear age

The experience of war between 1939 and 1945 could be taken to have validated established thinking about naval warfare. The Pacific campaign, in particular, certainly lent weight to those whose vision of maritime power was informed by Mahan, with a series of major battles setting the conditions for successful and decisive US power projection. Typically, in 1947 Admiral Chester Nimitz USN submitted a report to Congress which restated the validity of the traditional principles, with an additional focus on how air power had enhanced the reach and role of navies.¹⁷ In the years that followed, navies continued to focus on their traditional roles. However, the devastating potential of nuclear weapons posed challenges that were both tactical and strategic. At the tactical level the destructive potential of such weapons posed serious questions over the future viability of concentrated naval forces. Navies could (and did) point to their potential for manoeuvre and also dispersal as effective defensive countermeasures in addition to the value of the air defence and counter-strike potential provided by their aircraft carriers. Unfortunately, some maritime operations required a degree of concentration that would render them highly vulnerable. These included important activities that had been central to Allied success in the Second World War, such as large-scale amphibious operations and the delivery of convoys to congested ports. Over time, warship design adapted to make these vessels and their crew less vulnerable to the side-effects of nuclear attack and also to attack by chemical and biological weapons. The deployment of nuclear weapons (including bombs, missiles, depth charges and torpedoes) by some navies certainly enhanced their offensive potential. Nevertheless, the widespread use of nuclear weapons against maritime forces and their supporting infrastructure would have posed serious questions as to their ability to perform traditional duties, or indeed any co-ordinated role. The same, of course, applied to armies and air forces and was denied as vociferously by generals and air marshals as it was by admirals.

While questions remained about the role of navies in any major war, their value in situations short of all-out war soon became apparent. Western naval forces, in particular, fulfilled a variety of vital roles in the numerous lesser conflicts that characterised the Cold War. That the traditional roles of power projection and naval diplomacy remained as important as ever was demonstrated in a series of conflicts and crises from the Korean War in 1950 to Vietnam in the 1960s, the various Arab–Israeli disputes, the Malvinas/Falklands Conflict in 1982 and the Gulf War of 1990–1. Somewhat belatedly the Soviet Union came to recognise the value of this role and from the late 1960s developed both the size and reach of its navy. The presence of Soviet ships became a feature in waters that had previously been the preserve of the United States and its allies. Notably,

Box 8.5 The 1980s 'Tanker War'

The Tanker War occurred during the Iran–Iraq War (1980–8) when Iraqi air attacks on Iranian oil terminals eventually prompted Iranian strikes against ships heading to Kuwait (which provided Iraq with its only access to the sea). By 1987 the situation was sufficiently bad for the United States, Soviet Union and a number of European countries to have deployed warships to protect shipping operating in this vital area. The threat to shipping was posed by aircraft, missiles and unconventional fast craft armed with rockets and small arms. The dangers of operating in a war zone were illustrated when the USS *Stark* was hit by two Exocet missiles fired at a range of around 35km by an Iraqi aircraft (thirty-seven crew were killed). Iranian mine-laying posed a particular danger and resulted in serious damage to the cruiser USS *Samuel B. Roberts* and to numerous merchant ships. In retaliation the United States launched a number of offensive operations to degrade Iranian mine-laying capabilities and to punish the regime for its activities. In the midst of one such operation the USS *Vincennes* accidentally shot down a civilian airliner, with the loss of 290 innocent lives. By the end of the conflict around sixty Western warships and twenty-nine Soviet ones had been deployed to the region. Despite this it is estimated that 411 merchant ships were hit over the course of the Tanker War, with around a quarter lost. Nevertheless, neither side succeeded in cutting the flow of oil from the other.¹

¹ Martin S. Navias and E. R. Hooton, *Tanker Wars: The Assault on Merchant Shipping during the Iran–Iraq Conflict, 1980–1988* (London: I. B. Tauris, 1996).

for example, the United States, Soviet and some European navies all deployed warships to the Persian Gulf during the Tanker War in the 1980s, co-existing within the same operational space in a manner that was uncommon on land (see Box 8.5).

The years after 1945 saw the final demise of the battleship, most of which had been retired from active service by the 1960s. A handful of battleships remained in periodic service with the US Navy until 1992, providing shore bombardment with their 16-inch guns and, after a refit in the 1980s, with Tomahawk cruise missiles. Cruisers, destroyers and frigates remained important surface combatants although, by the 1970s, the gun was being replaced as the primary weapon system by missiles for anti-air and anti-surface warfare in addition to improved torpedoes, depth charges, rockets and mortars for anti-submarine warfare. The potency of anti-ship missiles was demonstrated in 1967 when the Israeli destroyer INS *Eilat* was sunk by Soviet-built Styx missiles fired by Egyptian missile boats. Israel was to gain revenge in successful missile exchanges with Egyptian and Syrian vessels in the 1973 War. The offensive potential of anti-ship missiles was also illustrated by successful Indian attacks on Pakistani shipping in the port of Karachi in 1971 and Indian sea control isolated the main theatre in East Pakistan (Bangladesh) from reinforcement by West Pakistan.¹⁸

The 1982 Falklands/Malvinas conflict provided a rare example of a limited war in which a sustained battle for sea control represented the key to eventual success or failure. The Argentine surface fleet was neutralised at an early stage after the loss of the cruiser ARA *General Belgrano* demonstrated a critical vulnerability to British nuclear-powered submarines. Argentinean submarines proved incapable of launching any effective attacks on the British task force or their vulnerable lines of supply and one of their three operational boats was disabled and captured off South Georgia. Mines could have posed a serious challenge, but none were deployed to the critical waters in Falkland Sound. The major battle for sea control was, therefore, between land-based aircraft, operating from the mainland c. 400 miles to the west of the islands and the ships and aircraft of the British task force. The British lost one destroyer and a merchant ship sunk by Exocet anti-ship missiles and three warships and one auxiliary sunk by conventional bombs. Numerous other vessels were damaged. These losses, while not surprising given the nature of the operation and the inadequacy of some of the British equipment, highlighted the continued requirement for effective air defence and, ideally, for a large aircraft carrier able to gain local air superiority. In the event the Argentinean Navy and Air Force were unable to disable any mission-critical assets (most obviously the light carriers HMS *Hermes* and *Invincible* or the two LPDs, HMS *Fearless* and *Intrepid*) and the Royal Navy secured enough control over the battlespace in order to land and support the military force in its successful campaign to liberate the islands. However, in many respects it was a close-run thing.¹⁹

The Falklands conflict reaffirmed the utility of aircraft carriers, particularly in operations beyond the range of friendly land-based fighters. Even before 1945 aircraft carriers had replaced the battleship as the capital ship of the future in the sense that they represented the most potent, and therefore the most desirable, symbol of maritime power. They provided a more flexible platform and more effective options for air defence, sea control and land-attack than did the battleship. Unfortunately, they, and their associated air complements, were also more expensive to operate and today only the US Navy can afford the largest and most capable vessels. Numerous other navies have maintained smaller carriers and the value of these was enhanced by the introduction of short take-off and vertical landing (STOVL) aircraft like the Harrier (which are able to operate from a smaller vessel and do not need the assistance of a catapult to launch). Unfortunately light carriers tend to embark a relatively small air group. The 27,000 ton Italian aircraft carrier *Cavour*, for example, can accommodate around twenty aircraft (helicopters and AV-8B Harriers). At 70,000 tons the new British *Queen Elizabeth* class should have the capacity for twice this number while the 100,000 ton USS *George H. W. Bush* can accommodate ninety aircraft; clearly, size matters (see [Figure 8.3](#)). Nevertheless, and as was demonstrated by the two light carriers deployed by Britain to the South Atlantic in 1982, a small air group can sometimes achieve rather a lot. Often (as was the case in 1982) the choice may be between this or no air group.



Figure 8.3 The aircraft carrier USS *George H. W. Bush* and the amphibious assault ship USS *Makin Island* prepare for a vertical replenishment in 2014. Capable of a range of operations across the spectrum of conflict, the power and versatility of these ships reflects the advantages accrued by the ability to take air power to sea.

The employment of helicopters at sea has helped to increase dramatically the range and flexibility of even small vessels. Helicopters are useful for a variety of tasks including surveillance, reconnaissance, search-and-rescue, boarding operations, anti-surface and anti-submarine warfare, the insertion of special forces and the conduct of amphibious operations. Their use in large numbers operating from converted aircraft carriers and later from specialist helicopter assault ships helped to transform the conduct of amphibious operations from the 1950s. They also help to make modern amphibious ships, such as the *Canberra*-class currently being completed for the Royal Australian Navy, extremely versatile assets with utility in a wide range of scenarios including disaster relief and humanitarian support. Large helicopter-capable vessels also have utility in anti-submarine operations (a task for which helicopters are particularly suited) and they can also provide a sea base for attack helicopters, as did HMS *Ocean* and the French helicopter carrier *Tonnere* during the 2011 NATO operation in Libya. Many navies operate such ships, including the Japanese Maritime Self-Defence Force which, for political reasons, describes their vessels as ‘helicopter destroyers’.

In the 1990s with the end of the Cold War and the removal of the threat posed by the Soviet Navy, most Western navies moved away from strategies designed to cater for a major war in Europe to focus on more limited contingencies. These ranged from crisis response, sanctions enforcement and peace support operations through to power projection in limited war, expeditionary warfare and maritime strike. This change in priorities was reflected in the policy of the US Navy, in medium-sized navies such as the French and British, and even, on a smaller scale, in the coastal navies of Denmark, Norway and Sweden.

As ever, while some navies focused on sea control and power projection, others concentrated primarily on sea denial. Existing weaponry such as mines, torpedoes, coastal artillery and land-based aircraft have been subjected to incremental improvements and in the 1960s were supplemented by anti-ship missiles. Small, cheap, fast attack craft armed with such weapons have sometimes appeared likely to fulfil the prophesies of the *Jeune École*. These vessels pack a powerful punch but are, like nineteenth-century torpedo boats, vulnerable to enemy countermeasures. The fate of Iraqi fast attack craft in 1991, destroyed with ease by coalition aircraft, may be instructive. Indeed, during that conflict it was a very old weapon, namely traditional sea mines, that posed the greatest threat to coalition activity at sea, badly damaging the cruiser USS *Princeton* and the amphibious assault ship USS *Tripoli*. Ironically *Tripoli* was, at the time, acting as flagship for the aviation minesweeping operations taking place in the northern Gulf. Mines had been used extensively by all sides in both world wars and have continued to pose a challenge to navies and merchant ships through to the present day.²⁰

Below the surface, submarines advanced considerably from those employed during the Second World War. By the end of that conflict the Germans had introduced snorkel equipment that allowed their boats to run on diesel engines without having to surface, greatly reducing their vulnerability to air attack. More recently, the development of Air-Independent Propulsion (AIP) systems for diesel-electric submarines (SSKs) further reduces the vulnerability of these boats making them more like true submarines rather than the submersibles of the two World Wars. Advances in submarine technology have made SSKs quieter and thus more difficult to detect, significantly enhancing their potential, particularly in shallow water where large nuclear-powered boats are at a disadvantage. The US Navy developed the first nuclear-powered submarines (SSNs) in the 1950s. Such vessels boast high speed, unlimited endurance and have no requirement to surface in order to replenish batteries. Equipped with torpedoes, anti-ship missiles and, in some cases, land-attack cruise missiles, they are extremely potent weapons platforms. It is significant that the British have given their SSNs old battleship names. However, being larger and noisier than SSKs they are less well suited to shallow, coastal waters. They are also more expensive and technologically demanding than SSKs, meaning that, unlike diesel-electric boats which are now widely available, only a few navies currently possess this capability. When equipped with ballistic missiles such boats (SSBNs)

have proved to be an ideal means of deploying a nuclear deterrent due to their relative invulnerability to a pre-emptive strike.²¹

Conclusion

Even from this brief survey it is clear that there have been many changes in the tactics and technology of naval warfare since the nineteenth century. No doubt there will be many more in the years ahead. Despite this it is still possible to identify continuities across the decades. If he were transported into the present Admiral Togo would probably be astonished to inspect a modern fleet and he might struggle to adjust to the range and tempo of operations enabled by modern systems and sensors. Nevertheless, he would probably recognise the continued importance of the battle to secure, maintain or challenge sea control and also the requirement to do something useful with such control once you have it. The manner in which battle is conducted has evolved, and will continue to do so, but the importance of achieving domain dominance remains. The things that one can do with such dominance and the range at which one can do this, have expanded considerably with the advent of long-range artillery and missiles, organic aircraft and modern amphibious forces. However, a commander from the Russo-Japanese War would be well placed to recognise the value of joint operations and be surprised only by the expanded range and scope of twenty-first century naval capabilities. The [next chapter](#) will focus on such capabilities, and on the challenges facing navies today and tomorrow.

RESEARCH QUESTIONS

1. What did the battle of Trafalgar achieve that Jutland did not?
2. How important is technology as a driver of change in naval warfare?
3. To what extent has air power enabled or inhibited naval operations since the First World War?
4. Is it true to say that, while tactics and technology have changed, the broad principles of maritime strategy remain valid?

FURTHER READING

Black, Jeremy, *Naval Power: A History of Warfare at Sea from 1500 Onwards* (Basingstoke: Palgrave, 2009). As the title suggests, Black provides a brief study of the history of war at sea over the past 500 years. The book offers a useful introduction to the relevant history.

Friedman, Norman, *Network Centric Warfare: How Navies Learned to Fight Smarter through Three World Wars* (Annapolis, MD: Naval Institute Press, 2009). Friedman provides an overview of 'picture centric warfare' over the course of the last century and analyses its importance today.

- Murfett, Michael, *Naval Warfare 1919–1945: An Operational History of the Volatile War at Sea* (Abingdon: Routledge, 2009). Murfett provides a valuable single-volume history of war at sea from 1919 until the end of the Second World War.
- Sondhaus, Lawrence, *Naval Warfare 1815–1914* (Abingdon: Routledge, 2001). This is a short but authoritative survey of the evolution of navies and of naval warfare from the Napoleonic Period to the First World War.
- Rose, Lisle A., *Power at Sea*, 3 vols. (Columbia, MO: University of Missouri Press, 2006). Over three volumes Rose examines naval warfare and sea power more generally from the 1890s to the present day, offering a challenging and sometimes controversial analysis that charts the rise of the US Navy.
- Till, Geoffrey, *Understanding Victory: Naval Operations from Trafalgar to the Falklands* (Santa Barbara, CA: Praeger, 2014). Till examines four major battles/campaigns from the last 200 years in an attempt to assess whether there emerges any consistent set of factors that help to explain victory and defeat. His conclusions are original and insightful.
- Vego, Milan, *Operational Warfare at Sea: Theory and Practice* (London: Routledge, 2010). Vego offers an analysis of the planning and conduct of naval campaigns and operations from the Napoleonic period to the present day.
- Wilmott, H. P., *The Last Century of Sea Power*, 2 vols. (Bloomington, IN: Indiana University Press, 2010). The first two volumes of Wilmott's history of sea power in the twentieth century cover the period 1894–1922 (vol. I) and 1922–45 (vol. II).

ONLINE RESOURCES

- The US Naval History and Heritage Command website includes a range of useful resources relating to the history of the US Navy, www.history.navy.mil/.
- Naval History Online provides a wide variety of links to records, documents and commentary relating to naval history (and with a particular emphasis on the British Royal Navy), www.naval-history.net/index.htm.
- The archives of the US Naval War College Review offer articles on a wide range of historical topics, www.usnwc.edu/Publications/Naval-War-College-Review.aspx.
- The online International Journal of Naval History includes articles on a range of historical topics, www.ijnhonline.org/.
- British Naval History is a website devoted to discussion of that topic, and includes articles and blogs, www.britishnavalhistory.com/about/.

NOTES

1. For an overview of developments during this period see Lawrence Sondhaus, *Naval Warfare 1815–1914* (London: Routledge, 2001) and Robert Gardiner (ed.), *Steam, Steel and Shellfire: The Steam Warship, 1815–1905* (London: Conway Maritime Press, 1992).
2. See Richard Harding, *Seapower and Naval Warfare 1650–1830* (London: Routledge, 1999).
3. Sondhaus, *Naval Warfare 1815–1914*, pp. 171, 189.
4. William F. Sater, *Andean Tragedy: Fighting the War of the Pacific, 1879–1884* (University of Nebraska Press, 2009), esp. chs. 4 and 5. Also see Charles Callwell, *The*

Effect of Maritime Command on Land Campaigns since Waterloo (London: William Blackwood & Sons, 1897), pp. 311–28.

5. Craig L. Symonds, *The Civil War at Sea* (Oxford University Press, 2012). Andrew Lambert, *The Crimean War: British Grand Strategy Against Russia, 1853–56*, 2nd edn (Farnham: Ashgate, 2011).
6. For a classic analysis of this war see Julian S. Corbett, *Maritime Operations in the Russo-Japanese War 1904–05* (Annapolis, MD: Naval Institute Press, 1994).
7. Sondhaus, *Naval Warfare*, ch. 7; Paul Halpern, *A Naval History of World War 1* (London: UCL Press, 1994), p. 318; Samuel Eliot Morison, *Leyte: June 1944–January 1945* (Edison, NJ: Castle Books, 1958), pp. 223–4.
8. Norman Friedman, *Network Centric Warfare: How Navies Learned to Fight Smarter through Three World Wars* (Annapolis, MD: Naval Institute Press, 2009), esp. ch. 1.
9. Lance E. Davis and Stanley L. Engerman, *Naval Blockades in Peace and War: An Economic History* (Cambridge University Press, 2006), ch. 5.
10. See Halpern, *A Naval History of World War 1*, Lawrence Sondhaus, *The Great War at Sea* (Cambridge University Press, 2014) and Norman Friedman, *Fighting the Great War at Sea. Strategy, Tactics, Technology* (Annapolis, MD: Naval Institute Press, 2014).
11. Tristan Lovering (ed.), *Amphibious Assault. Manoeuvre from the Sea* (Woodbridge: Seafarer Books, 2005) chs. 2–4.
12. For an overview of this period see Michael Murfett, *Naval Warfare 1919–1945: An Operational History of the Volatile War at Sea* (Abingdon: Routledge, 2009).
13. Friedman, *Network Centric Warfare*, passim.
14. For further details see Williamson Murray and Allan R. Millett (eds.), *Military Innovation in the Interwar Period* (Cambridge University Press, 1996).
15. See Clay Blair, Jr, *Silent Victory: The US Submarine War against Japan* (Annapolis, MD: Naval Institute Press, 2001).
16. Samuel Eliot Morison, *History of United States Naval Operations in World War II*, 15 vols. (Boston, MA: Little Brown and Co., 1947–62). For a more concise account see Samuel Eliot Morison, *The Two-Ocean War* (Boston, MA: Little Brown and Co., 1963).
17. Admiral Chester Nimitz, ‘Report to the Secretary of the Navy’, December 1947.
18. Frank Uhlig, *How Navies Fight: The US Navy and its Allies* (Annapolis, MD: Naval Institute Press, 1994), pp. 352–62; Vice Admiral G. M. Hiranandani, *Transition to Triumph: Indian Navy, 1965–75* (New Delhi: Lancer Publ., 2000), ch. 10.
19. See Admiral Sandy Woodward and Patrick Robinson, *One Hundred Days: The Memoirs of the Falklands Battle Group Commander* (London: Harper Press, 2012).
20. For example, see Scott C. Truver, ‘Taking Mines More Seriously: Mine Warfare in China’s Near Seas’, *US Naval War College Review*, 65(2) (Spring 2012).
21. For an analysis of current capabilities see the annual *IHS Jane’s Fighting Ships* (London: Jane’s Information Group).



9

Naval warfare in the twenty-first century

Contents

Naval warfare: changing roles?

Modern navies and post-modern navies

Naval warfare: changing forms?

Plus ça change plus c'est la même chose

Contemporary naval policy

Conclusion

KEY THEMES

- The political, military and economic importance of the sea appears to be enhanced rather than diminished by recent developments. The emergence of a globalised world economy, in particular, has prompted a growing appreciation that states have a shared interest in the security of the global commons.
- This has been reflected in heightened awareness of the importance of maritime security operations and of the value of multinational co-operation required to deal with transnational threats. This has encouraged collaboration between navies.
- On the other hand competitive tendencies remain. It seems likely that new technology and new techniques, often employed in conjunction with some distinctly old capabilities, will challenge the kind of access and freedom of manoeuvre than many navies had previously taken for granted. This will present challenges to some navies and opportunities to others.
- It may be that new navies gain in power and capability and old navies decline. There is nothing new or unusual in this. What does appear likely is that the ability to use the sea will remain important in both war and peace, that there will be many ways for adversaries to challenge such usage and that navies, in close co-operation with other joint forces, will need to adapt in an appropriate and timely fashion in order to meet such challenges.

Chapters 7 and 8 argued that despite continuing change at the tactical level of naval warfare there has been a considerable degree of continuity in the roles that navies fulfil. It is not necessarily true that this will always remain the case. It is possible that naval roles might change to reflect political, economic or

technological developments. Even if existing roles do remain, navies might be forced to change their forms in order to meet future challenges. Thus, new roles may evolve (and old ones disappear) in the face of new challenges while alternative means may be needed in order to sustain those roles that endure. This chapter will examine these issues and will discuss the nature of naval warfare today and into the future.

Naval warfare: changing roles?

Traditional interpretations suggest that naval warfare revolves around the use or denial of the use of the sea for military or economic purposes. Thus, almost forty years ago Ken Booth argued that people were interested in using the sea for three general reasons: the passage of goods and people; the passage of military resources; and, the exploitation of the resources in and under the sea.¹ An analysis of current maritime thought and practice suggests that, if anything, each of these three factors is now more important than they appeared in the past. The globalised world economy is entirely reliant on maritime trade. The viability of the system and the prosperity and stability of the states that comprise it depend on the timely arrival of goods that can only be carried in sufficient bulk by sea. Additionally, the sea remains a vital source of food and of raw materials (notably oil and gas) for many peoples; it is certainly not declining in importance economically. This implies that navies will continue to have a role in protecting economic interests at sea.²

The sea remains an avenue available for the projection of military force in peace and in war. That much was demonstrated by NATO use of the sea to project power against Libya in 2011. It was reflected again in the summer of 2013 when both the United States and Russia deployed naval assets to the eastern Mediterranean during the crisis caused by the use of chemical weapons in the Syrian civil war and, more recently, with Russian and US sea-launched missile strikes, and also US carrier air attacks, against Islamist and rebel (Russian) targets in Syria. Despite ongoing fears of creeping territorialisation of the seas, the area beyond the 12-mile limit remains free to navies as manoeuvre space and even territorial waters are available for 'innocent passage'. This manoeuvre space continues to be available (and to be used) for a wide variety of activities intended to support, deter or coerce. Further, it seems that the use of the sea for the passage of military resources is likely to be of growing significance as the costs (economic, political, military) of deploying force ashore appear to be increasing. Thus, and in the aftermath of their costly engagements in Iraq and Afghanistan, the US armed forces have emphasised the need to find means of projecting power and influence overseas without placing a footprint on foreign soil.³ Threats ashore and concerns about diminishing access have provided added impetus to the idea of sea basing (see [Box 9.1](#)).

In addition to the above, the many ongoing disputes over sovereignty at sea appear to suggest that navies will continue to have a role in protecting

Box 9.1 Sea basing

Sea basing has been defined by the US Department of Defence as the ‘deployment, assembly, command, projection, sustainment, reconstitution and reemployment of joint power from the sea without reliance on land bases within the operational area’. Sea basing aims to exploit the flexibility associated with manoeuvre at sea, and the protection that can be provided by a naval force, in support of an ability to operate simultaneously and seamlessly on the seas, ashore and in air and space across the range of military operations. The concept was developed in the 1990s but its roots go back much further. The ‘fleet train’ developed during the Second World War reduced reliance on land bases and contributed to an early form of sea basing. Aircraft carriers and amphibious ships have provided sea bases for joint assets for many decades, and cruisers, destroyers and other ships can (and do) sea base fire support, command and control, ballistic missile defence etc. The concept here is different insofar as the range and scale of those joint assets that may be sea based could increase, allowing the maritime force to provide a capability similar to a regional land base, but one that can move. It includes the projection ashore of joint expeditionary forces. The concept demands sea control; there will be little point sea basing assets in waters dominated by an enemy. However, assuming that control can be secured and maintained, the concept may offer one means of overcoming enemy countermeasures. The sea base should be harder to attack than its land-based equivalent and, unlike a land base, can move closer to the action or further from a threat depending on need. In some respects the idea represents a logical extension of those attributes discussed in [chapter 7](#) (mobility, flexibility, access etc.) and should enhance the US Navy’s ability to achieve its stated core capabilities by, for example, enabling forward presence or providing a capacity for humanitarian assistance and disaster relief (HA/DR). Still, it is worth noting that sea basing does demand heavy investment in enabling equipment and it remains untested in the most demanding circumstances.¹

¹ USMC, *Seabasing Required Capabilities Annual Report, 2013* (www.mccdc.marines.mil/Portals/172/Docs/Seabasing/Reports/POM15_AR2013.pdf). Also see Sam Tangredi, ‘Sea Basing: Concepts, Issues, Recommendations’, *US Naval War College Review*, 64(4) (Autumn 2011), 28–41.

(or challenging) rights of use and of passage and in supporting national claims to jurisdiction. The numerous disputes over sovereignty and jurisdiction in the South China Seas have seen the navies and other maritime agencies of many of the claimants engaging in posturing, confrontation, acts of violence and intimidation, and even occasional bouts of armed conflict. In May 2014, for example, Chinese and Vietnamese coastguard vessels and other boats collided in the disputed Paracel Islands, with the loss of at least one Vietnamese fishing boat, as Vietnam sought to hinder the progress of a Chinese drilling rig to an area that both states consider sovereign.⁴ Similarly, there are many competing claims in East Asian waters. The ongoing dispute between Japan and China over control of the Diaoyu/Senkaku islands is perhaps the most high profile of these and has seen both parties using their navies proactively (some might say aggressively) in support of their national claims.⁵ Here, as elsewhere, politics combines with

Table 9.1 *The span of naval tasks**

* *Australian Maritime Doctrine: RAN Doctrine 1* (2010), p. 100.

economics and strategy as sovereignty of rocks and islands promises jurisdiction over wider sea areas and the resources within them.

Writing during the Cold War, Ken Booth and Eric Grove argued that naval roles could be defined broadly as being military, diplomatic and constabulary.⁶ This remains a useful way of thinking about naval warfare, or at least of identifying and categorising naval activities. The Royal Australian Navy published an updated version of this 'trinity' in their 2010 *Maritime Doctrine*, listing a span of maritime tasks that ranges from the benign to the coercive. It is important to remember that the boundaries between these categories are porous and rather arbitrarily drawn. There is considerable overlap between them. The dividing line between coercive diplomacy (diplomatic) and power projection (military), for example, can sometimes be hard to spot. Many of the tasks identified in Table 9.1 fit traditional assumptions about the roles of navies; others, such as humanitarian assistance and disaster relief (HA/DR) or environmental

protection, appear rather new. In truth most 'new' tasks are far from unprecedented historically but it is fair to say that many navies now place a greater emphasis on these than was once typical. The US Navy, for example, has added 'maritime security' and 'humanitarian assistance/disaster relief' to its list of 'Core Capabilities', joining the established concepts of forward presence, deterrence, sea control and power projection.⁷

One of the features of recent discourse on naval policy has been the focus on broader security challenges posed by potential disorder at and from the sea and on the need to protect and police the global maritime commons. As a policing operation this is quite a task given that the seas cover over 139 million square miles and provide a medium that is opaque and under-regulated. Threats may arise as a result of terrorist or criminal activity at sea or ashore. They include the challenge of piracy, recently resurgent off the coast of Africa (see [Box 9.2](#)). Challenges to good order at sea may be linked to poor maritime practice and include environmental pollution and resource degradation (especially of fisheries). They may often involve the transportation of illegal cargo, including narcotics, arms or people, and the need to counter illegal immigration by sea is a notable task for many maritime agencies. These challenges are unconventional and transnational. They do not respect international boundaries and to solve them one may need to look beyond the pages of Mahan and Corbett.

Traditional ideas about maintaining good order at sea have broadened to include an additional focus on Maritime Security Operations (MSO), activities that blur the lines between the military, diplomatic and constabulary roles. MSO go beyond traditional constabulary tasks, such as resource protection within national waters, to include a wider spectrum that, according to *British Maritime Doctrine*, may incorporate 'a wide range of operations from defence (short of war fighting) through to security to development and relieving human suffering by utilizing the full spectrum of maritime forces'.⁸ MSO may thus involve robust enforcement actions, including those required to support Peace Operations. These may blur the lines between military and constabulary roles, and can be more coercive than traditional approaches to the latter. The emphasis is often as much on promoting good order *from* the sea as it is on activity *at* sea.

Associated with the interest in MSO has been an increased emphasis on the need for international co-operation for, as the US Navy has emphasised, '[n]o one nation has the resources required to provide safety and security at sea throughout the world' (see [Figure 9.1](#)).⁹ This has been reflected in recent American policy. Thus, the US has emphasised the importance of Global Maritime Partnerships as one means of promoting co-operative engagement to police the global commons in pursuit of what Admiral Mullen (US Chief of Naval Operations, 2005–7) described as a '1,000-ship navy' concept. Mullen's vision was not for the US to bankrupt itself trying to maintain a fleet of this size but rather for the navies of the world to work together to meet their mutual self-interest in policing the seas. More recently the current Chief of Naval Operations (Admiral Jonathan Greenert) has talked of forging a Global Network of Navies to create

Box 9.2 Contemporary piracy

Piracy has always existed and has often flourished in areas where there has been disorder ashore and insufficient control at sea. A resurgence of piratical activity in South-east Asia in the 1980s and 1990s was eventually suppressed by the concerted actions of the regional states. In the 2000s instability in Somalia and an associated lack of governance offshore resulted in a burst of piratical activity in the Gulf of Aden that prompted a major international response to protect this vital sea lane. The response included a variety of measures including the creation of a Recommended Transit Corridor in which ships could best be protected, the adoption by industry of Best Maritime Practice designed to inhibit attack and the issuing of naval guidance to merchant ships through the Maritime Security Centre – Horn of Africa. Many navies contributed ships and personnel to anti-piracy patrols. Some, such as the Chinese Navy, preferred to operate alone or through bilateral arrangements with others; a significant number contributed to multinational missions such as NATO's Operation *Ocean Shield* or the European Union's Operation *Atalanta*. As the threat increased some shipping companies invested in private armed security contractors to protect their ships, raising interesting questions about maritime law and safety.

Unfortunately the pirates proved resilient and adaptive. Legal ambiguities, naval restraint and the difficulty of distinguishing pirates from legitimate fishermen meant that they could generally only be engaged when actually in the process of attacking a ship. Once in possession of a vessel it was difficult to attack them without endangering the lives of the original crew, now held as hostages. Over time the pirates developed an impressive capacity to operate at great distance, broadening the area under threat. However, after a peak of 237 reported attacks off Somalia in 2011 the number of piratical incidents there fell back dramatically in 2012. This was partly a result of the activity at sea, including strikes from the sea designed to destroy the pirates' boats and supporting infrastructure. It was also a result of increased stability ashore; better governance on land represents the real long-term solution to the problem. Perhaps typically, as piracy declined off the east coast of Africa there was an increase in attacks in the Gulf of Guinea, off the west coast.¹

¹ IMB Maritime Piracy Reporting Centre (www.icc-ccs.org/piracy-reporting-centre/piracynewsfigures); The World Bank, *The Pirates of Somalia: Ending the Threat, Rebuilding a Nation* (Washington, DC: The World Bank, 2013).

an open and adaptive framework that would allow both long-term co-operation and short-lived collaboration on a 'plug-in and play' basis.¹⁰

Modern navies and post-modern navies

In light of the recent focus on maritime security operations and on international collaboration Geoffrey Till has suggested the existence of two different tendencies in current naval policy. He characterises these as 'modern', which emphasises traditional missions focused narrowly on the national interest, and



Figure 9.1 The Singapore navy frigate RSS *Formidable* (foreground) and the Indian Navy frigate INS *Brahmaputra* conducting exercises in the Bay of Bengal in 2007 as part of a multinational exercise that also included American, Australian and Japanese vessels. Such exercises provide valuable training in multinational operations and also serve a useful diplomatic purpose.

‘post-modern’, which emphasises a broader range of security concerns and recognises the need to collaborate in support of the common good. According to Till these categories are rather roughly drawn and most navies will reflect elements that are both modern and post-modern. Certainly many navies do now stress approaches that fit into Till’s post-modern categories and some – for example, the navies of Denmark and Sweden – have reconfigured themselves so that they can provide more effective support to international collaborative enterprises.¹¹

It may be the case that post-modern approaches allow friends and also potential rivals to co-operate; promoting common security and building confidence. On the other hand, one does not have to look too far to find examples of decidedly ‘modern’ thinking in naval planning. Navies remain the tools of states and are designed to protect the national interest. Post-modern approaches may flourish in a situation where the state does not face an immediate military threat and where national and international interests clearly align. Navies with a major rival close by have tended to be more focused on national defence and may see

international engagement in 'modern' terms of building relationships that support their traditional defence needs. Even those navies that have emphasised a post-modern agenda, in both word and deed, have continued to argue that their primary role is still national defence and, presumably, when rivals challenge their own national interests, the response may be less post and more modern.

Naval warfare: changing forms?

While one can argue that despite significant shifts in emphasis and priority there has been a considerable degree of continuity in the general roles that navies undertake, even a cursory glance through history demonstrates that navies have had to change their forms in order to meet new challenges or to seize opportunities. A century ago there were no aircraft carriers; today there are no battleships, and so forth. Change can be the result of many factors. Technology may be significant, but so also are a wide variety of other things. Changing societal values, for example, may influence decisions about whether, when and how to use force and, of course, about which technologies to develop. Changes in threat perception can be important as shifting priorities may imply different approaches and new force structures even in the absence of any radical technological advance. Such advances do occur, but they have tended to be evolutionary rather than revolutionary in their impact and are often less about the technology per se than about changes in the ways that technology is understood and used. It is wise to be sceptical of the all-too-frequent claims by some commentators that this or that new technology has changed everything. It may be true, but it almost never is. Nevertheless, of course, it is important to understand that change does occur, its sources and its impact can be hard to predict and there may be severe costs associated with a failure to adapt to it in ways that are timely and appropriate.

One of the developments that has excited particular attention recently has been the apparent proliferation of advanced technology designed to enable adversaries to deny access to and/or to restrict freedom of manoeuvre within an operational area. Some commentators believe that the recent Western lead in high-technology weapons and systems will be degraded as other countries (notably China) catch up. This may contribute to a process where very high-technology weapons systems, previously available only to major powers, become more widely accessible through arms transfers to client states or are bought on the open market. The successful attack on the Israeli corvette INS *Hanit* off the coast of Lebanon, conducted by Hezbollah in 2006 using a Chinese designed C-802 anti-ship missile, showed that even sub-state groups can now gain access to effective sea-denial weaponry; this may be a sign of things to come. The concern now is that a combination of advanced systems and also the innovative use of cheaper low-technology capabilities might allow a wide range of potential adversaries to deny the seas to even the most powerful navy. For some this represents a major problem; for others it is an opportunity. For

Box 9.3 Anti-access/area-denial (A2/AD)

'Anti-access' refers to actions and capabilities, usually long range, that are designed to slow or prevent opposing forces from entering an operational area. 'Area-denial' relates to shorter-range measures that are designed to limit freedom of action within the operational area. There are a wide range of actions and capabilities that can be employed. The threat comes not just from new technology but also in the ability of adversaries to adopt innovative tactics, often using older technologies, to maximise their strengths and to exploit enemy weaknesses.

Anti-access approaches might include long-range cruise and ballistic missile attacks to strike at forces or support infrastructure, and the use of submarines to interdict sea lines of communication. These may be supported by long-range reconnaissance and surveillance systems that can enable targeting. Mounting bases and support infrastructure might be subject to cyber-attacks and to strikes by special forces. Key space-based systems (important for command and control, communications, navigation, surveillance and targeting) could suffer attack by kinetic and non-kinetic anti-satellite systems including attacks on ground stations.

Area-denial might involve sophisticated air defences to deny air superiority or use of ships, missiles or submarines to deny sea control. Naval mines might be used to close straits or to deny a stretch of coastline, and landmines could be used to inhibit operations ashore. Within the littoral land-based rockets, mortars and artillery might pose a threat to inshore operations and also to landing forces. Low-technology threats (such as armed and explosive-laden small boats and craft) add another challenge, particularly if they are able to exploit cluttered or restricted waters to 'sneak up' on an adversary. Divers equipped with explosive charges may pose a threat to ships at anchor. Small, fast, rocket-armed vessels may exploit swarming tactics to swamp an adversary's defensive systems. Chemical or biological weapons could be used to degrade operational efficiency within a given area.¹

¹ US Department of Defence, *Joint Operational Access Concept* (2012), pp. 8–9.

those wishing to use the sea as a platform for their global presence it is the latter rather than the former that applies. The US Navy has thus identified what they describe as anti-access/area-denial (A2/AD) capabilities as a key challenge to current and future US naval policy (see [Box 9.3](#)).

New and emerging technology may exacerbate the threat, particularly to surface forces, by enhancing the ability of sensors to detect assets at sea and thus to cue attacks by a new generation of fast, stealthy and accurate weapons that may be increasingly difficult to detect, defeat or destroy. New missiles may exploit nano-technology to become smaller, faster, smarter, stealthier and more deadly at greater range. Mines may become more discriminating and harder to detect. Old-fashioned contact mines will continue to be cheap and easy to deploy but will remain both deadly and difficult to clear quickly. There will continue to be a battle between submarine and anti-submarine technology, and the wider availability of Air-Independent Propulsion systems will enhance the potency of



Figure 9.2 The US Navy's experimental unmanned X-47B lands aboard the aircraft carrier USS *Theodore Roosevelt* in August 2014. UAVs are already widely employed to provide persistent surveillance and strike options. Future systems are likely to have increased range, endurance and payload and may be less reliant on the maintenance of constant real-time communications, reducing their vulnerability to countermeasures such as jamming.

conventional boats. A growing number of states may deploy nuclear-powered vessels, giving their submarine forces enhanced speed, range and endurance. Torpedoes will continue to pose a deadly threat, and may gain yet more speed, range and accuracy. Enhanced ISR will make the sea less opaque than previously, opening up possibilities for targeting and attack, and increasing the need for robust countermeasures. Aircraft will continue to pose a deadly danger, particularly when equipped with the latest anti-ship missiles and supported by effective sensors. Unmanned systems, now common in the skies, may become more widespread on and under the surface, providing new options for persistent surveillance and for covert attack (see [Figure 9.2](#)).¹²

Of course it is important to remember that developments in one area are likely to be matched by others elsewhere. For example, the construction of effective anti-ship ballistic missiles may be countered by an enhanced capacity for long-range strike, designed to take out the missile sites or supporting infrastructure before they can be used against friendly assets. Equally, of course, just as anti-ship cruise missiles led to the deployment at sea of effective systems designed to jam, spoof or shoot down incoming threats, the evolution of anti-ship ballistic missiles will promote increased investment in ship-based

ballistic missile defences, although the challenges here are significant. The US Navy has been at the forefront in developing Ballistic Missile Defence (BMD), providing a sea-based component to the national plan for missile defence using the Aegis-combat system and SM-3 missiles. In future they may need such defences to protect themselves, and also to protect deployed forces ashore. For the time being it is worth noting that there is little sign that anyone has yet overcome the technical difficulties associated with getting a long-range ballistic missile to actually hit a moving target at sea; static targets, such as land bases and airfields, are much easier to hit.

The US Navy, and indeed all branches of the US armed forces, now recognise that access is likely to be challenged in all of the war-fighting domains; on land and sea, in the air, in space and also cyberspace. Some adversaries may be able to challenge in all domains, most will have access to a more limited range of capabilities. To meet this challenge the United States has emphasised a requirement for Globally Integrated Operations, aiming for previously unaccomplished levels of joint integration to create 'cross-domain synergies' to allow strengths in one area to mitigate weaknesses in others to achieve 'all-domain access'. In support of this, the US Navy and US Air Force have collaborated to develop the Air-Sea Battle concept. This focuses attention on the requirement for US joint forces to seize the initiative in the battle for air and sea control, developing networked and integrated forces capable of attacking adversary A2/AD capabilities in depth and thus fostering the level of control required for the conduct of joint operations. It requires strikes against the full depth of the enemy system, targeting all domains, and extends far beyond the sea. Air-Sea Battle appears offensive in nature, seeking to take the battle to the enemy. This could be problematic in a crisis. Strikes against land-based A2/AD capabilities might be necessary to ensure sea control, but pre-emptive action could be escalatory and might often be ruled out on political grounds.¹³

Plus ça change plus c'est la même chose

The notion that the challenge to sea control could become global as well as regional may be an uncomfortable realisation for some navies, including the most powerful, but it is not something that will surprise any historian. The requirement to fight to achieve domain dominance is nothing new even if, in recent years, the threat has tended to be both limited and localised. What may be new is that in future a combination of technology and innovative techniques will give weaker adversaries defensive options that are much more effective than those available to date, and this may affect all military domains. Of particular concern is the notion that major surface ships might become critically vulnerable to new and cheaper weapons systems. This has been a recurrent theme in thinking about warfare for the past century (and before). Typical of such approaches was the lament, over a decade ago, by two American commentators who argued that '[t]he US Navy is making a few well-armoured knights and

they will face a forest full of peasants armed with longbows'.¹⁴ One could note the similarities between perspectives such as these and the ideas of the *Jeune École* in the nineteenth century.

To date, claims that advances in weapons technology have made surface vessels obsolete have proven unfounded. A balanced naval task force, containing a mix of anti-submarine, anti-surface and anti-air assets, represents a powerful offensive and defensive force with relevance across the spectrum of warfare. The sheer utility of large surface ships suggests that major navies will continue to need them to fulfil a variety of roles and that they will therefore invest heavily in their protection. If the US Navy, or others, were to abandon their large ships they would also have to abandon many roles; they have shown no willingness to do either. Perhaps a more reasonable response to the challenge would be the recognition that there are some places that one cannot safely send major surface vessels, or at least not until major operations (across all domains) have significantly degraded enemy capabilities. Once again, this is the norm historically.

The diffusion of sophisticated A2/AD capabilities may require navies to be more circumspect in their use of large, expensive and mission-critical vessels. It may increase the desire to keep assets such as cruisers or aircraft carriers further offshore, operating over the horizon, where threats may be more manageable. Of course, enhanced anti-access capabilities may push the notional 'horizon' much further back than it has been previously. This may raise the value of more covert capabilities, including submarines that are able to operate closer to the source of the danger. It may enhance the utility of unmanned systems whose loss could be tolerated. It may also imply an increasing reliance on smaller and cheaper ships whose vulnerability might be compensated for by increased numbers and by a reduced investment (human, military and economic) in each individual platform. The ship itself might be a relatively basic host for a variety of more sophisticated remote weapons systems (including unmanned air, surface and under-water vehicles), with investment focused on the systems rather than the ship.¹⁵ Network-centric operations may encourage the continued dispersal of offensive and defensive capabilities into a wider range of assets that could be tied together into a cohesive whole (a 'system of systems') by a co-operative engagement capability enabled by their data-links despite the ships being spread over a wide area. This could foster a more distributed approach to operations, potentially reducing vulnerability, enhancing the reach and range of a force's sensors and weapons and making its actions less predictable. In the face of a sophisticated A2/AD challenge, concentration, identified by Mahan as a key principle in naval warfare, may now increase rather than reduce vulnerability. Certainly it now makes more sense to think of the concept in terms of concentrated effect rather than as the physical proximity of ships to each other. On the other hand, it may remain the case that ships will best be able to maximise their combined potential for self-defence when they operate in reasonable proximity to each other. The potential for an adversary to disable satellite communications, and to disrupt the connectivity on which network-centric operations

is predicated, poses an additional challenge to the notion of distributed operations.¹⁶

It is not impossible that new technology allied to new techniques will pose such a threat that navies will be forced to change their form in some way. This is not unprecedented. The evolution of the submarine and the torpedo, of aircraft at sea and on land, the development of anti-ship guided missiles etc. have all prompted navies to adapt: developing new weapons, systems or even entirely new ship types, and abandoning others, to meet new challenges or to seize new opportunities. One should also remember that navies have often had to fight for control, sometimes against peer rivals and sometimes against an asymmetric threat. Indeed this could be viewed as one of the defining characteristics of naval warfare. That fight must now be multidimensional with navies facing challenges in all of the war-fighting domains (land, sea, air, space, cyber). These challenges can only be met if navies co-operate effectively with joint forces, and particularly with land-based air forces. Once again, however, while the methods may have changed the principle is not new at all. An examination of the battle of the Atlantic or of the Solomon Islands campaign in the Second World War, for example, would reveal the need for joint integration to achieve cross-domain synergies in order to defeat enemy anti-access and area-denial capabilities.

Contemporary naval policy

It is difficult to generalise about current naval policy. The pressure of events means that anything written today will inevitably become dated very quickly. It is even more difficult to predict the future with any accuracy. Just as importantly, generalisations must inevitably fail to take account of the very great variation that exists between the policies of different navies. The roles that they fill will reflect a variety of factors associated with the general strategic context, national priorities and perceptions, likely adversaries and available resources. Put simply, there is no template that all navies must follow. The US Navy may still be the largest and most powerful in the world, but it makes little sense to judge others according to this standard unless those others aspire to maintain the same range and scale of capabilities and to fulfil exactly the same roles, and none does. Even the Chinese Navy, touted as an emerging rival to the USN, will face a different strategic context and thus a different set of challenges and will be asked to achieve objectives that may not reflect those in Washington. It is unlikely to mirror its larger rival exactly and certainly has not done so to date. The grammar of naval warfare may impose a certain discipline (and therefore promote some commonality between navies) but it does not impose its own logic.

Some navies focus on constabulary duties and/or international engagement, particularly if they operate in a benign international context. Others who are less fortunate and who have cause to fear their neighbours (or extra-regional intervention) may focus more on military tasks associated with sea control or denial. A smaller but still significant group may be interested in local, regional

or global power projection, and will develop the forces to support this. Most navies fall into the first two categories. The Iranian Navy, for example, must counter the threat of attack by the United States or its allies and has focused particularly on sea denial in the Strait of Hormuz and Persian Gulf although it does retain a capacity for limited force projection. Facing a different set of challenges the El Salvador Navy concentrates on constabulary duties with an emphasis on counter-narcotics operations. Similarly, on the other side of the Atlantic the tiny navy of the Republic of Benin is also devoted to constabulary tasks and operates a small number of patrol boats for this purpose. In the Baltic the Swedish Navy responded to the opportunity created by the extinction of the Soviet threat by refocusing on international engagement in multinational maritime security operations, although growing fears about a resurgent Russia may yet bring another change of direction. Navies such as these are more typical of world navies than are their larger counterparts and the roles that they fill can either contribute to or threaten maritime safety and security – something that has been emphasised in recent US policy.¹⁷

For two decades since the end of the Cold War the US Navy enjoyed a period of almost monopolistic naval power. Unchallenged by a peer rival they did not have to contend with a sustained global challenge to sea control such as occurred during both World Wars and was anticipated during the Cold War. Attention shifted away from the now defunct Soviet Fleet and instead the navy placed a growing emphasis on the projection of power overseas in support of US interests. This was reflected in a number of key statements including: ‘... From the sea’ (1992), ‘Forward... from the sea’ (1994) and, from the US Marine Corps, *Operational Maneuver from the Sea* (1996). The overt emphasis on activity ‘from’ the sea was significant. This was maintained into the twenty-first century, and was reflected in yet another vision statement, *Sea Power 21* (2002), which placed a noticeable weight on joint integration to meet conventional and asymmetric threats.¹⁸ The US Navy also began to prioritise a range of broader maritime security concerns and the need to sustain partnerships with other navies and agencies in order to police and protect the maritime commons. This lay at the heart of the 2007 Maritime Strategy published in partnership with the US Marines and Coast Guard. Tellingly entitled *A Cooperative Strategy for 21st Century Seapower*, this document argued that US interests would best be served by ‘fostering a peaceful global system comprised of interdependent networks of trade, finance, information, law, people and governance’.¹⁹

Since 2007 there has been a growing tendency for the US Navy and naval commentators to emphasise the challenges posed by adversary navies. Such adversaries might be small or large, conventional, asymmetric or hybrid but, as has been discussed, it is suggested that they will be able to pose a real challenge to the kind of unfettered use of the sea that has underpinned America’s global posture in recent years. This was reflected in the 2015 revision of the Maritime Strategy, which placed a much greater emphasis on such things than had the 2007 original. The revised Strategy continues to stress the importance of a

global network of navies but, tellingly, emphasises the value of such networks in enhancing war-fighting effectiveness between allies and partners.²⁰ This comes at a time when the United States has had to cut back its defence budget as a result of the economic down-turn and when other states appear to be catching up in terms of wealth and power. The rapid economic growth experience by the so-called BRIC countries (Brazil, Russia, India and China) has seen each of these increasing the investment in their navies, with a concomitant expansion in size and ambition. China has been identified by many as a future superpower, a state that may be able to challenge US dominance at sea. Brazil, Russia and India do not currently match this potential, but they (and perhaps others, such as South Africa) may continue to develop as regional powers with navies to match.

There has been a great deal of speculation about the possibility of China emerging as a new superpower and this has led to suggestions that the Chinese Navy may become a peer rival to the US Navy. The interest that this has caused was illustrated by the furore associated with the commissioning in 2012 of China's first aircraft carrier, the *Liaoning*. The growing reach and ambition of the People's Liberation Army Navy (PLAN) may be reflected in the commissioning of new and more capable ships and also in the first tentative out-of-area operations, such as the deployment of ships to participate in anti-piracy patrols off the coast of Somalia. China's growing dependence on seaborne trade, as well as a rather robust approach to issues of offshore sovereignty and jurisdiction, has provided an imperative towards naval expansion. Additionally, it is important to remember that Chinese decision-makers are very aware that their country suffered a century of humiliation at the hands of adversaries who came by sea. Such humiliations are not a thing of distant memory. In 1996 the United States responded to Chinese pressure on Taiwan by deploying two carrier groups into the Taiwan Strait, interfering in something that the Beijing government would prefer to think of as an internal affair and posing a challenge that the People's Republic was ill-placed to meet. The lesson appears to have been learnt.

China has concentrated on developing a range of joint capabilities, across all domains, that are designed to challenge US access and freedom of manoeuvre, aiming to keep US forces away from areas of Chinese interest and to threaten the viability of US bases in the region, not least by making it obvious to the host that such bases might be subject to attack. Chinese attempts to project power beyond the so-called 'first island chain' are thus less about an offensive mission to seize territories in the manner of Japan in 1941–2 and more about a capacity to threaten adversaries at greater range. Tactically and operationally the policy may be offensive; strategically it is probably defensive (at least from the perspective of Beijing). Of course, China's neighbours are less likely to see it this way and, perhaps inevitably, have responded with their own naval build-up and, in many cases, have sought to develop or maintain ties to the United States. The United States, for its part, has announced an overt 'pivot' of focus to the Asia-Pacific region and has reacted to the apparent A2/AD challenge with its own concept for Air-Sea Battle (see [Figure 9.3](#)).²¹



Figure 9.3 The guided-missile destroyer USS *Arleigh Burke* launches Tomahawk cruise missiles to conduct strikes against Islamic State targets in September 2014. Weapons such as these give navies the capacity to strike far inland without necessarily having to establish a base ashore.

Following on from the above, one of the more interesting developments in recent years has been the shift in naval spending away from the traditional maritime powers (i.e. US and Europe) and towards the emerging navies in the Asia-Pacific region. As Geoffrey Till has noted, 'For the first time in 400 years, the East will be spending more on its navies than is the "old west" – a truly momentous development'.²² The Japanese Maritime Self Defence Force, for example, has enhanced its own capabilities including the expansion of its submarine arm and the construction of new 'helicopter-destroyers', vessels that share most of the characteristics of a light aircraft carrier except for the name. The South Korean Navy has also benefited from recent investment, and exists in a difficult place, sandwiched between two powerful states with which it has a number of jurisdictional disputes (China and Japan) and next to a rogue regime that has launched numerous sneak attacks by sea (North Korea). Within South-east Asia many navies view the expansion of Chinese ambitions with alarm, and have sought to foster links with the United States as a counter to this. Thus, in April 2014 two US warships participated in six days of 'non-combat' exercises with the Vietnamese Navy, the fifth in a series of annual exercises with that country. That same month the United States and the Philippines announced a new deal to allow the United States access to bases within that country. Before

this the ten-month deployment to Singapore of the Littoral Combat Ship USS *Freedom* in 2013 reflected an enduring American interest in supporting maritime security operations, and provided a visible sign of US engagement in the region.²³ Within Europe, both Britain and France retain medium-sized navies with a capacity for significant power projection far from home and other navies have invested in major amphibious vessels (Spain, Italy and Netherlands) or light aircraft carriers (Spain and Italy). Still more have proven able to contribute to multinational operations including the European Union's first naval mission, Operation Atalanta. The leading role played by European countries, most importantly Britain, France and Italy, during the 2011 NATO operation in Libya may reflect a growing willingness on the part of such countries to take responsibility for security within their own backyard, although a general reluctance to do anything substantive during the Syrian civil war may indicate otherwise. Europe still lacks many key enabling capabilities that can still only be provided by the United States and the EU has tended to lack the will or the cohesion to act as a credible actor in terms of hard security. However, as the US pivot to Asia sees the reallocation of resources from the Atlantic to the Pacific it is possible that America's allies in the Old World may be forced to look after themselves a little more than has been the case in the past.

Those making naval policy face considerable challenges, whether they are based in Beijing, London, New Delhi, Washington or Porto Novo (Benin). They must make decisions about current threats and emerging challenges in an environment that is always constrained by cost and where other agencies also make compelling calls on a limited budget. They must also anticipate the actions of others and plan in the knowledge that it will be many years before the systems that are ordered today actually become operational. A desire for sophisticated systems may compete with the need to maintain numbers, providing a quality versus quantity dynamic. An apparent need to prioritise one role may imperil funds available for another. Difficult choices must be made. How many destroyers should one sacrifice (if any) in order to keep an aircraft carrier? Which is more important: mine countermeasures or anti-submarine operations? Is it acceptable to reduce assets focused on securing control of the seas in order to increase spending on those designed to exploit such control? The answers to such questions are not fixed and it is important to remember that they are not given by navies alone. Even if they know what they want, and think that they know what they need, navies must convince the government and the people to support their vision. Sometimes they may do this in co-operation with the other services, sometimes in direct opposition. It is not an easy process. The results are usually less than ideal.

Conclusion

The discussion above (and in the previous chapters) seems to confirm that, while the tactics and technology of naval warfare have changed and will continue to change over time, navies continue to fulfil their traditional roles

and that these remain focused upon achieving, exploiting or denying sea control in order to influence events on the land. A capacity to utilise the manoeuvre potential of the sea will remain as important in the future as it was in the past. The ability of naval forces to exploit their inherent attributes, and those of the medium in which they operate, in order to provide access that can be turned into leverage is certainly central to current US policy. The denial of such access will figure prominently in the minds of potential opponents and navies represent only one of the means by which denial may be achieved. This is normal. It is impossible to anticipate fully how the future will pan out. Some new technologies (such as anti-ship ballistic missiles) may threaten naval forces; others will enable them (e.g. rail guns or directed-energy weapons). It seems probable that the imperative to think more jointly will continue and that navies of the future will co-operate even more closely with other services and agencies than in the past. The complexity of contemporary security challenges and the multi-dimensional nature of the twenty-first century battlespace demand this, as does the need for cost-cutting synergies. What is clear is that the ability to use or to deny the use of the sea will continue to be critical in a wide range of circumstances and thus navies will remain important tools of policy in times of war, peace and in all of the grey areas between. An understanding of naval warfare will be as important in the twenty-first century as it was in the twentieth and, indeed, has been for centuries.

RESEARCH QUESTIONS

1. How new are current fears about anti-access/area-denial challenges? Do they represent anything more than 'old wine in new bottles'?
2. Is the sea today more or less important to global prosperity and security than it was when Mahan first wrote on the subject?
3. Does the current emphasis placed by many navies on multinational collaboration in support of maritime security operations signal a significant change in naval policies and priorities?
4. Given the importance today of joint operations, is there any point in continuing to think about naval warfare as a category distinct from land and air warfare?

FURTHER READING

- Dutton, Peter, Robert Ross and Oystein Tunsjo (eds.), *Twenty-first Century Seapower: Cooperation and Conflict at Sea* (London: Routledge, 2012). This edited collection provides a valuable insight into the challenges facing navies today.
- Goldirck, James and Jack McCaffrie, *Navies of South-east Asia: A Comparative Study* (London: Routledge, 2013). This book provides a useful survey of the navies in a particularly interesting and important part of the world.
- Mulqueen, Michael, Deborah Sanders and Ian Speller (eds.), *Small Navies: Strategy and Policy for Small Navies in War and Peace* (London: Ashgate, 2014). Examines how smaller navies meet the challenges of contemporary strategy and policy.

- Saunders, Philip *et al.* (eds.), *The Chinese Navy: Expanding Capabilities. Evolving Roles* (Washington, DC: National Defence University Press, 2012). A collection of essays that examines the rise of the Chinese Navy.
- Speller, Ian, *Understanding Naval Warfare* (London: Routledge, 2014). This book expands on many of the issues raised here, particularly in [chapter 9](#).
- Tangredi, San J., *Anti-access Warfare: Countering A2/AD Strategies* (Annapolis, MD: Naval Institute Press, 2013). Tangredi examines one of the key challenges facing the US Navy today.
- Till, Geoffrey, *Asia's Naval Expansion: An Arms Race in the Making* (London: Routledge, 2012). As the title suggests this work examines the expansion of navies in the Asia-Pacific region.
- Werthiem, Eric, *The Naval Institute Guide to Combat Fleets of the World, 16th edn* (Annapolis, MD: Naval Institute Press, 2013). This book, published annually, provides an accessible guide to current world navies.

ONLINE RESOURCES

The following sites offer an interesting perspective on contemporary security and naval policy. In addition, most navies and coastguards have websites where much information can be gained regarding role, force structure etc.:

- Australia, *Future Maritime Operational Concept – 2025: Maritime Force Projection and Control*, www.navy.gov.au/media-room/publications/future-maritime-operating-concept-2025.
- UK Ministry of Defence, *The Future Character of Conflict* (2010), www.gov.uk/government/uploads/system/uploads/attachment_data/file/33685/FCOCReadactedFinalWeb.pdf.
- USA, *A Cooperative Strategy for 21st-century Seapower: Forward, Engaged, Ready* (2015), www.navy.mil/local/maritime/150227-CS21R-Final.pdf.
- USA, *The Capstone Concept for Joint Operations* (2012), www.dtic.mil/futurejointwarfare/concepts/ccjo_2012.pdf.
- USA, *Naval Operations Concept* (2010), www.navy.mil/maritime/noc/NOC2010.pdf.
- USA, *The Air Sea Battle Concept* (2013), www.defense.gov/pubs/ASB-ConceptImplementation-Summary-May-2013.pdf.

A number of navies, including those of Australia, Britain and the USA, have made their key doctrine publications available online. In particular, see the following:

- Australian Maritime Doctrine* (2010), www.navy.gov.au/media-room/publications/australian-maritime-doctrine.
- British Maritime Doctrine* (2011), www.gov.uk/government/uploads/system/uploads/attachment_data/file/33699/20110816JDP0_10_BMD.pdf.
- Maritime Doctrine for the South African Navy* (2006), www.navy.mil.za/.
- UK *National Strategy for Maritime Security* (2014), www.gov.uk/government/publications/national-strategy-for-maritime-security.
- US *Naval Doctrine Publications, No. 1* (2010), www.dtic.mil/dtic/tr/fulltext/u2/a282471.pdf.

NOTES

1. Ken Booth, *Navies and Foreign Policy* (London: Croom Helm, 1977), p. 15.
2. Ian Speller, *Understanding Naval Warfare* (London: Routledge, 2014), esp. ch. 9.
3. For example, see US Joint Chiefs of Staff, *Capstone Concept for Joint Operations* (2012).
4. 'Vietnam Boat Sinks after Collision with Chinese Vessel', BBC News (online), 27 May 2014 (www.bbc.com/news/world-asia-27583564).
5. Christian Miere, *Maritime Diplomacy in the 21st Century* (London: Routledge, 2014), pp. 126–38.
6. Booth, *Navies and Foreign Policy*, p. 16; Eric Grove, *The Future of Sea Power* (Annapolis, MD: Naval Institute Press, 1990), pp. 232–6.
7. See ch. 7.
8. JDP 0–01, *British Maritime Doctrine* (2011), pp. 2–16.
9. US Navy, *Naval Doctrine Publication 1, Naval Warfare* (March 2010), p. 15.
10. Admiral Jonathan Greenert and Rear Admiral James M. Foggo III, 'Forging a Global Network of Navies', *US Naval Institute Proceedings* (May 2014).
11. Geoffrey Till, *Seapower: A Guide for the 21st Century*, 3rd edn (London: Routledge, 2013), ch. 2.
12. For a useful examination of emerging trends see the Royal Australian Navy, *Future Maritime Operational Concept – 2025: Maritime Force Projection and Control* (www.navy.gov.au/media-room/publications/future-maritime-operating-concept-2025).
13. *The Air Sea Battle Concept* (2013) (www.defense.gov/pubs/ASB-ConceptImplementation-Summary-May-2013.pdf). Also see Jonathan F. Solomon, 'Maritime Deception and Concealment: Concepts for Defeating Wide-area Oceanic Surveillance-Reconnaissance-Strike Networks', *US Naval War College Review*, 66(4) (Autumn 2013), 87–116.
14. Joseph Gattuso and Lori Tanner, 'Set and Drift: Naval Forces in the New Century', *Naval War College Review* (Winter 2001).
15. For example, see the UK *Black Swan* sloop concept (www.gov.uk/government/uploads/system/uploads/attachment_data/file/33686/20120503JCN112_Black_SwanU.pdf).
16. See Norman Friedman, *Network Century Warfare: How Navies Learnt to Fight Smarter through Three World Wars* (Annapolis, MD: Naval Institute Press, 2009).
17. For commentary on the size, state and policy of world navies see the annual reference book *IHS Jane's World Navies* (London: Jane's Information Group) or the *US Naval Institute Guide to Combat Fleets of the World* (Annapolis, MD: Naval Institute Press).
18. Admiral Vern Clark, 'Seapower 21: Projecting Decisive Joint Capabilities', *US Naval Institute Proceedings* (October 2002) (www.navy.mil/navydata/cno/proceedings.html).
19. *A Cooperative Strategy for 21st Century Seapower* (2007) (www.navy.mil/maritime/maritimestrategy.pdf).
20. *A Cooperative Strategy for 21st Century Seapower: Forward, Engaged, Ready. March 2015* (2015) (www.navy.mil/local/maritime/150227-CS21R-Final.pdf).
21. Bernard D. Cole, *The Great Wall at Sea: China's Navy in the Twenty-first Century*, 2nd edn (Annapolis, MD: Naval Institute Press, 2012); Bernard D. Cole, *Asia*

- Maritime Strategies: Navigating Troubled Waters* (Annapolis, MD: Naval Institute Press, 2013); Carnes Lord and Andrew S. Erickson (eds.), *Rebalancing the Force: Bases and Forward Presence in the Asia-Pacific* (Annapolis, MD: Naval Institute Press, 2014). Thomas Mahnken, 'China's Anti-access Strategy in Historical and Theoretical Perspective', *Journal of Strategic Studies*, 34(3) (2011), 299–323.
22. Geoffrey Till, 'NATO: War Fighting, Naval Diplomacy and Multilateral Cooperation at Sea', in Peter Dutton, Robert Ross and Oystein Tunsjo (eds.), *Twenty-first Century Sea Power: Cooperation and Conflict at Sea* (London: Routledge, 2012), p. 182.
23. 'US–Vietnam Naval Exercises Begin Amid Sea Tensions', Bloomberg News, 8 April 2014 (www.bloomberg.com/news/2014-04-07/u-s-vietnam-naval-exercises-begin-amid-se-asian-tension.html). 'Philippines to Give US Forces Access to Up to Five Military Bases', Reuters (US edition), 2 May 2014 (www.reuters.com/article/2014/05/02/us-philippines-usa-idUSBREA4107020140502).





Part IV

Air and space warfare

David Jordan



10

Concepts and characteristics of air and space warfare

Contents

Introduction

The air environment

 The attributes of air power

Concepts

 Control of the air

 Strategic air operations

Information dominance

Joint operations

Command and control

KEY THEMES

- The air and space environments are very complex. This complexity is reflected in the sustained and vigorous debates on the roles, use and utility of air and space power.
- Air and space power, while potent and flexible, are subject to limitations which can reduce their effectiveness. Technology has helped to offset a number of obstacles to the efficacy of air and space power.
- Air power has become a vital factor in most forms of modern warfare. Some degree of control of the air is required for armies or navies to perform effectively. Air power creates powerful synergies in combination with surface forces.
- The 'strategic effect' that air power can deliver is much broader than the simple delivery of firepower against targets beyond the battlefield. Air power is a flexible tool, although this flexibility can be undermined by trying to impose its use on a single conceptual model.

Introduction

Despite the perception that air power originated with the development of the aeroplane, some of the key aspects of modern aerial warfare can be traced to the late eighteenth century. They illustrate not only that some of the premises regarding modern air power had their origins in concepts that remained within

the institutional memory of armies and navies (even if the memories were rather dim) but also that some of the basic notions underpinning modern air power thinking are closely related to long-standing principles of war itself, rather than being revolutionary and unrelated to conflicts of centuries past. The First World War saw a pronounced quickening of aviation technology, and by the conclusion of the conflict in 1918 the fundamental roles and missions of air services had been established. The success of aircraft in the conflict – particularly in the vital role of correcting the fire of artillery, the dominant weapon in a war of static lines – also led to deeper thinking about military aviation, and from this stemmed a range of controversies and disputes over the fundamental purpose of air power, many of which remain relevant today.

These debates show that, despite its relative youth compared with the history of conflict within the land and maritime environments, the nature of air power is a contested area. It swiftly became the subject of intense debate as to its precise utility and its political, ethical and technological dimensions, a discourse complicated by contending perspectives regarding the degree of subservience or otherwise of air services to surface forces. This was further complicated by arguments between air power theorists propounding the notion that air forces can be the decisive factor in conflict with little or no involvement of the other services and those who thought such ideas ludicrous. The picture was further complicated by subtle variations in the views of the leading air power thinkers and the fact that some nations – notably France and Germany – took different views from these theorists with regard to their own air services. (See [Box 10.1](#).)

As will be seen, while the controversies began in the immediate aftermath of the First World War, they have not all been resolved, in spite of technological advances and different perceptions about the nature of war itself. Debate – often heated – has ebbed and flowed over the ensuing decades, frequently as practitioners, politicians and analysts use the most recent conflict to espouse a particular view of air power and the ‘right way’ to use it. Such contentions are influenced by service loyalties and the dictates of the prevalent fiscal climate – often inextricably linked factors as armed services fight for limited funding. Endeavouring to understand air power in modern war can, therefore, sometimes be a challenge, and comprehension can be made more difficult unless conceptual debates of air power’s early years are recognised. Inherent in any assessment of the evolution of air power is the constant of technological development. From the start, this has driven air power theorists to postulate views of revolutionary or near-revolutionary changes to the nature of warfare, sometimes despite the fact that their calculations have depended on nascent technology working perfectly from the outset, something that has rarely occurred.

Notions that air power and its associated technology have rendered the views of Carl von Clausewitz obsolete have all too frequently been disproved when subjected to Clausewitz’s ideas of ‘friction’ and ‘the fog of war’. This, in turn, has meant that cynics have been far too willing to criticise the quality of advocacy for air power, reducing often complex issues to simplistic sound bites.

Box 10.1 Air power defined

There are a number of ways of defining air power. These began with relatively broad views of the exploitation of the air; the leading American air power theorist 'Billy' Mitchell defined air power as 'the ability to do something in the air. It consists of transporting all sorts of things by aircraft from one place to another.'

Nearly sixty years later, two senior British airmen, Air Vice-Marshal Tony Mason and Air Marshal Sir Michael Armitage, interpreted air power as 'the ability to project military force by or from a platform in the third dimension above the surface of the earth.'¹ Moving into the twenty-first century, the Indian Air Force definition is as follows:

Air power, in a classic sense is defined as the total ability of a nation to assert its will through the medium of air. It includes both civil and military aviation, existing and potential. In the modern sense, air power which has evolved into aerospace power is defined as the product of aerospace capability and aerospace doctrine. Air power is the strength of an air force as opposed to an attendant capability.²

Although separated by almost ninety years, the three definitions encompass the wide range of activities that can be conducted by the exploitation of the air environment, all of which have, at their heart, issues of national power and its expression from aerial platforms, along with the often-forgotten elements (personnel, organisation, engineering and manufacturing facilities and technology) which support them.

¹ William Mitchell, *Winged Defense: The Development and Possibilities of Modern Air Power – Economic and Military* (New York: G. P. Putnam's Sons, 1925), p. xii; M. J. Armitage and R. A. Mason, *Air Power in the Nuclear Age* (London: Macmillan, 1983), p. 2.

² Indian Air Force, *Basic Doctrine of the Indian Air Force, 2012* (New Delhi: Air War Strategy Cell, 2012), p. 5.

This chapter examines the development and refinement of the basic principles established by interwar theorists and the imperatives underpinning their ideas. The association of bombing with strategic air power has become so entrenched within air forces that some of the other strategic applications of air power have been overlooked, and the chapter addresses some of the areas where air action other than that by bombers has been of importance. Air power has been vital when employed in conjunction with surface forces (used in this chapter as shorthand for armies and navies). It could be argued, therefore, that the views of the theorists regarding the strategic application of air power have been disproved by the successful application of air power in the land and maritime environments, but such contentions have often been based on a simplistic analysis of events and do not prove that air power is always subordinate to land and naval units, as some observers might suggest.

While the major concepts of control of the air (and space), strategic air operations and operations in conjunction with other forces remain valid, there has

been much evolution in both theory and practice, which has made air power a particularly dynamic environment in its short history. This and the following chapters analyse these factors, beginning with an examination of the notable characteristics of the air and space power environment before moving on to study the key concepts of air and space power and the way in which they have been employed since the First World War. Finally, Part IV of this book concludes with some considerations regarding air and space power in the twenty-first century, raising some of the major issues which have arisen to date, and which are likely to present challenges for policy-makers and those involved in delivering air and space power in a bid to achieve the political and strategic objectives laid down for them.

The air environment

The air is perhaps the most challenging environment in which humans can fight. As altitude increases, thinner air means that the supply of oxygen decreases to the point where the ability of a human being to function declines progressively until unconsciousness and then death ensue. The drastic fall in temperature as altitude increases adds to the problems of survival. Aircrew in the First World War operated aeroplanes that could, by the conflict's end in 1918, reach heights of approximately 20,000 feet. The bitter cold experienced even at relatively low altitudes could be partially overcome by wearing heavy clothing, but this hampered movement and dexterity, making the aircraft more difficult to fly. The cold and a lack of oxygen meant that operating at above 10,000 feet was a deeply fatiguing experience.

These physiological effects were soon recognised by doctors; by the outbreak of the Second World War, most – although by no means all – military aircraft were provisioned with an oxygen system, with the aircrew supplied through a face-mask. As technology advanced, pressurisation systems were developed for certain military aircraft to keep the pressure experienced in the cockpit or passenger area at that experienced several thousand feet lower than the actual altitude at which the aircraft flew. A further challenge facing aircrew, and one not regularly encountered by sailors or soldiers, is that of gravitational force. As an aircraft manoeuvres, *g*-forces may increase. Blood drains from the pilot's head, increasing the risk of unconsciousness. If *g* is sustained for any length of time, 'black-out' may follow. If the pilot fails to recover quickly, the aircraft can depart from controlled flight and crash. In combat, a semi- or unconscious pilot is highly vulnerable to attack. The crews of high-performance military aircraft are at risk of so-called *g*-induced loss of consciousness (GLOC). This factor has led to the need to provide aircrew with equipment to mitigate the effect of *g*, in the form of garments that inflate as *g*-force increases, compressing the lower limbs to restrict the flow of blood away from the head and reducing the onset of the loss of consciousness.

Physiological challenges presented by temperature and altitude are not the only issue. Platforms that aircrew operate need to be designed to meet the challenges of operating at a variety of altitudes and speed ranges. As a generalisation, it may be said that basic physics and aerodynamic principles complicate the design process for military aircraft rather more than for the equipment to be found in the other two environments. A further consideration for military aircraft designers is the need to provide the crew with the means of escape should their aircraft be fatally damaged or suffer a mechanical failure. After the First World War, air forces began to issue their pilots with parachutes to provide them with a chance of escaping from a fatally damaged aeroplane. The jet age added further complications to crew escape for designers. The increasing speeds of aircraft made escape more difficult, leading to the development of the ejection seat – but even this presented challenges, since the force required to expel the seat from the aircraft is such that it can cause spinal compression, while there is a risk of traumatic limb amputation caused by wind blast should the pilot be compelled to eject at particularly high speeds.

While these factors may not appear, at first glance, to have a great deal of relevance to understanding modern air warfare, they are of importance. Unlike sailors and soldiers, military aviators operating combat aircraft that operate at altitude require an array of personal survival equipment simply to ensure that they stay alive whether or not they encounter the enemy.

The attributes of air power

The challenges of operating in the air environment are not air power's only distinct characteristic. Proponents of maritime power sometimes note that 70 per cent of the earth's surface is covered by water. Airmen are not averse to responding that *all* the earth's surface is covered by their operating environment.¹ Air forces can, at least in theory, range across the entire globe, giving air power one of its main characteristics – *reach*.

Reach

The movement of armies can be hindered by terrain, while navies have historically been unable to attack deep into the heart of enemy territory, constrained by the need for navigable waterways for their warships. Aircraft are unaffected by such vagaries and can engage the enemy at depth, through strikes against rear-echelon forces or by attacking war industries or communications infrastructure. This characteristic has influenced much thinking about air power. It underpinned the concept that strategic bombardment could win wars without surface forces and played a broader part in the consideration of operational art – perhaps best evinced by the examples of Soviet thoughts on air power's part in deep battle, the German use of air power with land forces in blitzkrieg and NATO's development of AirLand battle doctrine in the 1970s and 1980s.

'Reach' is also relevant in terms of the range over which air and space power can operate. It has permitted the employment of aircraft (and space-based platforms) for the purpose of gaining information about opponents during both war and peace, and for the deployment of both air and land forces for the purpose of crisis response. This has included the dispatch of forces in a bid to deter opponents or to contain a situation, such as in response to the Iraqi invasion of Kuwait in 1990 and the subsequent deployment of a large multinational force to Saudi Arabia, or to deliver humanitarian assistance (such as the use of aircraft to deliver food during the Ethiopian famine of 1984). In terms of air power, 'reach' is significant because of the speed with which aircraft can be employed in response to an emerging crisis situation – that is to say, the *rapidity* of air power is a key complementary factor to *reach*.

Rapidity

Aircraft can generally deploy more quickly to a trouble-spot than either ships or vehicles. Where a swift response to a crisis is required, aircraft enable the deployment of troops and equipment. This can be of vital importance at a strategic level, since the ability to engage a desired target within a matter of hours as opposed to a matter of days may be critical in shaping an adversary's behaviour. Cases in point include the rapid reinforcement of Saudi Arabia in 1990 after Iraq's invasion of Kuwait noted above. Following the Saudi king's formal request for military assistance in the face of a perceived threat from Iraq, Operation Desert Shield was launched. Although an international response, the vanguard of this deployment came in the form of American fighter and attack aircraft, followed by warplanes from other nations. Desert Shield also saw the delivery of large numbers of troops and equipment, increasing the strength of the defences in Saudi Arabia within the space of a few days. However, despite the deployment, had Saddam Hussein decided to invade, his numerical superiority in terms of both troops and aircraft would have made defence difficult. Only when shipping carrying heavy armour and other equipment too large to be carried in any numbers by air arrived in Saudi Arabia did the balance swing decisively in favour of the defences. Thus it is important to note that while air power offers reach and rapidity this is not a panacea to all crises, as the military forces and materiel which can be delivered in this manner are relatively limited in terms of their fighting power, in terms of both the weight of their equipment and the supplies which may sustain them.

Furthermore, the concepts of reach and rapidity of action do not apply exclusively to land-based air power, since the ability to project air power from the sea has been a significant factor in warfare. Aircraft carrier operations began in the First World War, but carrier-based air power came into its own during the Second World War, most notably in the Pacific theatre. The Pacific War began with the projection of Japanese air power from its carrier fleet against the US naval base at Pearl Harbor, and ended in 1945 with targets in Japan being attacked from American and British carrier-based aircraft.

More recently, the importance of combining ships and aircraft was seen in the Falklands Conflict of 1982. Long-range air power from land bases in the form of transport operations, air-to-air refuelling and reconnaissance played a significant, and usually underrated, part in that operation, while the carrier-based Sea Harriers of the Royal Navy provided the fleet with the necessary measure of protection against Argentine attacks on shipping, and a small number of Royal Air Force (RAF) Harriers aboard the carriers provided air support for British forces once they had landed upon the islands. Without the carriers, it would have been impossible for fighter aircraft to cover the fleet and support the landings, since there were no land bases within range that could be used for this purpose. The difficulties facing the use of land-based air power were demonstrated in attacks by a single Vulcan bomber against the runway and Argentine positions around Port Stanley airfield. The attacks achieved much of the effect desired by the British chiefs of staff in inflicting damage sufficient to guarantee that the airfield could not be used by the most advanced aircraft within the Argentine inventory and in communicating intent to the Argentines, but at a price. It took more than a dozen air-to-air refuellings to get a single aircraft to and from the only available airfield on Ascension Island, and this ensured that the air-to-air refuelling of maritime patrol and transport aircraft supporting the naval task force sent to retake the islands could not be carried out. The chiefs of staff regarded this as a price worth paying, demonstrating that the provision of air power – its *apportionment* – is a difficult business, with planners having to decide upon the correct prioritisation of often scarce assets. These long-range attacks on the Falklands (known as Operation Black Buck) also highlighted the fact that the aircraft carriers were essential; without them operating fighter aircraft would have been impossible, ceding control of the air to the Argentine forces. As discussed below, being forced to grant an opponent almost guaranteed freedom of action in the air is regarded as one of the most dangerous positions to be in, with often-disastrous consequences resulting in instances where this has occurred.

Although the Falklands campaign was a prime example of the limits to air power's reach (even with air-to-air refuelling), the conflict served to demonstrate a number of critical aspects of air power. Most notable was the importance of air power to the success of joint operations. As discussed below, numerous instances exist of the contribution of air power to the efforts of the surface forces, either by *shaping* the conditions in which they operate, or through direct support of their operations, be that on land or at sea. The *flexibility* of air assets, combined with the rapidity with which they can reach the required part of the battle area, enables commanders to concentrate air power and exploit the synergies from co-ordinating air and surface units to maximise the psychological and material impact upon the enemy.

While air power can operate independently of surface forces, the evidence suggests that independent air operations, while often contributing a great deal to the conduct of a particular war or campaign, do not have the same potency

as when they are synergised with land and/or maritime forces. This conflicts with the views of the classic air power theorists and raises considerable debate amongst commentators, and is considered in more detail in the [next chapter](#).

Height

A further critical attribute of air and space power is that of height, or elevation.² Air and space craft have the ability to operate far above the surface of the earth, and in the case of space platforms of course, outside the earth's atmosphere. This gives them significant ability to observe activities across the surface of the globe. Aircraft are constrained by international law, preventing over-flight of territory without permission, but this does not apply to satellites, which can be placed in orbit so that they can provide imagery of a particular target of interest in territory which would otherwise be impossible to survey. Since the development of the surface-to-air missile (SAM) in the 1950s, aircraft conducting over-flight have been vulnerable to air defences, most obviously demonstrated by the destruction of a U-2 reconnaissance aircraft flown by Francis Gary Powers while it was carrying out a sortie gathering intelligence over the USSR in 1960. Although this was by far the most publicised example of air defences negating the advantages of height, it was far from the only case where reconnaissance aircraft operating at some altitude were brought down. The challenges presented by air defences were to be partly offset by the development of long-range sensors which could survey an area of interest obliquely without the need for over-flight, and in the case of one aircraft type, the Lockheed SR-71, through the use of extremely high speed to make it almost impossible for defence systems to track and acquire the aircraft on their fire-control radars and creating almost insuperable difficulties for SAMs in reaching it. Even so, the use of SR-71s for over-flight was limited, and stand-off systems were much preferred.

The combination of air-breathing craft and satellite systems enables extensive imagery of areas of interest to be obtained. It should be noted that aircraft and satellites complement one another, since the predictability of satellite orbits and the challenges inherent in changing them at short notice mean that it is possible for those knowing that they are under surveillance to cease or camouflage their activities when a satellite passes overhead. This relates to another key characteristic of air and space power, namely *flexibility*.

Flexibility

Air and space power can be used in a variety of ways to achieve a number of desired outcomes or *effects*. This can range from the use of force – often referred to as ‘kinetic activity’ – through to the gathering of information to the delivery of supplies and/or personnel. Air and space power may either serve as the mechanism through which these desired outcomes are achieved directly, or as *enablers* which either create the conditions which permit the achievement of the desired result or provide key elements of support without which land or maritime forces would be less likely or even unable to achieve the outcome which they had been

employed to achieve, otherwise known as the desired *end-state*. This flexibility is not, of course, unique to air and space power, since (for instance) an army unit might find itself conducting combat operations, intelligence-gathering and providing humanitarian assistance all in the space of a day's operations, or even one patrol, but while not unique, flexibility is a significant *force-multiplying* aspect of air power. A flight of combat aircraft might be employed to attack a pre-planned target, then to carry out reconnaissance and finally to support a unit of troops on the ground during a sortie, with each of these tasks being carried out with many miles between the locations at which the aircraft perform these tasks. It is the ability to conduct several different tasks over a considerable area during a single sortie which is perhaps the most notable – but not only – feature of the flexibility of air power. Flexibility can be seen, then, as the result of synergy between the characteristics of reach and rapidity.

The above characteristics may all be seen as positive aspects of air and space power, but as with all forms of military power, there are limitations as to what can be achieved. Without understanding the limitations of air and space power, it is impossible to accurately comprehend what air power can and cannot achieve, or to make nuanced analysis of what air and space power has been used to do in previous conflicts and with what degree of success.

Impermanence

The most significant limiting factor is that of impermanence. Aircraft are incapable of occupying ground, which means that there are very few instances in which troops will not be required, unless a campaign is conducted in which the seizure or occupation of territory is not necessary. Aircraft cannot stay aloft permanently, despite the development of air-to-air refuelling. Technical limitations, such as the finite amount of lubricants which can be carried aboard an aircraft to keep its engines functioning will necessitate an aircraft's return to base, while the physiological challenges of flying fast-jet combat aircraft mean that unless spare crew-members and/or suitable rest facilities are provided (an impossibility on a tactical combat aircraft because of the compromises this would impose on aircraft performance), the endurance of a crewed aircraft of this type is limited. Even in the case of larger aircraft with greater capacity for fuel, lubricants and extra crew members and/or rest facilities, fatigue becomes an issue in the endurance of an aircraft. This, of course, means that the presence of aircraft over a battlefield, or an area of interest can only be guaranteed if there are sufficient numbers of them to enable continuous coverage. Even in the days of the Second World War, when the major powers had thousands of aircraft, such coverage proved impossible.

In the main, this was the result of a lack of aircraft numbers. Although the United States of America produced over 200,000 combat aircraft during the course of the Second World War, even this was insufficient to furnish enough aircraft to provide theoretical permanent air cover.³ The practicalities were even more formidable. Visually identifying surface targets in darkness was, of course,

extremely difficult, while poor weather not only made visual acquisition of targets more difficult but often prevented aircraft from flying at all. Technological developments since the end of the Second World War increased the ability of tactical aircraft to operate in night and in poor weather conditions, but brought about increased costs – which in turn limited the number of aircraft which could be purchased, bringing the issue of numbers to the fore once again.

Impermanence thus gives an enemy the opportunity to gain respite from attack as aircraft return to base to rearm. Given the likelihood that it will be impossible to provide sufficient aircraft numbers to keep an aerial presence overhead, such lacunae in air coverage may reduce the shock effect of an air attack. This can, however, be offset by closely co-ordinated action with the other components to ensure that the effect upon the enemy achieved by air action, be it physical or psychological, is exploited to the full. During the Normandy campaign in 1944, for instance, air attacks against German positions caused considerable psychological disturbance to German troops, frequently leading to the abandonment of vehicles and emplacements as the air attack was followed up by the advance of ground forces.⁴

It is not just in the realms of attack that impermanence matters, since the absence of reconnaissance aircraft permits enemy forces to move with a reduced risk of being detected and thus targeted; this was, for instance, a notable factor in the success of North Korean and Chinese forces in moving supplies during the Korean War, who threw temporary bridges across rivers during the hours of darkness to move men and supplies before disassembling and concealing them prior to daylight and the reappearance of reconnaissance aircraft.

Technology, though, has brought about a number of changes which may have profound implications for this particular limitation to air and space power. The development of remotely piloted air systems (RPAS), more popularly known as 'drones', has permitted both aerial reconnaissance and attack sorties to be conducted by an air system which has long endurance unconstrained by human factors; operated from 'cabins' at air bases often thousands of miles away, RPAS have demonstrated an ability to keep an area under near-continuous surveillance and, if armed, have been able to conduct attack operations against fleeting targets which would otherwise have been impossible because of the delay which would have been caused by the need to dispatch combat aircraft to 'prosecute' the target. RPAS operations are controversial (as discussed below), but the rise of this type of aircraft may alter one of the key characteristics of air power, and as their technological sophistication increases, making them less fragile and prone to accidental loss, this could have profound consequences. Furthermore, the widespread utilisation of space-based reconnaissance and intelligence-gathering platforms means that the level of impermanence of air and space power is far less than it once was.

Environmental characteristics, therefore, have significant implications for the effectiveness of air power in warfare. Some of the limiting or adverse characteristics can be mitigated by design, technological development and close

co-ordination with the other services, while the positive characteristics, such as reach, speed and flexibility, allow air power to be employed to telling effect. The best means of exploiting these characteristics are, however, a source of controversy. These debates and the manner in which air and then space power have been employed are discussed in the [next chapter](#), but before we analyse the practical application of air and space power, it is useful to explore the key concepts that underpin much air and space power thinking and which have thus influenced their employment.

It is fair to say that early conceptual thought was often lacking in rigour, not least because those conducting the thinking were doing so from an almost non-existent knowledge base. Nevertheless, there was a general appreciation within armies and navies that using aircraft could be of great utility in war within a decade of the Wright brothers' first flight in December 1903. The First World War represented the first opportunity for testing the pre-war speculation about the use of aircraft (in the term's widest sense, meaning both aeroplanes and dirigibles/airships). As the war progressed, a number of key concepts either emerged or evolved from their nascent pre-war state. It is worth noting here that today's key concepts relating to air and space power can all be traced back to an era when the basics of aerodynamics and flying were not fully understood.

Concepts

Control of the air

Perhaps the most important aspect of air power is the need to ensure that air forces have the freedom to operate in the face of hostile activity. Without having at least some degree of freedom of action, air forces are unable to function properly, and this has serious effects upon surface forces. In the absence of air defence, enemy aircraft are able to attack targets almost at will. Control of the air, therefore, may be defined in general terms as possessing the ability to ensure that the enemy is unable to use its air forces, or enemy ground-based air defences are unable to prevent one's own activities in the air or to use the air to prevent the successful operations of friendly surface forces (see [Figure 10.1](#)).

As noted above, the first role that air power undertook came in the form of reconnaissance and observing for artillery. Both these functions were of critical importance in the First World War, and these functions had to be carried out if armies were to function effectively, particularly once the war bogged down into static, trench warfare. It then became necessary for the warring sides to attempt to deny their opponent the ability to utilise their air assets effectively, which led to the development of air fighting.

Although we will examine the evolution of control of the air later, it is useful to briefly note one particular way in which the First World War helped to develop a critical element of thinking within the concept, namely the use of air power in an offensive role. It became clear that it was impossible for fighter



Figure 10.1 To date, the ultimate exemplar of control of the air, the F-22 Raptor also exemplifies a key challenge facing air forces, namely that capability comes at great cost.

aircraft to mount a truly effective defence of their own forces if they waited for the enemy to come to them, particularly as early warning of the approach of enemy aircraft was limited because of the lack of any systems designed to achieve this. This mattered, since unless aircraft were airborne, mounting standing patrols, defensive fighters were required to take off and to climb to height within a very short space of time, often well beyond the capabilities of even the best-performing fighters of the war. Consequently, the head of the French air service, Paul du Peuty, and his British counterpart Hugh Trenchard, came to the view that the offensive use of air power was most effective. This involved sending patrols of fighters out over enemy lines and dominating the airspace over enemy-held territory, thus preventing their fighter aircraft from interfering with operations. The sheer scale of airspace over the Western Front meant that it was impossible to prevent German aircraft from crossing the line and attacking French and British ground positions, particularly during the battle of Verdun during 1916. Protests from French ground units led to du Peuty being ordered to carry out far more defensive patrols, with the result that even more German aircraft came over French lines. Trenchard, already a proponent of offensive air action, became possibly the leading early advocate of offensive air power as a whole (as discussed below), particularly in gaining control of the air. Although early-warning technology enabled a much more effective defence to be

conducted, as we shall see, the basic principle of offensive air power as an integral part of control of the air, as established by du Peuty and Trenchard, was proven to be valid even if the initial approach was relatively unsophisticated. It was further adopted and adapted by theorists, notably the American William 'Billy' Mitchell, in their thinking about the use of air power in war.⁵ Although he would later downplay the role of fighter aircraft – or 'pursuit' in the American lexicon of the time – Mitchell was clear about the value of the offensive in the gaining of control of the air:

Our aviation doctrine should be to find out where the hostile air force is, and to destroy it as rapidly as possible . . .

Pursuit aviation . . . is the kind designed to take and hold the offensive in the air against all hostile aircraft, . . . it is with this branch of Aviation that air supremacy is sought and obtained.

Pursuit squadrons are essentially an offensive element, and to enact their role successfully, they must take and maintain the offensive. They should seek the enemy and wherever found, attack and destroy him.⁶

It is not unfair to suggest that such thoughts provide much of the conceptual underpinning of the concept of control of the air today. By the end of the First World War, British and French fighter squadrons proved remarkably effective in keeping German fighters away from friendly aircraft conducting artillery-spotting and reconnaissance missions, with a concomitant benefit to the operations of the British and French armies, suggesting that the advocates of offensive action had a point.

The early experiences of the First World War have been replicated ever since. Nations and coalitions that have been at a serious disadvantage in the air – or, in the case of non-state actors such as insurgents, without any air power of their own – have struggled to succeed. It is important, though, to make a distinction between nation states or the alliances in which they fight and insurgents, since the characteristics of an insurgency (as described in chapter 13) mean that while possessing the freedom of action to exploit air assets is of great utility to counterinsurgents, it does not represent the difference between success and failure given the complex, multifaceted elements of a counterinsurgency (COIN) campaign which extend well beyond the military domain and their contribution to its outcome. Finally, although control of the air is a vital part of modern military operations, it is important to remember that it is rarely, if ever, an end in itself, but a critical enabler for other air operations and the prosecution of operations by surface forces.

Strategic air operations

The notion that air power can have an effect at the strategic level of conflict may have been first seen prior to the First World War, and remains a source of considerable debate today.⁷ The matter is complicated by the fact that the



Figure 10.2 Originally designed at the height of the Cold War to deliver nuclear weapons against targets in the Soviet Union by penetrating sophisticated air defences, the USAF's B-1B bomber fleet came to the fore in delivering conventional munitions in the wars of the early years of the twenty-first century.

term 'strategic air power' is open to many interpretations, most of which revolve around the use of aircraft to bomb targets that are generally not to be found on the battlefield or at sea but within the enemy homeland (see [Figure 10.2](#)). The use of bombers in an attempt to effect the strategic levels of war during both the First and Second World Wars has created an over-generalised concept of what strategic air power is.

It does not simply include attack against key targets in enemy territory but should be seen in its broadest context, namely the employment of air power to achieve an effect against the opponent. This may not involve fighting at all and in fact may *prevent* conflict (see [Box 10.2](#)). Complicating the picture is the fact that some of the thought about strategic air power owes much to initial concepts underpinning notions of control of the air, particularly the offensive use of air power, which expanded into broader theories of the use of air power for direct attack at the strategic level of war.

A further, and important, consideration is that 'strategic bombing' as understood in the context of the First and Second World Wars is associated with attacks on cities and thus civilian targets, giving rise to profound considerations about the legality and ethical use of air power. This is a complex area, and a work of this nature can only address the matter in broad terms, but those

Box 10.2 Air power and security

More central to the study of international security is the question, often conflated with the preceding one, of whether air power facilitates offense as a whole – whether investment in air power tends to make aggression and conquest easier to a greater degree than it protects against them – and of whether it encourages aggression as a result. Air power can certainly be a potent offensive instrument, and during the interwar years longer-range aircraft, and bombers in particular, were prominently cited as likely tools of aggression whose restriction or elimination would help avert war.¹

¹ Karl P. Mueller, 'Air Power' (www.rand.org/content/dam/rand/pubs/reprints/2010/RAND_RP1412.pdf), p. 6.

wishing to understand the use of air power for attacking strategic targets which may lie within cities, or in close proximity to areas where large numbers of civilians may live, have to consider the issue in great depth. It forms an integral part of the modern process of targeting – a process profoundly influenced by the experience of the Second World War – and considerations about the use of new technology in air power such as remotely piloted air systems and putative developments using so-called 'autonomous' air systems which are able to make targeting decisions through the use of artificial intelligence and sophisticated programming.⁸

More importantly, as will be seen, air and space power have demonstrated an ability to achieve effects at the strategic level without recourse to the use of force, but it is an unfortunate fact that much air power thinking, although recognising this, tends to suggest that bombardment of enemy targets at depth is the sole strategic function of air power.⁹ This is to overlook the powerful *deterrent* and *coercive* effects that the mere threat of the use of air power may have on an adversary. This is, of course, particularly applicable when one thinks of the role of the US, Soviet, British and French air forces as part of national deterrents during the Cold War, but the potency of conventional air attack, particularly in the modern era, deserves consideration. Conventional deterrence, a useful tool in *coercive diplomacy*, draws upon the characteristics of reach and rapidity to help shape adversary thinking.¹⁰ This is considered in more detail in the following chapters.

Offensive air power and morale bombing

Although the Italo-Turkish War (1911–12) saw the first use of air power in an offensive manner, the concept was not fully articulated until September 1916, by the commander of the RFC in France, Brigadier General Hugh Trenchard. He produced a memorandum entitled 'Future Policy in the Air' in which he laid down the key principles for gaining and maintaining control of the air.

Trenchard set out to address concerns from ground units that they should enjoy direct protection from their air service, friendly aircraft remaining overhead at all times. This concern remained a factor throughout the twentieth century, and, as will be discussed below, air forces have been subjected to complaints that they pay inadequate attention to the demands of the army and navy in pursuit of operations well beyond the immediate battle area. For Trenchard, such complaints ignored the fact that it was much better to prevent enemy aircraft from reaching friendly forces in the first place than having to seek to engage them while they were in the process of attacking or, worse still, after they had completed their attack. Trenchard argued: '[A]n aeroplane is an offensive and not a defensive weapon . . . The aeroplane is not a defence against the aeroplane; but it is the opinion of those most competent to judge that the aeroplane, as a weapon of attack, cannot be too highly estimated.'¹¹

This not only underpinned the approach that Trenchard and the RFC (and from 1 April 1918, its successor, the RAF) would employ for the remainder of the war but helped to form wider views upon the way in which air power could be best exploited. As well as taking the view that air power was best employed offensively, Trenchard added his voice to those who had suggested that one of the effects of air power might be psychological:

The mere presence of a hostile machine in the air inspires those on the ground with exaggerated forebodings with regard to what the machine is capable of doing . . . The sound policy then which should guide all warfare in the air would seem to be this: to exploit this moral effect of the aeroplane on the enemy, but not to let him exploit it on ourselves. Now this can only be done by attacking and continuing to attack.¹²

Two strands of thought emerged from this line of thinking (which was not unique to Trenchard). The first was that of using aircraft to attack the enemy more widely through hitting targets beyond the front line, and possibly in the enemy homeland; the second was that of the psychological effects of air power that could be exploited to bring about the collapse of enemy morale, thus bringing about the end of the war. When Trenchard was called upon to put such theories into action with his appointment in spring 1918 to command the so-called Independent Force that was specifically tasked with bombing Germany, he was singularly unimpressed, believing that this was a diversion of effort away from the crucial arena of war, the Western Front itself.¹³ However, as Chief of the Air Staff in Britain from 1919 to 1929, Trenchard took the idea of carrying air power to the enemy's homeland as the core building-block that would maintain the RAF's independence from the other two services. Trenchard offered the British government two distinct reasons for maintaining the RAF as an independent force. The first was that of policing the British Empire, which he claimed – correctly, as it transpired – could be done at less cost if aircraft were employed in some regions to support locally raised forces instead of garrisons of British troops. The second reason was the role of strategic bombardment of the enemy

in a major conflict. Both relied upon exploiting the offensive nature of air power to bring about moral collapse on the part of the opposition through bombardment.

Information dominance

The role of air and space power in joint operations has been a key concept since aircraft were employed in support of land and maritime forces during the First World War. The primary function during the period 1914–18 was to obtain information about the enemy through both visual and photographic reconnaissance. As technology evolved, it became possible to garner intelligence through the use of aircraft with equipment that could intercept radio traffic and signals and process electronic signatures given off by equipment (SIGINT and ELINT), a form of intelligence-gathering which could not only be of value to the maritime and land components, but of significance at the strategic level as it permitted analysis of and reasonably accurate conjecture about the capabilities of potential adversaries – notably the Soviet Union – during the Cold War period (approximately 1945–90). This is an excellent illustration of the way in which air power has developed from providing support at a purely tactical level to using the generic capabilities of gathering information to inform political decision-making. As technology has evolved in armies and navies, the use of detailed information for the purposes of targeting the enemy, or gaining *information dominance*, enabling detailed and effective planning of military operations to be conducted, has become critical, particularly in terms of the precise targeting of military force from air, land and sea – the construct of *network-centric warfare*. The rise of space power and intelligence-gathering platforms in orbit has only added to the importance of air and space power, both in joint operations and at the political level.

Joint operations

Air power, though, quickly developed to provide more than just information for land and maritime forces. The protection offered by fighter/pursuit aircraft, as noted above, was an important function above the battlefield, and in terms of protecting a fleet from enemy attack, particularly from the air. (See [Box 10.3](#).) Furthermore, the use of aircraft to attack enemy targets grew in importance; the use of *air-delivered fires* (terminology for the use of weapons by aircraft against ground or sea-surface targets) as a means of bringing greater firepower to bear against an opponent, or to shape the battlefield, has become of critical importance, perhaps best exemplified in the use of air power in a preparatory role prior to Operation Desert Storm in 1991. The concepts here are very straightforward, although as will be seen in the [next chapter](#), a debate about the most efficacious means of providing firepower in support of the land component has been ongoing since the end of the First World War; in simplified terms, armies

Box 10.3 Thoughts on air power and joint operations

There is much that air power can do to enable the ground element of joint operations to succeed. But for those tasks that require friendly forces to be continuously up close and personal to people – enemy, allies and the undecided – one needs soldiers present. Sensibly understood, air power does not aspire to perform all strategic missions unaided in all forms of warfare.¹

¹ Colin S. Gray, *Airpower for Strategic Effect* (Maxwell, AL: Air University Press, 2012), p. 314.

seek the provision of *close air support*, whereby aircraft are called in to deliver weapons against enemy positions blocking an advance, or which are bringing down fire on friendly forces, or otherwise presenting a clear risk which cannot be dealt with by the use of land-based systems such as artillery. Air forces, on the other hand, prefer to employ *air interdiction*, which sees aircraft operating at depth, attacking enemy supply dumps, troop concentrations and targets such as railheads or bridges, with the aim of destroying or inflicting serious attrition upon opponents before they reach the battlefield, thus rendering them much less effective. Furthermore, air interdiction is conducted away from friendly forces, thus reducing the risk of *fratricide* as a result of confusion over where enemy positions are and the accidental targeting of one's own side.

Command and control

The command and control (C²) of air power has been a lively subject for debate since the early years of air power development in the aftermath of the First World War. A concept often encountered in discussion of air power is that of the 'indivisibility' of air power, a concept under which the leaders of air forces and key air power proponents have argued that all aircraft should be under the command and control of air forces, rather than there being a situation in which armies and navies have their own air arms as an integral part of their orders of battle. This has been a controversial claim, and one which has not gained wide acceptance. In the first years of the RAF's existence, particularly during the 1920s, both the Royal Navy and British Army argued vigorously for the disestablishment of the new service and the division of air assets between the two of them.¹⁴ This was in part because of a genuinely held belief that air power was subordinate to maritime and land power with its primary purpose being to support surface units (thus rendering the existence of a separate service a non-sense), and partly because the existence of a third service meant that the reduced amount of funding available for defence purposes was even more thinly divided. The counter-argument held that the division of air forces between services and their subjection to the C² structures of the navy and army led to the 'penny

packetting' of aircraft, reducing the ability to use them flexibly, or to concentrate them at a decisive point.

The RAF ultimately lost the practical argument over the concept of indivisibility in its purest form with the decision to place the Fleet Air Arm under the control of the Royal Navy (in 1937), followed by the creation of an Army Air Corps (1957), and many – but far from all – nations now have their own air forces along with air arms under the direct control of their armies and navies. The most potent example of this, of course, is in the United States, with US Army Aviation, US Naval Aviation and the US Marine Corps (USMC) air elements being amongst the biggest air forces in the world in their own right. Another aspect of C² related to notions of indivisibility, but with rather less associated controversy, is that of *centralised control and decentralised execution*.

This concept may be understood in simple terms by examining British and American doctrine from the Second World War. The comparison is useful because the United States did not have an independent air service at the time, although the United States Army Air Force (USAAF) might be said to have enjoyed a degree of autonomy which made it appear on occasion to be an independent service in all but name. While British airmen argued strongly for all air power to be placed under the central control of an air force officer, the United States Army's doctrine did not accord with this principle. Following the serious reverse at the battle of the Kasserine Pass (February 1943) in which it became clear that the way in which the USAAF had been employed had been deficient, the United States urgently issued Field Manual (FM) 100-20, *Command and Employment of Air Power*, which held that:

The inherent flexibility of air power is its greatest asset. This flexibility makes it possible to employ the whole weight of the available air power against selected areas in turn; such concentrated use of the air striking force is a battle winning factor of the first importance. Control of available air power must be centralized and command must be exercised through the air force commander if this inherent flexibility and ability to deliver a decisive blow are to be fully exploited.¹⁵

The implications of this were considerable, not least since the doctrine went on to make clear that only in exceptional circumstances were air forces to be placed under the command of ground forces and that a *joint* approach to command was necessary. The air force commander, who was held to be best-placed effectively to apportion aircraft against competing demands from the ground forces, would be responsible to the overall commander in the operational theatre. Such a step would ensure that a sufficient concentration of air assets was available to gain and maintain control of the air and to deliver appropriate attacks against enemy formations.¹⁶ Not without reason has this been described as the US Army Air Force making a declaration of independence, a separation which was to be realised in 1947 with the creation of the US Air Force.¹⁷

The US was not alone in this view; the idea that army commanders should seek to exercise command of air force units was strongly deprecated by Field Marshal Bernard Montgomery as a result of his experiences in North Africa. Montgomery argued:

I hold that it is quite wrong for the soldier to want to exercise command over the air striking forces. The handling of an Air Force is a life-study, and therefore the air part must be kept under Air Force command.¹⁸

At the heart of Montgomery's contention lay concern that if army officers, unfamiliar with the complexity of the air environment, attempted to exercise close control of their air forces, then it would see a dilution of effort ('penny-packeting') and a reduction in the flexibility that could be obtained through the use of the air environment.¹⁹ Although Montgomery's relationship with his air commander, Air Vice-Marshal Arthur Coningham, was to collapse in acrimony, Coningham concurred with Montgomery's analysis, noting that the complexity of warfare in a technological age meant that it took years 'for a sailor, soldier or airman to learn his profession'.²⁰

This convergence of opinion has remained key to concepts of command and control within the field of air power. It is complemented by decentralised execution, which grants subordinate commanders the authority to decide upon the best way in which to carry out the task(s) assigned to them. This approach means that in a joint operation, air assets will fall under the command and control of air force officers working under the overall direction of the theatre-level commander (the Joint Force Commander or 'JFC' in modern parlance, who may themselves come from the air force), with the senior air officer – the Joint Force Air Component Commander (JFACC, pronounced 'Jay-Fac') – overseeing the integration of all air assets. This construct has not been without challenge, as we shall see below, but in its broadest sense remains valid. Finally, concepts of command and control are those which draw together the other key air-power concepts highlighted in this chapter when they are employed on operations. The way in which this has been done, as air and space power have evolved, is where we turn next.

RESEARCH QUESTIONS

1. To what extent have commentators taken the concept of control of the air for granted?
2. Why has effective co-operation between the air, land and maritime been so difficult?
3. Have the early air power theorists been vindicated in their belief that the nature of air power is inherently strategic?
4. Evaluate the view that air power can only ever be subordinate to armies and navies.

5. To what extent has space power been overlooked as a key factor in modern strategy?

FURTHER READING

- Biddle, Tami Davis, *Rhetoric and Reality in Air Warfare: The Evolution of British and American Ideas about Strategic Bombing, 1914–1945* (Princeton University Press, 2002).
- Cox, Sebastian and Peter Gray, *Air Power History: Turning Points from Kitty Hawk to Kosovo* (London: Frank Cass, 2002).
- Cooling, Benjamin Franklin, *Case Studies in the Development of Close Air Support* (Washington, DC: Office of Air Force History, 1990) (www.afhso.af.mil/shared/media/document/AFD-100924--035.pdf).
- Cooling, Benjamin Frankling, *Case Studies in the Achievement of Air Superiority* (Washington: Air Force History and Museums Program, 1994).
- Dolman, Everett C., *Astropolitik: Classical Geopolitics in the Space Age* (Abindgon: Frank Cass, 2002).
- Douhet, Giulio, *Command of the Air* (University of Alabama Press, 2009 [1921]).
- Jordan, David, 'Sir Hugh Trenchard', in Matthew Hughes and Matthew Seligman (eds.), *Personalities in Conflict* (Barnsley: Pen and Sword, 2000).
- Meilinger, Phillip S. (ed.), *Paths of Heaven: The Evolution of Air Power Theory* (Maxwell: Air University Press, 1997).
- Mitchell, William, *Winged Defense* (1925; University of Alabama Press, 2010).
- Olsen, John Andreas (ed.), *A History of Air Warfare* (Dulles, VA: Potomac Books, 2010).
- Warden III, John A., *The Air Campaign: Planning for Combat* (London: Brasseys, 1989).

ONLINE RESOURCES

- Air and Space Power Journal (US): www.airpower.maxwell.af.mil/.
- (Indian) Centre for Air Power Studies: capsindia.org.
- Joint Force Quarterly (US): <http://ndupress.ndu.edu/JFQ.aspx>.
- RAND Project Air Force: www.rand.org/paf.html.
- Royal Air Force Centre for Air Power Studies: www.airpowerstudies.co.uk/.
- Royal Australian Air Force Air Power Development Centre: <http://airpower.airforce.gov.au/>.
- Strategic Studies Quarterly (US): www.au.af.mil/au/ssq/.
- US Air Force Historical Support Division: www.afhso.af.mil/.
- (US) Air University Press: <http://aupress.maxwell.af.mil/>.

NOTES

1. See the United Kingdom's former joint air and space power doctrine publication, *AP3000: British Air and Space Power Doctrine*, 4th edn (Shrivenham: Ministry of Defence, 2009), p. 16.
2. See Phillip S. Meilinger, *Ten Propositions Regarding Air Power* (www.au.af.mil/au/awc/awcgate/au/meil.pdf).

3. See United States Army Air Forces Office of Statistical Control, *Army Air Forces Statistical Digest: World War II* (No place of publication given, 1945), p. 112 (www.dtic.mil/dtic/tr/fulltext/u2/a542518.pdf).
4. See Ian Gooderson, *Air Power at the Battlefront: Allied Close Air Support in Europe 1943–45* (Abingdon: Frank Cass, 1998) for extensive coverage of the effectiveness of air-launched weapons against German armour in terms of destruction and psychological effect.
5. See, for instance, Arthur Gould Lee, *No Parachute* (London: Jarrolds, 1969); David Jordan, 'The Army Cooperation Missions of the Royal Flying Corps/Royal Air Force 1914–1918' (unpublished PhD thesis, University of Birmingham, 1997); James Pugh, 'The Conceptual Origins of the Control of the Air: British Military and Naval Aviation, 1911–1918' (unpublished PhD thesis, University of Birmingham, 2014); Air Vice-Marshal Peter Dye, 'France and the Development of British Military Aviation', *Royal Air Force Air Power Review*, 12(1) (Spring 2009), 1–13; Richard P Hallion, *Control of the Air: The Enduring Requirement* (Washington, DC/Bolling Air Force Base: Air Force History and Museums Program, 1999).
6. William Mitchell, Testimony to Congress, January 1921, 'Statement Regarding the Necessity of the Air Service'; Staff Paper, 'Air Service Tactical Application of Military Aeronautics', 5 January 1919; *Memoirs of World War I: From Start to Finish of Our Greatest War* (New York: Random House, 1928), p. 82.
7. See, for example, Anon., 'Air Power', *Naval Review*, 1(2) (1913), 57–75; Brett Holman, *The Next War in the Air* (London: Ashgate, 2014); Michael Paris, *Winged Warfare: The Literature and Theory of Aerial Warfare in Britain 1859–1917* (University of Manchester Press, 1992).
8. For wide-ranging discussion of the legality and ethics of bombing, see, *inter alia*, Mark Clodfelter, 'Aiming to Break Will: America's World War II Bombing of German Morale and its Ramifications', *Journal of Strategic Studies*, 33(3) (June 2010), 401–35; Anthony C. Grayling, *Among the Dead Cities: The History and Moral Legacy of the WWII Bombing of Civilians in Germany and Japan* (London: Bloomsbury, 2007); Stephen A. Garrett, *Ethics and Airpower in World War II: The British Bombing of German Cities* (London: Palgrave Macmillan, 1993); W. G. Sebald, *On The Natural History of Destruction* (London: Hamish Hamilton, 2003); Peter R. Faber, 'The Ethical-Legal Dimensions of Strategic Bombing during WWII: An Admonition to Current Ethicists', Paper to Joint Services Conference on Professional Ethics XVII Washington, DC, 25–26 January 1996 (<http://isme.tamu.edu/JSCOPE96/faber96.html>); Ronald Schaffer, *Wings of Judgment: American Bombing in World War II* (Oxford University Press, 1985); Louis A. Manzo, 'Morality in War Fighting and Strategic Bombing in World War II', *Air Power History*, 39 (Fall 1992); Hermann Knell, *To Destroy a City: Strategic Bombing and Its Human Consequences in World War II* (Cambridge, MA: Da Capo Press, 2003); Matthew Evangelista and Henry Shue (eds.), *The American Way of Bombing: Changing Ethical and Legal Norms, from Flying Fortresses to Drones* (Ithaca and London: Cornell University Press, 2014).
9. See Richard Lock-Pullan, 'Redefining "Strategic Effect" in British Air Power Doctrine', *RAF Air Power Review*, 5(3) (Autumn 2002), esp. 60–3, for consideration of this (despite the title, the article is applicable to all air forces, particularly the USAF).
10. See Mueller, 'Air Power', pp. 6–8.

11. The National Archives (TNA), Air 1/718/29/1, 'Future Policy in the Air', 22 September 1916.
12. *Ibid.*
13. David Jordan, 'The Battle for the Skies: Sir Hugh Trenchard', in M. Hughes and M. Seligmann (eds.), *Leadership in Conflict 1914–1918* (Barnsley: Leo Cooper, 2000), p. 85.
14. There is voluminous literature on the fight for the survival of the Royal Air Force between 1919 and the early 1930s; H. Montgomery Hyde, *British Air Policy between the Wars, 1918–1939* (London: Heinemann, 1976); Malcolm Smith, *British Air Strategy Between the Wars* (Oxford University Press, 1984); Neville Parton, *The Evolution and Impact of Royal Air Force Doctrine, 1919–1939* (London: Continuum, 2013) all offer useful insights.
15. Field Manual 100–20, *Command and Employment of Air Power* (USAAF, 1943), p. 1:2.
16. *Ibid.* For a full discussion of the evolution of US doctrine in terms of the development and application of centralised command and decentralised execution, see Daniel R. Mortensen (ed.), *Air Power and Ground Armies* (Maxwell, AL: Air University Press, 1998), esp. David R. Mets, 'A Glider in the Propwash of the RAF?', pp. 30–63.
17. See Richard P. Hallion, *Strike from the Sky: A History of Battlefield Air Attack 1910–1945*, 2nd edn (Tuscaloosa: University of Alabama Press, 2010), p. 173.
18. Montgomery made this point, albeit sometimes with different wording, in a number of short pamphlets he issued during his command of British Eighth Army and later 21st Army Group; see, for instance, *Some Notes on High Command in War* (Tripoli, 1943).
19. David Syrett, 'The Tunisian Campaign', in Benjamin Franklin Cooling (ed.), *Case Studies in the Development of Close Air Support* (Washington: Air Force History Office, 1990), p. 173. Vincent Orange, *Coningham* (London: Methuen, 1990), pp. 181, 194–210.
20. Syrett, 'The Tunisian Campaign', p. 173.



11

The evolution of air and space power

Contents

The evolution of control of the air

Contemporary aspects of control of the air

Strategic air power

Theory, practice and controversy

Coercive air power

A loss of joint effort?

Joint air power in the nuclear age

Space power

The development of space power

Conclusions

KEY THEMES

- Air power has evolved rapidly, with enormous technological advances.
- Much early air power thinking was overly optimistic, placing too much faith in what could be achieved by aerial bombardment. Some proponents of air power claimed that air forces would swiftly make armies and navies irrelevant in future war.
- The state of technology meant that aerial bombardment lacked precision. This led to 'area' bombing of cities during the Second World War, raising profound moral, legal and ethical issues about targeting. Despite the development of precision weapons, concerns about targeting remain. The development of remotely piloted air systems and the way in which they are used has created further concern.
- The concentration upon bombing by leading proponents of air power often obscured the part air power played in other roles and in conjunction with land and maritime forces.
- The exploitation of space has had a profound effect on modern life. As a result, theories of space power have evolved, but international treaties have, to date, limited the prospect of out-of-atmosphere combat operations. The use of space for intelligence-gathering and navigation has grown exponentially.

The evolution of control of the air

Once the First World War began, it became clear that those commentators who had argued that fighting between aircraft would occur had been correct. Denying the enemy the ability to conduct aerial reconnaissance missions was desirable, and by the end of 1914, all the air services engaged in the war in France and Belgium had begun to realise this. The difficulty was how to achieve this. Aircraft were not routinely fitted with weapons, so pilots and observers armed themselves with pistols and rifles in an improvised attempt to bring down enemy aircraft. This proved to be ineffective, since aiming weapons from the cockpit of an aircraft was not an easy task.

The ability of aircraft to manoeuvre in three dimensions at some speed complicated aiming. The need to take account of the speed and angle through which the opposing aircraft was moving when engaged required whoever was firing the weapon in the aircraft to take account of such issues as deflection of aim and making necessary adjustments as the enemy aircraft manoeuvred. It soon became clear that the most effective way of attacking an enemy aircraft was with a machine gun. The rate of fire from this type of weapon increased the probability of obtaining hits on a manoeuvring enemy, but the best means of aiming the machine gun involved pointing the aircraft at the opponent. This, in turn, demanded that the gun be fitted to the centreline of the aircraft, but over-wing mountings that fired above the propeller were unwieldy, and mounting machine guns outside the propeller arc meant that they could not be attended to by the pilot if the weapons jammed, a common failing with First World War machine guns, which had not been designed to function in the freezing air encountered even at relatively low altitudes.

The French aviator Roland Garos fitted his aircraft with a machine gun on the centreline and overcame the problem of bullets shooting through the propeller blades by fitting them with deflector plates. Although successful, this was an interim solution. The development of interrupter gear by the Dutch designer Anthony Fokker and his team was a far more elegant answer, preventing the gun from firing when the propeller passed in front.

Fitted to one of Fokker's own aircraft designs used by the Germans, the Eindekker, the interrupter gear changed the dynamics of the air war over the Western Front. An Eindekker-equipped special fighting unit inflicted heavy casualties on the French and British to the extent that the period from June 1915 through to early 1916 became known as 'the Fokker Scourge'. The British Royal Flying Corps (RFC) found its operations badly hindered as a result of the threat posed by the Fokkers. This demonstrated one of the effects of gaining some form of control of the air, namely that of compromising the ability of the opponent to conduct aerial missions. New British and French designs, all using their own interrupter gear after mid 1916, tipped the balance in favour of the Allies again – but the period of the Eindekker's dominance marked the point at which control of the air could be thought of as a proper dynamic contest with

implications for the success or failure of air operations, and potentially of those operations on the ground that aircraft were trying to support.

The early experiences of the First World War have been replicated ever since. Without establishing some form of control of the air, the operations of surface forces become ever more difficult. The Second World War further demonstrated the point. In the opening rounds of the war, the Luftwaffe enjoyed considerable success against Allied forces, and their role contributed significantly to both the attack in the west and, in June 1941, the invasion of the Soviet Union. However, there were already warning signs for the Germans that they could not guarantee that they would maintain control of the air: in 1940 at Dunkirk, the RAF managed to wrest some degree of control from the Germans, which prevented full-scale attacks on the evacuation beaches and the ships that were withdrawing the British Expeditionary Force and parts of the French Army from the encircled port.

The subsequent Battle of Britain ended in failure for the Luftwaffe. Unable to defeat a well-organised force supported with radar, the Luftwaffe was unable to establish control of the air, preventing an effective coercive bombing campaign to bring the British to terms, and denying what had been set as an essential precondition for an invasion (which, thanks to the Royal Navy, would have been a spectacularly difficult and probably disastrous undertaking had it been mounted).

The Germans themselves realised the dangers of losing control of the air over one's homeland in the face of the Allied strategic air offensive. The ever-confident head of the Luftwaffe, Hermann Goering, once told the German public that no enemy aircraft would fly over the Reich, but this was soon proved incorrect. While the British bomber offensive had moved to night operations to reduce losses to fighters, the American philosophy of attempting highly accurate bombing required daytime operations, leading to large-scale air battles. The bombers suffered heavy losses, but from late 1943, there was a shift in emphasis which saw the bombing raids being used to wear down the Luftwaffe fighter force as well as to inflict damage on key targets. The Germans could not ignore the challenge posed by the escorted bomber formations and were forced to engage. This meant that the focus of German fighter efforts was over Germany itself as far as fighting the Western Allies was concerned. This militated against the deployment of large numbers of fighter aircraft to the occupied territories in Western Europe, and the pressure imposed by fighting against the numerically superior Soviet air forces in the East meant that the Luftwaffe was unable effectively to oppose Anglo-American air operations in North-west Europe from June 1944 onwards.

Control of the air was also key to the Pacific War. Initial Japanese successes were achieved thanks to the destruction of American, British and Commonwealth air assets, notably in Philippines, Malaya and Singapore. This success was transitory. The United States' ability to produce vast numbers of aircraft and the men to fly them eventually overwhelmed the Japanese air forces. The battle of Midway cost the Japanese numerous experienced pilots and their ability to



Figure 11.1 The ultimate in strategic bombardment during the Second World War, the B-29 Superfortress was the world's first nuclear-capable bomber and the backbone of the American bomber fleet in the immediate post-war era.

challenge American carrier-based air power. American carrier aircraft inflicted heavy and ongoing attrition upon the Japanese, as did long-range USAAF fighters. The nadir for the Japanese came in June 1944 with the American assault on the Marianas Islands, in which the Japanese Army and Naval Air Forces suffered crushing losses in what became known as the 'Great Marianas Turkey Shoot'. The inability to provide effective air support to the army from this point onwards meant that the Japanese were placed at a grave disadvantage during the last year of the war. Allied aircraft, by contrast, became particularly important in adding to the striking power of land forces, while gaining control of the air enabled the use of transport aircraft to support advancing Allied forces, particularly in Burma, where General Sir William Slim commented that the ability to resupply troops in the jungle had been a critical factor in the defeat of the Japanese in this theatre.

Once the Americans had captured island bases from which their B-29 bomber force could attack the Japanese home islands, the importance of defending the homeland from air attack was brought home to the Japanese leadership, just as it had been to the Nazi regime in Germany (see [Figure 11.1](#)).

Although the Japanese made a robust defence of their territory, they could not stop the raids. Consequently, the B-29s inflicted massive damage. The ultimate demonstration of what could occur with the loss of control of the air over the homeland came in August 1945, when the Americans delivered two atomic weapons against Hiroshima and Nagasaki: both attacks occurred without the interference of Japanese aircraft. While the importance of control of the air had been established beyond any doubt by 1945, the development of nuclear weapons had an interesting effect upon conceptions of air superiority during the Cold War. Initially, it was clear that the only means of delivering nuclear weapons was from aircraft. This prompted both sides in the Cold War to invest in their air defence assets. In the United States, this led towards a concentration upon fighter aircraft that could engage the enemy at long range using guided missiles, and the traditional fighter aircraft, designed to win control of the air over the battlefield, fell out of favour. The limitations of such a policy were seen in Vietnam and, most notably, in the Middle East. The success of the Israeli pre-emptive strike on Egyptian airfields in the 1967 Six Day War and the subsequent freedom of operational manoeuvre this granted the Israeli Defence Force (IDF) highlighted the fact that being able to fight and win against enemy fighters that were designed for combat with similar types remained as important as it had always been. The Israelis went on to demonstrate this again in 1982 in their invasion of Lebanon, ensuring that they destroyed the Syrian integrated air defence system in the Bekka Valley as a precursor to operations.

Contemporary aspects of control of the air

Despite the constant reminders of the importance of maintaining control of the air, recent conflicts have led to a debate about the concept. The success of coalition operations against the Iraqi Air Force in 1991 highlighted the possibility that lesser powers might adopt an asymmetric approach towards control of the air. Asymmetric responses were not new. The North Vietnamese used surface-to-air missiles (SAMs) and anti-aircraft artillery (AAA) extensively, and the 1973 Arab–Israeli Yom Kippur/Ramadan War had a clear strand of asymmetric thinking by the Egyptians, who relied upon SAMs against the Israelis to offset enemy air superiority.¹ The Israelis seriously damaged the Egyptian Army once it moved beyond the protection of its SAM coverage, but the lesson here was that the nature of control of the air had changed: the enemy would no longer seek to engage only in attritional air battles but would use a variety of different techniques to achieve the desired aim. The examples of making a limited air effort against the enemy before falling back on the use of SAMs reinforced the importance of another dimension to control of the air, namely suppression of enemy air defences (SEAD). The Americans made a start in Vietnam with ‘Wild Weasel’ fighter-bombers fitted with additional electronic equipment and anti-radar missiles duelling with SAM sites in a bid to protect strike packages.²

The change was further shown during Operation Allied Force, NATO's response to the Kosovo crisis in 1999. The Yugoslav Air Force opposed the NATO air attack, but their losses suggested that the flying would only bring about the destruction of most Yugoslav air assets. This saw the burden of the air defence effort switch to a small, but capable, SAM force, which was never fully suppressed. In 2003, the Iraqi Air Force chose not to fly at all in the face of the invasion by the US-led coalition. Operation Enduring Freedom, the response to the terrorist attacks on New York on 11 September 2001 (9/11), was conducted against a state (Afghanistan) which had no credible air force at all. These experiences have given rise to a debate amongst some commentators (mainly in Western nations) as to whether control of the air remains as important as it once did; this is discussed further in the [next chapter](#).

Strategic air power

The promise of a new arm bringing about a rapid victory proved enticing in the interwar era. The huge losses caused by the static warfare between 1914 and 1918 led to a determination that the experience should not be repeated, and this in turn led to the notion of exploiting the potential reach of bombers to carry the war to the enemy's homeland and in effect bombing the enemy population into a rapid surrender. The degree of destruction envisaged from bombing (which the Second World War showed to be wildly optimistic in terms of both the precision which could be achieved and the effect on civilian morale) also served as a deterrent.³ Three key theorists – who might equally be termed visionaries – led the way in this school of thought.

The Italian theorist Giulio Douhet's publication *Command of the Air* demonstrated his view that it was necessary to control the air, but his vision relied not upon the fighters of the First World War but upon heavily armed 'battleplanes'. Douhet argued that bombing population centres would shatter the enemy's will, while contending that it was impossible to defend effectively against attacking aircraft. Control of the air would be achieved by attacking logistic and manufacturing centres for enemy air power, enabling subsequent attacks on population centres. 'Battleplanes', capable of defending themselves against interception, would destroy the opposing air force's fighting power at the outset of war, preventing an enemy counter-stroke. Possessing a sufficiently large arsenal of 'battleplanes' would obviate the requirement for fighter aircraft to defend the homeland against enemy attack. The employment of an offensive doctrine, then, would bring about the enemy's utter defeat, particularly through undermining enemy social cohesion by bombing the civilian population.

Douhet's apocalyptic – and profoundly legally questionable – vision was flawed. He assumed every aeroplane reached its target and that every bomb dropped hit the target and exploded. Given the technological standards of the time, this was an unrealistic view of air power's capabilities.

In contrast to Douhet, Trenchard felt that directly targeting civilians was unjustifiable. Undermining enemy morale came from directly targeting not the people, but 'vital centres,' such as the factories that employed them and the transport network that took them to and from work. Trenchard's tenure as Chief of the Air Staff saw the production of several doctrine manuals, the most important of which was Air Publication (AP)1300, *Royal Air Force War Manual*. AP1300 suggested: 'Objectives should be selected the bombardment of which will have the greatest effect in weakening the enemy resistance and his power to continue war.'⁴ The manual noted that it was important there was a full understanding of the enemy country.⁵ The inference was clear. Without such knowledge, identifying the key targets for attack would be difficult. This raised a difficulty: it might be that the British conception of what was an enemy 'vital centre' was not the same as that of the opponent, and striking something that the enemy did not value would not be effective. This applied, of course, not only to the British, but to all air forces – could they be sure that they were bombing the right targets to affect enemy morale?

This fundamental point remains critical today. Intelligence and deep cultural understanding of the enemy may be difficult to obtain, but it is vital if bombardment is to have any effect at the strategic level of war.

Trenchard's outlook on air power was dominated by considerations of the offensive; he conceded that it was necessary to retain a small fleet of air defence aircraft largely to maintain the morale of the civilian population, even if the majority of funding for the RAF should be spent on bomber aircraft. Over-investment in fighters would be:

[R]ather like putting two teams to play each other at football and telling one team they must only defend their own goal . . . the defending team would certainly not be beaten, but they would certainly not win, nor would they stop the attack on their goal from continuing.⁶

This was entirely in keeping with the notion that air power could not have a truly strategic effect if used purely for defensive purposes. This was wrong, particularly once defences could use radar to gain early warning to launch fighter aircraft against incoming attacks – as demonstrated in the Battle of Britain.

The debate was no less heated in the United States where Colonel William 'Billy' Mitchell and later the US Army Air Corps (USAAC) Tactical School took forward US air doctrine. Following the First World War, 'pursuit' [fighter] aviation was seen as the main effort of the USAAC, since its duty was to maintain air superiority. From the mid 1920s, though, the USAAC saw bombardment as its most important task. Pursuit aviation remained significant, being employed to escort friendly bombers or to carry out sweeps against the enemy fighter force. By the early 1930s, the position had changed yet again. The development of high-speed bombers convinced many USAAC officers that bombers would be almost impossible to intercept in a future conflict. Advocates of pursuit aviation, notably Captain Claire L. Chennault, felt that this faith in the bomber was misplaced, but were ignored.⁷

The USAAC then developed a concept of targeting key elements in the enemy's war economy. The precision needed to do this would be delivered by sophisticated bombsights, while careful understanding of the enemy would allow a list of targets to be drawn up. This was the 'key-node' approach through which the USAAC (later renamed the US Army Air Force) developed thoughts about strategic air power. Rather like Trenchard, the USAAC's bomber advocates saw the erosion of enemy civilian morale stemming from the attack against vital centres and, combined with physical destruction, this would bring rapid victory. When war came again in 1939, 'strategic' air power did not deliver in the way in which the interwar theorists had envisioned.

Theory, practice and controversy

The nature of air bombardment in the Second World War has been the source of much controversy. The extensive historiography has produced much debate, which can obscure some of the key issues when it comes to understanding air power; certainly, it can suggest that strategic air power is only that delivered by bombers. Although there are dangers in compressing the analysis of such a voluminous corpus of literature into a few short sentences, some key conclusions regarding the strategic bombing campaign between 1939 and 1945 can be made. First, the *nature* of the Second World War did not meet theorists' expectations. Surface forces were heavily engaged from the outset, while the financial constraints of the interwar years meant that the British and American bomber fleets needed to conduct the sort of bombing campaign envisaged did not exist. Secondly, capability was lacking – the early years of the war were marked by a lack of navigational and bombing accuracy, driving the RAF to an 'area' policy of attacking German cities, increasing the chance of striking industrial targets; the corollary of this, of course, was that no discrimination could be made between civilian housing and industrial areas. Despite the introduction of navigation devices and improved bombsights, the commander of RAF Bomber Command, Sir Arthur Harris, remained convinced that area bombing would bring success, refusing to change target sets. Bomber Command *did* strike targets other than cities in the latter part of the war, but the Air Staff found it extremely difficult to persuade Harris to launch his forces against these alternative targets.

The US Army Air Force's daylight bombing policy gave some precision, but despite the sophistication of the Norden bombsight, the degree of accuracy desired could not be achieved. There were several reasons for this, including the depredations of German fighters, anti-aircraft fire, and the weather, which meant that it was often impossible to see the targets through heavy cloud. Bombers had to release their ordnance using radar targeting, which was inaccurate, making the realisation of key-node targeting theory impossible.

Weather conditions in the Pacific were better, but the dispersal of Japanese industry was such that firebombing of area targets – with enormous civilian casualties – resulted; the bombing concluded with the use of two nuclear weapons, which could hardly be said to fit in with the notion of precise targeting

of key nodes. The US Army Air Force remained committed to the notion that it should seek to target key elements of the enemy's war-making capability, exemplified by the dispute in 1944 over the employment of the bomber forces in the run-up to the invasion of Europe. While General Eisenhower, as Supreme Allied Commander, supported the concept of targeting German transportation put forward by several senior airmen, General Carl Spaatz, commander of the American bombing forces, argued vehemently for attacking oil production, knowing that the Germans had always been painfully short of oil. Spaatz was overruled, with the proviso that oil might become a major target once Allied forces were established ashore. The end result was to intensify the debate regarding 'strategic' air power.

The failure of the bomber offensive to achieve what had been promised by the interwar theorists damaged the credibility of air power in the eyes of many observers, who felt that their view that air power was better used in direct support of ground and naval forces had been vindicated. However, the atomic bomb meant that conceptions of air power operating in a strategic role had to be rethought. The Americans and Soviets established large forces of bomber aircraft to carry nuclear weapons, while the British and French developed smaller bomber forces to carry their own nuclear deterrent.

The appearance of nuclear weapons was particularly important in the United States, since it helped convince the Truman administration that while the US Army should still have its own air power, the bulk of American combat aircraft should be under the auspices of an independent air force, and so the US Air Force was created in 1947. The US Air Force remained wedded to the notion that air power could win wars alone, and this perspective was articulated through the refinement of two concepts that defined the nature of considering strategic air power, but which remain controversial. Building upon theories pertaining to nuclear strategy, air power thinkers developed the concept of employing air power for coercive purposes; this was then followed by attempting to take the oft-expressed idea that strategic air power had 'effect' and defining what, exactly, constituted effect at the strategic level, and the means by which it could be achieved. We shall return to the issue of coercion shortly.

One of the other problems which arose from the Second World War lay in the fact that the use of the term 'strategic' came to be associated with bombing, but this was a misnomer. Air power showed that it could have effect at the strategic level without a large-scale bombing offensive. The battle of the Atlantic was of great importance to Britain, as it was reliant upon shipping to bring in vital supplies; Churchill would observe that the threat from German submarines was the one thing which truly worried him during the war. The efforts of RAF Coastal Command and US naval aviation, in conjunction with surface units escorting convoys, defeated the U-boat threat; this, surely, was the use of air power to achieve strategic ends? It came in conjunction with surface forces, but could not be said to have been a case where air power was somehow subordinate to the surface units, not least since the maritime patrol aircraft were used for far more

than carrying out convoy escort. Some bombing of targets related to the U-boat force was also carried out, and if we piece together the events of the battle of the Atlantic, we begin to see that there is a danger in 'stove-piping' considerations of air power; attempting to define air power by roles such as bombing or air-fighting ignores the overlap which occurs between these functions, and they cannot be easily pigeon-holed into categories such as 'independent' or 'subordinate' action. The whole point was that air power was used as a critical tool to achieve a vital strategic end-state. Some air operations may have been independent of Allied naval units, but they were never independent of the strategic purpose for which they were applied.

An additional illustration of air power's strategic effect beyond simply bombing came in 1948 with the Berlin Airlift, in which Soviet aspirations to bring about revisions in the control of Berlin through blockade were thwarted by the massive airlift conducted by the Western allies to sustain the city. We might also cite the use of reconnaissance over Cuba in 1962, revealing the construction of Soviet missile sites on the island and setting off the chain of events which brought about the removal of these weapons as being an example of air power achieving a strategically significant role without the use of weapons. Further demonstrating the overlap of roles, however, is the fact that in both 1948 and 1962, the intent of the Western allies was underpinned by the possession of credible deterrent capabilities, giving us cause to think about the coercive effect of air power.

Coercive air power

Coercion is a complex and multi-layered concept (as described in chapter 3), but Robert A. Pape – drawing upon Thomas Schelling and Robert J. Art – provides a good definition:

Coercion is like deterrence in that both focus on influencing the adversary's decision calculus. However, coercion differs from deterrence in that deterrence is concerned with maintaining the status quo by discouraging an opponent from changing his behaviour, while coercion seeks exactly the opposite: to force change in the opponent's behaviour.⁸

This was demonstrated in the attempts by the Johnson administration in the 1960s to apply air power against North Vietnam, with the intent of coercing the Hanoi government into ending its support of the insurgency in South Vietnam conducted by the National Liberation Front (more commonly known as the Viet Cong) (see Figure 11.2). However, the attempt at coercion, an aerial bombing campaign – Operation 'Rolling Thunder' – failed. The North Vietnamese remained resolute in their defiance of American efforts. There were several reasons for this.

The American approach applied air power in an incremental fashion, with the threat of greater destruction in subsequent attacks being seen as the means



Figure 11.2 F-105D Thunderchief aircraft during a mission over Vietnam. The Thunderchief bore the brunt of the USAF's air efforts over North Vietnam during Operation Rolling Thunder, suffering heavy losses.

to make the North Vietnamese modify their behaviour.⁹ This ignored the determination of the government in Hanoi and the insurgents in the south of the country to bring about the unification of their nation. They were quite prepared to wait until the United States tired of the conflict, which in turn meant that Vietnamese resilience was greater than anticipated.

The Americans also failed to take account of the nature of the conflict: notions of strategic air power were based upon attacks against significant military, political or industrial targets, yet the first phase of American involvement in the conflict was in a guerrilla war, in which the key targets were those against which air power could not be employed effectively. Both military and civilian leaders in the United States thought that targeting North Vietnamese industry would prove an effective means of coercion, failing to appreciate that the only vital target was the transportation system, which itself could sustain considerable damage without having a major effect on supplies to forces in South Vietnam.

Constraints on targeting, inspired by a fear of Soviet or Chinese intervention further hampered Rolling Thunder, and it came to a dismal close in 1968. Four

years later, coercive air power had more effect. This came when the Americans launched Operation Linebacker, a much less restrained employment of bombing against targets throughout the North in response to Hanoi's 1972 Easter Offensive into the South. The culmination of Linebacker II – a short, brief assault against key targets in December 1972 – came with the signing of the Paris Peace Accords enabling the United States to disengage from the war. Air power proponents contend that Linebacker demonstrated that if air power was employed against key targets without the political constraints and gradual escalation seen in Rolling Thunder, then it could be decisive; indeed, they argued, if Rolling Thunder had been conducted in the same manner, the result of the conflict might have been different. The alternative perspective contends that this ignores the fact that the war had changed by 1972, since the Tet Offensive in 1968 saw enormous losses amongst the Viet Cong, forcing Hanoi to embark upon a more conventional form of warfare.

This, in turn, made them vulnerable to air power, as demonstrated by the telling effect of air attack on their forces during the invasion of the South during spring 1972. Targets chosen for attack in the Linebacker campaigns were important to the sustenance of a conventional army and had a notable effect. Mining Haiphong harbour ensured that the North Vietnamese Army began to run short of supplies.¹⁰

The debate over the nature of the employment of air power in Vietnam highlights several issues. The first is the long-understood need to know the enemy, so as to target effectively. The second is public opinion, both national and international. A key legacy of the Second World War was awareness of air attack's destructive capacity, creating aversion within Western societies towards anything that had the appearance of targeting civilians. Allegations that American bombing was wilfully targeting civilians not only gave the USSR a propaganda weapon but undermined public support for the war in the United States.

The air campaign in Vietnam led to more thought about air power. The Cold War paradigm in which 'strategic' meant the use of nuclear weapons began to unravel as the USAF looked at how it operated.

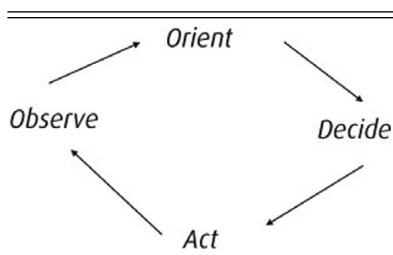
The introduction of AirLand battle doctrine during the mid 1970s led to a brief period during which the view that air power was simply a supporting arm to surface forces returned; this did not last long, as new perceptions of strategic air power developed. The new thinking drew heavily upon ideas postulated by a retired US Air Force officer, Colonel John Boyd, and taken to a new level by Colonel John A. Warden III (see [Box 11.1](#)). Warden, in particular, has had a significant influence on air power thinking because of his contribution to the way in which air power has been used after the Cold War. This is discussed in more detail in the [next chapter](#).

As noted earlier, air power theorists have been loath to concentrate upon the utility of air power in support of the surface battle, often for fear of the implications that this might be perceived to have on the validity of the existence of independent air forces. This has led to practical problems, air forces and

Box 11.1 John Boyd and John A. Warden III

Boyd's thought covered a wide range of subjects, and his major contribution to the development of strategic air doctrine was not designed with air power alone in mind. Boyd developed the notion that conflict could be won through 'decision cycle dominance'. The decision cycle laid out by Boyd of 'Observe – Orient – Decide – Act' became known as the 'OODA loop', the aim of decision cycle dominance being to 'get inside the opponent's OODA loop' (see [Table 11.1](#)).

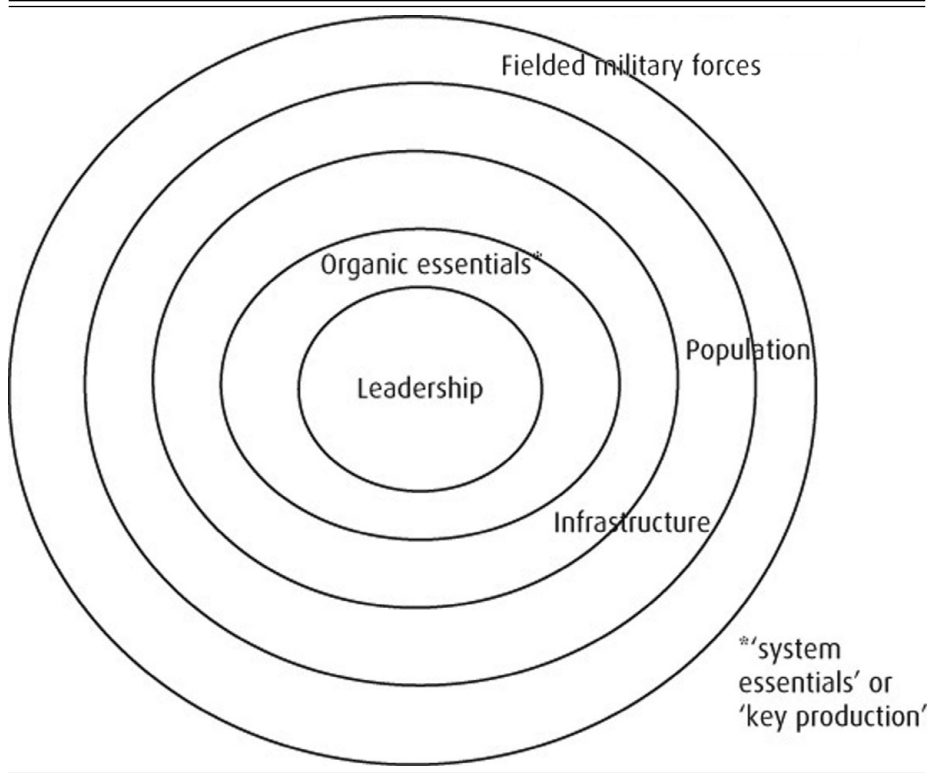
Table 11.1 *Boyd's OODA loop*



Conducting the decision cycle more quickly than the adversary meant that the opponent would lose situational awareness, becoming more confused as to what was being done to him; as confusion mounted, the enemy would become paralysed and unable to resist. Warden built upon these ideas during the 1980s and was presented with an opportunity to articulate them following the Iraqi invasion of Kuwait in August 1990. Warden's concept of strategic air power envisaged the use of air forces to attack targets vital to the enemy's ability to prosecute war without having to engage the enemy army, a concept that Mitchell and Douhet would have recognised. Warden did not consider the enemy's fielded forces completely irrelevant, since they formed one of the five key target sets in his conception. The target sets were represented visually by a series of concentric rings. Leadership lay in the centre of the rings, moving outward through key production, infrastructure and the population to the outer ring, military forces. This representation demonstrated Warden's view that the enemy state could be regarded as a system – or a 'system of systems' – all aspects of which were equally vulnerable to attack by air power at any time (see [Table 11.2](#)).

While Warden's vision might at first sight appear to be a simple refinement of earlier thinking, his concept of air power was enabled by technological developments. The visions of destruction held by the early theorists had fallen foul of the apparently insuperable problems of accurate weapons delivery and navigation to the target, particularly in poor weather. The problems of navigation and poor-weather flight were substantially overcome by technological development by the 1960s, but despite much-improved bombsight technology, striking targets remained problematic. The solution came through guided weapons. Their development began in the 1920s, and there had been limited use of guided bombs during the Second World War, although they had not enjoyed a high success rate. Further work in the United States led to the creation of the Paveway series of laser-guided bombs and electro-optically guided ordnance. These demonstrated their worth in the Vietnam War, notably in attacking bridges in North

Table 11.2 Warden's 'five ring' model



Vietnam. By the late 1980s, the range of precision-guided munitions (PGMs) available meant that it was possible for such theorists as Warden to have confidence in the ability of a relatively small number of aircraft not only to locate their targets but also to be able to attack them with a high probability of success. With the need to send large bomber fleets against individual key targets obviated by technological advances, a range of targets could be attacked simultaneously, giving air power the ability to cause the strategic paralysis that Boyd and Warden advocated. The 'parallel warfare' that simultaneity of targeting through the use of PGMs permitted appeared to offer the tantalising prospect that the long-held vision of air power bringing about swift victory without the need to engage hostile fielded forces might be realised.

armies almost wilfully choosing not to co-operate as closely as they might (while complaining about the attitude of the other service) when the two services have been required to co-operate in wartime.

Nevertheless, air power has been of critical importance when used alongside both armies and navies almost from the outset of the creation of organised air arms. In the First World War, even with the limited aviation technology at

hand, air power played a significant supporting role. As well as reconnaissance and artillery spotting – fundamental aspects of a static war where heavy guns dominated the battlefield – the French, British and Germans made growing use of aircraft to provide support to their ground forces, notably at the battle of Cambrai in 1917 and then in the German Spring Offensives of 1918 and the subsequent battles of ‘The Hundred Days’ which brought the war to an end.

A loss of joint effort?

The desire of airmen to stress the importance of their contribution to modern warfare, coupled with the creation of independent air forces (and the need to maintain them in financially constrained times in the face of arguments that air power was solely subordinate to armies and navies) meant that there has always been a tendency for air force leaders and air power proponents to downplay those roles which may be seen as ‘supporting’ and thus – in theory – which could be conducted by air services controlled by a navy or an army.

This approach, while usefully highlighting the fact that the flexibility, range and versatility of air power can be badly hindered if constrained in its use, and understandable in the face of hostility towards air power from elements of the other components, has been unhelpful, and ignores what has, in fact, occurred.

During the interwar period, the RAF, despite its published doctrine, expended much effort in colonial policing, working in conjunction with ground forces. Although the ground units were often local troops rather than British Army formations garrisoning territory, the fact that ‘boots on the ground’ – to employ a familiar, if over-used phrase – mattered in such operations was recognised by the RAF, particularly when the policing being conducted was of an urban area.¹¹ The US Marine Corps made much use of air–land co-operation during its interwar operations, although given the nature of the USMC as a potentially self-contained force with its own supporting arms, this was only to be expected. The use of air power in these situations also saw the effective use of aircraft in transport and liaison roles, all of which contributed to the effect that was achieved. This continued as a key theme after 1945. While the ground component was of key importance in a counterinsurgency, the role of air power was significant, if sometimes undersold.¹²

The use of transport aircraft and helicopters – as the technical capabilities of these machines increased – was of considerable importance, particularly in such campaigns as those fought in Malaya (in the 1950s) and Algeria (in the 1960s). The value of aircraft for transport has remained constant, particularly in fighting insurgencies in areas with limited communications – as seen in the importance placed on such activity by the Soviets and then the anti-al-Qaeda/Taliban coalition in Afghanistan, with air transport used for the swift deployment of troops to areas almost inaccessible by land.

Aerial reconnaissance was particularly valuable for the French when fighting the Front de Libération Nationale (FLN) in the Algerian countryside, but

once the insurgency moved into the cities its value lessened as the insurgents exploited the cover provided by an urban environment – exactly the same issue which had affected the RAF in Palestine in the 1920s and 1930s.

It was not just in terms of COIN where considerations of ‘independent’ air power did not apply across the board. The French air service remained dedicated to supporting the land component. The Soviets saw aeroplanes as an integral part of the land battle, notions of independent air action being replaced by the use of air power for deep attack. The Germans, assessing the reasons for their defeat in 1918, concluded that it was impossible for land forces to function against a serious opponent without air power and developed doctrine placing great emphasis on supporting ground forces, even if suggestions that they abandoned notions of strategic bombardment entirely are inaccurate – the German approach was to achieve effects, which thus involved gaining control of the air, interdiction of supplies and attacks on war production as well as serving as ‘aerial artillery’ to support the troops on the ground.

This proved one of the keys to German success in the early part of the Second World War. The failure of the French and British to develop an effective counter to this, not least because of an inability to gain control of the air, led to disaster in 1940. Complaints by the British Army regarding the inadequacy of air support grew louder in the face of defeats by the German Afrika Korps in the desert campaigns, but by this point steps were in hand to overcome the difficulties. The RAF, largely through an ad hoc approach by local commanders, rather than as a result of centrally formulated doctrine generated by the recently formed Army Co-Operation Command, fell back on the lessons learned in 1914–18. The quality of air–land integration improved immeasurably as close liaison was developed between air and ground formations at all levels, most notably between senior commanders. Similar improvements occurred as the US Army Air Force developed and refined its own doctrine for air support. The Afrika Korps commander, Erwin Rommel, complained that his forces could not compete in the face of co-ordinated action by enemy air and land forces. Lessons from North Africa were taken to the Italian campaign, and then to Western Europe after D-Day.

The system of air–land support was particularly significant, if not perfect. Heavy bombers, employed above the battlefield, did not always have the desired effect, and often caused ‘friendly fire’ incidents.¹³ Yet air power was effective. Air attack dented the morale of German soldiers, while the Allies’ control of the air reduced the number of German air attacks on their own forces. The campaign to interdict German supplies was highly effective, thus denying the Germans the ability to respond as rapidly to the Normandy invasion as might otherwise have been the case, while attempted counter-attacks were often brought to a standstill before the necessary units had reached their starting-point for the attack.

A similar situation pertained on the Eastern Front, where the Soviets used air power on a massive scale to support land operations. Careful co-ordination of air attacks in support of the objectives of the various Red Army fronts as they

advanced meant that a combination of firepower from both land and air assets, combined with sheer weight of numbers, first rebuffed the Germans and then forced them into retreat.

In the Far East, the use of transport aircraft was critical to success in the China–Burma–India (CBI) theatre of operations. The lack of communications infrastructure in many of the locations in which campaigning was conducted meant that aerial resupply was the only means by which rapid transport of men and supplies could be guaranteed, particularly given the considerable geographical area that the CBI theatre covered. Air supply was essential during the battles of Kohima and Imphal, for although the Japanese successfully cut off the British positions there, the regular delivery of supplies from the air and evacuation of casualties meant that controlling the land communications links was not as significant as it might have been; in the absence of control of the air, the Japanese were unable to prevent supplies from getting through. The commander of the British XIVth Army, Viscount Slim, observed that the victory in Burma belonged as much to the air forces involved as to the troops who had fought on the ground.¹⁴

In the maritime environment, the importance of combined air–maritime interaction was demonstrated on numerous occasions, in addition to the battle of the Atlantic. The Royal Navy's air attack on Taranto in 1940 played a significant part in the maritime campaign in the Mediterranean. Conversely, the Japanese attack on Pearl Harbor in December 1941 failed to destroy American aircraft carriers, and this gave the Americans the opportunity to strike a devastating blow six months later at Midway. The Americans went on to employ their maritime air power to telling effect in the advance on the Japanese home islands, notably in the destruction of the last real vestiges of Japanese air power during the Marianas campaign in June 1944 and in the provision of close air support to invasion forces as the 'island-hopping' campaign progressed.

Joint air power in the nuclear age

Although the post-war period was dominated by nuclear weapons and the creation of the strategic air forces that would employ them in a Third World War, air–surface co-operation remained an essential part of modern warfare. The Korean campaign (1950–3) was notable for the 'hard sloggling' of the troops on the ground, reliant upon air support to contain the North Koreans and then to offset the numerical superiority of Chinese forces. Marshal of the Royal Air Force Sir John Slessor (Chief of the Air Staff 1950–2) observed that there would be 'more Koreas', implying that visions of air power winning wars swiftly and independently were misplaced.¹⁵ The Vietnam War strengthened the notion that air power was inextricably linked with land and maritime power, a combination of American air-mobile tactics, aerial resupply and close air support by US Air Force, US Naval and US Marine Corps aircraft playing a critical part in the conflict, even if historians later tended to concentrate upon Rolling Thunder

and Linebacker as exemplars of the air war. The Israeli employment of air power in support of land forces was a significant – although sometimes overstated – factor in their successes in 1967 and, after heavy casualties, in 1973. In both wars, air power delivered considerable support to the Israeli army, both in the form of protection from air attack, and through the effects of close air support and interdiction.¹⁶ The original function of air power, namely the gathering of information, has remained critical throughout – and in an era where precise, proportionate targeting is required, information has become a vital area. The gathering of information has been transformed by the rise of space-based platforms, and it is now necessary to consider the evolution of this critical component of air and space power.

Space power

Space power is still a relatively new form of military power with its origins in the late 1950s, yet it rapidly became of great significance in modern war and to modern life as a whole. Furthermore, its importance to operations by major powers has given rise to a ready stream of doctrinal thinking as to how space power may be exploited in the future. There is some debate as to whether space power should be considered an adjunct of air power or whether it should be regarded as a completely separate entity altogether. Attempting to produce a universal definition of space power is difficult but the following premise put forward by David E. Lupton in his book *On Space Warfare: A Spacepower Doctrine* is amongst the best: ‘Spacepower is the ability of a nation to exploit the space environment in pursuit of national goals and purposes and includes the entire astronautical capabilities of the nation.’¹⁷

Like air power, space offers the characteristics of height, speed and reach discussed in the [previous chapter](#), although at different orders of magnitude. Conventional aircraft operating at up to 20 miles above the earth’s surface enjoy a considerable vantage point, which pales into insignificance in comparison with the field of vision from 100 miles above the earth. The speed at which space-based assets move is far greater than that of conventional aircraft. Depending on a space platform’s orbit and/or propulsion system, it is possible for it to travel around the world several times in twenty-four hours, providing regular coverage of an area of interest, or for the space asset to be moved above such an area from its extant orbit. Space power has become an increasingly important aspect of military planning and operations, and the potential of the operating environment is considerable. It does, perhaps, suffer from problems of perception, since notions of military operations from space seem to be almost inevitably coloured by perceptions drawn from the *Star Trek* television series (and associated franchises) and George Lucas’s *Star Wars* series of films. This perhaps obscures the fact that space technology, while not as advanced as portrayed on screen, is evolving rapidly and cannot be divorced from the military operations conducted by major industrialised powers today.

The development of space power

Space power's origins lay very much in the Cold War, the so-called 'space race' between the United States and the Soviet Union being a display of the two nations' rivalry as each sought to demonstrate the superiority of its technology – and perhaps subliminally the supremacy of the political system that had developed it. The success of the Soviets in placing the satellite Sputnik in orbit in 1957 caused uproar in the United States, the public fearing that the Soviet Union might be able to place nuclear weapons in space, rendering America vulnerable to attack. While such visions were far-fetched at the time, they inspired vast investment in the American space programme, the 'race' culminating with the landing of Apollo 11 on the moon in July 1969. The manned space programmes obscured major developments in satellite technology, much of which was of considerable utility to the armed services of both major powers and their allies.

After Sputnik, the Americans developed similar technology. From the early 1960s, both the Soviets and the Americans placed satellites in orbit to obtain information about their adversary. For the Americans, this was particularly desirable, since the use of a space platform meant that it was possible to obtain photographic evidence of military installations within the Soviet Union without recourse to dangerous (and potentially embarrassing) over-flights by manned aircraft. Although scientists and engineers recognised that satellites could be placed in a geostationary orbit (that is to say, remaining over a fixed point, rotating at the same speed as the earth), early satellites could not provide the twenty-four-hour coverage given by this approach, because they recorded their information on film cartridges which had to be jettisoned over friendly territory for subsequent analysis by intelligence services. This demanded a geosynchronous orbit pattern, the satellite circling the earth frequently and giving coverage of a wide area of the earth's surface as it did so. As satellite technology improved, it became possible to use recording devices which transmitted their information to an operating centre, where the images could be interpreted and disseminated to the agencies that required them. Throughout the Cold War, increasingly sophisticated reconnaissance satellites were developed. These employed a range of imaging capabilities, including radar, infra-red and photography, while the resolution and clarity of images obtained became almost legendary.

However, while the satellites provided large amounts of information about the disposition of opposing forces, such information was carefully controlled. The information gathered was of relatively little significance to the wider military, since it was highly classified and disseminated to a relatively small number of agencies, with the aim of supporting the deterrence strategy in place between the two rival superpower blocs. This meant that the wider military application of space power was relatively limited in scope, if not in scale. During the Vietnam War, satellites were predominantly used for the purpose of weather

forecasting and communications, with some navigational information transmitted in addition. Although the utility of space platforms became widely recognised, the high-security classifications applied to their use meant that the potential was not fully understood, nor was it integrated effectively into the wider military environment. By the late 1980s, the United States held a commanding lead in military space power – but much of the American military remained unaware of the capabilities that were on offer. The American interventions in Grenada (1983) and Panama (1989) saw the exploitation of information from satellites being used to inform military operations, even though the high levels of classification pertaining to the intelligence product obtained from space again meant that there were difficulties in ensuring that information gained from space assets was provided to those who needed to know about it. Having been a product of the Cold War, space power has become even more significant in its contribution to warfare (see [Figure 11.3](#)), and this is explored in more depth below.

Conclusions

Air power evolved rapidly during the First World War, and the rate of development in terms of the application of technology to aircraft and their systems has not slowed since. From the late 1950s, the exploitation of space became an important element in military power, leading to the rise of a number of theories and concepts as to how to exploit this domain for military use, albeit within the constraints of a number of international treaties. Despite a number of transformations in the way in which air warfare has been conducted, the fundamental roles of air power remained the same. Air power did not render land or maritime power obsolete, instead making major contributions to the way in which warfare was conducted. In peacetime, air power also played a significant part in the Cold War as an instrument of deterrence, as well as delivering strategically important outcomes during the Berlin Airlift and the Cuban missile crisis. During this period, the debate over the nature of strategic air power remained lively, and tended to focus upon bombardment rather than the wider application of air power to achieve effects at all levels of war. The evidence of numerous conflicts suggests that although air power evolved considerably in terms of the capability of the aircraft employed and the degree of accuracy that could be obtained in weapons delivery, its utility in supporting the land and maritime components was undiminished; if anything, it became an important enabler to both environments. Yet despite this clear evidence of air power's contribution to warfare shown in conflicts ranging from the global conflagration of 1939–45 through a variety of insurgencies to the Arab–Israeli war in 1982, it was not possible to dismiss the idea that air power offered policy-makers an extra dimension, being able to deliver rapid effect at a strategic level without necessarily being involved in the support of the other components. Events since the end of the Cold War have not resolved this tension, and it is to the post-1990 era that we now turn.

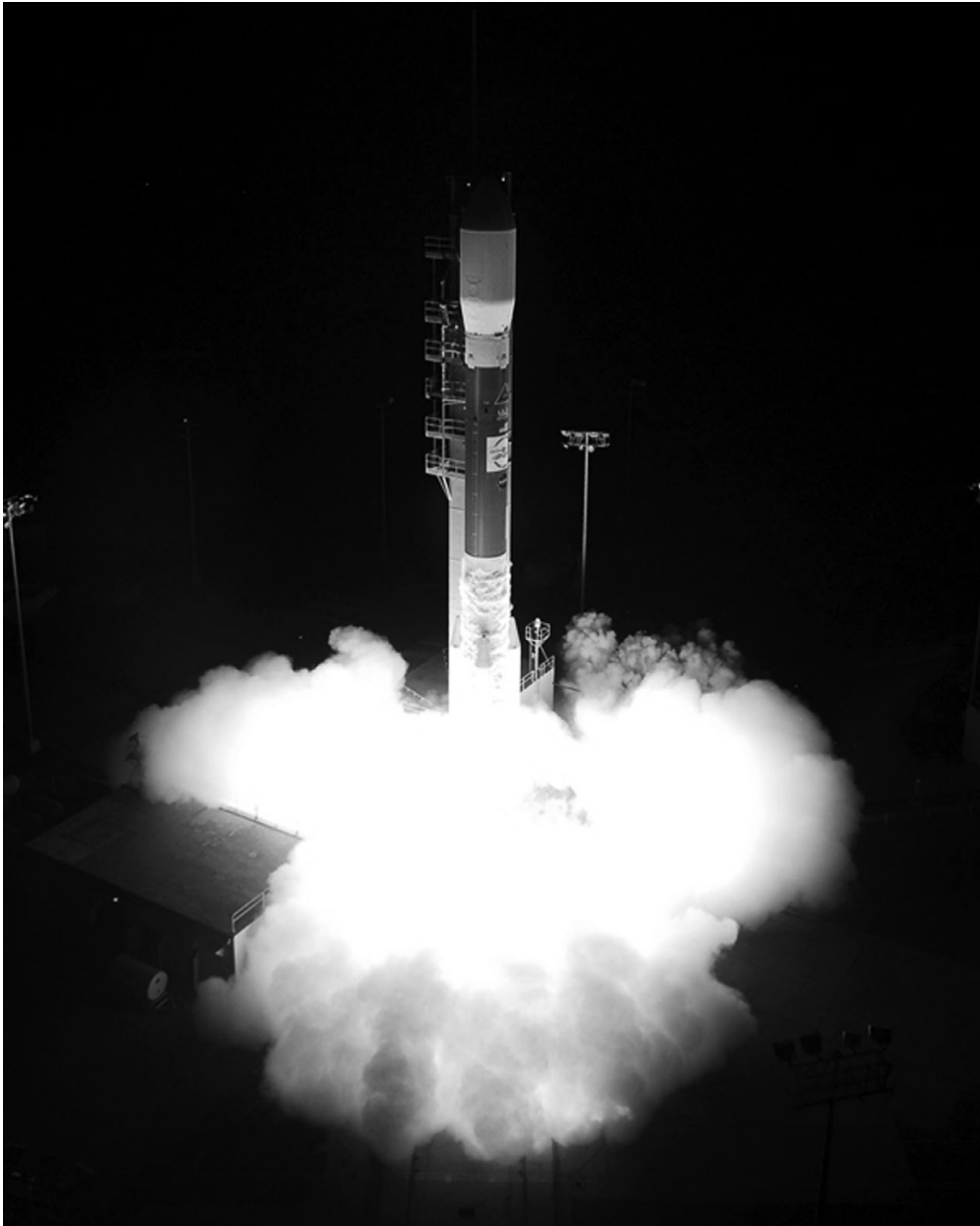


Figure 11.3 A Delta II Rocket launches with an earth observing satellite payload, January 2015. The exploitation of space for both military and civilian purposes makes it a critical aspect of modern security and defence thinking.

RESEARCH QUESTIONS

1. Why did strategic bombardment retain such an important position in the thinking of air forces during the Second World War, even in the face of evidence that it did not deliver rapid victory?
2. To what extent was co-operation between armies, air forces and navies driven by the circumstance of war and then forgotten after 1945?
3. Analyse the view that the Vietnam War demonstrated enduring limitations of air power which no amount of technological advancement will overcome.
4. To what extent did the rise of space as an operating environment mark a revolution in military affairs?
5. Evaluate the idea that the use of air power since 1939 has demonstrated that its strategic significance lies not in bombing but in its key characteristics of reach, flexibility and rapidity.

FURTHER READING

- Boyne, Walter J., *The Influence of Air Power on History* (Gretna, LA: Pelican, 2003).
- Clodfelter, Mark, *The Limits of Air Power: The American Bombing of North Vietnam* (New York: The Free Press, 1989).
- Gray, Colin S., *Airpower for Strategic Effect* (Maxwell, AL: Air University Press, 2012).
- Klein, John J., *Space Warfare: Strategy, Principles and Policy* (London: Routledge, 2006).
- Mason, R. A., *Air Power: A Centennial Appraisal* (London: Braseys, 1994).
- Meilinger, Phillip S., *Airwar* (Abingdon: Frank Cass, 2003).
- Meilinger, Phillip S. (ed.), *Paths of Heaven: The Evolution of Air Power Theory* (Maxwell: Air University Press, 1997).
- Olsen, John Andreas, *John Warden and the Renaissance of American Air Power* (Dulles, VA: Potomac Books, 2007).
- Olsen, John Andreas (ed.), *A History of Air Warfare* (Dulles, VA: Potomac Books, 2010).

ONLINE RESOURCES

See [Chapter 10](#).

NOTES

1. Shmuel L. Gordon, 'The Air Force and the Yom Kippur War: New Lessons', *Israel Affairs*, 6(1) (1999), 221–37.
2. See Bernard C. Nalty, *Tactics and Techniques of Electronic Warfare: Electronic Countermeasures in the Air War against North Vietnam* (Washington: Air Force History Office, 1977).
3. See Brett Holman, *The Next War in the Air* (Farnham: Ashgate, 2014); the same author's 'World Police for World Peace: British Internationalism and the Threat

- of a Knock-out Blow from the Air, 1919–45', *War in History*, 17(3) (2010), 313–32; George H. Quester, *Deterrence before Hiroshima* (New Brunswick and Oxford: Transaction Publishers, 1986).
4. Phillip S. Meilinger, 'Trenchard and Morale Bombing', in Phillip S. Meilinger, *Air War: Theory and Practice* (London: Frank Cass, 2003), p. 49.
 5. *Ibid.*
 6. TNA, AIR 19/92, Minutes of a Meeting in Chief of the Air Staff's Room, 19 July 1923.
 7. *Ibid.*, p. 78.
 8. Robert A. Pape, 'Coercive Air Power in the Vietnam War', *International Security*, 15(2) (Fall 1990), 106.
 9. Mark Clodfelter, *The Limits of Air Power* (New York: Free Press, 1989), pp. 45–72.
 10. *Ibid.*, p. 196.
 11. Air Ministry, *Air Staff Memorandum 46: Notes on Air Control in Undeveloped Countries* (1930).
 12. See, for example, Harry Kemsley, 'Air Power in Counter-insurgency: A Sophisticated Language or Blunt Expression?' and David Jordan, 'Countering Insurgency from the Air: The Post-War Lessons', both in *Contemporary Security Policy*, 28(1) (2007), 96–111, 112–26; Andrew Mumford, 'Unnecessary or Unsung? The Role of Air Power in Britain's Colonial Counter-insurgencies', *Small Wars and Insurgencies*, 20(3/4) (2009), 636–55; James S. Corum and Wray R. Johnson, *Airpower in Small Wars: Fighting Insurgents and Terrorists* (Lawrence, KS: University of Kansas Press, 2003).
 13. Ian Gooderson, 'Heavy and Medium Bombers: How Successful Were They in the Tactical Air Support Role during World War II?', *Journal of Strategic Studies*, 15(3) (1992), 367–99.
 14. Field Marshal Viscount Slim, *Defeat Into Victory* (1956; London: Pan, 2012), p. 624.
 15. Quoted in David MacIsaac, 'Voices from the Central Blue: The Air Power Theorists', in Peter Paret (ed.), *Makers of Modern Strategy: From Machiavelli to the Nuclear Age* (Princeton University Press, 1986), p. 644.
 16. See Kenneth M. Pollack, 'Air Power in the Six-Day War', *Journal of Strategic Studies*, 28(3) (2005), 471–503.
 17. David E. Lupton, *On Space Warfare: A Spacepower Doctrine* (Maxwell, AL: Air University Press, 1988), p. 7.



12

Air and space power in the contemporary era: 1990–2030

Contents

An air power revolution?

Theory into practice

Air power after 9/11

Operation Iraqi Freedom

The Age of Insurgency

Air power in a post-Afghanistan world?

The expansion of space power

Space vulnerabilities

Space control and force application

The ultimate frontier

Air and space power – ongoing challenges and concluding thoughts

Conclusions

KEY THEMES

- The contested concept of air power as a revolution in military affairs.
- Technology as a key enabler for air power to achieve effect.
- Air power's role in the post-Cold War world.
- The importance of maintaining access to space and the growing threat to space platforms.

As the preceding chapters have shown, the development of air and space power has been swift. In little more than a century of heavier-than-air flight, aircraft have moved from being capable of spending little more than five minutes in the air to possessing the ability to spend more than a day aloft; the speeds which military aircraft can attain have gone from under 100 miles per hour (160 kilometres per hour) to over 2,000 mph (3,218 kph) and the loads they are capable of carrying have increased dramatically.¹ By the end of the Vietnam War, it was possible for fighter aircraft to routinely engage enemy aircraft with missiles (although the success rate of such weapons in Vietnam was low), while attack aircraft with precision-guided munitions (PGMs) could use one weapon to destroy a target when it might have previously taken several



Figure 12.1 An X-47 unmanned combat air vehicle alongside an F/A-18 Super Hornet during carrier trials aboard the USS *Theodore Roosevelt*.

aircraft using over a dozen bombs to achieve the same result. Between the early 1900s and 1989, air and space power had a clear role, directed against potentially hostile nation states or alliances, with regular diversions into ‘small wars’. It became something of a mantra, particularly after Vietnam, to note that high-speed fighter-bombers and attack aircraft, designed to fight a possible Third World War, were not best-suited for this sort of ‘brushfire’ war, not least since their speed and relative lack of endurance over a target area meant that it was difficult for aircrew to locate targets. Although a prominent and repeated charge – and not without reason, particularly in the 1950s and 1960s, prior to the development of targeting pods and PGMs, as well as technology which allowed ground troops to accurately direct aircrews’ attention to the target which needed to be struck – the fact that air transport and reconnaissance were critical contributions to such campaigns often went unremarked. The constant fear of a major state-on-state war, notably between NATO and the USSR, meant that the development of high-technology fighter aircraft largely went unquestioned. When doubts were raised in the 1970s as the result of the escalating cost of combat aircraft, the solution (aircraft such as the General Dynamics – now Lockheed Martin – F-16 and McDonnell Douglas – now Boeing – F/A-18 (see [Figure 12.1](#))) came in the form of aircraft designed to be

as light and as cheap as possible, but which steadily added more technology so that they, too, have become sophisticated and costly weapon systems in their own right.²

The ending of the Cold War in the final years of the 1980s, followed by the collapse of the USSR in 1991, dramatically changed the strategic landscape. The United States was left as the world's one true superpower, while there was optimistic talk of a 'new world order' in which democracy and peaceful co-operation would be the norm.³ In Western nations, politicians seized upon what became known as the 'Peace Dividend', imposing large cutbacks in defence spending; as aircraft were (and are) amongst the most expensive items of equipment to buy and operate, air forces suffered considerable cuts; in some nations, it appeared that cutbacks in their air power began in around 1990 and continue to this day.⁴ Unfortunately, the envisaged stable, more peaceful world did not materialise. The first challenge came even as the concept of a New World Order was being framed, when Saddam Hussein invaded Kuwait on 2 August 1990. Air and space power played a leading part in the response.

An air power revolution?

The Iraqi invasion of Kuwait came as a surprise, and prompted the formation of a large American-led international coalition. The Soviet Union, formerly a prominent supporter of Iraq, did a great deal to try to ensure that force was not required to solve the crisis, but without success. Under a United Nations Security Council Resolution, Saddam was given until 15 January 1991 to withdraw his troops from Kuwait, otherwise UN member states were permitted to use 'all necessary means' to bring about the withdrawal. This set the stage for war.

The first response to Saddam's invasion, as noted in the [previous chapter](#), had come in the form of a rapid airlift of troops and combat aircraft from coalition nations – Operation Desert Shield. As it became clear that force would be required to remove Saddam from Kuwait, the airlift, accompanied by the arrival of heavy equipment moved by ship, transitioned into a preparation for conflict. Planning to meet a possible Iraqi invasion of Saudi Arabia, and then the eviction of Iraqi forces from Kuwait drew heavily on AirLand battle concepts. These ideas, tightly linking the USAF and ground forces to the point where some felt that the air force had been subordinated in a manner contrary to FM100–20, were not fully implemented, since it was at this point that Colonel John A. Warden had influence upon the way in which the war was to be fought.

Theory into practice

Warden believed that with the appropriate use of air power, an Iraqi invasion attempt might be stopped in its tracks; if none came, then the Iraqis might

be compelled to leave Kuwait without the need for a ground war. As part of the USAF's Air Staff Plans Directorate, Warden established an informal planning cell known as 'Checkmate' and, on his own initiative, began to draw up plans for an American aerial response to the invasion, just as General Norman Schwarzkopf, commanding US Central Command (CENTCOM, the American command leading the anti-Saddam coalition) asked that the air force do just this. This brought Warden's planning team and its ideas to the fore. The plan, 'Instant Thunder', was the antithesis of Rolling Thunder. Warden envisaged an air campaign against command-and-control facilities in Iraq itself, power stations and communications infrastructure. This would be enabled through control of the air. We can see here an echo of the interwar era and the view that air attack might be decisive, particularly if directed at vital points of the enemy's infrastructure. Although Schwarzkopf was enthused, the air component commander, Lieutenant General Charles Horner, was not convinced. He thought Warden's approach impractical, with its focus upon targets in Iraq rather than upon the Iraqi units occupying Kuwait.

The end result was Instant Thunder's modification. The plan that emerged involved four key phases: attacking strategic targets; a campaign against Iraqi air forces in Kuwait; neutralising the Iraqi Republican Guard and cutting off Iraqi forces in Kuwait; and supporting the ground assault into Kuwait to evict the Iraqis. All of these phases were to be enabled by winning air superiority, and the ground offensive would begin only when Iraqi forces in Kuwait were reduced to 50 per cent combat effectiveness by air attack. When the war began on 17 January 1991, the opening air campaign was considered a spectacular success.

The first three phases of the campaign occurred simultaneously, the concept of *parallel air operations*. Under this construct, the use of precision weapons, coupled with 'stealth' aircraft, enabled multiple targets to be struck; this offered the prospect of the application of rapid, overwhelming force which even if it did not bring about the immediate downfall of the enemy, would cause such disruption and confusion that the opponent would be unable to recover (see [Box 12.1](#)). This served as a precursor for notions of 'Shock and Awe' and 'Rapid Dominance', concepts which came to the fore over a decade later in another war against Saddam.

As a result of the air campaign, the Iraqis ceased to contest control of the air with anything other than SAMs and anti-aircraft artillery (AAA). The air campaign was a key enabler for the short ground war which followed, driving Saddam out of Kuwait. The effectiveness of precision munitions, particularly when employed in combination with the F-117 'stealth' aircraft that were almost impossible for Iraqi radar to detect (see [Box 12.2](#)), gave rise to the argument that the strategic air campaign had worked as planned. However, some observers contended that the attack against Saddam Hussein's command-and-control system had not worked as planned, despite the fact that it had a considerable effect on dislocating the ability of the Iraqi dictator to effect



Figure 12.2 Tornado GR4 strike aircraft of 12 Squadron, RAF. Developed as part of a tri-national project, the Tornado has been in service since 1981 and has seen operational use in Iraq and Kosovo.

command and control.⁵ John Andreas Olsen contends that the coalition failed fully to comprehend the nature of the Iraqi regime, which meant that targeting its command-and-control system to fullest effect could not be achieved, although he makes the important point that air power still achieved a great deal.⁶ Furthermore, there was an unexpected element to the air campaign of strategic significance, namely what became known as ‘the great Scud hunt’ in the Iraqi western desert.

Saddam Hussein sought to break the coalition by detaching its Arab members by bringing Israel into the war against him. He did so by firing SS-1 ‘Scud’ (and Iraqi-built derivatives) into Israel. The Israelis were dissuaded from intervening by considerable diplomatic pressure from the United States, the provision of Patriot SAMs and the diversion of a large amount of air power in a bid to locate and destroy the Scud mobile missile launchers. This was a diversion which brought little in terms of destruction of the launchers, but a great deal in terms of keeping the Israelis out of the war, since the Israelis recognised that they could not do any more than the coalition was attempting to do to suppress the threat. The indiscriminate nature of the missile attacks did little for

Box 12.1 A precision revolution?

The 1991 Persian Gulf War came to be seen in the media, and later in some historical accounts, as a war dominated by the use of precision-guided munitions. Many of the press conferences given by senior coalition leaders, particularly General Norman H. Schwarzkopf, included footage from the aircraft targeting systems used to guide the weapons, demonstrating the degree of accuracy which could be achieved. In fact, the 1991 war was far from a precision revolution – only 9 per cent of the weapons dropped were PGMs; the rest were unguided ('dumb' bombs and rocket projectiles). Although the degree of accuracy which could be achieved in 1991 with an unguided weapon was greater than it had been in any previous war thanks to the weapons aiming systems on aircraft, the degree of precision achieved was far lower than that of the PGMs. One of the key factors in the relative lack of use of PGMs lay in the fact that only 10 per cent of the American strike aircraft involved were capable of using these weapons, while of the other air forces engaged, only the RAF could match this – but used two of the three strike aircraft types deployed (the Tornado GR1 and Jaguar GR1) to deliver, for the most part, unguided weapons, with the Tornado moving to the use of PGMs only after Buccaneer S2 aircraft arrived in-theatre in February 1991 to provide designation for the Tornados' laser-guided weapons as well as employing their own PGMs.

The Kosovo campaign was presented as another precision war in the media, but again, the figures are instructive – while the amount of PGMs used was unprecedented, 71 per cent of the weapons released by NATO aircraft were unguided.¹

It was not until the 2003 war against Iraq that the overwhelming majority of weapons used by aircraft could be considered precision-guided. The presentation of air power as a tool of overwhelming precision was not corrected in the aftermath of the 1991 war, and this created an image which has become a problem for air forces ever since. The fact that PGMs can malfunction, or lose guidance, or can be mis-targeted even with the most sophisticated equipment has not been widely understood, making errors and technical failures leading to bombs striking the wrong targets and causing civilian casualties extremely controversial events, often portrayed by the media as little more than examples of callous targeting or wilful incompetence.

¹ Benjamin S. Lambeth, *NATO's Air War for Kosovo* (Santa Monica: RAND, 2001), p. 88 (www.rand.org/pubs/monograph_reports/MR1365.html).

Saddam's image, but the coalition suffered similar problems when the al-Firdos command bunker was attacked.

Coalition planners were unaware that it had been abandoned and pressed into service as a shelter rather than as a command-and-control facility. Several hundred civilian deaths resulted, with extremely negative media coverage. The US Joint Chiefs of Staff (JCS) then reviewed targeting, a process which dramatically reduced the number of attacks against targets in Baghdad – not least since it became clear that a number of targets were being revisited long after they

Box 12.2 Stealth aircraft

The growing sophistication of air defences long exercised air commanders, who recognised the need to find some means of defeating them, so as to allow attack aircraft through. The first approach came during the Second World War with the use of electronic warfare, and this was built upon with the development of the Suppression of Enemy Air Defences (SEAD) role during Vietnam. Another approach considered during the Second World War was that of making aircraft less visible to the human eye, particularly through the use of devices known as Yehudi lights which reduced an aircraft's luminance against the sky.¹ These measures in fact built on early – and largely unsuccessful – experiments in the First World War. They were not implemented, but not forgotten.

As technology developed, thoughts turned to the prospect of developing aircraft with an extremely low radar signature, through the use of radar-absorbing material (RAM) and shaping an aircraft's structure so that radar energy was not reflected in such a manner as to make it obvious that the radar had detected an approaching aircraft. The reduction of the aircraft's radar cross-section (RCS) was but one approach to what became known as 'stealth' or, more precisely, low observable (LO) aircraft.²

The United States, with its considerable research and development capability led the way in this field, and by the early 1970s, it became possible to consider the production of an LO aircraft, with the main aim being to penetrate Soviet air defences. In 1975, Lockheed began a programme known as 'Hopeless Diamond', to develop a technology demonstrator; this in turn gave rise to the Advanced Technology Bomber (ATB) programme from 1979. The Hopeless Diamond project ultimately led to the F-117 Nighthawk, while the ATB became the B-2 Spirit.³

The F-117 gained fame during Operation Desert Storm thanks to its successful penetration of Iraqi air defences and the use of laser-guided bombs against an array of key targets; no aircraft were lost. Some of the claims for the success of the F-117s were later challenged by the US General Accounting Office, but the public perception of 'stealth' aircraft, invisible to radar and delivering their weapons with absolute precision, was set.⁴

In fact, 'stealth' aircraft are not invisible; as the term 'low observability' suggests, such aircraft are more difficult to detect rather than impossible to locate. The cost of stealth aircraft is prodigious, because of the technology involved in terms of both design and the RAM used to build them.⁵ The cost makes such aircraft controversial, a particular feature during conflicts post 9/11, where critics argue that the stealthy properties of this type of aircraft have not been required. It is also argued that stealth is a wasting advantage as detection technology increases; proponents counter that non-stealthy aircraft would be at an even greater risk of detection and destruction.⁶

¹ Office of Scientific Research and Development, *Visibility Studies and Some Applications in the Field of Camouflage* (www.dtic.mil/dtic/tr/fulltext/u2/221102.pdf), esp. ch. 6.

² An excellent brief survey can be found in Arend G. Westra, 'Radar versus Stealth: Passive Radar and the Future of U.S. Military Power', *Joint Force Quarterly*, 55 (4th Quarter 2009) (www.dtic.mil/doctrine/jfq/jfq-55.pdf).

³ For the development of stealth aircraft, see Thomas Withington, *B-2 Spirit Units in Combat* (Oxford: Osprey, 2006), pp. 6–29; Paul F. and Alison J. Crickmore, *Nighthawk F-117 Stealth Fighter* (St Paul, MN: Motorbooks, 2003); Steve Pace, *B-2 Spirit* (McGraw Hill, 1999).

⁴ See US General Accounting Office, *Operation Desert Storm: Evaluation of the Air Campaign* (Washington: GAO, 1997) (www.gao.gov/archive/1997/ns97134.pdf). The GAO was renamed the Government Accountability Office in 2004.

⁵ GAO, *B-2 Bomber Cost and Operational Issues* (Washington: GAO, 1997) (www.gao.gov/assets/230/224502.pdf). RAM also cost more to maintain than conventional aircraft construction materials such as metals and composites, although as technology evolves, the cost differential has declined.

⁶ Jan van Tol et al., *AirSea Battle: A Point of Departure Operational Concept* (Washington: Center for Strategic and Budgetary Assessments, 2009) makes the case for the value of stealthy platforms, and not just aircraft.

had ceased to be used by the Iraqis. As the chairman of the JCS, General Colin Powell, put it, it seemed that the attacks were now simply ‘making the rubble bounce’ for no obvious reason.⁷

This revisiting of targets suggested that the notion of effects, which did not necessarily demand the destruction of a target, merely rendering it inoperable or ineffective for the duration of the conflict, had not fully entered the planning process.

While the 1991 Gulf campaign demonstrated the efficacy of air power against strategic targets, it helped further to demonstrate that the notion that strategic air power meant only the destruction of targets vital to the prosecution of the enemy’s war effort was an unsatisfactory concept.

Using air power against the Scuds had a major strategic effect, despite the negligible amount of destruction of the missile launchers. Prior to the war starting, the effort to deter an Iraqi invasion of Saudi Arabia had been bolstered by the rapid reinforcement of the kingdom by air, which was also of strategic significance, just as it was during the 1948–9 Berlin crisis.

While such examples demonstrate that the strategic applicability of air power is far broader than that envisaged by the early air power theorists, this is not to say that strategic attack should be disregarded as a concept. This was brought home in 1999 with Operation Allied Force, the bid to bring about the end to the ethnic cleansing campaign against Kosovar Albanians by the Federal Republic of Yugoslavia, led by President Slobodan Milosevic. The campaign was conducted by NATO, against a backdrop of considerable concern from certain alliance members over the wisdom of the operation, and considerable opposition from Russia. Political concerns saw the prospect of an invasion by NATO troops being ruled out, and air power was employed as the means to bring about Milosevic’s compliance with international demands for a settlement of the Kosovo crisis.⁸

There were expectations, even within the US administration, that the campaign would last a few days, rather than the seventy-eight it actually took.

Without the presence of a credible ground threat to Slobodan Milosevic’s regime, air power shaped the political conditions by which Milosevic felt it necessary to capitulate to NATO’s demands to end ethnic cleansing in Kosovo. The

air campaign was hindered by initial confusion over targeting and bad weather, which militated against the use of laser-guided weapons. NATO's Supreme Allied Commander Europe (SACEUR), General Wesley K. Clark, argued that the air campaign should be conducted against fielded forces in Kosovo itself. His air component commander, Lieutenant General Michael Short, disagreed, contending that air power should 'go for the head of the snake', in keeping with classic strategic air-attack theory. Clark's view – in fairness, a perhaps more politically aware one than Short's – prevailed initially, and the combination of bad weather and poor results from targeting forces in Kosovo led to a re-evaluation of the campaign.

The strategic target sets that Short had advocated began to be attacked, particularly targets that were of significance to supporters of the Milosevic regime. This 'crony attack' approach undermined support for Milosevic, while the apparent ability of NATO air assets to attack almost at will created an impression within the Serbian regime that the weight of bombing could leave Serbia utterly devastated if the government failed to comply with the demands of the international community.⁹ Combined with the realisation that expected support from Russia would not be forthcoming, indictment as a war criminal and growing talk of a credible threat of an invasion, Milosevic chose to capitulate. The perceived success of air power in Kosovo enhanced consideration of the use of air attack to bring about rapid victory, through use of precision weapons. Once again, though, there was a need for caution. While critics of the air war could be found proclaiming that something other than air power was responsible for victory, it seems reasonable to conclude that without the air campaign, Milosevic would not have been forced to accede to the demands of the international community.¹⁰ Yet even allowing for this, there were questions about targeting. The Chinese embassy was bombed after it was misidentified, causing considerable political difficulties for NATO, while the headquarters of the state broadcaster in Yugoslavia were bombed, with the death of a number of journalists. This was a controversial attack, held by a number of commentators to be a violation of international law, although the International Criminal Tribunal established to deal with war crimes in the former Yugoslavia disagreed.¹¹ Furthermore, the appalling weather meant that it was difficult to employ laser-guided weapons, as laser designators could not be used to 'paint' the target to produce the reflected 'basket' of laser light which laser-guided weapons need to strike their target accurately.¹² Casualties caused amongst the civilian population by cluster bombs and unguided weapons increased political pressure to move to a more precise form of weaponry and demands for a ban on cluster munitions.¹³

As the twenty-first century began, the results of the 1991 Gulf War and the Kosovo campaign seemed to suggest that air power might become the 'weapon of choice' in future conflicts, at least for Western powers, anxious to avoid the political risks that casualties amongst ground troops might bring in what were referred to as 'wars of choice'.¹⁴ In addition to the prospect of using air power in

short, decisive conflicts, the ongoing containment of Saddam Hussein through the use of 'no-fly zones' added to the view that air power would become the dominant mechanism of crisis intervention. This dynamic was to change on 11 September 2001.

Air power after 9/11

The attacks on the United States on 11 September 2001 led to a US-led intervention in Afghanistan, initially in a bid to bring Osama bin Laden, leader of the al-Qaeda (AQ) terrorist organisation to justice and to remove the Taliban government which was hosting him. After the removal of the Taliban regime was complete, the United States and many of its allies embarked upon the reconstruction of Afghanistan, but against a backdrop of continued resistance from AQ and Taliban fighters in what became a protracted and deadly insurgency. In the opening phases of the campaign, carrier-based attack aircraft and USAF bombers supported the anti-Taliban Northern Alliance (aided by Western special forces) with considerable success. There were several controversial incidents involving targeting errors (particularly against parties of civilians mistaken for insurgents), but the support from air power was of great value in removing the Taliban.

As the number of conventional troops in Afghanistan increased to meet the growing insurgency, a number of issues relating to air power became apparent. During an operation in the Tora Bora mountains – Operation Anaconda – there was considerable consternation created when it appeared that Bin Laden had managed to slip away during particularly heavy fighting in which a number of American soldiers were killed. This created some inter-service tension, particularly when senior US Army officers criticised the USAF for failing to provide adequate support; the air force, irritated, criticised the army for failing to include it fully in the planning process. This prompted a swift review and the development of improved procedures to enable better co-operation between the air and land components.¹⁵

As well as Afghanistan, the United States, now led by President George W. Bush, became increasingly focused upon the matter of Saddam Hussein. Although largely contained by the use of no-fly zones over the north and south of Iraq, Saddam remained a regional problem, particularly since he refused to make clear whether or not he had disposed of his weapons of mass destruction (WMD) as required after the 1991 war. His behaviour had seen a number of air strikes made against him as a result, notably the four-day Operation Desert Fox in 1998, in a fruitless bid to make him comply with a weapons inspection regime.¹⁶

Operation Iraqi Freedom

The Bush administration's decision-making over Iraq after 2000 has been the source of enormous controversy, outside the scope of this book. For our

purposes, though, it is worth noting that air power was to play a critical role even though the desired result of removing Saddam from power would require the use of a large invasion force.

The role envisaged for air power during what became known as Operation Iraqi Freedom (OIF) was based upon the evolving concept of 'Rapid Dominance'. This required air power to play a key part in overwhelming the enemy with great rapidity. The authors of the 'Rapid Dominance' concept presaged the final version of their study with a pamphlet called *Shock and Awe*, a title which became closely associated with air power in OIF.¹⁷ John Warden's ideas of targeting the enemy leadership were taken a stage further, with time-sensitive intelligence information used to launch an attack on the Dora Farm complex, where Saddam was thought to be, on the night of 19 March 2003.¹⁸ Saddam was not killed in the attack, but this was the first of a number of time-critical air attacks against enemy command-and-control systems with the aim of quickly achieving the desired level of dominance over the Iraqi regime, with another (equally unsuccessful) example of 'time-sensitive targeting' applied against Saddam on 7 April.¹⁹

The 'shock and awe' concept was poorly articulated in public, however, and television coverage of attacks on leadership targets in the middle of Baghdad were taken – erroneously – by some sources as a sign that air power was being used in an indiscriminate fashion. Once again, a failure to explain the complexities of air power, particularly the exceptionally detailed targeting process which requires estimates of (amongst other things) potential 'collateral damage', the legality of the choice of target and the proportionality of the use of force against it, laid air power open to criticism because of the visually spectacular nature of many of the attacks on key targets. The concentration of the media on the bombing in and around Baghdad obscured the fact that air power was being used in other ways.

Battlefield mobility was enabled by the use of both fixed- and rotary-wing aircraft, while a number of types of reconnaissance aircraft, in conjunction with space-based platforms, delivered vast amounts of information for commanders, providing 'information dominance' over the Iraqi forces, even when they were blanketed by sand-storms which they thought gave them protection from coalition air attack. The sheer threat presented by the opposing fighter aircraft, coupled with psychological warfare operations against its commanders prompted the Iraqi air force to remain on the ground, leaving the opposition to coalition air action to come from SAMs.²⁰ Air power's most notable effect was arguably in terms of providing close air support and interdiction; in the latter role, coalition aircraft inflicted devastating levels of damage and disruption upon Iraqi forces.²¹ An effective combination of air and land power brought about enormous combined effects against the enemy, just as had been expected and predicted by Bernard Montgomery in his observations about air-land integration during the Second World War.

Although air power had been of great utility in bringing about the desired initial outcomes in both Afghanistan and Iraq, the challenge which now faced

those thinking about and delivering air power came not from state-on-state warfare, but insurgencies. In both cases, the long insurgencies suggested that notions of rapid dominance and swift victory were consigned to the past; coupled with the Israeli experience in Lebanon, it seemed that it might be the case that air power's role was to be not dissimilar to that in the First World War, namely supporting troops on the ground. Superficially, this appeared correct – but this was to overlook the exceptional force-multiplying effects of air power and, crucially, to ignore the possibility that insurgencies were not the only sort of wars which Western nations would have to fight.

The Age of Insurgency

As noted in previous chapters, air power has often been seen as a peripheral factor in counterinsurgency campaigns, there to provide support to troops on the ground. This remained largely true during the campaigns in Iraq (2003–11) and Afghanistan (2001–14), since air mobility was of critical importance in moving troops around both countries in the face of extremely dangerous road conditions created by improvised explosive devices (IEDs). Helicopters in particular were used for this task, and were not invulnerable, but their presence enabled movement that would otherwise have been conducted on the ground to be carried out at lower risk; it was, of course, impossible for air mobility to replace the need for troops on the ground.

The use of air power to deliver fire support to troops remained controversial, even though the majority of casualties amongst the civilian population in both Iraq and Afghanistan were inflicted by the insurgents, and then by heavy weapons such as artillery, rather than by air attack. The use of air-delivered fires, exploiting precision weapons in a manner not previously seen in COIN operations, made a considerable contribution to ground operations, although there were suggestions that an over-reliance upon air power had crept into land-component thinking, even though the new American COIN doctrine, FM3–24, gave little detailed coverage to the use of air power in its first edition.²² In contrast, Lieutenant General Karl Eikenberry, commander of the Combined Forces Command Afghanistan, was absolutely clear on air power's contribution:

Without air and space power, 500,000 to 600,000 troops would be needed in Afghanistan to achieve the same effects as the 40,000 soldiers, sailors, marines and airmen we have there today. Air and space power provides the asymmetric advantage over the Taliban such that no matter where they choose to fight, coalition forces can bring to bear overwhelming fire-power in a matter of minutes. Moreover, putting 5–600,000 troops into the country may achieve the same military effect, but it could also have a negative impact on the population; such numbers could appear as an occupying force rather than a security assistance force. In short, there is no substitute for effective air and space power.²³

Air power was also a source of controversy because of the dramatic increase in the use of remotely piloted air systems (RPAS), particularly to conduct attacks against targets over the Afghan border in Pakistan. These attacks, conducted by remotely piloted aircraft operated by the Central Intelligence Agency, struck at supposed Taliban and Al-Qaeda safe havens, but the implications for Pakistani national sovereignty, as well as frequent alleged instances of civilian casualties alongside the individuals targeted has been a source of great controversy.

This has created a number of difficulties for air forces using RPAS, with criticisms of 'drone wars' conflating CIA operations with those of regular forces. RPAS have provided regular persistent surveillance over large areas of Iraq and Afghanistan, and their extended endurance has been critical in allowing the development of detailed intelligence pictures. 'Weaponised' RPAS, such as the MQ-9 Reaper, have added an extra dimension in terms of their ability to provide overwatch for ground forces and fire support when required, often ensuring immediate support has been provided rather than seeing troops have to wait for aircraft to be called in to deliver an attack. RPAS also offer a greater opportunity for the crews to increase their situational awareness before deciding to release weapons, reducing – but not eliminating – the danger of collateral damage or mistakes in identification which might occur if the decision to release has to be made by aircrew travelling at notably higher speeds in a jet.²⁴

The importance of air power, both as a force multiplier and as a means of providing critical support to the land component may have been overlooked and sometimes misrepresented in popular reporting, but a study of the evidence means that it is difficult to disagree with Anthony H. Cordesman's analysis:

Much of the reporting on the Iraqi and Afghan wars focuses on the ground dimension... The fact remains, however, that Iraq and Afghanistan are air wars as well, and wars where air power has also played a critical role in combat.²⁵

Air power in a post-Afghanistan world?

The drawing down of coalition forces in Afghanistan took place against another change in the strategic context in which air power has been used. The idea that states would not fight against other states, popular in the mid 2000s, came to be replaced by an understanding that the types of conflict in which air power might be used would be just as varied as before, but likely to be more complex. This was exemplified by the NATO-led operation to enforce UN Security Council Resolution 1973, aimed at protecting the population of Libya from the efforts of Colonel Muammar Gaddafi's regime to quell an insurgency against his rule. The operation was notable for the relative lack of American participation beyond critical enabling capabilities such as air-to-air refuelling and heavy attacks upon the Libyan air defence system during the opening phases of the campaign. Air power was used to destroy Libyan regime targets, since Gaddafi

had been identified as the centre of gravity of the whole Libyan effort – reducing his power over his own forces and his prestige through attacks on targets closely associated with Gaddafi. A further apparently novel feature came in the use of air power to support indigenous Libyan forces, aided by a small number of special forces soldiers from the wider anti-Gaddafi coalition; this created complications in terms of air–land integration, but it is perhaps worth reflecting upon the fact that air support to indigenous units is not entirely new, being a feature (albeit to a much lesser extent) in the colonial policing operations of the interwar period.

Tensions between NATO and Russia, coupled with an increase in the number of long-range flights by the Russian air force requiring interception and escort by a number of NATO air forces, suggested that the need to consider wider possibilities than COIN campaigns remained valid, particularly given the fact that the procurement times for aircraft and their associated technologies mean that there is a risk of procuring extremely limited platforms if just one type of war is envisaged – a multi-role fighter (such as the Dassault Rafale or Eurofighter Typhoon) might not be seen as the ideal close air support (CAS) aircraft compared to the USAF's A-10, but today's multi-role fighters can perform the CAS role nearly as well as dedicated aircraft types – and it may well be the case that concerns over affordability dictate the decline in the production of specialised combat aircraft designed to undertake one distinct set of missions in a distinct type of war. Concerns over affordability also affect the area to which we turn now, namely the further rise of space power as a critical component in modern warfare.

The expansion of space power

The end of the Cold War and the Iraqi invasion of Kuwait changed the way in which the United States exploited space power; with the latter being the leading nation in the exploitation of space for military purposes at this point, the events of the early 1990s informed perceptions of space power generally around the world. When Operation Desert Shield began in August 1990, the Americans were notably unprepared to use space power to full effect. Although some doctrine regarding the effective integration between space assets and terrestrial-based forces existed, there were areas in which the command and control of space assets, hampered by the previously mentioned over-classification of information product, did not function as well as they might. While national command authorities were provided with much useful information from space platforms, General Schwarzkopf, as theatre commander, did not receive information that would have been extremely useful.

Nevertheless, space power played a vital role in operations in the Persian Gulf. Approximately 80 per cent of communications to and from CENTCOM came via space-based communications assets. The Global Positioning System (GPS), a relatively new piece of technology, proved an essential asset to the

coalition, giving precise navigational co-ordinates to army formations manoeuvring across featureless deserts, while aircrews purchased their own hand-held GPS terminals and attached them in their cockpits with velcro, preferring to trust GPS ahead of the less accurate navigation equipment fitted to their aircraft.

Multi-spectral sensor information (that is to say, using radar, infra-red and electro-optical sensors) provided an array of highly detailed images that aided in battle damage assessment (BDA), although there were complaints from within the theatre of operations that BDA information that would have been of considerable utility in planning had not reached the units that required the information the most. Perhaps most importantly, space power's most dramatic contribution to Desert Storm came with its use in the campaign against the Iraqi Scud missile attacks against Israel. Satellites with infra-red sensors were employed to detect the heat 'bloom' generated by the firing of the Scuds' rocket motors. From this information, which provided the co-ordinates of the launch site, it was possible to make a rough calculation as to where the missile was heading. This information was then sent to the US Air Force's Space Command in the United States, which then sent the information directly to the crews manning Patriot SAMs based in Israel and Saudi Arabia, bringing them to alert so that they could use their own radar systems to search for and detect the incoming missile. As Gregory Billman observes, the 1991 Gulf War 'provided space power with its first large-scale opportunity to demonstrate its capabilities'.²⁶

The Americans reflected deeply upon the use of space power in Desert Storm and made amendments to the way in which space assets were tasked and information disseminated. The significance of GPS increased dramatically after Desert Storm as the US Air Force pursued the development of GPS-guided munitions, notably with the Joint Direct Attack Munition (JDAM) series of weapons, where a GPS guidance kit is fitted to standard unguided bombs. JDAM is not as precise as laser-guided weapons, but its accuracy is impressive, bombs usually landing within thirty feet or less of their designated aim point. The need for such a system was illustrated during the NATO operation against Slobodan Milosevic's regime in 1999, when many aircraft sorties had to be aborted because bad weather prevented aircrew from seeing the ground, preventing them from designating targets for their laser-guided weapons.

Space vulnerabilities

The increasing importance of space power during the 1990s in terms of both GPS and information-gathering created new concerns about the exploitation of space. These are particularly acute in the United States, which relies upon space assets like no other nation; in turn, many of America's allies are dependent upon US space systems for their militaries – for instance, the RAF now employs the 'Enhanced Paveway' series of bombs, which combine laser and GPS guidance to make the weapons employable in all weathers. The concern stems from the

fact that it is evident that an adversary may seek to deny the Americans (and thus their allies) the use of space assets in future conflict. It is possible for GPS signals to be jammed. Indeed, in Operation Iraqi Freedom in 2003, the Iraqis employed GPS jamming systems purchased from Russia; in this case, the investment proved unsound, since the Americans destroyed the jammers – ironically, using JDAMs – but the point that an enemy may seek to mitigate American space superiority was well made.

The potential for adversaries to make use of the widely available commercial satellite systems has also become an issue. In Operation Desert Storm, the coalition made sure that the Iraqis were unable to make use of the French commercial SPOT imaging satellite system (*Système, probatoire pour l'observation de la terre*), which might have given Saddam information about the disposition of coalition forces; in the insurgency in Iraq after Saddam's deposition, and in the intervention in Afghanistan, concerns have been expressed about the ability of insurgents to use packages such as Google Earth to gain images of the layout of coalition bases, enabling them to make effective indirect fire attacks on the most vulnerable parts of those facilities. To an extent, this was addressed by Google Earth responding positively to requests to obscure the imagery of bases, but commercial concerns offering similar services may not be so accommodating in future. Furthermore, it is possible that adversaries with access to suitable technology may seek to 'blind' satellites with the use of ground-based laser systems that dazzle the satellite's equipment. A US Army test in 1997 employing a low-powered laser blinded a US Air Force satellite flying 300 miles above the earth's surface.

Subsequently, there have been suggestions that the Chinese, embarking upon a space programme of their own, may have utilised similar low-powered laser blinding devices to deny the United States intelligence information about sites of interest in Chinese territory. Another threat may emanate from adversaries launching a cyber-attack on space-based platforms, interrupting communication between the control room and the platform itself. Such an attack could be used to render the space asset useless, either by sending it tumbling out of orbit or shutting down key systems.

Space control and force application

American concerns about such developments have increased discussion of the concept of space control. Like control of the air, this seeks to ensure freedom of action of friendly assets without interference from the enemy, while denying the opposition access to the benefits provided by space. This represents a horrendously complex issue for the United States, since blinding or destroying a civilian-owned, -launched and -operated satellite system if the owner refuses to ensure that either information product or communications facilitation provided by their space asset is unavailable for enemy exploitation is fraught with legal difficulties. The Americans developed an air-launched

anti-satellite (ASAT) weapon in the 1980s which was successfully tested before the project was cancelled over concerns that it violated international treaties (see Figure 12.3); a Chinese test of an ASAT in 2007 caused considerable consternation, particularly in the United States. US military war games have illustrated the threat posed by an adversary capable of employing ASAT or blinding technologies. As a participant noted at the end of one such war game, '[the enemy] took out most of our space-based capabilities. Our military forces just ground to a halt.'²⁷ Observers suggest that the only means by which the United States can guarantee space control is not only to invest in anti-satellite capabilities deployed from earth-based locations but also actively to fight for control in space itself; the US Space Commission report of 2001 concluded that the threat to American space capability was a 'virtual certainty', making the need to invest in some form of space control technology seem even more important. The degree of space control considered necessary by some proponents would involve placing weapons in space, an issue of considerable political sensitivity. Weaponisation of space, even for defensive purposes, is controversial; the Soviets fitted a 30 mm cannon to the Salyut-3 space station in the 1970s, with the aim of protecting the station from a possible attack by an American platform, but such was the sensitivity about placing weapons in orbit that this was not revealed at the time, details emerging only as the result of information provided from those who served aboard Salyut-3. This makes another potential space concept, that of force application, an even more controversial area.

The concept of space force application takes the possibility of weaponising space a stage further, in that it would see the development and use of weapons systems against terrestrial targets. Such a step would involve the use of directed-energy and kinetic-energy weapons (that is to say, lasers and ultra-high-speed projectiles that destroy the target through their kinetic energy rather than with a warhead) against both missiles and enemy ground targets; enemy targets on earth might also be attacked using space-based conventional weapons. The value of the kinetic-energy weapon from space against such targets as hardened bunkers would be that the energy inherent in such a weapon descending from space would be far greater than anything that could be obtained by a conventional weapon launched from within the earth's atmosphere.

The ultimate frontier

While space power is still, in many ways, a nascent form of warfare, the critical nature of space-based platforms means that there is a widespread belief that issues of space control and the weaponisation of space will need to be addressed, even if not in the short term.²⁸ Space power shares many features of air power, notably in the way in which it is a key enabler for operations. The former commander-in-chief of both North American Aerospace Defense Command and the USAF's Air Force Space Command, General Howell M. Estes III, has observed: 'The potential of aircraft was not recognised immediately. Their



Figure 12.3 An F-15 Eagle launching an ASM-135 anti-satellite missile (ASAT). The ASAT was cancelled by Congress prior to service introduction for fear that it would breach international laws regarding the weaponisation of space. The importance of space assets to American military power means that such weapons systems are once again under consideration.

initial use was confined to observation . . . until one day, the full advantage of applying force from the air was realised, and the rest was history. So too with the business of space.²⁹

While Estes's view may involve too linear a description of the way in which the utility of using air power for the projection of military force was realised, the comparison with air power (and arguably land and naval power) is valid. As soon as the value of the operating environment was realised, it became clear that future conflict between powers of roughly comparable technical proficiencies would involve the struggle for control of that environment. This has become the underpinning rationale of air power, and it appears that space power will increasingly be subjected to the same dynamics, albeit only as long as the competing powers possess an approximately symmetrical degree of technical advancement and aptitude. For asymmetric operations, where one power enjoys overwhelming technological superiority against the opponent, the value of space power as a key enabler remains unchallenged, hence the notion that 'space superiority has joined air superiority as a *sine qua non* of global reach and power'.³⁰

Air and space power – ongoing challenges and concluding thoughts

The rise of technology in both the air and space environments, and their associated cost has been an increasing challenge for the traditional, mainly Western, air power nations. Further complications lie in the fact that many states now have growing access to sophisticated equipment. The days when countries had to accept downgraded 'export' versions of combat aircraft, equipped with previous-generation missiles, bombs and lower-quality avionics are long gone. Advanced Russian aircraft such as the Su-27 are readily available to those who have the money to purchase them, along with advanced air-to-air missiles and PGMs. China, in its bid to develop air forces which contain a large number of high-quality aircraft, rather than relying simply on mass, has produced aircraft which are attractive to export partners; the JF-17 'Thunder', while not in Chinese service, has been the subject of an enthusiastic partnership programme with Pakistan to replace some of its obsolete aircraft, while the Chinese J-10 fighter, again armed with a panoply of advanced weapons, compares favourably with American and European designs. Although it seems unlikely that the Chinese J-20 'stealth' fighter will be exported in the near future, the possibility of it entering service with nations unfriendly towards Western (and particularly American) interests is a source of possible concern. Nations generally well-disposed to the traditional air powers, such as Qatar and the United Arab Emirates have small but exceptionally well-equipped air forces; the F-16s employed by the UAE are probably the best in the world, their avionics surpassing those currently fitted to the F-16s in use by the USAF.

Finally, the trend for investment in advanced SAMs continues. Although there is a lack of clarity over the exact dangers presented by many of the latest

Russian SAMs – the so-called ‘Double Digit SAMs’ (the name deriving from the NATO reporting designations ‘SA-10’ and upward) – there is little doubt that they represent a potentially serious threat to control of the air. Fond notions of permissive air environments that were floated in the early part of the twenty-first century look likely to be dashed in many possible future conflicts; the likelihood of a nation equipped with modern air defences following the approach of the Iraqi Air Force in 2003 in any conflict they face seems very low indeed.

Conclusions

As has been discussed over the last three chapters, air power has grown from being purely subordinate to surface forces to playing a significant part in the delivery of military power; the rise of space power has only added to this. The air is a complex operating environment, and visions over how it might best be exploited have led to considerable debate. Nowhere has this discussion been more heated in terms of the potential contribution of air power at the strategic level, through attacks against targets deep in the enemy heartland. As has been shown, limiting consideration of air power’s possible strategic effect to ‘kinetic’ operations has been short-sighted; also, the critical point that air, maritime and land power are all subordinate (or at least should be) to national strategic objectives is sometimes in danger of being missed, often for parochial reasons as service chiefs seek to promote the value of their arm in the seemingly constant struggle over budgets.

Despite this debate as to where air power is best employed, a few general observations can be made. No army or navy can operate truly effectively in the absence of some form of control of the air, particularly over the course of a long conflict. Possessing control of the air for a limited period of time or in a limited area may be all that is required for the success of a particular operation, but prolonged activity in the face of enemy air superiority makes the achievement of success increasingly unlikely. That said, control of the air is an enabler for operations rather than an end in itself.

A second key point is to note how important air power is to surface operations. The Second World War demonstrated that surface forces could not function to full effect in the face of enemy air superiority and that the efficacy of their operations was greatly enhanced by being able to exploit the key characteristics of air power discussed in the first section of this chapter so as to project power beyond the range of the surface forces. Synergy between air and surface units greatly enhances the effect of what can be achieved, particularly in an era when precision weaponry enables targets to be attacked with considerable discrimination. The value of intelligence gained from air and space platforms has become a fundamental part of military planning in Western nations, while some space technology, such as GPS, has become so embedded in the conduct of day-to-day operations that it is difficult to envisage how Western militaries might operate without it.

Air and space power's attributes make it essential to conducting modern conflict. Now that the veil of secrecy of the Cold War era has been lifted and some of the mystique of what space-based assets can achieve, it is difficult to argue against the view that air and space are critical enablers to the prosecution of modern war. While the time has perhaps not yet arrived for air and space to be blended seamlessly into a concept of 'aerospace power', it is evident that the synergies of air and space power, and the way in which these contribute to modern war, make them vital to the business of war.

Although these chapters have sought to give an insight into the complex realm of air and space power, it is worth ending with a caveat – and this is that air and space power are immature forms of warfare. They have proved their worth in a comparatively short space of time, and are still developing. Colin Gray suggests:

The grand strategic historical narrative of air power is complete in the record to date, but it is seriously unfinished. Air power is still in the process of becoming something different in character, though not in nature.³¹

This sage observation is an appropriate note on which to end: air and space power have developed at a tremendous rate, enabled by technological advancement of which the early aviators could not have dreamed. Yet while it seems that technology has enabled some of the early visions of key air power thinkers to come to fruition, we are nowhere near witnessing the end of air and space power development.

RESEARCH QUESTIONS

1. To what extent has the development of remotely piloted air systems revolutionised air power?
2. Evaluate the view that space has become so important that we should now talk about aerospace power rather than talking about air and space power as separate environments.
3. Do the wars in Afghanistan and Iraq demonstrate that air forces should be configured to perform COIN operations as their main role, or that they should be configured so as to enable intervention without 'boots on the ground'?
4. Does the proliferation of high-technology equipment mean that the advantages traditionally enjoyed by Western air power will inevitably come to an end by the end of the twenty-first century?
5. To what extent is the value of close co-operation between air forces, armies and navies an enduring lesson of air and space power history? And why has this co-operation been so difficult to achieve?

FURTHER READING

- Corum, James S. and Wray R. Johnson, *Airpower in Small Wars* (Lawrence: University of Kansas Press, 2003).
- Gray, Colin S., *Airpower for Strategic Effect* (Maxwell, AL: Air University Press, 2012).
- Hallion, Richard P., Roger Cliff and Phillip C. Saunders, *The Chinese Air Force: Evolving Concepts, Roles and Capabilities* (Washington, DC: National Defence University, 2012).
- Johnson, David E., *Learning Large Lessons: The Evolving Roles of Ground Power and Air Power in the Post-Cold War Era* (Santa Monica: RAND, 2006).
- Lambeth, Benjamin S., *The Unseen War: Allied Air Power and the Take Down of Saddam Hussein* (Annapolis, MD: Naval Institute Press, 2013).
- Meilinger, Phillip S. (ed.), *Paths of Heaven: The Evolution of Air Power Theory* (Maxwell: Air University Press, 1997).
- Olsen, John Andreas (ed.), *Global Air Power* (Dulles, VA: Potomac Books, 2011).
- Olsen, John Andreas (ed.), *European Air Power: Challenges and Opportunities* (Dulles, VA: Potomac Books, 2014).
- Pape, Robert A., *Bombing to Win: Air Power and Coercion in War* (New York: Cornell University Press, 2006).
- Sadeh, Eligar (ed.), *Space Strategy in the 21st Century: Theory and Policy* (London: Routledge, 2012).
- Sheehan, Michael, *The International Politics of Space: No Final Frontier* (London: Routledge, 2007).
- Singer, P. W., *Wired for War: The Robotics Revolution and Conflict in the 21st Century* (New York: Penguin, 2009).
- Strawser, Bradley J., *Killing by Remote Control: The Ethics of an Unmanned Military* (New York: Oxford University Press, 2013).

ONLINE RESOURCES

See [Chapter 10](#).

NOTES

1. The maximum speed of combat aircraft has, in fact, decreased since the retirement of the Lockheed SR-71 'Blackbird' reconnaissance aircraft, which reached a top speed of just over 2,000 mph, three times the speed of sound (Mach 3). Experience from the Vietnam War demonstrated that supersonic speed was useful but rarely reached in combat, and the fastest aircraft since that time have a maximum speed of around 2.5 times the speed of sound.
2. Robert Coram, *Boyd: The Fighter Pilot Who Changed the Art of War* (Little, Brown, 2002), p. 307.
3. See, amongst others, George H. W. Bush and Brent Scowcroft, *A World Transformed* (New York: Alfred A. Knopf, 1998); 'New World Order Galloping Into Position', *Washington Post*, 25 February 1990; Odd Arne Westad, *The Global Cold War: Third World Interventions and the Making of Our Times* (Cambridge University Press, 2007); T. V. Hall and John A. Paul, *International Order and the Future*

of *World Politics* (Cambridge University Press, 1999). The concept of 'New World Order' is not new: President Woodrow Wilson spoke of one during the First World War.

4. As an example, the Royal Air Force ended the Cold War with thirty squadrons of fighter and attack aircraft. At time of writing (mid 2014) it has six. This reduction was not matched with a reduction in the number of commitments to conflicts.
5. Daryl G. Press, 'The Myth of Air Power in the Persian Gulf War and the Future of Warfare', *International Security*, 26(2) (Fall 2001), 5–44.
6. John Andreas Olsen, *Strategic Air Power in Desert Storm* (London: Frank Cass, 2003), pp. 269–74.
7. Rick Atkinson, *Crusade: The Untold Story of the Persian Gulf War* (Boston: Mariner Books, 2004), pp. 288–9.
8. See Ivo H. Daalder and Michael E. O'Hanlon, *Winning Ugly: NATO's War For Kosovo* (Washington: Brookings Institution, 2000) for a contemporary analysis and David L. Phillips and Lawrence Burns, *Liberating Kosovo: Coercive Diplomacy and U.S. Intervention* (Cambridge, MA: MIT Press, 2012).
9. Julian H. Tolbert, 'Crony Attack: Strategic Attack's Silver Bullet?' (Master's Thesis, School of Advanced Air and Space Studies, 2002) (www.dtic.mil/dtic/tr/fulltext/u2/a462291.pdf); Stephen T. Hosmer, *Why Milosevic Decided to Settle When He Did* (Santa Monica: RAND, 2001), p. 75 (www.rand.org/content/dam/rand/pubs/monograph_reports/2001/MR1351.pdf).
10. As an extreme example, shortly after the end of the campaign, the author was told by a British Army officer that the talk of a land option had been the cause of Milosevic's capitulation, thus proving that land power is *always* the decisive factor in war, and showing air power's strategic irrelevance. Even with the benefit of a much greater evidential base today, this seems excessively parochial.
11. Human Rights Watch, 'The Crisis in Kosovo' (www.hrw.org/reports/2000/nato/Natbm200/01.htm) and 'Final Report to the Prosecutor by the Committee Established to Review the NATO Bombing Campaign against the Federal Republic of Yugoslavia' (ICTY, June 2000), paras. 71–8 (www.icty.org/x/file/Press/nato061300.pdf).
12. Michael Russell Rip and James M. Hasik, *The Precision Revolution: GPS and the Future of Aerial Warfare* (Annapolis, MD: Naval Institute Press, 2002), pp. 210–11.
13. Human Rights Watch, 'The Crisis in Kosovo'.
14. The phrase was first used in 1982 by the Israeli prime minister, Menachem Begin – a 'war of necessity' being one fought to maintain a state's existence, or where a vital national interest could only be defended by the use of force.
15. Benjamin S. Lambeth, *Air Power against Terror: America's Conduct of Operation Enduring Freedom* (Santa Monica: RAND, 2005), pp. 204–31 (www.rand.org/content/dam/rand/pubs/monographs/2006/RAND_MG166-1.pdf).
16. Mark J. Conversino, 'Operation DESERT FOX: Effectiveness with Unintended Effects', *Air and Space Power Journal, Chronicles Online Journal* (July 2005) (www.airpower.maxwell.af.mil/airchronicles/coj/ccconversino.html).
17. Harlan K. Ullman *et al.*, *Shock and Awe: Achieving Rapid Dominance* (Washington: National Defense University, 1996); Harlan K. Ullman and James Wade, 'Rapid Dominance: A Force for All Seasons', *RUSI Whitehall Paper Series*, 43(1) (1998).
18. Donald Rumsfeld, *Known and Unknown: A Memoir* (New York: Sentinel, 2011), pp. 459–61.

19. Anthony H. Cordesman, *The Iraq War: Strategy, Tactics and Military Lessons* (Washington: Center for Strategic and International Studies, 2003), p. 281.
20. Lt. Col. Stephen Collins, 'Mind Games', *NATO Review* (Summer 2003) (www.nato.int/docu/review/2003/issue2/english/art4.html).
21. Carl Conetta, 'Catastrophic Interdiction: Air Power and the Collapse of the Iraqi Field Army in the 2003 War' (www.comw.org/pda/fulltext/0309bm30.pdf); David E. Johnson, *Learning Large Lessons: The Evolving Roles of Ground Power and Air Power in the Post-Cold War Era* (Santa Monica: RAND, 2007), pp. 105–36.
22. Lara M. Dadkhah, 'Close Air Support and Civilian Casualties in Afghanistan' (<http://smallwarsjournal.com/blog/journal/docs-temp/160-dadkhah.pdf?q=mag/docs-temp/160-dadkhah.pdf>); Phillip S. Meilinger, 'Airpower and Collateral Damage: Theory, Practice and Challenges' (www.ausairpower.net/APA-EMEAC-2013-02.html).
23. 'NATO's Future Joint Air And Space Power', NATO Joint Air Power Competence Centre, April 2008, p. 4 (www.japcc.org/research/Documents/080424_NFJASP_Final.pdf).
24. Damian Killeen and David Jordan, 'RPAS: Future Force or Force Multiplier? An Analysis of Manned/Unmanned Platforms and Force Balancing', *RAF Air Power Review*, 16(3) (Autumn/Winter 2013), 10–29 (www.airpowerstudies.co.uk/sitebuildercontent/sitebuilderfiles/APRVol16no3).
25. Anthony H. Cordesman, 'Air Combat Trends in the Afghan and Iraq Wars', Center for Strategic and International Studies, March 2008 (www.csis.org/media/csis/pubs/080318_afgh-iraqairbrief.pdf).
26. Gregory Billman, 'The Inherent Limitations of Space Power: Fact or Fiction?', in Bruce M. DeBlois (ed.), *Beyond the Paths of Heaven: The Emergence of Space Power Thought* (Maxwell, AL: Air University Press, 1999), p. 511.
27. Benjamin S. Lambeth, *Mastering the Ultimate High Ground: Next Steps in the Military Uses of Space* (Santa Monica, CA: RAND, 2003), p. 101.
28. *Ibid.*, p. 118.
29. *Ibid.*
30. Department of the (US) Air Force, *Global Reach – Global Power* (Washington, DC: Department of the Air Force, 1992), p. 8.
31. Colin S. Gray, *Airpower for Strategic Effect* (Maxwell, AL: Air University Press, 2012), p. 315.





Part V

Irregular warfare

James D. Kiras



13

Key concepts and terms of irregular warfare

Contents

Introduction

Definitions

Irregular warfare

Types of irregular warfare

KEY THEMES

- Irregular warfare is primarily about politics and organisation. Violence plays a role in irregular warfare, as in all forms of war, for political purposes. The immediate purpose of violence is to demonstrate the political ineptitude of the ruling government and as a tool to intimidate and coerce populations.
- The ultimate goal of irregular warfare, regardless of its type, is political power for the purposes of political, social, economic and/or religious change.
- The character of irregular warfare is dynamic but its nature, outlined above, is not. This character is shaped by social, environmental and technological factors. Insurgents, terrorists and revolutionaries have adopted irregular warfare methods for a key pragmatic reason: to offset their military and organisational weaknesses.
- Insurgency and terrorism will likely continue to be the most prevalent forms of irregular warfare now and in the future for a number of reasons.

Introduction

In order to achieve victory in warfare, most political leaders have relied historically upon their military leaders to use their various armed forces – armies, navies and, later, air forces – as effectively and efficiently as possible. As has been discussed in preceding chapters, the destruction of enough of an adversary's armed forces or core capacity in a decisive operational campaign is the traditional manner of convincing political leaders to seek an end to hostilities. US or other Western-led offensive campaigns throughout the last decade are textbook examples of how to fight a joint military campaign effectively and economically and include: Operation Enduring Freedom (Afghanistan, 2001),

Operation Iraqi Freedom (Iraq, 2003) and Operation Unified Protector (Libya, 2011). The initial phases of all three operations made use of available and in some cases highly limited, land, sea, air, space, cyberspace and special operations capabilities in an integrated manner to unbalance and unhinge Taliban, Iraqi and Libyan forces. The net result of all three joint applications of military power was the toppling of the various regimes in all three cases in a matter of days or months at a minimal cost in friendly casualties.

Despite this demonstrated conventional military prowess, coalition forces in Iraq, Afghanistan, Libya and elsewhere seem hard pressed to deal effectively or efficiently with the irregular threats that have appeared once the ruling regimes were removed. Irregular violence in Iraq, for example, characterised as terrorism, insurgency, sectarian violence and civil war, lasted more than eight years. During that time, coalition forces suffered almost 5,000 deaths and the United States alone had more than 30,000 wounded. In Afghanistan, NATO forces have taken almost 30,000 casualties combined (dead and wounded). These numbers do not factor in the casualties of local security forces, private contractors or civilians. The financial costs for Western nations involved in prolonged actions against irregular threats over the past decade are staggering, with some estimates ranging as much as US\$4 trillion.¹ Despite the payment in national blood and treasure, the results or benefits of such actions are hardly encouraging. Only three years after US forces withdrew from Iraq, irregular violence not only continued, but the insurgents of the Islamic State of Iraq and Syria (ISIS) swept away most local armed forces and security units in Western Iraq trained by US and coalition forces in a matter of weeks in June and August 2014. The scheduled withdrawal of US forces from Afghanistan will also likely signal a return to civil war that has plagued the country for many decades.

Interest in irregular warfare has been extremely high over the past decade. For some, serious consideration of irregular warfare is long overdue. They have argued hidebound military officers and organisations continue to organise, train and equip for conventional wars that are increasingly unlikely and anachronistic. They add that Western militaries have a long tradition of fighting irregular opponents and irregular warfare has been the norm, rather than the exception historically. Irregular warfare is fundamentally different from conventional warfare, such advocates charge, in that force plays a greatly diminished role compared to security, development and good government. One of the paradoxes of such warfare is 'some of the best weapons for counterinsurgents do not shoot' as the struggle is largely political and cognitive, as opposed to military and physical.² More effective approaches to irregular warfare begin with understanding of the specific cultural or religious dimensions of the 'human terrain' in which friendly forces will be operating (see [Figure 13.1](#)). An inability or unwillingness of Western military forces to understand or value ethnic and religious differences as well as local language, culture or norms, such as the Afghan tribal code of Pushtunwali, suggests that any potential responses to irregular threats cannot succeed.³ Another cultural explanation offered relates to how



Figure 13.1 Understanding the 'human terrain' in irregular warfare often is the difference between success and failure. Obtaining sufficient understanding quickly enough, particularly local information about insurgents or terrorists, is often the greatest challenge for external powers assisting host nations and their security forces in dealing with internal threats. This photograph, taken on 24 March 2002, shows Philippine Scout Ranger Carmen Binasang Ambrocio, Jr, a member of the Philippine Citizen Armed Forces Geographical Unit (CAFGU) at Harvest City, Basilan. This unit has citizens who volunteered to join the Armed Forces of the Philippines Scout Rangers in combating the Abu Sayyaf Rebels in their fight against terrorism.

organisations innovate and adapt. Irregular opponents are characterised by their speed of decision-making, adaptability and flexibility in the field. In order to prevail against irregular opponents, Western militaries should embrace their organisational characteristics as embodied in the counterterrorism phrase 'It takes a network to defeat a network.'⁴

The unsatisfying and indecisive results of the past decade have led others to critique not only the conduct of Western militaries in irregular warfare but the very concept itself. War is war they argue; adding adjectives such as 'conventional', 'irregular' or 'unconventional' is unnecessary and unhelpful. The problem with specific forms of irregular warfare, such as counterinsurgency, is that they become operational concepts or, even worse, tactics elevated to the level of strategy.⁵ Critics charge that such wars have no conceivable end in sight, a point reinforced by policy statements alluding to 'the long war' and vaguely worded US authorisation to use military force against terrorists globally. At the heart of such criticism is that irregular wars are largely local wars fought for local grievances that outsiders cannot possibly understand. They constitute wars of choice, as opposed to wars of necessity, and divert attention from preparation or uses of force that really matter given much higher stakes of failure.

Yet another group sees irregular warfare as a signal of the change of the nature or character of war itself. Authors in this camp argue irregular warfare is effective in achieving desired political effects in the most economical way possible. Insurgent or terrorist groups leverage information technologies to reach out to constituents locally, regionally and globally to recruit manpower, gain support and obtain resources. Group members can hide amongst local populations, operate in ungoverned spaces and move seamlessly across borders. The net effect of these social and technological changes, combined with the power of modern media, creates either 'fourth-generation warfare' (4GW), in which insurgents leverage technologies and asymmetric tactics against the political, social and economic weaknesses of post-industrial states or an 'insurgent archipelago' difficult for Western democracies to defeat.⁶ Such warfare can be exceptionally cheap to wage but expensive to counter. Members of Al-Qaeda in the Arabian Peninsula, for example, bragged that their plot to destroy UPS aircraft through bombs contained in printer cartridges cost the organisation \$4,200 against the cost of millions in additional cargo-screening measures.⁷ Irregular warfare is not limited to sub-state terrorist or insurgent groups as a cost-effective means of accomplishing objectives. Russia was able to annex the Crimean Peninsula from Ukraine in a matter of weeks in 2014, using a deft mix of local proxy forces and militias calling for self-determination backed by Russian special operations and other military forces.

Do the conflicts in Ukraine, Yemen, Iraq, Syria and Afghanistan herald a fundamental change in the nature of war? Is irregular warfare the only form of war that will be waged in the twenty-first century? Why are the armed forces of democracies, which have a well-demonstrated ability to project combat power, frustrated and seemingly powerless in the face of irregular adversaries? The

three chapters on irregular warfare in this volume suggest there is nothing mystical or even uniquely new about the rationale for adopting irregular warfare.

In order to demystify irregular warfare, this chapter first defines and distinguishes between its key terms of reference. This straightforward task is much more difficult than it appears at first glance. Indeed much of the confusion related to irregular warfare results from the misunderstanding and misapplication of its core terminology. With the core terms and concepts firmly in hand, [chapter 14](#) surveys various theories of irregular warfare from the perspective of those who seek to exploit insurgent and terrorist violence. The key theme of this review is that irregular warfare methods themselves have rarely been sufficient to achieve meaningful change. Historically, insurgents and terrorists have taken one of two approaches depending on the specific context of the conflict. The first approach is simply to outlast and force the eventual withdrawal of an occupying adversary, often after the latter has experienced considerable political frustration or exhaustion. The second approach has been to translate internal support from the population, as well as external support from sponsors, into political and military capabilities that overwhelm the forces of its adversaries. From this perspective on irregular warfare [chapter 15](#) changes the perspective and looks next at the challenges confronting those who seek to defeat insurgent and terrorist groups. The ‘science’ of countering insurgents and terrorists has been established historically through a number of well-known and, some might argue, self-evident principles. Difficulties in counterinsurgency and counterterrorism, as in all other forms of warfare, lie largely in attempting to apply such principles thoughtlessly without accounting for the unique context and changing circumstances of the conflict – in other words, failing to grasp the nature of the struggle at hand or the ‘art’ of strategy.

Definitions

Irregular warfare

Defining irregular warfare, much less its component elements, is no easy task. A number of problems plague attempts to define irregular warfare clearly and concisely. In some cases, authors are inclined to lump everything together under a concept to the point where a term describes everything but explains nothing. Examples include ‘insurgency’ and ‘asymmetric warfare’. At the other end of the spectrum are scholars and soldiers who create categorical definitions as they try to separate the terminology of irregular warfare with etymological and mechanical precision, such as the canon of US Joint Doctrine. Such definitions suit the purposes of specialised disciplines but have little recognition or application outside of their individual communities. In addition, new terms are created or revived in an attempt to capture the realities of the most recent manifestations of violence. According to US doctrine, the preferred term has changed from ‘counter guerrilla operations’ through ‘counterinsurgency’⁸ and ‘stability

operations' back to 'foreign internal defense'.⁹ On the surface there appears to be little that distinguishes them apart. The term 'foreign internal defense', however, was coined in the wake of Vietnam and it captures two essential realities associated with US or other Western countries' involvement in dealing with insurgencies: military operations play a secondary role in countering insurgents and US and Western actions are restricted largely to training, advising and equipping host nation forces; and US and Western freedom of action and ability to direct the course of operations is not only subject to domestic political considerations but likely subordinated to those of the leaders of the nation trying to defeat the insurgents. Organisations, professional communities and influential individuals potentially become wedded to terms that describe their understanding of a phenomenon (the '3 Ds' of counterinsurgency, for example) and reflect their particular functional focus or biases. As a result, seemingly innocuous language can lead to either misunderstanding, vociferous disagreement in policy and planning circles or confusion amongst allied nations.

Language, and in particular specific terms, presents other challenges in attempting to define irregular warfare. The two most daunting obstacles are the changing connotations of specific terms and the subjective and often emotional qualities associated with them. For example, Bruce Hoffman has charted the evolution in meaning behind the term 'terrorism' from state violence against the population to the violence used by sub-state groups for political purposes.¹⁰ Insurgency was first used to describe insurrections, or violence with limited aims and duration, but the term now conveys motivation, organisation and a degree of popular support. Although this might appear to be an exercise in definitional hair-splitting, words have meanings that in turn can influence how and what actions are taken. For several months after the fall of Baghdad in 2003, senior leaders within the office of the secretary of defense forbade the use of the term 'insurgency' to describe the violence lest it appear to be anything more threatening than the last vestiges of a dead regime. More recently, the national security debate has focused on the implications of a counterterrorism approach to local or global irregular threats as opposed to a counterinsurgency one.¹¹ The subjective value associated with specific terms can also lead to misunderstandings and disagreement over collective action that should be taken between governments or agencies. Differences of opinion over what should be labelled 'terrorism' have nothing to do with the form of violence employed. What colours perceptions is the fact that innocent civilians are often the victims of indiscriminate violence that strikes an emotional chord. In addition, scholars and decision-makers are likely to disagree over whether such violence should be rationalised and legitimised. One consequence has been an inability to forge meaningful consensus on what constitutes terrorism which, in turn, has impeded attempts to proscribe it internationally.¹²

One method of defining irregular warfare and its associated concepts relates to the purpose for which the violence, regardless of its form, is used. Ultimately, irregular warfare has at its core a cause based on grievances. The range of potential grievances and motivations behind them are numerous and can

Table 13.1 *Types of insurgencies**

Anarchist	Egalitarian	Traditionalist
Apocalyptic-Utopian	Pluralist	Secessionist
Reformist	Preservationist	Commercialist

* Bard E. O'Neill, *Insurgency and Terrorism: From Revolution to Apocalypse*, rev. edn (Washington, DC: Potomac Books, 2005), pp. 19–29.

include ethnic and religious persecution, foreign occupation or domination, economic disparity and other forms of injustice that are real or merely perceived. Bard O'Neill has offered a comprehensive categorisation of types of causes for which groups are willing to commit violence (see Table 13.1).

As Table 13.1 suggests, some groups conduct irregular warfare to weaken the existing order and destroy it (anarchists), or profit from chaos (commercialist) or prefer to break away from the existing order and establish one of their own (secessionist). Others believe the existing order can be saved (reformist) or changed to serve traditional norms (traditionalist) or privileges (preservationist). Still others believe a more just and equitable society (egalitarian) can be created, while some believe that they are anointed to mete out religious rewards or punishments (apocalyptic-utopian). Groups can espouse a single cause, blend together one or more of the above and shift from one cause to another over time. Almost all groups seek to change the existing system whether on a national, regional or global basis. The only way for most groups to change the status quo is establish their own territory, or seize control of the existing system by some means and redistribute power or remake society according to their ideology. For the Taliban in Afghanistan, control means enacting Sharia law along strict Wahhabi lines and replacing Pushtunwali and other tribal codes. Power and control are ultimately political objectives. The violence within irregular warfare is far from random in nature. Instead, it should be subordinated to a political purpose to the maximum extent possible.¹³

Another trait that irregular warfare has in common with other forms of warfare is the idea of dynamic created as a result of the struggle between adversaries. On a theoretical level, each adversary is seeking to impose its will upon the other while avoiding having another will imposed upon them – what Clausewitz termed 'polarity'. What distinguishes war in theory from war in practice is the dynamic that exists within and between adversaries and other actors, the level of information available to each and other contextual variables that give each war its own unique traits and characteristics.¹⁴ The interaction between adversaries, in turn, creates friction, chance and uncertainty that influences the outcome of events and disrupts even the soundest and most elegant strategy.¹⁵ Within irregular warfare, the role played by friction, chance and uncertainty is magnified due to the political nature of the struggle, the character it takes and the resources involved. The key distinction between irregular and other forms of warfare, and different types of irregular warfare, rests on resources and the

ability to translate them into effective capabilities. Groups conducting irregular warfare are attempting to defeat or overcome adversaries that possess significantly more powerful and numerous resources. Most often, sub-state groups are fighting against a state that not only possesses superior resources but also has a legitimate monopoly on violence within its borders. In order to have a reasonable chance of success in any type of irregular warfare, groups must keep their activities hidden from their adversary for as long as possible so as not to be detected, tracked and destroyed.¹⁶ Those who conduct irregular warfare rarely do so as a matter of preference or because they find a semi-clandestine and potentially dangerous lifestyle appealing. Most groups adopt irregular warfare because other, more decisive, forms of political violence are unavailable to them. If given a choice, some insurgent or terrorist leaders would prefer to have the nuclear or conventional resources and capabilities of their adversary to achieve their objectives in order without the need for prolonged struggle.¹⁷ To some extent, the appeal of a group's cause with and the access to a particular support base determines the resources it has available to it. Groups can also forge alliances with others to pool together their resources, as well as seek out state sponsors willing and able to finance, train, equip and support them.¹⁸ Yet few groups are co-ordinated, sophisticated or competent enough, at least in the early stages, to turn resources into capabilities that can overwhelm or coerce their adversary. Due to the nature of their cause, or extremist views, other groups are unable to translate their call for social justice into anything more than a handful of fanatical supporters and modest resources. The quality and quantity of resources a group enjoys have a significant bearing on the type of irregular warfare it can wage, the character of the violence employed and the time it takes to achieve its objectives.¹⁹ To sum up, *irregular warfare* is defined as the use of violence by sub-state actors, groups within states or state actors working by, with or through such groups or actors, for the political purposes of achieving power, control and legitimacy, using unorthodox or unconventional approaches to warfare contrary to prevailing norms, due to a fundamental weakness in resources or capabilities or the desire to limit the escalation of conflict. Other definitions of irregular warfare are contained in [Box 13.1](#).

Types of irregular warfare

The types of irregular warfare dictated by resources can be broken down into five main categories or forms: coup d'état, terrorism, revolution, insurgency and civil war. Each will be explored briefly on the basis of their centre of gravity, mechanism for success, strategic and tactical orientation and duration. [Table 13.2](#) summarises the key attributes of the different types of irregular warfare.

Coup d'état

As a form of irregular warfare, coups d'état are conducted to change the system by overthrowing key facilities of government as quickly and with as little initial

Box 13.1 Definitions of irregular warfare

The term 'irregular warfare' is only one of many coined to describe conflict between those fielding conventional forces and capabilities against an opponent who refuses battle, uses hit-and-run tactics and even targets non-combatants indiscriminately in order to achieve their objectives. W. E. D. Allen wrote on the subject of irregular war prior to the Second World War and his definition is shaped and limited by his studies and experience gained while serving with Lieutenant Colonel Orde Wingate in 1941. Wingate and Allen advised, trained, equipped and fought with Abyssinian irregulars against occupying Italian regulars. The other definition of irregular warfare below was in response to guidance from the 2006 Quadrennial Defense Review to develop an 'umbrella' term within the Department of Defense and other agencies for all its activities that captures both the use of and defence against irregular opponents.

Irregular war is in fact as old as the hills which offer it its best terrain; older, clearly, than 'regular' war which has grown out of it, as the city grew out of the village. It would be difficult to define precisely where irregular war ends and regular war begins. It would be possible to say that irregular war is warfare carried out by forces other than the regular armies of a belligerent power. But this definition is not exact... Irregular and regular war are clearly two aspects of the same activity, but it should be taken as axiomatic that irregular action in the field should always be ancillary to parallel regular action... in no instance have guerrilla operations proved ultimately successful without the regular forces of a friendly power. (W. E. D. Allen, *Guerrilla War in Abyssinia* (New York: Penguin, 1943), p. 18.)

IRREGULAR WARFARE (IW): A violent struggle among state and non-state actors for legitimacy and influence over the relevant populations. IW favors indirect and asymmetric approaches, though it may employ the full range of military and other capabilities, in order to erode an adversary's power, influence, and will...

the central problem of which is:

Adaptive adversaries such as terrorists, insurgents, and criminal networks as well as states will increasingly resort to irregular forms of warfare as effective ways to challenge conventional military powers. Advances in technology and other trends in the environment will render such irregular threats ever more lethal, capable of producing widespread chaos, and otherwise difficult to counter. These threats are enmeshed in the population and increasingly empowered by astute use of communications, cyberspace, and technology, such that their impact extends regionally and globally. Many of these conflicts are essentially contests for influence and legitimacy over relevant populations. (Joint Staff, *Irregular Warfare Joint Operating Concept*, version 2.0 (Washington, DC: The Joint Staff, May 2010), pp. 4, 9.)

Table 13.2 *Attributes of types of irregular warfare**

Type	Resources	Centre of gravity	Mechanism	Strategic orientation	Tactical orientation	Duration
Coup d'état	Few	Elites (organise)	Seizure of power	Offensive	Offensive	Immediate
Terrorism	Few	Elites (influence)	Coercion or provocation	Offensive	Offensive	Lengthy
Revolution	Vanguard, growing to many	Population** (stimulate)	Popular support (provocation and uprising)	Defensive	Defensive	As quickly as possible
Insurgency	Varied, but often significant	Population (control)	Denial leading to victory over or withdrawal of opponent	Defensive, switching to offensive	Offensive given local superiority	Lengthy
Civil war	Varies, positive correlation of forces	Population (mobilisation)	Denial or negotiated settlement	Varies	Varies	Most often lengthy

* I freely acknowledge the intellectual debt I owe to Ariel Merari for this chart. His comparison of forms of insurgency provided the inspiration for my own analysis that differentiates between forms of irregular warfare. See Merari, 'Terrorism as a Strategy of Insurgency', *Terrorism and Political Violence*, 5(4) (Winter 1993), 220.

** I use population throughout in its most general sense, which can include specific segments or diasporas of like-minded groups of individuals bound by cultural, political, economic and/or religious ties.

bloodshed as possible. Coups tend to be instigated by small numbers within the system to ensure the secrecy of the plot and prevent its discovery. For these reasons, conducting a coup d'état requires nerve, timing, planning and luck but little in the way additional resources, as the coup in Mali in 2012 suggests. Often the instigators of coups d'état come from the officer corps, as the military is the one government institution that can quickly crush such acts of rebellion. Those who conduct coups seek to gain power in order to change national policies or correct ethnic or religious exclusion – and gain wealth in the process.²⁰ In some cases, coup plotters seek the active support or tacit approval of other state leaders before they conduct their actions, often to achieve tacit approval or access key resources such as funding, equipment or intelligence.

Terrorism

Terrorism also involves small numbers of individuals, operating in cells and groups, with relatively few resources in order to function. Groups or cells face resource constraints for a number of reasons, including effective security services which constrain group members' freedom of action and causes which appeal only to a slim minority within a population. Acts of terrorism deliberately target the non-combatant civilian population and other symbols associated with the state (military and police forces, government buildings, monuments etc.) or the ruling government's power base in order to spread fear disproportionate to the damage it causes. Groups choose terrorism as a means of using the severely limited resources at their disposal in unorthodox and dramatic attacks, which often are accompanied by self-generated or external media reports, in order to compel or persuade one or more target audiences. Against adversaries, terrorism in its various forms can convince a target population and its leadership that the stakes of a conflict are not worth the current and potential future costs through coercion given their willingness to conduct wanton acts of mayhem and destruction.²¹ For domestic audiences, sympathisers and supporters, including states, acts of terrorism demonstrate a group's power, reach and competence and can facilitate access to resources or even ownership of a particular cause between competing terrorist groups as was the case between Hamas and Fatah in Palestine in between 2005 and 2006.²² In addition, acts of terrorism can influence segments of the domestic population into either action (fleeing or overreacting) or inaction, as was the case with the publicised ISIS execution of captured Iraqi soldiers and policemen in May 2014. Terrorism often ends in one of the following ways: decapitation of the terrorist leadership; a negotiated settlement; success of the terrorist campaign in achieving its objectives; failure by terrorist groups imploding, provoking a backlash or becoming marginalised; repression; or by reorientation of their goals.²³

Revolution

Revolution bears many similarities to a coup d'état, in that it requires an intellectual or technological elite or 'vanguard' to organise and guide its ultimate

success, but it also possesses a number of crucial differences. The primary goal of most revolutions is to paralyse and collapse or overthrow an existing government, in order to redistribute power and resources. There are two critical components to a revolution: a cadre of political organisers that work within a population, and a population that is oppressed, exploited and frustrated enough that they just might turn to protest or violence if the right spark is provided. The cadre of organisers, under the direction of the vanguard, seeks to demonstrate the incompetence of the existing regime, conduct political agitation overtly and covertly and elevate the consciousness of the oppressed population. At the appropriate time, or perhaps as the result of an action either by the revolutionaries or the regime overreacting in response, a critical mass of the population will rise up, overpower the control elements of the regime and allow the vanguard to assume power. Without a sufficient mass of the population behind it, revolutions are doomed to fail. If the vanguard does not impose control should revolution succeed, the result is often prolonged insurgency or civil war between competing factions.

Insurgency

Insurgency has been the subject of recent scrutiny and revision largely as a result of the Global Struggle against Violent Extremism (formerly the Global War on Terrorism), and its characterisation of violent Islamic extremism as a 'global insurgency', and confusion surrounding the character of the violence in Iraq. Some authors suggest that our concept for this type of irregular warfare requires substantial revision and rethinking given the change in context in which insurgencies now occur.²⁴ Insurgency shares a number of characteristics in common with other forms of irregular warfare. Successful revolutions place primary emphasis on organising, educating and proselytising amongst the population to prepare them for the inevitable conflict against the system. Maoist and other agrarian-based insurgencies, for example, also stress the need to develop a 'shadow government', or parallel political structure, which controls the rural population and extracts resources without alienating its crucial support base. The level of support required for insurgencies to sink roots and operate successfully is context-specific, as in the case of clan- or tribal-based societies, in which these social units can be self-supporting. Insurgencies of this type may not require the support of the general population, but rather the spreading of enough instability, uncertainty and fear to prevent the population from assisting the insurgents' adversaries.

While theorists agree that the centre of gravity for an insurgency is popular support, a fundamental difference between revolutions and insurgencies is how resources gained from the population or other sources are utilised. Insurgencies harness resources in order to conduct attacks using guerrilla tactics that are designed to inflict ever-increasing losses on government or occupying forces and tip the balance of forces in the insurgents' favour. Such tactics include hit-and-run raids, ambushes and, more recently, remote attacks using

mortars, rocket launchers and improvised explosive devices. Guerrilla tactics are conducted by small units organised along conventional military lines, such as squads, platoons or companies, on a scale and intensity designed to achieve operational or strategic effects over time: the weakening of the resolve of their political adversaries and the withdrawal of competing, occupying and/or government forces. The insurgent political structure fills the subsequent vacuum and attempts to consolidate its hold over the population by a mixture of intimidation, coercion, co-option or terror. Insurgent groups also recruit amongst the population, levy taxes and convince the local residents that the government or other groups cannot protect them and it is in their best individual and collective interest to support the insurgents actively or passively. Successes for insurgent groups allow them to translate support into larger militias and battalions, regiments and divisions of guerrillas with which to conduct even more widespread and ambitious attacks. These attacks further put their adversaries on the tactical, operational and even strategic defensive. Should the circumstances permit, insurgent groups translate time and resources into conventional or other capabilities that assume the strategic offensive to escalate the conflict into a civil war and end the conflict once and for all.

Insurgency is also one of the most forgiving and flexible forms of irregular warfare in terms of defeat. Local and even regional defeat of guerrilla forces does not spell the end of most insurgencies. The insurgent shadow network structures embedded within the population, including a confederation of informers and supporters, mean even if government or other forces return, the insurgent threat and presence remains. For example, Taliban forces have sent 'night letters' to specifically identified locals in Afghanistan warning them of future consequences of collaboration with coalition or Afghan national forces once they return. Time and space, as well as an external sanctuary such as Pakistan for the Taliban, can allow insurgent groups to recover and recoup their losses. Given the time it takes to weaken its adversaries to the point of collapse, or to tip the balance of forces and translate resources into meaningful capabilities, insurgencies are almost always prolonged in nature. The character of and popularity of the cause championed by insurgent leaders, as well as willingness to compromise on their desired goals, is often critical in determining an insurgency's duration and likelihood of success. Where adversaries are unwilling or unable to compromise or negotiate, the insurgency is almost always likely to end only after one of them submits, is eradicated or collapses.

Civil war

Civil wars differ from other forms of irregular warfare in that they occupy a space between the boundary of conventional and irregular warfare. Within civil wars the struggle is between competing and roughly matched factions within a territory for the control, recognition and legitimacy to govern a state. What makes these wars difficult to categorise is their wildly varying causes, scope and scale. Factions may espouse any one or a number of the causes that Bard O'Neill

suggests (see [Table 13.1](#) above), for example: within societies over differences in policies (the American Civil War, 1861–5) and form of governance (the English Civil War, 1642–51 or the Syrian Civil War, 2011–present); a desire for self-governance (Eritrea from Ethiopia, 1974–91); or disagreement over a negotiated settlement to a conflict (the Irish Civil War, 1922–3). The scope and scale of civil wars are also diverse; a few last for short periods of time while most stretch over several decades and casualties can range from tens of thousands into the millions. In some cases former insurgents fight one another whereas in the most devastating civil wars, factions fielding large armies fight more or less conventionally to impose their control upon one another through the destruction of their adversary's fielded forces. Some reject civil wars as a form of irregular warfare for this reason; others object to its categorisation as such because the adversaries possess nearly the same capabilities. The key distinguishing aspect of civil wars is geographic in nature; such wars exist within identifiable territories between two or more factions that may or may not employ civilian militias, paramilitaries and unorthodox tactics. A contemporary illustration of the problem is the Syrian Civil War, which features the incumbent Ba'athist regime of Bashar al-Assad fighting a loose constellation of groups ranging dramatically in size, capability and cause.

In order to avoid potential confusion and bound the review of the subject, subsequent chapters focus on two forms of irregular warfare: insurgency and terrorism. One reason for this is historical in nature. Revolution, for example, has largely fallen out of favour as a form of irregular warfare over the last half century, the events of the Arab Spring notwithstanding, due to changes in the international economic and political system. Stripped of their former political ideological impetus, and desire to provoke a conflict within the state, the vast majority of revolutions over the past two decades have been largely non-violent in nature.²⁵ The globalisation of information and trade has only increased the economic divide between the post-industrial 'north' and the geographic 'south'. Although working conditions within the nations of the south can be appalling, manufacturing nevertheless creates economic opportunities for local entrepreneurs and workers alike. Organising and maintaining a clandestine mass movement for several years is exceptionally difficult but this has been mitigated, to a degree, by the use of social media, cell phones and peer-to-peer networks. The ideologies which fuelled the bulk of the revolutions over the past three centuries – egalitarianism and Marxism – have either been discredited or discarded partially due to the collapse of their sponsors but also as a result of political and economic progress and reform. The most likely potential ideological heir to the throne of revolution as a form of irregular warfare, as opposed to social or economic protest, is theological: a movement led by militant Salafi or other religious radicals that exploit conditions in a state that is sliding into the abyss of political collapse, as was the case in Iran in 1979. Revolution, in this case, does not occur as a result of careful planning and meticulous preparation. Instead, revolutions of this stripe are opportunistic in nature and rely heavily

on chance and an ability to exploit the unfolding situation. The group that will prevail is the one that is sufficiently organised and that is supported either by enough of the population or by force of arms to seize and consolidate its power base.

The second reason to focus on insurgency and terrorism, as opposed to other forms of irregular warfare, is related to time and geographic scope. Coups d'état occur quickly if they are properly organised and luck favours the plotters. There is little that potential intervening powers can do to halt a coup that is taking place. If the plotters succeed, they become the de facto legitimate governing authority of the state and should be dealt with according to international rules and norms. If the plotters fail, they are likely to be executed by the state as both traitors and a potential future threat. Although morally repugnant to some, coups d'état are viewed by many outside political and business leaders pragmatically as a source of stability in troubled states when military leaders assume power. The coup that brought Pervez Musharref to power in Pakistan, for example, stabilised a weak political system characterised by a succession of corrupt or inept political leaders who were democratically elected. Civil wars, in contrast, are difficult to identify for many of the reasons outlined above. As suggested above, they vary so wildly in geographic and military scope, scale and cause as to be problematic as a definable category of irregular warfare. Most often, civil wars feature an incumbent regime and an insurgent (a single or many groups) vying for power and control. One method of exploring the complexities and identifying the continuities of irregular warfare, to make sense of the present and the future, is to look at its theory and practice throughout history, which is the subject of the [next chapter](#).

RESEARCH QUESTIONS

1. What differentiates irregular warfare from regular or conventional warfare? Does this differentiation matter?
2. Are terrorism and insurgency unique and separate phenomena or are they related? If so, how? If not, why?
3. Can terrorism form the basis of a strategy or is it merely a tactic?
4. Why is legitimacy such an important concept in irregular warfare – and to whom?
5. Should civil wars be considered a type of irregular warfare or are they sufficiently unique?

FURTHER READING

Porta, Donatella Della, *Clandestine Political Violence* (Cambridge University Press, 2013) identifies a number of causal mechanisms that help explain why social groups move towards, persist or move away from clandestine political violence that has a level of applicability across the spectrum of irregular warfare.

Hoffman, Bruce, *Inside Terrorism*, rev. edn (New York: Columbia University Press, 2006) is the best single-volume treatment of various aspects of terrorism, including definitions, motivations, forms of violence and the role of the media.

Kalyvas, Stathis, *The Logic of Violence in Civil War* (Cambridge University Press, 2006) offers a powerful analysis, as well as several theoretical frameworks across different levels of analysis, to determine where selective and indiscriminate violence are most effectively applied in irregular warfare.

O'Neill, Bard E., *Insurgency and Terrorism: From Revolution to Apocalypse*, rev. edn (Washington, DC: Potomac Books, 2005) remains one of the best single-volume analyses of the difference between forms of irregular violence and the factors which shape its character, conduct and response.

ONLINE RESOURCES

The Center for Irregular Warfare and Armed Groups (CIWAG) website (www.usnwc.edu/ciwag/) features a number of useful resources and links, including a number of informative case studies and reports.

The RAND Corporation website (www.rand.org/search/advanced-search.html) has a searchable index of reports on various aspects of irregular warfare, including insurgency and terrorism, going back to 1948. Most contemporary reports are available free for download in several formats optimised for mobile devices.

Various Department of Defense organisations have compiled bibliographies of sources on irregular warfare that are accessible online, including the US Air Force Muir S. Fairchild Research and Information Center (www.au.af.mil/au/aul/bibs/irregular.htm), the US Army War College (www.carlisle.army.mil/library/bibs/IWnov09.pdf), and the US Army Special Operations Command (www.soc.mil/ARIS/13-01654%20ARIS%20IW%20Bibliography-small.pdf).

The online *Small Wars Journal* (<http://smallwarsjournal.com/>) features short articles and blog postings by subject matter experts and practitioners that are highly informative and provocative.

NOTES

1. Daniel Trotta, 'Cost of War at Least \$3.7 Trillion and Counting', Reuters.com (29 June 2011) (www.reuters.com/article/2011/06/29/us-usa-war-idUSTRE75S25320110629).
2. Field Manual (FM) 3-24/Marine Corps Warfighting Publication (MCWP) 3-33.5, *Counterinsurgency* (Washington, DC: Department of the Army, 2008), para. 1-153.
3. See, for example, Montgomery McFate, 'The Military Utility of Understanding Adversary Culture', *Joint Force Quarterly*, 38 (3rd Quarter, 2005), 42-8; 'Honor Among Them', *The Economist*, 381(8509) (23 December 2006), 36-40.
4. Stanley McChrystal, 'It Takes a Network: The New Front Line of Modern Warfare', *Foreign Policy* (22 February 2011) (www.foreignpolicy.com/articles/2011/02/22/it_takes_a_network).

5. Emile Simpson, *War From the Ground Up* (New York: Columbia, 2012), p. 144.
6. 4GW is the central concept within T. X. Hammes, *The Sling and the Stone* (New York: Zenith, 2006). On the concept of the insurgent archipelago, see John Mackinlay, *The Insurgent Archipelago* (New York: Columbia University Press, 2009), esp. pp. 221–36.
7. The notional costs of the attack, in terms of supplies, is broken down in Yahaya Ibrahim, '\$4,200', *Inspire* (November 2010), p. 15.
8. FM 3-24, *Counterinsurgency* (Washington, DC: Headquarters, Department of the Army, December 2006).
9. AFDD 2.3–1, *Foreign Internal Defense*, Coordination Draft (15 May 2007), supercedes AFDD 2–3.1 (10 May 2004).
10. Bruce Hoffman, *Inside Terrorism*, revised and expanded edn (New York: Columbia University Press, 2006), pp. 3–20.
11. A 2008 debate regarding these choices for Afghanistan is contained in Bob Woodward, *Obama's Wars* (New York: Simon and Schuster, 2010), pp. 165–9.
12. A key US policy goal is to build a global antiterrorist environment or consensus, in which terrorism is recognized as an international scourge. Analogies used to inform this policy are piracy and slavery; both still exist, but are no longer seen as credible or legitimate. See, for example, the recent collection of essays in Joseph McMillan (ed.), *'In the Same Light as Slavery': Building a Global Antiterrorist Consensus* (Washington, DC: National Defense University Press, 2006).
13. See esp. ch. 10, 'Intimacy', in Stathis Kalyvas, *The Logic of Violence in Civil War* (Cambridge University Press, 2006).
14. This point is unfortunately misconstrued by Emile Simpson in his otherwise exceptional work. See Simpson, *War From the Ground Up*, pp. 54–66.
15. Carl von Clausewitz, *On War*, ed. and trans. Michael Howard and Peter Paret (Princeton University Press, 1976), pp. 89, 119–21.
16. James Kiras, 'Irregular Warfare: Terrorism and Insurgency', in John Baylis *et al.* (eds.), *Strategy in the Contemporary World*, 2nd edn (Oxford University Press, 2007), p. 179.
17. Osama bin Laden repeated the desire to obtain legal sanction for and acquisition of nuclear weapons. See, for example, 'Jihad against Jews and Crusaders' first published in *al-Quds al-Arabi* in London on 28 February 1998 (www.fas.org/irp/world/para/docs/980223-fatwa.htm) as well as the 2003 fatwa cited by bin Laden. Nasir Bin Hamd Al-Fahd, 'A Treatise on the Legal Status of Using Weapons of Mass Destruction against Infidels' (May 2003).
18. Daniel Byman does justice to this exceptionally difficult subject in *Deadly Connections: States that Sponsor Terrorism* (Cambridge University Press, 2005).
19. Weinstein concludes resource-rich rebels are likely to abuse local populations because they can afford to do so. See Jeremy Weinstein, *Inside Rebellion* (Cambridge University Press, 2006).
20. See Edward Luttwak, *Coup d'État: A Practical Handbook* (New York: Knopf, 1969).
21. Robert Pape, *Dying to Win* (New York: Random House, 2005), pp. 20–3.
22. Mia Bloom, *Dying to Kill* (New York: Columbia University Press, 2005), pp. 76–85.
23. Pathways identified and explored in Audrey Kurth Cronin, *How Terrorism Ends* (Princeton University Press, 2009).

24. See Mackinlay, *The Insurgent Archipelago*; David Kilcullen, *The Accidental Guerrilla* (Oxford University Press, 2009), esp. 1–38.
25. Examples include Czechoslovakia's 'Velvet Revolution' (1989), Serbia's 'Bulldozer Revolution' (2000), Georgia's 'Rose Revolution' (2003), Ukraine's 'Orange Revolution' (2004–5), Lebanon's 'Cedar Revolution' (2005), the various revolutions of the 'Arab Spring' (2010) and, most recently, the abortive Hong Kong 'Umbrella Revolution' (2014).



14

The historical practice of irregular warfare

Contents

Understanding modern irregular warfare through the lens of the past

Johann Ewald and partisan war

Karl Marx, Louis Auguste Blanqui and socialist revolution

Charles Callwell and imperial soldiering

The First World War, Paul von Lettow-Vorbeck and T. E. Lawrence

Mao Zedong and protracted people's war

The Second World War

KEY THEMES

- Understanding the continuity and differences between historical rationale and practice, and not just the most recent era of lessons learned, is the key to success in current and future irregular wars.
- Striking a balance between gaining the willing and demonstrated support of the population, versus simply controlling and coercing it, is a consistent theme historically. Brute force has been effective in the short term, but its use creates more problems than it solves.
- Traumatic events for those conducting or countering it, particularly failure, are often the catalyst for reflection and an upsurge in interest in and writing on irregular warfare.
- The most effective practitioners of irregular warfare historically accurately assess the subjective and objective conditions for success within its specific context, or provide a comprehensive roadmap that links tactical action to strategic purpose.

Understanding modern irregular warfare through the lens of the past

The study of irregular warfare historically presents a number of unique challenges. Although the history of irregular warfare, in the form of bandit raids or other unconventional tactics, arguably existed long before more organised, conventional warfare by fielded armies, there are relatively few memoirs, much less theories of violence, by those conducting irregular warfare against militarily superior opponents prior to the nineteenth century. Classic texts on ancient

warfare from across the globe, including Maurice's *Strategikon* (Byzantium), Sun Tzu's *The Art of War* (China) and Kautilya's *Arthashastra* (India), addressed unconventional methods of fighting such as raids, ambushes, stratagems and ruses as a method of gaining an advantage over an opponent prior to or during battle.¹ Other ancient texts, including epic poems, theological texts and historical narratives, suggest that irregular tactics were used to overcome formidable defences. The most famous example remains the Trojan Horse in Homer's *Iliad*, but far more accounts describe a common subterfuge: taking walled cities by using traitors inside to open gates for armies waiting outside.² In addition, some texts discuss the specific fighting qualities, or what scholars now call 'strategic culture' or 'way of war', of irregular or barbarian opponents.³ Indeed, some scholars have gone so far as to suggest that civilisations have distinct styles of fighting, either as a result of geography or as an explanation for their political and economic success.⁴ Ancient and medieval texts, however, must be approached with caution. Amongst their problems are author bias, hearsay or fabricated evidence, lack of supporting evidence and, in many cases, few other surviving corroborating or contradictory accounts.⁵ Scholars and practitioners are forced to either read between the lines of existing accounts and manuals on the art of war or make inferences from histories of powers that were confronted by and defeated irregular threats.⁶ With the benefit of hindsight and contemporary knowledge, authors are also inclined to interpret portions of ancient texts as evidence of the historical practice and theoretical continuity of irregular warfare. For example, contemporary scholars suggest that the public and dramatic assassinations conducted by Jewish Zealots against leading Roman citizens and subjects, or by the Nizari Ismaili Assassin sect against medieval Christian and Muslim rulers, were not just acts of terrorism but the first recorded acts of suicide terrorism.⁷ Not only was the context of such attacks completely different, but so too were the targets and intended effects. It is crucial not to cast modern connotations onto ancient texts without explaining key differences and providing sufficient background information.

Out of necessity, this chapter will survey the most important developments in irregular warfare theory or doctrine historically. Works by a number of other authors provide comprehensive surveys of irregular warfare campaigns and thinkers.⁸ For the purposes of this chapter, irregular warfare as it is understood in the modern context begins with the end of limited war in the West in the eighteenth century. A number of social, economic, political and military changes spurred the development of irregular warfare from the eighteenth century until the end of the Second World War. The first change, associated with the ideas of the Enlightenment, was the desire to replace feudal monarchies with a more representative, equitable and just form of government. Western armies during the early and mid eighteenth century were comprised of limited numbers of professional, uniformed soldiers, organised into battalions and regiments, whose drill, tactics and battlefield manoeuvres were highly stylised and scripted, and little had changed since the first manual-of-arms for pike,

harquebus and musket was published in 1607.⁹ Monarchs feared arming the citizenry, which was comprised largely of peasants. This fear was based on the thought that armed peasants would reject the status quo, organise and overwhelm the small standing forces of the monarchs. Such philosophers as John Locke questioned the moral and ethical basis of non-representative forms of government including monarchies. Locke's libertarian ideas influenced thoughts on democracy amongst revolutionaries such as Alexander Hamilton and Thomas Jefferson, amongst others. Hand in hand with the development of the idea of representative government, such as a democracy, was the rise of one of the most potent forces to mobilise the population for war: nationalism. The rise of nationalism, and threats to nascent revolutionary governments in the American colonies and in France in the late eighteenth century from standing professional armies, led in turn to the development of conscription and the concept of 'the nation in arms'.

The American Revolutionary War (1775–83) is important in the modern history of irregular warfare. The details of this war, including the role that the citizen-soldier militia of 'Minutemen' played in securing victory, are well known.¹⁰ Most Americans are familiar with the basic details of the conflict but time, popular culture and national myth have shaped the public understanding. The war had its fair share of brutality, including massacres committed by Continental and British forces alike.¹¹ Some firebrands at the time claimed that the British should have been more brutal in dealing with the upstart colonial population.¹² The war featured elements of irregular warfare familiar to the modern reader: unorthodox tactics used by colonial troops that had been shaped in fighting against native American tribes and in the North American theatre during the Seven Years War against the French (1756–63);¹³ and unconventional campaigns, including the pivotal insurgency conducted in South Carolina by partisan leaders such as Andrew Pickens and the most famous guerrilla leader during the campaign, Francis Marion.¹⁴ These aspects of the American Revolutionary War are not what make it unique. Indeed, the Austrian Hapsburg Empire fought campaigns against irregulars on its eastern borders. The English and French, as well as the Austrians, developed light infantry and other skirmishing forces almost a century prior to the American revolutionary war, and Robert Rogers established 'Standing Orders' for his famed irregular Rangers in 1759.¹⁵ What marks the American Revolutionary War as unique is the mark the fighting left on a Hessian officer, Johann Ewald.

Johann Ewald and partisan war

Johann Ewald served as an officer in an elite *jäger* regiment recruited to fight on the side of the British Crown against the rebels. *Jäger* were light infantry skirmishers who operated in advance or independent of main force combat units and were responsible for screening their movements, gathering information, and conducting probing attacks and raids against enemy camps and units. The

Box 14.1 What is doctrine?

Scholars agree on the purpose of doctrine, but not on its form. The purpose of doctrine is to transmit and share 'best practices' or 'lessons learned' about the conduct of warfare. But how are these practices and lessons codified, understood and transmitted? Andrew Birtle suggests that US Army counterinsurgency doctrine does not have to be published but can consist instead of 'custom, tradition, and accumulated experience that was transmitted from one generation of soldiers to the next through a combination of official and unofficial writings, curricular materials, conversations, and individual memories'.¹⁶ This chapter suggests that doctrine must be written down and published to be considered as such.

fighting in British North America made a deep impression on many officers during the conflict, including the remarkably introspective Ewald.¹⁷ Upon his return to Cassel in 1784, Ewald penned his *Abhandlung über den kleinen Krieg*, or *Treatise on Partisan War*, one of the first manuals, or what we would call today 'doctrine' (see Box 14.1) on the unique character of and requirements for conducting and countering irregular warfare in the context of his day. This work, as well as Ewald's other published works, including his diary, offer considerable insights into both the continuities of and changes in modern irregular warfare from the eighteenth century until today.¹⁸

As in works by other officers of his generation, Ewald offers a number of prescriptions on how to prepare and organise for, and conduct, partisan operations. However he insists that different skills and a different mindset and approach are necessary to succeed in partisan warfare. Anticipating more recent critiques of British performance in Iraq and Afghanistan, Ewald suggests that British troops 'were not really suitable for the small war, because they did not have sufficient patience for this difficult and laborious kind of warfare'.¹⁹ Unlike the conventional units of his day, partisan leaders require independent thought and action in order to identify and seize opportunities, prevent ambush, retain mobility and make the best possible use of the terrain.²⁰ Although Ewald's advice was bound by the norms and codes of conduct of his day, many of his recommendations at the tactical level for light infantry or special operations forces commanders either still ring true in modern doctrine or remain applicable today, including the commander's need to identify patterns in rebel attacks and quickly adapt and develop innovative countermeasures against them.²¹

Ewald differs from his contemporaries in his passing discussions on the nature of irregular warfare. The revolutionary ideals that inspired the American colonists had a significant impact on their operations. For example, Ewald was amazed when he observed the suffering American forces were willing to endure, and the minimal supplies they required to fight effectively, for the cause of 'Liberty and Independence'.²² As colonial powers were to discover after the end of the Second World War, and the United States and its coalition

partners discovered in Iraq, ideals such as nationalism and self-determination are potent motivating forces and extremely difficult to defeat unless one is willing to either apply relatively unrestricted force or provide an equally powerful, contrasting set of ideas. On the use of force and relations with the local population, Ewald's works highlight the dilemma that confronts all powers fighting irregular wars. To suggest that Ewald discusses 'winning the hearts and minds' of the population is to read too much into his observations and prescriptive advice. The dilemma confronting Ewald is how to gain potentially valuable support, including information about local irregular forces, from the population without alienating them. His works are filled with references and incidents where acts of kindness towards the population, even known revolutionary leaders, yielded valuable information about the condition or location of American forces.²³ Trusting the population blindly, without verifying information gained through reconnaissance and other sources, was a formula for potential betrayal and disaster.²⁴ But Ewald also did not shirk from using force against the population – 'the place was ransacked and plundered because all the inhabitants were rebellious-minded' – either to punish known rebel sympathisers or when necessity dictated.²⁵ He understood, however, that force used out of necessity came at the cost of alienating support from segments from the population that either had supported or were willing to support his forces at the time.²⁶ The need to strike a balance between gaining the willing and demonstrated support of the population, versus simply controlling and coercing it, is a consistent theme historically that separates those who have succeeded in irregular warfare and those who have failed.

The history of irregular warfare from the end of the eighteenth century until the First World War (1914–18) is filled with considerable political, social, economic and military change. The most violent episode in Western history during the nineteenth century, the Napoleonic Wars (1798–1815), saw the concept of 'the nation in arms' lead inexorably to much more destructive, total warfare and the harnessing of popular sentiment in raising armies against the occupying armies of Napoleon. The combination of weather, inadequate French supplies, vast geographic space and the incessant attacks by partisan forces in Russia under the command of such leaders as Denis Davidov led to the decimation and retreat of Napoleon's *Grande Armée*.²⁷ In Spain, French forces faced a more favourable climate and small geographic space that contained much more rugged terrain and a population that had grown weary of and exceptionally hostile to French looting, reprisals and other depredations. The subsequent campaign waged by Spanish irregulars or *guerrilleros*, operating in the countryside while French field armies were occupied with garrison duties and contending with a small British expeditionary force, harassed and bled the army of Marshal Nicholas Soult white in a campaign noted for its duration and brutality.²⁸ It is from this conflict that the term 'guerrilla' was derived.

Other significant changes in the history of irregular warfare at this time include a number of revolutions and the development of theories of violence,

followed by colonial conquest and writings reflecting on pacification campaigns and experience throughout Africa, Asia and America. The most noteworthy revolutions during this period were the French Revolution (1789–98) and events during the so-called ‘Year of Revolutions’ which swept Europe and parts of Latin America in 1848. The leaders of the French Revolution sought to change from a monarchy to a more equitable form of government, a republic. Republican ideals were ultimately corrupted by personal power and institutional paranoia, which both set the stage for Napoleon’s rise to power and contributed to the lexicon of irregular warfare the term ‘terrorism’, which was first used to describe systemic and deliberate violence by the state to instil fear and preserve order amongst its citizenry.²⁹ The range of causes that spurred the Year of Revolutions relates to societal change stemming from the economic revolution in modes and means of production associated with the Industrial Revolution which figure prominently in the theories of Karl Marx and Louis Auguste Blanqui. Although the theories of both are largely footnotes to history today, the disagreements between them over the mechanism and methods for success of the revolution are worth exploring for one key reason. The intellectual points of departure between Blanqui and Marx have been echoed by Latin American revolutionaries such as Che Guevara and Carlos Merighella, and most recently, between radical Salafist terrorist authors such as Ayman al-Zawahiri and Abu Musab al-Suri, as the [next chapter](#) suggests.

Karl Marx, Louis Auguste Blanqui and socialist revolution

From the perspective of those seeking to overthrow the existing political system using irregular violence, the writings of Karl Marx and Louis Auguste Blanqui provided two contrasting yet related rationales for violence. Marx, who was decidedly the more intellectual of the two writers, contributed the theory of ‘historical materialism’ the central tenets of which are well known but have also been misconstrued. Capitalism, according to Marx, contained internal contradictions that would lead to its own collapse. The pace of economic change outstrips the ability of the capitalist system controlling society to adjust to such change; the class inequalities inherent in the capitalist system and the social disruption caused by cycles of economic growth and collapse would impel those in control of the means of production, the workers, to revolt. In addition to this objective condition inherent in the system, Marx also wrote about the need to develop the subjective conditions for revolution. In particular, Marx saw the socialist workers’ party as performing two vital tasks. The first task was to educate the masses of workers, or proletariat, about the inequities in the system through pamphlets, speeches and other means so that the party could exploit crises and foment revolution. The second task was to build the party infrastructure to organise the workers, increase the size of the organisation and seize the initiative and power when the revolutionary moment presented itself. Revolution, in Marx’s mind, was a historical inevitability that could be exploited by

the revolutionary. The key to success in Marx's mind, however, was organisation and, above all else, patience that the internal contradictions would lead to revolution.³⁰

Although Louis Blanqui largely agreed with Marx on the economic foundations of revolution, he differed in opinion on its social and organisational aspects. That the workers were exploited was unquestionable. However, Blanqui believed that the workers would rebel anyway and, as a result, they could not be educated and should not be informed of the plans for revolution. Only the vanguard, or the intellectual elite which was freed from class bonds and formed into its own general staff, could truly understand the purpose and direct the course of the revolution. The best method for organisation, therefore, was the creation of secret societies that would be difficult for the state security apparatus to monitor and track. Proper organisation, thorough planning, a correct reading of the conditions and boldness of action in the right place and time would create the subjective conditions for revolution.³¹ In addition, Blanqui urged his co-conspirators to read about military tactics and organisation so that revolutionaries understood their purpose during the insurrection, organised the masses and fought in a more cohesive manner.³² When the conditions were deemed ripe, the vanguard would lead the attacks against the key organs of the state. Revolutionary ideals were a necessary precondition but not sufficient to guarantee success of the insurrection. The oppressed masses would not only follow the example set by the insurgents, but their revolutionary zeal could be harnessed in an effective military manner against any counter-attacks by the remaining forces of the state.³³ It was up to the leaders of the vanguard to organise the masses once they were committed to the struggle. Put simply, according to Blanqui's scheme the insurgent vanguard could create and manipulate the subjective conditions so that the revolution could succeed. Once the time had arrived for insurrection, the vanguard must not wait and continue to talk about revolution: it must seize the initiative and lead the revolution to victory.³⁴ Although Blanqui's technique for insurrection grew more sophisticated after each unsuccessful attempt, his appreciation for and assessment of the social and economic factors of revolution remained virtually unchanged. In fact, one author has concluded that 'parts of Blanqui's thinking were frustratingly vague and immature' and his understanding of socialism was 'ideological chop suey'.³⁵ As Blanqui and those who followed after him, such as Mikhail Bakunin, were to discover, perceiving that the conditions are right, assessing the chances for success objectively and accounting for differences in the context of irregular warfare are not easy tasks when revolutionary fervour colours one's perceptions.

Given the broad range and diverse perspectives that characterised socialist theories of revolution throughout the nineteenth and early twentieth centuries, it is surprising that there was relatively little by way of theory, doctrine or treatises written about how to counter the threat of irregular warfare. Indeed while the leading military theorists of the nineteenth century, Carl von Clausewitz and Antoine Henri de Jomini, acknowledged the underlying power of and

difficulties associated with facing a people in arms, they nevertheless paid scant attention to the subject in their works.³⁶ With the possible exception of Clausewitz, who grappled with a theoretical understanding of the nature of war and the place that battle played within it, the vast majority of military officers who put pen to paper were concerned less with irregular warfare than with fighting and winning conventional battles decisively to win wars quickly. At their worst, works on 'strategy' and 'the art of war' were tedious studies on lines of manoeuvre and battlefield formations.³⁷ Other works provided historical narratives of battles past, often those conducted with such skill by Napoleon, to illustrate principles of war and other maxims.³⁸

In addition to divining the secret of decisive battle, military writers during this time were preoccupied with the changes to the art of war that technology was rendering. In 1815, the accurate range of a smoothbore musket was less than 100 yards and cannons fired solid iron shot at targets in their line of sight. Lack of accuracy meant commanders relied on mass to make up for the difference. Over the next century, the range of rifles increased almost tenfold due to the development of rifling, cartridges, primers and the cylindro-conoidal bullet. The introduction of smokeless powder made the source of rifle fire difficult to pinpoint. Accurate fire and breech loading meant men could disperse, seek cover, and still inflict numerous casualties. Advances in metallurgy and engineering led to the development of the first automatic weapons that could sustain rates of fire of several hundred rounds per minute, making the ground in front almost impassible, as well as long-range, breech-loading, indirect-fire artillery. Artillery could also fire hollow shells designed to burst upon impact, increasing their lethal radius. Breech loading and powder cartridges allowed for high rates of sustained fire.³⁹ All of these advances created a 'storm of steel' on the battlefield, in which the number of casualties grew significantly, and against which frontal assaults were all but suicidal.⁴⁰

Charles Callwell and imperial soldiering

In addition to significant advances in the range and lethality of military weapons, the nineteenth and early twentieth century was characterised by the race by European powers to secure overseas colonies at the expense of the local population throughout Asia and Africa. Although the sheer range of campaigns makes generalisations about them difficult, they all shared one thing in common. Nationalism, combined with feelings of divine purpose and a sense of racial or ethnic superiority, led military commanders from a number of nations to consider that they were fighting opponents who were mere 'savages' and therefore undeserving of courtesies and treatments due their civilised opponents. While some concepts and ideas from this era continue to be discussed in the current context of counterinsurgency, such as the oil-spot or ink-blot theory of French officer Joseph-Simon Galliéni (whereby the territorial control of adjacent garrisons combines and spreads like blots of ink on a sheet of paper),



Figure 14.1 Religion, zeal and numbers were no match for Western technology and military discipline at the close of the nineteenth century. In this popular depiction of imperial soldiering from 1900, greatly outnumbered US Marines fight against rebel Chinese ‘Boxers’ while attempting to relieve besieged legations in Peking.

the brutality with which some of these campaigns were waged seemed reasonable to the participants at the time but appears cruel and inhuman today.⁴¹ In South West Africa, for example, German General Lothar von Trotha waged a vicious campaign over three years (1904–7) that sought to annihilate the Herero tribe.⁴² Occasionally ‘savages’ inflicted significant defeats on imperial armies, as occurred during the disastrous First Afghan War (1839–42) fought by the British or at Little Big Horn in 1876, but defeats on this scale were a rarity.⁴³ A more common scenario involved small numbers, relative to the local population, of European or American forces conducting operations against the natives and relying on a combination of discipline, firepower, pluck and superior will to turn the tide of the campaign (see [Figure 14.1](#)). More often than not, ‘savage armies’ obliged them by making mass charges against Western forces armed with rifle, artillery and, in some cases, machine guns. Perhaps the defining incident of this type of warfare was the battle near the town of Omdurman in the Sudan on 2 September 1898. British forces and auxiliaries, numbering no more than 25,000, were confronted by an Ansar, or dervish, army of some 50,000. The Ansar army believed that their spiritual leader, Abdullah al-Taashi, was the Mahdi or imam prophesied to return and restore justice and greatness according to the Shi’ite interpretation of Islam. The results of the battle

confirmed that religious zeal and courage, combined with primitive weapons and tactics, was no match for modern firepower in a set-piece battle. Almost half of the Ansar army was killed or wounded, while the British and their auxiliaries suffered just over 400 casualties.

Although military handbooks, or doctrine, largely reflected preoccupations with decisive battle in the face of destructive, modern technology, imperial and colonial campaigns in far-off lands were bound to make their presence felt within the literature. Irregular warfare, especially against tribes, was nevertheless largely considered as either an adjunct to major war at best or an aberration fraught with danger but with little opportunity for honour or glory. As is often the case in the history of irregular warfare, events that cause national trauma and reflection provide the catalyst for an upsurge in interest and writing in the subject. In Ewald's time, it was the defeat inflicted upon the British forces and their Hessian allies despite the tactical prowess and success of the *Jäger*. The defining irregular war of the end of the nineteenth and the beginning of the twentieth century was fought against Dutch farmers, or Boers, who were living in what is now called South Africa.

The Second Boer War (1899–1902), in common with other irregular wars, contained numerous different forms of war in different phases. The pretext for war was a failed coup d'état in 1895 backed by British immigrants and mine owners, including the prime minister of the Cape Colony, Cecil Rhodes. The goal of the failed coup was the overthrow of the Boer government in the Transvaal. After four years of uneasy truce, during which time Boer governments in the Transvaal and the Orange Free State made preparations for war, British demands and ambitions, and the rejection of Boer concessions, led to the issuing of ultimatums. The war officially began on 11 October 1899 with mixed results. The initial Boer offensive got bogged down with sieges of Kimberley, Mafeking and Ladysmith, but nevertheless the Boers were able to inflict significant defeats on British forces especially during 'Black Week' (10–15 December 1899) at the battles of Stormberg, Magersfontein and Colenso. The arrival of British reinforcements from throughout the Commonwealth resulted in the relief of the besieged towns. Faced with the defeat of their army, Boer leaders shifted to a mobile guerrilla-warfare phase which the authors of the British official history of the war labelled as 'the gadfly of regular armies, finding its natural prey in railways, convoys and isolated posts, and born exponents in the horse-men of the Free State'.⁴⁴ Operating in independent commandos, small numbers of Boer farmers used horses for mobility and relied on accurate fire using the latest magazine-fed Mauser rifles to conduct hit-and-run raids against isolated British detachments. The guerrilla phase lasted until the end of the war on 31 May 1902. Initially British commanders, including Horatio Herbert Kitchener, had difficulty dealing with Boer tactics. The combination of tactics intended to deny the Boers mobility and support, including forced resettlement of the Boer population into concentration camps, the destruction of Boer farmsteads, the construction of reinforcing blockhouses, and the use of

mobile columns to intercept Boer forces once detected, led to British victory, but not without domestic and international criticism. The Second Boer War was one of the first irregular wars that featured the prominent impact of timely media reporting and battlefield photography. The reporting of aspects of the war related to moral issues, augmented by photographs of dead British soldiers and emaciated Boer civilians, shocked and outraged the British public at how the war had been handled.⁴⁵

In addition to calls for reform of the British Army's outdated commissioning, officer development and training regimens, the Second Boer War stimulated public and professional interest in irregular wars. As a result, a number of handbooks and manuals were published on the subject. They ranged from Ernest Swinton's innovative and practical *Defence of Duffer's Drift*, designed as a primer for junior officers, to works that described functional dimensions of irregular warfare, illustrated by examples, such as Thomas Miller Maguire's *Guerrilla or Partisan Warfare* and William Heneker's *Bush Warfare*.⁴⁶ The most detailed and influential work, however, was written by Colonel Charles Callwell in 1896 with updated editions appearing in 1899 and 1906. What sets *Small Wars: Their Principles and Practice* apart from other contemporary works is Callwell's attempt to analyse comprehensively the realm of irregular warfare and present this assessment in a logically consistent manner, including classes of campaigns, types of adversaries and the influence of different types of terrain upon operations, amongst other matters. Some of his advice is dated and can be offensive to modern readers, especially on the subject of the pacification of local inhabitants. Callwell suggests that harsh punitive measures against villages, crops and livestock are required as a way of 'bringing such foes to reason . . . for they understand this mode of warfare and respect it'.⁴⁷ One must appreciate that Callwell's work was a product of its time and to dismiss it outright on these grounds would be remarkably short-sighted. He was one of the first authors to recognise the need for conventional forces to use their advantages against the key strengths of irregulars. For example, Callwell appreciated that irregular opponents did not have to fight decisively in order to win. Irregulars possessed the strategic advantages of time, manoeuvre, intelligence, limited supply requirements and the initiative to accept battle at the time and place of their own choosing.⁴⁸ In addition, Callwell also identified that irregular warfare was not limited to tribes who preferred such fighting culturally but was adopted by enemy commanders as a form of warfare out of either choice or necessity.⁴⁹ In any event, he understood the stresses that guerrilla war imposed upon conventional forces, including the requirement to fight accordingly to methods that ran contrary to well-established military principles and maxims such as mass and never dividing one's force. Although now taken as common sense, Callwell struck upon a key dimension of irregular warfare in 1896: 'The more irregular and the less organised the forces of the enemy are, the more independent do they become of strategical rules.'⁵⁰ In other words, irregular enemies organise and fight according to their own preferences and not according to yours.

Conventional forces could defeat irregular adversaries but not by trying to fight them in a conventional manner. Such adversaries preferred to fight protracted campaigns and Callwell also recognised that a number of 'preventable causes' might allow conventional forces to play into their adversary's hands, such as:

bad organization or they may follow from insufficient preparation . . . lack of zeal among subordinates, or they may be due to want of energy in high places. But one of the commonest causes of operations being unduly prolonged, is to be found in their having been allowed to drift into a desultory form of warfare, and this is a question of strategy and tactics.⁵¹

Fighting irregular enemies required flexibility on the part of the commander, constant vigilance and self-assessment, as well as an understanding of the enemy's preferences and the practical difficulties associated with irregular warfare – considerations that field commanders confronting irregular enemies now and in the future would do well to adopt.

The close of the nineteenth century until the start of the First World War saw a number of developments in the field of irregular warfare but little in the way of theory or doctrine development. Having quickly and decisively defeated Spanish forces in Cuba and the Philippines for the ostensible reason of freeing the local populations from Spanish oppression, the United States found itself engaged in struggles against local insurgents in the Philippines instead. Led by Emilio Aguinaldo, the Philippine insurgents fought a guerrilla campaign with the goal of prolonging the conflict long enough to influence the peace lobby in the United States. Instead of achieving this goal, guerrilla attacks led to the deployment of more troops and the adoption by the US forces of harsh tactics and policies very similar to those adopted by the British in South Africa.⁵² The US declared the war over with the capture of Philippine General Miguel Malvar in 1902. As in Great Britain, the harsh conduct of military forces against the civilian population, including charges of wartime atrocities, was reported in the press (see [Figure 14.2](#)). Such reports led members of Congress to form the Lodge Committee in January 1902 to investigate the charges, although little came of the proceedings. Throughout Europe, the socialist revolutionary movement saw numerous debates, disagreements and schisms between rival claimants over when to take action, the form that action should take and who should be allowed to join each specific national social democrat political party. The more radical elements, dubbed anarchists, believed in 'propaganda by the deed' against the state and its supporting elements.⁵³ Although some of the tactics used by such groups in Britain, France and Russia were already well established, such as kidnapping and assassination, anarchists were amongst the first to conduct bombings using stable, high-yield explosives for their attacks. The successful assassination of Russian Tsar Alexander in 1881 by Russian anarchists prompted other groups to imitate their success, a pattern that would be repeated amongst various Marxist-Leninist terrorist groups throughout the 1960s and 1970s and amongst terrorist and insurgent groups today.



Figure 14.2 Popular notions of the noble aspects of imperial soldiering, such as civilising 'savage' races, were challenged during the Second Boer War (1899–1902) and Philippine War (1899–1902). Newspaper reports of atrocities by British and American soldiers, and graphic photographs taken by reporters and soldiers that were subsequently published, drove home the brutal realities of irregular warfare to the public. According to the original caption, this photograph is of 'insurgent dead just as they fell in the trench near Santa Ana, February 5th, 1899'.

The First World War, Paul von Lettow-Vorbeck and T. E. Lawrence

Although it is a historical stretch to suggest that an act of terrorism started the First World War more than a century ago, there is little doubt that the assassination of Archduke Franz Ferdinand and his wife by members of a Serbian nationalist group provided the pretext for the event that would set the chain of mutual defence pacts and mobilisation orders in motion and led to German troops violating Belgian neutrality. Once the front lines between armies stabilised in the West and East in 1915, there was little scope for irregular warfare on the Eurasian continent. British and German military and political leaders concocted a number of schemes designed to draw enemy attention and resources by conducting irregular warfare by proxy in each other's various

colonial possessions or the territory of their enemy's allies. Some schemes seemed sound in theory but were executed ham-fistedly or were simply impractical. Amongst these were German attempts to foment *jihad* and other plans against Egypt. These attempts were designed to threaten British control of the Suez Canal, and without exception they failed for reasons of national chauvinism, inadequate resources, insufficient knowledge and competing local and national interests.⁵⁴ Other independent rebellions took place. In the case of Ireland, the uprising in Dublin during Easter Week 1916 failed to achieve the objectives envisioned by the leaders of the Irish Republican Brotherhood but paved the way for the more successful urban insurgency planned and led with daring and skill by Michael Collins from 1918 to 1921.⁵⁵

Other irregular campaigns during the First World War were driven by necessity or were accidental in nature. For example, the guerrilla campaign led by Paul von Lettow-Vorbeck in East Africa resulted from conventional victories that significantly depleted his force despite their success. For the next four years, Lettow-Vorbeck and his initial force of 3,000 Germans and 11,000 native *askaris* threatened the British East Africa Protectorate (Kenya) and largely gave battle at the time and place of their choosing while evading Commonwealth forces on a trek through Portuguese Mozambique, German East Africa (Tanzania) and British Rhodesia.⁵⁶ Three features of Lettow-Vorbeck's campaign stand out in particular. First, Lettow-Vorbeck and his officers conducted the campaign in mobile columns with a mixed force of Germans and Africans that retained a high level of unit cohesion until the end of the war, which is evident in the almost equal rates of casualties the force suffered.⁵⁷ Secondly, Lettow-Vorbeck's force largely subsisted off the land and captured supplies from 1916 onwards. Thirdly, and perhaps most importantly, Lettow-Vorbeck and his force managed to frustrate a Commonwealth force that contained a number of South Africans who fought the British during the Second Boer War, demonstrating that even those well-versed in irregular warfare face much greater challenges when trying to defeat an insurgency.⁵⁸ As impressive as Lettow-Vorbeck's achievements were, they were eclipsed by the actions of a British officer fighting as part of the Arab Revolt in the Transjordan.

Thomas Edward Lawrence, better known by his sobriquet 'Lawrence of Arabia', can be considered as the progenitor of modern irregular warfare in the West. This consideration is based on his ability to assess objectively the conditions that existed within his theatre of operations, intuit the strength of the Arab forces and the weaknesses of the Turks, and understand the local culture to the extent he did. Nowhere are these qualities exhibited than in his account of the Arab Revolt published in the inaugural edition of the *Army Quarterly*. This account is stripped of both the prose and literary pretensions that fill his other accounts of the campaign, *Seven Pillars of Wisdom* and its abridged version, *Revolt in the Desert*. In 'The Evolution of a Revolt' Lawrence provides a brief background sketch of the Arab Revolt and the reasons for its early setbacks.⁵⁹ In particular, Lawrence suggests that British attempts to force the Arabs to fight according to

modern European battlefield practices ran contrary to their cultural strengths and was doomed to fail. While recovering from an illness Lawrence claims to have discovered a formula to assess the Arab Revolt strategically. This formula consisted of the following factors: the algebraic, or the area of the terrain and number of Turks required to control it; the biological, or the resources that the Turks needed to sustain their forces; and the psychological, or the morale of both the Turks and the Arabs.⁶⁰

The weakest link in the Turkish garrison chain was the biological factor and, using superior Arab mobility in the desert where the Turks rarely ventured, Lawrence sought to stress it even further. Rather than battle Turkish forces in outposts or attempt the liberation of the holy city of Medina, he believed that the Arabs should strike against Turkish supplies: 'Our cards were speed and time, not hitting power, and these gave us strategical rather than tactical strength.'⁶¹ Tactically his forces would fight 'tip and run, not pushes, but strokes.'⁶² Lawrence also discriminated the strategic purpose for the violence – namely, forcing the Turks to abandon the Transjordan and freeing the Arabs from Turkish rule – from the tactical limitations and motivations of the tribes that were to conduct the operations. Individual tribes would fight their neighbours just as easily as the Turks. In order to solve this problem, Lawrence adopted two solutions. The first, and base motivation, was personal gain and he secured sums of gold with which to maintain the loyalty of tribal chiefs even though some of the funds were diverted and payment did not necessarily guarantee numbers for duty.⁶³ As much as Lawrence waxed eloquently about Arabs fighting for the idea of freedom, wealth and plunder sustained the Arab Revolt throughout the campaign. The second solution was to use a number of tribes individually so that they formed a 'ladder' on the way to Damascus.⁶⁴ To increase the range and destructiveness of the Arabs, and by extension the pressure that the tribes could put on Turkish supply lines, Lawrence secured modern rifles, machine guns, explosives and experts trained in their use and tinned rations.⁶⁵

Whether or not Lawrence deserves the laurels piled at his feet is a matter for debate. T. E. Lawrence remains an enigma to this day, filled with contradictions, not beyond pranks, and not always absolute with the truth.⁶⁶ Whether or not Lawrence could have succeeded in forcing the Turks to withdraw by guerrilla action alone is also a matter of conjecture. As it was, Lawrence's Arab forces secured the right flank of the three corps comprising the Egyptian Expeditionary Corps under the command of General Edmund Allenby and continued to harass Turkish supply lines, drawing off manpower to guard and repair them.⁶⁷ Yet Lawrence continues to be held in high esteem by modern irregular warfare specialists as worthy of emulation by design or accident. One counter-insurgency specialist imitated Lawrence's famous '27 Articles' as a method of transmitting important information to company commanders who deployed to Iraq.⁶⁸ A white paper entitled 'One Tribe at a Time', written by an American Special Forces officer and circulated with great fanfare in 2009, offered a

strategy for success in Afghanistan that bore a number of striking parallels with Lawrence's approach in Arabia.⁶⁹ Lawrence was one of the first authors to outline how to conduct a successful unconventional campaign against a conventional opponent as well as emphasise the psychological dimensions of irregular warfare over battle. Lawrence's written account of the objective and subjective conditions in Transjordan in 1916, regardless of whether it was real or fabricated after the fact, serves as one of the few examples of how such conditions can and should be accurately assessed.

Mao Zedong and protracted people's war

One of the few socialist theorists who also read the subjective and objective conditions strategically after early failure was Mao Zedong (Tse-Tung). As with Lawrence, a number of Mao's claims about the revolution and his role in it are open to question.⁷⁰ A rival socialist predecessor of Mao, Vladimir Ilyich Ulyanov, or Lenin, had written extensively on revolution, including partisan warfare. But his impact on the revolution in Russia in 1917 had more to do with chance, support from the Imperial German government at a crucial juncture of his exile, and the incompetence of the Tsarist regime than reading and influencing the subjective conditions within the country.⁷¹ As a young revolutionary dedicated to the socialist cause, Mao slavishly attempted to follow the then-accepted formula for Leninist proletariat revolution in his native China in 1921. Its failure, primarily due to the lack of meaningful numbers of oppressed urban workers fuelling the machine of capitalism, caused Mao to reflect upon the nature of revolution in China. Over the course of the next sixteen years, Mao succeeded in consolidating his hold on power of the Chinese Communist Party and expanding its base amongst the rural peasantry, despite aggressive attempts by the Chinese Nationalists to crush it. In 1937, however, the Japanese invaded China and nationalists and communists were faced with a hostile, competent adversary that initially swept away all attempts to stop it. The following year, Mao produced two complementary works on irregular warfare: the pamphlet entitled *Yu Chi Chan*, or 'On Guerrilla War', and a series of lectures at the Yen'an Association for the Study of the War of Resistance against Japan, one of which was entitled 'On Protracted War'. No work by an author on or practitioner of irregular warfare has before or since so accurately read most the subjective and objective conditions for success within its specific context, or provided a comprehensive roadmap that links battle to revolutionary victory, as Mao did in 'On Protracted War'.

As with most other significant military theorists, including Clausewitz, Mao's ideas are quoted, cited, and too often utterly mischaracterised. For example, Mao's metaphor of the guerrilla fish swimming in the sea of the population is well known, and variations of it were cited in early speeches related to the Global War on Terrorism.⁷² Glibly citing the metaphor and understanding the difficulties and problems associated with isolating guerrillas from the population

are two completely different things as most counterinsurgency practitioners understand. Mao's three-phase plan for revolutionary victory, for example, has been rigidly interpreted to provide the foil to advance arguments for why approaches to counterinsurgency plans for Iraq were in need of significant revision.⁷³ Part of the reason Mao has been misinterpreted is that his theory of warfare, like Clausewitz's, is descriptive rather than prescriptive. And like Clausewitz, Mao provides a framework for understanding the often contradictory relationship that exists between battles and campaigns in achieving the overall political aim of the war.

In 'On Protracted War', Mao sets out to prove that notions of quick victory are impossible and maintaining a line of tactical defences against the Japanese will only result in defeat of Chinese communist forces. The Japanese are simply too well equipped and trained to be fought conventionally. Much like Lawrence, Mao understands that there are too few Japanese troops to conquer and occupy China.⁷⁴ China's strengths are its vast geographic space and size of its population.⁷⁵ These strengths alone, however, are insufficient to defeat the Japanese. The Chinese peasants must be educated, trained, organised and equipped to fight, and this process will take time. In order to buy the required time, Mao suggests trading geographic space and conducting offensive action at the tactical level instead of slavishly trying to blunt every Japanese offensive and prevent the occupation of major Chinese cities.⁷⁶ Trading space, however, does not imply retreating before the Japanese. Tactical offensive action will take place in the spaces occupied by the Japanese in a 'war of jig-saw pattern' designed to threaten their supply lines.⁷⁷ In order to deal with such threats, the Japanese will be confronted with a dilemma: cede territory in their rear area to the guerrilla cadres, who would exploit the situation to recruit amongst the population, or divert units to garrison towns and secure supply lines, weakening the strategic offensive and making such units tactically vulnerable to attack and annihilation.⁷⁸ Mao states much of what is obvious and intuitive in irregular warfare at the tactical and strategic level: cause the enemy to disperse his forces, strike where the enemy is weak and inflict casualties, avoid pitched battles unless you have overwhelming force, time is a weapon that favours the guerrillas etc.

Three related features distinguish 'On Protracted War' and its companion work for the communist cadres, *On Guerrilla Warfare*, from other volumes on irregular warfare. The first feature is Mao's distinction between tactical action and its relationship to the overall strategic effort. Mao's discussion is aimed at those who rigidly advocate all-out offence or all-out defence. Both views, in Mao's estimation, are wrong and what is required instead is a more nuanced appreciation of the relationship between offence and defence at the tactical and strategic levels of war. The second unique feature of Mao's theory is his framework for victory, which places tactical actions into an overall plan for victory. The three phases outlined by Mao – strategic defensive, stalemate and strategic offensive – outline an achievable vision of victory provided the members of the Chinese Communist Party are patient and diligent enough.⁷⁹ Unlike

other irregular-warfare theorists before or since, Mao grasped that tactical guerilla actions alone, however cleverly executed or destructive, were insufficient to drive the Japanese from China. Guerrilla action was not only designed to weaken the Japanese forces, improve the quality of Chinese forces in the 'university of war' and lead to the balance of forces tipping in the favour of the Chinese communists but also to allow the development of more large-scale military units to conduct mobile warfare in more conventional offensives.⁸⁰ The third and final distinctive feature of Mao's theory of irregular warfare, related to the last point, is the primacy of politics and the role that military action plays in support of it. Guerrilla action was designed not just to achieve military objectives. Such action first and foremost shielded the political activities of the communist cadres, providing them with the breathing-space to organise and receive the support of the population. Mao devotes a portion of both works to the object of war, the role that politics plays and the need to mobilise the population for the war.⁸¹ In order to do this, the population must be educated about why they are fighting and how victory will improve their lives. In addition, the Chinese communists must demonstrate their moral superiority over the Japanese in deed and in action by following Mao's 'Three Rules and Eight Remarks'.⁸² In this way the local peasants would see that the Chinese communists were worthy of support. Military action alone could not achieve victory unless a critical mass of the Chinese population actively and willingly sustained and swelled the number of the guerrilla forces. The pivotal role that the population plays in irregular warfare, outlined so clearly by Mao, has figured prominently in almost all modern theory and doctrine, but the relationship between violence, ideology and politics continues to remain elusive, misunderstood and open to disagreement.

The Second World War

Although there were a number of other important irregular wars and incidents prior to 1939, including the Russian (1918–22), Irish (1922–3) and Spanish (1936–9) Civil Wars, US Marine interventions in the so-called 'Banana Wars' in Central and South America, the bombing of the financial district of Wall Street in New York (1920) and Arab and Jewish guerrilla and counter-guerrilla warfare in British-mandated Palestine, these events were overshadowed historically by a seemingly unstoppable string of German and Japanese victories at the start of the Second World War. Within Great Britain, the prospect of German invasion appeared all too real by the late summer of 1940. The British Army had lost most of its equipment earlier that year in battle or on the beaches of Dunkirk. To defend the island nation, Prime Minister Winston Churchill approved plans to create a cadre of civilian irregulars eventually labelled the 'Home Guard'. Amongst those selected to train the volunteers were Tom Wintringham and Bert Levy, who had fought together in the Spanish Civil War on the Loyalist side.

Both Wintringham and Levy wrote short, mass-produced works designed as tactical training aids for civilian irregulars based on their experiences in Spain. Wintringham's *New Ways of War* and Levy's *Guerrilla Warfare* discussed guerrilla history, organisation, propaganda requirements, ambush techniques and methods to defeat German panzers.⁸³

The German invasion of Great Britain was postponed and then later cancelled and Wintringham and Levy's techniques were not put to the test there. The populations of other countries in Europe and Asia were not so fortunate. Harsh German and Japanese policies within the countries they had occupied, including brutal reprisals against villages suspected of supporting guerrillas or partisans, dealt with a tactical military problem but strategically proved ruinous. As the true nature of the German and Japanese occupations were revealed, a percentage of the local population in countries as diverse as Poland, Czechoslovakia, France, the Soviet Union, Malaya and French Indochina conducted active and passive resistance in response. Reprisals and indiscriminate sweeps, especially by German and other national auxiliaries behind their lines on the Eastern Front, did not cow locals into submission but instead swelled partisan ranks.⁸⁴ For the allied powers, resistance, guerrilla and partisan movements in occupied Europe and Asia were a potential source of offensive support to plans to liberate countries, particularly given the power ascribed to so-called 'Fifth Column' efforts as an explanation for the rapid Axis conquests.⁸⁵ The actual contribution made by partisan and resistance movements, operating with the assistance of the British Special Operations Executive, the American Office of Strategic Services and the Soviet STAVKA, remains controversial due to lost or missing records, inflated claims by partisan and resistance groups and the difficulties in assessing the impact of subversion, sabotage and ambush as a drain on the German and Japanese war effort.⁸⁶ What is certain, however, is that the ideals articulated by the Allies set the stage for the [next chapter](#) in modern irregular warfare.

RESEARCH QUESTIONS

1. Why are there so few theories of irregular warfare by practitioners?
2. Why have theorists disagreed historically on how to conduct irregular warfare?
3. Why were so few insurgent campaigns successful during the nineteenth century? Did this change during the twentieth century and, if so, why?
4. What is the relationship of social context and technology to the theory and practice of irregular warfare?
5. Do Mao and Lawrence deserve the praise for their thoughts on insurgency and guerrilla warfare? If so, why? If not, why?

FURTHER READING

- Asprey, Robert, *War in the Shadows*, rev. edn (New York: Little, Brown & Co., 1994) continues to be the most comprehensive and accessible survey of irregular warfare from ancient to modern times.
- Beckett, Ian F. W., *Modern Insurgencies and Counter-insurgencies* (London: Routledge, 2001) is the most lucid and condensed single-volume survey on insurgency and counterinsurgency from 1750 until 2000.
- Boot, Max, *Invisible Armies* (New York: Liveright, 2013) is the most recent narrative history of guerrilla warfare from ancient times until the present.
- Neuberg, A. (pseud.), *Armed Insurrection* (New York: St Martin's, 1970) is a translation of a 1928 manual for insurrection from a Marxist perspective. The work remains useful for its relative accessibility and insights into Marxists on the role violence plays to achieve political effect.

ONLINE RESOURCES

- The Marxist Internet Archive website (www.marxists.org/archive/index.htm) provides a veritable treasure trove of online primary source documents, including the writing and correspondence of all the major – and most of the minor – theorists, leaders and luminaries of socialist thought, including Blanqui (www.marxists.org/reference/archive/blanqui/index.htm) and Mao Zedong (www.marxists.org/reference/archive/mao/selected-works/index.htm).
- The US Navy Department library has posted, on a single page, the entire text of Charles Callwell's classic *Small Wars* (www.history.navy.mil/library/online/small_wars.htm), including the original margin headings.
- The T. E. Lawrence Studies website (www.telstudies.org/index.shtml), offers a range of resources, including a bibliography and maps. The website, which is currently undergoing significant revision and update, offers a collection of Lawrence of Arabia's writings (www.telstudies.org/writings/writings_introduction.shtml), which are arranged chronologically.

NOTES

1. See, for example, Maurikios, *Maurice's Strategikon: Handbook of Byzantine Military Strategy*, trans. George T. Dennis (Philadelphia, PA: University of Pennsylvania Press, 1984), pp. 52–7, 93–105; Harro von Senger, *The Book of Strategems: Tactics for Triumph and Survival*, ed. and trans. Myron Gubitz (New York: Viking, 1991); *The Seven Military Classics of Ancient China*, trans. Ralph Sawyer (Boulder, CO: Westview, 1993). The ancient Indian text by Kautilya, *Arthashastra*, provides little guidance on the tactics of fighting as it was meant to be a manual on statecraft. Like Sun Tzu's *The Art of War*, Kautilya places some emphasis on spies, intelligence networks and using proxies in enemy states to foment unrest. See T. N. Ramaswamy, *Essentials of Indian Statecraft: Kautilya's Arthashastra for Contemporary Readers* (New York: Asia Publishing House, 1962), pp. 55–8, 60–1.

2. See, for example, acts of treachery during the Peloponnesian Wars in Robert Strassler (ed.), *The Landmark Thucydides: A Comprehensive Guide to the Peloponnesian Wars* (New York: The Free Press, 1996), pp. 33, 250, 303, 515; Paul Bentley Kern, *Ancient Siege Warfare* (Bloomington, IN: Indiana University Press, 1999), pp. 21, 33–4, 59–60, 119.
3. Maurikios, *Maurice's Strategikon*, pp. 113–26.
4. Archer Jones, *The Art of War in the Western World* (New York: Barnes and Noble, 1987), pp. 54–7; Victor Davis Hanson, *Carnage and Culture* (New York: Doubleday, 2001).
5. In particular, Flavius Josephus's works, including his account of the Jewish uprising in AD 70, *The Jewish War*, have been the subject of considerable scholarly inquiry and criticism. See, for example, S. J. D. Cohen, *Josephus in Galilee and Rome: His Vita and Development as a Historian* (Leiden: E. J. Brill, 1979); Tal Ilan and Jonathan Price, 'Seven Onomastic Problems in Josephus' "Bellum Judaicum"', *The Jewish Quarterly Review*, 84 (2/3) (October 1993–January 1994), 189–208.
6. Robert Asprey, for example, begins his historical exploration of guerrilla warfare from the Persian empire under the rule of Darius in 531 BC; John Ellis, in contrast, goes back even further to the actions of the Israelites against the Hittites as described in the Book of Joshua in the Bible. Asprey, *War in the Shadows*, rev. edn (London: Little, Brown and Co., 1994), pp. 3–4; Ellis, *From the Barrel of a Gun: A History of Guerrilla, Revolutionary and Counter-insurgency Warfare, from the Romans to the Present*, rev. edn (London: Greenhill, 1995), p. 17.
7. Such works include Bruce Hoffman, *Inside Terrorism*, revised and expanded edn (New York: Columbia University Press, 2006), pp. 83–4; Robert Pape, *Dying to Win* (New York: Random House, 2005), pp. 12–13. Critical scholarly treatments of the Zealots and Assassins are: Martin Hengel, *The Zealots: Investigations into the Jewish Freedom Movement in the Period from Herod I until 70 AD*, trans. David Smith (Edinburgh: T&T Clarke, 1989); Farhad Daftary, *The Assassin Legends: Myths of the Isma'ilis* (London: I. B. Taurus, 1995).
8. See, for example, Asprey, *War in the Shadows*; Ellis, *From the Barrel of a Gun*; Walter Laqueur, *Guerrilla: A Historical and Critical Study* (Boston, MA: Little, Brown and Co., 1976); Anthony James Joes, *Guerrilla Conflict Before the Cold War* (Westport, CT: Praeger, 1996); Max Boot, *Invisible Armies* (New York: Liveright, 2013).
9. Christer Jörgensen et al., *Fighting Techniques of the Early Modern World: Equipment, Combat Skills, and Tactics* (New York: St Martin's, 2006), p. 25.
10. Mark Kwasny, *Washington's Partisan War, 1775–1783* (Kent State University Press, 1996).
11. For details of Indian raids and massacres, see John Ferling, *Almost a Miracle* (Oxford University Press, 2007), pp. 203, 436–7, 495–6.
12. *Ibid.*, pp. 400–2.
13. See Fred Anderson, *Crucible of War: The Seven Years' War and the Fate of Empire in British North America, 1754–1766* (New York: Knopf, 2000).
14. For details of Marion's exploits, see Daniel E. Fitz-Simons, 'Francis Marion the "Swamp Fox": An Anatomy of a Low Intensity Conflict', *Small Wars and Insurgencies*, 6(1) (Spring 1995), 1–16. To put the 'Swamp Fox's' exploits in the context, see Walter Edgar, *Partisans and Redcoats* (New York: Perennial, 2003).

15. Rogers's 'Standing Orders' consist of nineteen aphorisms and common-sense advice such as 'Don't forget nothing' and 'If we strike swamps, or soft ground, we spread out abreast, so it's hard to track us.' The Standing Orders continue to form part of the heritage of the American 75th Ranger Regiment, which draws its lineage from Rogers's Rangers, and are posted on the regiment's website. See www.soc.mil/75thrr/75thrrorders.shtml.
16. Andrew Birtle, *U.S. Army Counterinsurgency and Contingency Operations Doctrine, 1860–1941* (Washington, DC: U.S. Army Center of Military History, 1998), p. 5.
17. Laqueur, for example, summarises the 1789 work by Andreas Emmerich entitled *The Partisan in War or the Use of a Corps of Light Troops to an Army*. Walter Laqueur, 'The Origins of Guerrilla Doctrine', *Journal of Contemporary History*, 10(3) (July 1975), 344–5.
18. Johann Ewald, *Diary of the American War*, ed. and trans. Joseph Tustin (New Haven, CT: Yale University Press, 1979).
19. Walter Laqueur, 'The Origins of Guerrilla Doctrine', *Journal of Contemporary History*, 10(3) (July 1975), 346.
20. Ewald, *Treatise on Partisan Warfare*, pp. 68, 80, 114, 183. J. F. C. Fuller provides a concise summary of, and considerable praise for, this work in his *British Light Infantry in the Nineteenth Century* (London: Hutchinson, 1925), pp. 137–51.
21. See, for example, Ewald, *Diary of the American War*, pp. 57–8.
22. *Ibid.*, pp. 355, 341.
23. See, for example, Ewald, *Treatise on Partisan Warfare*, pp. 69, 70, 76, 78 and the anecdote about the Reverend Dr William Smith related in Ewald, *Diary of the American War*, pp. 92–3.
24. Ewald, *Treatise on Partisan Warfare*, p. 88.
25. Ewald, *Diary of the American War*, p. 57.
26. *Ibid.*, pp. 335–6.
27. See Denis Davidov, *In the Service of the Tsar against Napoleon*, ed. and trans. Gregory Troubetzkoy (London: Greenhill, 1999), pp. 83–161.
28. Ellis, *From the Barrel of a Gun*, pp. 73–6.
29. Hoffman, *Inside Terrorism*, pp. 3–4.
30. Marx's seminal works on the subject remain *Das Kapital* and the volume he penned with co-author Friedrich Engel, *The Communist Manifesto*. This section draws upon Tucker's work that separates the wheat from the Marxist intellectual chaff. Robert Tucker, *The Marxian Revolutionary Idea* (New York: Norton, 1969).
31. Samuel Bernstein, *Auguste Blanqui and the Art of Insurrection* (London: Lawrence and Wishart, 1971), p. 309.
32. Auguste Blanqui, 'Manual for an Armed Insurrection', trans. Andy Blunden (1866) (www.marxists.org/reference/archive/blanqui/1866/instructions1.htm).
33. Bernstein, *Auguste Blanqui and the Art of Insurrection*, pp. 309–10.
34. Auguste Blanqui, 'Manual for an Armed Insurrection'.
35. Bernstein, *Auguste Blanqui and the Art of Insurrection*, p. 299.
36. See, for example, Clausewitz, *On War*, pp. 479–83 and Antoine Henri de Jomini, *The Art of War*, ed. and trans. G. H. Mendell and W. P. Craighill (Westport, CT: Greenhill, 1971), pp. 31–6.
37. Illustrations of the type of diagrams that Clausewitz scorns are contained in Jomini, *The Art of War*, pp. 188–94 and J. B. Wheeler, *A Course of Instruction in the*

Elements of the Art and Science of War (New York: Van Nostrand, 1879), pp. 16–18, 92, etc.

38. See, for example, the discussion of the three principles of war and nine maxims contained in the Sandhurst primer by P. L. MacDougall, *The Theory of War*, 3rd edn (London: Longman, Green, Longman, and Roberts, 1862), pp. 51–2, 98–115.
39. For details see Trevor Dupuy, *The Evolution of Weapons and Warfare* (Fairfax, VA: HERO Books, 1984), pp. 190–202.
40. The phrase from the title of Ernst Junger's memoir entitled *Storm of Steel* (London: Chatto and Windus, 1930).
41. Douglas Porch, 'Bugeaud, Galliéni, Lyautey: The Development of French Colonial Warfare', in Peter Paret (ed.), *Makers of Modern Strategy* (Princeton University Press, 1986), pp. 376–407.
42. Von Trotha's statements reproduced in Gil Merom, *How Democracies Lose Small Wars* (Cambridge University Press, 2003), pp. 36–7.
43. For the First Afghan War, see Patrick Macrory, *The Fierce Pawns* (New York: Lippencott, 1966), esp. pp. 241–90.
44. Frederick Maurice and Maurice Grant, *History of the War in South Africa, 1899–1902*, vol. III (London: Hurst & Blackett, 1908), p. 93.
45. For details on media reporting of the war, see Denis Judd and Kevin Surridge, *The Boer War* (New York: Palgrave Macmillan, 2003), pp. 251–6.
46. Swinton's work remained in print until 1991, which is a testament to its quality and utility. Ernest Swinton, *The Defence of Duffer's Drift* (London: Leo Cooper, 1991). See also T. Miller Maguire, *Guerrilla or Partisan Warfare* (London: Hugh Rees, 1904); W. C. G. Heneker, *Bush Warfare* (London: Hugh Rees, 1907).
47. Charles Callwell, *Small Wars: Their Principles and Practice*, 3rd edn (London: HMSO, 1906), p. 41.
48. *Ibid.*, pp. 85–107.
49. *Ibid.*, pp. 126–7.
50. *Ibid.*, p. 52.
51. *Ibid.*, p. 99.
52. For details on the guerrilla war, see Brian Linn, *The Philippine War, 1899–1902* (Lawrence, KS: University Press of Kansas, 2000), pp. 185–321.
53. The phrase has been attributed to a number of anarchists, including Carlo Pisacane, Mikhail Bakunin and Paul Brousse. The latter used the phrase as the title for an article published in 1887. Paul Avrich, *Anarchist Portraits* (Princeton University Press, 1988), p. 244.
54. German strategic objectives are from Tilman Lüdke, *Jihad Made in Germany: Ottoman and German Propaganda and Intelligence Operations in the First World War* (Münster: Lit Verlag, 2005), pp. 33–8, 90–105. See also Hew Strachan, *The First World War, I: To Arms* (Oxford University Press, 2001), pp. 700–4.
55. On the rationale for violence and the conduct of the campaign, see Michael Collins, *The Path to Freedom* (Boulder, CO: Roberts Rinehart Publishers, 1996), pp. 63–74.
56. Paul von Lettow-Vorbeck, *My Reminiscences of East Africa*, 6th edn (Nashville, TN: Battery Press Reprints, 1990), p. 72.
57. When Lettow-Vorbeck's force surrendered on 25 November 1918, it numbered 155 Germans and 1,156 *askaris*. Charles Miller, *Battle for the Bundu: The First World War in East Africa* (London: Purnell Book Services, 1974), pp. 325–6.

58. Byron Farwell, *The Great War in Africa: 1914–1918* (New York: Norton, 1986), pp. 80–1.
59. T. E. Lawrence, 'Evolution of a Revolt', *Army Quarterly*, 1 (October 1920), 55–6.
60. *Ibid.*, pp. 59–62.
61. *Ibid.*, p. 63.
62. *Ibid.*, p. 64.
63. A summary of monies paid early in the revolt is contained in 'Sherif Hussein's Administration', in Malcolm Brown (ed.), *Secret Dispatches from Arabia and Other Writings by T. E. Lawrence* (London: Bellew, 1991), pp. 61–4 (originally in *Arab Bulletin*, 26 November 1916).
64. Lawrence, 'Evolution of a Revolt', p. 63.
65. *Ibid.*, pp. 63, 65–6.
66. See, for example, the assessments contained in Linda Tarver, 'In Wisdom's House: T. E. Lawrence and the Near East', *Journal of Contemporary History*, 13 (1978), 585–608; Lawrence James, *The Golden Warrior: The Life and Legend of Lawrence of Arabia* (New York: Paragon House, 1993); Suleiman Mousa, *T. E. Lawrence: An Arab View* (New York: Oxford University Press, 1966).
67. Matthew Hughes, *Allenby and British Strategy in the Middle East, 1917–1919* (London: Frank Cass, 1999), pp. 77–8.
68. David Kilcullen, 'Twenty-eight Articles: Fundamentals of Company-level Counter-insurgency', (March 2006) (www.d-n-i.net/fcs/pdf/kilcullen_28_articles.pdf).
69. Major Jim Gant, *One Tribe at a Time* (Los Angeles: Nine Sisters Imports, 2009).
70. Jung Cheng and Jon Halliday, *Mao: The Unknown Story* (New York: Knopf, 2005), pp. 233, 243.
71. V. I. Lenin, 'Partisan Warfare', in *Modern Guerrilla Warfare*, ed. Franklin Mark Osanka (New York: Free Press, 1962), pp. 65–79.
72. Mao Tse-Tung, *On Guerrilla Warfare*, trans. Samuel B. Griffith (New York: Praeger, 1961), p. 93. A variation of Mao's metaphor, 'draining the swamps', was used by senior Department of Defense officials in speeches just after the 11 September 2001 attacks.
73. Steven Biddle, 'Seeing Baghdad, Thinking Saigon', *Foreign Affairs*, 85(2) (March/April 2006), 2–14.
74. Mao Tse-Tung, 'On Protracted War', in *Selected Military Writings* (Peking: Foreign Languages Press, 1966), pp. 196–7 (www.marxists.org/reference/archive/mao/selected-works/volume-2/mswv2_09.htm).
75. *Ibid.*, p. 201.
76. *Ibid.*, p. 255.
77. *Ibid.*, pp. 219–22.
78. *Ibid.*, pp. 248–50.
79. *Ibid.*, pp. 210–19.
80. Mao, *On Guerrilla Warfare*, p. 73.
81. Mao, 'On Protracted War', pp. 226–31, 257–61; Mao, *On Guerrilla Warfare*, pp. 88–93.
82. Contained in Mao, *On Guerrilla Warfare*, p. 92.
83. T. Wintringham, *New Ways of War* (Harmondsworth: Penguin, 1940); 'Yank' Levy, *Guerrilla Warfare*, 2nd edn (Boulder, CO: Panther, 1964). The other work used by

the Home Guard at this time was S. J. Cuthbert, *We Shall Fight in the Streets*, 2nd edn (Boulder, CO: Panther, 1965).

84. See Ben Shepherd, *War in the Wild East* (Cambridge, MA: Harvard University Press, 2004), amongst others.
85. Explored in Louis de Jong, *The German Fifth Column in the Second World War* (University of Chicago Press, 1956).
86. A point emphasized in Gordon Harrison, *Cross Channel Attack: The United States Army in World War II* (Washington, DC: US Army Center of Military History, 1989), p. 207.



15

Current irregular warfare

Contents

Insurgent strategic after Mao

Developments in counterinsurgency

Doctrine

Organisational adaptation

The principles of counterinsurgency

Why is modern irregular warfare so difficult?

The importance of context

Misunderstanding the character of violence

Mistaking means and methods for purpose

'Mirror-imaging'

SOF and the golden age of counterterrorism?

Conclusion

KEY THEMES

- Given their resource constraints and the character of irregular warfare, insurgents and terrorists have relatively few choices regarding the organisation or strategies they pursue. There is more continuity in insurgent strategy and organisation than there is change.
- Doctrine for countering insurgency, whether historical or contemporary, agrees in kind but differs only in preferred terminology, degree or specific approach. The principles of countering irregular warfare, therefore, are largely immutable. What matters regarding doctrine, however, is the ability of organisations and their leaders to adapt to the environment, learn faster than their opponent and connect their actions to the overall strategic effort, and not become the strategy itself.
- Irregular warfare is a complex phenomenon that is difficult to understand as different types of violence can be used individually or simultaneously. This can lead to confusing the method or tactics of irregular warfare for its strategy or its purpose. Fixating on the tactics of violence, or improving one's own performance, as opposed to tackling your opponent's organisation and rationale, leads to operational frustration and strategic failure.

- Special operations forces are the preferred instrument of policy-makers now and in the future but they are best suited to tackle immediate irregular threats and only establish the conditions for the future success of others to exploit, but are not a solution to irregular warfare in and of themselves.

The current era of irregular warfare begins with the end of the Second World War and decolonisation or, rather, had its genesis during the Second World War. American President Franklin Roosevelt and British Prime Minister Winston Churchill could not possibly have known how truly strong and problematic the future whirlwind would be when they issued the Atlantic Charter in August 1940. Although most of its points appeared innocuous, the third point – the right of all peoples to self-determination – would cause the greatest difficulties. In particular, the leaders of socialist or nationalist movements in colonial territories interpreted the Atlantic Charter as the basis for declarations of independence once the war was over. The most famous example occurred in French Indochina in September 1945, where Ho Chi Minh declared Vietnam independent using language borrowed from the US Declaration of Independence.¹ Within the first three years of the end of the Second World War, France, the Netherlands and Great Britain were fighting insurgent groups that used guerrilla and terrorist tactics in their colonies including Malaya (Malaysia), Palestine, French Indochina (Vietnam, Laos and Cambodia) and the Dutch East Indies (Indonesia). In other cases, the struggle for post-war predominance between different groups raged during the war in former monarchies such as Yugoslavia. In the case of occupied Greece, the struggle for supremacy between a number of rival nationalist and socialist factions started as early as 1942 and developed into a civil war that concluded only in 1949.

To outside observers, and within the broader context of the Cold War between the West and the Soviet Bloc, the majority of groups waging guerrilla or terrorist campaigns appeared to be communist-inspired or -directed. This observation, combined with increasing tensions with and territorial gains by the Soviet Union, suggested that the wave of irregular warfare was part of a vast, socialist conspiracy to defeat the West. In addition, both the Western and Soviet powers were looking at ways to compete with one another and expand their own influence in ways that would limit the possibility of escalation of conflict into conventional or nuclear war. One way to expand this influence was by providing support to proxies, including friendly states and insurgent or other groups. Nikita Khrushchev's speech in January 1961 implied Soviet support for those groups fighting 'wars of national liberation' and this speech only cemented the idea of a global conspiracy. Arguably the perception of a global communist conspiracy was a key element in the American decision to intervene in Vietnam.

Insurgent strategies after Mao

Up to the American commitment of combat troops to Vietnam in 1965, irregular wars conducted against colonial powers, newly independent governments or other 'puppet' regimes followed four basic strategic approaches regardless of the group's political or religious ideology. In some cases, groups used these approaches independently, in parallel or in sequence depending on their success (or lack thereof). The first attempted to imitate Mao's success in China by following his advice in 'On Protracted War'. The success of Mao's approach was based on the assumption of large geographic space and a sizeable rural peasant population. Vietnamese General Vo Nguyen Giap is the most famous student of Mao who applied the formula to great success against the French. Giap oversaw the growth of a large Viet Minh army that contained regular forces and guerrillas. The siege and defeat of the French base at Dien Bien Phu in 1954, although not precisely Mao's envisioned strategic offensive phase, was enough to convince the political leaders in France that the war could not be won.² Since Vietnam, a number of insurgent groups have also attempted to apply Mao's ideas in their own country. Although Maoist-inspired insurgencies continue to be waged to this day in India, the formula has been less successful in countries such as Thailand, Sri Lanka, Peru and the Philippines as groups have emphasised the military aspects of the campaign, including acts of terrorism, at the expense of political considerations.³

The second approach that irregular groups used involved attempts to convince the colonial power that the costs of maintaining their hold over the country were unacceptable without fighting major battles. In most cases, geography dictated this choice; this is especially true where population density was high and massed guerrilla movement was impractical or unwise. Zionist groups in the British-mandated Palestine, for example, conducted independent operations and some co-ordinated attacks ranging from pipeline sabotage and assassination to car bombings. Two incidents in particular characterised the type of attacks conducted by groups that included the Palmach, Lehi, the Stern Gang and Irgun: the King David Hotel bombing in July 1946 which housed numerous British-mandate offices, including that of the Secretariat and the Criminal Investigation Division and the kidnapping and hanging of two British sergeants in reprisal for the death sentences carried out against three Irgun members in July 1947.⁴ Factors that contributed to the British decision to turn the territory over to the United Nations included awkward questions in the British parliament about the costs of operations within the mandated territories; media reporting of significant incidents, the loss of credibility after security measures led to an increase in attacks in circumstances similar to those in Iraq in 2006 and the shocking nature of some of attacks.

The third strategic approach was to seek diplomatic resolution because the conflict had reached a stalemate. Insurgent groups prevailed upon outside parties and sought to gain the attention of the member states of the United Nations

in the hope their cause would be raised as an issue for resolution by the General Assembly. The leaders of the Algerian *Front de Libération Nationale* (FLN), for example, which was losing ground militarily against French forces, sponsored a general strike in 1957, as well as bombings and other attacks, timed to maintain interest within the General Assembly on the subject of Algeria's future.⁵ Indonesian leaders routinely prevailed on outside powers, including Britain, the United States and the United Nations to resolve their on-again, off-again war to secure their independence from the Dutch (1945–9).

The fourth and last strategic approach was something entirely different which recalled the nineteenth-century anarchist motto of 'propaganda by the deed'. Writers including Ernesto 'Che' Guevara, Régis Debray and Carlos Marighella made a fundamental break with established socialist revolutionary theory by suggesting that action take the place of organisation. Guevara and Debray, for example, suggested that the objective conditions within Latin America were ripe for revolution and guerrilla action by a revolutionary vanguard could itself create the subjective conditions.⁶ Guevara's experience in the Cuban Revolution (1956–8) only confirmed this.⁷ The peasants and workers did not need to be educated or enlightened; the actions of a few would inspire, awaken and attract the masses to join in the insurgency. Marighella, on the other hand, suggested that urban guerrilla action would reveal to the masses the true nature of the ruling regime. Urban guerrillas could find all of the resources they needed within cities, including money, arms, media outlets and targets to attack. The regime would have difficulty locating the guerrillas. The vain attempts by the regime to do so would only demonstrate its incompetence as the guerrillas continued their assault, leading to reprisals and security measures that would impose hardships upon the population.⁸ The rural population would eventually be involved in the struggle but Marighella was vague on precisely how.⁹ In their zeal, Guevara and Marighella convinced themselves of their own revolutionary prowess. Guevara, for example, failed to take his own advice about when one should not attempt guerrilla warfare when he departed for Bolivia in 1966: 'Where a government has come into power through some form of popular vote, fraudulent or not, and maintains at least an appearance of constitutional legality, the guerrilla outbreak cannot be promoted since the possibilities of peaceful struggle have not yet been exhausted.'¹⁰ Considered the 'Guerrilla Enemy Number One' by the United States, perhaps even the Osama bin Laden of his generation, Guevara was tracked down by Bolivian Rangers and killed in October 1967. Although both Guevara and Marighella were killed by state security forces, their writings and example inspired a generation of Marxist-Leninist terrorists from 1967 until the collapse of the Soviet Union.

Developments in counterinsurgency

The current generation of irregular warfare experts drew much of their early knowledge from the classic works written between 1960 and 1973 by Filipino,

British, French and American authors with first-hand experience in the Philippines, Malaya, Kenya and Algeria.¹¹ Works by such authors as Robert Thompson, Frank Kitson, David Galula, Roger Trinquier, Napoleon Valeriano and Edward Lansdale are read and still held in high regard for their insights.¹² After President John F. Kennedy officially made counterinsurgency a high national priority in 1962, most of these authors shared their experiences in sponsored conferences and written works or acted as advisors to the United States government.¹³ French Lieutenant Colonel David Galula, for example, produced both of his written works, *Pacification in Algeria* and *Counterinsurgency Warfare*, while sponsored by the RAND Corporation and Harvard's Center for International Affairs in 1963 and 1964.¹⁴ Robert Thompson, considered one of the foremost experts on pacification during the Malayan Emergency and author of works including *Defeating Communist Insurgency*, served as head of the British Advisory Mission in South Vietnam from 1961 until 1965 and later as a personal advisor on counterinsurgency to President Richard Nixon.¹⁵ Much of what these authors have to say is still useful today provided its context is understood (see [Figure 15.1](#)). Given their range of experience and the quality of the works that these authors produced, it is not surprising that this era, which began with the Malayan Emergency in 1948 and ended with the US withdrawal of its combat forces in 1973, is considered the 'golden age of counterinsurgency'.

Doctrine

The 'golden age of counterinsurgency' was marked by two significant developments in modern irregular warfare: an explosion of doctrinal publications and the creation of specialised counter-guerrilla forces. As the [previous chapter](#) has outlined, authors as far back as the American Revolutionary War had attempted to distil lessons learned in fighting irregular adversaries for the benefit of officers faced with this threat in the future. Interest in partisan and small wars waxed during such conflicts but waned as military officers focused their efforts on problems of conventional warfare on land, sea and, later, in the air. Given the wealth of experience that a number of French and British officers had accumulated, it is surprising how few formal attempts were made to capture hard-won lessons in doctrine.¹⁶ The mere existence of doctrine, however, does not guarantee success. Doctrine provides guidelines for leaders to review, digest, use and modify provided it is read and understood. If used improperly and unimaginatively, doctrine can instead be misconstrued as a set of inviolate rules never to be broken, becoming a set of intellectual handcuffs. The absence of doctrine, however, places an undue burden on the leader and his subordinates to improvise and make up techniques and procedures as they go along. As David Galula lamented, the lack of common doctrine to pacify regions in Algeria led to completely different practices being followed even in neighbouring regions and these often changed as officers came and went.¹⁷ Much the same



Figure 15.1 The French experience with irregular warfare in Indochina and Algeria led Roger Trinquier and David Galula to different conclusions about how to combat irregular foes. Here French Foreign Legionnaires, backed by US-supplied tanks, conduct large-scale and ineffective sweeps against guerrilla Viet Minh forces in the Red River Valley in the year of the French defeat at Dien Bien Phu, 1954.

situation existed on a much larger scale within and between the various geographic regional commands in Afghanistan prior to General Stanley McChrystal's appointment as commander of the International Stabilization Force (ISAF) in 2008.

Published American doctrine on irregular war had existed as early as 1921 and was refined into a work that is still useful today: the Marine Corps's 1940 *Small Wars Manual*.¹⁸ Amongst the lessons incorporated into this manual, which had to be relearned recently, were the need for liaison with diplomats (US State Department officials), the role played by aviation in small wars and guidelines for disarming the population, holding elections, establishing local security forces and withdrawing forces.¹⁹ A quarter of a century after the release of the *Small Wars Manual*, the Department of the Army and Headquarters, United States Marine Corps released a catalogue of field manuals and pamphlets from 1963 until 1973 which included: FM 31-15, *Operations Against Irregular Forces*; FM 31-16, *Counter guerrilla Operations*; DA-PAM 550-100, *US Army Handbook of Counterinsurgency Guidelines for Area Commanders: An Analysis of Criteria*; DA-PAM 550-104, *Human Factors Considerations of Undergrounds in Insurgencies*; FMFMRP 12-40, *Professional Knowledge Gained from Operational Experience in Vietnam 1965–1966*; and FMFM 8-2, *Counterinsurgency Operations*. As Colin Gray has noted, the American defence community has a short attention span, is captivated by the most recent intellectual fad and is largely ahistorical in its outlook.²⁰ A decade after the withdrawal of US forces from Vietnam, and with the refocusing of defence priorities back on problems of potential conflict with overwhelming Warsaw Pact forces in Europe, most of these doctrinal publications were shelved and forgotten.

Organisational adaptation

Doctrine, no matter how prolific, is only as useful as the soldiers who put it into practice. As Johann Ewald noted, soldiers who excelled at conventional warfare had difficulties in adapting to the less rigid, more individualist style of fighting in irregular war. To help address this problem, authors including Ewald, Charles Callwell and William Henecker suggested conventional forces should be augmented by forces skilled in partisan and guerrilla methods. Not surprisingly, this era saw the formation or development of a number of specialised military units skilled in guerrilla methods. The British in Malaya had developed the Ferret Force and the Malaya Scouts, as well as deployed 21 Special Air Service Regiment.²¹ The French in Indochina in turn fielded the *Groupe ment de commandos mixtes aéroportés* (GCMA), in which French commissioned and non-commissioned officers led groups of local tribesmen. The commanding officer of the GCMA during the battle of Dien Bien Phu was one Roger Trinquier.²² Within the United States, President Kennedy elevated the status of the fledgling US Army Special Forces, or Green Berets, as the tip of America's spear in the fight against communist insurgencies. Other service commanders saw the way in which the political winds were blowing and Air Force General Curtis LeMay authorised the formation of a squadron, the 4400th Combat Crew Training Squadron or 'Jungle Jim', to provide advice and training to foreign personnel in the use of air power against guerrillas.²³ As with doctrine, the uniquely

qualified personnel of such special operations forces (SOF) could not overcome military and political incompetence and misuse at the operational and strategic level.²⁴

The principles of counterinsurgency

The legacy of the ‘golden age of counterinsurgency’ continues to be a blessing and a curse for those facing current irregular threats. Most doctrine or works from the era, such as Thompson, Kitson, Galula, Trinquier, Valeriano or Lansdale, agree in kind on counterinsurgency but not on the degree or specific approach. These works have led to a set of fairly well-understood principles of countering irregular warfare. Such principles incorporate aspects of, but are not limited to:²⁵

- controlling the population and tracking their movements, through the use of checkpoints, identification checks, forced relocation and/or intimidation and coercion
- gaining the support of the most important population, which is often expressed in the phrase ‘winning the hearts and minds’, which can mean their active participation in local government, self-defence militias and providing information to authorities, usually by providing local security and stability; at the very least, the population should be prevented from supporting insurgents and terrorists (see [Figure 15.2](#))
- obtaining precise information on the organisation and location of insurgent and terrorist groups so they can be killed, captured or turned against their comrades
- creating or improving the quality and quantity of military and paramilitary forces and law enforcement bodies so that they are competent enough to provide local security and stability, with limited outside assistance, as well as deny the insurgents and terrorists sanctuary; these forces must be able to accomplish these objectives in such a manner as to not be perceived by the population as tools of government oppression
- developing, or assisting in the development of, just and responsive governance and co-ordinated responses across all government agencies at the local, regional and national levels to address the political, social and economic conditions which led to violence in the first place and inform ongoing and future actions against the insurgents and terrorists.

As with doctrine and forces trained in defeating terrorists and insurgents, knowledge of such principles alone is insufficient. The curse of this era, and largely of attempts to counter irregular warfare and write doctrine over the past decade, can be summed up in two main problem areas: the ‘big idea’ and stressing performance over politics.



Figure 15.2 The support of the population, which can include providing information on insurgents, terrorists and revolutionaries or taking direct action against them, is based in part on effective local governance. Such governance must be credible, responsive, honest and based on intricately linking local needs with available resources. Such understanding often requires specific cultural information that can only be provided by direct contact with the population. Here an American governor is photographed with locals in Zamboanga province, Mindanao in the Philippines in 1900.

The first problem area, the ‘big idea’, begins while searching for solutions to deal with irregular threats. Practitioners rediscover, repackage or are introduced to campaign narratives and analysis, the details of a specific historical programme or works from the previous generations. For example, several of the doctrinal publications mentioned in a previous section from the Vietnam era were amended or republished recently as part of the Assessing Revolutionary and Insurgent Strategies (ARIS) project.²⁶ Such analogues are appealing as they appear to provide the answer to the current questions. For example, the iconic ‘success’ story of this era, the British campaign in Malaya from 1948 until 1960, was mined for lessons of victory in counterinsurgency applicable today. Most current practitioners have since drifted away from Malaya, given the unique contextual attributes of that conflict such as a racially distinct ethnic-minority

insurgent group and the existence of an integrated, functioning colonial civil administration, and have focused instead on the failed French campaign in Algeria (1954–62). Algeria was appealing in that the works of Trinquier and Galula are accessible and the conflict featured a tapestry of religious and cultural elements similar to those found in Iraq, Afghanistan and elsewhere. Also ‘rediscovered’ was Alastair Horne’s narrative of the Algeria war, *A Savage War of Peace*. While there is a sizeable body of literature on the subject from this period that was rediscovered, other conflicts for which there are few published details or accounts that occurred after ‘the golden age of counterinsurgency’ are apt to be overlooked. The various ‘bush wars’ of Africa and counterinsurgency and counterterrorism campaigns in Central and Southern America during the 1970s and 1980s are rich with insights about insurgent and counterinsurgent adaptation, organisational learning and the interactions between adversaries at the level of policy and strategy which changed the character of those wars.²⁷ More recent ‘big ideas’ to which future practitioners may be drawn, in the historical cycle of interest in and abandonment of counterinsurgency, are the much-publicised and lengthy doctrine manuals, such as the US Army Field Manual 3–24, *Counterinsurgency* and British Army Field Manual Volume 1, Part 10, *Countering Insurgency*, as well as the works of various noteworthy practitioners including David Petraeus, Rupert Smith, Peter Mansoor, H. R. McMaster, David Kilcullen and John Nagl, amongst others.

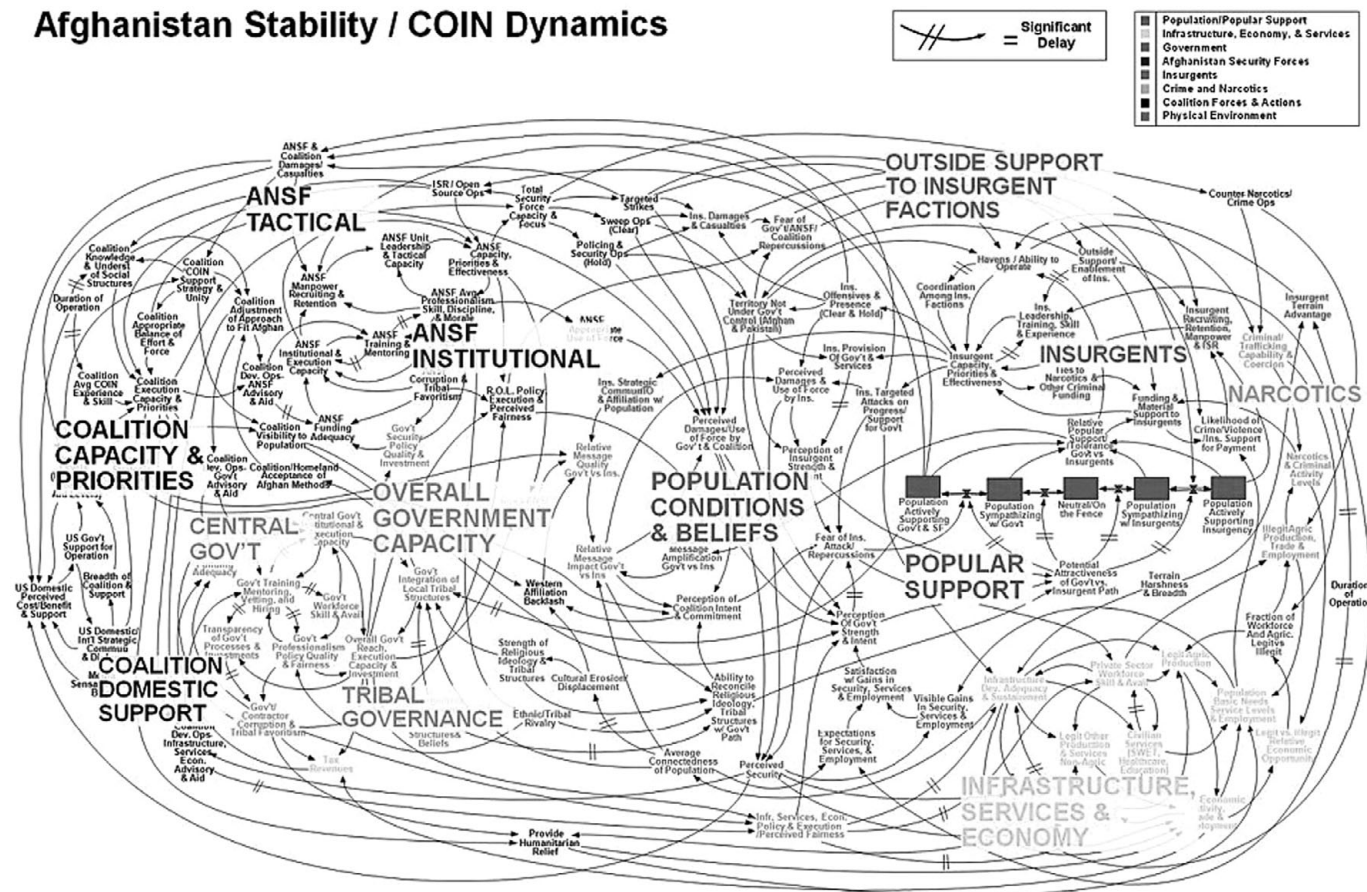
As several scholars have pointed out, reasoning by analogy and looking to history for answers to contemporary problems are fraught with pitfalls.²⁸ One such pitfall is simplifying the past. In reviewing works produced during this time, those looking for answers are apt to take the principles, insights and conclusions they find at face value. What may be overlooked is the specific context of the conflict in which the works were written.²⁹ More recently, the attempt to apply the template for success in Iraq from 2007 onwards, embodied in ‘The Surge’, proved problematic when transferred to Afghanistan from 2008 onwards. Some of this can be put down to time and resource constraints, excessive haste or intellectual laziness. More often than not other errors are based on sins of omission or commission by the original authors or architects. Some authors, including Thompson and Kitson, and more recently David Petraeus and Stanley McChrystal, compressed significant details and omitted nuances and the complexities of the campaigns to save space and ease readability. Contemporary accounts also suffered because participants often related the details of the conflict as they knew and understood them from their specific vantage point or the subjects painted themselves in a positive light. Sins of commission include deliberately playing down negative aspects or contrary evidence of the concepts and ideas put forth. Such accounts are often distilled down even further as they are read for the immediately applicable or ‘actionable’ wisdom contained within. Kitson’s and Thompson’s works, in particular, depict a version of counterinsurgency that appears kinder and more humane to that painted by their French counterparts, Trinquier and Galula. More recent scholarship on

British counterinsurgency challenges the traditional image of 'winning hearts and minds' and suggests 'where they won they did so by being nasty, not nice, to the people' through harsh, coercive methods.³⁰

In simplifying the past, other details can be overlooked as well. For example, the most basic depiction of irregular warfare identifies three primary actors: the insurgent or terrorist group, the population and the government (i.e. counterinsurgent or counterterrorist) forces. The insurgent or terrorist group and the government seek to influence the population to gain their support. On a conceptual level, this is understandable. But for reasons that Clausewitz well understood, what separates war in theory from war in reality, regardless of the form it takes, are the details, frictions and interactions between actors. Few campaigns during the 'golden age of counterinsurgency' involved monolithic insurgent groups, populations or governments. Petty jealousies, rivalries and struggles for predominance between competing factions, shifting allegiances and indifference and incompetence between all three, as well as the practical difficulties of comprehending these much less deal directly with them, is the stuff of irregular warfare. What is often difficult for outsiders to understand in contemporary irregular wars, such as Syria, Iraq and Afghanistan, is the purpose of seemingly brutal and discriminate violence actually reflects the complex tapestry of motivations and uses of political violence at the organisational and local level between numerous competing groups, for which there are no clear-cut or easy solutions (see [Figure 15.3](#)).³¹ Those directly dealing with Mali, Syria, Iraq and Afghanistan have been faced by a range of difficulties, such as competing functional and political requirements, minimising their own casualties, dealing with local tribal and militia leaders, focusing on different problems (insurgent or terrorist groups as in Mali, or corruption in Afghanistan) and making decisions based on incomplete information in areas that have no clear moral dividing line.

The labels 'complex' and 'difficult', however, can be applied to almost all irregular wars, especially Vietnam and Algeria. The war in Vietnam was not a simple 'us versus them' conflict, with the population in between that could have been resolved by invading the North and destroying its alleged centre of gravity, its army. As Harry Summers points out, this 'purely military' option was completely divorced from political reality and therefore unwise and impractical.³² The war in Vietnam was comprised of a number of nested and interrelated problems spiced with the vested interests of various groups. These groups included the US military, whose leaders acted according to their own individual and organisational imperatives, as well as competing social, political, economic and religious elements within South Vietnam such as the Montagnard hill tribes, an influential Buddhist minority and a social elite of Catholic Vietnamese. In Algeria, the French also did not face a single enemy. The conduct for which French forces are mostly remembered, involving torture and forced confessions, must be placed against the backdrop of a caustic sectarian war taking

Afghanistan Stability / COIN Dynamics



WORKING DRAFT – V3

Figure 15.3 No single graphic better depicted the complexity of the operating environment in Afghanistan than the so-called ‘spaghetti chart’ produced in the International Stabilization Force Headquarters and circulating in 2010. For its supporters, the chart demonstrates the complex web of connections between the numerous actors characteristic of an irregular warfare environment. For its detractors, however, the chart was a perfect illustration of bureaucratic process run amok (demonstrating in particular the negative influence of software tools such as Microsoft PowerPoint on strategic thinking) or the ultimate futility associated with bringing stability to Afghanistan.



Figure 15.4 Technology, such as computers, biometrics and data analysis, allows US and coalition forces to manage vast amounts of information today in ways undreamt of by practitioners from the 'golden age of counterinsurgency'. Technology allows new lessons to be learned but contains its own challenges, such as the need to collect, update, manage and analyse mountains of data along with other legal, moral and ethical issues. Here soldiers of the 10th Mountain Division process applicants of a local militia in Iraq as part of the controversial 'surge' of forces in 2007.

place within Algerian society. What is often overlooked in reviews of the campaign is the bitter rivalry, torture, executions and purges conducted between various leaders and groups, as well as factions within the FLN, that claimed tens of thousands of lives.³³ If viewed at the time, the situation confronting French commanders and politicians bears little resemblance to the contemporary perception. Another aspect of oversimplification in the face of complex problems is jumping to snap judgements that the defeat of major powers in irregular wars is inevitable.³⁴

The second and final problem associated with learning from the era can be summarised as the tendency to learn military lessons at the expense of political ones. All of the coalition members involved in the campaigns of Iraq and Afghanistan learned from and adapted to their adversaries. Current practitioners look back into the 'golden age of counterinsurgency', and future leaders will look back to Afghanistan and Iraq, to identify potential techniques and tricks of the trade to improve their overall performance (see [Figure 15.4](#)). Reaching back to the past is laudable, especially if it saves lives and defeats the

irregular threat. The more recent trend, however, stresses improving the performance of the means used (counterinsurgent or counterterrorist forces) while divining little about the ends to be achieved – elevating the conduct of the campaign to the level of strategy by losing sight of the political goal. Countering irregular warfare has little to do with fighting more effectively against insurgents and terrorists oneself, as the principles outlined above suggest, but rather assisting local forces to create the conditions for security and stability from the bottom up while assisting the host nation government in its ability to govern credibly from the top down. Studying Nixon's policy of 'Vietnamisation' for potential political insights applicable to Iraq or Afghanistan, for example, is not to equate the latter with the former. The end result, the collapse of the Army of the Republic of Vietnam, was not due primarily to the removal of US air power from the equation. The assumptions behind the policy of Vietnamisation were sound, but its execution left something to be desired.³⁵ The bottom line was that the host nation, South Vietnam, neither possessed military forces competent or trustworthy enough to protect the nation nor did it have a political system worth defending. And yet, as Harry Summers related to a North Vietnamese colonel years after the fall of Saigon: 'You know you never defeated us on the battlefield' to which the response was '[t]hat may be so . . . but it is also irrelevant'.³⁶ As recent experience in Iraq suggests, in which large parts of the military evaporated in the face of an irregular offensive by forces of the Islamic State of Iraq and Syria (ISIS), all of the military successes in Iraq prior to the withdrawal of coalition forces in 2011 could not offset a fundamentally untenable political situation – inadequate power-sharing between different religious, ethnic and tribal groups within the country.

Why is modern irregular warfare so difficult?

Teasing out the details of irregular warfare, including the forms used and the reasons for success and failure, is not difficult after such conflicts end. Academics can state with certainty that the Irish Civil War began on 28 June 1922 and ended on 30 April 1923.³⁷ Usually the start and end of specific conflicts is tied to crucial events with the benefit of hindsight.³⁸ When faced with such wars now and in the future, however, even the most basic information is unclear and the subject of speculation. The cardinal error a political leader or military commander could commit, according to Clausewitz, is misunderstanding the kind of war about to be undertaken and ensuring that the means applied are suitable to desired ends.³⁹ Problems occur for the political leader and military planner when they are confronted by insurgency or terrorism, or any other form of irregular warfare, for a number of related reasons: confusing differences in the most recent context of political violence for a change in the nature of warfare itself; conceiving the character of irregular warfare in absolute and static terms; mistaking the means and methods used by the adversary for the purpose of the violence; and uncertainty associated with the purpose of irregular violence.

The importance of context

At the risk of repeating a claim made throughout this work, even the most cursory reading of military history and strategy conveys the clear message that all conflicts are unique to and shaped by their specific context. The Chechens, for example, have fought three irregular wars against Russia over more than 150 years. Although the geography was the same in each case, the character of each conflict was shaped by changes in society, technology, culture, information and a host of other factors. In order to understand the root causes and motivations behind irregular conflicts, some scholars suggest that cultural and language studies are crucial. Few would dispute this sound advice. Unfortunately, other authors have interpreted this to mean that cultural and religious reasons alone explain violence, irregular or otherwise, as an end in and of itself. The implication of this conclusion is clear and disturbing: there is no room for negotiation or bargaining with those for whom violence forms an integral part of their religion and/or culture. Violence serves a purpose that suits the social context – a reflection of a ‘sacred value’ – that makes bargaining or negotiation exceptionally difficult.⁴⁰ When cultural or religious warriors fight professional soldiers, the former are willing to adopt methods that the latter cannot or will not embrace. This advantage is crucial in explaining why militias, tribes and others using irregular tactics have frustrated and defeated more powerful adversaries with professional modern armies.

The kernel of truth behind such claims mistakes individual and group motivation, organisation and fighting style – in other words the technique of violence – for its purpose. Irregular warfare is no different from its regular counterpart in terms of the intended outcome. Those who adopt irregular means are looking to obtain power of one sort or another. That power is political in nature and its possession allows the wielder to achieve control as well as conduct systemic change – a redistribution of power and wealth within society all the way to recasting it according to a specific political or religious ideology. The purpose of violence, and the rationale for its use, has been remarkably similar from Auguste Blanqui to the current generation of violent Islamic extremists. Che Guevara, for example, suggested that guerrillas were ‘the *armed vanguard* of the great popular force that supports them’ in which ‘the guerrilla fighter is a social reformer . . . [who] fights in order to change the social system that keeps all his unarmed brothers in ignominy and social misery’ in order to ‘achieve an ideal, establish a new society . . . and to achieve, finally, the social justice for which they fight’.⁴¹ Egyptian Sayyid Qutb, considered one of the most influential modern thinkers of the radical Salafi *jihād* movement, used very similar language in 1966 to describe the catalyst for change: ‘It is necessary that there should be a *vanguard* which sets out with this determination [creating a new system] and then keeps walking on the path.’⁴² The ultimate purpose of this vanguard is almost identical to that spelled out by Guevara: ‘Our foremost *objective is to change the practices of this society*. Our aim is to change the *Jahili* system

at its very roots.⁴³ Much of this same rationale is visible in the statements made by other group leaders or spokesmen, such as Abubakar Shekau, the leader of Boko Haram in Nigeria, or the late Anwar al-Awlaki in Yemen. Not much separates the ultimate purpose of today's insurgents and terrorists than those who struggled 50 and 150 years ago.

Although some militias and other groups limit their scope of their power to traditional ethnic and tribal lands, other groups have sought complete control within state borders. The reason for this is that power continues to be exercised most effectively at the state level in the international system. State leaders have a legitimacy that sub-state leaders and groups do not. Irregular warfare differs from conventional warfare in terms of the form that violence takes and its functional purpose. As the preceding section suggests, heavy emphasis is placed on the organisation for and exploitation of the political aspects of conflict regardless of the form of irregular warfare chosen. Violence serves the purpose of removing a threat to a group's control over a population, as well as permitting the group to expand sympathy for its cause and consolidate its support network and power base. Even violent Islamic extremists, who have as their goal the remaking of all society according to the guidelines contained exclusively in the Qur'an, understand that global revolution must occur one step at a time: 'When God restrained Muslims from Jihaad for a certain period, it was a question of strategy rather than of principle; this was a matter pertaining to the requirements of the movement and not to belief.'⁴⁴ For these very pragmatic reasons, bin Laden and Zawahiri repeatedly stated that the first step in the realisation of this goal is the establishment of one such state (caliphate). The caliphate will serve as a beacon to the faithful by demonstrating that God's will is being realised through governance and social reforms according to radical Salafi norms, as Qutb so clearly indicated: 'The beauty of this new system cannot be appreciated unless it takes concrete form. Hence it is essential that a community arrange its affairs according to it and show it to the world. In order to bring this about, we need to initiate the movement of Islamic revival in some Muslim country.'⁴⁵ The decision of Abu Bakr al-Baghdadi to declare an Islamic caliphate in Iraq and Syria in June 2014 is as much a statement of the confidence of the leadership of ISIS as it is the next perceived step by violent Islamic extremists to remake the political and social landscape of the region.

Misunderstanding the character of violence

In addition to misunderstanding the purpose of irregular violence, political and military leaders are prone to misread the character of the violence. There is little likelihood of this occurring when groups are limited in size, purpose and issue, as was the case with terrorist groups such as the Palestine Liberation Organization, the Popular Front for the Liberation of Palestine, the Red Army Front and the Japanese Red Army. Other cases of irregular violence, such

as coups d'état, only succeed if the plot is clandestine until the moment that violent action is taken. Military and political leaders can misunderstand the character of irregular warfare in any one of a number of ways. In some cases, rather than viewing irregular warfare as a potential set of complex, embedded problem, the phenomenon is viewed instead in absolute terms or as a static phenomenon. In other words, instead of recognising that they might be faced with a series of intertwined irregular problems, or that its character is dynamic and may change or evolve over time, policy-makers and military planners may believe that they are confronting one phenomenon instead. Civil war, insurgency or terrorism each have 'prescribed' solutions. The thought that a civil war can occur within or at the same time as an insurgency or terrorism, that the conflict in Iraq or Syria can remain uncoupled from the global phenomenon of violent Islamic extremism, or that a humanitarian mission can morph into a robust manhunt as was the case in Somalia, is potentially inconceivable.⁴⁶ Similar problems occur when the response to irregular warfare is conceived almost exclusively in military terms. At first glance, such misperceptions may appear trivial or academic until they are linked to policies, military operations and public perception in democracies.⁴⁷ The seemingly endless variety of labels used to encompass all of the irregular violence within Iraq, Syria, Somalia, Yemen, Nigeria and Afghanistan – insurgency, terrorism, civil war and sectarian violence – without sufficiently explaining the changing character and complex nature of each has led to military missteps, public confusion and domestic and international opposition.

Mistaking means and methods for purpose

Irregular warfare can be difficult to understand due to its character, or the means and methods of violence used. There is a tendency to equate the type of irregular violence according to the methods and means used while overlooking the purpose it is designed to achieve. Put simply, the tactic is mistaken for its strategic utility and purpose of the violence (see [Figure 15.5](#)). For example, there is a common misperception that insurgents only use guerrilla tactics. Acts of terrorism, such as bombings, assassinations, hijacking and kidnapping, are strictly the realm of terrorists. Noteworthy theorists, including Mao Zedong (Tse-Tung) and Ernesto 'Che' Guevara, would rail against such simplistic assessments of the use of violence within an insurgency. Both authors acknowledged the role that terrorism, as a tactic, can play in their overall campaign.⁴⁸ Other methods and techniques such as sabotage, subversion and passive resistance, which have considerable tactical utility but are exceptionally difficult to identify and evaluate, have been confused as unique forms of irregular warfare with their own purpose.

Insurgents and terrorists are not limited to specific means or methods but instead make the best use of the available tools to suit the environment. During the portion of the French campaign in Algeria known as 'the battle of the



Figure 15.5 In an effort to grasp the nature of the war in which one is involved, there is a danger of oversimplifying complex aspects of irregular warfare such as confusing means used by insurgents, terrorists and revolutionaries for the ends they are trying to achieve. Those who conduct irregular warfare are first and foremost pragmatic opportunists when it comes to using resources at hand. Viet Cong leaders, for example, paid little mind to the conceptual distinction between acts of terrorism or insurgency during the bombing campaign it conducted in the city of Saigon.

Casbah', Algerian revolutionary Abane Ramdane believed that attacks by guerilla units in the city were senseless and impractical given the terrain and available resources. The complex urban terrain, range of potential targets and presence of international media outlets favoured the use of terrorist tactics for a variety of reasons. Tactically such attacks were an efficient use of terrorist resources and were difficult for the French to prevent.⁴⁹ From a strategic perspective, the terrorist bombings were an effective means of achieving the wider goals of the insurgent cause: generating international recognition and sympathy for their cause or even driving the French out on their own.⁵⁰ Unlike insurgents, those who utilise terrorism exclusively do so not by choice but out of necessity. Insurgents have more options in the method and means of violence they use than does the terrorist, due largely to their ability to harness local discontent and grievances and turn available resources into a greater range of capabilities. That may not stop insurgents from using terrorism as a means of strategic communication or violent 'political theatre'. ISIS beheadings of captured Western

hostages in 2014 served a variety of organisational purposes, including: expanding the conflict by provoking the West to respond with airstrikes; elevating the stature of the group from a perceived minor to a major threat; and horrifying a variety of audiences and bolstering the morale of others.

Some writers draw linkages between the tactic of violence and the cultural arguments mentioned previously when they try to make sense out of suicide bombings conducted in London, Bali and elsewhere. Some of the Western curiosity surrounding so-called 'martyrdom operations' relates to the individual purpose and rationale for killing others and dying in the process. Only by understanding such motivations of individuals such as Umar Farouk Abdulmutallab, the failed Nigerian underwear bomber, can states develop responses to address them. Much of the same logic has been applied to more recent 'lone wolf' attacks at Fort Hood (2009), London (2013), Montreal (2014) and Ottawa (2014). What is missing from such interpretations is a broader context or organisational purpose for the violence. According to Ayman al-Zawahiri, one of the co-founders of al-Qaeda and a frequent spokesman for it, the types of attacks conducted in the name of al-Qaeda and its cause do more than just spread fear and test the faithful.⁵¹ Terrorism is the only one of a number of means in a broader strategy of cumulative effects over time that will eventually achieve the goals of the violent Islamic extremism.⁵² In the absence of a clear and concise 'game plan' along the lines of Mao's 'On Protracted War', some writers, journalists and media pundits attribute a level of strategic genius and divine the specific intent, as well as purpose, of every successful terrorist attack or plot.

The reason policy-makers and others confuse means with ends is related to the nature of organisations that conduct irregular warfare. Leaders of such groups rely on a variety of measures to prevent their detection and tracking by security forces and obtain the resources necessary to conduct their operations. One measure is limiting the amount of information as well as the means by which it is disseminated. For example, Osama bin Laden limited his travel and contact to personal couriers; locating him at a compound in Abbottabad, Pakistan took almost ten years of intelligence-gathering and analysis. Group leaders also often change the scope and tone of their messages and tailor them for specific audiences.⁵³ In his early published statements and audio and video recordings, bin Laden provided lengthy theological justification and historical evidence in support of his call for a global Islamic uprising.⁵⁴ The target audience was not Islamic scholars and imams, who would question his interpretation of theology and history, but rather those with more limited knowledge who favoured action over debate. The earliest speeches in 1996 and 1998 spoke of American support for Israel and other apostate governments, as well as the presence of American forces in the land of the two mosques (Saudi Arabia), amongst other issues.⁵⁵ In a taped statement in 2004, bin Laden suggested that the goal was economically weakening and bankrupting the United States.⁵⁶ In 1998 bin Laden exhorted his followers to kill the Americans and their

allies – civilians and military, whereas the 2006 much-publicised truce statement was directed at the American public.⁵⁷ In other words, al-Qaeda's goals and targets have changed, based on the speaker and the intended audience. Changes in the means of communication have also influenced terrorist and insurgent messaging. ISIS leaders, for example, are adept at using social media and the shock value of content to reach and influence a much younger generation of followers without the need for the more long-winded theological justification associated with statements by bin Laden and Zawahiri.

Other elements complicating understanding of a group's goals result from internal changes and interactions between opponents at the strategic level over time. Given that terrorist and insurgent groups rely on a variety of clean or illicit means to support their activities, such as front companies, extortion, racketeering and smuggling, there is always a risk that the original goals of a group are subsumed by greed and a relish for local influence and power. The sums of money involved are not insubstantial. The loyalist Ulster Volunteer Force in Northern Ireland, for example, conducted activities such as fuel scams, video piracy, drug dealing, counterfeiting and running legitimate laundry and taxi services that netted the group's 350-odd members an annual income of over £3 million.⁵⁸ According to some estimates, ISIS currently benefits not only from external donations but from the sale of petroleum illicitly which provides a *daily* income of US\$2 million. As evidence of the corrupting influence of substantial income, the group *Fuerzas Armadas Revolucionarias de Colombia*, or FARC, originally espoused a Marxist-Leninist platform of social reform but now provides protection to drug cartels and others benefiting from the country's narcotics trade.⁵⁹

Other internal or external events, such as struggles for power within organisations, the death or capture of influential leaders or catalytic events can morph terrorist and insurgent objectives or goals, or even the form of irregular warfare, into something much more complicated. The attack by al-Qaeda in Iraq against the Al Askariya mosque in Samarra in 2006 is one example. Prior to the attack, coalition and Iraqi forces focused their efforts in a conventional counterinsurgency campaign characterised by patrolling operations and large-scale sweeps of known or suspected insurgent sanctuaries.⁶⁰ Shi'ite groups and militias were left largely untouched by coalition forces or utilised in the fight against Sunni groups. The deliberate, systematic attack approved by Abu Musab al-Zarqawi, an al-Qaeda affiliated leader, targeted one of the holiest shrines in Shia Islam.⁶¹ Zarqawi allegedly intended the attack to spark a civil war between Shia and Sunnis. US and coalition commanders prior to the attack were beginning to understand better the context, and nature of conflict and the enemy, in Iraq. Widespread sectarian violence, roving Shia death squads, continued car bombings and tit-for-tat executions challenged the commanders' understanding of and ability to respond to this evolution in irregular violence in the country. In addition, the Samarra attacks, subsequent sectarian violence and the death of Zarqawi opened a rift between the uncompromising foreign fighters of

al-Qaeda in Iraq and other indigenous Sunni insurgent groups and led to US arming and support of the latter against the former.⁶² The so-called 'Sunni Awakening' in al Anbar province, combined with a range of other measures that comprised 'The Surge', contributed to the ability of US and coalition forces to stabilise Iraq and withdraw from the country in 2011.

'Mirror-imaging'

Understanding irregular warfare is also difficult because policy-makers and military planners are prone to 'mirror-image' their irregular adversaries. Part of the reason for this is understandable. If the terrorist and insurgent violence is undertaken for a purpose, then surely the leaders of such movements must have a plan to translate tactical action into achieving stated ends. In addition, as the true scale of the insurgent and terrorist movement is understood through such tools as social network analysis, it is easy to be convinced of a 'master plan' guiding it. In reality, there may be less of an initial plan for victory than the hope that action will somehow translate into purpose. One of the challenges posed by ISIS is the lack of a centralised leadership and the largely decentralised nature of the organisation, which places a premium on the initiative of a number of individual militia commanders. Such measures are useful in the short term. Given enough time, however, most practitioners from Lawrence to Mao have developed a concept of operations that takes the current context into account. As the preceding review has demonstrated, revolutionary theorists have often disagreed on the appropriate method or mechanism for success in irregular warfare. Unfortunately for the recent generation of practitioners dealing with the violent Islamic extremism, these debates continue to rage amongst their opponents. The debates encompass almost all aspects of strategy, including whether to fight the 'near' enemy (where the violent Islamic extremists reside, most often states which are ostensibly Muslim but follow secular laws, such as the Ba'athist Syrian regime of Bashar al-Assad or the southern Christian-dominated democratically elected government of Nigeria) or the 'far' enemy (the United States and other Western nations which represent the purported source of all things wrong and evil).⁶³ In addition, writers and organisers such as Abu Mus'ab al-Suri (born Mustafa Setmariam Nasar), a former member of the Syrian Brotherhood with ties to Zarqawi's al-Qaeda in Iraq, has argued that the violent Islamic extremists conduct what was once labelled 'propaganda by the deed' in an entirely decentralised and independent manner. Victory will eventually result from the cumulative effects of such activities.⁶⁴ Other writers argue that a more focused, sequential plan of action is necessary. Abu Bakr Naji outlines an approach that, in keeping with the principles and practices of irregular war, emphasises both the political and military dimensions of the struggle:

This plan requires: A military strategy working to disperse the efforts and forces of the enemy and to exhaust and drain its monetary and military

capabilities; A media strategy targeting and focusing on two classes. [The first] class is the masses, in order to push a large number of them to join the *jihad*, offer positive support, and adopt a negative attitude toward those who do not join the ranks. The second class is the troops of the enemy who have lower salaries, in order to push them to join the ranks of the mujahids or at least flee from the service of the enemy.⁶⁵

A core component of the Naji's vision of success is a hierarchy of highly capable administrators who guide and manage the revolutionary masses harking back to the earlier generation of socialist revolutionary theorists.⁶⁶ Insurgent and terrorist groups are not immune to the same squabbles and incompetence as their opponents, nor can they escape the logic of organisation or strategy.

SOF and the golden age of counterterrorism?

The declared withdrawal of US forces from Afghanistan in 2014, combined with the declared strategic pivot to Asia, led many in the West to breathe a sigh of relief regarding irregular warfare. Military forces would not have to train or prepare for irregular warfare, but return instead to conventional core competencies such as combined arms warfare or cross-domain warfare outlined in AirSea Battle. The frustrations of irregular warfare would fade into memory in pursuit of new technologies and cracking new conceptual problems such as 'anti-access/area denial' (A2/AD). Political leaders, as well as the domestic political base, have little apparent stomach or enthusiasm to counter irregular warfare or tackle nation-building on the previous scale of Iraq and Afghanistan, especially given the fiscal restraints under which their nations are operating. Terrorist groups threatening the West or Western interests in countries including Mali, Nigeria, Yemen, Somalia and elsewhere still need to be combated, given the extreme acts of political violence they or their agents undertake, such as kidnappings, beheadings, abductions of school girls amongst others.

Unlike counterinsurgency, there is no modern deep well of experience from which to draw for counterterrorism to avoid the pitfall of the 'big idea'. Prior to 11 September 2001, terrorism was viewed in most countries, including the United States, as crime best dealt with by law-enforcement agencies. The shocking scale of attacks first in New York and Washington (2001) and later in London (2005), Mumbai (2008), Paris (2015) as well as the ambitious nature of failed or thwarted plots emanating from Yemen and elsewhere has led increasingly to the use of the military in response, turning the response to terrorism from a police matter into warfare. This in turn has led to increasing emphasis on specialised military forces to combat terrorism.

For Western nations the economical solution to address the current irregular threat of terrorism, as during the 'golden age of counterinsurgency', is to counter them with irregulars of their own: special operations forces (SOF). Comprised of highly motivated, specially selected, designated and equipped operators, SOF



Figure 15.6 Special operations forces (SOF) lower political risk for decision-makers through their high degree of tactical performance based on their specialised selection, designation, equipment and joint training. SOF assume higher personal and tactical risk than conventional forces, and often have effects disproportionate to the small size of their units by doing what others think is impractical or impossible. Here US Navy SEALs train on a transportation system with pilots and crew of the US Army's 160th Special Operations Aviation Regiment.

are appealing strategically for a variety of reasons, including their ability to access or threaten remote regions, conduct risky operations with a high likelihood of success, operate across and between legal authorities, experiment and innovate on a small scale and resolve problems economically and quietly compared to other military forces (see [Figure 15.6](#)). The selection of special operators, however, comes at the price of limited numbers, and therefore an inability to mass or sustain strategic effects for long periods of time.⁶⁷ To add numbers or mass, or as a prerequisite for access, SOF must work by, with and through others, which can include conventional forces or other government agencies as well as proxies such as host-nation armed or security forces or local militias and armed groups. SOF are most successful when they improve the operational performance and strategic effectiveness of other forces or proxies through their unique skills and training, access to equipment and information, or utilise their connections to other forces or agencies.⁶⁸ This has led some authors to herald the combination of SOF, proxies and precision air power as a new American way of war.⁶⁹ In addition, SOF equipment, procedures and processes, such as

Box 15.1 SOF: force of choice

The idea that SOF will be the acceptable 'force of choice' to address current and future irregular threats, within an entirely unique domain of warfare, is apparent in the following quotes from the strategic direction of the Commander, United States Special Operations Command (USSOCOM):

- 'The Global SOF Enterprise will become a globally networked force of special operations forces, services, interagency, allies and partners able to rapidly or persistently address regional contingencies and threats to stability.'
- 'SOCOM must not only continue to pursue terrorists wherever we may find them, we must rebalance the force and tenaciously embrace indirect operations in the Human Domain – the totality of the physical, cultural and social environments that influence human behavior in a population-centric conflict.'
- 'We must think differently, seek greater understanding of local, regional and global contexts and strengthen trust through interagency and partner cooperation.'
- 'While SOF is designed to contribute to or support efforts in every domain of warfare, the vast majority of SOF expertise lies in the Human Domain of competition, conflict and war. The Human Domain is about developing understanding of, and nurturing influence among critical populaces. Operating in the Human Domain is a core competency for SOF and we are uniquely suited for successful operations or campaigns to win population-centric conflicts.'

Quoted in *ARSOF 2022, Part 1* (Fort Bragg, NC: US Army Special Operations Command, April 2013), p. 7

the F3EA methodology (find, fix, finish, exploit, analyse), have migrated to conventional and other coalition forces.

SOF are not only appealing culturally, as evidenced by a number of books and films about their exploits including the raid that killed Osama bin Laden in 2011, but also strategically and politically. Politically SOF are 'more cost effective . . . and less likely to cause attendant political and diplomatic complications' than larger conventional forces.⁷⁰ One measure of the confidence political leaders place in SOF has been the creation of specialised commands, along with increased budgets, manpower and support functions, in the UK, Australia, Canada, Russia and Norway, as well as within NATO. Another indication of SOF's political appeal has been their use in more recent military operations to bridge the gap between conventional forces and proxies on the ground, as occurred in Libya in 2011 and more recently in Iraq in 2014, or in 'global pursuit' of terrorist and insurgent groups and leaders in a number of countries (see [Box 15.1](#)). In addition to being economical relative to other military forces, SOF are seen as reliable by political leaders, in the sense they achieve results with manageable political risk.

SOF appear as the cheap solution to the seemingly intractable problems irregular warfare poses for Western nations. For political leaders or other military

forces to place the burden of countering irregular threats exclusively on the shoulders of SOF is to miss several of the problems attendant on irregular warfare. First, SOF can often achieve immediate results in pressing irregular problems but this is akin to curing the immediate symptom at the expense of treatment of the underlying disease. Secondly, SOF act as enablers for other forces by enhancing their performance strategically or operationally over time. One of the greatest challenges about working with host-nation forces, or through proxies, is the inability to control them in the direction you prefer strategically, much less politically. Thirdly, by operating across and between legal authorities, SOF lower political risk at the expense of greater legal and ethical risk. In order to pursue irregular opponents who flaunt international norms and laws SOF often operate in normative and legal grey areas that can be morally hazardous, especially in the covert or clandestine realm. Put simply, although SOF are best suited to tackle immediate irregular threats, they only establish the conditions for the future success of others to exploit and are not a solution to irregular warfare in and of themselves. Given the difficulties inherent in irregular warfare, the proposed solution may seem appealing and cheap now but such preferences are apt strategically to miss the contextual forest for the trees.

Conclusion

To sum up the three chapters on the subject, irregular warfare is difficult to understand for a variety of reasons. One reason is the belief that irregular violence takes unchanging, absolute forms. Those conducting irregular warfare ultimately have to be pragmatic about the violence they use in order to survive, and leaders choose the violence that best suits the local conditions at the time. Different forms of violence can be used simultaneously, or in sequence, in an overall national or global campaign. The purpose of the violence can also change over time. Another reason why irregular warfare is difficult to understand has to do with confusing the method of violence employed for its purpose. This creates enough problems on its own, but intuiting the overarching purpose of groups conducting irregular warfare, much less the competing groups and agendas, is not as simple as it may appear. Insurgent and terrorist leaders may not reveal true goals at the time to their followers or in statements. In addition, published and available statements are often contradictory and the message can change based on the intended audience or over time. Finally, some mistake the complexity of irregular wars for changes in the nature of war. The very character of the irregular wars can change as enemies adapt to one another, alliances are forged, rifts form between leaders and groups, or other actors join in the fight. In other words, irregular warfare is difficult to understand for precisely the reasons that Carl von Clausewitz suggested make all wars difficult: 'The consequences of the nature of War, how ends and means act in it, how in the modifications of reality it deviates sometimes more, sometimes less, from its strict original conception, fluctuating backwards and forwards.' To this Clausewitz added an important qualifier. No matter which form war takes, or how

puzzling it appears to be to the participant and observer, war cannot escape its nature: 'always remaining under that strict conception [the nature of War] as under a supreme law'.⁷¹ Despite its complexities and changes, these three chapters have argued that many aspects of irregular warfare have remained and will remain remarkably consistent due the nature of all wars that Clausewitz so keenly deduced almost two centuries ago.

RESEARCH QUESTIONS

1. Is irregular warfare primarily a problem of resources, organisation or information? For whom is the specific problem most acute and why?
2. What changes and what stays the same in irregular warfare? What drives change in irregular warfare, and what will its future look like?
3. Why is context so important in irregular warfare? Are lessons from the specific contexts during the 'golden age of counterinsurgency' transferrable to other contexts or not?
4. How do those confronting irregular warfare avoid the traps of 'the big idea' and 'mirror-imaging'?
5. Do special operations forces herald a new chapter in the history of irregular warfare? What problems do SOF address and what do they not?

FURTHER READING

- Atran, Scott, *Talking to the Enemy* (New York: Ecco, 2010) reads like popular fiction but examines terrorism at both the local group level as well as the cultural group level based on the author's research, travels and experience interviewing terrorists.
- Cronin, Audrey Kurth, *How Terrorism Ends* (Princeton University Press, 2009) is compelling not only for the strength of its categorisation and analysis but for the clarity and accessibility of the author's writing.
- French, David, *The British Way in Counter-insurgency, 1945–1967* (Oxford University Press, 2011) provides a useful corrective, based on solid research and scholarship, to a number of the myths associated with the British operations during the 'golden age of counterinsurgency'.
- Galula, David, *Pacification in Algeria, 1956–1958*, MG-478–1 (Santa Monica, CA: RAND, 2006). Unlike Galula's more famous analysis of what succeeds and fails against insurgents, *Counterinsurgency Warfare*, this reprinted monograph provides a battalion-level view of the practical challenges associated with putting counterinsurgency theory into practice. In particular, Galula offers insights into the moral hazards and other problems small-unit leaders face when they conduct counterinsurgency.
- Merom, Gil, *How Democracies Lose Small Wars* (Cambridge University Press, 2003) suggests that the character of violence in irregular warfare can contribute to a divergence in opinion between political elites in democracies and those with significant influence who bear the costs of the conflict.
- Simpson, Emile, *War From the Ground Up* (New York: Columbia University Press, 2012) is a thoughtful and insightful attempt by a contemporary former practitioner to

put his experiences and observations into a broader theoretical and philosophical context.

Trinquier, Roger, *Modern Warfare: A French View of Counterinsurgency* (New York: Praeger, 1964) is a concise monograph written after the author's experiences leading partisan forces in Indochina and combating insurgents and terrorists in Algeria. This work remains influential for two reasons: the author's argument that the character of modern warfare demands a pragmatic, modern, response and no other work better captures the practitioner's expression that the ends justify the means, regardless of their consequences.

ONLINE RESOURCES

The Joint Command and Staff College bibliography on counterinsurgency (www.da.mod.uk/Portals/0/Documents/JSCSC%20Library%20-%20Bibliographies/Research%20Guides/Counter%20Insurgency.pdf) is both comprehensive on the subject as well as current, compared to other online bibliographies.

The US Army Special Operations Command's Assessing Revolutionary and Insurgent Strategies (ARIS) website (www.soc.mil/ARIS/ARIS.html) offers a number of classic counterinsurgency studies and reports to download, including the two-volume 'Casebook on Insurgency and Revolutionary Warfare', 'Human Factors Considerations of Underground in Insurgencies', as well as an irregular warfare bibliography.

The Combating Terrorism Center at West Point website (www.ctc.usma.edu/) offers several useful and informative resources, including an online journal (the *CTC Sentinel*) as well as a number of worthwhile reports on different aspects of regional and global terrorism. A number of the reports are based on captured and translated documents, making their insights exceptionally valuable.

The blog 'War Is Boring' (<https://medium.com/war-is-boring>) features short, informative articles on a wide range of topics related to irregular warfare. For example, recent article titles include 'Even Islamic State Can't Get Baghdad and Erbil to Cooperate', 'U.S. Commandos Are Flying Around Iraq', and 'Bitcoins Kind of Suck at Funding Terrorism'.

To keep track of daily, newsworthy developments, the Canadian Forces College operates a website, the Spotlight on Military News and International Affairs (www.cfc.forces.gc.ca/254-eng.html), that provides daily links to articles and analyses from a wide range of English- and French-language media outlets and journals.

NOTES

1. The text of the speech is available online through a number of websites, including http://coombs.anu.edu.au/~vern/van_kien/declar.html and www.fordham.edu/halsall/mod/1945vietnam.html.
2. For details see Martin Windrow, *The Last Valley* (London: Weidenfeld and Nicolson, 2004); Bernard Fall, *Hell in a Very Small Place: The Siege of Dien Bien Phu* (Philadelphia, PA: Lippencott, 1967), esp. pp. 415–25.
3. A conclusion reached by Thomas Marks in *Maoist Insurgency since Vietnam* (London: Frank Cass, 1996), pp. 285–9.

4. Menachim Begin, *The Revolt: The Story of the Irgun*, trans. Samuel Katz, 7th edn (Jerusalem: Steinmatzky, 1977), pp. 212–37, 283–90.
5. Alistair Horne, *A Savage War of Peace: Algeria, 1954–1962* (New York: Penguin, 1985), pp. 245–7.
6. Brian Loveman and Thomas Davies, Jr, *Che Guevara: Guerrilla Warfare*, 3rd edn (Lanham, MD: SR Books, 2004), pp. 52–5.
7. Regis Debray, *Revolution in the Revolution? Armed Struggle and Political Struggle in Latin America* (London: Pelican, 1968), pp. 22–5.
8. Carlos Marighella, 'Minimanual of the Urban Guerrilla' (n.p., 1969) (www.baader-meinhof.com/students/resources/print/minimanual/manualtext.html).
9. Carlos Marighella, *For the Liberation of Brazil*, trans. John Butt and Rosemary Sheed (Harmondsworth: Penguin, 1971), pp. 49–50.
10. Loveman and Davies, *Che Guevara*, p. 51.
11. The year 1960 marked the end of the Malayan Emergency while 1973 marked the withdrawal of US forces from Vietnam.
12. Works include Frank Kitson, *Low Intensity Operations: Subversion, Insurgency, Peace-keeping* (London: Faber and Faber, 1971); Roger Trinquier, *Modern Warfare: A French View of Counterinsurgency* (New York: Praeger, 1964); Napoleon Valeriano and Charles Bohannon, *Counter-guerrilla Operations: The Philippine Experience* (New York: Praeger, 1962); Edward Lansdale, *In the Midst of Wars: An American's Mission to Southeast Asia* (New York: Harper and Row, 1972).
13. See, for example, the list of participants at a symposium hosted by RAND in 1962. The report from the symposium has been recently reissued. See Stephen Hosmer and Sibylle Crane, *Counterinsurgency: A Symposium, April 16–20, 1962*, R412–1-ARPA (Santa Monica, CA: RAND, 2006) (www.rand.org/pubs/reports/2006/R412-1.pdf).
14. David Galula, *Counterinsurgency Warfare: Theory and Practice* (New York: Praeger, 1964) and *Pacification in Algeria, 1956–1958*, MG-478–1 (Santa Monica, CA: RAND, 2006).
15. Thompson's most noteworthy published works include *Defeating Communist Insurgency: Experiences in Malaya and Vietnam* (London: Chatto & Windus, 1966) and *Revolutionary Warfare in World Strategy, 1945–1969* (New York: Taplinger, 1970). The subtitle of Thompson's autobiography gives some idea of his sense of self-worth. *Make for the Hills: The Autobiography of the World's Leading Counterinsurgency Expert* (London: Leo Cooper, 1989).
16. The exceptions occurred in but were specific to the conflict in Malaya: 'Despatch 5, Intelligence Summary: Lessons of the Malayan Emergency' and 'The Conduct of Anti-terrorist Operations, Malaya' or 'ATOM'. A summary of the development of these documents and their contents is contained in Tim Jones, *Postwar Counterinsurgency and the SAS, 1945–1952: A Special Kind of War* (London: Frank Cass, 2001), pp. 134–7.
17. Galula, *Pacification in Algeria*, pp. 64–8.
18. Headquarters, United States Marine Corps, *Small Wars Manual* (Washington: Government Printing Office, 1940) (www.smallwars.quantico.usmc.mil/sw_manual.asp). For details on the development of Marine Corps irregular warfare doctrine, see Keith Bickel, *Mars Learning: The Marine Corps' Development of Small Wars Doctrine, 1915–1940* (Boulder, CO: Westview Press, 2001).
19. HQ, USMC, *Small Wars Manual*, chs. IX, XI–XIV.

20. Gray's assessment of the American way of war is both keenly insightful and disheartening at the same time. Colin Gray, *Irregular Enemies and the Essence of Strategy: Can the American Way of War Adapt?* (Carlisle Barracks, PA: Strategic Studies Institute, 2006), pp. 30–49.
21. The creation and disbandment of Ferret Force, the Malayan Scouts and 21st SAS Regiment are recounted in Alan Hoe and Eric Morris, *Re-enter the SAS: The Special Air Service and the Malayan Emergency* (London: Leo Cooper, 1994).
22. For details, see Howard Simpson, *Dien Bien Phu: The Epic Battle America Forgot* (Washington: Brassey's, 1994), pp. 170–1; Roger Trinquier, *Les Maquis d'Indochine: les missions spéciales du service action* (Paris: Albatros, 1976).
23. For details on Jungle Jim see Edward Westermann, 'Relegated to the Backseat: Farm Gate and the Failure of the Air Advisory Effort in South Vietnam, 1961–1963', in Donald Stoker (ed.), *Military Advising and Assistance* (London: Routledge, 2007), pp. 127–50.
24. A problem discussed in depth in James Kiras, *Special Operations and Strategy: From World War II to the War on Terrorism* (London: Routledge, 2006), pp. 58–82.
25. Anthony James Joes, for example, suggests a much more comprehensive list of principles and practices in a phased plan in *Resisting Rebellion: The History and Politics of Counterinsurgency* (Lexington, KY: University Press of Kentucky, 2004), pp. 232–55.
26. See the ARIS website for details (www.soc.mil/ARIS/ARIS.html).
27. Such works include John Cann, *Counterinsurgency in Africa: The Portuguese Way of War, 1961–1974* (Westport, CT: Greenwood, 1997); Jose Bracamonte and David Spencer, *Strategy and Tactics of the Salvadoran FMLN Guerrillas: Last Battle of the Cold War, Blueprint for Future Conflicts* (Westport, CT: Greenwood, 1995); David Spencer, *From Vietnam to El Salvador: The Saga of the FMLN Sappers and Other Guerrilla Special Forces in Latin America* (Westport, CT: Greenwood, 1996).
28. Peerless works on the subject remain Richard Neustadt and Ernst May, *Thinking in Time: The Uses of History for Decision Makers* (New York: Free Press, 1988); Yuen Foong Khong, *Analogies at War: Korea, Munich, Dien Bien Phu, and the Vietnam Decisions of 1965* (Princeton University Press, 1992).
29. This obvious, but often overlooked point, is the basis of David Kilcullen, 'Counterinsurgency Redux', *Survival* 48(4) (Winter 2006–7), 111–30.
30. David French, *The British Way in Counter-insurgency, 1945–1967* (Oxford University Press, 2011), p. 251.
31. Stathis Kalyvas, *The Logic of Violence in Civil War* (Cambridge University Press, 2006), pp. 146–72.
32. See Harry Summers, Jr, *On Strategy: The Vietnam War in Context*, 4th edn (Carlisle Barracks, PA: Strategic Studies Institute, 1983), p. 75.
33. A concise summary of the rivalries and religious dimension of the conflict is contained in Abder-Rahmane Derradji, *The Algerian Guerrilla Campaign: Strategy and Tactics* (Lampeter: The Edwin Mellen Press, 1997), pp. 213–17. For an inside account of these rivalries and the subsequent purges that took place, see Rémy Madoui, *J'ai été fellagha, officier et déserteur: biographie du FLN à l'OAS* (Paris: Seuil, 1994), pp. 117–92.
34. Dale Walton does an exceptional job of outlining the strategic roads open to, but ultimately left unchosen by, US decision-makers during the Vietnam War in *The Myth of Inevitable U.S. Defeat in Vietnam* (London: Frank Cass, 2002), pp. 151–8.

35. For a comprehensive assessment of the policy see James Willbanks, *Abandoning Vietnam: How America Left and South Vietnam Lost its War* (Lawrence, KS: University Press of Kansas, 2004).
36. Summers, *On Strategy*, p. 1.
37. Bill Kissane, *The Politics of the Irish Civil War* (Oxford University Press, 2005), p. 1.
38. For different dates see *ibid.*, p. 93 and J. Bowyer Bell, *The Secret Army: The IRA, 1916–1979* (Cambridge, MA: MIT Press, 1980), p. 38.
39. Carl von Clausewitz, *On War*, ed. and trans. Michael Howard and Peter Paret (Princeton University Press, 1976), p. 89.
40. Scott Atran, *Talking to the Enemy* (New York: Ecco, 2011), pp. 373–401.
41. Loveman and Davies, *Che Guevara*, pp. 52, 55.
42. Sayyid Qutb, *Milestones* (New Delhi: Islamic Book Service, 2002), p. 21.
43. *Ibid.*
44. *Ibid.*, p. 76.
45. *Ibid.*, p. 11.
46. The failure of senior US policy-makers to anticipate reactions by Somali warlords to the humanitarian relief mission is contained in Jonathon Stevenson, *Losing Mogadishu: Testing U.S. Policy in Somalia* (Annapolis, MD: Naval Institute Press, 1993).
47. The central argument in Gil Merom, *How Democracies Lost Small Wars* (Cambridge University Press, 2003), pp. 33–60.
48. See, for example, Mao Tse-Tung, *On Guerrilla Warfare*, trans. Samuel B. Griffith (New York: Praeger, 1961), pp. 21, 87; Loveman and Davies, *Che Guevara*, pp. 60–1. It should be noted that Mao does not refer to terrorism specifically but his statements on how to deal with traitors, collaborators and other counter-revolutionary elements, couched with euphemisms, suggest his intent.
49. Derradji, *The Algerian Guerrilla Campaign*, p. 161.
50. Derradji suggests that Ramdane first developed the idea for an urban terror campaign, a claim supported by Alistair Horne. Aussaresses suggests that Larbi Ben M'Hidi, who received orders from Ramdane, saw the bombing campaign as a method of coercing the French to leave. *Ibid.*; Horne, *A Savage War of Peace*, p. 184; Paul Aussaresses, *The Battle of the Casbah: Terrorism and Counter-terrorism in Algeria, 1955–1957* (New York: Enigma, 2002), p. 65.
51. Lawrence Wright suggests that Zawahiri, far from being a subordinate to bin Laden, formed one-half of the whole which allowed the formation of al-Qaeda: 'Each man filled a need in the other. Zawahiri wanted money and contacts, which bin Laden had in abundance. Bin Laden, an idealist given to causes, sought direction; Zawahiri, a seasoned propagandist, supplied it.' See Lawrence James, *The Looming Tower: Al-Qaeda and the Road to 9/11* (New York: Knopf, 2006), p. 127.
52. See, for example, translations of excerpts from Ayman al-Zawahiri's *Knights under the Prophet's Banner* printed in *Al-Sharq al-Aswat*. 'Al-Sharq Al-Aswat Publishes Extracts from Al-Jihad Leader Al-Zawahiri's New Book', translated and published by the Foreign Broadcast Information Service, Version 2, GMP20020108000197 (2 December 2001), 72–5.
53. I owe an intellectual debt to Bard O'Neill, whose succinct analysis of the problems associated with insurgent goals remains unsurpassed. He identifies five

- problems: changing goals, conflicting goals, misleading rhetoric, ambiguous goals and confusion of ultimate and intermediate goals. Bard E. O'Neill, *Insurgency and Terrorism: From Revolution to Apocalypse*, rev. edn (Washington, DC: Potomac Books, 2005), pp. 29–31.
54. See, for example, the extensive and selective quotations from the Qur'an and radical interpretations of it in bin Laden's 1996 fatwa, published by the *Al-Islah* newspaper in London on 2 September 1996, entitled 'Declaration of War against the Americans Occupying the Land of the Two Holy Places' (www.pbs.org/newshour/terrorism/international/fatwa_1996.html).
 55. See *ibid.* and 'Jihad against Jews and Crusaders' first published in *al-Quds al-Arabi* in London on 28 February 1998 (www.fas.org/irp/world/para/docs/980223-fatwa.htm).
 56. See 'Bin Laden: Goal Is to Bankrupt the U.S.', CNN.com (1 November 2004) (www.cnn.com/2004/WORLD/meast/11/01/binladen.tape/).
 57. See "'Jihad against Jews and Crusaders" and Text: "Bin Laden Tape"', *BBC News*, 19 January 2006 (http://news.bbc.co.uk/2/low/middle_east/4628932.stm).
 58. Andrew Silke, 'Drink, Drugs, and Rock'n'Roll: Financing Loyalist Terrorism in Northern Ireland – Part Two', *Studies in Conflict and Terrorism*, 23(2) (April 2000), 107–27.
 59. For FARC funding amounts see Gabriel Marcella, *Plan Columbia: The Strategic and Operational Imperatives* (Carlisle Barracks, PA: Strategic Studies Institute, April 2001), pp. 3–4.
 60. The most notorious Sunni sanctuary was Fallujah, which was the object of two major operations in April and November 2004. The first operation, Vigilant Resolve, ended in the premature withdrawal of forces after a negotiated settlement. For details of the second operation, Phantom Fury, see John Ballard, *Fighting for Fallujah: A New Dawn for Iraq* (Westport, CT: Praeger, 2006).
 61. The Al Askariya mosque is also known as the 'Tomb of the Two Imams' as it houses the remains of two of the twelve Shi'a imams, Ali al-Hadi and Hasan al-Askari. According to one report, the perpetrators of the attack took six hours to place charges in selected points throughout the structure to inflict maximum damage. 'TRITON Quick Look Report: IED Attack on the Al Askariya Mosque in Samarra, 22 February 2006' (Faringdon: Hazard Management Solutions, 26 February 2006), p. 4.
 62. John Ward Anderson, 'Sunni Insurgents Battle in Baghdad: Residents of Western Neighborhood Join Groups' Fight against Al-Qaeda in Iraq', *Washington Post* (1 June 2007), A11; John Burns and Alissa Rubins, 'U.S. Arming Sunnis in Iraq to Battle Old Qaeda Allies', *New York Times* online edition (11 June 2007) (www.nytimes.com/2007/06/11/world/middleeast/11iraq.html?ex=1339214400&en=7c69df022224828e&ei=5090&partner=rssuserland&emc=rss).
 63. Marc Sageman, *Understanding Terror Networks* (Philadelphia, PA: University of Pennsylvania Press, 2004), pp. 25–48.
 64. al-Suri is perhaps best known for his 1,600-page manual, the title of which is roughly translated as 'The Call to Global Islamic Resistance'. Murad Batal al-Shishani, 'Abu Mus'ab al-Suri and the Third Generation of Salafi-Jihadists', *Terrorism Monitor*, 3(16) (11 August 2005), 1–3.

-
65. Abu Bakr Naji, *The Management of Savagery: The Most Critical Stage through which the Umma Will Pass*, trans. William Cants (West Point, NY: Combating Terrorism Center, 2006), p. 21 (www.ctc.usma.edu/Management_of_Savagery.pdf).
 66. *Ibid.*, pp. 23–36.
 67. For details see Kiras, *Special Operations and Strategy*, pp. 58–82.
 68. James Kiras, 'Special Operations and Strategies of Attrition', *Infinity Journal*, 2(4) (Fall 2012), 18–21.
 69. Richard Andres, Craig Wills and Thomas Griffith, 'Winning with Allies: The Strategic Value of the Afghan Model', *International Security*, 30(3) (Winter 2005–6), 124–60.
 70. Linda Robinson, *One Hundred Victories* (New York: Public Affairs, 2013), p. 261.
 71. Carl von Clausewitz, *On War*, trans. J. J. Graham (London: Routledge and Kegan Paul, 1956), vol. I, p. 45.





Part VI

Weapons of mass destruction

C. Dale Walton



16

Weapons of mass destruction: radiological, biological and chemical weapons

Contents

Introduction: defining weapons of mass destruction (WMDs)

Radiological, biological and chemical weapons: a brief primer

 Radiological weapons

 Biological weapons (BW)

 BWs and terrorism

 BWs and states

 Chemical weapons (CW)

 Military use of CW

Why states acquire and retain chemical and biological weapons today

Conclusion: CBWs in the present and future

KEY THEMES

- Weapons of mass destruction (WMDs) are treated as distinct from 'conventional' weapons, and their stockpiling and use are particularly controversial.
- WMDs are divided into four general types: radiological, biological, chemical and nuclear weapons.
- For a well-equipped modern military, the battlefield utility of radiological and current biological weapons (BWs) is modest. Nonetheless, terrorists might find such weapons useful in inciting fear and causing economic and other damage.
- There is an existing norm, accepted by the great majority of states, against the stockpiling and use of chemical and biological weapons (CBWs). However, this norm could break down in the future.

Introduction: defining weapons of mass destruction (WMDs)

While most of the chapters in this book focus chiefly on how warfare is conducted, and only address weapons within that context, any discussion of 'weapons of mass destruction' (WMDs) must have a somewhat different focus.

This is because these weapons have been placed in a special category, marking the weapons themselves as being extraordinary and making their use uniquely controversial. Indeed, the term conventional weapons, which generally is applied to virtually all non-weapons of mass destruction, is telling, as it implies that such weapons are 'normal' and thus usable, unlike their WMDs counterparts. This focus on the weapon itself is unusual, and is in substantial contrast to the attitude towards most conventional weapons, which generally are treated more like various tools in a tool chest, some of which are appropriate for particular jobs but not for others, but all of which are ethically acceptable in a general sense, even if many of them ethically cannot be used in every situation.

Any reasonable recounting of the overall WMDs 'story', therefore, must address *why* these weapons have not been used in various situations where they would have promised military advantage. This is critical because, in general, one should expect that a weapon which is likely to yield military advantage will be used in warfare, and that its use will not be limited or prohibited by international agreement. Indeed, the very notion of prohibition seems slightly bizarre in regard to most conventional weapons. Few individuals would call for tanks, artillery, helicopters or most other 'conventional' weapons systems to be banned globally (although there have been efforts to ban or control the use of landmines, cluster munitions, white phosphorous and certain other weapons that are perceived as being especially cruel in character and/or dangerous to civilians). Instead, it is simply, and almost universally, assumed that for so long as these systems are militarily useful they will be fielded, and when and if they cease to be useful they will be consigned to museums, joining longbows, chariots, bronze cannon and thousands of other outmoded weapons.

There is, however, no question that at least some WMDs continue not only to be potentially useful on the battlefield, but, as the term WMDs itself implies, potentially devastating. This raises the question of why limitations have arisen on the use of these weapons. Nor, it should be noted, have these limits only been legal in character. The legality of the use of nuclear weapons under international law is debatable, but a good case could be made that, if used against military targets (particularly ones well removed from substantial civilian populations), their use could be lawful, at minimum for those states recognised as 'legitimate' nuclear possessors by the Non-Proliferation Treaty. However, despite the numerous wars fought by nuclear-armed states, none has used these weapons in combat since 1945. (Possible reasons why this has been the case are addressed in the [next chapter](#).)

Even the term 'weapons of mass destruction' is problematic, as it describes a wide variety of weapons and there is no single, authoritative definition of what constitutes WMDs. However, the term generally is used to describe any and all nuclear, chemical, biological and radiological weapons (see [Figure 16.1](#)), contrasting such devices with 'conventional' weapons – which essentially are all weapons that are *not* WMDs. Nonetheless, WMDs vary enormously in their characteristics and destructive capabilities.



Figure 16.1 US Navy sailors during an exercise in which yellow smoke is used to simulate exposure to chemical, biological or radiological weapons.

The term WMDs itself can be deceptive, as the phrase ‘mass destruction’ evokes images of devastation over a broad area. Some WMDs, certainly, are capable of inflicting such cataclysmic damage – thermonuclear warheads, for example, can have a lethal radius measured in miles, with a large area around ground zero (the point where the warhead is detonated) being utterly destroyed, buildings obliterated and humans and animals instantaneously killed. However, a canister of mustard gas (a chemical weapon) is also considered a weapon of mass destruction. This is despite the fact that such a modest quantity of mustard gas, even if released on a busy city street, possibly would kill no one (although a fair number of people likely would be hospitalised, with some being grievously injured); moreover, there would be no structural damage to the buildings near the point of release.

Similarly, in late 2001, WMDs were used in terrorist attacks in the United States; anthrax toxin (a biological weapon) was mailed to a number of targets, including various media outlets and the Capitol Hill offices of then-Senate Majority Leader Tom Daschle. Five individuals died and numerous others were hospitalised, but no other direct damage was inflicted, although a thorough (and expensive) cleaning of certain locations through which the anthrax had passed was required. Needless to say, the damage inflicted on the United States by these WMDs terrorist attacks was far smaller than that inflicted a short time before

in the September 11 terrorist attacks, which involved no WMDs (at least as the term generally is used).

This chapter examines, in turn, three major categories of WMDs, in approximate order of their general tactical utility in warfare between states in the current era, beginning with the militarily least useful WMDs, radiological weapons. Nuclear weapons will be addressed separately and in greater detail in the [next chapter](#), because those uniquely ominous strategic tools have profoundly shaped international politics since the end of the Second World War, and can be expected to continue to do so for decades to come.

Radiological, biological and chemical weapons: a brief primer

Radiological weapons

Until recently, radiological weapons were by far the least-discussed WMDs category – indeed, whether they even are WMDs is debatable, as many authors omit them altogether, restricting the WMDs ‘nameplate’ to nuclear, biological and chemical weapons. However, they are included here because in recent years, particularly since the September 11 terrorist attacks in the United States, public discussion of radiological weapons has increased greatly, and it has become common for defence professionals to treat radiological weapons as WMDs.

The first thing that one must understand in any discussion of radiological devices is that they are *not* nuclear weapons, even though they do contain radioactive materials. Unfortunately, many press accounts use the phrase ‘dirty nukes’ to describe radiological weapons – or, even worse, interchangeably to describe both radiological weapons and relatively crude nuclear devices. Actual nuclear weapons use fissile material, such as Uranium²³⁵ or Plutonium, to create a nuclear reaction and resulting explosion.

In a radiological weapon, no nuclear explosion occurs. A radiological device simply is a bomb which uses a conventional explosive to scatter radioactive material. This material is not necessarily fissile material, which is expensive to produce and extremely difficult for terrorists and similar actors to acquire. More likely, the radioactive matter in question would be something such as Cobalt⁶⁰, which is used for cancer radiotherapy and other legitimate purposes worldwide. Radioactive isotopes have a wide variety of uses in medicine, food processing and packaging (using radiation to kill bacteria) and other industries.

It should be noted that different radioactive substances have enormously varied levels of radioactivity. Even the slightest exposure to very ‘hot’ – that is, highly radioactive – material can result in fatal radiation poisoning. On the other hand, exposure to low levels of radiation might not adversely impact an individual’s health. Indeed, human beings constantly receive small doses of ‘background radiation’ from their surroundings – sunlight, radioactive elements naturally present in soil and water and so forth.

Depending on the materials used and their quantity, exposure to the nuclear material scattered by a radiological bomb would present a variable health risk. Given that stealing (or otherwise surreptitiously obtaining) and handling extremely radioactive materials is very difficult, while less radioactive – though still quite dangerous – materials are used in medicine and industry throughout the world, often with few safeguards to prevent their theft, it seems most likely that a terrorist group's radiological weapon would present only a limited threat to the population of a targeted city. Exploding a conventional bomb is a far-from-optimal way to scatter radioactive material, and most likely only a small area would be very dangerously contaminated. Depending on the circumstances, the actual number of fatalities directly attributable to a radiological device could be quite small, even zero, although those close to the explosion possibly would face long-term health risks, such as a greatly increased likelihood of developing cancer. However, tiny quantities of radioactive material would be spread over a much wider area, and that raises a potentially even more serious prospect: public panic and likely long-term economic consequences related to cleaning up contaminated areas.

If a radiological device exploded in the centre of a metropolis, chaos surely would soon follow. Most people – whether residents of Shanghai, Chicago, São Paulo or any other great city – have little or no knowledge of radiation poisoning, and would find it difficult to calmly and accurately assess the risks which they and their loved ones faced, especially in an environment in which they would be receiving very incomplete, and no doubt often inaccurate, information from panicky media reports. The number of fatal injuries occurring in the resulting mass exodus from the city probably would be far greater than the number of individuals directly killed by the bomb blast itself.

Long after the attack itself, local and national governments could be faced with difficult economic and political questions. Some former residents never would be willing to return, while others would demand a clean-up so comprehensive that its cost would be exorbitant, potentially with part of the city being abandoned for several years while it was ongoing. The long-term local, and even national, economic repercussions might potentially be severe, with the direct and indirect costs mounting into the trillions, and it even is possible that a wide section of the victimised city simply would be abandoned, as prohibitive clean-up costs and public fear made reconstruction problems irresolvable. Although the 2011 Fukushima Daiichi nuclear power plant disaster was an accident – and the technical challenges related to the clean-up are unusually complex – the ongoing difficulties and costs that Japan faces as a result of it provides a useful illustration of the political and economic 'price of radioactivity'.

While radiological weapons might present a serious danger in the hands of terrorists, they generally would be of little use to a traditional military force. During the Cold War, the United States and the Soviet Union demonstrated little interest in radiological weapons, as there essentially was no military task that could not be better performed by nuclear weapons. Unlike in a nuclear attack,

few enemy troops would be immediately killed or incapacitated by a radiological weapon, even if they might face long-term health risks. Moreover, if one's own troops were to subsequently operate in the polluted area, they might also face long-term health risks.

Today, no state is known to maintain a radiological arsenal. Thus, there have been little in the way of international efforts to create and implement treaties banning these weapons – there is not much point in creating legal agreements to prevent the stockpiling of weapons that states have no interest in possessing. However, terrorists might find such weapons very useful for furthering their goal of inflicting political and economic damage on a targeted society. Given their simple design and the very real possibility that a terrorist group would be able to acquire radioactive material suitable for use in such a device, it is easily imaginable that at some point in the future terrorists will mount one or more radiological attacks, most likely, as discussed above, in densely populated urban areas.

Biological weapons (BW)s

As noted above, biological warfare (BW), also known as biowarfare, has been practised for thousands of years, but the use of modern BWs has been very limited. The most significant attempt to practise BW was Japan's in the Second World War, and it largely was aimed at the Chinese civilian population. The most notable efforts were those of the infamous Unit 731, based in Manchuria. Unit 731 conducted horrific experiments on human test subjects and devised a number of means to spread disease through the Chinese population, such as the release of fleas infected with bubonic plague.

In essence, biological weapons include a living pathogen of some type – a virus, for example – or a toxin produced by living organisms (toxins invented by humans generally are treated as chemical weapons). Weapons including a living pathogen tend to be more difficult to store and deliver than are toxins, because measures must be taken to ensure that the pathogen does not die before its intended target is exposed to it. In any event, the robustness of the agent is important – a living virus that will die very quickly when exposed to sunlight, for example, probably would have little utility as a biological weapon.

Another important characteristic of a BW agent is infectivity. Most actors using BWs would want even a very tiny quantity of an agent to be capable of infecting individuals; the smaller the amount of agent required to infect an individual, the higher its infectivity. Closely related to this is pathogenicity, which relates to the percentage of individuals who actually will manifest the symptoms of a disease to which they have been exposed. A biological weapons user would desire very high pathogenicity, with the overwhelming majority of individuals manifesting disease symptoms after a bioattack. Similarly critical is virulence – in this context, essentially the severity of a particular disease. A biological weapon designer generally would desire great virulence, although, in some

cases, he or she might not want the agent actually to kill exposed individuals, preferring instead that they be incapacitated for a time with symptoms severe enough to render them unable to perform military or other duties.

The incubation period of a particular disease is another key characteristic of a biological weapon. In military operations a shorter incubation period usually is more desirable, as one would wish to disable or kill enemy troops as quickly as possible. A terrorist group, however, might find a longer incubation period desirable, as this might allow a communicable disease to spread far, infecting a great number of people, before public health authorities became aware of its existence.

Effective delivery of BWs presents many challenges. For instance, placing a BW agent on a ballistic missile may result in the destruction of most of the agent when the missile's warhead impacts its target. Moreover, a biological agent generally must be dispersed over a wide area, if it is to be used effectively. Again, placement of a large quantity of an agent in a cruise or ballistic missile warhead may not be an effective way in which to spread the agent, as this will contaminate only a relatively small area, and much of the agent will be destroyed when the missile impacts its target. Similar problems confront efforts to place an agent in an artillery shell. In general, the most effective way to deliver BWs is in the form of an aerosol (for a liquid) or a dust cloud (for a solid). However, under combat conditions delivering BWs in this fashion could prove extremely difficult, as the most plausible delivery vehicles for the agent would be slow-moving aircraft, and the latter inherently are highly vulnerable in a modern battlespace. In addition, targeted troops might possess protective gear that would guard them against exposure. The civilian residents of a city at peace, on the other hand, would be far more likely to fall victim to such a mode of delivery. In such an environment, ground vehicles also could be used to disperse an agent, albeit far less efficiently.

Biological weapons can have targets other than humans. In fact, in nature many of the illnesses caused by biological weapons primarily afflict animals, not humans, and BW aimed at an enemy's animals is possible. Germany attempted this in the First World War, seeking to infect the draft animals of its enemies with anthrax and glanders. Today's modern armies obviously do not rely on draft animals for transport, but some experts have expressed concern about a possible terrorist attack intended to inflict economic damage by causing an epidemic amongst animals raised for food. Also, biological weapons can target plants, and a state could attempt to destroy its enemy's food supply by spreading diseases such as rice blast. Indeed, when it maintained an active BW programme, the United States developed considerable expertise in this area and maintained a significant anti-crop biological arsenal, though it never actually was used. The United States did, however, make extensive use of chemical defoliants, most famously Agent Orange, during the Vietnam War. (Although not a chemical weapon in any usual sense, Agent Orange did have undesired long-term side effects on the health of individuals exposed to it, including US troops.)

As noted above, not all BWs must be intended to kill – because such weapons cause illness, a user may match the agent used to the effects desired and, for one reason or another, may not desire that targeted individuals actually die as a result of the biological attack. One could develop a biological agent intended to cause temporary incapacitation, but not death. For example, Q fever, a highly infectious, but generally not fatal, illness can be used in BW. However, *all* biological weapons intended for use against humans are banned by the Biological Weapons Convention (BWC), which was opened for signature in April 1972; non-lethality is irrelevant to legality under the Convention. Similarly, the use of non-lethal chemical agents in warfare is banned under the Chemical Weapons Convention (CWC), which was opened for signature in January 1993. This creates some intriguing ethical questions: one might ask whether it is reasonable that biological and chemical weapons having temporary incapacitating, but generally non-fatal, effects are banned. After all, use of such agents in a military operation could decrease the number of deaths and debilitating wounds suffered by combatants, which generally would be considered to be ethically desirable.

In any case, most offensive BWs research over the years has been aimed at the development of lethal agents. Indeed, high lethality generally is considered a critical characteristic of BWs. It is, however, very easy to exaggerate the potential effectiveness of these weapons. This is because tiny quantities of most such substances are fatal. Thus, one often sees statistics indicating that a small quantity of a given biological weapon could kill the entire population of the earth. While, in a literal sense, such a statement might be true, such a calculation assumes the efficient delivery of that agent to every person on earth – a practical impossibility, of course.

Conversely, one should not forget that biological weapons are indiscriminate – when actually used in military operations, the user must be careful to ensure that its own troops are not placed in danger. Depending on the weapon, this may require injections of a protective vaccine, and some vaccines – the anthrax and smallpox vaccines, for example¹ – have a non-negligible rate of serious side effects. Of course, if there is no vaccine or other effective method to create immunity for a given infectious disease, most potential BWs users – apocalyptically minded terrorists being a critical exception – would regard that disease as one unfit for use in biological warfare.

BWs and terrorism

Not all BWs facilitate the spread communicable diseases – indeed, most do not. Biological agents such as the anthrax and botulinum toxins do not cause diseases that typically are spread by personal contact. Thus, an outbreak of the disease is inherently limited. From a military standpoint, this may be highly desirable, as it ensures that no uncontrollable pandemic will occur that might impact the population of one's own state. However, in the past states certainly

have weaponised communicable diseases, albeit generally low-lethality ones such as tularemia and the aforementioned Q fever.

Amongst oft-lethal diseases, smallpox has received particular attention in the United States, where there has been great concern about the possibility that a terrorist group might obtain samples of the smallpox virus and undertake a bioterrorist attack intended to create a mass epidemic. In the simplest scenarios, a terrorist group with access to the virus simply could infect a number of 'suicide bioterrorists' with the virus and use them to spread the disease. Because there is a considerable time (typically weeks) between the point at which a human becomes infectious and when he or she is immobilised by the disease, the terrorist volunteers would have ample time to travel on airliners, wander through busy shopping malls, and undertake other activities which would bring them into close contact with others, who might then be infected and, in the period before they became symptomatic, unwittingly infect others. How well such a scheme – or a more sophisticated variant – might work is debatable, but it conceivably could cause a devastating multi-country smallpox outbreak: unless treated promptly, smallpox has a high rate of lethality.

Decades ago, great numbers of people worldwide routinely were inoculated against smallpox, but as the disease was 'conquered' the decision was made to cease routine inoculation, because of the serious (even lethal) side effects in a small percentage of patients. Supposedly, two institutions – the US Center for Disease Control and Prevention and Russia's Vector State Research Centre of Virology and Biotechnology – hold the only samples of the smallpox virus anywhere in the world, though there are suspicions that there may be other, 'outlaw' stocks of smallpox held by other states.

While some have argued that the danger of terrorists obtaining access to smallpox virus is sufficiently great that routine smallpox inoculation should be reintroduced, thus far the counter argument, that 'smallpox terror' is too unlikely to justify the human and financial costs of reintroducing mass inoculation of civilian populations, has prevailed in countries throughout the world.²

The debate over whether routine smallpox inoculation should be reintroduced is only one example of a larger problem related to biological weapons, particularly those that spread communicable diseases: how countries should prepare for the possibility of BWs terrorism. A number of factors complicate such discussions. First, there has never been a truly large-scale BWs terrorist attack using a readily communicable agent, so even experts rely on scenarios of uncertain accuracy. For example, depending on one's assumptions, the smallpox scenario cited above might result in a few dozen fatalities – or, at the opposite extreme, tens of millions of deaths may occur. A great number of variables would impact the seriousness of the outbreak: how competent the suicide bioterrorists were in spreading the disease; when and under what circumstances public health authorities became aware that a smallpox outbreak was occurring; how effective the procedures are for quarantining infected persons and vaccinating vulnerable populations; and so forth.



Figure 16.2 Italian firefighters participating in a WMDs exercise. They are standing near a container suspected of carrying WMDs.

Where the next bioterrorist attack will occur and which agent will be used obviously remains unknown. Nevertheless, if they are to be prepared properly for biological terrorism, governments must decide long before an attack how much money and effort they will devote to preparing for a bioterrorist attack, and how they are going to spend those resources (see [Figure 16.2](#)). Choices concerning public health infrastructure and the stockpiling of vaccines, which diseases will be vaccinated against universally and many other decisions are made in the hope that they will prove to be adequate in the event that a bioterrorist attack did occur.

The way in which biological weapons are produced makes it possible for terrorists or 'rogue' states to create fairly substantial quantities of a BWs agent secretly and at minimal cost. A biological weapons laboratory is equipped similarly to a legitimate medical research facility and may follow a similar routine. Moreover, access to knowledge relevant to the creation of BWs is not controllable – the basic principles of the creation of BWs are derived from biology and medicine, and every country with a sizeable population contains many individuals who possess the intellectual tools necessary to create BWs. Given these facts, a knowledgeable terrorist literally could operate a small-scale BWs weapons laboratory in his or her apartment, while a state could operate a decent-sized national BWs programme in a warehouse. Given the practical impossibility of controlling access to basic medical equipment, BWs research

by terrorists, much less states, is not preventable, although successfully restricting access to certain extremely dangerous pathogens may be possible – the best example of this perhaps being the very tight control over access to the smallpox virus noted above.

While it is fairly simple to create crude BWs, however, such weapons might lack the characteristics necessary to make them effective. Ricin, for instance, is a highly lethal toxin that is readily derived from castor beans, which are easily obtained legally – and in 2003, British police uncovered an alleged plot by a number of Islamists of North African origin to use ricin as a terror weapon (only one individual, however, subsequently was convicted of planning to spread ricin).³ Delivering ricin in a fashion that would kill large numbers of people is exceedingly difficult, however, because of both the difficulty of effective dispersal and the fact that the agent is not very robust.⁴ Thus, even if the British bioterrorist plot had reached fruition, it is entirely possible that few, or even no, fatalities would have resulted.

In general, anthrax toxin is far superior to ricin as a biological weapon, although effective delivery requires that it be inhaled into the lungs, as this will result in pulmonary anthrax, which has a high fatality rate. (Gastrointestinal or cutaneous anthrax result from the entry of the toxin into the digestive system or skin, respectively, and both, particularly the latter, tend to be far less deadly.) This is difficult, as it requires that the toxin be milled to a very specific size optimal for inhalation and lodging in the respiratory system. Moreover, inhalation of the anthrax is only likely to occur while the agent actually is floating in the air; once it is on the ground, it presents only a minimal danger to those around it.

Given these limitations, the most effective means to deliver the agent is to spray it over the targeted population using a complex spraying system that disperses the toxin in a very precise fashion. Also, ‘anti-clumping’ additives are required to prevent the finely milled anthrax from bunching together – and combining the latter with the anthrax requires considerable sophistication. In any case, even if the terrorists were capable of creating anthrax spores of a high quality, it would be difficult for terrorists (unless they had help from a state) to obtain the equipment necessary to precisely mill and spray the toxin, although they might be able to deliver anthrax using a relatively crude spraying system, such as that on a crop duster. Such a method would be of uncertain effectiveness, although any such large-scale biological attack surely would succeed in sowing terror in the targeted population.

When assessing the threat of a BW attack by terrorists, one should consider their level of sophistication and financial resources, the agents that might be used, how those agents might be dispersed, what the incubation time and level of pathogenicity and lethality are for the disease caused by the agent and whether effective treatments exist for that disease. Potentially most important of all is whether the biological weapon in question causes a communicable disease and what the characteristics of that disease are, as an uncontrolled outbreak of a very dangerous communicable disease could have worldwide effects and cause many millions of deaths (see [Box 16.1](#)).

Box 16.1 The Ebola virus and bioterrorism

Recently, as a result of the most severe outbreak to date of the disease Ebola, a type of haemorrhagic fever, there has been considerable speculation in Western media – and even on the part of some legislators – regarding the possibility that terrorists will succeed in weaponising the disease to strike one or more Western countries. Such concerns starkly illustrate both the fear that infectious disease arouses and, relatedly, the degree to which far-fetched scenarios (i.e. a massive Ebola outbreak resulting from bioterrorism) can distract from far more plausible threats. In reality, absent substantial modification by a BWs designer, the virus would make a rather poor biological weapon.

At the time this chapter was written, the lethality of Ebola remained very high, particularly amongst individuals who receive only limited or substandard medical care shortly after becoming symptomatic. However, the manner in which the disease incubates would be problematic from a terrorist's perspective – most importantly, there is only a brief time in which individuals both are able to function effectively *and* simultaneously present a major danger of infecting others. Asymptomatic individuals present little threat, while symptomatic ones soon become entirely incapacitated.

Most importantly, the disease does not readily spread between individuals who only have casual contact – caregivers who are exposed to a patient's bodily fluids are at risk, but even an infected terrorist who was intentionally attempting to spread the disease in, say, a crowded airport would find it difficult to infect others without acting in a manner that would soon result in his or her apprehension. While it is conceivable that terrorists might succeed in intentionally infecting themselves by accessing Ebola patients or their bedding, presumably at a clinic or hospital with an overwhelmed and under-resourced staff, the likelihood that the individuals would then succeed in causing a large-scale outbreak in an affluent society with a strong public-health infrastructure would be extremely low.

It is also unlikely that a terrorist group would be able to take a sample of the Ebola virus and modify it so as to 'fix' its shortcomings as a biological weapon. While it is plausible – given sufficient time and resources – that a state could achieve this goal,¹ such an effort at present would likely be far beyond the financial and technological capabilities of even the most sophisticated of terrorist groups, including the self-proclaimed Islamic State. While technological progress eventually may have the unfortunate side-effect of making such a scenario more plausible (see Box 16.2), at present it can be regarded as extremely unlikely.

Such terrorists surely would, however, achieve the more modest goal of causing fear in Western populations and compelling their governments to expend enormous resources to address a minimal threat. Nevertheless, while that limited victory for the terrorists would be unfortunate, it would be enormously less tragic than the outbreak of a 'manufactured' epidemic.

¹ The Soviet Union attempted to weaponise Ebola, though it ultimately abandoned the effort in favour of the related Marburg virus. See Andrew Pollack, 'Stabbing with Syringe in Nigeria Raises Concerns of Ebola as Weapon', *New York Times* online edn, 10 September 2014 and Joby Warrick, 'Ebola Crisis Rekindles Concerns about Secret Research in Russian Military Labs', *Washington Post* online edn (www.washingtonpost.com), 24 October 2014.

BWs and states

This discussion heretofore has focused mainly on the possible use of BWs by terrorist groups. However, these weapons do have potential military utility and, while no states currently acknowledge possession of them, it is near-certain that a number of them do possess a biological arsenal – although precisely *which* states have BWs remains a matter of contention.

Nonetheless, the aforementioned problems with effective dispersion of BWs, protection of one's own troops and other issues go far in explaining why BWs have not played a significant role on the modern battlefield. Thus, when President Richard Nixon decided in 1969 unilaterally to renounce US stockpiling and use of BWs, his declaration created relatively little controversy: most Western observers of military affairs agreed that there were no critical roles in which BWs would be superior to other weapons, particularly nuclear ones, and therefore that there was little need to maintain a biological arsenal.

Under international pressure because of the US abandonment of BWs, the Soviet Union also announced that it would destroy its biological stockpile, and both countries eventually acceded to the BWC. The convention has been joined by the great majority of the world's countries, although about two dozen states have not ratified it. The BWC bars states from researching or stockpiling BWs for offensive purposes. However, it does not forbid research focused on improving defences against BWs, and thus states may create very small quantities of biological agents for research purposes. Some critics, however, contend that this provision allows states to skirt the BWC's ban on offensive research, as the line between offensive and defensive research can be hazy, given that even defensive research may yield knowledge useful to the development of BWs (see [Box 16.3](#)). Various rogue states, including North Korea, Iran and Syria, are considered likely to have a research programme and/or a BWs arsenal, and even such major powers as China and, as noted, Russia (see [Box 16.2](#)) may possess these weapons. At present, given the difficulties involved and the international opprobrium that would accompany their use, it would seem unlikely that a major power will soon use BWs in warfare. However, the possibility that a rogue state or other relatively small actor might use such a weapon, either against an outside enemy or in an effort to commit genocide against some segment of its own population, should not be dismissed casually.

Although today's BWs may offer their users only limited military advantages, the possibility that they will again be used in combat should not be ignored. The section below on the use of chemical warfare also addresses many of the issues that a force defending against BWs must confront.

Chemical weapons (CWs)

Chemical weapons are the only form of WMDs to be used with some regularity in warfare during the twentieth century. Although most of that century's wars

Box 16.2 Cheating on arms control: the Soviet Union and the Biological Weapons Convention (BWC)

Although the Soviet Union was one of the original signatories to the BWC, Moscow proceeded to break that agreement in spectacular fashion, running an enormous biowarfare industry, costing the equivalent of billions of dollars a year, which produced enormous quantities of BWs. Nevertheless, for decades Moscow was fairly successful in keeping these efforts secret. Many outsiders believed that the Soviet Union was maintaining a BWs arsenal, particularly after a strange incident in the town of Sverdlovsk, in which numerous residents died of anthrax poisoning. The Soviets claimed that the epidemic was caused by anthrax-contaminated bone meal in livestock feed and the consumption of meat from infected animals. However, after the fall of the Soviet Union it was confirmed that the deaths were, in fact, the result of an accidental discharge of weaponised anthrax. It was not until Kenneth Alibek (Kanatjan Alibekov), a major figure in the Soviet BW programme, moved to the United States in 1992 and openly discussed it that substantial evidence of its size became available publicly. To this day, however, there are significant questions about the programme; particularly troubling is the fact that the Soviet BWs programme was clearly very advanced – almost certainly the most sophisticated in the world – and although Moscow claims to have fully disclosed the details of its programme and to have ceased all research into offensive BWs, some critics doubt this and contend that Russian scientists may be continuing research into new and very deadly BWs.¹

¹ In Congressional testimony, an Assistant Secretary of State warned in 2002 that, 'Key components of the former Soviet programme remain largely intact and may support a possible future mobilization capability for the production of biological agents and delivery systems. Moreover, work outside the scope of legitimate biological defense activity may be occurring now at selected facilities within Russia. Such activity, if offensive in nature, would contravene the BWC, to which the former Soviet government is a signatory. It would also contradict statements by top Russian political leaders that offensive activity has ceased.' Statement by Carl W. Ford, Jr, Assistant Secretary of State for Intelligence and Research before the Senate Committee on Foreign Relations Hearing on Reducing the Threat of Chemical and Biological Weapons, 19 March 2002, accessed from the Avalon Project at Yale Law School (http://avalon.law.yale.edu/sept11/ford_002.asp). Russian BW efforts remain opaque today.

saw no use of CWs, many tons of chemical weapons were used in the First World War, and they also appeared in a small number of subsequent conflicts.

The Hague Convention of 1899 banned 'the use of projectiles the object of which is the diffusion of asphyxiating or deleterious gases'.⁵ However, less than two decades later, in 1915, Germany – seeking a military 'edge' that would allow it to break the tactical deadlock on the Western Front – began using chlorine gas; this quickly was followed by the use of CWs by other belligerents. In total, probably less than 100,000 troops, a majority of them Russian (poison gas fatalities on the Western Front actually were relatively small), were killed by CWs during the war. However, a much greater number of troops – probably over 1 million – were wounded by CWs. Interestingly, the survivors of chemical

Box 16.3 'Super-BWs'?

Unfortunately, the world may be on the cusp of a technological revolution that would exponentially increase the military utility and lethality of biological weapons. The discussion herein mainly addresses 'known' BWs – biological weapons that have been or almost surely could be created using current technology. However, in recent years there has been an explosion of knowledge in the life sciences. Of particular importance to the current discussion is the rapidly advancing field of genetics (including the first complete mapping of the human genome) and the related development of biotechnology.

Increasing knowledge about the human body and how it counters disease should bring about great advances in medicine in coming years, but, unfortunately, the ongoing biotechnological revolution also creates a very real possibility that biological weapons far more dangerous than those fielded today may be created. For example, it may be possible to create weapons that cause new and extraordinarily lethal diseases against which there is no current medical defence; there has even been speculation concerning the possible creation of 'ethnic bioweapons' tailored to attack only individuals having genetic markers associated with a particular group of humans. Indeed, it eventually might be possible even to craft a biological agent intended to harm only a single specific individual.¹

Although such grim possibilities are purely speculative at this point, they should be kept in mind when considering the possible future of BWs. In particular, if it clearly became feasible to develop 'super-BWs', powerful states might re-examine their decision to dismantle their biological arsenals. Also, of course, if such weapons ever should come into the possession of bioterrorists, either provided by a state sponsor or developed by the terrorists themselves, the results could be catastrophic.

¹ See Andrew Hessel, Marc Goodman and Steven Kotler, 'Hacking the President's DNA', *The Atlantic* online edn (www.theatlantic.com), 24 October 2012.

attack included an Austrian corporal, Adolf Hitler, who was blinded temporarily by mustard gas in 1918.

The horrible effects of the chemical weapons used in the First World War – including death by asphyxiation and, for many survivors, terrible burns, long-term severe lung damage and other debilitating problems – resulted in renewed revulsion for CWs. However, this did not prevent CWs use between the World Wars. Notably, in the course of its conquest of Abyssinia (modern Ethiopia) in the mid 1930s, the Italian military used considerable quantities of CWs against its opponents, including by aerial bombardment.

Techniques for delivery of CWs are generally similar to those used for BWs. During the First World War, these included artillery shells and chemical weapons projectors (essentially mortars that fired a canister filled with a chemical agent); chemical artillery shells continue to be stockpiled by some countries. Chemical-tipped warheads also can be placed on cruise and ballistic missiles. Effective dissemination of a chemical agent using such techniques

does, however, present major challenges. When explosives are used to 'aerosolise' and disperse a liquid chemical agent, they tend to destroy much of the agent; moreover, some agents are flammable and may catch fire, destroying them.⁶ As noted above, aircraft also can be used for the delivery of chemical weapons. For safety and other reasons, many devices for the delivery of chemical weapons are 'binary'. Rather than the agent itself, a binary device – such as an artillery shell or bomb – will contain two or more chemical precursors, which are separated from each other while the weapon is in storage. After the munition is fired, the precursors mix together, creating the deadly chemical agent.

Difficulties in effective dispersion are one of the key disadvantages of both biological and chemical weapons. As with biological agents, regardless of how chemical weapons are delivered, climatic factors, particularly wind direction and speed, are critical to their effective diffusion. Indeed, on many occasions during the First World War, troops were exposed to chemical agents dispersed by their own side when the wind direction changed, blowing the CWs back to their lines.

During the Second World War, both Axis and Allied forces maintained CWs stockpiles and declared their willingness to retaliate in kind against enemy use of such weapons. In the European theatre both sides refrained from first-use of CWs, although Germany did seriously consider their use against Allied troops. (There was, however, substantial use of CWs by Japan in China.) While the Third Reich did not use CWs against Allied troops, however, it did practise another, far more repellent, form of chemical warfare (CW), which is discussed below.

Saddam Hussein's Iraq repeatedly used chemical weapons as a terror weapon against civilian populations. In the Iran–Iraq War of 1980–8, Iraq launched missiles armed with chemical weapons against Iranian cities, killing thousands and causing public panic; Iran is alleged to have retaliated with its own chemical attacks on Iraqi forces. In the latter part of that conflict, Saddam's government also used CWs against Iraqi Kurdish civilians as part of a genocidal campaign against them; thousands of Kurds died as a result.

During the Cold War, the United States and the Soviet Union amassed large CWs stockpiles, and both superpowers were prepared to use them in the course of a Third World War, although chemical weapons were decidedly less important in military planning than were nuclear weapons. As with BWs, there were serious questions concerning how great the military utility of these weapons actually would be, especially given the unpredictability of their dispersion and, again like BWs, there were few military missions for CWs which would not have been better performed by nuclear weapons.

Nevertheless, chemical weapons may be judged as having real potential military utility under the right circumstances. While they also have important shortcomings, against a poorly prepared opponent modern CWs may be devastating, combining very high lethality with rapidity of action. This in practice means that chemical weapons are likely to be far more useful on a battlefield

than are currently available biological weapons. There are five main classes of chemical weapon: choking (or pulmonary), blood, blister, nerve and non-lethal agents.

The first modern chemical weapon used in combat, chlorine, was a choking agent, as was phosgene, which caused far more fatalities in the First World War than did any other chemical weapon. Choking agents cause damage to the lungs, and exposure to phosgene can result in pulmonary edema – a build-up of fluid in the lungs that can be fatal. In general, the choking agents are regarded as ‘outmoded’ chemical weapons, but choking agents may be in the arsenals of a small number of states. Most notably, chlorine gas has been used extensively in the Syrian Civil War by pro-government troops and insurgents. Also, as chlorine has many legitimate uses (such as in sewage treatment), it is a readily available chemical that has been used in attacks by both Syrian and Iraqi insurgents. Chlorine gas is no longer a particularly useful weapon against ‘top-tier’ militaries, but it recently has inflicted considerable numbers of fatalities and other casualties amongst both civilians and poorly equipped fighters.

Blood agents prevent the body from utilising oxygen properly, causing suffocation. (The term ‘blood agent’ is somewhat confusing, as the agent does not actually poison the blood; rather, it is distributed throughout the body by the blood.) Hydrogen cyanide and cyanogen chloride (generally referred to as CK) initially were used on the battlefield in the First World War. Moreover, hydrogen cyanide, commonly known by the German trade name Zyklon B, was used in the Holocaust, including at the enormous Auschwitz camp complex, located near Kraków, Poland, to expedite industrialised mass murder. Like the choking agents, blood agents generally are considered to be obsolescent. However, there are reports indicating that hydrogen cyanide was used by Iraq in the Iran–Iraq War against both Iranians and Iraqi Kurds, as well as in 1982 in the city of Hama during the Syrian government’s brutal campaign to put down an uprising by Muslim Brotherhood-affiliated insurgents.

Blister agents, as the name implies, have gruesome effects on human skin, with exposure causing blistering and burns; if inhaled, they damage the respiratory system. Of the blister agents, mustard is by far the most well known and, although it caused relatively few of the fatalities in the First World War, it was perhaps the most feared chemical weapon because of the excruciatingly painful wounds that it inflicted on victims. During the 1930s and 1940s, Japan made extensive use of choking and blister agents in China. Several countries are likely to have CWs arsenals that include blister agents.

Nerve agents currently are the most advanced chemical weapons known to exist. These weapons attack the human nervous system, causing rapid death unless the victim receives an antidote such as atropine. The best-known classes of nerve agent are the G- and V-series. The first of the G-series agents GA (more commonly referred to as tabun) was developed by German scientists in 1936, and the last of them, GF (cyclosarin), was initially synthesised in 1949. However, G-series agents continue to be regarded as highly lethal modern

chemical weapons. The even more advanced V-series agents were invented somewhat later, with the first and best-known of them, VX, being developed by the United Kingdom in 1952.

Later still, the Soviet Union secretly developed the even more sophisticated *Novichok* nerve agents in the 1980s and early 1990s; the existence of these weapons was not publicly known until 1992, when the programme was exposed in the Russian media. Interestingly, Russia had not previously informed the United States of the existence of the *Novichok* agents despite the existence of a 1989 agreement between the two countries, known as the Wyoming Memorandum of Understanding (MOU), to share information of their respective chemical weapons programmes. Moreover, a Russian scientist who 'blew the whistle' on the programme, Vil Mirzayanov, subsequently was prosecuted for releasing state secrets, although charges against him eventually were dropped.⁷

Because of their relatively late development, nerve agents have been used less frequently than have other lethal CWs. However, such agents were used extensively by Iraq during the Iran–Iraq War, as well as against Iraqi Kurds. More recently, sarin gas has been used several times during the Syrian Civil War, most notably in an attack near Damascus, at Ghouta. While the Syrian government disputes responsibility, the United States and the European Union, amongst others, contend that the attack was undertaken by pro-government forces.

Military use of CWs

While they present some of the same difficulties, such as ensuring that the agent is dispersed properly and preventing one's own troops from being exposed to the agent, most chemical weapons very quickly impact on the health, and thus military effectiveness, of exposed individuals. While an agent such as sarin can kill within minutes of exposure, the long time between exposure and death that is usual for biological weapons allows the exposed individual to continue performing his or her duties without hindrance for days or even weeks. Moreover, given that advanced military forces carefully monitor their troops for exposure to BWs, it is likely that endangered personnel would quickly receive appropriate treatment that, depending on the agent, might well prevent the overwhelming majority of them even from falling ill, much less dying. In addition, as noted above, for numerous biological agents, including smallpox and yellow fever, vaccines are available that allow individuals to be inoculated before exposure.

Protecting troops against CWs may require that they wear bulky and uncomfortable protective gear which impedes their capability to conduct military operations effectively. The same is true of operating in areas where BWs may be, or have been, used (an environment heavily contaminated with the radioactive by-products of nuclear weapons use presents even greater challenges). Chemical/biological protective gear is bulky, uncomfortable – particularly in warm

climates and/or when a mask is worn – and can make it more difficult for soldiers to perform tasks ranging from fighting to driving, to equipment maintenance. However, when such gear is used properly, it is highly effective in protecting troops from exposure to dangerous agents.

Given the challenges relating to the use of protective gear, one of the most important effects of both chemical and biological weapons is the inconvenience created by these arms. Such weapons create a hostile physical environment that presents special operational problems. Thus, even if CBWs have little direct effect in terms of casualties, a relatively weak state facing a much stronger opponent may be tempted to use these weapons. Indeed, simply by possessing (or being presumed to possess) such weapons, a state complicates its enemy's operational and tactical planning. In both the Persian Gulf War of 1991 and the 2003 invasion of Iraq, the United States and its allies took careful measures to protect troops from possible CBWs exposure. It is notable that one of the great difficulties for commanders confronting possible use of CBWs is determining how much emphasis should be placed on defensive preparations, for the simple reason that time and energy devoted to possible chemical or biological attack could be spent on other important tasks.

Advanced militaries spend an enormous amount of capital on chemical (and biological) defence and devote valuable training time to mastering appropriate defence techniques, and much can be done to hamper the lethality of chemical warfare.⁸ For example, detection and early-warning devices, as well as masks, suits and other protective gear, can be made available to units, and training time devoted to proper use of such equipment; vehicles can be outfitted with air-filtration systems that protect occupants from contaminated air; a naval vessel can be constructed so that there is a capacity to 'pre-wet' the ship so as to prevent agents from settling, while the ship's core is a 'citadel' with filtered air (see [Figure 16.3](#)). Additionally, it should also be noted that the sort of defences that protect military forces from chemical and biological exposure also are useful in guarding against nuclear contamination. Indeed, one of the few things that the various weapons of mass destruction do share in common is that they all present unusual and difficult contamination problems that require substantial forethought and planning if they are to be addressed successfully.

As the above indicates, even though CWs have considerable potential military utility, battlefield defence against such weapons is quite feasible. Like BWs, CWs potentially are most effective not against well-equipped advanced military forces, but as a terror weapon aimed at civilian populations. However, the fact that CWs do not spread a communicable disease strictly limits their ability to inflict casualties. While, hypothetically, an epidemic resulting from the use of certain BWs could cause hundreds of millions of deaths, only individuals directly exposed to chemical weapons suffer harm. With that said, however, if a large quantity were dispersed in a suitable fashion, it is entirely conceivable that a highly lethal agent could inflict tens of thousands of fatalities in a compact and densely populated area.



Figure 16.3 The guided-missile destroyer USS *Boxer* performs a test of the ship's countermeasure water wash-down system, which coats the ship with water so as to prevent chemical, biological or radioactive materials from settling.

Why states acquire and retain chemical and biological weapons today

As noted above, the possession, much less the use, of CBWs is banned outright by the Chemical and Biological Weapons Conventions, treaties which most countries have signed and ratified. Even Syria, under great international pressure, in 2013 agreed to be bound by the CWC. Slightly more than twenty states, most notably Egypt, Israel and Syria, currently are not party to the BWC. Only a handful of states are not party to the CWC (Angola, Egypt, Israel, North Korea and South Sudan). However, as noted above, not all parties to these Conventions actually obey their treaty obligations – and this group may include one or more of the great powers.

Why, one might ask, are biological and chemical weapons regarded as different and their use, presumably, somehow less ethical than purportedly conventional weapons, even though, in total, the weapons lumped in the latter category have killed an enormously greater number of human beings? Indeed, the conventional arsenals of many of the more militarily sophisticated states now include devastating devices such as high-yield thermobaric weapons and the (non-thermobaric) US GBU-43, nicknamed the ‘Mother of All Bombs’, a behemoth filled with 8,500 kilograms of high explosive that has a blast yield about two-thirds as large as that of the nuclear device dropped on the Japanese city of Hiroshima. Such a weapon would seem to merit placement in the WMDs category far more than, for instance, chlorine gas. Yet, the most powerful states have made it clear that they do not intend to reclassify such weapons, much less remove them from their arsenals.

Part of the answer to the above question surely lies in the fact that CBWs touch emotional ‘triggers’ in many observers – and, while BWs as we know them today are a comparatively recent innovation, disease and poison have been used in warfare for millennia. For example, poisoning wells by throwing animal or human corpses into them is a very old (and sometimes devastatingly effective) practice in warfare. Yet, in many cultures such an action is seen as very wicked, and therefore as violating a taboo. There are many reasons why such a taboo may develop – the indiscriminate fashion in which communicable disease strikes both warriors and non-combatants, the fear engendered by epidemics, and so forth – but, in any case, ‘war by disease’ has unquestionably been held in disrepute in many times and places, and this certainly influences perceptions of BWs.

Despite the disapproving attitude towards CBWs, numerous countries clearly see them as offering advantages significant enough to make it worthwhile spending significant sums obtaining and maintaining a stockpile of such armaments. A particularly notable example is provided by Iraq. As part of the armistice agreement ending the 1991 Gulf War, Iraq agreed to surrender its CBWs arsenal (as well as end its effort to obtain nuclear weapons) and submit to UN inspections that would ensure that its WMDs stockpile was destroyed and its capability to make further weapons dismantled. For more than a decade

thereafter, an elaborate game ensued in which Iraq would sometimes appear to co-operate partially with inspectors but would also endeavour to thwart their efforts and stymie their investigations. Many governments, intelligence agencies and other observers worldwide believed that Iraq was secretly maintaining a chemical and biological weapons (CBWs) arsenal, and seizing Saddam Hussein's supposed arsenal of WMDs was a major motive for the 2003 American-led invasion of Iraq.⁹ However, it now appears that Iraq both surrendered the great bulk of its CBWs arsenal in the months following the end of the 1991 conflict and never rebuilt a significant CBWs production capability. (After the war, UN inspectors seized the bulk of Baghdad's arsenal; Iraq retained possession of some of its CWs, but by the time of the 2003 invasion of that country, virtually all of these weapons were severely decayed and of little or no military utility.¹⁰)

Why would Saddam undertake an elaborate hoax to convince outsiders that his country was defying an international agreement – a ruse that encouraged the UN Security Council (UNSC) to maintain economically ruinous sanctions on Iraq and inclined American political leaders to believe that Iraq's dictator was incorrigible and must be overthrown? It now appears that his chief motive was his fear of Iran and belief that a CBW arsenal was critical to discouraging Iranian military aggression against an Iraq weakened by UN sanctions that remained in place after the 1991 Persian Gulf War.¹¹ The Iraqi leader hoped to 'bluff' potential enemies into believing in a non-existent CBW arsenal while not so flagrantly violating the 1991 ceasefire that the UNSC, or a US-organised coalition acting without UNSC authorisation, would invade Iraq. Eventually, of course, Saddam lost his dangerous wager, but the fact that he made it is indicative of how important a leader might believe a CBW arsenal to be.

The Iraqi example highlights a characteristic notable in regard to all WMDs (and, for that matter, conventional weapons): their psychological impact on leaders and populations. States desire particular weapons for their direct military potential (i.e. what they would do on the battlefield) but also for what they indirectly promise, which can include factors as varied as their possible deterrence value and projected positive economic consequences for particular communities that would accompany acquisition of a given system. Moreover, intangible factors, such as prestige and a weapon's value as a 'statement' of national greatness, also play a role in acquisition and retention decisions.

These numerous factors may be of greatly varying importance depending on the weapon under consideration, and every government will weigh them according to its own unique calculus. In the case of CBWs, unlike nuclear weapons, 'prestige value' presently is quite low. They have been repudiated formally by the great powers and therefore (at least publicly) have no place in their arsenals. Thus, today CBs mainly are seen as the weapons of relatively weak actors such as rogue states and terrorist groups. CBWs – unlike the aircraft carriers, fifth-generation aircraft and nuclear-tipped ballistic missiles possessed by more powerful states – are distinctly 'unfashionable'. They do, however, evoke

fear very effectively, and a sizeable CBWs arsenal can be acquired for less than the cost of a single F-22 Raptor fighter aircraft.

Clearly, a simple distinction between 'good' and 'bad' weapons is insufficiently nuanced to explain fully why countries acquire, or do not acquire, CBWs. Given the broad assortment of pressures and threats present in the international security environment, as well as the variety of strategic cultures worldwide, it is hardly surprising that diverse states demonstrate a broad range of attitudes towards CBWs acquisition, retention and use.

Conclusion: CBWs in the present and future

It is difficult to realistically ascertain how great the threat presented by BWs is to global security. Today, interest in biological weapons *appears* to be minimal in most of the world's major and medium powers, but one or more states may maintain secret research programmes that, potentially, will bear very deadly fruit – a new generation of BWs that overcomes the limitations of the agents of previous decades. Given the opacity of several technologically sophisticated powers, China and Russia in particular, it is a possibility that cannot be dismissed summarily.

States that are currently both strongly suspected of fielding BWs and possessed of an apparent willingness to use them tend to be relatively weak. Merely by maintaining their arsenals they risk the censure of the international community – particularly since most such states have acceded to the BWC and thus are legally obligated not to maintain a biological arsenal. In any case, biological weapons clearly are a poor substitute on the battlefield for nuclear or even, in most cases, chemical weapons. BWs do, however, appear far more potentially threatening in the context of terrorism, as at present they are most likely to succeed in inflicting terrible casualties if used against a vulnerable civilian population.

The number of fatalities that would occur as a result of a major bioterror attack would vary by several orders of magnitude, depending on a number of variables, including the biological agent used, how the terrorists attempted to spread the agent and how successful they were in doing so, the level of preparedness of public health officials and so forth. In any case, however, bioterrorism clearly presents a threat in the twenty-first century, particularly if one or more states should prove willing to arm terrorists with sophisticated biological weapons and the means to deliver them effectively.

While the danger presented by BWs is largely speculative, the fact that CWs remain 'on the battlefield' in some conflicts has been established clearly in recent years. Thus far, the use of chemical weapons has been relatively limited, and confined to international outlaws and semi-outlaws: rogue regimes, terrorists and insurgents who either are in dire straits or actually desire to shock global opinion. There appears to be little prospect that CWs will be openly possessed

and used by ‘respectable’ states in the medium-term future. However, it is conceivable that the (perhaps somewhat brittle) norm against the use of CWs in combat could break down, most likely as a result of either of two general circumstances.

First, a great power actor might be presented with an opportunity or danger so large as to justify, in the minds of its policy-makers, risking loud and prolonged international outrage. For example, if Beijing were ever to conclude that Taiwan *must* be reabsorbed militarily within a specific timeframe, the use of chemical weapons against Taiwanese forces might appear to be little more than a ‘diplomatic footnote’. After all, the risks – including war with the United States – would be grave even if China avoided CWs use, and the economic and political consequences would be serious even if China won. If CWs appeared to offer a significant prospect of speeding the consolidation of control over the island before US forces arrived in large numbers, the temptation to use them might prove irresistible to China’s leaders. Secondly, the (presumably secret) development of a new generation of CWs whose characteristics radically enhanced their utility on the battlefield might convince the state possessing such weapons to revise its doctrine and ‘normalise’ the powerful new weapons.¹² Of course, the combination of both of these conditions would maximise the likelihood of a breakdown in the anti-CWs norm.

RESEARCH QUESTIONS

1. How might terrorists use radiological, biological and/or chemical weapons as a tool to obtain their political objectives?
2. How great a threat do radiological, biological and chemical weapons present to the global security environment? How does this threat compare to that presented by nuclear weapons?
3. Has the heretofore rather minimal international response to the use of chemical weapons in the ongoing Syrian Civil War significantly weakened the international norm against the use of such arms?
4. How likely is it that the norm prohibiting CBWs possession will survive even if future technological developments make these weapons more useful militarily to major states?
5. On balance, should we regard the efforts to prevent the stockpiling and proliferation of chemical weapons as essentially successful? Why or why not?

FURTHER READING

Barenblatt, Daniel, *A Plague upon Humanity: The Secret Genocide of Axis Japan’s Germ Warfare Operation* (New York: Harper, 2004). A detailed history of the Japan’s BWs research programme of the 1930s and 1940s, and the extensive use of such weapons against civilians by its military.

- Croddy, Eric A. with Clarisa Perez-Armendariz and John Hart, *Chemical and Biological Warfare: A Comprehensive Survey for the Concerned Citizen* (New York: Copernicus Books, 2002). This readable guide provides much useful information on both chemical and biological weapons.
- Harris, Robert and Jeremy Paxman, *A Higher Form of Killing: The Secret History of Chemical and Biological Warfare*, rev. and updated edn (New York: Random House, 2002). Co-written by a popular novelist and a well-known journalist, an accessible history of the development of CB warfare.
- Koblentz, Gregory D., *Living Weapons: Biological Warfare and International Security* (Ithaca, NY: Cornell University Press, 2011). Discusses the dangers posed by BWs in the context of the global security environment.
- Leitenberg, Milton and Raymond A. Zilinskas, *The Soviet Biological Weapons Program: A History* (Cambridge, MA: Harvard University Press, 2012). A carefully researched academic history of the Soviet biological weapons programme that provides a comprehensive summary of the publicly available knowledge concerning that effort.
- Smith III, Frank L., *American Biodefense: How Dangerous Ideas about Biological Weapons Shape National Security* (Ithaca, NY: Cornell University Press, 2014). An informed critique of the US Department of Defense's attitude and policy regarding BWs defence.
- Tucker, Jonathan, *War of Nerves: Chemical Warfare from World War I to Al-Qaeda* (New York: Pantheon, 2006). A history of the development and use of chemical weapons, with a particular focus on nerve agents.
- Zubay, Geoffrey, *Agents of Bioterrorism: Pathogens and their Weaponization* (New York: Columbia University Press, 2012). A detailed examination of thirteen potential BWs agents which assesses the dangers that each would pose if used in a terrorist attack.

ONLINE RESOURCES

- www.opcw.org: The website for the Organisation for the Prohibition of Chemical Weapons (OPCW), the implementing body for the CWC, contains information about the treaty, as well as on chemical agents.
- www.un.org/disarmament/WMD: The United Nations Office of Disarmament Affairs (UNODA) website contains information of UN counterproliferation activities regarding WMDs of all types.
- <http://armscontrolcenter.org>: The Center for Arms Control is an American 'think-tank' that studies proliferation and counterproliferation of WMDs, with a particular emphasis on US government policymaking.
- www.nonproliferation.org/: The website of the James Martin Center for Nonproliferation Studies, which is part of the Monterey Institute of International Studies, includes a variety of videos and other resources related to WMDs of all kinds.
- www.nonproliferation.eu/: The EU Non-proliferation Consortium includes many think-tanks located in a variety of countries, and its website contains documents related to EU institutions, as well as links, organised by country, to the organisations that together comprise the Consortium.
- www.nti.org/: The website of the US-based Nuclear Threat Initiative, which focus on the prevention of proliferation of WMDs of all kinds, includes a variety of resources including profiles providing country-specific information related to WMDs.

NOTES

1. On possible risks related to the smallpox vaccine, see John Schwartz, 'Soldier's Smallpox Inoculation Sickens Son', 18 May 2007; Nicholas Bakalar, 'Risks of Smallpox Vaccine Vary by Virus Used as Base', 5 September 2006, both in *New York Times* online edn (www.nytimes.com).
2. On the danger that smallpox may again be released in nature, see Richard Preston, *The Demon in the Freezer* (New York: Random House, 2002); Jonathan B. Tucker, *Scourge: The Once and Future Threat of Smallpox* (New York: Atlantic Monthly Press, 2001).
3. Alan Cowell, 'Just One of Nine Suspects Convicted in 2003 British Poison Plot Case', *New York Times* online edn, 14 April 2005.
4. For a discussion of ricin's use as a terror weapon, see Donald G. McNeil, Jr, 'Ricin on Capitol Hill: The Poison', *New York Times* online edn, 4 February 2004.
5. 1899 Hague Convention, Declaration II: Declaration on the Use of Projectiles the Object of Which is the Diffusion of Asphyxiating or Deleterious Gases, 29 July 1899, accessed from the Avalon Project at Yale Law School (http://avalon.law.yale.edu/19th_century/dec99--02.asp).
6. See Federation of American Scientists, Chemical Weapons Delivery (<http://fas.org/programs/bio/chemweapons/delivery.html>) for a more detailed discussion of difficulties related to delivery of CWs.
7. See David Wise, 'Novichok on Trial', *New York Times* online edn, 12 March 1994.
8. On the US military's counter-CBW capabilities, see Department of Defense Chemical and Biological Defense Program, 2014 *Department of Defense Chemical and Biological Defense Annual Report*, March 2014 (www.acq.osd.mil/cp/cbd_docs/home/Final%202014%20DoD%20CBDP%20ARC_signed%2021%20Mar%202014.pdf).
9. For examples of the large literature on Iraqi WMDs released in the period between 1991 and the 2003 invasion of Iraq, see Richard Butler, *The Greatest Threat: Iraq, Weapons of Mass Destruction, and the Growing Crisis of Global Security* (New York: Public Affairs, 2001); Robert W. Chandler, *Tomorrow's War, Today's Decisions: Iraqi Weapons of Mass Destruction and the Implications of WMD-armed Adversaries for Future U.S. Military Strategy* (McLean, VA: AMCODA Press, 1996); Anthony H. Cordesman, *Iraq and the War of Sanctions: Conventional Threats and Weapons of Mass Destruction* (Westport, CT: Praeger, 1999).
10. See C. J. Chivers, 'The Secret Casualties of Iraq's Abandoned Chemical Weapons', *New York Times* online edn, 14 October 2014.
11. Glenn Kessler, 'Saddam Hussein Said WMD Talk Helped Him Look Strong to Iran', *Washington Post* online edn, 2 July 2009.
12. On possible future developments in CWs, including the 'merging' of biological and chemical warfare technologies, see Jonathan B. Tucker, 'The Future of Chemical Weapons', *The New Atlantis*, 26 (Fall 2009/Winter 2010), 3–29 (www.thenewatlantis.com).



17

Weapons of mass destruction: nuclear weapons

Contents

Introduction: the 'nuclear club'

Nuclear weapons

The nightmare of nuclear terrorism

Nuclear deterrence

The First and Second Nuclear Ages: past and future possibilities for conflict

 Nuclear conflict in a Cold War context

 The Second Nuclear Age

 Nuclear conflict: future possibilities

Conclusion: nuclear weapons in the present and future

KEY THEMES

- The period from 1945 to the end of the Cold War, the 'First Nuclear Age', was defined by the relations between the two superpowers. The world now is in a 'Second Nuclear Age' in which an increasing nucleus of actors possess nuclear weapons.
- Capability, credibility, severity and surety are vital factors in determining the success or failure of a deterrence threat.
- Given the increasing number of states possessing nuclear weapons and the diversity of their strategic cultures, the 'deterrence lessons' of the First Nuclear Age may be of limited applicability in the current era.
- There are two distinct types of nuclear proliferation. When the number of states possessing nuclear weapons increases, this constitutes horizontal weapons proliferation. An increase in the number of nuclear weapons is vertical proliferation.
- It is not probable that terrorists will obtain one or more nuclear weapon in the medium-term future, but the likelihood of such an outcome is not negligible. The questionable security of the Pakistani nuclear arsenal is a matter of particular concern.

Introduction: the 'nuclear club'

The Nuclear Non-Proliferation Treaty (NPT) restricts legal possession of nuclear weapons to just five states – China, Great Britain, France, Russia, and

the United States. All other signatories to the NPT legally renounce the right to build or possess nuclear weapons. However, some of these signatories, such as Iran, may be attempting to break the Treaty's restrictions. Moreover, three countries that never signed the NPT are known to possess nuclear weapons (India, Pakistan and Israel), while North Korea – which apparently has conducted four nuclear tests and can be presumed to have a nuclear arsenal – is a former party to the Treaty but withdrew from it in 2003. Indeed, even though the number of states that actually possess these weapons is small, the three most populous states in the world all have them; altogether the nuclear powers are home to roughly half of humanity – so, at least in one respect, nuclear possession is not really unusual.

Chemical, radiological and nuclear weapons all are viewed negatively, but perceptions of the last of these, as the [previous chapter](#) noted, are decidedly mixed. Although many voices worldwide denounce the very possession of nuclear weapons as being immoral, it is also clear that enormous numbers of people – both leaders and regular citizens – associate these devices with international standing and great-power status. Indeed, notwithstanding their diplomatic and economic repercussions, the 1998 nuclear tests by India and Pakistan were popular domestically in both countries. Many millions of Indians and Pakistanis saw their country's successful nuclear tests as proof of its technological prowess and status on the world stage – in short, as a source of national pride.

Clearly, great numbers of people worldwide regard nuclear weapons differently from chemical, biological or radiological weapons. While there is a general view that nuclear devices are morally problematic – and it is obvious that the effects of their use can be horrifying – they are far from universally despised. Rather, this 'absolute weapon', as Bernard Brodie famously dubbed the atomic bomb,¹ is feared and hated for its killing power but, at the same time, seen as granting a special status in the international community. (If nuclear weapons did not, and simply brought disrepute upon a state, it is difficult to see why Britain and France, neither of which is likely in the foreseeable future to face critical security threats of the sort that can be addressed by nuclear possession, and both of which, in any case, are members of NATO and therefore protected by the American 'nuclear umbrella', do not surrender their small and expensive national arsenals.) The nuclear club is an exclusive one, even if it is growing (see [Box 17.1](#)).

Nuclear weapons

The first nuclear devices developed were the type known as fission weapons, but they were of two different designs – the ultimate products of the secret US Manhattan Project that was undertaken during the Second World War.² In a simple 'gun-type' design, a conventional explosive is used to drive two separate sub-critical masses together, forming a critical mass and setting off a nuclear chain reaction. The first nuclear weapon used in warfare, 'Little Boy', dropped

Box 17.1 Strategic culture and nuclear weapons

States have distinct strategic cultures that shape profoundly how they 'do strategy' – culture influences human behaviour generally, including strategic decision-making. Leaders' values and preferred methods for obtaining goals, the preferred national 'style' of war-making and many other areas all are influenced by culture. Like any other individuals, leaders are shaped by their life experiences, and culture plays an enormous influence in shaping those experiences. Humans can strive to be aware of how culture has shaped their views and behaviour, but actually escaping the bonds of culture is impossible; even individuals who reject the culture of their formative years were shaped profoundly by it.

Strategic culture certainly reflects the overall culture of a state's population, including the interpretation of historical experiences and other shared beliefs that shapes public attitudes towards strategy and war, but it is also influenced by the specific life experiences of the individuals who make strategy (and strategic culture certainly can change over time, though how rapidly and profoundly it may do so is debated). Therefore, it is not possible to assess strategic culture simply by looking at the general culture of a country – it is also necessary to study which classes and regions a state's political and military leaders come from and their education, their religious and ideological beliefs as well as other attitudes (and how they may differ from those of the general population) and many other factors which shape decision-making. Understanding the influence of strategic culture, in turn, can provide valuable insights into why a state might acquire nuclear weapons, as well as chemical and biological weapons, and under what circumstances it might use such weapons aggressively.¹

¹ Examples of the literature of strategic culture include Lawrence Sondhaus, *Strategic Culture and Ways of War: An Historical Overview* (London and New York: Routledge, 2006); Haiyun Feng, *Chinese Strategic Culture and Foreign Policy Decision-making: Confucianism, Leadership, and War* (London and New York: Routledge, 2007); Colin S. Gray, *Nuclear Strategy and National Style* (Lanham, MD: Hamilton Books, 1986); Alastair Ian Johnson, *Cultural Realism: Strategic Culture and Grand Strategy in Chinese History* (Princeton University Press, 1995); Forrest E. Morgan, *Compellence and the Strategic Culture of Imperial Japan: Implications for Coercive Diplomacy in the Twenty-first Century* (Westport, CT: Praeger, 2003).

on Hiroshima, Japan, on 6 August 1945, was a gun-type device. The more complex 'implosion' warhead uses a conventional explosive to implode a sub-critical mass, which looks like a round ball with a hollow core. This design uses fissile material more efficiently than the gun-type device and therefore is 'cleaner': a greater percentage of the fissile material is consumed in the nuclear explosion, leaving less of this highly radioactive matter behind as fallout. The first nuclear weapon ever tested, in the 'Trinity' test on 16 July 1945 at Los Alamos, New Mexico, was an implosion device, as was 'Fat Man' – the second, and heretofore last, nuclear weapon to be used in warfare. The latter was dropped on Nagasaki, Japan, on 9 August 1945.

The period of American nuclear monopoly was brief. Even before the first nuclear test, Soviet intelligence had already infiltrated the highly secret US

nuclear development programme, known as the Manhattan Project, and was receiving stolen technical data which was used to advance the Soviets' own secret nuclear programme. In August 1949, the Soviets exploded their own atomic device.

A few years later, the next major development in nuclear weapons technology occurred when the United States conducted the first test of a thermonuclear weapon, in October 1952.³ Thermonuclear weapons use an initial fission explosion to create the conditions for nuclear fusion. Thus, thermonuclear weapons sometimes are called fusion or, more accurately, fission-fusion, devices; also, these weapons are often referred to as 'hydrogen bombs' or 'H-bombs'. Theoretically, there is no limit to the potential explosive yield of a thermonuclear warhead.

Nuclear weapons' yield is measured in kilotons and megatons. A kiloton is equal to the explosive force of 1,000 tons of TNT (dynamite); a megaton is the equivalent of 1 million tons of TNT. Nuclear weapons have been created with an enormous range of yields. The largest nuclear device ever tested (nicknamed the 'Tsar Bomba' in the West) was a Soviet warhead with a estimated yield of at least 50 megatons. By contrast, some tactical nuclear weapons have a yield measured as a fraction of a kiloton. It is important to understand that nuclear weapons can vary enormously not only in yield but in many other essential characteristics. Weapons designers are tasked with creating warheads that can accomplish particular military objectives, and those objectives, in turn, are shaped by the political goals of a state and its military doctrine.

Many authors erroneously refer to nuclear weapons with very high yields, particularly those in the megaton-range, as 'city-busters', assuming that they are designed to be used primarily against civilian targets. In fact, such high-yield weapons are more likely to be used against very 'hard' targets, such as concrete missile silos, command bunkers and similar installations. Urban-industrial targets, by contrast, are 'soft': it requires relatively little explosive pressure to destroy office buildings and similar structures. Such considerations are significant for war planners for numerous reasons, including the fact that the size of a nuclear device is related to its yield (the higher the yield, the more fissile material is required); how many nuclear warheads a missile may carry is determined essentially by the size and weight of those warheads. In general, nuclear devices have become progressively smaller in yield over the decades as accuracy has improved. This, of course, is because when a warhead lands on, or very close to, a target, that target is exposed to much more explosive force than when the warhead misses by a considerable distance.

Nuclear weapons often are divided into two major categories: tactical nuclear weapons (TNWs) and strategic nuclear weapons. The distinction between the two can be fuzzy but, in general, it would be accurate to say that TNWs are delivered at shorter ranges and are intended for battlefield use. Possible means of delivery for TNWs include tactical aircraft (such as fighter aeroplanes), artillery shells, short-range missiles, cruise missiles and so forth. The United States

and the Soviet Union even developed nuclear torpedoes and depth charges for use in naval warfare. The superpowers each built enormous tactical nuclear arsenals, with thousands of these devices kept in Europe for possible use in a Third World War. Today, the great majority of the American TNWs have been removed from Europe, although a modest number of weapons that could be delivered by tactical aircraft remain. Russia has not revealed in detail the current status of its tactical nuclear arsenal, although it is quite possible that several thousand TNWs remain in European Russia. The overall number of TNWs maintained by Russia and the United States is not limited by arms control treaties between those two countries; rather, existing treaties focus on strategic nuclear weapons.

Strategic nuclear warheads generally have a larger yield and are delivered at long range, using intercontinental ballistic missiles (ICBMs), submarine-launched ballistic missiles (SLBMs) or heavy bombers. These three means of delivery together constitute the classic Cold War triad. The basic logic underlying the triad is that it is desirable for a nuclear state to possess multiple methods of delivering its strategic nuclear warheads, as this helps to ensure that it may do so reliably, particularly if it is the victim of a devastating first strike inflicted by a nuclear-armed enemy. Each of the three 'legs' of the Cold War triad was developed because it possessed specific advantages.

Bombers were the first means used for nuclear delivery. Originally, bombers carried simple, unguided (and thus relatively inaccurate) gravity bombs, such as those dropped on Hiroshima and Nagasaki, but nuclear-armed bombers now can carry advanced weapons systems, such as nuclear-tipped cruise missiles, which contain highly accurate modern guidance systems. Importantly, bombers may be recalled before their mission is completed. Thus, for example, if a state believes that a nuclear attack may be imminent and sends bombers out with the intent of delivering a retaliatory strike its leaders may cancel the bombing mission if they soon thereafter discover that the attack will not, in fact, occur. Using a ballistic missile is more akin to firing a pistol: once the trigger is pulled, the bullet cannot be recalled. Moreover, bomber pilots may be tasked with the job of locating and destroying targets whose precise position is not known (see [Figure 17.1](#)).

The great advantage of SLBMs is the difficulty that an enemy is likely to encounter in locating and sinking the ballistic missile submarines (colloquially known in the US Navy as 'boomers') on which they are placed. A single ballistic missile submarine may contain many nuclear warheads. For instance, *Ohio* class ballistic missile submarines, which at present are the only American submarines tasked with delivering strategic nuclear weapons, originally were designed to carry twenty-four Trident II SLBMs, each with up to fourteen warheads (the number of launch tubes and warheads subsequently has been reduced because of US–Russian arms control agreements). For most of the Cold War, ICBMs enjoyed a great advantage over SLBMs in their accuracy, as measured by circular error probability (CEP), which essentially is the size of the circle



Figure 17.1 The YB-52, a prototype version of the B-52 bomber, lands on a dry lake at Edwards Air Base, 1953. The B-52 is still used today by the US Air Force to carry nuclear weapons.

surrounding a target in which half of the weapons fired at that target may be expected to land. (For example, if 50 per cent of the warheads fired at a target fell within 1 kilometre of it, but 50 per cent landed more than 1 kilometre from it, then the CEP would be 1 kilometre.) However, the development in recent decades of very precise SLBM warheads has eroded enormously the accuracy advantage previously associated with ICBMs (see [Figure 17.2](#)).

Unlike bombers, which are vulnerable to air defences (such as fighter aircraft), ballistic missiles (either land- or sea-based) and the warheads that they carry are very difficult to destroy. This is especially true of long-range ballistic missiles after the initial boost phase of their flight. Nevertheless, the United States has deployed ballistic missile defence (BMD) systems intended to intercept long-range missiles, and numerous countries have expressed interest in acquiring BMD defences. It also should be noted that the speed with which ballistic missiles deliver their weapons payload would be a particularly important factor in a first strike, as the targeted country would have only a very brief period to make critical strategic decisions before the warheads struck.



Figure 17.2 A Standard Missile Three (SM-3) is launched from the US Navy Aegis cruiser USS *Lake Erie* in a ballistic missile defence test.

During the Cold War, the United States and the Soviet Union developed enormous nuclear arsenals: by the late 1970s, they possessed tens of thousands of tactical and strategic warheads. The sheer destructive power of these arsenals, and the fear that they would actually be used in warfare, resulted in a very strong emphasis throughout the Cold War on deterrence theory.

The nightmare of nuclear terrorism

Particularly since the 9/11 attacks, there has been enormous concern about the possibility that terrorists might acquire nuclear weapons. Certainly, terrorist acquisition of a nuclear device would present a 'nightmare scenario'. As noted above, certain chemical or biological agents could be used by terrorists to inflict an enormous number of casualties, although there is considerable reason to doubt whether a terrorist group would be able to deliver an agent in a fashion efficient enough to cause truly large numbers of casualties. However, if a nuclear weapon were smuggled into the centre of a major urban area and detonated, it would surely cause a horrifying number of casualties. Actually acquiring nuclear weapons has, however, thus far proven difficult for terrorists, and it is improbable that even a wealthy and sophisticated group would actually be able to build a nuclear device, assuming it somehow acquired the necessary fissile material (which itself would be no small feat). While there is some dispute amongst experts about whether a terrorist group plausibly could build a nuclear weapon, it would appear, on balance, quite unlikely. While the basic principles underlying the construction of a nuclear device are well known, actually building such a device would require advanced engineering that probably would be beyond a terrorist organisation's capabilities. Thus, the more likely route for terrorist nuclear acquisition would be obtaining an already existing nuclear weapon, either by theft or by gift.

Since the fall of the Soviet Union there have been various rumours that Soviet nuclear weapons have 'disappeared', presumably having been stolen by greedy military officers seeking to resell the devices, but these claims have never been confirmed and probably are unfounded.⁴ However, it is not impossible that a nuclear weapon might one day be acquired by terrorists. For instance, the case of Pakistan, a state that contains a very substantial number of individuals (including powerful ones) who are sympathetic to al-Qaeda and similar groups, is especially worrisome.

Pakistan's political order is extraordinarily troubled; the country is in a 'permanent crisis' of indefinite duration. While today it is no longer ruled formally by its powerful and secretive military, the latter remains extraordinarily powerful. Its public politics are chaotic and bloody – riots and terrorist bombings are sufficiently commonplace that most such events receive little press attention outside Pakistan's immediate 'neighbourhood'. The country's problem with terrorism and insurgency is so severe that it would not be excessive to say that Pakistan is in a state of civil war. (Indeed, in early 2009, the Pakistani Taliban had come to dominate enough of north-west Pakistan that for a time areas disturbingly close to Islamabad were under its control.) The failure of the Pakistani state is a distressingly plausible future event.

Even absent collapse, the security of Pakistan's sizeable (and growing) nuclear arsenal is questionable. Some sources claim that the procedures for guarding the warheads are grossly inadequate, with the army even moving them around

the country in unmarked vans in the regular flow of traffic.⁵ However, the Pakistani Army denies the allegation of inadequate nuclear security, and numerous US and other foreign officials have publicly expressed confidence in the security of the Pakistani arsenal.⁶ Such alleged security measures, which would be shocking in the most pacific of countries, are reportedly a response to the fear that the United States will execute a military operation to seize the Pakistani arsenal. The most plausible motivation for such an operation, of course, would be to eliminate the threat that a terrorist group or other sub-state actor might acquire some of Pakistan's warheads. It would be unpleasantly ironic if Pakistan has responded to fears of US seizure by instituting practices exponentially increasing the likelihood that terrorists will obtain one or more nuclear warheads.

Given Pakistan's unique circumstances, its arsenal appears the most vulnerable in the world to 'leakage' in the foreseeable future. However, other countries ruled by brittle and unpredictable governments, such as North Korea and (if it obtains a nuclear capability) Iran, also present a non-negligible danger in this regard.

It is very difficult to estimate with precision the number of casualties that might be inflicted by a nuclear terrorist attack, as these would vary greatly depending on the explosive yield of the weapon, the layout and population density of the urban area in which it was detonated, and many other factors; meteorological conditions such as wind speed and direction and post-attack rainfall are important in determining the spread of radioactive fallout. Clearly, though, any such attack would be a monstrous event with substantial international political repercussions.

Nuclear deterrence

Deterrence is not a concept unique to the nuclear era; indeed, deterrence has been practised throughout human history. At its core, the concept is a simple one: using threats, explicit or implicit, to prevent a potential foe from doing something undesirable, such as starting a war over a disputed piece of territory. In order to prevent the 'deteree' from starting a war, the 'deterer' must convince the deteree that doing so is likely to be unprofitable.⁷

There are two main ways in which the deterer may do this. First, it may try to impress upon its counterpart that any war that the latter launched to gain the territory would be unsuccessful. This is deterrence by denial: if the deterrence attempt is successful, the deteree believes that its goals would not be achieved, and thus it will not launch a war. Alternatively, the deterer may attempt to convince the deteree that even if the latter did achieve its goals, it would pay a prohibitive price to do so: it may win the territory, but it would be a victory costing more than it is worth. This is deterrence by punishment. These two forms of deterrence are not mutually exclusive, and, indeed, the most potent

deterrence threats generally combine aspects of each. For instance, the deterrent may try to convince its opponent that it would both fail to gain the disputed territory *and* pay a high price – in military, financial and/or other losses – for its failed effort.

The flip side of deterrence is compellence, convincing an opponent to perform an action that he or she believes is not in his or her best interests and, thus, would not agree to except under duress.⁸ For example, in the period immediately after 9/11 the United States attempted to convince the Taliban government of Afghanistan to hand over Osama bin Laden, leader of the al-Qaeda terrorist network. Bin Laden had been living in Afghanistan for several years under the protection of that country's Taliban government. Washington warned Taliban leaders that if they did not turn bin Laden over, it would invade Afghanistan and overthrow their regime. In this case, compellence was not successful – the Taliban did not surrender bin Laden. In response, the United States carried out its threat, invading Afghanistan and displacing the Taliban from power in Kabul.

Both deterrence and compellence are forms of coercion.⁹ If deterrence or compellence is successful, the deterred or compelled party will shape its behaviour so that it is in accordance with the demands placed upon it. In either case, the state whose behaviour is to be shaped may refuse to co-operate – and in such a case deterrence or compellence has failed. At that point, the would-be coercing power must either carry out its threat – which, in any scenario involving nuclear armaments, carries many dangers, including the possibility (or, depending on what the deterrence or compellence threat was, even the certainty) of catastrophic violence – or 'back down', accepting that it has failed to deter or compel its opponent.

While deterrence has always been a key aspect of international politics and diplomacy, during the Cold War the concept was absolutely critical: particularly in the West, scholars and policy-makers obsessed over how to ensure that neither superpower would make a misstep that would result in a Third World War (see [Box 17.2](#)). As a result, a sophisticated body of deterrence theory soon developed.

Because nuclear weapons are capable of inflicting overwhelming damage and can be delivered very rapidly, they presented deterrence theorists and policy-makers with novel problems. While states always had to worry about surprise attacks, technological limitations placed constraints on the ability of aggressors to launch speedy and devastating assaults. Moreover, geography provided many states with a substantial degree of protection. However, the ability of aircraft – and, particularly, ballistic missiles – to deliver nuclear weapons could make geographic factors almost irrelevant in preventing, or even slowing, offensive success. A nuclear war could be fought, and a state or states utterly destroyed, in an afternoon.

How easy, or difficult, it will be to deter a given country is a matter of conjecture, not an exact science. As noted above, in order for deterrence to work,

Box 17.2 The first-strike quandary and the Cuban missile crisis

One of the great difficulties of nuclear strategy is coping with the challenges presented by the possibility of a first strike. Because of the speed with which nuclear war could be fought, there might be no time to mobilise reserve forces, much less build new weapons – states would rely on those which they had available. Thus, there is a natural, and very dangerous, tendency for states in a crisis to wish to ‘land the first punch’, devastating the enemy polity and, hopefully, damaging it so severely that it is unable to retaliate effectively. This encourages crisis instability – even if neither side wants war and would prefer that the crisis be resolved peacefully, the threat of a first strike may inspire such fear in one side that it chooses to launch its own first strike, even if it does not wish to be aggressive.

During the Cuban missile crisis of October 1962, many observers feared precisely this outcome. The crisis resulted from the decision of Soviet leader Nikita Khrushchev secretly to deploy nuclear-armed missiles in Cuba, a state ruled by Soviet ally Fidel Castro and located only a short distance from Florida. At the time, the United States enjoyed a very considerable advantage over the Soviet Union both in its number of nuclear weapons and in its ability to deliver them to the soil of the enemy superpower. The Soviets hoped to decrease the US advantage by placing medium-range ballistic missiles in Cuba. Not only would this put missiles within range of a multitude of US targets, but, given Cuba’s location, targets in the Washington, DC metro area could be struck with little time between missile launch and warhead impact. Moreover, Khrushchev hoped that once the deployment was publicly announced it would discourage the United States from possibly invading Cuba in the future and overthrowing its communist government (in the previous year, the United States had sponsored the disastrous Bay of Pigs invasion undertaken by anti-Castro Cuban exiles).

Once it had discovered that the deployment was under way – thanks to intelligence flights over Cuba by US reconnaissance aircraft – the United States considered a number of options, including an invasion of Cuba, but President John F. Kennedy soon decided on the relatively cautious course of confronting the Soviets publicly while blockading Cuba. The crisis continued for about two weeks, Washington and Moscow eventually agreeing to a secret compromise in which the Soviets removed nuclear weapons from Cuba, while Kennedy pledged not to invade Cuba in the future and to remove US nuclear-armed missiles deployed in Turkey.

the deterred party must ‘consent’ to deterrence – that is to say, it has the option of accepting the possible consequences of its action and of *not* being deterred. Thus, the success of deterrence can never be guaranteed. However, various factors – credibility and capability – are likely to impact the success of a specific deterrence threat. Credibility, in turn, is determined largely by the past actions of the would-be deterrent and the perceived appropriateness of the threat to the situation at hand.¹⁰ Thus, a state with a long record of blustering, making extravagant but idle threats that are not carried out, is likely to have much less credibility than one with a reputation for always carrying out threats if the demands that it puts forth are not met.

This highlights one of the important limitations faced by nuclear powers: the political and military consequences of using nuclear weapons would be so serious that possession of these weapons is relevant only in a very limited set of circumstances. For instance, in 1982 Argentina invaded the Falkland Islands, a British possession claimed by both countries. Britain was nuclear-armed, and Argentina had good reason to worry that Britain would defend the islands militarily, although Argentina appears to have greatly underestimated the likelihood that Britain's Thatcher government would do so. But Britain's nuclear status was not a major factor in either Argentine or British decision-making. Simply put, Argentina, rightly, assumed that Britain did not consider the Falklands to be worth the enormous diplomatic and other costs that would accompany a nuclear threat, much less the actual use of nuclear weapons. As the Falklands example illustrates, possession of nuclear weapons is not even a reasonably reliable guarantee against military attack in many circumstances.

Capability is a relatively simple concept: a state either has or does not have the ability to carry out a threatened action. In some cases, capability is not at issue, and the would-be deterrent clearly has the ability to inflict a particular punishment, the only question being whether it has the will to do so. Sometimes, however, capability may be questionable. This is particularly notable in regard to second-strike deterrence threats. The purpose of such threats – which played a very large role in Cold War deterrence – is to deter a foe from attempting a nuclear first strike. Essentially, the state threatening a second strike is trying to convince its potential foe that even a very well-organised nuclear first strike would not eliminate its ability to retaliate. Thus, a first strike would be a 'murder-suicide', with the attacked state, even as it was in its death throes, lashing out at its aggressor.

Clearly, such a threat can have a high credibility: a state that had suffered a nuclear strike would wish to take revenge on its enemy. However, whether it could actually do so would depend on many factors. A successful first strike would enormously damage the nuclear arsenal, communications network and other relevant assets of the victim state. Thus, to preserve second-strike capability, the superpowers spent vast sums building and maintaining both large and diverse arsenals and highly robust command, control and communications networks, as well as otherwise endeavouring to ensure that its peer believed that even the most successful first strike would not preclude devastating retaliation.

Credibility and capability are central to deterrence because of their influence over severity and surety of punishment. Deterrence is, ultimately, based on threatened punishment, and in order to succeed the would-be deterrent must issue a threat that is properly 'balanced'. The punishment threatened must be severe enough to dissuade the other party from an action but not be so severe as to be implausible. Thus, the more centrally a deterring state's interests are threatened – particularly if its population, form of government, territory and other most vital assets are placed in danger – the more severe a punishment

may be while still remaining credible. The greater the severity and surety of punishment, the greater the likelihood of deterrence success; if one of these critical factors is missing – the threatened punishment is too minor or it seems unlikely that the threat actually would be carried out – deterrence can be expected to fail.

As noted above, however, it is the would-be deteree who ultimately casts the deciding vote in whether a deterrence threat will succeed. If a leader is risk-averse and places a high value on the avoidance of punishment, deterrence threats are likely to succeed. If, however, a leader is reckless and places a much higher value on potential gains than on potential losses, successful deterrence is much less likely. Moreover, if a leader simply believes that there is some overriding imperative – religious, political or otherwise – that makes it utterly and totally unacceptable to do as the would-be deterrent wishes, deterrence failure is certain.

In this regard, such rogue states as Iran and North Korea are of particular interest. It is generally assumed, probably rightly, that the leaders of most nuclear-armed states, including the five ‘legitimate’ nuclear powers acknowledged by the NPT, place a tremendously high value on avoiding nuclear conflict and are both sensitive to and likely to correctly interpret deterrence threats. Therefore, even if they are willing to be aggressive and take calculated risks – as was Russian President Vladimir Putin in absorbing Crimea and stirring conflict in eastern Ukraine – they are presumed unlikely to take actions so reckless as to be likely to result in nuclear conflict. ‘Cold War-style’ deterrence strategy might not, however, be effective when used against nuclear-armed rogues whose strategic priorities and threat perceptions may be highly unorthodox.

The First and Second Nuclear Ages: past and future possibilities for conflict

Nuclear conflict in a Cold War context

The phrase ‘nuclear war’ often is used casually to mean an essentially unlimited conflict: a simple exchange of warheads, with combatants showing no regard for civilian life and property, whose ultimate outcome is, inevitably, the destruction of the human species, or at least a very great portion of it. This, indeed, is one possible type of nuclear engagement and during the Cold War the possibility of such a conflict – not unreasonably – aroused fear worldwide. A full-scale nuclear exchange between the superpowers could have involved tens of thousands of warheads, the targeting of major cities in Asia, Europe, North America and perhaps elsewhere, and severe – indeed, potentially globally devastating – environmental damage. However, this sort of clash is only one of the possible types of nuclear war.

At the other extreme, nuclear weapons could – at least in theory – be used in a fashion that would cause no civilian casualties and only minimal, mostly local, environmental consequences. For example, use of a relatively low-yield

tactical nuclear weapon against a command bunker located in a very rural area or a ship at sea fits into this category. Of course, given the international outrage that would surely follow after even the most discriminating use imaginable of a nuclear weapon (and the less discriminate nuclear retaliation that might follow if a nuclear-armed state were attacked), there are relatively few plausible scenarios in which such limited nuclear use might occur.

In discussing nuclear conflict, it is useful to make a distinction between counterforce and countervalue targets. The first category includes targets of military/political value to a government, such as military forces (including, even especially, nuclear forces), certain government facilities, communications networks, the leadership itself and similar targets. The second is composed of things that a government would value for their own sake, aside from their military value, such as the population, industrial and transportation infrastructure and so forth. It should be noted that the two categories are not mutually exclusive: many things – such as the leadership and the communications network, for example – may be considered both counterforce and countervalue targets. However, speaking generally, most counterforce targets are of a military/political character, while countervalue ones are not. A carefully constrained nuclear attack would therefore probably be focused on counterforce targets, avoiding collateral damage insofar as possible.

Between the most extreme and limited types of nuclear war falls a broad variety of possible conflicts. It is certainly possible to imagine conflicts involving only the limited use of nuclear weapons, perhaps even to send a 'signal' to the opponent. During the Cold War, the United States and Soviet Union contemplated many possible war-fighting scenarios involving a limited use of nuclear weapons. Examining the United States' attitude towards nuclear weapons is particularly interesting for the purposes at hand, because its nuclear strategy was essentially built on the belief that the Warsaw Pact possessed overall military superiority in Northern Europe, which was thought to be the likely main theatre in a future war. A major fear was that the Soviet Union and its allies might launch an invasion, centred on what was then West Germany, which would punch rapidly through NATO's front-line troops, throwing the defenders into disarray and permitting a rapid Soviet advance to the English Channel. A possible solution to this threat was seen in the use of nuclear weapons.

The notion of extended deterrence is critical to understanding the NATO nuclear deterrence problem of the Cold War, as well as many possible crises involving rogue states or other actors. The chief immediate deterrence goal of the United States during the Cold War was not to protect its own homeland against Soviet attack. Rather, Washington was trying to protect its NATO allies from invasion, and a potential foe is always likely to question a state's willingness to suffer massive damage on behalf of its allies. Thus, maintaining a high level of credibility was critical, because it was feared that if Soviet leaders became convinced that the United States would not accept enormous damage on behalf of its allies, they would proceed to invade Western Europe. NATO's

European members worried that the United States would actually desert them in their time of need, accepting (at least temporarily) a Soviet conquest of Western Europe in exchange for being spared the horror of a nuclear attack on its own homeland.

If, however, the United States did not quickly abandon its European allies, it would then have confronted all the problems described previously in regard to managing escalation, including the very real possibilities either of accidentally allowing the escalation process to slip out of control, the result being a massive nuclear exchange, or of failing to escalate quickly and boldly enough to convince Soviet leaders (assuming, of course, that they could be convinced) to abandon their invasion and evacuate NATO territory.

Fortunately, we do not know with certainty how well NATO armies would have been able to cope with a nuclear environment. However, it is easy to imagine the struggles that war-fighters would have encountered after even a limited nuclear exchange. First, operating in a radioactive environment, which would require the use of stifling protective suits, would be exceedingly difficult; goods (including food and water) might be contaminated with radioactive materials; both mass media and military communications networks would be damaged and perhaps fatally undermined; enormous numbers of civilian and military casualties would overwhelm medical facilities of all kinds; panicked civilians, desperate to escape dangerous areas, would clog the transportation infrastructure, delaying troop movements and degrading, if not collapsing, the military logistic network; and there probably would be severe damage to the public utility infrastructure, with electricity, safe drinking water and other necessities suddenly unavailable to soldiers and civilians alike. Many soldiers, confronted with immense stress, would be strongly tempted to throw off their own uniforms and assist their loved ones in fleeing from the conflict. Moreover, this probably would have occurred in a very compressed timeframe, with highly complex societies reduced to Hobbesian chaos – and all of this occurring in a situation where an even more terrifying large-scale nuclear exchange loomed as a distinct possibility.

The vision of precisely how nuclear weapons would be used in the defence of Europe, however, changed over time, and it should be emphasised that there was never an entirely concrete, unalterable vision of what form a nuclear defence would take. In general, however, it would be accurate to say that the early US vision of how to defend NATO-Europe relied on the capability of the bombers in the US Air Force's Strategic Air Command (SAC) to 'leap-frog' past Soviet forces in the field and strike directly at the Soviet homeland, destroying the government, communications and military infrastructure of the Soviet Union. In short, in these early years many American officials envisioned a quite brief (and, critically, very one-sided) nuclear war.

The thinking of this era is generally associated with the concept of massive retaliation, although that term was not coined until 1954, when it was used in a speech by Secretary of State John Foster Dulles. Massive retaliation is

commonly misrepresented as a very simplistic doctrine in which any provocation by the Soviet Union would be met by a nuclear response, but the actual concept was rather more nuanced. Dulles actually referred to a US ability to respond to threats 'at places and with means of our own choosing'.¹¹ This ambiguous formulation did not rule out the possibility of an all-out nuclear attack, but it did not clarify under precisely what circumstances the United States would use nuclear weapons. Indeed, the critics of massive retaliation largely focused on its perceived lack of credibility in all but the most extreme situations; certainly, the Soviet Union had good reason to fear that the United States actually would resort to nuclear force to defend Western Europe, but it also had reason to doubt that Washington would do the same in response to a provocative act in a more marginal strategic region.

A threat that is excessively severe may be perceived as being incredible. Moscow did many things of which the United States disapproved – for example, attempting to increase its influence in South-east Asia, Africa and elsewhere – but even during the years in which massive retaliation was espoused, Washington did not really attempt to use its nuclear arsenal to dissuade the Soviets from such actions. This cautious stance was prudent, since casual use of nuclear threats would have been extremely dangerous, surely leading to many crises, any one of which might have resulted in nuclear war. Even aside from the possibility that the Soviet Union would have been goaded into a nuclear first strike, however, there was the probability that, sooner or later, Soviet leaders would have refused to believe American threats, especially regarding relatively petty issues. Thus, they would have been tempted to 'call the bluff', forcing Washington either to carry through its threat, with the result being nuclear war, or accept humiliation and the loss of credibility.

Criticisms of massive retaliation, as well as the growing availability of relatively small TNWs – and the growing size of the Soviet nuclear arsenal, which increasingly undermined massive retaliation's credibility – encouraged a shift in focus over time, with an increasing interest in limited nuclear alternatives. Under the right conditions, nuclear weapons could allow numerically inferior NATO forces to defeat an enemy of greater size, and, beginning in the 1950s, TNWs began to be dispersed throughout the US military. Academics, military officers, government officials and others debated intensely whether a nuclear war could be very limited in scope.

Perhaps, some argued, nuclear weapons could be used to signal seriousness to an opponent, making clear the willingness to escalate to nuclear use and encouraging the opponent to back down. If a state can control its opponent's behaviour by 'ratcheting up' the level of potential risk and violence in a conflict, it is said to enjoy escalation dominance; this puts it in a position to coerce its opponent into surrendering. Of course, dominance is dependent on the opponent actually being willing to accept defeat. To use a metaphor made famous by the nuclear strategist Herman Kahn, one can envision a military conflict as a ladder with many rungs – an escalation ladder.¹² If NATO had been the first

to employ nuclear weapons in a conflict with the Warsaw Pact (in the West, it was a common, although very questionable, assumption that this would be the case), it could have limited their use in various ways – only using nuclear weapons against military targets, for example, or even refusing to use such weapons except against invading enemy troops operating on NATO soil. The hope was that such a constrained exercise of military force would convince the Soviet Union that further aggression would be unprofitable. However, the actual Soviet response could have been very different. Indeed, if it had wished to do so, Moscow could have responded with its own escalation, leaping further up the ladder – which, in turn, would have left NATO with the decision either to back down or continue the process of escalation.

For a time, nuclear weapons were seemingly being integrated into the mainstream of US forces: in the late 1950s, the US Army even reorganised into ‘pentomic’ divisions, a force structure constructed with fighting and surviving on the nuclear battlefield in mind. However, TNWs never did become a fully and uncritically accepted part of the US arsenal (indeed, the pentomic division structure itself was soon abandoned), and NATO leaders did not achieve a common understanding of when defensive nuclear use should occur. NATO policy remained vague: a ‘no-first-use’ policy regarding nuclear weapons was never announced, but NATO also never stated explicitly when nuclear weapons would be employed. As time passed, and the Soviet arsenal grew ever larger, there was considerable worry in NATO-Europe that the Soviet Union might be able to sever the ‘transatlantic link’ with the United States.

During the early period of the Kennedy administration, the United States attempted to maximise its freedom of action through the doctrine of flexible response, which was shaped largely by criticisms of massive retaliation. Flexible response was intended to provide a president with a long menu of conventional, tactical nuclear and strategic nuclear options that could be implemented as appropriate. Thus, the thinking went, the United States would be better prepared to cope with situations, such as Soviet adventurism in the Third World, in which the use of nuclear weapons would be inappropriate.

Over time, however, the United States moved away intellectually from flexible response and towards mutually assured destruction (MAD), which was essentially based on a belief that a full-scale nuclear war could not, in any meaningful sense, be ‘won’: that Soviet forces would be able to inflict such massive damage on the United States even in a second strike that both societies would simply be destroyed. This of course made a first strike, at least against the Soviet homeland, unacceptable.

As a strategic planning concept, MAD grew out of a budgeting exercise conducted by Secretary of Defense Robert S. McNamara (he actually referred to ‘assured destruction’; the catchier term ‘MAD’ was coined by a critic of the exercise). If there was no plausible first-strike role for US forces, then their core purpose would be to avert a Soviet strike against the United States. McNamara believed that a Soviet first strike could be averted if the United States possessed

a nuclear arsenal sufficiently large and secure to inflict devastating counter-value damage on the USSR. His stated requirements were that the surviving US nuclear arsenal should be able to destroy 30 per cent of the Soviet Union's population, 50 per cent of its industrial capacity and devastate its 150 most important cities. However, the United States never really implemented 'MAD assumptions' into its nuclear war planning, always maintaining a far larger and more diverse nuclear arsenal – as well as a more complex targeting scheme – than the very simple requirements of MAD-based planning would have dictated.

One can understand, given the increasing size of the Soviet nuclear arsenal, why US leaders began to think in terms of mutual, rather than one-sided, nuclear devastation. However, this MAD-centred vision also had some curious side-effects, encouraging, for example, many to argue against ballistic missile defences because they feared that such defences might undermine the mutual character of assured destruction and thus upset the 'nuclear balance'. Those who made this argument had, in essence, accepted the notion that mutual vulnerability was the best possible condition in a world with two nuclear-armed superpowers.

MAD, however, never offered a satisfactory answer to many of the critical military questions facing the United States and NATO (most importantly, perhaps, it did not offer a clear role for nuclear weapons if the effort to deter a Soviet conventional invasion of Western Europe failed). Over time, American leaders tried to move away from MAD, the Nixon administration pivoting slightly away from it and the Carter administration putting forward the concept of counter-vailing strategy, a carefully stated doctrine that did not reject MAD outright but clearly was not based on its core premises. The Reagan administration moved still further away from MAD in its nuclear doctrine. Moreover, in 1983 Reagan proposed the Strategic Defense Initiative (SDI), more commonly known as 'Star Wars', a BMD system that would, it was hoped, be so robust as to make it very difficult for an aggressor to be confident of its ability to carry out a successful first strike. SDI challenged the essential premise of MAD, which was that stability resulted from the *mutual* character of assured destruction.

The evolution in US thinking about nuclear weapons reflected the difficulties inherent in attempting to build a coherent strategy that accounts for both deterrence and possible war-fighting. Risks accompany any nuclear strategy; each possible nuclear doctrine involves trade-offs, some dangers being lessened and others increased, and nuclear weapons do not offer a simple solution to all of a state's security needs. Moreover, it should be noted that the great swings in American thinking about nuclear war occurred in the relatively stable security environment of the Cold War, and decisions were made primarily in reference to a single potential foe, the Soviet Union. In the relatively freewheeling security environment of the twenty-first century, with its great variety of actors, including potentially unpredictable rogue states and terrorist groups, the creation of a satisfactory nuclear doctrine may prove to be even more challenging.

The Second Nuclear Age

Some academics and other observers argue that the world is in the early phase of a 'Second Nuclear Age', which is distinct from the 'First Nuclear Age' (lasting, approximately, from 1945 to the fall of the Soviet Union in 1991) in a number of key respects.¹³ One of the most important of these characteristics is the likelihood of an increasing unreliability of deterrence.

There are two basic forms of nuclear proliferation. When weapons proliferate horizontally, the number of states possessing nuclear weapons increases, but when vertical proliferation occurs, the total number of nuclear weapons increases. In general, the Cold War era was one of relatively modest horizontal proliferation; the number of states possessing these weapons grew slowly over the decades. However, the total number of nuclear weapons increased dramatically over the early decades of the Cold War because both Washington and Moscow engaged in rapid vertical proliferation. In recent years, however, the actual number of nuclear weapons on earth has decreased: we are experiencing 'vertical de-proliferation' as Russia and the United States allow their nuclear arsenals to constrict in size. However, horizontal proliferation is proceeding, with India and Pakistan testing nuclear weapons in 1998 (although both possessed nuclear weapons capabilities well before that year and India even conducted a 'peaceful' nuclear test in 1974), while North Korea appears to have tested warheads in 2006, 2009 and 2013. Iran may be well along in its own nuclear programme. It is impossible to say how many, if any, new possessors of nuclear weapons will appear over the next few decades, but there are some reasons to believe that it will be a substantial number.

First, it should be kept in mind that nuclear weapons have now existed for about seventy years. Basic nuclear weapons technology is not, in any reasonable use of the term, 'cutting-edge'. There are many thousands of individuals on earth who possess at least some of the knowledge and skills required to build a nuclear device, and many of them are citizens of – or simply are willing to work for – states that may desire nuclear weapons. The career of a single individual, Pakistani scientist and engineer A. Q. Khan, illustrates this fact well. Khan worked in Europe during the 1970s and stole technical information related to uranium enrichment. He became central to Pakistan's nuclear weapons development effort and eventually built up an international network that provided nuclear weapons-related parts and technology to various would-be proliferators, apparently including Iran and North Korea. The government of Pakistan contends that it was unaware of Khan's nuclear sales to foreign governments, although the huge scope of Khan's activities – and his frequent visits to countries, such as North Korea, that are not known chiefly for their tourist attractions – casts some doubt on this claim. In any event, however, Khan's career demonstrates how an individual with the right knowledge and resources can contribute greatly to horizontal proliferation (see [Box 17.3](#)).¹⁴

Box 17.3 The 'Libya quandary'

Over his decades in power, Libyan dictator Muammar Gaddafi had a reputation for unpredictability and a long record of 'roguish' behaviour. Libya's combination of a small population and substantial oil revenues, however, provided revenue sufficient to permit Gaddafi to remain in power for decades, as he combined domestic terror with ample financial support to favoured tribal leaders. This prevented the formation of a domestic coalition capable of overthrowing the mercurial and sadistic autocrat. Relatively late in his reign, however, Gaddafi attempted to improve relations dramatically with the West in general, and Great Britain and the United States in particular. While he began to inch in this direction in the 1990s, after the 9/11 attacks on the United States his efforts became more serious. Fear that the United States might invade Libya may have been a key factor motivating this foreign policy shift.

The G. W. Bush administration presented Gaddafi with a stark choice in the negotiations regarding his 'rehabilitation': part of the price would be prompt and complete disclosure of all Libyan WMD programmes.¹ He agreed in late 2003, and subsequent disclosures shocked observers globally. Libya's documentation made it clear that the A. Q. Khan proliferation network was far larger and more sophisticated than Western intelligence agencies had supposed. Even more shockingly, Libya – which was not believed to have a truly serious nuclear weapons programme – was surprisingly close to achieving a nuclear capability.

For a time, Gaddafi enjoyed the benefits of an enhanced international standing, but when the Libya Civil War began in February 2011, he was surprised that Western states, including the United States, soon came to support the rebellion against his rule. By March an air operation was launched by NATO and several non-NATO states, with most of the combat sorties flown by a combination of France, Great Britain and the United States. The foreign intervention soon proved decisive, and he was captured and killed by rebels in October 2011.

While NATO intervention in the Libyan Civil War was eminently defensible from a humanitarian standpoint, it was questionable from a counterproliferation perspective. Gaddafi's surrender of his WMDs ultimately did not buy protection from outside intervention, and left him in a weakened position when fighting the war itself – while of limited utility against a sophisticated military, the Libyan chemical arsenal might have been devastating against rebel forces, particularly in urban combat. If Libya had obtained a known nuclear capability by 2011 – a believable outcome, if not for the 2003 agreement – NATO intervention in Libya would have been unthinkable. Repressive authoritarians who observed Gaddafi's downfall may have learned some very unhelpful lessons regarding the desirability of obtaining WMDs in general, and nuclear weapons in particular.

¹ On the negotiations leading to the surrender of Libya's WMD programme, see Robert G. Joseph, *Countering WMD: The Libyan Experience* (Fairfax, VA: National Institute Press, 2009).

Secondly, if one or more particular states obtain, or appear to be close to obtaining, nuclear weapons, this may convince certain other states that they should build nuclear weapons. We have seen this phenomenon to a limited degree in the past – for example, India's determination to have a nuclear capability was clearly a major driver of Pakistan's nuclear programme, while India's own nuclear efforts were largely driven by its concern about China's nuclear arsenal – and we may see more extreme cases of it in the future. For instance, there has been speculation that Iranian acquisition of nuclear weapons might drive several other Middle Eastern states, possibly including Saudi Arabia, Turkey and others, to seek nuclear weapons.

Thirdly, and most importantly, nuclear weapons continue to have great utility both on and off the battlefield. Diplomatically, nuclear weapons bring (potentially considerable) benefits in terms of prestige, ability to practise nuclear deterrence and coercion and in other areas. Indeed, a nuclear-armed state may be able to dissuade a more powerful opponent even from taking military action against it. Many observers, for example, have speculated that if Iraq had possessed nuclear weapons when it invaded Kuwait in 1990, the US-led force that eventually liberated Kuwait would never have been assembled. Indeed, it is quite possible that today Saddam Hussein could still be alive, in office, and more affluent and powerful than ever. This, in turn, ultimately flows from the fact that nuclear weapons remain the most powerful weapons yet devised by humankind.

Warfare as practised by the United States and other militarily advanced states, the kind of combat that often is discussed in relation to the arguably ongoing Revolution in Military Affairs, or RMA (see [chapter 2](#)), places states with less technologically advanced equipment and less well-trained personnel in an unenviable position. As numerous conflicts, particularly the 1991 Gulf War, have demonstrated, a mediocre conventional force using conventional tactics is no match for a first-rate military, even if the latter is smaller. Under current conditions, and regardless of whether one is fighting at land, on sea or in the air, quality of personnel and equipment generally trumps quantity. If, however, nuclear weapons are brought into the equation, the situation can be radically altered, and even an otherwise weak force may be able, at least temporarily, to enjoy a critical advantage over a far more powerful enemy.

Nuclear conflict: future possibilities

Unfortunately, while concerns over a US–Soviet confrontation in Europe are safely relegated to the past, the occurrence of a limited nuclear conflict somewhere in the world remains entirely possible. Some of the many possibilities include:

- an Indian–Pakistani nuclear exchange
- an Israeli–Iranian nuclear war (assuming, of course, that Tehran obtains nuclear weapons)

- North Korean nuclear use against South Korea and/or Japan, which probably would result in American nuclear retaliation against North Korea.

Any of these events would result in an enormous humanitarian disaster, although, again, the exact circumstances of a conflict would play a key role in determining the extent of the damage.

Most importantly, there would be the issue of what is targeted. It is possible, as previously noted, that nuclear weapons might be used – probably out of desperation – in a strictly battlefield role, and perhaps it would be possible to limit a conflict to tactical nuclear use. Alternatively, however, it is possible that a state may use its nuclear arsenal in a much different manner, perhaps even undertaking a first strike intended to maximise civilian casualties. Such an action would not fit the vision of the escalation ladder, which presumes that intentional attacks on civilian populations would be near the ladder's top – and even strikes on military and other government facilities that happened to be located in or near cities would be very high on the ladder, and thus most probably avoided. However, the ladder – and similar Cold War deterrence and war-fighting concepts – is built on the assumption that foes are reasonably similar in what they value and fear. However, as an increasingly diverse group of states acquires nuclear weapons, there is no guarantee that this will continue to be the case (see [Box 17.4](#)).

Fortunately, at least for now, the danger of a civilisationally devastating nuclear war appears to have receded greatly. However, the possibility in a given year of a nuclear war occurring *somewhere* in the world is much higher than it was during most of the Cold War. Some of the reasons for this are obvious, such as the fact that more states possess nuclear weapons and, presumably, the more nuclear-armed actors there are, the greater the likelihood that one of them will actually decide to use its weapons. However, there also are more subtle reasons: the actors involved may be different in critical ways.

Ultimately, what we know about the avoidance of nuclear conflict is based on the behaviour of a small sample of actors over, historically speaking, a rather brief time. We do not have a sufficiently large body of historical evidence to make highly trustworthy predictions about the future use, or non-use, of nuclear weapons. With that in mind, however, it does appear that horizontal proliferation is a continuing trend. While there was considerable optimism immediately after the Cold War that a 'nuclear-free' world was possible, most observers now treat this as, at best, a distant hope. There is substantial reason to believe that the nuclear club will continue to increase in numbers over time, although the rapidity with which it will grow is uncertain.

Given this uncertain, and possibly quite hazardous, international security environment, ballistic missile defence will likely become progressively more important in the future. When SDI was announced, its would-be architects were faced with the daunting task of discovering how to build a highly reliable defensive shield that would destroy the great majority of many thousands of

Box 17.4 Different leaders, different goals

Leaders may vary enormously in what they most value, what risks they are willing to take and other attitudes relevant to WMD use. Even leaders within the same broad political tradition have diverse priorities and beliefs – Bill Clinton and George W. Bush are both recent US presidents, but they are quite distinct personalities. The differences between two leaders from very dissimilar political and strategic cultures – for example, a US president and a North Korean despot or Iranian supreme leader – may be so great that it is difficult for either to comprehend accurately what motivates the other. The possibility for misunderstanding, missed signals and other communication breakdowns is high when two such leaders are interacting.

This is especially the case when leaders engage in ‘mirror-imaging’. That term, which came into common use during the Cold War, relates to the tendency to assume that another individual shares one’s own general value hierarchy, reasons in a similar fashion and is otherwise quite similar to oneself. Western diplomats and other leaders often, and arguably quite justifiably, were criticised for tending to mirror-image Soviet leaders, paying insufficient attention to the specifics of Marxist-Leninist ideology and the Russian-Soviet historical experience and culture. Of course, there was no US–Soviet nuclear conflict, so, presumably, deterrence ‘worked’ sufficiently well overall. However, there is no guarantee that it will continue to work in the future.

Leaders – like any other human beings – ultimately decide what they value, and those choices will not always appear reasonable and responsible to outside observers. One can think of many historical examples of states, even great powers, whose leaders had unsavoury or bizarre goals – Hitler’s Germany is an obvious twentieth-century example, but Stalin’s Soviet Union and Mao’s China also both acted murderously towards perceived internal enemies and pursued various objectives that were in line with state ideology but that an outside observer might have judged imprudent or abnormal. If great powers can act unpredictably, one can readily imagine that North Korea – with its paranoid secrecy, murky internal power struggles and bizarre state ideology – might act in a surprising fashion.

nuclear warheads. Even with today’s far more advanced computer-processing power and other technological advances, this would be an extremely difficult and expensive task. However, against the small arsenals of most of the current and likely future nuclear states, missile defence might prove quite feasible, and missile defences can enormously complicate nuclear war planning. When a state launches a nuclear attack on a country with BMD, the attacker cannot know which of its warheads will break through the defence and land on target; when the attacking country has only a small number of nuclear weapons, they might all be intercepted by the defence. Given these potential defensive advantages, spread of missile defences to an increasing number of states may be expected in the coming years.

Of course, the possibility of nuclear conflict does not preclude the possible use of chemical and biological weapons (CBWs). It is likely that we will again

see the use of CBWs by states that are not nuclear-armed. Indeed, it is even possible that the use of CBWs may be the catalyst for the next use of nuclear weapons, as a nuclear-armed state could decide to retaliate for CBW-use with nuclear weapons. Some nuclear-armed states implicitly or explicitly maintain the option to retaliate in this fashion.¹⁵

Overall, given continuing horizontal proliferation and the military usefulness of nuclear weapons, it is likely that the norm prohibiting nuclear use will be broken at some point in the Second Nuclear Age. Although some authors believe that there genuinely is a strong taboo that will prevent nuclear weapons from being used again in warfare,¹⁶ it is notable that since the invention of these devices no nuclear power has been confronted with a situation in which *not* using nuclear weapons would have immediate and catastrophic consequences. For instance, no nuclear-armed state has chosen to allow its military to be utterly destroyed in conventional battle, much less permitted conquest by a foreign invader, rather than resort to the use of nuclear weapons. Such extreme situations would provide real tests of whether the norm against nuclear use truly is strong enough to be described as a taboo. Moreover, as discussed above, there is no guarantee – especially given the continuing horizontal proliferation of nuclear weapons – that all future leaders of nuclear-armed powers will have a very high threshold for nuclear use.

The occurrence of one nuclear conflict would not ensure that there would be many such wars. However, the possibility of a progressive ‘normalisation’ of nuclear weapons usage should not be dismissed casually.¹⁷ Perceptions concerning the military usability of nuclear weapons are shaped continuously by the behaviour of the actors possessing such weapons; if a number of state and non-state entities prove willing to use nuclear weapons, it may be that the high ‘wall’ separating conventional weapons from nuclear weapons slowly will break down, leading to an increasing willingness to use the latter in combat.

Conclusion: nuclear weapons in the present and future

At present, nuclear weapons are ‘dormant’ in regard to actual war-fighting, having remained unused since the Second World War. They do, however, play an important role in shaping the international security environment, and their existence, and particularly the possibility of nuclear proliferation, drives many political events. For example, the nuclear programmes of North Korea and Iran have created two of the most significant ongoing diplomatic predicaments facing the United Nations Security Council (UNSC), amongst other bodies.

With regard to nuclear weapons, the period from the end of the Cold War to the present might reasonably be described as a transitional stage – the early period of the Second Nuclear Age, whose characteristics are becoming clearer over time. While there were numerous states that came to possess nuclear weapons, the First Nuclear Age was defined by an icy standoff between two heavily armed superstates, with humanity’s fate resting on the decisions made by

the two opponents. The Second Nuclear Age that we have entered, by contrast, surely will not have such a simple, straightforward narrative. Rather, an assortment of relationships amongst various states that possess nuclear weapons – the number of which is, of course, growing over time – will be significant. Most probably, various scenarios will unfold in different parts of the world. In some cases, countries will use nuclear weapons to menace their less well-armed neighbours, with a regionally destabilising effect; however, other states that obtain nuclear weapons will use these devices in a more defensively orientated manner. Sometimes, one or more great powers – whether or not acting under the aegis of the UN – may choose to use force to prevent a specific instance of proliferation; on other occasions, proliferation will proceed without meaningful interference by outsiders. Some of the states that today are at odds with each other may find that mutual possession of devastating weapons ‘cools the passions,’ and their leaders will find it prudent to seek compromise and the diminishment of tensions rather than military victory. Other states might not be so accommodating, careering down a path leading to a nuclear war.

RESEARCH QUESTIONS

1. How strong is the norm against nuclear weapons amongst states? Could it survive the severe ‘weight’ of, for example, a devastating CB terrorist attack on a major American or British city, even if intelligence indicated strongly that the CBs in question had been provided to the terrorists by a state actor?
2. In what ways are the ‘First’ and ‘Second’ Nuclear Ages similar? In what respects are they distinct?
3. Is there a realistic way to ‘roll back’ the horizontal proliferation of nuclear weapons even to countries, such as North Korea and Iran, that appear to desire them desperately?
4. Was the invention and vertical proliferation of nuclear weapons the key reason why the Cold War did not become a Third World War?
5. Do the advantages accompanying membership in the nuclear ‘club’ still outweigh the disadvantages for France and Great Britain?
6. Would the continued proliferation of BMD make nuclear weapons potentially less useful as a strategic instrument, especially for states with small arsenals?
7. Would further reduction in the nuclear stockpiles of states belonging to the ‘NPT Five’ significantly enhance the global security environment?

FURTHER READING

Barnaby, Frank, *How to Build a Nuclear Bomb: And Other Weapons of Mass Destruction* (New York: Nation Books, 2004). Explains in accessible language the steps that a state or terrorist group must take if it is to build nuclear weapons.

- Cirincione, Joseph, Jon B. Wolfsthal and Mirium Rajkumar, *Deadly Arsenals: Nuclear, Biological, and Chemical Threats*, 2nd rev. and expanded edn (Washington, DC: Carnegie Endowment for International Peace, 2007). Addresses many proliferation-related issues and includes detailed discussion of numerous states and their arsenals.
- Croddy, Eric A. and James J. Wirtz (eds.), *Weapons of Mass Destruction: An Encyclopedia of Worldwide Policy, Technology, and History* (Santa Barbara, CA: ABC-CLIO, 2004). Contains information on a variety of topics and issues related to nuclear armaments and other WMDs.
- Freedman, Lawrence, *The Evolution of Nuclear Strategy*, 3rd edn (New York: Palgrave MacMillan, 2003). Offers an excellent, detailed discussion of how the nuclear strategy of various states has developed over time.
- Freedman, Lawrence, *Deterrence* (London: Polity, 2004). An examination of the evolution of the concept of deterrence.
- Gray, Colin S., *The Second Nuclear Age* (Boulder, CO: Lynne Rienner, 1999). Addresses the character of the Second Nuclear Age, including the possibility that WMDs will be used in warfare.
- Langewiesche, William, *The Atomic Bazaar: The Rise of the Nuclear Poor* (New York: Farrar, Straus and Giroux, 2007). Discusses many of the key issues related to nuclear proliferation, particularly the possibility that terrorist groups will obtain nuclear weapons.
- Sagan, Scott D. (ed.), *Inside Nuclear South Asia* (Stanford University Press, 2009). An examination of the factors driving nuclear acquisition by Pakistan and India and the military doctrine of the two powers as it relates to nuclear weapons.
- Sagan, Scott D. and Kenneth N. Waltz, *The Spread of Nuclear Weapons: A Debate Renewed*, 2nd edn (New York: W. W. Norton, 2002). Two scholars of international relations debate what the likely impact of the spread of nuclear weapons will be on the global security environment.
- Walton, C. Dale, 'Navigating the Second Nuclear Age: Proliferation and Deterrence in this Century', *Global Dialogue* 8(1) (Winter/Spring 2006), 22–31 (www.worlddialogue.org). Briefly discusses horizontal proliferation in this century and how Western states can best address the dangers presented by this phenomenon.

ONLINE RESOURCES

- <http://fas.org/>: The Federation of American Scientists (FAS) is an American group opposed to nuclear weapons and other WMDs. Its website contains information on WMDs of all kinds.
- www.acq.osd.mil/ncbdp/nm/nm_book_5_11/index.htm: The Nuclear Matters Handbook Expanded Edition was put together by the US Department of Defense as a resource for US government professionals working on issues related to nuclear weapons and power.
- www.nrdc.org/nuclear/: The US-based National Resources Defense Council (NRDC), which opposes both nuclear arms and energy, maintains this website, which includes a variety of reports and other materials.
- www.nuclearfiles.org/: The Nuclear Age Peace Foundation, which favours global nuclear disarmament, maintains this website, which includes a timeline of historical developments related to nuclear weapons and other resources.

www.rusi.org/globalsecurity/nuclear/: The Royal United Services Institute (RUSI) is a British think-tank whose website includes regularly updated analysis of events relating to nuclear weapons and their proliferation.

NOTES

1. Bernard Brodie, *The Absolute Weapon: Atomic Power and World Order* (New York: Harcourt, Brace, 1946).
2. See Leslie R. Groves, *Now It Can Be Told: The Story of the Manhattan Project* (New York: Da Capo Press, 1983 [1962]); Robert S. Norris, *Racing for the Bomb: General Leslie R. Groves, The Manhattan Project's Indispensable Man* (South Royalton, VT: Steerforth Press, 2002); Richard Rhodes, *The Making of the Atomic Bomb* (New York: Simon and Schuster, 1987).
3. For a history of the US thermonuclear weapon programme, see Richard Rhodes, *Dark Sun: The Making of the Hydrogen Bomb* (New York: Simon and Schuster, 1995).
4. For an example of the fear of 'loose nukes' displayed by some observers during the early post-Soviet period, see A. M. Rosenthal, 'Only a Matter of Time', *New York Times* online edn (www.nytimes.com), 22 November 1996.
5. Jeffrey Goldberg and Marc Ambinder, 'The Ally From Hell', *The Atlantic* online edn (www.theatlantic.com), 18 October 2011.
6. See Paul K. Kerr and Mary Beth Nikiton, *Pakistan's Nuclear Weapons: Proliferation and Security Issues*, Congressional Research Service Report for Congress # RL34248, 19 March 2013 (<http://fpc.state.gov/documents/organization/134991.pdf>).
7. On nuclear deterrence, see Stephen J. Cimbala, *The Past and Future of Nuclear Deterrence* (Westport, CT: Praeger, 1998); Stephen J. Cimbala (ed.), *Deterrence and Nuclear Proliferation in the Twenty-first Century* (Westport, CT: Praeger, 2000); Herman Kahn, *On Thermonuclear War*, new edn (Edison, NJ: Transaction, 2007 [1960]); Herman Kahn, *On Escalation: Metaphors and Scenarios* (Westport, CT: Greenwood Press, 1986 [1965]); Thomas C. Shelling, *Arms and Influence* (New Haven, CT: Yale University Press, 1966).
8. Morgan aptly describes compellence as 'the use of threats to manipulate the behavior of others so they stop doing something unwanted or do something they were not previously doing'. Patrick M. Morgan, *Deterrence Now* (Cambridge University Press, 2003), p. 2.
9. The concept of coercion – particularly non-nuclear coercion – is addressed in detail in Robert A. Pape, *Bombing to Win: Air Power and Coercion in War* (Ithaca, NY: Cornell University Press, 1996).
10. On the concept of credibility, see Daryl G. Press, *Calculating Credibility: How Leaders Assess Military Threats* (Ithaca, NY: Cornell University Press, 2005).
11. Quoted in David M. Kunsman and Douglas B. Lawson, *A Primer on U.S. Strategic Nuclear Policy*, Sandia Report SAND2001-0053 (Albuquerque, NM: Sandia National Laboratories, January 2001), p. 99 (<http://prod.sandia.gov/techlib/access-control.cgi/2001/010053.pdf>).
12. On Kahn's life and career, see Sharon Ghamari-Tabrizi, *The Worlds of Herman Kahn: The Intuitive Science of Thermonuclear War* (Cambridge, MA: Harvard University Press, 2005).

13. See Paul B. Bracken, *The Second Nuclear Age: Strategy, Danger, and the New Power Politics* (New York: Times Books, 2012); Colin S. Gray, *The Second Nuclear Age* (Boulder, CO: Lynne Rienner, 1999); Keith B. Payne, *Deterrence in the Second Nuclear Age* (Lexington, KY: University Press of Kentucky, 1996).
14. Khan's activities and proliferation network are discussed in depth in Gordon Corera, *Shopping for Bombs: Nuclear Proliferation, Global Insecurity, and the Rise and Fall of the A. Q. Khan Network* (New York: Oxford University Press, 2006); William Langewiesche, *The Atomic Bazaar: The Rise of the Nuclear Poor* (New York: Farrar, Straus and Giroux, 2007).
15. The United States long maintained an ambiguous policy regarding nuclear retaliation for CB attack, but the Obama administration formally rejected this option in 2010, except in regard to states that are not party to the NPT or are in violation of its provisions. US Department of Defense, *Nuclear Posture Review*, April 2010, pp. xii–xiii (www.defense.gov/npr/docs/2010%20Nuclear%20Posture%20Review%20Report.pdf).
16. See John Mueller, *Atomic Obsession: Nuclear Alarmism from Hiroshima to Al-Qaeda* (New York: Oxford University Press, 2009).
17. The possible future use of all forms of WMD is examined in Peter R. Lavoy, Scott D. Sagan and James J. Wirtz (eds.), *Planning the Unthinkable: How New Powers Will Use Nuclear, Biological, and Chemical Weapons* (Ithaca, NY: Cornell University Press, 2000).





Part VII

Conclusion

Christopher Tuck



18 The future of warfare

Contents

The future of war

Future conflict

The conflict environment

Warfare and technology

Future developments

The singularity

Technological challenges

The limits of technology

The operational level of war and operational art

Operational art

The operational level

Joint warfare

The challenges of jointery

What is victory?

'Modern' victory

Conclusion

What is the future of warfare? The preceding chapters of this book have highlighted the contentious nature of modern warfare: central themes and concepts have often remained open to debate; elements of change in the theory and practice of warfare have been accompanied by countervailing aspects of continuity. Developing these themes, we conclude this book by asking the broad question: 'in what direction is modern warfare travelling?' This is a question of fundamental importance. Ideally, we need to prepare now for the sort of warfare that will confront us in future years. But, compounding the contentious nature of the debates discussed in the main body of this volume, one of the recurring *leit-motifs* in any examination of the future is uncertainty. The strategist Colin S. Gray puts this bluntly: 'we know nothing, literally zero, for certain about the wars of the future, even in the near-term'.¹ We will structure this discussion of the future by focusing first on the general issue of the future of war and then on the debates surrounding four key areas that might shape the nature of that future: the impact of technology; joint warfare; the operational level of war and operational art; and finally the vexed question 'what is victory?'

The future of war

We begin with the question ‘what will future war look like?’ It is, of course, the largest of understatement to assert that predicting the future of war is difficult. Indeed, as the chapters of this book have demonstrated, militaries have historically often had to fight wars very different from those they had anticipated. Predicting the future is simply a problematic activity: how far forward must one predict? Who will one be fighting? How will they fight? Will new means and methods be available? What will be government policy? Answering these questions is complicated by the problems of strategic intelligence-gathering, the difficulties in using historical analogies, and the fallibilities associated with human decision-making such as groupthink and organisational culture. But governments and their militaries cannot ruminate indefinitely: eventually they must make decisions about what the future will look like in order to inform defence policy.

Future conflict

There is no shortage of Western, especially US, analysis on the issue of future war. Shaped by a combination of interventionist foreign policies, geopolitical change (especially the rise of China), technological developments, fiscal retrenchment and the consequences of the wars in Iraq and Afghanistan, Western militaries have evinced a particular interest in identifying the future character of war. Western analyses tend to be pessimistic. The US, for example, sees a future global environment driven by the influence of two key factors: globalisation and the information revolution. US analyses argue that globalisation will continue to intensify the creation of worldwide financial, political, societal and cultural networks. The information revolution, reflected in such phenomena as the continuing growth in social media and the spread of mobile technology, will give people unparalleled access to information. While these factors may have many positive consequences they are also likely to interact negatively with such factors as political extremism, climate change, resource shortages, state failure, regional conflicts, the proliferation of weapons and advanced technology, the empowerment of criminal and/or terrorist non-state actors and the rise of strategic competitors.²

Collectively, these features promise a future marked by unprecedented global connectedness, rapid and unpredictable change and growing complexity. ‘Armed conflicts will be inevitable in such an environment’, argues US military doctrine.³ In consequence of the fluidity and dynamic nature of this change, the US military acknowledges that: ‘One of the great problems that confronts American strategists and military planners is the conundrum of preparing for wars that remain uncertain as to their form, location, level of commitment, the contribution of potential allies, and the nature of the enemy.’⁴ Increasingly, Western militaries have focused on the need for military planning that is ‘capabilities-based’ rather than ‘threat-based’: in other words, rather than

designing military forces to meet a specific adversary, armed forces will need to be developed on the basis of the general capabilities required for the future.

The conflict environment

From a Western military perspective, this uncertain vision of the future has a number of consequences for the character of the coming conflict environment. The likely consequences are believed to include:

- the creation of adaptable new enemies through the empowerment of such non-state actors as militias, insurgents and terrorists, and the proliferation of advanced technology
- the erosion of Western technological superiority over future adversaries through the spread of precision-guided munitions, ballistic missile technology and anti-access/area denial capabilities
- continuing intense media scrutiny of military operations and the rapid dissemination of the human consequences of conflict
- the reshaping of civil–military relations thanks to further developments in command-and-control technology that allows strategic level decision-makers to influence tactical activities
- new ethical difficulties as the result, for example, of the problems of determining who is or is not a combatant
- the importance of the internet and mass media, and the consequent importance of winning the ‘battle of the narratives’ – the competing struggle to project ideas and influence
- the need to understand the future battlespace as something more than a battlefield; the need to understand it, instead, in terms of an environment shaped by PMESII-PT factors (political, military, economic, social, information, infrastructure, physical environment and time)
- the growing importance of the cyberspace and space environments – while each of these will become an increasingly important tool of national power, society’s reliance on them will create new and compelling vulnerabilities.⁵

For example, speculating on the character of future armed conflict, the Australian Army has identified a future conflict environment marked by five overlapping and converging features or ‘meta trends’ (for an alternative, see [Box 18.1](#)) that will define the conflict environment:⁶

- **Crowded:** Increasing urbanisation will be compounded by increasingly busy sea lanes and air space. The electro-magnetic and information environments will also become choked with traffic. Military operations will focus increasingly on littoral environments, and urban operations may become a necessity. Capabilities to disrupt the space and information spheres will proliferate.
- **Connected:** Developments in telecommunications will lead to the growing interconnectedness of political, social, economic and communications networks. Non-state and semi-state actors will be empowered. Militarily,

Box 18.1 The future operating environment

British thinking focuses on a future of growing instability caused by state failure, strategic competition, struggles for access to resources, climate change, the effects of globalisation and extremist political ideologies. The operating environment for military forces will be characterised by five attributes:

- Congested:** The battlespace is likely to be densely populated by civilians and/or by the vessels and aircraft of neutral or friendly powers.
- Cluttered:** Adversaries will be able to exploit the environment to conceal themselves, exploiting better local knowledge and the constraints imposed on us by the need for legitimacy and discrimination.
- Contested:** Exploiting the proliferation in sophisticated technology, enemies will be able to broaden the dimensions of conflict, exploiting the political, social, cyber and information spheres.
- Connected:** Future conflict will focus increasingly on the key nodes that connect together the various geographic, logistic, technological and communication networks.
- Constrained:** Legal and moral requirements will demand greater discrimination in the use of force.¹

¹ *Strategic Trends Programme: Future Character of Conflict* (Development, Concepts and Doctrine Centre, Shrivenham, February 2010), pp. 1–6, 21–5.

the land, sea, air, cyber and space environments will become increasingly interlinked: jointery, effective information management and interagency co-operation will become even more critical; dependency on space-based assets will grow as will potential space and cyber threats.

- **Lethal:** Non-state actors will be able increasingly to obtain and use new military and dual-use sensors and munitions to inflict, or threaten to inflict, more damage at greater ranges. Reducing the physical and political risk in such environments may require a greater focus on such countermeasures as hardening and dispersal. It is likely to mean a greater use of increasing automation: robotics, artificial intelligence, unmanned vehicles and autonomous systems.
- **Collective:** The trend since the end of the Cold War towards co-operative activity will continue. Coalition and joint and interagency military operations will become increasingly central features. Interoperability across services and agencies and between allies will become increasingly necessary.
- **Constrained:** Financial and recruitment constraints are likely to persist; political, social, legal and ethical considerations are also likely to predominate given the sensitivity of domestic and international audiences to questions surrounding the legitimacy of the use of armed force. Militaries will need to consider further ways to use technology to substitute for manpower; consideration will need to be given to force regeneration. Issues relating to discrimination and proportionality in targeting will become even more significant.

Box 18.2 The sources of future peace

In his book *The Better Angels of Our Nature* Steven Pinker sees a more peaceful future built on the effects of six key developments:

- the emergence of the state system, which has centralised and ordered the use of violence
- the continued growth in international commerce with its positive-sum logic
- the 'feminisation' of society's values, in which the traditional prioritising of such masculine ideals as honour and aggression are less prevalent
- increasing 'cosmopolitanism' through the growth in literacy, mobility and the mass media
- the 'escalator of reason': the increased application of knowledge and rationality to human affairs.¹

Thus the general historical trend has been, and will continue to be, a reduction in the incidence of war because these factors change the rational logic of violence, changing the physical, political but also, crucially, the psychological costs and benefits of armed conflict.

¹ Steven Pinker, *The Better Angels of Our Nature* (London: Penguin, 2011).

But whether warfare is likely to be a more or less prominent feature of the international system is still a matter of debate. For some, war is a fact of the international system. Mankind's long history of resorting to warfare, combined with the slew of high-profile contemporary conflicts such as those in Syria, Ukraine and Iraq, seems to establish a firm foundation upon which to predict a future marked by armed conflict. War, argues the contemporary strategist Colin S. Gray, is 'a condition of political life'.⁷ But it is important also to note that, in the view of some other commentators, war is likely to decline in the future, making warfare a less relevant feature of the international system. The political scientist Steven Pinker concludes that, despite perceptions that the world has become increasingly violent, the empirical evidence suggests exactly the reverse. Trends – including the sustained historical reduction in rates of homicide, the relative lack of great-power and even state-on-state warfare since 1945 and, since 1989, the decline in organised conflicts throughout the world – suggest a more peaceful future (see [Box 18.2](#)). Statistical analysis of conflict over time has been used by others to identify a likely decline in the number of states embroiled in armed conflict from 14 per cent at the present to only 8 per cent by 2050 with the strongest reduction in the Middle East and North Africa.⁸

Moreover, we cannot assume that the Western experience in Iraq and Afghanistan, or Western thinking on the future threat environment in general, reflects the totality of the debate. Some states share some of the United States' concerns about the future. Russia, for example, also sees the future as one marked by great dynamism, shaped by the forces of globalisation and interdependence. There are similarities also in the character of the threats that it sees as emerging from this – threats of terrorism, weapons proliferation,

migration, demographics, environmental struggles and the increasing stratification between rich and poor. But some states are more optimistic. China, for example, sees a complex but different future, its latest Defence White Paper arguing that 'Countries are increasingly bound together in a community of shared destiny. Peace, development, cooperation and mutual benefit have become an irresistible tide of the times.'⁹

Moreover, even where other states might see a future marked by rising challenges, those challenges do not necessarily align with those of the United States or its European allies. Instead of a focus on the threats posed by failing states and intra-state conflict, Japan, for example, is focused on a range of future state-on-state difficulties, including quite traditional disputes over sovereignty and territory – with Russia, and, especially, China.¹⁰ For both Russia and China, the United States itself is a major source of future challenges. Where China sees threats, they are from hegemonic interventions, an implicit identification of perceived threats posed by an activist US foreign policy. Russia also identifies concerns relating to 'a number of leading foreign countries', for which we can read the United States and its allies, in relation to the consequences of interventionism, military competition and the aggressive expansion of alliances (in essence, NATO).¹¹ But for other political actors, while the international situation may be dynamic, there is a strong continuity in the nature of the expected challenges of the future. India, for example, sees major threats in terms of themes extant well before the end of the Cold War – external threats from China over disputed borders; internal threats from insurgent groups; and closely linked internal and external challenges from Pakistan and insurgent groups with links to Pakistan, such as Lashkar-e-Taiba. Future developments, such as the continued growth in China's military power, or the prospects post-2014 for renewed Indo-Pakistan competition in Afghanistan might change the character of the challenges for India, but not their origins.¹²

And of course, the characteristics of future conflict may depend to a considerable degree on context: that context may be different for the Islamic State organisation in Iraq, Russian separatists in Ukraine, Bashar al-Assad in Syria or China in Xinjiang.

Warfare and technology

In debates about the future of war, technology often looms large. One of the recurring themes in the chapters of this book has been the idea that warfare is not a decisively technological activity. Technology is not an independent variable: war has a technological dimension, but success in war is as much about *how* technology is used *in relation* to an enemy and in the *context* of such factors as strategy, logistics and generalship as it is about weapons and equipment per se. History is littered with states that have lost battles and wars against technologically inferior opponents. But will this continue to be the case in the future?

For some commentators, we are moving swiftly in human affairs towards a new watershed that is fundamentally technological in origin. The root of this shift lies in the fruits of exponential technological development: while technology has always developed over time, we seem now to be part of an age in which that change is occurring at unprecedented and increasing rates. For example, in the decade of the 1990s there was more change in technology than in the whole of the ninety preceding years. This development is reflected, for example, in 'Moore's Law': Gordon Moore, co-founder of Intel Corporation, observed in 1965 that the number of transistors on a micro-chip was doubling every two years. This process has actually speeded up, with the doubling occurring now every eighteen months. This pace of change is also evident in a wide array of other technologies: for example, wireless capacity is also doubling every eighteen months.¹³ Technological change can usefully be conceptualised in terms of 'sustaining' or 'disrupting' developments, the former enhancing existing ways of doing things, the latter undermining them.¹⁴ What is significant about current technological change is not just its exponential quality but also the disruptive nature of its implications. Developments in such spheres as nano-technology (working with matter at the nano-scale of one-billionth of a metre), bioscience, robotics, artificial intelligence (AI), computing and information science have established trajectories which might have profound implications for the conduct of warfare.

Certainly, many states regard investment in future technology as a key way to sustain a qualitative advantage in future armed conflicts. For example, the US 2014 *Quadrennial Defense Review* highlights the significance to US military superiority of the need to maintain an advantage not just in the techniques of war but also in the technology as a key means of ensuring that its armed forces, even if they become smaller, can maintain or even enhance their effectiveness.¹⁵ But the United States is hardly unique in this regard. The Indian Prime Minister Narendra Modi has argued that 'we need to further speed up our defence programme and keep pace with the world'.¹⁶ China's latest defence white paper also argues for the need to 'strengthen the development of new- and high-technology weaponry and equipment'.¹⁷

Future developments

That technology has become a recurrent element in discussions on the future of warfare is unsurprising given the kinds of trends that have already begun or which seem feasible in principle. Key areas of military technological development include:

- **Enhanced human performance:** These include technological and/or biological enhancements for military personnel, making them faster, stronger, more resilient and more networked – enhanced armour made of liquids or lightweight nano-material, for example; or powered exo-skeletons (Human

Universal Load Carriers, HLUCs); biological markers to help track, tag and locate friendly and enemy forces; or contact lenses that might transmit data directly into a soldier's retinas.

- **Directed energy weapons:** These offer the prospect of more accurate and rapid firepower, at longer distances and with virtually limitless ammunition. These weapons might include weaponry harnessing millimetre-wave technology, microwaves, lasers, electro-magnetic pulses, including lasers and electro-magnetic rail guns.
- **New power systems:** These offer a potential revolution in logistics, mobility and sustainability. Possible developments include wearable power systems, high-energy batteries, air-independent propulsion and Electrolytic Cat-ion Exchange Module (E-CEM) to produce fuel from seawater.
- **Robotic Systems:** With unmanned aerial, ground and maritime systems (UAVs, UGVs and UUVs respectively) already deployed, advances in nano-technology and AI may promise increasingly sophisticated and autonomous robotic systems: bio-mimetic robots, for example, that mimic the attributes of insects, snakes or dogs; weapons, reconnaissance or logistics platforms that can be controlled as groups, rather than as individual platforms, or that may be able to perform some tasks without human supervision.
- **Cyber capabilities:** New software and hardware, the latter including advanced cellular base-stations, mesh networking and new data extraction and processing technologies promise more secure, reliable and enormously more capable networks. Devolved problem-solving and development becomes even easier with these technologies.
- **New manufacturing techniques:** 'Additive', or 3-D printing, techniques using a variety of materials including plastics, metals and concrete could allow personnel to build parts in situ with local resources, making military logistics cheaper, tailored and more flexible. This could benefit disaster-relief activities and promote more rapid tactical adaptation of military equipment in the field of kit. 'Additive' manufacture could be used to make drones, warheads and even human skin for medical purposes.

The singularity

Many see these current trends in technology as leading inevitably, perhaps even within the next two decades, to 'the technological singularity': the advent of genuine artificial intelligence. The result of this might be a momentous paradigm shift in human affairs that would have huge consequences for warfare. What this new paradigm might look like unsurprisingly is the subject of much debate. Just as with the works of such writers as H. G. Wells and Jules Verne at the turn of the twentieth century, visions of the impact of technology have been shaped in the contemporary period amongst other things by popular culture: films, literature, digital media, video games. These have helped shape competing utopian

and dystopian visions of a future machine age; of ubiquitous household robots, for example, or lethal Terminators, or Matrix-style virtual reality.

If, for example, developments in AI play out as some believe, then the possibility of developing artificial consciousness is a very real one. Combined with further developments in mechanical and electrical engineering, and computer science, the potential military implications of a shift from remotely operated to genuinely complex autonomous robotic systems is huge. The importance of unmanned platforms has been evident for many years. President George W. Bush asserted in 2001 that: 'We're entering an era in which unmanned vehicles of all kinds will take on greater importance – in space, on land, in the air, and at sea.'¹⁸ In consequence, the United States already deploys a wide range of unmanned systems including Predator, SWORD and PackBot.¹⁹ Autonomous weapons are already being deployed, with 'fire and forget' missiles, such as the Brimstone missile, able itself to choose between targets.²⁰ Autonomous systems may mitigate even further such problems as the rising cost of military personnel, difficulties in recruitment and the political costs of human casualties. At the same time, without human physical and emotional limitations, autonomous robots may offer a leap forward in battlefield effectiveness.

Equally, developments in the cyber domain may elevate cyber warfare from one dimension of conflict to *the* key dimension: a shift from a focus on 'information-in-warfare' to 'information-as-warfare'.²¹ For some, cyberwar will represent a twenty-first-century form of blitzkrieg: belligerents will be able to dislocate or destroy their opponent's information and communication systems, finding new ways of collapsing enemy military systems and new non-military methods of attacking an opponent's critical nodes. The use, possibly by Israel and the United States, of the Stuxnet computer 'worm' in 2009–10 to attack Iran's nuclear programme has provided one possible marker for the future, as does Russia's use against Georgia in 2008 of computer attack in concert with conventional military operations.

Technological challenges

If the singularity occurs then, by definition, predictions regarding its consequences become difficult because all of the rules of the warfare 'game' will change. But whether or not the singularity comes to pass, new technology may pose a wide array of new challenges:

- Will new forms of weaponry privilege the attack over the defence?
- Will increases in weapon lethality result in a return to the need for more mass in warfare?
- How might traditional concepts of deterrence and compellence function in an environment of realisable cyberwar?
- To what extent might counter-technologies emerge that might constrain the revolutionary implications of new technology?

- Will the proliferation of the knowledge and means to create new technologies and pathogens benefit non-state actors more than states?
- What are the ethical difficulties surrounding the use of increasingly autonomous military systems?
- What implications might robotic and cybernetic systems have for such traditional building blocks of military cohesion as camaraderie and service ethos?
- If warfare becomes increasingly 'non-corporeal', then are we in danger of it becoming game-like: a situation in which we are cognisant of our actions but not of the costs and consequences?
- What kind of personnel might be required to fight future war? Will their skills, outlook and values need to change radically? Will future militaries become 'civilianised'?

Beyond such uncertainties, however, this is another school of thought that argues that this technologically focused perspective on the future is mistaken.

The limits of technology

Critics argue that we have a tendency to be seduced by the exciting possibilities inherent in advanced technology. History, however, suggests a number of correctives. First, technology operates in a context. As the strategist Michael Handel notes: 'technology is not a panacea that exists in a political vacuum, and when treated as such . . . it cannot succeed'.²² This wider context embodies limits on technology imposed by ethics, politics and organisational factors: the US use of drone strikes in Pakistan, for example, has proved to be hugely controversial politically, whatever the benefits militarily. Secondly, there are many technological problems that have still to be overcome. In relation to AI, for example, the nature of warfare, with its friction and uncertainty, poses huge challenges for the development of autonomous systems. One defence analyst has commented that: 'The autonomy thing is f'ing hard. All the little decisions build up, especially in a chaotic situation like war.'²³ With regards to nano-technology, the possibility of deploying huge numbers of smaller systems poses enormous problems for command and control.²⁴ Indeed, the latter is but one reflection of the more general 'human bandwidth' problem – the growing differential between the complexity of humans and the complexity of the environment around us – that we have yet to resolve. It may be the case that new technology might act to reduce military flexibility because it may require greater human specialisation in order to manage it; or that human limitations will cap our ability to leverage the theoretical capabilities of machines. Thirdly, warfare is still likely to generate scenarios that place a premium on human–human interaction. In counterinsurgency, stabilisation, military engagement activities – anywhere where the goal is to influence rather than simply to destroy – having human 'boots on the ground' is likely to remain crucial.

Enamoured with the possibilities posed by developments in tank technology, the British soldier and strategist J. F. C. Fuller in 1928 argued for the growing

irrelevance of infantry and mass armies, commenting that: 'Tools, or weapons, if only the right ones can be discovered, form 99 per cent of victory.'²⁵ Fuller was wrong then, and the extent to which his statement may become more accurate in the future is still subject to debate.

The operational level of war and operational art

In addition to debates about the impact of future technology, there are also serious debates about the continued relevance to the future of certain key contemporary doctrinal concepts, not least operational art and the operational level of war. These concepts remain central to contemporary Western military doctrine. The concept of the operational level as the vital connection between the strategic and tactical levels is well entrenched. General Martin Dempsey, in 2014 the US Chairman of the Joint Chiefs of Staff, noted the importance for the military to 'simultaneously think ahead at the strategic level, stay current at the operational level, and be informed by tactical level developments.'²⁶ But the concepts of the operational level of war and of operational art have spread beyond Western militaries: the Indian Army, for example, began from the late 1980s informally to recognise an operational level of command, sitting at the corps level.²⁷ Even the Chinese had, by the late 1990s, engaged in a much more focused examination of the specific requirements of fighting at the level of the campaign: what in the West would be associated with the operational level.²⁸

That the operational level and operational art are likely to be of growing doctrinal importance in the future does not mean, however, that they are uncontroversial.

Operational art

For the most part, critics of operational art do not focus on the value of the concept itself; rather, they argue that Western militaries have over-complicated operational art, underplaying the difficulties in implementing it, and overstating its effectiveness as a force multiplier. For example, while many Western commentators have seen Soviet successes in the latter part of the Second World War as stemming from their growing virtuosity in operational art, others have been more prosaic in their assessment of the sources of German defeat. One Soviet general commented in the 1970s that: 'the Soviets did not win the great patriotic war because Soviet generalship and fighting skills were superior to those of the Germans. The Soviet armed forces simply overwhelmed the Germans with superior numbers.'²⁹

Looking forward, however, one particular point of contention is the relevance of operational art in unconventional conflicts. As [chapter 6](#) has noted, there are some who argue that future armed conflict is likely to be marked by brutal, low-intensity struggles along the 'new wars' model. Can current doctrinal conceptions of operational art fit into unconventional counterinsurgency or

stabilisation scenarios in which there are likely to be multiple actors, many of them paramilitary or non-military; in which the context is likely to be highly political in character; and in which domestic and international scrutiny is likely to invest many tactical decision with strategic consequences? As [chapter 5](#) has noted, the concept of operational art was born from developments in conventional warfare and was associated with such factors as the growing scale of warfare and a focus on restoring mobility. While there remains a need in unconventional operations to co-ordinate the activities of tactical actions to ensure that strategic-level goals are achieved, does this really equate to operational art in the way that it was conceived historically? By trying to assert that operational art can be used in complex unconventional scenarios are we in danger, as some have argued, of simply ‘confusing operational art and purposeful action’?³⁰

The operational level

Important criticisms have also been levelled at the concept of the operational level of war. For some, technological trends may make it increasingly difficult to sustain the idea of a distinct level of war that exists between strategy and tactics. Here, the argument is that developments in networking and communications technology may make it increasingly easy for strategic-level decision-makers to interact directly with tactical-level commanders and vice versa. In such circumstances it may be that the operational level is transformed into a bureaucratic or managerial level, without real operational power; or it may be that it becomes difficult to sustain a specific operational level of command at all. How one might fit autonomous robotic systems into traditional concepts of the operational level may also be problematic.

But beyond the difficulties of making the operational level work, there are larger questions surrounding whether the concept might hinder, rather than help, the translation of strategic intent into tactical action. In particular, critics have argued that reserving for the operational level of war the functions of ‘planning and conducting campaigns’³¹ appropriates activities that have been traditionally performed by strategy. But why should this matter?

One danger is that operational-level commanders will interpret their role as one that requires autonomy from higher-level political interference. In essence the fear is that the operational level risks locking out the strategic level: creating in doctrine the notion that the role of strategic-level actors is simply to set objectives which the operational-level commander should then be free to achieve by the means that they see as appropriate, free from further political interference.³² That this danger is not just a theoretical one is evident from Iraq in 2003 when the coalition operational-level commander, General Tommy Franks, told the Secretary of Defense’s deputy at the Pentagon to ‘keep Washington focused on policy and strategy. *Leave me the hell alone to run the war.*’³³ Instead, critics argue that the planning and conduct of campaigns should be

activities that are shaped by the demands of strategy and policy. Indeed, the so-called 'compression of the levels of war', far from being a problem, might simply be a reflection of strategy trying to reassert its proper role. Far from connecting strategy and tactics, then, a formal doctrinal concept of the operational level of war may act to separate the military conduct of a campaign from its proper political context.

Looked at from another perspective, the concept of an operational level poses another potential problem. For this group of critics, the difficulty is that decision-makers at the strategic level may see the operational level of command as a substitute for proper strategy and strategic direction. In circumstances where there is no clear strategy or policy, the operational level may be forced to act as a 'strategic alibi' for the higher level of war, providing the objectives and strategy that have not been articulated effectively from above.³⁴ What a formal operational level of war may do, therefore, is to mask the dangerous lack of effective strategy-making, forcing the operational-level commander to 'fill in the blanks' with all the attendant problems that this may create for the coherence and relevance of the military campaign itself.

Thus, while the concept of a distinct operational level of war is well-entrenched in Western military doctrine going forward, some argue that it will hinder rather than help effective military activity either because it will discourage in the future strategic political involvement in the planning and execution of military campaigns, or because it will allow higher-level decision-makers to absolve themselves of the need to develop coherent over-arching strategy and policy – or both.

Joint warfare

A recurrent theme in this book has been the growing importance since 1900 of joint warfare. The significance of this mode of warfare was recognised in 1946 by General Dwight D. Eisenhower, who asserted that '[s]eparate ground, sea, and air warfare is gone forever. If we ever again should be involved in war, we will fight with all elements, with all services, as one single concentrated effort'.³⁵

If anything, joint warfare has become even more important. For the United States and its allies, the campaigns in Iraq and Afghanistan have forced the armed forces into better joint co-operation. Non-Western countries have also come to focus on the importance of joint warfare as a foundation for military effectiveness. China notes the need in its latest defence White Paper for forces that are 'lean, joint, multi-functional and efficient'. Moreover, the Indian Chief of Staff of the Army noted in 2000 that 'The Armed Forces must be fully joint: doctrinally, institutionally organisationally, intellectually and technically'.³⁶

This widespread importance attached to joint warfare reflects the demands of modern military effectiveness, especially in relation to operational art, networking, modularity and functional integration:

- **Operational art:** Operational level campaigns are unlikely ever to require only one of the services. Skill at joint warfare is thus one of the prerequisites for effective operational art.³⁷ For example, joint command structures can prevent unnecessary duplication of effort and create a more coherent campaign.
- **Networking:** The advantages of networking can only really be fully realised if the 'system of systems' extends across, and links together, the different services. Maintaining rigid single-service structures and doctrine builds in formidable obstacles to the fluidity, responsiveness and rapidity of action that networking should deliver.
- **Modularity:** The shift towards decentralising the structures and decision-making within the separate services places a premium on joint approaches in order to impose unity of effort on what might otherwise become a host of disparate sub-elements.
- **Functional integration:** Jointery has been facilitated by the growth in tasks that can only be performed effectively with inter-service co-operation, including air-lift, close air support, reconnaissance, special forces operations, anti-submarine operations, air defence and so on.³⁸

The importance of jointery has been reflected in the move towards the creation of joint structures and an increasing volume of joint doctrine, and is evident in the use in the United States of the term 'the Joint Force' collectively to describe the armed forces. In the United States, for example, the Goldwater-Nichols Act of 1986 had already ended operational control of single-service military forces by their own service chiefs. But the United States has also set about turning its regional combatant commands (RCC) into effective substantive joint headquarters organisations. Individually, the US armed services have also made improvements: for example, the US Army has increased the joint element in its tactical training regime. Conceptually, the significance of joint warfare has been reflected in the US capstone concept of globally integrated operations (GIO), defined as 'the integration of emerging capabilities – particularly special operations forces, cyber, and intelligence, surveillance and reconnaissance (ISR) – with new ways of fighting and partnering'.³⁹ The GIO concept will require a focus on such features as partnering (developing better capabilities to work across different government agencies and with other countries), flexibility (in terms of organising joint forces around specific security challenges rather than along functional or geographic boundaries) and the use of more lower-signature capabilities (such as cyber and special forces) (see [Box 18.3](#)).

Individual services have also continued to develop their joint warfare concepts. The US Army, for example, has developed a doctrine of unified action based on the absolute need to work inclusively with other US government agencies and with other partners, viewing the battlefield as 'a human-based network that is beyond a military-only ability to understand, visualize, and influence'.⁴⁰

Box 18.3 The key elements of globally integrated operations

- Mission command
- Seizing, retaining and exploiting the initiative
- Global agility
- Partnering
- Flexibility in establishing Joint Forces
- Cross-domain synergy
- Use of flexible, low-signature capabilities
- Increasing discrimination to minimise unintended consequences.¹

¹ *Capstone Concept for Joint Operations: Joint Force 2020* (Joint Chiefs of Staff, 10 September 2012), p. 4.

The challenges of jointery

The challenge for the future is that, despite the manifest importance of joint warfare, modern militaries have continued to struggle with its implementation. History demonstrates that joint warfare skills are often ephemeral. Thus the US effectiveness in joint warfare reached something of a peak in the Second World War, with the extraordinary achievements of the Normandy and Pacific campaigns; but these skills were quickly lost once the war ended, leading to a mixed performance in a joint context in the Korean War (1950–3) and a lamentable performance in Vietnam (1965–73). Thus, despite the declaratory commitment to jointery there remains a range of issues that are likely to challenge the implementation of jointery in practice.

The foundation problem in joint warfare is the impact of organisational factors: service-specific identities, allegiances, procedures, values, mindsets, definitions of interest and visions of the future that create fundamental differences in outlook between armies, navies and air forces. The implications of the power of single-service identity include: a tendency to define problems and solutions from a single-service perspective, the defence of vested service interests, the protection of single-service ‘boundaries’ and an intense sense of ‘them-and-us’.⁴¹ These organisational factors are impossible to eliminate because (a) they are inherent to all forms of human organisation, and (b) they are often essential to military effectiveness in terms of promoting within units a sense of identity, belonging, cohesion and commitment.

Another obstacle to effective jointery is the question of resources, especially where budgets are being squeezed. In theory, one might think that austerity would promote greater jointery because of the need for greater efficiency. If this were the case then the global financial crisis of 2007–8, coupled with the end of the war in Afghanistan in 2014, should act as motors for an even wider

application of jointery. But history suggests that peace can be problematic for sustaining joint approaches to warfare, especially since the end of wars is often associated with heavy cuts to defence spending. As one US admiral commented: 'In peacetime I could just never, never bring myself to do anything to help the Air Force or Army.'⁴² In financially constrained environments services often tend to prioritise core single-service capabilities and training at the expense of those for joint warfare since the latter are often viewed as a second order commitment.

Particular operational contexts have also been problematic for joint operations. Operations in Afghanistan from 2001 and Iraq from 2003 onwards seemed to involve increasingly complex scenarios in which jointery had to be practised in complex multinational and multi-agency environments.

Two final points also explain the problematic nature of joint operations. First, jointery is a dependent variable: it depends for its success on other factors such as an effective articulation of strategic-level objectives. Thus, one of the problems for British jointery after 1945 was the lack of a long-term policy that would inform joint concepts. Secondly, the preceding problems are mutually reinforcing. For example, where resources are very constrained, struggles between the services over the division of defence budgets can further politicise the identities of armies, navies and air forces.

In addition to these problems, the implementation of jointery is also influenced by a paradox of effectiveness: too zealous a focus on jointery may result in some circumstances in reduced effectiveness. One legitimate fear is that rigid joint structures can create inflexibility and over-centralisation; this was the fear of Winston Churchill who preferred a committee system for joint command rather than formal military structures.⁴³ More fundamentally, too rigid an insistence on the application of jointery can erode the single-service expertise and cohesion upon which effective joint operations depend. Distinct service cultures facilitate mastery of the demands of each separate environment: without this mastery, joint operations are built on foundations of sand.

The problem with joint warfare is that, as one author notes 'jointness is as much a state of mind as a method of prosecuting war'.⁴⁴ For this reason, institutional or structural innovations to promote jointery can have only a limited practical effect unless the services themselves 'buy in' to joint warfare enthusiastically. Historically, joint warfare concepts and techniques honed in one conflict have quickly been forgotten in the aftermath. A key question for the future, then, is whether the current moves to embrace greater jointery will succeed in overcoming this historical pattern of gradual decay in peacetime.

What is victory?

Finally, we consider an issue of fundamental future importance: success in war. Speculating upon the nature of future warfare, leveraging technology, enhancing our understanding of operational art and the operational level and

developing our capacity for joint warfare: these are enterprises all directed ultimately towards trying to improve our ability to win wars. But recent experiences have seemed to demonstrate that even where military victory has been forthcoming, overall success in war can be elusive.

The US General Douglas MacArthur commented famously in 1951 that: 'War's very object is victory, not prolonged indecision. In war there is no substitute for victory.'⁴⁵ A protest against the close political constraints placed on his operations, MacArthur's comments reflected a very traditional view of the relationship between military activity and political victory. From this perspective, military success and political victory were the same things: military victory, achieved through the destruction of the enemy's fielded forces, would deliver strategic political success and victory overall. What followed from this was that the metrics for success in war were clear: progress could be measured through such tangibles as casualties and ground taken. In 1944–5, for example, Allied progress towards the defeat of Germany could be measured by the destruction of German units, the rate of Allied advance and the shrinking distance to Berlin.

That this view of success in war has never been fully adequate is evident from [chapters 13–15](#) of this book, which have noted the great problems caused in counterinsurgency campaigns by the ruthless pursuit of narrowly military objectives. The need to prioritise such activities as winning the hearts and minds of the local population gave counterinsurgency crucial political, economic and social dimensions that could not be serviced through military success alone. Moreover, as MacArthur found out, the advent of nuclear weapons had focused Cold War rivalries on more limited means and ways to prosecute armed conflict and also more limited objectives. The problems associated with conceptualising victory in a traditional way were also evident in the emergence after the Second World War of the concept of peacekeeping, in which military forces were deployed under the auspices of the United Nations for tasks designed not to win military victory but instead to prevent or ameliorate armed conflict. Over time, peacekeeping developed into the broader concept of peace operations (often also called peace support operations, or PSO) in which military forces were utilised for the goals of conflict prevention, conflict management, peace-building and conflict resolution through the conduct of such tasks as monitoring ceasefires; the demobilising, disarmament and reintegration of warring factions (DDR); security sector reform (SSR); infrastructure development; and maintaining order.

'Modern' victory

The trend towards allocating to military forces more complex political tasks continued in the first decade of the twenty-first century with the emergence of the concept of stabilisation. Stabilisation had its origins in a number of factors:

Box 18.4 Stabilisation defined

'[V]arious military missions, tasks and activities conducted outside the United States in co-ordination with other instruments of national power to maintain or re-establish a safe or secure environment, provide essential government services, emergency infrastructure reconstruction, and humanitarian relief'.¹

¹ Joint Publication 3-0, *Joint Doctrine*, 11 August 2011, p. V-5.

- **Threat assessments:** In the 1990s there was increasing concern over the threat posed by failing or failed states, especially with regards to their negative effects on regional neighbours. After 9/11, this threat seemed magnified because of the way in which emerging 'ungoverned spaces' could become base areas for terrorist groups.
- **Normative change:** Particularly in democratic states, popular sentiment focused increasingly on the issue of the legitimacy of military operations. Domestic and international concern over the moral, ethical and legal context in which war was fought made it increasingly important for many states that the objectives and conduct of warfare were shaped by consideration for the short- and long-term welfare of non-combatants.
- **Failure:** In Somalia in 1992–5, Haiti in 1994 and Bosnia in 1994–5, Western forces often found themselves floundering when confronted by the need to conduct complex nation-building tasks in concert with military operations. These difficulties were further exposed during the extended campaigns in Iraq (2003–11) and Afghanistan (2001–14).

The impact of these factors was to shape the emergence of doctrines of stabilisation: a concept that became associated with the military's role in wider nation-building (see [Box 18.4](#)).

The consequences of these developments were potentially profound. In complex nation-building scenarios traditional definitions of victory made little sense: as early military victories in Afghanistan and Iraq demonstrated in 2001 and 2003 respectively, the comprehensive military defeat of the enemy in no way guaranteed the creation of a stable post-war political order.

For many, then, victory in warfare needed to be redefined. As US doctrine argued, in contemporary and future conflict scenarios 'the margin of victory will be measured in far different terms from the wars of our past'.⁴⁶ A wider concept of 'modern' victory was needed to replace 'pre-modern' definitions that were too short term and too narrowly military. An alternative is to reconceptualise military victory as 'strategic success': a concept of victory that encompasses 'inter-related informational, military, political, economic, social, and diplomatic elements'⁴⁷ (see [Box 18.5](#)), a multidimensional concept that puts a priority on non-military indices of success and measures performance against

Box 18.5 Strategic success

	Pre-modern total war success	Modern limited war success
Informational dimension	Coerce formal surrender	Influence cessation of hostilities
Military dimension	Destroy and subjugate enemy	Neutralise and deter enemy
Political dimension	Impose foreign-dominated government	Allow self-determined government
Economic dimension	Exploit economic resources	Develop economic reconstruction
Social dimension	Enforce hierarchical social order	Promote progressive social transformation
Diplomatic dimension	Accomplish war aims on one's own	Invite third-party conflict involvement

The problem of modern victory: strategic success requires shifting objectives in a broad range of dimensions from the absolute results on the left to more ambiguous results on the right. But how much success is required in each category; and is progress in one dimension always compatible with progress in another?¹

¹ Robert Mandel, *The Meaning of Military Victory* (Boulder, CO: Lynne Rienner, 2006), p. 16.

long-term political criteria. Rational though this development seems to be, does it promise in the future a more effective approach to achieving overall success in war?

The evidence suggests that clear problems are likely to remain. Attempts to reconceptualise victory were evident in the campaign in Afghanistan from 2001, in which explicit reference was made to the inadequacy of such traditional metrics of success as body counts and ground taken and the need instead to measure progress against other criteria such as protecting the population, creating effective governance, socio-economic development and building up the Afghan security forces. But what Afghanistan also demonstrated is that, while such concepts as 'modern victory' and 'strategic success' may seem more relevant to the sorts of goals pursued in many contemporary conflicts, wider definitions of victory have often proved in practice problematic to implement. The manifold difficulties include:

- **Metrics:** There is no consensus on the sorts of metrics that we should use to measure progress towards the achievement of these broader concepts of victory, with advocates arguing for the need to focus on everything from the number of government officials who serve and live in a district (as an indicator of the strength of local government) through to relative changes in school attendance or the mood of the local population as ascertained by polling them.⁴⁸

- **Attainability:** Many critics have argued that ambitious nation-building objectives are simply difficult or impossible to achieve. Rejecting the 'planning school' approach that sees success as a matter of developing the right doctrine and applying the necessary resources, these critics argue that the liberal assumptions that underpin nation-building objectives have led external interventions to try and impose alien political, economic and social models on indigenous societies.
- **Compatibility:** Multidimensional definitions of victory are difficult to implement. Since the dimensions are interrelated, it is likely that a minimum level of success in each may be necessary for victory. But it may well be the case that success in one dimension appears incompatible with success in others. For example, military objectives may be achieved only at the expense of political objectives or vice versa.
- **Suitability:** Are military organisations, with their war-fighting ethos, properly suited to achieving the highly political, non-war-fighting requirements of modern victory? Evidence would seem to suggest that, even in complex stabilisation environments, organisational imperatives will often tend to drive military organisations back to a focus on body counts and the execution of major military operations.⁴⁹

In the future conduct of warfare, then, we are left with a crucial unresolved problem: while we can recognise that traditional concepts of military victory might be inadequate, the alternatives are contested and often difficult in practice to implement.

Conclusion

As this brief survey of key issues for the future demonstrates, the challenges associated with modern warfare are unlikely to dissipate. Predicting the future is simply too difficult to do. In general, there is a tendency to take familiar elements from the present and to extrapolate them forwards; as one commentator has noted: 'The future we envision can only be an extrapolation of present trends taken to a logical and therefore often illogical conclusion.'⁵⁰ Irrespective of the confidence often displayed by futurologists, we really know very little, in any objective sense, about the future of war. For this reason, notes the strategist Colin Gray, the future is unlikely to be the one that we expect: 'Surprise is not merely possible, or even probable, it is certain.'⁵¹

But another reason is that the practical, organisational and conceptual tools that we have at our disposal to meet this uncertain future are themselves contested. How will technology affect military capabilities? Will operational art and the operational level provide an appropriate framework for fighting in future conflicts? Can joint warfare escape from the deeply ingrained historical patterns of decay? Can we deliver in practice on more sophisticated conceptual approaches to military victory? As we have seen in this final chapter, all of

these issues are subject to debate. As the chapters of this book collectively have demonstrated, looking at current and past military operations seems to show that modern warfare comprises powerful strands of interlocking continuity and change, but the consequences of this interaction are often difficult to foresee and challenging to adapt to.

RESEARCH QUESTIONS

1. What will be the likely features of the future conflict environment?
2. Which technologies are likely to have the most impact on future warfare, and why?
3. Are we better placed now to overcome the historical challenges of joint warfare?
4. How useful is the concept of an operational level of war in the modern strategic environment?
5. Will the term 'military victory' have any meaning in the future?

FURTHER READING

- Angstrom, Jan and Isabelle Duyvesteyn (eds.), *Understanding Victory and Defeat in Contemporary War* (Abingdon, Routledge, 2007). This volume provides a thought-provoking collection of chapters on the difficulties associated with defining and achieving military success in the contemporary era.
- Arquilla John and David Ronfeldt, 'Cyber War Is Coming', in John Arquilla and David Ronfeldt (eds.), *In Athena's Camp* (Santa Monica: RAND, 1997). This is a seminal piece that argues for the inevitable rise of cyber warfare.
- Cocker, Christopher, *Warrior Geeks: How 21st-century Technology Is Changing the Way We Fight and Think about War* (London: Hurst, 2013). A tour-de-force examination of the human consequences of new military technology.
- Rid, Thomas, *Cyber War Will Not Take Place* (London: Hurst, 2013). As the title suggests, this is a more sceptical piece on the prospective emergence of cyber warfare.
- Singer, P. W., *Wired for War: The Robotics Revolution and Conflict in the 21st Century* (New York, NY: Penguin, 2010). Singer argues for the revolutionary impact on warfare of future developments in robotic technology.

ONLINE RESOURCES

Interesting assessments of the future threat environment can be found by looking at the following documents:

- The US *Capstone Concept for Joint Operations: Joint Force 2020*: www.dtic.mil/doctrine/concepts/ccjo_jointforce2020.pdf.
- The British Ministry of Defence's future *Global Strategic Trends* paper: www.gov.uk/government/publications/global-strategic-trends-out-to-2045.

The US National Intelligence Council's *Global trends 2030: Alternative Worlds* publication: <http://globaltrends2030.files.wordpress.com/2012/11/global-trends-2030-november2012.pdf>.

China's 2015 Defence White Paper, *China's Military Strategy*: <http://eng.mod.gov.cn/Database/WhitePapers/>.

Russia's 2009 *National Security Strategy to 2020*: <http://rustrans.wikidot.com/russia-s-national-security-strategy-to-2020>.

A wide selection of national security strategies, and defence White Papers from around the world is available here: www.isn.ethz.ch/Digital-Library/Publications/Series/Detail/?lng=en&id=154839.

For a discussion of the origins and problems of the operational level of war, see Justin Kelly and Mike Brennan, *Alien: How Operational Art Devoured Strategy*, Strategic Studies Institute (2009): <http://www.strategicstudiesinstitute.army.mil/pdffiles/pub939.pdf>.

For an analysis of the complexities inherent in modern definitions of victory, see J. Boone Bartholomees, *Theory of Victory*, Parameters (Summer 2008): <http://strategicstudiesinstitute.army.mil/pubs/parameters/Articles/08summer/bartholo.htm>.

NOTES

1. Colin S. Gray, 'War: Continuity in Change, and Change in Continuity', *Parameters* (Summer 2010), 5.
2. *Quadrennial Defense Review 2014* (Washington, DC: Department of Defense, 2014), pp. 3–9.
3. *Capstone Concept for Joint Operations: Joint Force 2020* (Joint Chiefs of Staff, 10 September 2012), p. 2.
4. *The Joint Operating Environment 2010*, US Joint Forces Command (February 2010), p. 62.
5. See, for example, *Capstone Concept for Joint Operations: Joint Force 2020* (Joint Chiefs of Staff, 10 September 2012), p. 2; Army Doctrine Publication 3-0, *Unified Land Operations*, Headquarters, Department of the Army (10 October 2011), p. 2; *Insights and Best Practices: Joint Operations*, 4th edn (Joint Training Staff, March 2013), p. 5; *The Joint Operating Environment 2010*, US Joint Forces Command (February 2010), pp. 34–7.
6. *The Future Land Warfare Report 2013* (Canberra: Australian Army Headquarters, May 2013), pp. 5–14.
7. Colin S. Gray, *Another Bloody Century: Future Warfare* (London: Weidenfeld & Nicolson, 2005), p. 340.
8. See Håvard Hegre *et al.*, 'Predicting Armed Conflict, 2010–2050', *International Studies Quarterly*, 57(2) (2012), 250–70.
9. China's 2015 Defence White Paper, *China's Military Strategy* (<http://eng.mod.gov.cn/Database/WhitePapers/>).
10. Japanese Ministry of Defense Annual White Paper, *Defense of Japan 2014*, Part I (www.mod.go.jp/e/publ/w_paper/pdf/2014/DOJ2014_Digest_part1_1st_0730.pdf).

11. Russia's 2009 *National Security Strategy to 2020* (<http://archive.kremlin.ru/eng/text/docs/2008/07/204750.shtml>).
12. See, for example, Major General Afsir Karim, 'India's Security Concerns', *Indian Defence Review*, 4 May 2014 (www.indiandefencereview.com/news/indias-security-concerns/).
13. P. W. Singer, *Wired for War: The Robotics Revolution and Conflict in the 21st Century* (New York, NY: Penguin, 2010), pp. 96–101.
14. Shawn Brimley, Ben FitzGerald and Kelley Sayler, *Game Changers: Disruptive Technology and U.S. Defense Strategy* (Disruptive Defense Papers, Center for a New American Security, September 2012), p. 11.
15. *Quadrennial Defense Review 2014* (Washington, DC: Department of Defense, 2014), p. I.
16. 'Modi Calls for Faster Development of Defense Technology', *India Real Time*, 21 August 2014 (<http://blogs.wsj.com/indiarealtime/2014/08/21/indias-modi-calls-for-faster-development-of-defense-technology/>).
17. 'The Diversified Employment of China's Armed Forces' (http://news.xinhuanet.com/english/china/2013-04/16/c_132312681_2.htm).
18. 'US President George W. Bush Addresses the Corps of Cadets', 11 December 2001 (www3.citadel.edu/pao/addresses/presbush01.html).
19. Special Weapons Observation Reconnaissance Detection is a system that allows the remote firing of small-arms weaponry. PackBot is a light back-pack portable system that can be used for such functions as surveillance and bomb disposal.
20. John Markoff, 'Fearing Bombs That Can Pick Who to Kill', *New York Times*, 11 November 2011 (www.nytimes.com/2014/11/12/science/weapons-directed-by-robots-not-humans-raise-ethical-questions.html?_r=0).
21. Christopher Cocker, *Warrior Geeks: How 21st-Century Technology Is Changing the Way We Fight and Think about War* (London: Hurst, 2013), p. 116.
22. Michael I. Handel, *Masters of War: Classical Strategic Thought* (London: Frank Cass, 1996), p. 16.
23. Quoted in P. W. Singer, *Wired for War: The Robotics Revolution and Conflict in the 21st Century* (New York, NY: Penguin, 2010), p. 75.
24. *Nanotechnology: The Future Is Coming to You Sooner than You Think* (Joint Economic Committee, United States Congress, March 2007), pp. 10–11.
25. Col. J. F. C. Fuller, *On Future Warfare* (London: Sifton Praed, 1928), pp. 90–2, 153.
26. Joint Publication 1 (JP1), *Doctrine for the Armed Forces of the United States* (25 March 2013), Forward.
27. G. D. Bakshi, 'Operational Art in the Indian Context: An Open Sources Analysis', *Strategic Analysis*, 25(6) (2001), 723–36.
28. See James Mulvenon and David Finkelstein (eds.), *China's Revolution in Doctrine Affairs: Emerging Trends in the Operational Art of the Chinese People's Liberation Army* (RAND, 2002), pp. 10–18 (www.defensegroupinc.com/cira/pdf/doctrinebook.pdf).
29. Jacob W. Kipp, 'Tsarist and Soviet Operational Art', in John Andreas Olsen and Martin van Creveld (eds.), *The Evolution of Operational Art: From Napoleon to the Present* (Oxford University Press, 2011), p. 88.
30. Justin Kelly and Mike Brennan, 'The Leavenworth Heresy and Perversion of Operational Art', *Joint Forces Quarterly*, 56 (1st Quarter, 2010), 116.

31. Field Manual 100-5, *Operations*, 1982, pp. 2–3.
32. Justin Kelly and Mike Brennan, *Alien: How Operational Art Devoured Strategy* (Strategic Studies Institute, 2009), p. 67.
33. Hew Strachan, 'Strategy or Alibi: Obama, McChrystal, and the Operational Level of War', *Survival*, 52(5) (October–November 2010), 166.
34. *Ibid.*, 168.
35. Stuart Griffin, *Joint Operations: A Short History* (London: Training Specialist Services HQ, 2005), p. 7.
36. Quoted in Vinod Anand, 'The Evolution of a Joint Doctrine for Indian Armed Forces', *Strategic Analysis*, 24(4) (July 2000) (www.idsa-india.org/an-jul-600.html).
37. Griffin, *Joint Operations*, p. 10.
38. Roger A. Beaumont, *Joint Military Operations: A Short History* (Westport, CT: Greenwood, 2003), p. 186.
39. *Capstone Concept for Joint Operations: Joint Force 2020*, p. 4.
40. *Insights and Best Practices: Joint Operations*, 4th edn (Joint Training Staff, March 2013), pp. 1–3.
41. Beaumont, *Joint Military Operations*, p. xv.
42. *Ibid.*, p. 190.
43. *Ibid.*, p. 193.
44. Griffin, *Joint Operations*, p. 7.
45. Quoted in Richard H. Rovere and Arthur Schlesinger, Jr, *General MacArthur and President Truman: The Struggle for Control of American Foreign Policy* (New Brunswick, NJ: Transaction, 1992), p. 276.
46. Field Manual 3-07, *Stability Operations*, October 2008, p. vi.
47. Robert Mandel, *The Meaning of Military Victory* (Boulder, CO: Lynne Rienner, 2006), p. 16.
48. Antony Adoff, 'Measuring Success in Afghanistan', 18 March 2010 (<http://news.change.org/stories/measuring-success-in-afghanistan>).
49. Anthony King, 'Understanding the Helmand Campaign: British Military Operations in Afghanistan', *International Affairs*, 86(2) (2010), 325.
50. Christopher Cocker, *Warrior Geeks: How 21st-century Technology Is Changing the Way We Fight and Think about War* (London: Hurst, 2013), p. xix.
51. Gray, *Another Bloody Century*, p. 25.



GLOSSARY

4GW	<i>Fourth-generation war.</i>
A2/AD	Anti-access and area denial: technologies and techniques designed to deny access and freedom of manoeuvre to adversaries within particular areas.
ABM	Anti-ballistic missile: a missile designed to counter ballistic missiles.
Air-Sea Battle	A US concept designed to allow joint forces to overcome challenges posed by enemy A2/AD capabilities.
Air superiority	The ability to use air power at a given time and in a given place without prohibitive interference from the enemy.
Air supremacy	<i>Air superiority</i> in which the enemy is incapable of effective interference.
Aircraft carrier	A warship designed to carry and operate numerous fixed-wing aircraft at sea.
Amphibious operation	The landing of military forces from the sea against a hostile or potentially hostile shore.
ASAT	Anti-satellite: weapons designed to destroy or disable satellite systems.
Asymmetric warfare	Warfare between dissimilar adversaries. A contested concept, asymmetric warfare is often associated with conflict against such unconventional adversaries as terrorists and insurgents.
Attrition	The reduction of the effectiveness of a military formation caused by the loss of equipment and/or personnel.
Ballistic missile	Missiles that follow a ballistic flight path (as opposed to flying, as does a <i>cruise missile</i>).
Battalion	A military unit normally consisting of 500 to 1,200 troops divided into four to six <i>companies</i> . Two or more battalions may be grouped into a <i>brigade</i> , or one or more may constitute a <i>regiment</i> .
Battlefleet	A fleet, usually composed of the most powerful warships, designed to contest <i>command of the sea</i> in combat with the enemy.
Battlegroup	A flexible <i>combined-arms</i> force usually created around the nucleus of a <i>battalion/regiment</i> -sized infantry or armoured unit and augmented with other arms.
Battleship	A large, heavily armoured warship armed primarily with large calibre guns. Until the mid-twentieth century these were the largest and most powerful warships.

Battlespace dominance	A US concept that encompasses the ability to dominate the three-dimensional battlespace by establishing zones of superiority around deployed forces.
BCT	Brigade Combat Team.
Blitzkrieg	Second World War German operational-level mobile warfare.
Blockade	An attempt to deny access to a particular area. Often associated with attempts to cut off maritime trade to enemy ports.
Blue water	A colloquial term used to refer to the <i>high seas</i> .
BMD	Ballistic missile defence.
Brigade	A military unit consisting of two to five <i>battalions</i> or <i>regiments</i> . Two or more brigades may be grouped into a <i>division</i> .
Brigade combat team	The basic large deployable manoeuvre unit of the US Army, consisting of a combat <i>brigade</i> with attached supporting units.
Brown water	A colloquial term used to refer to inshore sea areas, estuaries and deltas.
C2	An acronym to represent the military functions of <i>command</i> and <i>control</i> . The process by which a commander exercises <i>command</i> and <i>control</i> of their forces in order to direct, organise and co-ordinate their activities to maximum effect.
C3I	An acronym to represent the following military functions: command, control, communications and intelligence.
C4ISR	An acronym to represent the following military functions: command, control, communications, intelligence, surveillance and reconnaissance.
C4ISTAR	An acronym to represent the following military functions: command, control, communications, intelligence, surveillance, target acquisition and reconnaissance.
Campaigns	A connected series of <i>tactical</i> engagements within a <i>theatre</i> designed to achieve a <i>strategic</i> objective.
Capital ship	The most powerful and influential type of warship, often valued for its symbolic as much as its military value.
Centre of gravity	Characteristics, capabilities or locations that are key to the ability of an organisation to function effectively.
CIA	[US] Central Intelligence Agency.
Close air support	Air attacks launched against enemy targets in close proximity to friendly forces.
Close operations	Operations on or near the enemy front line, as opposed to <i>deep operations</i> , conducted far into the enemy's rear area.
COIN	Counterinsurgency operations. Military, paramilitary, political, economic, psychological and civic actions taken by a government to defeat insurgency.
Combined arms	The integration of different arms into a single system to achieve a complementary effect.

Combined operations	Traditionally, a term used to describe what would now be called <i>joint operations</i> . In modern nomenclature, the term is used to describe multinational operations.
Command	The authority to <i>control</i> the activity and organisation of armed forces.
Command of the sea	The ability to use the sea and to deny that use to the enemy.
Company	A military unit consisting of 100–200 troops, divided into two to five platoons.
Conflict prevention	Diplomatic and other actions taken in advance of a predictable crisis to prevent or limit violence, deter parties and reach an agreement short of conflict.
Control	The process by which a commander directs, organises and co-ordinates the activities of the forces under their <i>command</i> .
Corps	A military unit consisting of two or more <i>divisions</i> . Derived from the French term <i>corps d'armée</i> ('body of the army'), traditionally a corps was a <i>combined-arms</i> force capable of fighting independently for a limited period of time.
Corvette	A small warship, usually lightly armed.
Cruise missile	Guided-weapons designed to fly to their target on a non-ballistic trajectory, often at low altitude.
Cruiser	A powerful warship, less heavily armed and armoured than a battleship, often designed to conduct <i>scouting</i> and <i>screening</i> operations in support of the <i>battlefleet</i> or to protect/attack trade or project power overseas.
Culminating point	The point at which the power of an attack ceases to overmatch that of the defence.
Cyber warfare	The use of computers, the internet and other systems to attack an opponent's computers, websites and networks in order to cause harm.
Damage control	Controlling the damage caused to ships by enemy action in naval warfare.
DDR	Disarmament, demobilisation and reintegration of local armed groups conducted as part of <i>peace operations</i> or <i>stability operations</i> .
Deep operations	Large-scale, simultaneous attacks against enemy reserve and rear areas as well as the front line. Deep operations can be performed by remote firepower (air, artillery and missile attack) and by the rapid exploitation of breakthroughs by mobile forces.
Destroyer	A capable <i>flotilla</i> vessel designed to fulfil a range of missions that include anti-submarine operations, air defence and various escort duties, but is capable of operating in support of the battlefleet.

Distributed operations	A war-fighting concept where units are deliberately spread over a wide area and conduct separated but co-ordinated independent tactical actions.
Division	A large, usually <i>combined-arms</i> , military unit consisting of two or more <i>brigades</i> . Two or more divisions may be grouped together in a <i>corps</i> .
Doctrine	The principles that are codified and taught in order to guide the actions of military forces.
Double envelopment	Simultaneous movement around the flanks of the enemy designed to achieve, or to threaten to achieve, the encirclement of the enemy.
Dreadnought	British <i>battleship</i> launched in 1906 that, by its exclusive armament of heavy-calibre guns and new steam-turbine engines, revolutionised battleship construction, lending its name to subsequent vessels of this type.
Echelon	As a formation, describes the use of multiple reserve formations to create successive waves of attack. Generally, the term 'second echelon' is sometimes used to describe all of the forces that follow on from the initial attacking forces. The term 'rear echelon' refers generally to non-combat support elements.
EBAO	<i>Effects-based approach to operations.</i>
EBO	<i>Effects-based operations.</i>
Effects-based approach to operations	a 'way of thinking' about the relationship between military activity and political effect, and that prioritises the need for multi-agency approaches to operations. A less rigid form of <i>EBO</i> .
Effects-based operations	Rather than prioritising the destruction of enemy forces through the application of military power, EBO is an approach that focuses first on establishing the broad end-states that need to be delivered (the 'effect') and then choosing the most appropriate range of instruments to achieve it (of which military power may be only one).
Envelopment	Movement around or over the enemy.
Expeditionary operations	Operations where military force is projected and sustained at some distance from the home bases, usually across the seas.
Fire control	The control and direction of artillery fire, particularly from warships.
Fire team	The building block of modern tactics, the fire team is, in general, a four-man grouping of soldiers. Two or more will comprise a <i>squad</i> or section.
Fleet-in-being	The maintenance of one's own fleet intact in the hope that its continued existence will deny options to the enemy.
Flotilla	A formation of smaller warships, sometimes included as part of a larger fleet.

Fourth-generation war	A perspective that sees future warfare as an evolved form of networked insurgency.
Friction	Those factors that make the practice of warfare more difficult than its theory including fear, chance, uncertainty and physical exertion.
Frigate	A <i>flotilla</i> vessel, usually optimised to focus on either anti-submarine warfare or air defence.
Future Combat System	An ambitious programme between 2003 and 2009 to create a family of sophisticated, networked, manned and unmanned vehicles. These would form a key part of the US army's <i>transformation</i> agenda.
Global maritime partnership	A US Navy concept for formal and informal partnerships between navies in order to promote maritime security.
Global network of navies	An US Navy concept for both long-term and short-term co-operation between navies, particularly within the context of <i>maritime security operations</i> .
Globally integrated operations	US defence concept that aims to develop 'a globally postured Joint Force to quickly combine capabilities with itself and mission partners across domains, echelons, geographic boundaries, and organizational affiliations.'
Grand strategy	The application of national resources to achieve national/alliance policy objectives.
Green water	A colloquial term used to refer to coastal waters, archipelagos and sea areas above the continental shelf.
<i>Guerre de course</i>	An attack on enemy merchant shipping at sea.
Gunboat diplomacy	The use of naval forces in support of diplomatic activity. It is often associated with the coercive use of naval forces in situations short of armed conflict.
High seas	The area of the sea beyond <i>territorial seas</i> . Navies are free to navigate through the high seas without restriction.
Hybrid warfare	Warfare characterised by a fusion of regular and irregular techniques by the same organisation.
ICBM	Inter-continental ballistic missile
IDF	The Israeli Defence Force.
Innocent passage	The right provided for under <i>UNCLOS</i> for ships to transit through the territorial seas of other states providing they do not engage in military activity.
Insurgency	An organised movement aimed at the overthrow of a constituted government through the use of subversion and armed conflict. Like <i>terrorism</i> , however, insurgency is a contested concept. Its essence is the combination of irregular warfare techniques and political ideology.
Integration	The development of closer, complementary interaction between elements within a military system. Generically, integration is

	required for effective performance in all aspects of warfare whether <i>command and control</i> , <i>logistics</i> , <i>combined arms</i> or <i>joint operations</i> . The term has acquired specific importance in relation to contemporary debates regarding the creation of a <i>system of systems</i> .
Intelligence	In the military context this involves the collection and analysis of information in order to guide actions.
Interdiction	Actions to divert, disrupt or destroy the enemy before they reach the area of battle.
IRBM	Intermediate-range ballistic missile.
ISR	Intelligence, surveillance and reconnaissance.
ISTAR	Intelligence, surveillance, target acquisition and reconnaissance.
Joint operations	Operations involving the integration of land, air and/or naval forces.
Linear tactics	A tactical system based upon the use of troops deployed in line, designed to maximise firepower and minimise problems of command and control. Characteristic of warfare in the nineteenth and early twentieth centuries.
Littoral	The coastal region. Sometimes identified as coastal sea areas and the area of land that is susceptible to influence and support from the sea.
Logistics	The art of moving armed forces and keeping them supplied.
Manoeuvre	Movement in relation to the enemy to occupy advantageous positions.
Manoeuvre warfare	An approach to warfare developed in the 1980s and 1990s. Rather than focusing on <i>attrition</i> , manoeuvre warfare leverages <i>tempo</i> to achieve success. Surprise, deception, speed, mobility and the seizing of the initiative are used to confuse and dislocate the enemy and to invalidate their decision-making.
Manoeuvrist approach	A philosophy of war based on the principle of defeating the enemy by attacking such intangibles as cohesion and the will to fight rather than focusing on destroying the enemy's material. Despite its label, manoeuvre is not necessarily a prerequisite for a manoeuvrist approach to war.
Maritime security operations	Operations designed to enhance maritime security and good order at sea. These can range across a broad spectrum from the benign through to robust action short of war-fighting.
Maritime strategy	The principles that govern a war (or conflict) in which the sea is a substantial factor.
Mechanised forces	Mobile forces where the transport is usually composed of armoured, armed and (often) tracked vehicles. The transport is designed to contribute to combat operations by providing protection and fire support.

Military strategic level	The application of military resources to help grand strategic objectives.
Military transformation	Generically, 'major change' in the way warfare is conducted. More specifically, it refers to the contemporary idea of US military reform focusing on such ideas as NCW, modularisation and an <i>expeditionary</i> focus.
Mission command	A command philosophy based on the principles of decentralisation of responsibility and the use of initiative. Classically, subordinates are informed of what they must achieve (the mission) but are allowed to use their own judgement on how this should best be achieved given local conditions.
Motorised forces	Mobile forces where the transport is usually composed of wheeled and unarmoured vehicles.
Multidimensional peacekeeping	Sometimes known as 'new peacekeeping' or 'second-generation peacekeeping'. Complex operations with wider remits, multiple participants and complex intra-state contexts.
NATO	North Atlantic Treaty Organisation.
Naval diplomacy	The use of naval forces in support of diplomatic activity.
NCO	Non-commissioned officer such as a corporal, sergeant or warrant officer. A military officer without a commission (i.e. they are still enlisted soldiers). NCOs provide the crucial interface between commissioned officers and the men and women under their command.
NCW	Network-centric warfare. A concept in which intense networking creates a <i>system of systems</i> that will, in turn, deliver information superiority over an adversary.
NEC	Network-enabled capability. A less ambitious British version of NCW.
'New wars'	A perspective on contemporary and future war that argues that a paradigm shift has occurred in warfare reflected in such characteristics as new participants, new objectives, an increased blurring between war and criminality and the obsolescence of much of traditional military thinking.
NGO	Non-governmental organisation.
OODA loop	The Observe-Orient-Decide-Act (OODA) process believed by Col. John Boyd to underpin effective decision-making.
OOTW	Operations other than war.
Operational analysis	The employment of advanced analytical methods, and other problem-solving techniques, to make better decisions.
Operational art	The execution of the <i>operational level</i> of command.
Operational level	The level at which campaigns and major operations are planned, sequenced and directed.
Outflank	Movement around the enemy flanks. Although often used synonymously with the term <i>envelopment</i> , envelopment often

	implies a much deeper outflanking movement designed to pin and encircle the enemy.
Paradigm	A worldview, conceptual model or framework of thinking.
Peace-building	A long-term process designed to create a self-sustaining peace, to address the root causes of conflict and to prevent a slide back into war.
Peace enforcement	<i>Peace operations</i> that are coercive in nature and rely on the threat of the use of force.
Peacekeeping	A term sometimes used generically to mean <i>peace operations</i> , but more properly an operation undertaken with the consent of all major parties to a dispute, and one designed to monitor and facilitate implementation of an agreement to support diplomatic efforts to reach a long-term political settlement.
Peace-making	A diplomatic process aimed at establishing a cease fire or an otherwise peaceful settlement of a conflict that is already ongoing.
Peace operations	Crisis response and limited contingency operations to contain conflicts, redress the peace, and shape the environment to support reconciliation and rebuilding and to facilitate the transformation to legitimate governance. Sometimes also termed <i>peace support operations</i> .
Peace support operations	see <i>Peace operations</i> .
PGM	Precision-guided munitions such as laser-guided bombs and cruise missiles.
PLA	The Chinese People's Liberation Army: the Chinese armed forces.
PLAN	The Chinese People's Liberation Army Navy.
PMC	Private military contractor.
Policy	In relation to strategic theory, policy represents political objectives. Policy differs from <i>strategy</i> in that policy sets the ends to which strategy is directed.
Power projection	The ability to project military power overseas.
PSO	<i>Peace support operations</i> .
Radar	An acronym for 'radio detection and ranging'. The use of electromagnetic waves to detect the range, altitude, bearing and speed of moving objects such as aircraft, ships and vehicles and also the position of fixed installations.
Regiment	A tactical or administrative grouping of one or more battalions.
Revolution in Military Affairs	An RMA is a <i>paradigm</i> shift in war; 'an observable breaking point between two recognisably different types of warfare'. The RMA refers to a concept that emerged in the 1990s that future war would be revolutionised by combinations of <i>PGMs</i> , sensors and networking.
RMA	<i>Revolution in Military Affairs</i> .
Scouting	Activity designed to locate the enemy.

Screening	Activity designed to protect your forces from enemy observation and enemy action.
Sea basing	The basing of military capabilities, such as troops, fire support, logistics and command and control, on ships offshore.
Sea control	The ability to use an area of the sea for one's own purposes and to deny that use to the enemy for a given period of time.
Sea denial	The attempt to prevent an opponent from gaining use of the sea.
Sea lines of communication	Commercial shipping routes and sea routes connecting military forces to their sources of supply and reinforcement. Such lanes are notional and are determined by the movement of ships. Unlike roads on land they do not exist in any physical sense.
Shock	A psychological state marked by fear and disorientation. Shock is a desirable state to induce in an enemy because it may result in slowed, unco-ordinated and irrelevant activity. Translated through military organisations, 'systemic shock' undermines the capacity of a military force to operate effectively and may result in heavy defeat without the infliction of large amounts of physical destruction.
Sonar	An acronym for 'sound, navigation and ranging'. The use of sound waves to detect objects under water.
Space force application	The use of space for military purposes, employing weapons systems as opposed to reconnaissance and communications platforms. Defined by US Strategic Command as encompassing 'combat operations in, through and from space to influence the course or outcome of events'.
Squad	A small infantry unit comprising two or more <i>fire teams</i> . In some armies the squad is known as a section. Two or more squads will be combined to form platoons.
Squadron	A small formation of aircraft, ships or mobile ground forces such as cavalry or tanks.
SSR	Security sector reform. Activities designed to build capacity in the target nation's military, police and judicial structures.
Stability operations	From a US perspective, tactical activities designed to maintain or re-establish a safe or secure environment, provide essential government services, emergency infrastructure reconstruction and humanitarian relief. But other states have alternative definitions.
Strategic effect	In air power doctrine, the use of air power against designated <i>centres of gravity</i> to produce higher order political effects. In strategy, any action that has an impact on the relationship between means and ends.
Strategic performance	The degree to which an action moves one measurably towards the achievement of the desired political goals.

Strategy	The planning and application of military resources to help achieve <i>grand strategy</i> .
Stryker	A US wheeled armoured fighting vehicle introduced in 2002. One element of the <i>Future Combat System</i> .
Suppression	The neutralisation, rather than destruction, of forces through the psychological effect of firepower. Typically, 'suppressive fire' forces an enemy to remain in cover, pins them, and prevents them from firing themselves by rendering them unable (through <i>shock</i> or disorientation) or unwilling (through fear) to expose themselves.
System	A set of interrelated parts that together constitute a whole. Military organisations are made up of a multitude of systems, such as administrative systems, logistic systems and training systems.
'System of systems'	A concept in which very high degrees of <i>integration</i> between different military systems, facilitated by improvements in data transfer and <i>CAISR</i> , produce something more resembling a single <i>system</i> . In theory, this should dramatically improve the co-operation, co-ordination, speed and flexibility of a military organisation.
Tactics	The conduct of battles and engagements.
Tempo	The rhythm or rate of activity of operations, relative to the enemy.
Territorial seas	Under <i>UNCLOS</i> these are defined as areas up to 12 miles of the adjacent coast. States enjoy sovereignty within this region and the activities of outside naval forces are circumscribed, notwithstanding a right to <i>innocent passage</i> .
Terrorism	A basic definition would be 'the use of violence, or the threat of violence, to create fear'. Terrorism is a <i>strategy</i> in which violence is important for the psychological effect that it produces. However, terrorism is a contested concept, not least because it has associated moral, political and pejorative connotations and a multitude of different definitions exist.
Theatre	Large geographically defined areas of land, water and air that bound the conduct of military operations.
Transformation	A US vision of defence reform embodying a radical vision of the demands of future warfare including the need for smaller, modular, rapidly deployable and highly networked forces.
UAV	Unmanned aerial vehicle, also known as a drone.
UNCLOS	The Third UN Convention on the Law of the Sea that was signed in 1982 and came into effect in 1997.
USAF	US Air Force.
USN	US Navy.
USV	Unmanned surface vehicle (usually referring to a vessel at sea). Also known as autonomous surface vehicles (ASV).

UUV	Unmanned underwater vehicle, including remotely operated underwater vehicles (ROV) and autonomous underwater vehicles (AUV).
Vertical envelopment	Moving over the enemy in order to threaten their flanks or rear. This may be executed through air-mobile (helicopter borne) or airborne (parachute or glider) forces.
War	Organised violence conducted by political units against each other.
Warfare	The military conduct of war; it is concerned centrally with fighting.
WMD	Weapons of mass destruction. A collective term for nuclear, radiological, biological and chemical weapons focusing on their capacity for large-scale indiscriminate damage. Often also referred to as 'weapons of mass effect' because of the potential political, coercive and moral impacts associated with their use or threat of use.



INDEX

- 3D printing, 444
- 9/11 attacks, 62, 67, 282, 365
- Abdulmutallab, Umar Farouk, 362
- Abu Sayyef, 303
- Abyssinia, 393
- additive manufacture, 444
- Advanced Technology Bomber programme, 279
- aerial warfare, 130–1, 227–30
 - air interdiction, 244
 - air power
 - rapidity, 232–4
 - reach, 231–2
 - at sea, 190–1
 - coercive, 259–64
 - Cold War, 254
 - combat environment, 230–1
 - command and control, 244–6
 - control of the air, 237–9
 - flexibility, 234–7
 - height, 234
 - helicopters, 197
 - impermanence, 235–7
 - information dominance, 243
 - joint operations, 243–4
 - naval warfare, 190–1
 - stealth aircraft, 279–81
 - strategic operations, 239–41, 255–7
 - World War I, 251–2
 - World War II, 252–4
- Afghanistan, 9
 - 2001 invasion
 - success metrics, 455–6
 - counterinsurgency, 356
 - First Afghan War (1839–1842), 327
 - War (2001–), 8, 135, 255, 282, 302
 - NATO casualties, 302
- Agent Orange, 385
- Air-Independent Propulsion, 198, 210–11
- aircraft carriers, 190–1, 196
- AirLand battle doctrine, 121–2, 261, 275
- air-launched anti-satellite (ASAT) weapons, 288–9
- Air-Sea Battle, 212
- Alexander the Great, 42, 55
- Algeria, 346–7, 348, 354–6, 360
- Alibek, Kenneth, 392
- Allen, W. E. D., 309
- Allenby, Edmund, 333
- al-Qaeda, 135, 304, 362–3
 - in Iraq, 363–4
- American Civil War, 185, 314
- American Revolutionary War, 321
- amphibious operations, 172, 190, 197
- Anaconda, Operation, 282
- Ansar, 327–8
- anthrax, 381, 385, 389, 392
- anti-access/area denial, 365
- anti-ship missiles, 195
- Arab Spring, 314
- Argentina, 416
- Arleigh Burke* (USS), 217
- Aron, Raymond, 26, 43
- artificial intelligence (AI), 445
- artillery, 105, 109, 326
 - aerial spotting, 237
 - chemical weapons, 393–4
- ASDIC, 190
- Asprey, Robert, 339
- assassination, 320
- Assessing Revolutionary and Insurgent Strategies (ARIS), 352
- asymmetric warfare, 67
- Atlantic, Battle of, 214, 258
- Atlantic Charter, 345
- attrition, 94–5
- Auftragstaktik*, 97
- Auschwitz, 395
- Australia
 - navy, 170, 205
- autonomous weapons, 445, 446
- B-2 Spirit, 279
- Baghdadi, Abu Bakr al-, 359
- Bagration, Operation, 119
- Bakunin, Mikhail, 325
- ballistic missiles, 198–9
 - defence systems, 212
- Barbarossa, Operation, 118

- battle
 - Clausewitz on, 31–2
- Battle Damage Assessment, 287
- battlecruisers, 185
- battleships, 181–2
- Beaufre, André, 32
- Belgrano* (ARA), 196
- Berlin Airlift, 259
- Betts, Richard, 23, 26, 27
- Bin Laden, Osama, 282, 359, 362, 414
- biological weapons, 384–6
 - military use, 396–7
 - state retention, 399–401
 - state use, 391
 - super-, 393
 - terrorism, 386–90, 401–2
- Biological Weapons Convention, 386, 392, 401
- Black, Jeremy, 146, 150
- Black Buck, Operation, 233
- Blanqui, Auguste, 358
- Blanqui, Louis, 325
- Blister agents, 395
- blitzkrieg, 53–4, 111, 116–18
- blockade (naval), 169, 187
- Boeing B-29 Superfortress, 253
- Boer war 104–5, 328–9, 331
- Boko Haram, 358–9
- bombing, 229, 255–7
- Booth, Ken, 205
- Bosnia, 454
- Boyd, John, 262
- Brodie, Bernard, 23, 173, 406
- Browning, Peter, 25
- Bruchmüller, Georg, 109
- Bulge, Battle of the, 92
- Burma, 253
- Bush, George W., 41
- Byman, Daniel, 72
- Cable, James, 163
- Callwell, Charles, 55, 172, 326–30
- campaign, 122
- capitalism, 324–5
- Carthage, 48
- Castex, Raoul, 175–6
- Castro, Fidel, 415
- Cavour*, 196
- Central Intelligence Agency (CIA), 285
- centralisation, 96–7
- centre of gravity, 31
- CERT-UK, 69
- Chad, 147–8
- chemical weapons, 391–6
 - choking agents, 395
 - Iran–Iraq war, 394
 - military use, 396–7
 - protection against, 396–7
 - state retention, 399–401
 - World War I, 392–4
 - World War II, 394
- Chemical Weapons Convention, 386
- Cheney, Dick, 129
- Chennault, Claire L., 256
- Chile, 184
- China, 402
 - air force, 291
 - Diaoyu/Senkaku dispute, 204–5
 - future strategy, 442
 - irregular warfare, 334–6
 - military modernisation, 139–40, 148
 - navy, 207, 214
 - joint operations, 216
 - Sino-Vietnam war (1979), 147–8
 - space power, 288
 - Taiping rebellion, 136
- chlorine, 392–3
- choking agents, 395
- Churchill, Winston, 30, 336, 345
- circular error probability (CEP), 409
- civil war, 8, 303
- Clark, Wesley G., 281
- Clausewitz, Carl von, 28–32, 325–6, 368
 - definition of war, 3
 - on battle, 31–2
 - on concentration, 96
 - on friction, 52
 - on human factors, 52–3
 - on policy, 42
 - on primacy of combat, 50
 - on strategic theory, 25
 - polarity, 307
- close air support, 244
- close blockade, 169
- Cohen, Eliot, 40
- Cold Start doctrine, 140
- Cold War, 63
 - aerial power, 254
 - chemical weapons, 394–5
 - conventional arms race, 89
 - Cuban Missile Crisis, 415

- Cold War (*cont.*)
 nuclear deterrence, 414–15
 nuclear weapons, 417–22
 collateral damage, 283
 Colomb, Philip, 164–6, 169
 command and control
 aerial warfare, 244–6
 Clausewitz on, 30–1
 decentralised, 96–7
 land warfare, 89–90
 command of the sea, 164–6
 communication lines, 33
 radio, 117
 compellance, 64–5, 414
 compound wars, 136
 Congo, Democratic Republic of, 8
 Coningham, Arthur, 246
 consolidation, 95–6
 Coral Sea, Battle of the, 191, 192
 Corbett, Julian, 164, 166, 172
 Cordesman, Anthony H., 285
 Correlates of War project, 2, 8
 counter-attack, 106
 counterinsurgency (COIN), 43, 56, 284–5, 347–8
 Malaya, 56
 organizational adaptation, 350–1
 principles, 351–7
 counterterrorism, 365–8
 coups d'état, 308–11, 315
 Creveld, Martin van, 29
 Crimean War (1854–5), 185
 cruise missiles, 195
 cruisers, 186
 Cuba, 259, 347
 Cuban Missile Crisis, 415
 cyanogen chloride, 395
 cyberwar, 69–70, 445

 Darius, 339
 Daschle, Tom, 381
 Davidov, Denis, 323
 decentralisation, 96–7
 decisive battle, 166–7
 defence
 World War I land warfare, 105–8
 World War II land warfare, 118–20
 defence diplomacy, 65
 Delta II Rocket, 270
 Dempsey, Martin E., 41, 447

 Desert Shield, Operation, 232
 Desert Storm, Operation, 288
 destroyers, 182
 deterrence, 62–3
 nuclear, 413–17, 418–20
 Dien Bien Phu, 144
 directed command, 97
 directed energy weapons, 444
 dispersal, 102
 distant blockade, 169
 distributed operations, 213–14
 Douhet, Giulio, 255
 drones, 71, 211, 236, 285
 Dulles, John Foster, 419–20
 Dunkirk evacuation, 252
 Dupuy, T. N., 30

 Ebola virus, 390
 Effects-Based Organisations, 137
 Egypt, 146–7, 254
 Eikenberry, Karl, 284
Eilat (INS), 195
 Eisenhower, Dwight D., 50–1, 258, 449
 El Salvador, 215
 Ellis, John, 339
 enablers, 234–5
 Enduring Freedom, Operation, 255
 enhanced human performance, 443–4
 Enthoven, Alain, 23
 Estes, Howell M., 289–91
 ethics, 73–5
 Ethiopia, 147–8
 ethnic bioweapons, 393
 Ewald, Johann, 321–4
 Experimental Armored Force (EAF), 116
 Experimental Mechanised Force (EMF), 116
 exploitation, 95–6, 108, 118–19
 sea cover, 172

 Fabius Maximus, 48
 Falklands war (1982), 196, 233, 416
 Fat Man, 406–7
 Ferret Force, 350
Feuerwalze, 109
Flachen und Lücken, 109
 Flavius Josephus, 339
 fleet train, 204

- fleets-in-being, 168–9
 flotilla craft, 186
 Foch, Ferdinand, 102
 Fokker Eindekker, 251–2
 Fourth-Generation Warfare, 133, 304
 France
 North Africa, 10
 Algeria, 346–7, 348, 354–6
 World War I, 102–4
 World War II, 54
 Franz Ferdinand of Austria, 331
 fratricide, 244
 Freedman, Lawrence, 26, 63
 French revolution, 324
 friction, 51–2, 85, 228
Front de Libération Nationale (FLN), 346–7
Fuerzas Armadas Revolucionarias de Colombia (FARC), 363
 Fukushima Daiichi disaster (2011), 383
 Full Spectrum Operations, 137–8
 Fuller, J. F. C., 54–5, 114–16, 446–7
 Future Combat Systems, 137

 Gaddafi, Muammar, 285–6, 424
 Galliéni, Joseph-Simon, 326–7
 Galula, David, 348, 349
 Galvin, John R., 149
 Garos, Roland, 251
 GBU-43 bomb, 399
 General Dynamics F-16, 274–5
George H. W. Bush (USS), 196
 Georgia, 8
 Germany
 air force, 252
 Army, 103
 land warfare
 interwar, 115
 World War I, 90
 biological weapons, 385
 World War II, 32, 47–8
 blitzkrieg, 53–4, 116–18
 Holocaust, 395
 Kharkov offensive, 96
 Operation Barbarossa, 118
 Giap, Vo Nguyen, 346
 G-induced loss of consciousness (G-LOC), 230
 Global Financial Crisis (2007–8), 451
 Global Positioning System (GPS), 286–7
 globalisation, 132
 Goering, Hermann, 252
 Goldwater-Nichols Act, 450
 Gorshkov, Sergey, 173
 Grandmaison, Louis Loyzeau de, 102
 Gray, Colin, 40, 46, 63, 456
 dimensions of strategy, 47
 Greenert, Jonathan, 5
 Grenada, 269
 Grove, Eric, 205
 Guderian, Heinz, 54, 116
 guerillas, 323, *see also* irregular warfare
guerre de course, 175–6, 187
 Guevara, Che, 347, 358, 360
 Gulf War (1991), 53, 66–7, 129, 141, 232, 275, 278, 425

 Hagel, Chuck, 41
 Hague Convention (1899), 392–3
 Haig, Douglas, 41
 Haiti, 454
 Hamilton, Ian, 104
 Hammes, Thomas X., 133
 Handel, Michael, 446
 Hannibal, 48
 Harris, Arthur, 257
 Hawker-Siddely Harrier, 233
 Heidelberg Institute for International Conflict Research, 6
 height, 234
 helicopters, 197
 Heraro tribe, 327
 Hezbollah, 134
 Hiroshima, 406–7
 historical materialism, 324–5
 Hitler, Adolf, 393
 Ho Chi Minh, 345
 Hoffman, Bruce, 306
 Hoffman, Frank G., 136
 Holocaust, 395
 Horner, Charles, 276
 horses, 118
 Howard, Michael, 46
 human factors, 52–3
 Human Universal Load Carriers (HLUC), 443–4
 Hume, David, 13
 Hussein, Saddam, 8, 145–6, 277–8, 282, 400, 425
 hybrid warfare, 8–9, 134, 136
 hydrogen cyanide, 395

- imperial wars, 326–30
- India
 - insurgency, 346
 - joint operations, 449–50
 - military modernisation, 140
 - Naxalite insurgency, 148
 - nuclear weapons, 406, 423
 - Pakistan war (1971), 147
- Indo-China War (1946–1952), 144
- Indonesia, 347
- infantry, 87, 103
 - World War I, 109
- infiltration, 108–11
- information dominance, 243
- initiative, 95
- insurgency, 284–5, 312–13, *see also*
 - counterinsurgency; irregular warfare;
 - terrorism
 - types, 306–7
- intelligence, 191
- International Criminal Tribunal, 281
- International Stabilization Force Headquarters, 355
- inter-state wars, 8
- intra-state wars, 8, 303
- Iquique, Battle of, 184
- Iran, 417
 - chemical weapons, 394
 - navy, 215
- Iran–Iraq war (1980–88), 195, 394
- Iraq, 8, *see also* Hussein, Saddam
 - 2003 invasion, 91, 141, 278, 283
 - air force, 283
 - al-Quaida in, 363–4
 - chemical and biological weapons, 399–400
 - Gulf War (1991), 53, 66–7, 129, 141, 232, 275, 278, 425
 - Islamic State offensive (2014–), 149
- Iraq War (2003), 91, 141, 278, 283
- Ireland, 332
 - Civil War, 357
- ironclads, 181–2
- irregular warfare, 55, 284–5, 301–5
 - causes, 306–7
 - character of violence, 359–60
 - civil war, 303, 314
 - coups d'état, 308
 - definitions, 305–8
 - historical study, 319–21
 - Napoleonic Wars, 323
 - insurgency, 312–13
 - mirror-imaging, 364–5
 - partisan warfare, 321–4
 - past texts, 319–21
 - revolution, 311–12
 - terrorism, 311, 360
 - World War I, 332
 - World War II, 336–7
- Islamic State (IS), 87, 148–9, 302, 359, 364
 - beheadings policy, 361–2
- Israel
 - 2006 Hezbollah campaign, 134
 - air power, 267
 - Lebanon war (1982), 254
 - nuclear weapons, 425
 - Six Days war (1967), 254
 - targeted killing, 71, 72–3
 - Yom Kippur War (1973), 96, 146–7, 195
- Israelites, 339
- Italo-Turkish War, 241
- Italy, 393
- J-20 (aircraft), 291
- Jablonsky, David, 23
- Jäger*, 321–2
- Japan
 - air force, 252–4
 - chemical weapons, 394, 395
 - Diaoyu/Senkaku dispute, 204–5
 - Maritime Self-Defence Force, 197–8, 217
- wars
 - Russia (1904–5), 182, 185
 - WWII, 252–4, 394
- Jeune École*, 174
- Joint Direct Attack Munitions, 129
- Joint Force Air Component Commander, 246
- joint operations, 91–2, 214, 292
 - aerial warfare, 243–4, 264–6
 - future challenges, 451–2
- Jomini, Antoine Henri de, 33, 112–13, 325–6
- Junkers JU-87 Stuka, 117
- jus ad bellum, 74
- jus in bello, 74
- Just War tradition, 73, 74
- Jutland, Battle of, 187
- Kaldor, Mary, 4, 132
- Kasserine Pass, Battle of, 245
- Kautilya, 320
- Keegan, John, 29
- Kennedy, John F., 348, 350
- Khan, A. Q., 423

- Khrushchev, Nikita, 345, 415
Kitchener, Horatio Herbert, 328–9
Kober, Avi, 72
Korea, North, 406, 417, 426
Korean War (1950–3), 144, 451
 air power, 236, 266–7
Kosovo conflict (1999), 64, 66–7, 255, 278, 280, 281
Kursk, Battle of, 119
- land warfare
 attack and defence, 92–4
 centralisation, 96–7
 command and control, 89–90
 consolidation and exploitation, 95–6
 decisiveness, 88
 firepower, 89
 forces
 complexity, 86–7
 persistence, 87
 versatility, 87
 friction, 85
 joint operations, 91–2
 logistics, 90–1
 manoeuvre, 94–5
 political environment, 85
 post-nuclear, 419
 scale, 88–9
 terrain
 opacity, 86
 variety, 85
World War I
 defence in depth, 105–8
 infiltration, 108–11
World War II, 111–13
 defence in depth, 118–20
- Lawrence, T. E., 332–4
Lebanon War, 134
Lenin, 334
Lettow-Vorbeck, Paul von, 331–2
Levy, Bert, 336
Libya, 147–8, 367, 424
 2011 intervention, 203
Liddell Hart, Basil H., 116
Linebacker, Operation, 261
Lissa, Battle of, 183
Little Boy, 406–7
littoral region, 160, 162
Locke, John, 321
Lockheed F-105D Thunderchief, 260
Lockheed F-117 Nighthawk, 279
Lockheed Martin F-35A Lightning II, 7
Lockheed SR-71 Blackbird, 234
Lockheed U-2, 234
logistics
 blitzkrieg, 118
 land warfare, 90–1
London Treaties (1930, 1936), 189
Love Bug virus, 69
Lupton, David E., 267
Luttwak, Edward, 34
 on friction, 52
 on Pearl Harbour attack, 47
- MacArthur, Douglas, 453
Mahan, Alfred, 161, 164, 165
Malaya, 56, 352–3
Malaya Scouts, 350
Malaysian Armed Forces Staff College, 5
Maltzhan, Curt von, 167
Malvinas conflict (1982), 196, 233, 416
Manhattan Project, 406
manoeuvre, 94–5
manoeuvre warfare, 123
Mao Tse-Tung, 34, 334–6, 360
Marighella, Carlos, 347
maritime operating environment, 160–2
Maritime Security Operations (MSO), 206
Marx, Karl, 324–5
massive retaliation, 419–20
Materialschlacht, 105–6
McAfee, 69
McChrystal, Stanley, 348–9, 353
McDonnell Douglas F-15 Eagle, 290
McDonnell Douglas F/A-18, 274–5
McNamara, Robert S., 421–2
Metz, Steven, 56
 on asymmetric warfare, 67
Midway, Battle of, 192, 252–3
Mikasa (HIJMS), 181
Military Assistance to Civil Authorities (MACA), 65–6
military doctrine
 counterinsurgency (COIN), 348–50
 irregular warfare, 322
 United States, 92, 93, 140–1, 322
 Western culture and, 149–51
militias, 321
Milosevic, Slobodan, 280, 281
mines
 naval, 182, 210
Minutemen, 321

- mirror-imaging, 364–5, 427
- Mitchell, William ‘Billy’, 239, 256–7
- Mons, Battle of, 104–5
- Montgomery, Bernard, 246
- Moore’s Law, 443
- Munkler, Herfried, 132
- Murray, Williamson, 28
- Musharref, Pervez, 308–11
- muskets, 326
- mustard gas, 381
- Mutually Assured Destruction (MAD), 63, 418–20
- Nagasaki, 406–7
- Naji, Abu Bakr, 364–5
- nanotechnology, 443, 446
- Napoleon Bonaparte, 30, 326
- Napoleonic Wars, 323
- nationalism, 321
- NATO, 10
 - comprehensive crisis approach, 10
 - Kosovo conflict, 280–1
 - Libya intervention, 424
- naval diplomacy, 163
- naval warfare
 - aircraft, 190–1
 - amphibious operations, 172, 190, 197
 - anti-access/area-denial, 210–12
 - attributes of forces, 162–4
 - blockade, 169
 - contemporary policy, 214–18
 - convoys, 188
 - cover, 170
 - early 20th century, 185–6
 - fleets-in-being, 168–9
 - interwar and WWII, 189–94
 - network-centric, 213–14
 - operating environment, 160–2
 - post-modern, 207–9
 - post-WWII, 194–9
 - power projection, 189
 - strategy
 - Anglo-American, 164–73
 - command of the sea, 164–6
 - other approaches, 173–5
 - World War I, 186–9
- Navarino, Battle of (1827), 183
- Network-Centric Warfare, 131, 135–6
 - naval, 213–14
- New Wars model, 132–448
- Nixon, Richard, 348, 391
- Normandy campaign (WWII), 236
- North Korea, 406, 417, 426
- Norway, 150
- Novichok* agents, 396
- Nuclear Non-Proliferation Treaty (NPT), 405–6
- nuclear powered submarines, 198, 210–11
- nuclear weapons, 24, 121, 194, 406–11
 - aircraft delivered, 409–10
 - ballistic weapons, 409
 - Cold War, 417–22
 - deterrence, 418–20
 - future conflicts, 425–8
 - post-Cold War, 423–5
 - post-nuclear environments, 419
 - strategic, 409
 - submarines, 409–10
 - tactical, 408–9
 - terrorism, 412–13
 - yield, 408
- Obama, Barack, 41, 69
- Ocean Shield (operation), 207
- offence, 65
- Office for Force Transformation, 137
- Ogaden War (1977–8), 147–8
- O’Neill, Bard, 307
- OODA loops, 122–3, 262–3
- operational analysis, 191
- operational level, 122–3, 448–9
 - operational art, 447–8
- Owens, William, 129
- PackBot, 445
- Paget, Julian, 56
- Pakistan, 147, 315
 - nuclear weapons, 406, 412–13, 423
- Palestine, 346
- Panama, 269
- Panavia Tornado, 277
- Panzer divisions, 54
- Pape, Robert A., 32, 259
- paradigm army, 137–42, 149
- parallel air operations, 276
- partisan warfare, 321–4
- Patton, George S., 120
- Paveway bombs, 262–3
- peace dividend, 275
- peacekeeping, 453
- Pearl Harbor attack, 47, 191
- Persian empire, 339

- Peru, 184
- Petreaus, David, 353
- Peuty, Paul du, 238
- Philippines, 303, 330, 331, 352
- phosgene, 395
- Pinker, Steven, 441
- piracy, 207
- polarity, 307
- policy, 40–1, 42–4
- posturing, 65
- Powell, Colin, 280
- power projection, 189
- power systems, 444
- precision-guided munitions (PGM), 262–3, 273–4, 278–82
- private security, 207
- Punic War, Second, 48
- Pushtunwali, 302–4
- Putin, Vladimir, 417
- Q Fever, 386
- Qaddafi, Muammar, 285–6, 424
- Qutb, Sayyid, 358, 359
- radar, 190
countermeasures, 279
- radio communication, 117
naval, 186
- radiological weapons, 382–4
- rail transport, 90
- Ramdane, Abane, 361
- Rapid Dominance, 283
- Rapoport, Anatol, 4
- Reagan, Ronald, 422
- regional combatant commands (RCC), 450
- Remotely Piloted Air Systems (RPAS), 236, 241, 285, *see also* robotic systems
- revolution, 311–12, 314, 324–6
- Revolution in Attitudes to Military (RAM), 150
- Revolution in Military Affairs (RMA), 53, 66–7, 129–32
as airpower superiority, 130–1, 135
as information dominance, 131–2, 135–6
- Rhodes, Cecil, 328
- Richmond, Herbert, 172–3
- ricin, 389
- rifles, 326
- robotic systems, 444
- Rockwell B-1B, 240
- Rogers, Robert, 321
- Roman Republic, 48
- Rome Labs incident, 69
- Ropp, Theodore, 6
- Roskill, Stephen, 173
- Royal Air Force (RAF)
Falklands war, 233
strategic bombing in WWII, 256
- Rumsfeld, Donald, 67, 137, 141
- Russia, *see also* Soviet Union
military modernisation, 140, 148
wars
Georgia (2008), 8
Japan (1904–5), 182, 185
Ukraine (2014), 304
- Salyut-3 station, 289
- Samarra, 363–4
- Samuel B. Roberts* (USS), 195
- satellites, 268–9, 286–7
- Saudi Arabia, 232
- Schelling, Thomas, 64
on compellance, 64
- Schlieffen, Alfred von, 113
- Schwartzkopf, Norman, 87, 278
- sea basing, 204
- sea denial, 188
- sea mines, 182, 210
- Second Intifada, 72–3
- Seeckt, Hans von, 116
- Serbia
Kosovo conflict, 64
- Sheehan, Michael, 5–6
- Shekau, Abubakar, 358–9
- Shineski, Eric, 141
- shock and awe, 283
- Simpkin, Richard, 95
- singularity, 444–5
- Sinope, Battle of, 183
- Slessor, John, 266
- smallpox, 387, 396
- Smith, Rupert, 132
- Solomon Islands Campaign, 214
- Somalia, 147–8, 454
- Somme offensive, 105
- sonar, 190
- Soult, Nicholas, 323
- South China Seas, 204–5
- South Korea, 217
- Soviet New School, 174–5

- Soviet Union, *see also* Georgia; Russia; Ukraine
 biological weapons, 392
 chemical weapons, 394–5, 396
 Cold War, 63
 military doctrine, 121
 navy, 194–5
 space power, 268–9
 World War II
 land warfare, 114
- Spaatz, Carl, 258
- space power, 267–8
 force application, 288–9
 post-Cold War, 286–7
- spaghetti chart, 355
- Special Operations Forces (SOF), 365–8
 as force of choice, 367
- SPOT, 288
- Sputnik I, 268
- stealth aircraft, 279–81, 291
- strategic air operations, 239–41
- strategic bombing, 241, 257–9
- Strategic Defense Initiative (SDI), 422, 426–7
- strategic ethics, 75
- Strategic Studies, 22–7
- strategic success, 455
- Strategic Theory, 27–8
 Clausewitz, 28–32
 Jomini, 33
 modern, 34–5
 Sun Tzu, 33–4
- strategy
 definition, 40–2
 dimensions, 46
 levels
 grand strategy, 44
 harmony between, 46–8
 operational, 44–5
 policy, 42–4
 tactical, 44–5
 naval warfare, 164–73
 study of, 22–7
- submarines, 171, 182, 187–8, 198–9, 210–11
 nuclear weapons, 409–10
- suicide terrorism, 320
- Summers, Harry, 357
- Sun Tzu, 33–4, 64, 320
- Sunni Awakening, 364
- Suppression of Enemy Air Defences (SEAD), 279
- Surface-to-Air missiles (SAM), 234, 254, 276–8, 291–2
- Suri, Abu Mus'ab al-, 364
- Svechin, A.A., 114
- Sweden, 215
- SWORD, 445
- Syria, 364
 civil war (2011–), 133–49
 Yom Kippur war, 96, 146–7
- systems, 131–2
- Taashi, Abdullah al-, 327–8
- tactical nuclear weapons (TNW), 408–9
- tactics, 44–5
- Taiping rebellion, 136
- Taliban, 8, 307
- Tanker War, 195
- tanks, 54, 110, 150
- targeted killing, 70–3
- Terraine, John, 41
- terrorism, 306, 311, 360
 biological weapons, 386–90, 401–2
 nuclear weapons, 412–13
- Tet offensive, 44–5
- Theodore Roosevelt* (USS), 274
- thermobaric bombs, 399
- Thirty Years War, 136
- Thompson, Robert, 56, 348
- torpedoes, 182
- Torrington, Arthur Herbert, 1st Earl, 169
- trade defence, 172
- Trafalgar, Battle of, 161, 166–7
- Trenchard, Hugh, 238, 241–2, 256
- Trinity (Clausewitz), 29–30
- Trinquier, Roger, 349
- Trojan Horse, 320
- Trothat, Lothar von, 327
- Tsar Bomba, 408
- Tsushima, Battle of, 185
- Turner, Stansfield, 173
- Ukraine, 7, 304
- United Kingdom
 Air Force, 244–5, 256
 Army
 Experimental Mechanised Force, 116
 post-WWI, 68
 counterinsurgency
 Malaya, 56
 cyberwarfare, 69
 future conflicts, 440
 integrated warfare, 10
 joint operations, 452

- Navy, 193
- wars
 - Boer (1899–1902), 104–5, 328–9
 - Burma (1824–6), 184–5
 - First Afghan (1839–1842), 327
 - First Opium War (1838–42), 184–5
 - Malaya, 352–3
- United States
 - Air Force, 68, 245–6, 256–7
 - stealth aircraft research, 279
 - Army
 - as paradigm army, 137–42
 - pentonic divisions, 421
 - chemical weapons, 394–5
 - conflicts. Gulf (1991) *see* Gulf War,
 - Afghanistan: Afghanistan, Cold War:
 - Cold War, Iraq (2003): Iraq War
 - (2003), Vietnam: Vietnam War, World War II: World War II
 - Civil (1861–5), 185, 314
 - Kosovo conflict (1999), 64, 66–7
 - Mutually Assured Destruction, 63
 - Philippines (1902), 330
 - War on Terror, 67
- Marine Corps
 - Small Wars Manual*, 350
 - Warfighting*, 28, 52, 123
- military doctrine
 - Army Strategic Planning Guidance*, 140–1
 - counterinsurgency, 322
 - Principles of War, 92, 93
- National Security Strategy, 62
- Navy
 - contemporary policy, 215–16
 - core capabilities, 163–4
 - future development, 212
 - Maritime Partnerships, 206–7
- space power, 268–9
- targeted killing, 71
- Uppsala Conflict Database Project, 2
- vaccines, 396
- Verdun, Battle of, 120
- victory, 452–3
 - strategic success, 455
- Victory* (HMS), 181
- Viet Cong, 44–5
- Vietnam
 - declaration of independence, 345
 - navy, 217
 - Sino-Vietnam war (1979), 147–8
 - Vietnam war, 44–5, 254
 - aerial bombing, 259
 - insurgent strategies, 346, 354
 - satellite use, 268–9
 - targeted killing, 70–1
 - Vietnamisation policy, 357
- Vincennes* (USS), 195
- violence, 50
- Walzer, Michael, 74
- war
 - definitions, 2
 - legal state of, 2–3
 - nature of, 49–51
 - polymorphous character, 53–6
- War Convention, 74
- War on Terror, 67
- Warden, John A, 262, 275, 283
- warfare
 - definitions
 - US joint doctrine, 5
 - Washington Treaty (1922), 189
- Watts, Barry D., 52
- weaponry, 89
- Weapons of Mass Destruction (WMD), 55, 68, 379–82
 - biological, 384–6
 - radiological, 382–4
- Wegener, Edward, 173
- Wegener, Wolfgang, 167
- Wingate, Orde, 309
- Wintringham, Tom, 336
- World War I, 88–9
 - aerial warfare, 251–2
 - chemical weapons, 392–4
 - irregular warfare, 332
 - land engagements
 - Mons, 104–5
 - Verdun, 120
 - land warfare, 102–5
 - logistics, 90–1
 - naval warfare, 186–9
- World War II
 - aerial warfare, 252–4
 - chemical weapons, 394
 - Germany
 - war economy, 32
 - Japan
 - Pearl Harbour attack, 47
 - land engagements
 - Battle of the Bulge, 92

- World War II (*cont.*)
 - Kasserine Pass, 245
 - Kursk, 119
 - Normandy campaign (1944), 236
 - Operation Barbarossa, 118
- naval engagements
 - Atlantic, 214, 258
 - Coral Sea, 191, 192
 - Midway, 192, 252–3
- Normandy campaign, 236
- strategic bombing, 257–8
- Wylie, J. C., 43–4, 173
- Yalu, Battle of (1894), 183
- Year of Revolutions, 324
- Yehudi lights, 279
- yellow fever, 396
- Yom Kippur War (1973), 96, 146–7, 195
- Yugoslavia (former) conflicts, 132
- Zarqawi, Abu Musab al-, 363
- Zawahiri, Ayman al-, 359, 362
- zone of death, 102
- Zyklon B, 395